TRAUMA — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 68 chapters (1 from TO notes) –

#	Chapter	Section index
1	Acetabular Fractures 4	1 / 67
2	Adult Respiratory Distress Syndrome 21	2 / 67
3	Amputations 24	3 / 67
4	Ankle Fractures 36	4 / 67
5	Brachial Plexus Injuries 54	5 / 67
6	Calcaneus Fractures 63	6 / 67
7	Capitellum Fractures 76	7 / 67
8	Clavicle Fractures - Distal 81	8 / 67
9	Clavicle Fractures - Midshaft 86	9 / 67
10	Coronoid Fractures 94	10 / 67
11	Distal Femur Fractures 99	11 / 67
12	Distal Humerus Fractures 107	12 / 67
13	Distal Radial Ulnar Joint (DRUJ) Injuries 116	13 / 67
14	Distal Radius Fractures 121	14 / 67
15	Distal Radius Malunion 132	15 / 67
16	Domestic and Elder Abuse 137	16 / 67
17	Elbow Dislocation 141	17 / 67
18	Evaluation, Resuscitation & DCO 147	18 / 67
19	Femoral Head Fractures 161	19 / 67
20	Femoral Neck Fracture Nonunion 167	20 / 67
21	Femoral Neck Fractures 171	21 / 67
22	Femoral Shaft Fractures 183	22 / 67
23	Flail Chest 194	23 / 67
24	Foot Compartment Syndrome 198	24 / 67
25	Galeazzi Fractures 202	25 / 67
26	Geriatric Patient Trauma 206	26 / 67
27	Gun Shot Wounds 209	27 / 67
28	Gustilo Classification 215	28 / 67

#	Chapter	Section index
29	Hand & Forearm Compartment Syndrome 217	29 / 67
30	Hip Dislocation 222	30 / 67
31	Humeral Shaft Fractures 229	31 / 67
32	Humeral Shaft Nonunion 240	32 / 67
33	Intertrochanteric Fractures 245	33 / 67
34	Isolated Ulnar Shaft Fracture 255	34 / 67
35	Knee Dislocation 259	35 / 67
36	Leg Compartment Syndrome 269	36 / 67
37	Monteggia Fractures 277	37 / 67
38	Morel-Lavallee Lesion 281	38 / 67
39	Necrotizing Fasciitis 285	39 / 67
40	Neonatal Forearm Compartment Syndrome 290	40 / 67
41	Olecranon Fractures 293	41 / 67
42	Open Fractures Management 298	42 / 67
43	Osteomyelitis - Adult 306	43 / 67
44	Other Trauma Topics 313	44 / 67
45	Patella Fracture 315	45 / 67
46	Pelvic Ring Fractures 321	46 / 67
47	Proximal Humerus Fracture Nonunion and Malunion 334	47 / 67
48	Proximal Humerus Fractures 338	48 / 67
49	Proximal Third Tibia Fracture 348	49 / 67
50	Radial Head Fractures 353	50 / 67
51	Radius and Ulnar Shaft Fractures 360	51 / 67
52	Sacral Fractures 367	52 / 67
53	Scapula Fractures 372	53 / 67
54	Scapulothoracic Dissociation 379	54 / 67
55	Septic Arthritis - Adult 382	55 / 67
56	SI Dislocation & Crescent Fractures 390	56 / 67
57	Sternoclavicular Dislocation 393	57 / 67
58	Subtalar Dislocations 398	58 / 67
59	Subtrochanteric Fractures 403	59 / 67
60	Talar Neck Fractures 410	60 / 67
61	Talus Fracture (other than neck) 418	61 / 67
62	Terrible Triad Injury of Elbow 423	62 / 67
63	Tibial Plafond Fractures 430	63 / 67
64	Tibial Plateau Fractures 438	64 / 67

#	Chapter	Section index
65	Tibial Shaft Fractures 451	65 / 67
66	Trauma Scoring Systems 463	66 / 67
67	Wound & Hardware Infection 468	67 / 67
68	Ilizarov Principles 472	TO notes

Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO — verify" = TO figure that differs from standard teaching

– – Active space – –

ACETABULAR FRACTURES

– Trauma · 1 / 67 –

1 Basics

Epidemiology

Incidence	**~4 / 100,000 / yr**
Bimodal	**young + high energy** (MVC) · **elderly + low energy** (fall from standing)
Elderly volume	~4,000/yr in USA → may become the commonest age group

Age group	Commonest patterns
Young	**Posterior wall** · transverse "family" (transverse, T-type, transverse + PW) · associated both column
Elderly	**Anterior wall** · **anterior column (quadrilateral plate)** · **anterior column posterior hemitransverse (ACPHT)** · associated both column

Elderly low-energy fall onto greater trochanter	**anterior column + "medial wall"** (quadrilateral plate) ± intrapelvic head dislocation ✗ Wrong options: PC + PW · transverse · T-type · T + PW → young, high energy

Associated injuries

Associated injury	Rate
Lower extremity injury	**36%** (commonest)
Head injury	19%
Chest injury	18%
Nerve palsy	**13%** (3–33% in OB)
Abdominal	8%
Genitourinary	6%
Spine	4%

Nerve palsy pattern	**sciatic, peroneal division → foot drop**; highest with **transverse + posterior wall**
Nerve-injury share by pattern (OB)	PW 9.5–23% · ABC 15.6% · PC + PW 12.5–23% · isolated PC rarely
Root most often involved	**L5** (→ tibialis anterior, EHL, EDB) ✗ Wrong options: L4, S1, S2, S3
Isolated peroneal-division injury	commoner than complete sciatic injury

Posterior hip dislocation + deceleration think **traumatic thoracic aortic rupture**; CXR may be normal → chest pain + hypotension + new leg weakness (cord ischaemia) → **CT angiography of chest**
✗ Wrong options: CT head · CT pelvis · CT L-spine · leg angiogram

Anatomy – column theory

Columns **inverted Y**, linked to sacrum through the **sciatic buttress**

	Anterior column	Posterior column
Parts	Anterior ilium (gluteus medius tubercle), **anterior wall + dome**, iliopectineal eminence, lateral superior pubic ramus	**Quadrilateral surface**, **posterior wall + dome**, **ischial tuberosity**, greater/ lesser sciatic notches
Radiographic line	**Iliopectineal line**	**Ilioischial line**
Best Judet	**Obturator oblique**	**Iliac oblique**

TO anatomy (TO) anterior column = anterior ½ iliac crest, iliac spines, anterior ½ acetabulum, pubis · posterior column = ischium, ischial spine, posterior ½ acetabulum, dense sciatic-notch bone

Acetabular orientation lateral inclination **40–48°** · anteversion **18–21°**

Quadrilateral plate (TO) thin medial wall; involved in **ABC + ACPHT**; displacement → **medial head migration**

Iliopectineal eminence (TO) intra-operative landmark of **anterior column**

Foramen (TO)	Contents
Greater sciatic	**Piriformis** · **superior gluteal NV above piriformis** · **sciatic nerve below piriformis** · internal pudendal vessels + nerve
Lesser sciatic	**Obturator internus tendon** · internal pudendal vessels

Superior gluteal nerve (TO) high risk with **superior posterior** acetabular approaches

Corona mortis & vascular anatomy

Corona mortis ("crown of death") anastomosis **external iliac (inferior epigastric) ↔ obturator** (internal iliac system) behind **superior pubic ramus**

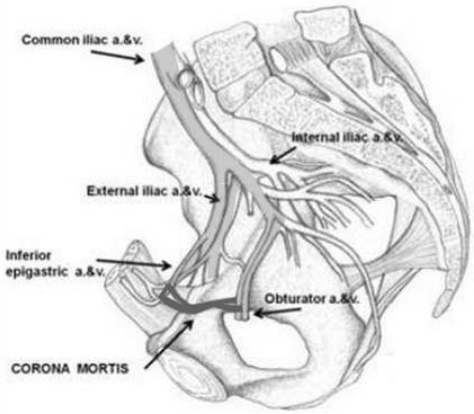
Prevalence (Tornetta) **84%** have an anastomosis · **arterial 34%**, **venous 70%**, both 20%

Distance from symphysis **mean 6.2 cm (range 3–9 cm)** → "brisk bleeding ~5 cm from midline along superior ramus"

At risk **ilioinguinal** (laceration **10–15%**) + **modified Stoppa** (must be **exposed + ligated**); retracts into pelvis → hard to control

✗ Vessel traps superior gluteal + pudendal · internal iliac + MFCA · external iliac + deep femoral · femoral branch

– – Active space – –



Corona mortis: inferior epigastric–obturator anastomosis

2 Mechanism

Pattern determined by	**force vector** + **femoral head position** at impact + **bone quality** (age)
Dashboard / posteriorly directed force	posterior wall ± posterior dislocation
Dashboard injury (TO)	**posterior wall + posterior column**; check **knee** (associated dashboard injury)
Axial load on abducted hip (TO)	**anterior column** fracture
Crescent fracture (posterior iliac wing on CT)	**lateral compression** (laterally directed force on anterior iliac wing) ✗ Wrong options: APC · VS · combined

3 Presentation

Foot drop / ↓ dorsiflexion after acetabular fracture	**peroneal division of sciatic** → most likely pattern **transverse + posterior wall**
Hip flexed, adducted, IR	posterior dislocation ± posterior wall
Sciatic nerve exam (TO)	**EHL**, ankle dorsiflexion, **first web-space** sensation
Narath sign (TO)	femoral pulse not palpable (head displaced posteriorly) → part of acetabular / dislocation exam
Os acetabuli marginalis superior	accessory ossification centre at superior acetabular rim in adolescents; **mimics fracture / AVN**; usually **fuses by age 20** ✗ Wrong options: fovea capitis · myositis ossificans · AVN · fracture

4 Diagnosis & Classification

Key radiology findings

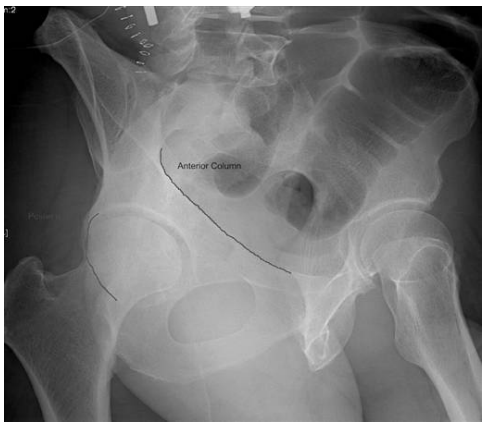
Study	Shows / use
AP pelvis	Screening; 6 landmarks: **iliopectineal line (AC)**, **ilioischial line (PC)**, anterior wall, posterior wall, **teardrop**, **weight-bearing roof** (+ Shenton's line)
Obturator oblique (Judet)	Obturator foramen in profile → **anterior column + posterior wall** (**"OOPWAC"**, TO)
Iliac oblique (Judet)	Iliac wing in profile, obturator foramen perpendicular → **posterior column + anterior wall**
Inlet / outlet	Only if pelvic ring involvement suspected
CT	**Gold standard**: pattern orientation, fragment size + position, **marginal impaction**, **loose bodies** (post-reduction), gap / step-off, surgical planning; 3D recon for planning
CT not for	**Pre-existing degenerative change** (EXCEPT answer)
Post-op CT	**Most accurate for PW reduction → best correlation with outcome**
EUA (dynamic fluoroscopy)	**Best test of posterior wall stability**
Duplex US	Delayed presentation → rule out **DVT**

Judet = 45° obliques — ipsilateral **iliac oblique = contralateral obturator oblique** (same image)

"Best view of LEFT posterior column?" — **left iliac oblique** (= right obturator oblique)
✗ Wrong options: left obturator oblique · right iliac oblique

Judet traps — PW on **obturator oblique** · OO does **not** show SI joint well
✗ Wrong options: iliac oblique · inlet · outlet · AP

Interobserver reliability (Beaulé) — Letournel substantial (κ > 0.70) **with or without CT** → CT not essential for classification, but adds **impaction + loose bodies**



Obturator oblique Judet view: anterior column + posterior wall

– – Active space – –

Sign / measure	Meaning	View
Spur sign	**Intact ilium** (most caudal part of intact ilium / posteroinferior ilium) exposed as acetabulum medialises → **pathognomonic for associated both column**	**Obturator oblique** (also AP)
Gull (seagull) sign	**Superomedial dome impaction → poor prognosis** (Orthobullets)	**Iliac oblique** (Orthobullets)
Gull sign – TO wording (TO)	**Anteromedial dome impaction**; "pathognomonic for **posterior column** fractures" → needs surgery	**Obturator oblique** (TO)
Roof arc angle (Matta)	Vertical line through head centre vs line to fracture; **stable if > 45° on AP + both Judets** (fracture exits outside weight-bearing dome)	AP + Judets
Roof arc – per-view cut-offs (TO)	**Medial > 30°** (AP) · **anterior > 40°** (obturator oblique) · **posterior > 50°** (iliac oblique); revised biomechanical values are higher	AP + Judets
CT subchondral arc (Olson & Matta)	**Superior 10 mm** of acetabulum intact = equivalent to **45° roof arc** (TO)	CT
Articular incongruity	**> 2 mm** = unstable	CT
Medial clear space widening on EUA	**Posterior wall instability**	Obturator oblique under stress
Double beak sign (TO)	Two parallel shadows: intact posterior column + elevated anterior wall → **anterior wall fracture**	AP / Judets
Double shadow of posterior wall (TO)	**Large displaced PW fragment**	AP / OO

Gull sign view	**iliac oblique** (Orthobullets) · **obturator oblique** (TO) – read the options
Gull sign – second TO lecture (TO)	**superomedial dome impaction**, seen on **iliac oblique**, suggests **posterior wall** involvement → elevate + graft
AP wall shadows (TO)	**anterior wall = concave** shadow · **posterior wall = straight** shadow · **teardrop** = quadrilateral plate + obturator canal · **sourcil** = weight-bearing dome
Axial CT orientation (TO)	column = **horizontal** · transverse = **vertical** · **wall = oblique**
Column fracture (TO)	**always crosses the quadrilateral plate** and usually the inferior pubic ramus · anterior vs posterior direction decides the column (Judet views)

Congruency – incongruency signs

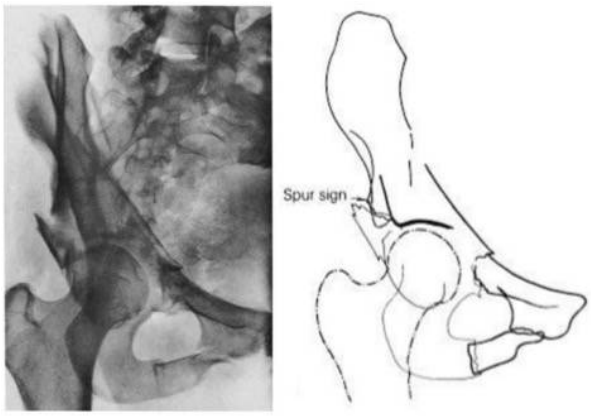
Order of assessment (TO)	**congruency first → stability → CT planning**; congruent + stable = only nonop candidate
Incongruency signs (TO)	**↑ medial joint space** · **superior joint-space widening** · **non-parallel joint surfaces** · **hinged abduction** · **loose intra-articular fragments** · persistent subluxation
↑ Medial joint space after reduction (TO)	**incarcerated fragment** (capsule or flipped PW fragment) → surgery

Four CT questions (TO)

Question (TO)	Implication
How close to the dome?	**Posterosuperior** dome involvement = highly unstable → **trochanteric flip** (abductors block view)
Comminuted wall / column?	**Spring plates** or **Allis T plate** as buttress
Intra-articular fragments?	Tiny **infracotyloid / foveal** → may be left · **superior dome fragments removed even if 3–4 mm**
Marginal impaction?	**Elevate** → graft (often from **greater trochanter**) → **mini-fragment screws**

Roof arc NOT applicable — **posterior wall** + **associated both column** (no intact acetabulum to measure)

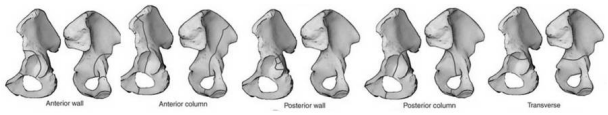
Spur sign absent — **nondisplaced** ABC → use CT to confirm acetabulum–axial skeleton continuity



Spur sign on obturator oblique (associated both column)

Letournel & Judet classification

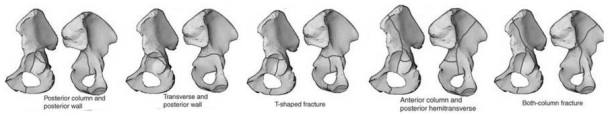
Structure — **5 elementary + 5 associated** = 10 types



Letournel elementary patterns

– – Active space – –

Elementary	Lines / views	Key fact
Posterior wall	PW outline broken on **OO**	**Commonest overall**; check **superior gluteal NV bundle**; only ~30% single large fragment
Posterior column	**Ilioischial line only** broken on AP; IO shows break, **anterior wall intact**; OO iliopectineal intact	Superior gluteal NV bundle (may be **entrapped** in fracture, TO); isolated PC rarely injures nerve; head migrates **medially + posteriorly** (TO)
Anterior wall	OO: **trapezoid fragment** · IO: **posterior column intact** (TO)	**Very rare**; **geriatric**; fragment = anterior acetabulum + quadrilateral plate (TO)
Anterior column	**Iliopectineal line broken on OO** with **undisturbed posterior acetabulum** · IO: iliac wing extension (TO)	3–5% of fractures; may extend to quadrilateral plate without exiting PC; subtypes by exit: **low · intermediate · high** (TO)
Transverse	**Iliopectineal + ilioischial** both broken; single line crossing acetabulum; posterior displacement > anterior (TO)	**Only elementary to involve both columns**; axial CT: **vertical (sagittal) A→P line**; articular surface still attached to intact ilium; subtypes by level vs dome: **infratectal · juxtatectal · transtectal** (TO)



Letournel associated patterns

Associated	Key fact
Associated both column (ABC)	**All articular segments detached from intact ilium** (which stays on sacrum via SI joint) → **no articular surface attached to axial skeleton** · **spur sign** · may show **secondary congruence**
Transverse + posterior wall	**Commonest associated** (Orthobullets) · ~**20%** of acetabular fractures (TO) · **highest nerve injury** · frequently with dislocation · high DJD risk
T-type	Transverse + vertical split into **obturator ring** · high DJD risk
ACPHT	**Commonest in elderly** · femoral head **follows the anterior segment** (TO)
PC + PW	**Only associated type NOT involving both columns** · wall fragment moves with femoral head (TO)

Commonest associated pattern	**transverse + PW** (Orthobullets) · **associated both column** (TO) – read the options
Commonest associated – second TO lecture (TO)	**ACPHT** called "most common associated"; T + PW "common and unstable" – read the options
T + PW reduction order (TO)	**transverse first**, then posterior wall
Column vs transverse on axial CT	**transverse = vertical (sagittal) line** · **column = horizontal (coronal) line**
"Only one column involved?"	**PC + PW** ✗ Wrong options: ABC · transverse · T + PW
Line-pattern traps	both lines + **iliac wing + obturator ring** = ABC or ACPHT · both lines + **obturator ring** = T-type · iliopectineal only = AC · ilioischial only = PC

Distinguishing ABC from ACPHT / T / T + PW — those keep **part of articular surface on intact ilium**

Posterior wall size & stability

PW involvement	Stability	Action
< 20%	Generally stable	Protected WB – but **EUA if fracture exits superiorly (cranial exit near dome)**
20–40%	**Indeterminate**	**EUA**
> 40–50%	**Unstable**	**ORIF**

Fragment-size methods (less accurate than EUA) — **Keith** (depth vs contralateral **at fovea**) · **Moed** (depth vs contralateral **at greatest involvement**) · **Calkins** (posterior **arc length** at greatest involvement)

EUA technique — hip **flexion + adduction + axial load**, **obturator oblique** view → **↑ medial clear space** = unstable

EUA – TO version (TO) — flexion + adduction + **internal rotation** + axial (telescoping) load, obturator oblique → **loss of parallelism** = unstable; called **gold standard** for stability

PW size cut-offs — 20–40% indeterminate · > 40–50% unstable (Orthobullets) · **20–50% indeterminate · > 50% unstable** (TO) – read the options

History of dislocation — **not** predictive of instability (Grimshaw & Moed: stable on stress exam → nonop fine)

Not instability predictors — fragment size alone, dislocation history, PW comminution, roof arc, MRI labral tear, iliac oblique

5 Management

Acetabular fracture

- Associated hip dislocation → urgent closed reduction (< 12 h) → **CT**
  - incarcerated fragment → traction pin → **ORIF ASAP**
- Stable / minimally displaced / high-risk patient → **protected WB 6–8 wk**
- Displaced roof > 2 mm, unstable, impaction, loose body → **ORIF < 3 wk**
  - elderly + impaction / comminution / head injury → **ORIF + acute THA**

Posterior wall fracture (post-reduction CT)

- < 20%, exits low → **protected WB**
- < 20% exiting near dome, or 20–40% → **EUA (obturator oblique)**
  - unstable → **ORIF**
- > 40–50% or intra-articular fragment → **ORIF**

– – Active space – –

Nonoperative criteria

Patient factors	Fracture factors
High operative risk (elderly, **DVT**)	**Minimally displaced < 2 mm**
Morbid obesity	**< 20% posterior wall** (EUA best test)
Open contaminated wound	**Head congruent with roof out of traction**
Late presentation > 3 wk	**ABC with secondary congruence**
	Roof arcs > 45° on AP + Judets

Nonop regimen — protected WB **6–8 wk**; **skeletal traction NOT needed** if stable outside dome; **DVT prophylaxis**; serial X-rays

Elderly ABC with secondary congruence — **nonop**
✗ Wrong options: ORIF + THA

Operative criteria

Patient factors	Fracture factors
< 3 wk from injury	**Roof displacement > 2 mm**
Physiologically stable	**Unstable pattern (PW > 40–50%)**
Adequate soft tissue	**Marginal impaction**
No local infection	**Intra-articular loose bodies**
Pregnancy is NOT a contraindication	**Irreducible fracture-dislocation**

Absolute surgical indication after dislocation — **osseous loose body in weight-bearing joint**
✗ Wrong options: sciatic palsy · labral tear · PW fracture per se · chondral defect

Fracture-dislocation – reduction, traction & ORIF

Posterior dislocation + small PW fracture — **reduce hip now, test stability**
✗ Wrong options: CT first · MRI · open reduction · transfer before reduction

After reduction — **CT mandatory** even if no fracture on X-ray → incarcerated osteochondral fragments · sciatic palsy ~10%
✗ Wrong options: exploration of nerve · EMG · traction

Reduction timing — **< 12 h** (Orthobullets; > 12 h delay = poor result, OB) · **< 6 h** to minimise AVN (one OB explanation) – read the options

Incarcerated fragment after reduction — **distal femoral traction pin + ORIF ASAP** (offloads cartilage)
✗ Wrong options: knee immobiliser + delayed ORIF · nonop

Incarcerated fragment + shock / unresuscitated + knee laxity — **distal femoral** pin
✗ Wrong options: proximal tibial pin with knee ligament injury · emergent ORIF · binder · EUA with intra-articular fragment

Marginal impaction on CT — must be **elevated + bone grafted + buttress plated**; otherwise incongruent joint → poor result

Loose fragments — remove, but **incorporate** large articular fragments into fixation

Pregnancy (12 wk, normal fetal HR, unstable hip)	**acute ORIF** ✗ Wrong options: wait till 15 wk · traction 6–8 wk
Fetal mortality ↑ with	**injury severity, mechanism of injury, maternal haemorrhage** ✗ Wrong options: trimester · approach · classification · fetal position

Timing of surgery

Worse outcome if fixed	**> 3 wk** from injury ✗ Wrong options: 48 h · 5 d · 1 wk · 2 wk
Delay > 11 days	↓ rate of anatomic reduction (OB)
↑ Time to surgery	↓ **chance of anatomic reduction** · infection trend ↑ · **no** effect on HO or neuro injury

Early fixation → ↓ LOS, ↓ **organ dysfunction**, ↑ function, ↑ discharge home, ↑ reduction quality (EXCEPT trap: "greater organ dysfunction")

Surgical approach selection

Approach	Indications	Risks
Kocher-Langenbeck (posterior)	**PW, PC, PC + PW, most transverse + T**	**Sciatic nerve (2–10%)**, **MFCA** (head blood supply), **↑ HO vs anterior**
Ilioinguinal (anterior)	**AW, AC, ABC (standard), ACPHT**	**Femoral nerve**, **LFCN**, femoral vessel thrombosis, **corona mortis 10–15%** · **lowest HO**
Modified Stoppa (anterior intrapelvic, AIP)	**Quadrilateral plate** buttress (medial wall comminution, ABC); ± lateral window ≈ ilioinguinal access	Corona mortis **must be ligated**
Extended iliofemoral	**Only single approach to see both columns directly**; T-type / transverse with **posterior comminution**, **transtectal T**, ABC with **posterior column comminution**, **SI joint involvement**, **> 3 wk** delay	**Massive HO (highest)**, **gluteal muscle necrosis**
Iliofemoral	**High anterior column** only	Not ABC
Modified Gibson	Gluteus max–TFL interval ± **trochanteric flip**; PW/PC, selected T + transverse	–
Trochanteric flip osteotomy + surgical dislocation	**PW + femoral head fracture (Pipkin IV)**	–

Stöckle (extended iliofemoral)	80% anatomic (< 1 mm) · loss of reduction 8% · **HO grade 3 13%** · AVN 4%
ABC with spur sign, intact SI + PW, acute	**ilioinguinal** ✗ Wrong options: extended iliofemoral · iliofemoral · KL · Gibson
74-yo, anterior column	**ilioinguinal** ✗ Wrong options: Watson-Jones = hip · Hardinge = lateral THA
Transverse + PW	**KL** (✗ Wrong options: trochanteric flip · extended iliofemoral) · **LC2 crescent** – anterior or posterior (✗ Wrong options: extended)

Approach does not affect outcome (Matta)

– – Active space – –

Ilioinguinal – TO extras (TO)	also **transverse displaced anteriorly** · **T-type with dominant anterior column** · **not** for PW, PC, PC + PW, **posteriorly displaced transverse**

Ilioinguinal window (TO)	Exposes / structures
Lateral	Iliac crest, internal iliac fossa, anterior ilium → anterior column plating
Middle (most critical)	**Inguinal canal contents**, **ilioinguinal nerve**, **LFCN**, **femoral vessels**; **iliopectineal fascia** must be incised (opens the whole anterior pelvis)
Medial	**Symphysis**, superior ramus, **quadrilateral plate**, posterior column (indirect); bladder retraction; **corona mortis ligated**

Ilioinguinal set-up (TO)	supine · **Foley mandatory** · lumbosacral bump · **hip flexed to relax iliopsoas** · drains in **space of Retzius** + internal iliac fossa
Modified Stoppa steps (TO)	**midline rectus split (linea alba)** → protect bladder → **ligate corona mortis** → incise iliopectineal fascia to **SI joint** → expose quadrilateral plate
KL prone vs lateral (TO)	**prone** improves ischiopubic segment reduction, **clamp through sciatic notch**, palpation of quadrilateral surface · KL exposes entire posterior column, both sciatic notches, ischial spine + tuberosity, retroacetabular surface
KL steps (TO)	split fascia lata → release **gluteus maximus tendon** → tag + detach **short external rotators** → expose sciatic notch
Transverse reduction tools (TO)	distal femoral traction · **Schanz pin in ischial tuberosity** · **angled Weber clamp through notch** · Farabeuf clamp · Schanz pin in iliac pillar (joystick) · ball-spike pusher
Reduction rule (TO)	**column first, then wall**; leave haematoma until full exposure
ABC reduction sequence (TO)	anterior column → **intact ilium** → plate **pelvic brim** → reduce posterior column → **PC screws through pelvic brim** → **intrapelvic buttress** plate

Posterior wall fixation (TO)

Buttress plate (TO)	**mandatory**; **under-contoured** plate resting close to the rim → plate loads the fragment, prevents displacement
Lag screws (TO)	through plate or independent; **never lag screws alone**
Comminuted wall (TO)	**spring plates** under buttress plate · divergent screws ↑ stability
Intra-articular screw check (TO)	**end-on (tangential) fluoroscopic view** of the joint line; AP + Judets alone **not sufficient**
Trochanteric flip (TO)	**posterosuperior wall involving the roof**; fragment must carry **gluteus medius + minimus**; **excise damaged gluteus minimus** (↓ HO)

Sciatic nerve – positioning & injury

Least sciatic tension during KL	**hip extended + knee flexed (90°)** ✗ Wrong options: hip flexed · knee extended · ilioinguinal (that risks femoral + LFCN)
Highest intraneural pressure	**hip flexed 90° + knee extended** (excursion from sciatic notch to leg)
KL position to relax sciatic	hip **extended** + knee flexed (Orthobullets) · hip **and** knee flexed, lateral position (TO) – read the options

Percutaneous column screws – fluoroscopic views

– – Active space – –

View	Shows
Obturator outlet ("teepee" / teardrop)	**Joint penetration** · **cranial–caudal** position · **supra-acetabular start point** (AIIS pin)
Iliac inlet (inlet iliac oblique)	**Anterior–posterior position in pubic ramus** → screw not exiting **posterior ramus** / not intrapelvic
Obturator inlet	Supra-acetabular screw **between inner + outer iliac tables**
Iliac oblique	Trajectory **above sciatic notch** · posterior column screw
Inlet	**SI joint penetration**
IO / OO with **hip + knee flexed**	**Posterior column screw**

Teardrop on obturator outlet column of bone **AIIS → PSIS** (ex-fix pins / screws)
✗ Wrong options: ASIS · sciatic buttress · roof

Antegrade AC screw start **gluteal pillar** → toward pubic tubercle · retrograde = ramus → iliac wing

Percutaneous screw (TO)	Start / trajectory
PC antegrade	**Iliopubic eminence → ischial tuberosity** or → ischial spine · iliac crest → sciatic notch
PC retrograde	**Medial half of ischial tuberosity** → posterior column
AC antegrade	Superior to acetabulum / iliac crest; narrow corridor at pubic root
AC retrograde	**Inferior + lateral to pubic tubercle**; flexible nail helps find corridor

AC screw dangers (TO) **femoral vessels**, **corona mortis**, **bladder** · views: inlet + obturator + iliac oblique

Percutaneous PC screw contraindication (TO) **comminuted posterior column**

Elderly – acute THA with / without ORIF

Acute THA indications elderly + **osteopenia / comminution**, **pre-existing arthritis**, **marginal impaction (gull sign)**, **PW comminution**, **femoral head fracture / damage**, **superomedial dome impaction**, **dislocation**, **delayed presentation**

Elderly PW + dislocation + impaction / comminution / head damage **acute ORIF + immediate THA** (single stage) → **superior function** vs ORIF + delayed THA

THA construct wall fractures → **buttress plate + multi-hole cup** · column fractures → **cage + cup** · columns must be stable first

78-yo, posterior column + femoral head fracture **PC ORIF + THA same sitting**
✗ Wrong options: cup screws alone → cup migration, LaFlamme · staged · cup-cage (only for protrusio / discontinuity)

– – Active space – –

74-yo neglected insufficiency fracture, 4 mo, head damage	**hip reconstruction (THA, flanged / custom cup)** ✗ Wrong options: ORIF · traction
Fracture-dislocation + articular injury + head impaction + loose bodies, ORIF alone	**~50% convert to THA**; THA → **better function, fewer complications**
Kreder	**> 50 yr + marginal impaction + PW comminution → 54%** early THA; residual displacement **> 2 mm** → arthritis
THA after acetabular fracture	**> 60 yr: ~30% late conversion** · 10-yr implant survival **75–80%**
Acute ORIF + THA not for	young patients · isolated AC · small (10–20%) PW · transverse with good bone · ABC with secondary congruence

Postoperative weight bearing & hip contact pressures

Activity	Hip contact force
Rising from chair on affected leg	**Highest** (> walking); **posterosuperior acetabulum**
Full WB gait	High
Isometrics, bed pan	Approach walking, not exceed
Cane	~60% of walking
Toe-touch WB + passive abduction	**Lowest** (< NWB — muscles contract in NWB)

Touch WB	**10–15 kg** (< 20% BW) · partial WB = 20–25 kg (30–50% BW) → after **transtectal transverse PW ORIF**

6 Complications

Complication	Fact
Post-traumatic DJD	**Commonest**; **80% native-hip survival at 20 yr** after ORIF; ~**20% need THA by 20 yr**; risk: **age > 40**, **T + PW and T-type**, **femoral head injury**, marginal impaction, malreduction → **THA or fusion**
Heterotopic ossification	**Highest extended iliofemoral** · KL intermediate · **lowest ilioinguinal / Stoppa**; ~25% incidence
Osteonecrosis	**6–7%** overall · **18% posterior patterns**
Nerve injury	Highest **T + PW**; sciatic (KL 2–10%), femoral / LFCN (ilioinguinal)
DVT / PE, infection, bleeding	–
Intra-articular hardware	Check obturator outlet
Abductor weakness	–

Prognostic factors

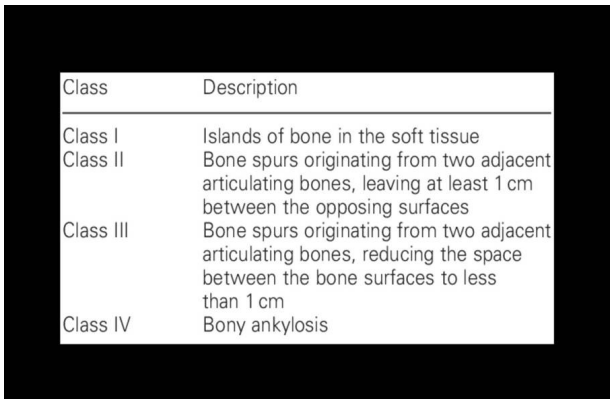
Poor outcome	multi-system trauma · **↑ age** · **poor articular congruency / malreduction** · **marginal impaction** · **femoral head injury** · post-traumatic arthritis · intra-op complications · delay > 3 wk

– – Active space – –

Strongest risk for THA	**articular malreduction** (linear relationship) ✗ Wrong options: > 50% THA at 20 yr · op time ↓ infection · delay ↓ malreduction · anterior column highest arthritis (PW is)
Protective vs early THA	**age < 40**, **simple pattern**, **initial displacement < 20 mm**, anatomic reduction, **no head injury** ✗ Wrong options: gender · approach
PW + age > 50 + non-anatomic reduction	**highest THA conversion** ✗ Wrong options: AC · PC · AW · ABC
NOT a negative factor	**involvement of both columns** (EXCEPT answer) · surgical approach
Matta results	excellent 40% · good 36% · fair 8% · poor 16%; clinical result tracks radiographic reduction
Young, simple transverse, anatomic, no head injury — 20-yr native hip survival	**> 75%**
PROs after PW ORIF	correlate with **hip muscle strength** + gait restoration ✗ Wrong options: age · arc of motion · immediate WB

Heterotopic ossification

Brooker classification used to grade HO



Class	Description
Class I	Islands of bone in the soft tissue
Class II	Bone spurs originating from two adjacent articulating bones, leaving at least 1 cm between the opposing surfaces
Class III	Bone spurs originating from two adjacent articulating bones, reducing the space between the bone surfaces to less than 1 cm
Class IV	Bony ankylosis

Brooker classification of heterotopic ossification

Prophylaxis	**indomethacin ×1 wk**, **low-dose radiation**, **debride gluteus minimus** (Orthobullets)
OB update	only **radiation 700–800 cGy within 72 h** clearly effective · **indomethacin unproven** in acetabulum + **prolonged use → ↑ nonunion** · gluteus minimus debridement now questioned
Natural history	~85% asymptomatic; ~15% disabling stiffness → **excision** (even early if sciatic compression); **low symptomatic recurrence** (technical ~30%); expect ↑ blood loss
✗ HO traps	generous gluteus **medius** debridement (→ Trendelenburg) · "most need excision" · "high symptomatic recurrence" · ISS / fracture pattern as cause · radiation causing HO
T-type via extended iliofemoral, HO at 3 mo despite radiation	cause = **surgical approach**

– – Active space – –

X vs Y

	Obturator oblique	Iliac oblique
In profile	Obturator foramen	Iliac wing
Shows	**Anterior column + posterior wall**	**Posterior column + anterior wall**
Signs	**Spur sign**, PW stress EUA	**Gull sign** (Orthobullets)

	Transverse	Associated both column
Elementary / associated	**Elementary**	Associated
Articular surface	Still attached to intact ilium → axial skeleton	**None attached**
Lines	Iliopectineal + ilioischial	Both + iliac wing + obturator ring
Sign	Vertical line on axial CT	**Spur sign**

	Iliac inlet	Obturator outlet	Obturator inlet
Screw check	**AP position in ramus**	**Joint penetration / cranio-caudal**	**Between iliac tables**

	Kocher-Langenbeck	Ilioinguinal	Extended iliofemoral
Pattern	PW, PC, PC + PW, transverse	AW, AC, ABC, ACPHT	Both columns, > 3 wk, posterior comminution
Nerve	**Sciatic**	**Femoral, LFCN**	–
HO	Intermediate	**Lowest**	**Highest**

How it's asked

"Elementary fracture involving both columns?"	→ **transverse** ✗ Wrong options: ABC · T-type · T + PW
"All articular segments detached from intact ilium attached to sacrum?"	→ **associated both column**
"Associated fracture NOT involving both columns?"	→ **posterior column + posterior wall**
"Commonest associated pattern?"	→ **transverse + posterior wall**
"Commonest pattern in elderly?"	→ **anterior column posterior hemitransverse** (low-energy: anterior column + medial wall)
"Spur sign – what and where?"	→ **intact ilium; obturator oblique → ABC** (TO)
"Spur sign indicates?"	→ **associated both column** ✗ Wrong options: transtectal transverse · vertical shear · H-type sacral · APC III
"Obturator oblique shows?"	→ **posterior wall + anterior column** (TO)

"Disruption of which line = posterior column?"	→ **ilioischial** (TO) ✗ Wrong options: iliopectineal = anterior column · Shenton · sacral arcuate
"Anterior column fracture — expected finding?"	→ **iliopectineal line broken on obturator oblique, posterior acetabulum intact**
"Only ilioischial line broken, anterior wall intact on IO?"	→ **posterior column**
"CT indicated for all EXCEPT?"	→ **pre-existing degenerative change**
"Most accurate test of PW stability?"	→ **dynamic fluoroscopic EUA, obturator oblique** ✗ Wrong options: Keith · Moed · Calkins · dislocation history
"PW < 20% but exits superiorly?"	→ **fluoroscopic EUA** ✗ Wrong options: bedside exam · consent ORIF · angiography · protected WB
"Post-reduction CT, PW fracture with intra-articular fragment?"	→ **traction pin + ORIF** ✗ Wrong options: EUA · protected WB
"Hip dislocation reduced, still foot drop?"	→ **CT hip** ✗ Wrong options: explore nerve · EMG
"Delay most linked to poor result after dislocation?"	→ **closed reduction > 12 h** ✗ Wrong options: > 3 attempts · traction · ORIF > 5 d
"Delay of fixation associated with poor outcome?"	→ **> 3 weeks**
"Single approach for ABC with segmental posterior column comminution?"	→ **extended iliofemoral**
"Approach for posterior column / PC + PW?"	→ **Kocher-Langenbeck**
"Modified Stoppa for?"	→ **quadrilateral plate (ABC with medial wall)**
"Trochanteric flip osteotomy indicated?"	→ **PW + femoral head fracture**
"Least sciatic tension?"	→ **hip extended, knee flexed 90°**
"Foot drop after acetabular fracture — which pattern?"	→ **transverse + posterior wall**
"Nerve root most affected?"	→ **L5**
"Corona mortis joins external iliac with?"	→ **obturator**
"Retrograde ramus screw — view to avoid posterior exit?"	→ **inlet iliac oblique**
"Obturator oblique-inlet view checks?"	→ **screw between inner + outer iliac tables**
"Obturator oblique-outlet view checks?"	→ **cranial–caudal position / joint penetration**
"Antegrade AC screw — avoid intrapelvic penetration?"	→ **iliac oblique inlet**

– – Active space – –

– – Active space – –

"Activity with greatest hip contact force?"	→ **rising from chair on affected leg** (posterosuperior acetabulum)
"Touch WB 10–15 kg most appropriate after?"	→ **ORIF transtectal transverse PW**
"Best predictor of outcome after PW ORIF?"	→ **displacement on post-op CT**
"Improved PROs after PW ORIF?"	→ **hip muscle strength**
"Negative prognostic factors EXCEPT?"	→ **involvement of both columns**
"Strongest risk factor for later THA?"	→ **articular malreduction**
"Protective factor vs early THA conversion?"	→ **age < 40**
"Pattern most linked to THA conversion > 50 yr with malreduction?"	→ **posterior wall**
"Fracture-dislocation with impaction + loose bodies treated with ORIF – THA conversion?"	→ **~50%**
"Acute ORIF + THA best indication?"	→ **elderly, PW + dislocation + marginal impaction / gull sign + femoral head fracture**
"Elderly posterior column + head fracture?"	→ **PC ORIF + THA same procedure**
"Approach with highest HO (of options)?"	→ **Kocher-Langenbeck** (extended iliofemoral highest overall)
"HO prophylaxis – true?"	→ **prolonged indomethacin ↑ nonunion**; only radiation clearly effective
"Acetabular ORIF in pregnancy – fetal mortality linked to?"	→ **mechanism of injury** / injury severity / maternal haemorrhage
"Posterior dislocation, then chest pain + hypotension + leg weakness?"	→ **CT angiography of chest** (aortic injury)
"Adolescent superior acetabular ossicle?"	→ **os acetabuli marginalis superior**
"Crescent fracture on CT – mechanism?"	→ **lateral compression**
"Increased medial joint space after reduction?"	→ **incarcerated fragment** (TO)
"How to confirm no intra-articular screw?"	→ **end-on fluoroscopic view of joint** (TO)
"Transverse + PW – sequence?"	→ **transverse first, then wall** (TO)

ADULT RESPIRATORY DISTRESS SYNDROME

– Trauma · 2 / 67 –

1 Basics

Definition

ARDS	acute lung injury after **polytrauma / long-bone fracture** → progressive respiratory distress, **refractory hypoxaemia**, **non-cardiogenic pulmonary oedema**

Causes – acute endothelial damage

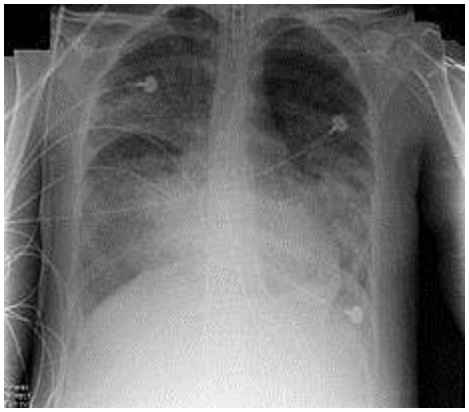
Triggers	**aspiration**, **infection**, **pancreatitis**, **multiple transfusions**, **lung injury**, **sepsis / shock**, **major trauma**, **large burns**, fat emboli, thromboembolism, MOF
Cascade	**non-cardiogenic pulmonary oedema** → respiratory distress → **refractory hypoxaemia** → **↓ lung compliance**

Genes & lab markers

IL-6 (interleukin 6)	inflammatory marker **most closely tied to systemic inflammatory response** after orthopaedic injury (**femur fracture**) **and its treatment (IM nailing)** ✗ Wrong options: IL-1 · IL-10 (no correlation with treatment) · TNF-α · D-dimer (↑ after trauma but don't track treatment)
Status	research marker, **not yet clinically applicable**

Phases

Phase	Timing	Histology
Exudative	Initial	**Hyaline membranes (fibrin)**
Proliferative	**~3 days**	Alveolar exudate resolves or organises
Fibrotic	**3–4 weeks**	Alveolar ducts + spaces fibrose



ARDS histology

– – Active space – –

3 Presentation

Onset	acute, **12–48 h** after injury
Symptoms	**dyspnoea**, fever, **mottled / cyanotic skin**
Exam	**resistant hypoxia**, intercostal retractions, rales / crackles, rhonchi, tachypnoea

4 Diagnosis & Classification

Diagnostic criteria

PaO_2 / FiO_2	Grade
< 300 mm Hg	**Mild** (= acute lung injury, ALI)
< 200 mm Hg	**Moderate** – **ARDS threshold**
< 100 mm Hg	**Severe**

Other thresholds	respiratory compliance **< 40 mL/cm H_2O** · PEEP **> 10 cm H_2O** · corrected expired minute volume **> 10 L/min**
Hallmark	hypoxaemia **refractory to O_2**

Key radiology findings

CXR	**diffuse bilateral infiltrates**, patchy air-space oedema, **normal-sized heart** (→ CHF less likely)
Serial CXRs	early identification in at-risk patients

Differential

Mimics	cardiogenic pulmonary oedema (CHF, MI), bilateral pneumonia, SARS

5 Management

Prevention	**early stabilisation of long-bone (femur) fractures** · monitor PEEP in at-risk patients · serial CXR
Treatment	**PEEP ventilation** + steroids · treat underlying cause

6 Complications

Pneumothorax	barotrauma from **high-PEEP ventilation**
Mortality	**~50%** even with ICU care

X vs Y

	ARDS	Cardiogenic oedema
Oedema	**Non-cardiogenic**	Cardiogenic
Heart size	**Normal**	Enlarged
Response to O_2	**Refractory**	Responds

	ALI	ARDS
PaO_2/FiO_2	**< 300**	**< 200**

How it's asked

"Inflammatory marker most closely tied to systemic response after orthopaedic injury and treatment?"	→ **IL-6** ✗ Wrong options: IL-1 · IL-10 · TNF-α · D-dimer
"PaO_2/FiO_2 defining ARDS?"	→ **< 200 mm Hg** (< 300 = ALI / mild, < 100 = severe)
"Prevention in polytrauma with femur fracture?"	→ **early long-bone stabilisation**
"Complication of high-PEEP ventilation?"	→ **pneumothorax**

– – Active space – –

AMPUTATIONS

– Trauma · 3 / 67 –

1 Basics

Goals & epidemiology

Goals	preserve **functional length** + **useful sensibility** · prevent **symptomatic neuroma** + **adjacent joint contracture** · **early prosthetic fitting** · early return to work
Prevalence	**1.7 million** amputees in USA · **80% for vascular insufficiency**
Commonest cause	**upper limb = trauma** · **lower limb = vascular disease**
Indications – 3 Ds (TO)	**dead** limb (crush, gangrene) · **deadly** disease (gas gangrene, malignancy) · **dis-useful** limb (congenital deficiency) · advanced PVD
MESS inputs (TO)	skeletal + soft-tissue injury · **limb ischaemia** · **age** · **blood pressure (shock)**
Closed vs open (guillotine) (TO)	closed = stump skin-covered · **guillotine** = left open, closed later (infection / contamination)

Wound healing predictors & level selection

Predictor	Threshold for healing
Serum albumin	**> 3.0 g/dL** – **best lab predictor** (✗ Wrong options: total protein · Ca · CRP · ESR)
Total lymphocyte count	**> 1500 /mm³**
ABI	**> 0.45**
Ischaemic index	**> 0.5** (Doppler pressure at level ÷ brachial systolic)
TcPO₂	**> 30 mm Hg** (ideally **45**)
Toe pressure	**> 40 mm Hg**; **will not heal < 20**

TcPO₂ cut-offs	**> 30** healing (Orthobullets, OB) · **> 40** normal healing, **< 20 → amputation is end result** (TO) – read the options
Wound healing depends on	**vascular supply + nutrition + immune status**

Hyperbaric oxygen

Indication **crush injury / acute open fracture** (✓ NOT a contraindication)

HBO contraindication	Why
Chemo (bleomycin) / radiation	Bleomycin + O_2 → **life-threatening pneumonitis**
Pressure-sensitive implanted device	**AICD, pacemaker, dorsal column stimulator, insulin pump** → malfunction
Undrained pneumothorax	Gas embolism, tension PTX, pneumomediastinum
COPD	Air trapping / blebs → PTX

Energy expenditure by amputation level

Level	↑ Energy over baseline
Syme	**15%**
Transtibial – traumatic	**25%** (long BKA 10%, short BKA 40%)
Transtibial – vascular	**40%**
Bilateral transtibial	**40%** (41%, OB)
Through-knee	Between BKA and AKA (habitus dependent)
Transfemoral – traumatic	**68%** (Orthobullets) · **65%** (Miller, OB)
Transfemoral – vascular	**100%**
AKA + BKA	118%
AKA + AKA	> 200%
Medium BKA (TO)	**~25%**
Hip disarticulation (TO)	**~100%**

Rule energy ↑ with **more proximal** level; inversely proportional to residual length → amputate at **lowest possible level**

Exception **Syme more efficient than midfoot** despite being more proximal

Vascular > traumatic at the same level

Unilateral AKA > bilateral BKA in energy cost

Walking speed, fastest → slowest trauma BKA → vascular Syme → trauma AKA → vascular BKA → **vascular AKA (slowest, highest cost)**

Children **through-knee = most proximal level keeping normal walking speed** without ↑ energy
✗ Wrong options: AKA · BKA · Chopart · Girdlestone

4 Diagnosis & Classification

Stump closure techniques (TO)

Technique (TO)	Description	Key fact
Myofascial closure	Fascial envelope over muscles	Minimal stabilisation; used in **severe ischaemia**
Myoplasty	Muscle sewn to **opposing muscle** over bone end	**Bursa, crepitus, pain** → not recommended alone
Myodesis	Muscle attached **directly to bone / periosteum**	Recommended; **contraindicated in severe ischaemia** (wound breakdown)
Tenodesis	Tendon fixed to bone	**Most secure + most physiologic**; **used in disarticulations** (knee: patellar tendon → cruciate origin)

– – Active space – –

Muscle closure choice	myodesis recommended, avoid in severe ischaemia; myoplasty not recommended alone (Orthobullets) · **myodesis preferred except PVD**, **myoplasty preferred in PVD** (poor bone vascularity) (TO) — read the options
Skin flaps (TO)	**skewed / long posterior flap "most preferred"** (weight-bearing surface) · **Scandinavian (equal) flaps in PVD** (better vascularity) · conflicts with other TO note (equal flaps non-ischaemic, long posterior ischaemic) — read the options

Neurovascular handling (TO)

Vessels (TO)	**double ligation**
Nerves (TO)	**stretch + cut sharply** → retracts away (↓ symptomatic neuroma)
Sciatic nerve (TO)	**ligate first, then cut** (bleeding from **vasa nervorum**)
Tourniquet (TO)	generally used; **no exsanguination** in **infection or malignancy**

Amputation nomenclature (TO)

Krukenberg (TO)	forearm **radius + ulna split** → functional **pincer** (sensate)
Ray amputation (TO)	complete digit removal (with metacarpal)

Syme variant (TO)	Technique
Syme	Through ankle; **talus removed**; malleoli retained (TO) · malleoli removed flush with plafond (Orthobullets) — read the options
Sarmiento	Bone cut **1.3 cm above malleoli**
Boyd	**Talectomy** + **calcaneus shifted forward + fused to tibia** · stump bulbous, only ~1 cm longer than Syme
Pirogoff	**Vertical section of calcaneus rotated + fused to tibia**

Gritti-Stokes (TO)	through distal femur; **prepatellar tissues** for closure
Disarticulation preferred when possible (TO)	**end-bearing stump**

Ideal stump length (TO)	Value
BK	**15 cm below knee**
AK	**20 cm above knee** (TO) · cut ~12 cm above joint (Orthobullets) — read the options
BE (transradial)	**15 cm below olecranon**
AE (transhumeral)	**18–20 cm** (TO) · 4–5 cm above elbow (Orthobullets) — read the options

Landmine / blast zones (TO)

Zone (TO)	Injury
Zone I	Nearest blast: traumatic amputation, widespread destruction + contamination → **nonviable**
Zone II	Focal muscle micro-lacerations + haemorrhage along NV bundles / fascial planes
Zone III	Arteriole avulsion, impaired venous return, axonal reactive change

Amputation level after landmine (TO) — **border of zone II and III** (muscle contracts, minimal oedema)

5 Management

Upper-extremity amputation

Indications — irreparable blood supply loss, severe soft-tissue loss, malignant tumour, smouldering infection, congenital

Upper vs lower — mangled **UE has far greater functional impact** than LE amputation; nerve repair **better in UE**; **replantation > UE amputation** for function (worse more proximally, esp. around elbow)

UE salvage vs amputation (civilian + METALS) — **no significant difference** in function or psychological outcome; **no broadly accepted indications**; **MESS is a poor predictor in UE** (MESI more reliable)
✗ Wrong options: blast = contraindication to salvage

Level	Key facts
Transcarpal	Divide finger flexors / extensors; **anchor wrist motors to carpus**
Wrist disarticulation	**Longer lever arm + better pronation / supination**; **preserve radial styloid flare** (suspension); needs intact **DRUJ**; **children – preserves distal radial + ulnar physes**; harder to fit high-function prosthesis
Transradial	**Middle third ideal**; better cosmesis; **easier myoelectric fit**; **≥ 5 cm ulna** needed for elbow motion (little pronosupination)
Transhumeral	Max length; ideal **4–5 cm above elbow**; ≥ **5–7 cm** residual for GH mechanics
Elbow disarticulation	**Children** – avoids **bony overgrowth** of transhumeral
Shoulder disarticulation	**Retain humeral head** for contour → **shoulder hiking** (unopposed girdle muscles) · child → **scoliosis**

Technique — identify + preserve **named motor + sensory branches** (myoelectric signal, TMR) · **myodesis** recommended

Return to work after UE amputation — **offer psychological counselling**
✗ Wrong options: FCE · wait for final prosthesis · full elbow ROM

Targeted muscle reinnervation (TMR)

Concept — transfer amputated major nerves into **expendable remaining muscles** → new discrete **myoelectric signal**

Proven for — **post-amputation neuroma pain** + better **myoelectric prosthesis control**
✗ Wrong options: brachial plexus neurapraxia · Parsonage-Turner · SCI · CTS

– – Active space – –

Transfemoral – adductor myodesis

Level	ideal cut **~12 cm (10–15 cm) above knee** (room for prosthetic knee; resect condylar flare)
Alignment	**5–10° adduction** ideal
Adductor myodesis	**adductor magnus → drill holes at distal femur end** (Gottschalk) → **most important step**
Benefits	**dynamic muscle balance** (vs unopposed abductors + flexors) · **soft-tissue envelope / cushion** · **better prosthetic (suction) fit** · efficient femur position
NOT a benefit	**preserving femoral length** (extra femur is actually resected) (EXCEPT answer)
Missed adductor myodesis	**femur drifts into abduction / subcutaneous lateral thigh** → side lurch, ↑ energy, persists despite good prosthesis ✗ Wrong options: flap length · failure to bevel · abductor myodesis
Transfemoral flaps	usually **equal anterior + posterior**
Gritti-Stokes	through femur at **adductor tubercle**; **excise synovium**; **patella arthrodesed to femur end** (end-bearing); better outcomes than transfemoral

Osseointegration

Concept	implant in residual bone → **transcutaneous stoma** → prosthesis (transfemoral, transtibial)
Benefits	↑ prosthesis wear, mobility, QoL, **proprioception** vs socket
Commonest complication	**low-grade soft-tissue infection**, managed without surgery (oral antibiotics) ✗ Wrong options: failed integration · deep infection · peri-implant fracture

Knee disarticulation vs other levels

Indications	ambulatory patients who **cannot have BKA** · **non-ambulatory** patients
Technique	**patellar tendon sutured to cruciates** in notch (**tenodesis**) · gastrocnemius padding

vs Transfemoral	vs Transtibial (LEAP)
Less energy	**Slower** walking speed
End-bearing limb, proprioception, sitting, lever arm	**Worse SIP** (worst function at 2 yr of all levels)
↓ hip flexion contracture, thigh muscles preserved	Physicians **less satisfied** (cosmesis, function – lost gastrocnemius padding)
Children – **no overgrowth**	**More dependence in transfers**
Asymmetry of prosthetic knee (sitting + gait), limited knee components, cosmesis	**Decreased prosthesis use**
No difference in prosthesis fitting	Pain similar to AKA + BKA

Through-knee in zone of injury	**poorest outcome** of LE amputations; AKA ≈ BKA outcome (BKA faster); gender, prosthesis sophistication, muscle anchoring **don't affect outcome**

Transtibial – flaps

Level — **12–15 cm below joint** (10–16 cm residual tibia); more distal → Achilles region (poor blood supply, padding); need **8–12 cm from ground** for modern prostheses

Flaps (TO) — **non-ischaemic → equal anterior + posterior flaps** · **ischaemic → long posterior flap** (better vascularity)
✗ Wrong options: long anterior flap

Dog ears — leave in place in long posterior flap → **preserve saphenous + sural arteries** (posterior flap supply)
✗ Wrong options: ATA · PTA · peroneal

Fish-mouth (equal coronal) flaps in diabetics / PVD — ↑ **anterior flap necrosis** (Manoli)

Burgess (modified) — standard **long posterior flap**; fibula cut **1–2 cm shorter** than tibia; gastrocnemius fascial closure

Ertl bone bridge & osteomyoplasty

Ertl — **tibia–fibula bone bridge** (original = corticoperiosteal flap; **modified = fibular strut**) → **stable platform for load transfer / end-bearing**; prevents **"chopsticking"**

Modified Ertl vs Burgess	Fact
Op + tourniquet time	**Longer**
Reoperation	**Higher** (hardware, **nonunion**, **bone-bridge dislocation**)
Wound / deep infection	Equivalent
Flaps	Any tension-free flap (posterior, skew, fish-mouth, medial, sagittal)
Function	**No definitive improvement** (small retrospective series; TAOS pending)
Early fitting / immediate WB	**Contraindicated** (needs union)

Osteomyoplasty vs myodesis — **enhanced end-bearing** · fibular abduction = complication of **myodesis** (syndesmotic instability)
✗ Wrong options: early fitting · immediate WB · less morbidity

✗ Wrong option: fibula-pro-tibia — not an amputation; fibular centralisation for tibial defects

Syme amputation – heel pad & contraindications

Syme = ankle disarticulation
✗ Wrong options: "hindfoot amputation"

Requirements — **patent posterior tibial artery** + **stable, viable heel pad (most important)**

Contraindication — **ABI < 0.5 in PT artery** (diabetic)
✗ Wrong options: albumin 3.4 · osteomyelitis unless hindfoot / tibia · $TcPO_2$ 45 · neuropathy

Technique — malleoli removed **flush with tibial plafond** + flares bevelled · heel pad secured to anterior tibia · **tibialis anterior + EDL tenodesed to anterior heel pad** (stop posterior migration)

Problems — **skin slough** (vascular) + **heel-pad migration / hypermobility**

– – Active space – –

Uses forefoot gangrene in diabetics; tumour, trauma · equinus is **not** its complication

Hindfoot & partial-foot amputation

Amputation	Level	Deformity / key fact
Pirogoff	Remove forefoot + talus → **calcaneotibial arthrodesis**; calcaneus rotated **50–90°**	Walk without prosthesis
Chopart	**Talonavicular + calcaneocuboid**	**Equinus (equinovarus) → TA lengthening / tenotomy + tibialis anterior → talar neck**
Lisfranc	**Tarsometatarsal** (2nd MT osteotomised to keep medial cuneiform stable)	**Equinovarus** – unopposed **gastroc-soleus + tibialis posterior** → keep **peroneus brevis** insertion (5th MT base) + TA lengthening; walking cast 4 wk
Transmetatarsal	MT shafts	**Almost all need TA lengthening**; appeals to patients refusing BKA; **rocker-sole shoe + toe filler + carbon plate**
Great toe	Keep **1 cm of proximal phalanx base**	Preserves **plantar fascia, sesamoids, FHB** → ↓ transfer ulcers

Chopart (TO) **forefoot gangrene to navicular** in smoker → **Chopart**: preserves heel support, shoe + AFO, little shortening, heel-skin end-bearing, muscle balancing possible (tibialis anterior transfer + TA lengthening) – easier than Syme / Pirogoff

Boyd grouped with **Chopart** as a hindfoot amputation (Orthobullets) · **talectomy + calcaneotibial fusion** (TO) – read the options

Tibialis anterior innervated by **deep peroneal nerve**

Not Chopart prevention posterior capsule release · TA → cuboid (cuboid removed) · FHL / peroneus brevis → calcaneus · Achilles shortening

Lisfranc supination prevention **preserve PB, peroneus tertius + plantar fascia around 5th MT base**
✗ Wrong options: release tib post · TA → cuneiforms · Achilles lengthening alone

Lisfranc varus TA + tib post overpower peroneus brevis · equinus from loss of **EDL + EHL** · fixes: TAL / gastroc recession, **STATT**, 4th–5th flexor-to-extensor tenodesis, PB → PL

Lisfranc energy ≈ BKA

Soft-tissue balancing required **Lisfranc**
✗ Wrong options: Gritti-Stokes · transtibial · TMA · knee disarticulation

Commonest deformity after midfoot amputation **equinovarus**

Only partial-foot amputation functional in a shoe **transmetatarsal** (others need ankle-controlling prosthesis)

Hindquarter amputation (TO)

Standard hemipelvectomy (TO) **posterior (gluteal) flap**; **disarticulates symphysis pubis + SI joint** + limb

Anterior flap hemipelvectomy (TO)	buttock / posterior thigh lesions; **quadriceps myocutaneous flap** on **superficial femoral vessels**
Extended hemipelvectomy (TO)	+ sacrum / lumbar spine
Mortality (TO)	once > 50% → now **< 10%**

Limb salvage vs amputation – decision factors

Highest impact on decision (LEAP)	**severity of soft-tissue injury** ✗ Wrong options: reduction quality · smoking · insurance · time to debridement
2nd highest	**absent plantar sensation** — **not an absolute contraindication**; often recovers
High late-amputation risk	high-energy foot trauma with **1st MT fracture** or **all 5 metatarsals** (up to 20% at 1 yr) ✗ Wrong options: CRPS commonest
Late amputation after salvage associated with	**fracture nonunion**, LE periarticular fracture, limb dysfunction, chronic pain ✗ Wrong options: UE injury (lower rate) · nerve injury · flap failure · diaphyseal

Limb salvage vs amputation – outcomes

Study	Finding
LEAP (civilian)	**SIP + return to work not different at 2 yr**; ~**50% return to work**; infection 25%; salvage → longer to full WB + more rehospitalisation
LEAP — mangled foot / ankle with free flap or ankle fusion	**Worse (higher) SIP than BKA** (✗ Wrong options: faster walk · more RTW)
METALS (military)	Amputation → slightly better PROs, **lower PTSD**, **more return to vigorous sport / activity**; depression, pain, RTW same

Strongest predictor of patient satisfaction / PROs after LE amputation	**ability to return to work** (+ no depression, faster walking, less pain) ✗ Wrong options: age · gender · graft · plantar sensation · amputation vs salvage · cosmesis · counselling
Higher SIP (Sickness Impact Profile)	= **more disability**
Psychological counselling	improves outcomes (rehab team incl. psychiatrist, social worker)

– – Active space – –

6 Complications

Complication	Fact
Wound healing	See predictors
Contracture	Adjacent joints → early mobilisation, positioning
HO	More common in **trauma** amputations
Infection	**~34%** in trauma amputations
Neuroma	**20–30%**; prevent with nerve handling; treat with **TMR**
Phantom limb pain	**53–100%** of traumatic amputations → **mirror therapy**
Terminal bone overgrowth	**Commonest paediatric complication** (up to 50% of transosseous)

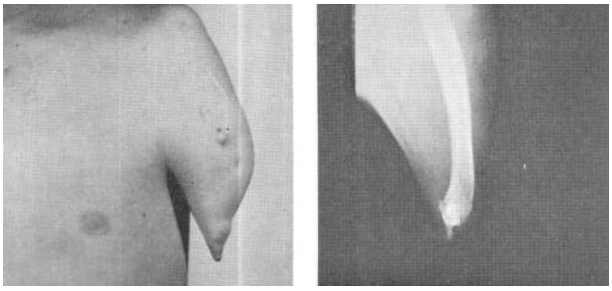
Paediatric amputation – terminal bone overgrowth

Mechanism — disorganised **appositional growth** at cut bone end (periosteal, endosteal, heterotopic) → **tapering spike** → skin ulceration, pain

Prevention — **disarticulation** (articular-cartilage-covered end never overgrows) · **epiphyseal / osteochondral cap**

Treatment — **revision with bone-end capping or Ertl**
✗ Wrong options: HO + NSAIDs · neuroma + TMR · remodelling · osteomyelitis

Knee disarticulation in child avoids — **bone overgrowth**; costs proximal tibial physis → shorter limb
✗ Wrong options: neurogenic pain · contractures



Terminal bone overgrowth after paediatric transosseous amputation

Krajbich principles – children (TO) — **preserve physes** · **prefer disarticulation** · prevent overgrowth with **epiphyseal caps**

Phantom & neuroma pain

Phantom sensation (TO) — **telescoping** (phantom part feels progressively closer to stump)

Neuroma (TO) — best treatment **excision** · best conservative option **TENS**

Prosthesis timing (TO) **pylon** = temporary endoskeletal prosthesis immediately post-op · **final prosthesis at ~3 months**

Level (TO)	Prosthesis
Above knee	**Quadrilateral socket**
Below knee	**PTB**
Syme	**Canadian Syme prosthesis**
Partial foot / hand	Cosmetic fillers

Mirror therapy for **phantom pain**: visual feedback corrects proprioceptive mismatch
✗ Wrong options: proprioception training · flexor tendon · plumbism · ACL

X vs Y

	Wrist disarticulation	Transradial
Lever arm	**Longer**	Shorter
Pronosupination	**Better**	Less
Children	**Preserves distal physes**	–
Prosthesis	Harder	**Easier myoelectric**, better cosmesis

	Chopart	Lisfranc	Syme
Level	TN + CC joints	TMT joints	Ankle disarticulation
Problem	**Equinus / equinovarus**	**Equinovarus / supination**	**Heel-pad migration**, skin slough
Fix	TAL + **TA → talar neck**	Keep **PB at 5th MT base** + TAL	Stable heel pad, patent PTA

	Myodesis	Myoplasty	Tenodesis
Attachment	Muscle → bone	Muscle → antagonist	Tendon → bone
Note	Avoid in severe ischaemia	Bursa, crepitus	**Most secure; disarticulations** (TO)

How it's asked

"Best lab predictor of wound healing?"	→ **serum albumin**
"$TcPO_2$ below which amputation is the likely end result?"	→ **< 20 mm Hg** (TO)
"NOT a contraindication to HBO?"	→ **acute open fracture / crush injury** ✗ Wrong options: COPD · bleomycin · PTX · insulin pump
"BKA flaps in non-ischaemic limb?"	→ **equal anterior + posterior** (TO); ischaemic → long posterior
"Removing dog ears risks?"	→ **saphenous + sural arteries**

– – Active space – –

"Greatest O_2 cost per metre walked?"	→ **transfemoral** ✗ Wrong options: BKA · through-knee · Syme · midfoot
"40% ↑ energy?"	→ **vascular transtibial** or **bilateral transtibial**
"Slowest walking, highest energy?"	→ **vascular AKA**
"Through-knee advantage over transfemoral?"	→ **less energy**
"Through-knee disadvantage vs transfemoral?"	→ **asymmetry of prosthetic knee**
"Through-knee vs transtibial – true?"	→ **decreased prosthesis use**
"Most proximal level in child with normal walking speed?"	→ **through-knee**
"AKA limb abducted at 3 months – forgotten step?"	→ **adductor myodesis**
"Adductor myodesis benefits EXCEPT?"	→ **preserving femoral length**
"Most important for Syme outcome?"	→ **viable, stable heel pad**
"Contraindication to Syme?"	→ **ABI 0.4 in posterior tibial artery**
"Gangrene to navicular in smoker – ideal amputation?"	→ **Chopart** (TO)
"Chopart without balancing – deformity?"	→ **equinus → TAL + tibialis anterior to talar neck**
"Transferred tendon after Chopart – nerve?"	→ **deep peroneal**
"Lisfranc – prevent supination?"	→ **preserve PB / tertius / plantar fascia at 5th MT base**
"Lisfranc deformity forces?"	→ **gastroc-soleus + tibialis posterior**
"Amputation needing soft-tissue balancing?"	→ **Lisfranc**
"Rocker sole + toe filler + carbon plate?"	→ **transmetatarsal**
"Nonunion + bone-bridge dislocation risk?"	→ **modified Ertl**
"Ertl purpose?"	→ **stable platform for load transfer / end-bearing**
"Modified Ertl vs Burgess – true?"	→ **lacks definitive functional improvement**
"Most secure closure in disarticulation?"	→ **tenodesis** (TO)
"Landmine amputation level?"	→ **border of zone II and III** (TO)
"Standard hemipelvectomy?"	→ **posterior gluteal flap; disarticulate symphysis + SI joint** (TO)
"Biggest factor in salvage vs amputation decision?"	→ **extent of soft-tissue injury**
"Absent plantar sensation?"	→ **2nd biggest factor; not absolute contraindication**
"Strongest predictor of satisfaction after amputation?"	→ **return to work**
"Free flap / ankle fusion salvage vs BKA at 2 yr?"	→ **higher SIP (worse)**

"METALS — amputation gives more?"	→ **return to vigorous sport** (lower PTSD)
"High-energy foot with all 5 MTs fractured?"	→ **high late amputation risk**
"Associated with late amputation after salvage?"	→ **fracture nonunion**
"UE salvage vs amputation — true?"	→ **wrist disarticulation = longer lever + better pronosupination**
"Glenohumeral disarticulation leads to?"	→ **shoulder hiking**
"Child, spike of bone at transtibial stump?"	→ **terminal overgrowth → revise with capping / Ertl**
"Paediatric complication avoided by disarticulation?"	→ **bone overgrowth**
"TMR proven for?"	→ **post-amputation neuroma pain**
"Mirror therapy for?"	→ **phantom pain**
"Commonest osseointegration complication?"	→ **low-grade infection managed without surgery**
"Return to work after UE amputation — recommend?"	→ **psychological counselling**
"Calcaneus shifted forward and fused to tibia after talectomy?"	→ **Boyd** (TO)
"Lowest energy expenditure amputation?"	→ **long BKA (~10%)** (TO)

– – Active space – –

– – Active space – –

ANKLE FRACTURES

– Trauma · 4 / 67 –

1 Basics

Epidemiology

Incidence	**187 / 100,000 adults / yr**
Bimodal	**young men 15–24 yr** · **elderly women 75–84 yr**
Pattern split	isolated malleolus **70%** · bimalleolar **20%** · trimalleolar **7%**
Risk factors	male, younger age, **obesity**, **smoking**, **alcohol**
Associated	open 2% · **syndesmotic injury 10%** · chondral injury · peroneal tendon tears 4%

Anatomy – ligaments

Complex	Components	Key fact
Deltoid	**Superficial** (MM → navicular, sustentaculum, talus) – resists **hindfoot eversion** · **Deep** (MM → talus) – resists **talar ER**	Deep deltoid = **primary restraint to anterolateral talar displacement**; inserts on **posterior colliculus**
Lateral	**ATFL** – restrains anterior displacement, IR, inversion; **most often injured** · **CFL** – deep to peronei; 2nd · **PTFL** – **strongest**, least injured	–
Syndesmosis	AITFL · **PITFL** · IOL · interosseous membrane · inferior transverse ligament	See below

Syndesmotic ligament anatomy

Ligament	Tibial origin	Fibular insertion
AITFL	**Chaput** tubercle (anterolateral tibia)	**Wagstaffe** tubercle (anterior fibula)
PITFL – **strongest**	**Volkmann** (posterior malleolus / posterolateral tibia)	Posterior lateral malleolus
IOL	Distal continuation of interosseous membrane	–
Inferior transverse	Runs with PITFL to posterolateral plafond	–

Pilon fragments	**Chaput (anterolateral) = AITFL** · **Volkmann (posterolateral) = PITFL** ✗ Wrong options: ATFL · IOL · ITL · tibionavicular (deltoid)
Avulsion eponyms	**Tillaux–Chaput** = AITFL off **tibia** · **LeFort–Wagstaffe** = AITFL off **fibula** (with SER)

Anatomy – tendons & nerves at risk

Structure	Location	At risk with
Peroneal tendons (PL, **PB**)	Posterior fibular groove	**Posterolateral (antiglide) fibular plate**
Tibialis posterior	Posteroinferior to MM	**Posteriorly placed MM screws**
ATA + **deep peroneal n.**	Between EDL + EHL	Anterior approach
PTA + **tibial n.**	Behind MM, between FDL + FHL	Posteromedial approach
Superficial peroneal n.	Crosses fibula **posterior → anterior ~5–7 cm** above tip (3–12 cm)	Proximal **direct lateral**, posterolateral, anterior / anterolateral
Sural n.	Posterolateral	Posterolateral + lateral approaches

SPN facts **L4–S1** · branches off CPN at fibular neck · motor to **PL + PB** (eversion) · subcutaneous in **lower third of leg** · **two terminal branches** (medial + intermediate dorsal cutaneous)
✗ Wrong options: 1st web space = deep peroneal · dorsiflexors = deep peroneal

Biomechanics

Talus trapezoidal, **wider anteriorly** → dorsiflexion = fibula ER + lateral translation; plantarflexion → narrow posterior talus, talar IR

Fibula **lateral buttress** against talar shift

1 mm lateral talar shift **↓ tibiotalar contact area 42%** (40–42%)

2 Mechanism

Usual **twisting injury**

Lauge-Hansen classification

Name first word = **foot position**, second = **force on foot relative to tibia**; predicts MRI ligament injury in **< 50%**

Type	Stages	Fibula pattern
Supination–adduction (SAD)	1 talofibular sprain / **distal fibular avulsion (tip)** → 2 **vertical MM** + **anteromedial plafond impaction**	**Low transverse**, infrasyndesmotic
Supination–external rotation (SER) – **commonest**	1 AITFL sprain → 2 **short oblique fibula (anteroinferior → posterosuperior)** → 3 PITFL rupture / **posterior malleolus** → 4 **transverse MM or deltoid**	At syndesmosis (Weber B)
Pronation–abduction (PAB)	1 **transverse MM / deltoid** → 2 AITFL → 3 **transverse comminuted fibula above syndesmosis** (butterfly)	Comminuted, above / at syndesmosis
Pronation–external rotation (PER)	1 **transverse MM / deltoid** → 2 AITFL → 3 **short oblique / spiral fibula (anterosuperior → posteroinferior) above joint** → 4 PITFL / posterior malleolus	High spiral (Weber C); **Maisonneuve**

PAB failure sequence medial → AITFL → lateral (medial-to-lateral disruption)

– – Active space – –

✗ Wrong options: "pronation-adduction" / "supination-abduction"	do not exist



Supination–adduction: vertical medial malleolus, low transverse fibula

Supination–adduction specifics

SAD = 5–20% of ankle fractures	**highest talar / hindfoot injury** (talar OCD + medial plafond impaction) → PTOA up to 80%
Vertical MM fracture on axial CT	SAD
Hyperdorsiflexion association	SAD with plafond impaction

3 Presentation

Symptoms	severe pain, unable to bear weight
Exam	swelling, ecchymosis; **medial tenderness / swelling / ecchymosis NOT reliable** for deltoid competence (Tornetta) · **palpate proximal fibula** (Maisonneuve) · fracture blisters, tenting, open wounds · PVD, diabetic neuropathy
On-field obvious fracture-dislocation in athlete (TO)	after ABC → **splint limb in situ as it lies**; assessment < 60 s; reduce after leaving field of play ✗ Wrong options: reduce on field · immediate shift · oral analgesic
Pulseless foot with fracture-dislocation	**reduce + splint → re-evaluate pulses** ✗ Wrong options: vascular consult · CTA · angiogram first

4 Diagnosis & Classification

Key radiology findings

View / study	Use
AP, lateral, mortise	Ankle series
Mortise (TO)	AP in **15° internal rotation** (clears fibular shadow) — best for articular surfaces
ER stress (manual)	**Mortise view**, foot **dorsiflexed + ER**, tibia stabilised → **deltoid competence**; more sensitive than tenderness / ecchymosis
Gravity stress	Lateral decubitus; **= manual stress**
Full-length tib-fib	Rule out **Maisonneuve** (any medial injury / isolated posterior malleolus / widened MCS without fibula fracture)
CT	**Trimalleolar** (posterior malleolus size + morphology — **axial + sagittal**; 25% of surgeons change plan) · **SAD** (anteromedial impaction, talar cartilage) · loose bodies, comminution · **suspected syndesmotic malreduction post-op** · **occult nonunion** · **not routine** — comminuted, posterior malleolar, suspected syndesmotic, pre-op planning (TO)
MRI	Deltoid, syndesmosis (**lambda sign**), lateral ligaments, peroneals, talar OCL — utility controversial

Radiographic parameters

Parameter	Normal	Abnormal / meaning
Medial clear space	**≤ 4 mm** (mortise or stress)	**> 5 mm with ER stress on dorsiflexed ankle → deep deltoid tear** · > 4 mm or **≥ 1 mm more than superior clear space** = deltoid incompetent (SER-4)
Tibiofibular clear space	**< 6 mm** on AP + mortise, measured **1 cm above joint**	Widened → syndesmotic injury (fixation if **> 5 mm on AP**, treatment section)
Tibiofibular overlap	**> 6 mm on AP**, **> 1 mm on mortise**	↓ → syndesmotic injury / **malreduction → CT** (e.g. 2 mm overlap on AP)
Talocrural angle	**83 ± 4°** (OB) · **75–86°** (one OB explanation)	Fibular **shortening → ↑ angle** → lateral talar tilt
Talar tilt	**< 5°**	–
Talar tilt / asymmetry	–	**> 2 mm** = instability marker
Ball / dime sign (TO)	**Unbroken curve** from recess at fibular tip to **lateral talar process**	**Absent → fibula shortened / malreduced** (✗ Wrong options: deltoid tear · ATFL · syndesmosis)

Talocrural angle — line ⊥ plafond (or tibial axis) vs **line between malleolar tips**; not 100% reliable for fibular length

Weber B stable fracture on AP — **tibiofibular clear space < 4 mm**
✗ Wrong options: overlap < 3 mm · Weber C · PER · MCS > 5 mm

Radiological signs of instability (TO) — ↑ medial clear space · ↓ tibiofibular overlap · **break in mortise arc** · **proximal fibular migration** · talar tilt / subluxation (TO)

– – Active space – –

TF overlap / clear space	overlap **> 6 mm AP** (Orthobullets) · **> 10 mm AP** (TO) · clear space **< 6 mm** (Orthobullets) · **< 5 mm** (TO) – read the options



Medial clear space

Posterior malleolus signs	double contour, misty mountains, **spur sign**
Spur sign (hyperplantarflexion variant)	**double cortical density at inferomedial tibial metaphysis**; **79% sensitive, 100% specific**; pathognomonic
Fleck sign (TO)	cortical avulsion off **lateral malleolus → superior peroneal retinaculum injury** → peroneal subluxation ✗ Wrong options: ATFL avulsion · Tillaux · Wagstaffe

Stability assessment – stress radiographs & deltoid competence

Best test of deltoid competence	**ER stress radiograph** (manual or gravity) ✗ Wrong options: anterior drawer = ATFL · IR / inversion stress · palpation · squeeze · Canale view
ORIF indication in isolated fibula fracture	**MCS widening on gravity stress** (= SER-4 / bimalleolar equivalent) ✗ Wrong options: medial tenderness / swelling · squeeze test · Cotton (intra-op)

Deep deltoid intact even with 8–10 mm stress widening (recent MRI studies) – MRI role controversial

Other classifications

Danis-Weber / AO-OTA 44	Level	Note
A / 44A	**Infrasyndesmotic**	Generally stable
B / 44B	**Trans-syndesmotic**	SER
C / 44C	**Suprasyndesmotic**	PER; syndesmotic injury common

Weber stability (TO)	**A** = most stable → conservative · **B** = variable → ORIF often required · **C** = most unstable → ORIF + syndesmotic fixation (TO)
Ring concept (TO)	ankle = closed ring · **single break = stable** · **≥ 2 breaks = unstable → surgery** (TO)
Anatomic	isolated medial / lateral / posterior · bimalleolar-equivalent · bimalleolar · trimalleolar
Variants	Bosworth · hyperplantarflexion (6%) · **curbstone** (posterior tibial avulsion from tripping) · LeFort–Wagstaffe · Tillaux–Chaput

Posterior malleolus classification (Mason / modified Haraguchi)

Type	Fragment
1	Small **extra-articular** posterior malleolar fragment
2	**Posterolateral triangle (Volkmann)**
3	**Coronal line involving whole posterior plafond**

Differential

Mimic	Clue
Ankle sprain	May weight-bear; + anterior drawer / talar tilt; no fracture
Syndesmotic injury	**+ squeeze (Hopkin)**, widening on stress, **lambda sign** on MRI
Achilles rupture	Gap, + Thompson, ↑ resting dorsiflexion
Pilon	High energy, axial load; Chaput, Volkmann, MM, central impaction
Subtalar dislocation	High energy; 25% open

5 Management

Rotational ankle fracture

- Stable (no talar shift on stress) → **splint / boot, early WBAT**
- Talar shift / bimalleolar / Maisonneuve / Bosworth / open → **ORIF**
  - after fixation → stress syndesmosis → **fix if unstable**

Syndesmosis after malleolar fixation

- Posterior malleolus fractured → **fix PM first (restores PITFL)**
- Cotton / ER stress still widens → **open reduction + screw or suture button**

Operative criteria

Nonoperative	ORIF
Stable isolated MM (no talar shift), avulsion tips	**Any talar displacement**
Stable isolated LM (stress negative)	**Bimalleolar / bimalleolar-equivalent**
PM < 25% and < 2 mm step-off, stable syndesmosis	**PM > 25% or > 2 mm** step-off, syndesmotic instability, posterior talar subluxation
Unfit for surgery	**Maisonneuve** · **Bosworth** · **open** · symptomatic malleolar nonunion
–	**LM > 3 mm displacement**

Stable fractures – nonoperative management

Stable Weber B (stress negative, < 3 mm, TCA 83°) — **below-knee splint 2–4 wk → WBAT in boot**
✗ Wrong options: TCC · MRI · CT · ORIF

– – Active space – –

Stable LM	**removable walking boot + progressive WB** ✗ Wrong options: strict NWB cast · fibula ORIF ± medial repair
Isolated MM, no talar shift	NWB 4–6 wk; **> 95% union**
Unstable fractures in elderly	operate on same indications as young; cast for unstable (esp. PAB) → **48–73% malunion / nonunion**

Medial malleolus fixation

Technique	Indication
Lag screws ⊥ fracture	Transverse MM
Antiglide / buttress plate + lag screws, screws parallel to joint	**Vertical shear MM (SAD)** – biomechanically superior to screws alone
Tension band	Fragment too small, poor bone

Bicortical 3.5 mm fully threaded (lag by technique) > unicortical 4.0 mm partially threaded

Supination–adduction – marginal impaction & fixation

Unique, must-recognise feature	**anteromedial plafond marginal impaction** ✗ Wrong options: syndesmotic diastasis (syndesmosis not disrupted in SAD) · deltoid tear · fibular overlengthening
Treatment	**anteromedial approach / arthrotomy → elevate impaction + bone graft → medial antiglide (buttress) plate** + fibular plate ✗ Wrong options: oblique MM screws · tension band · bridge plating
Unaddressed impaction	**malunion with anteromedial pain → surgical correction** ✗ Wrong options: syndesmotic fixation · steroid · PT · hardware removal

Lateral malleolus fixation

Construct	Fact
Lateral plate (1/3 tubular / anatomic)	**Distal bicortical screws may be intra-articular**
Posterolateral antiglide plate	**Biomechanically stronger** than lateral; screws exit anteriorly; **peroneal (brevis) irritation** – hardware removal up to 43%
Locking plate	**Stiffest / most rigid** (✗ Wrong options: fibular nail · antiglide · non-locked bridge · lag screws alone); risks prominence, excess stiffness
Retrograde IM fibular nail	Poor soft tissue / wound-healing risk

Peroneal tendon lesions	**low plate + prominent screw in most distal hole** ✗ Wrong options: trimmed · uncontoured · locked plates
Plate modes	**neutralisation** protects a lag screw · **buttress** = under-contoured plate + axilla screw (compression vs shear) · **antiglide** = uncontoured plate fixed at fracture apex (posterolateral in SER) · bridge = relative stability · tension band = tension side of eccentric load
Weber B reduction (TO)	fibula displaced **posteriorly + externally rotated → traction + internal rotation**; contour plate with an **internal-rotation bend** (TO)

Plate choice (TO)	anatomical distal fibular or 1/3 tubular · **hook plate if distal fragment small** · unstable Weber A → plate or **tension band** (TO)
Fixation sequence (TO)	**fibula first** (length + rotation) → posterior malleolus → medial malleolus (antiglide / tension band) → **Cotton test** for syndesmosis (TO)
Absolute stability	articular fractures: **PM > 25%** / large PM 45% → **ORIF with absolute stability** ✗ Wrong options: relative stability · cast · nail · ex-fix

Bimalleolar-equivalent & bimalleolar

Bimalleolar-equivalent	fibula ORIF ± syndesmosis; **deltoid repair not needed**; explore medially only if **mortise irreducible** (interposed deltoid)
Bimalleolar	ORIF any displacement; ORIF > closed; **early WB → better ROM, earlier return**

Syndesmosis – diagnosis & intra-operative testing

Incidence by fibula level	Weber A **< 10%** · Weber B **40–50%** · **Weber C > 80%**; fixation usually not needed if fibula fracture **within 4.5 cm of plafond**
Most unstable direction	**anterior–posterior → lateral stress view more reliable** ✗ Wrong options: medial–lateral
Cotton (hook) test	bone hook pulls fibula laterally under fluoro after fixation → **most reliable intra-op test**; best done in **sagittal plane, 100 N, lateral view** (van den Bekerom)
ER stress after fibular fixation widens	syndesmosis injured → **open reduction + fixation of syndesmosis** ✗ Wrong options: arthrotomy · close · revise fibula · ex-fix
Cotton test without widening	**no further fixation**
Best method to assess syndesmosis (Weber B, intra-op)	**dynamic ER stress (ankle at 90°)** ✗ Wrong options: MCS alone · overlap · anterior drawer · pre-op CT

Syndesmosis – reduction technique, fixation & outcomes

Most important for outcome	**anatomic syndesmotic reduction** ✗ Wrong options: immediate WB · OCD debridement · medial repair · neutral casting
Malreduction rate	**16–51%** (> 50% reported; up to 30% with screws / flexible fixation on CT)
Most accurate reduction check	**direct visualisation at anterolateral joint line** (plafond, talar dome, lateral malleolus) ✗ Wrong options: radiographic parameters · incisura view

– – Active space – –

Open reduction shown to guide anatomic reduction
✗ Wrong options: clamp 2 cm too proximal · bump + IR · plantarflexion = ER → malreduction · dorsiflexion (no effect)

Clamp rule	Detail
Lateral tine	**Directly on lateral fibular (malleolar) ridge** (✗ Wrong options: posterior to ridge → sagittal malreduction)
Medial tine	**Mid-medial tibia** = **anterior third of tibia on true lateral**
Axis	**Parallel to trans-syndesmotic angle** / joint line
Level	**1–2 cm above plafond** (incisura)
Force	Judicious – avoid over-compression
No	Clamp on malleolar tips

Fibular shortening first causes **syndesmotic malreduction** (lengthening → impingement; gap → nonunion) → restore **length, alignment, rotation** (dime sign / Shenton line)

Ankle position reduce at **neutral**; **full dorsiflexion not required**
✗ Wrong options: "complete dorsiflexion" → stays widened

Fixation fact	Value
Implant	**1–2 screws** or **suture button**, **2–4 cm above joint**
Angle	**20–30° posterior → anterior** (fibula posterior to tibia)
Cortices / size	3 or 4 cortices, 3.5 or 4.5 mm – **no proven difference**
Suture button	**Lowest malreduction + reoperation**
Screw retention	**≥ 8–12 wk** (Orthobullets) · **≥ 12 wk** (OB) – **NWB**, screws can't take gait loads

Syndesmotic fixation indications (TO) PER injuries · suprasyndesmotic fractures · **MCS widening > 2 mm after fibular fixation** · screw **3.5 mm, 3–4 cm above talar dome, 3 cortices, routine removal at 3 months** · tightrope needs no removal (TO)

Retained intact screw **may be worse than loosened / broken screw**; broken ≈ loose; removing broken screw doesn't help · remove intact screw if **painful prominence + dorsiflexion < 10°** (~9 mo)

Posterior malleolus fixation restores **PITFL** (strongest syndesmotic ligament) → **70% stiffness vs 40%** with syndesmotic screw alone → **obviates syndesmotic fixation in most cases**; outcomes ≥ screw fixation; still stress-test after

PM fixation EXCEPT trap (✗ Wrong options: "biomechanically **inferior** to trans-syndesmotic fixation") · PITFL usually **intact / partially injured** attached to fragment

PER-4 with PM fragment – INCORRECT statement "**PITFL is disrupted**" (syndesmosis fails through avulsed PM)

Posterolateral incision in trimalleolar to **reduce + stabilise syndesmotic attachments** via PM fixation
✗ Wrong options: protect SPN · ↓ wound problems · direct syndesmosis repair · see articular surface

Maisonneuve / proximal fibula injury

Definition (TO)	**proximal-third fibula fracture + distal syndesmotic disruption + MM fracture / deltoid → pronation–external rotation** ✗ Wrong options: ER-abduction · SAD
Missed rate	14–44% → **full-length tib-fib films** ✗ Wrong options: MRI · CT · bone scan · repeat ankle films
Isolated posterior malleolus fracture / medial injury without fibula fracture	**image whole tibia-fibula**
High ankle sprain with diastasis, "no fracture" at urgent care	missed **proximal fibula fracture** ✗ Wrong options: proximal TFJ dislocation · ATFL · CFL
Treatment	almost always operative: **reduce + stabilise syndesmosis** (open reduction best)

Posterior malleolus – fixation & approach

Posterolateral approach	interval **FHL (tibial n.) ↔ peroneals (SPN)** → true internervous plane; prone / lateral; reaches **PM + posterior fibula** through one incision · workhorse for PM + pilon ✗ Wrong options: deep peroneal / sural · deep peroneal / tibial · SPN / DPN = anterolateral
Posteromedial approach	PM with medial exit; needs separate fibular incision
Fixation	**posterior antiglide / buttress plate** · **A→P lag screws** (fibula first, must be reduced)
Hyperplantarflexion variant (spur sign)	**vertical shear of posteromedial tibial rim → buttress plate PM via posterolateral approach** + fibula ORIF ✗ Wrong options: A→P screws · syndesmotic stress only
40–90% of distal-third spiral tibia fractures	have a posterior malleolus fracture



Spur sign – hyperplantarflexion variant

– – Active space – –

Irreducible fracture-dislocation (Bosworth)

Bosworth	fibula **entrapped behind posterolateral ridge of tibia → irreducible** (foot held ER) ✗ Wrong options: AITFL · PITFL · PB · FHL · deltoid · EDB
Treatment	**open reduction via posterolateral approach** + fixation of fibula + posterolateral fragment
Risks	**compartment syndrome** (irreducible) · posteromedial rim extension = "posterior pilon variant" · **post-op loss of dorsiflexion** from **posterior fixation**
Interposition elsewhere	FHL blocks **lateral** subtalar dislocation · EDB blocks **medial** subtalar dislocation

Soft tissue – fracture-dislocation, external fixation, open fracture

Fracture blisters	appear **6–72 h**; delay definitive fixation until re-epithelialised; leave intact blisters, unroofed → **silver sulfadiazine**
Timing (TO)	urgent: dislocations, open fractures · operate before swelling or after **wrinkle sign (~10–14 days)** · delay if blisters on incision line · **deep dirty wounds > 12–24 h = contraindication** (TO)
Failed closed reduction + haemorrhagic blisters over incisions	**temporary spanning ex-fix → relative stability** ✗ Wrong options: repeat closed reduction · 1.6 mm K-wires (too thin) · definitive ORIF
Fracture-dislocation + blisters (well-controlled DM)	**closed reduction + ex-fix** least loss of reduction / skin necrosis ✗ Wrong options: TCC · splint · fibular pinning alone
Ex-fix vs splint in fracture-dislocation	**lower redislocation + skin complications**
Pilon (closed, high energy)	**spanning ex-fix → staged reconstruction** (1–2 wk) ↓ soft-tissue complications
Open ankle fracture	emergent debridement + ORIF if soft tissue allows; **primary closure** OK in clean **I, II, IIIA** in healthy patients
Type I open, clean – NOT true	"**wound VAC + delayed closure ↓ infection**" (delayed closure ↑ infection) · early antibiotics ↓ infection · **ASA 2 ↑ infection risk**
Allgöwer-Donati suture	proven **↑ incisional perfusion** ✗ Wrong options: faster · fewer complications · cosmesis
Smoking	↑ wound complications; **4 wk cessation → infection risk = non-smokers** ✗ Wrong options: MTX · gold · HCQ · TNF-α inhibitors

Diabetes & neuropathy – complications & fixation strategy

Neuropathy	**independent risk factor for complications after ankle fracture** (✗ Wrong options: DR · patella · metatarsal · distal femur) · **most important predictor** (✗ Wrong options: BMI · CKD III · age · sex)
Complication rates	absent pulses **83%** · neuropathy **92%** → **neuropathy > absent pulses** · surgical complications up to 40% · **deep infection up to 20%** · cast skin breakdown up to 50%
Commonest post-op complication in diabetics	**wound problems** · Charcot after neuropathy **> 5%** · neuropathy ↑ delayed / nonunion ✗ Wrong options: malunion

Nonop in diabetic	worst risk = **loss of reduction / fracture displacement** (> Charcot, ulcer, DVT); also Charcot, malunion, nonunion → **ORIF preferred** (may forestall Charcot)
No neuropathy / vasculopathy	**standard ORIF** ✗ Wrong options: TCC for fracture stability (TCC heals soft tissue only)
Enhanced fixation	**multiple quadricortical syndesmotic screws even if syndesmosis intact** · **tibiotalar Steinmann pins / retrograde hindfoot nail** · spanning ex-fix · IM fibular K-wires · **stiffer plates** (locking / compression vs 1/3 tubular) · bicortical MM screws · limited dissection, percutaneous where possible
WB	**NWB 8–12 wk** (vs 4–8 wk normal) → then protected WB ~6 wk → e.g. **CAM boot at 10 wk** ✗ Wrong options: 2 wk · 6 wk
Primary TTC nail candidate	**complicated diabetes** (neuropathy, nephropathy, PVD) + poor control (e.g. 71 yo, neuropathy, A1c 9.4%) ✗ Wrong options: smoker · controlled DM without neuropathy · PVD alone · osteopenic refracture
TTC fusion position	**neutral dorsiflexion, 5–10° ER, 5° hindfoot valgus, 0–5 mm posterior talar translation**
TTC salvage	locking screws, **long nail** (avoid proximal stress fracture) ✗ Wrong options: short nails
Failed fixation → PTOA in DM + neuropathy + PVD	**ankle arthrodesis** ✗ Wrong options: TAA (PVD contraindicates) · triple · re-ORIF
Not a supplemental adjunct	deltoid repair, extraperiosteal bridge plating, denosumab, above-knee immobilisation

Return to driving – braking time

After right ankle ORIF	braking normal at **9 wk** post-op (Egol) · **6–9 wk** (OB) · **6–8 wk** (one OB answer) – read the options
Lower-limb long bone / periarticular	**6 wk after starting WB**
Not proxies	full WB, radiographic union, plantarflexion strength, walking speed, full ROM

– – Active space – –

6 Complications

Complication	Fact
Wound problems	~5%
Deep infection	1–2% · **up to 20% in diabetics** (neuropathy biggest risk)
SPN injury	**10–15%**
Hardware removal	~20%; risk: young, women, long op time
Peroneal tendonitis	**5–40%**; posterolateral plate, low plate, prominent distal screw
Tib post tendonitis	Posterior MM screws
Stiffness	**Loss of dorsiflexion** after posterior fixation
PTOA	Rare if anatomic · **log-splitter 54–84%** (+ infection 15–17%, nonunion 15–17%)
Malunion	Missed SAD impaction; fibular shortening
Nonop	Cast ulcer, malunion, nonunion, **DVT 5%**, stiffness
CRPS	–

Prolonged immobilisation (TO) capsular adhesions · cartilage degeneration · myostatic contracture · bone atrophy · **CRPS** → early mobilisation (TO)

Rehab phase (TO)	Weeks	Focus
I	0–4	Immobilisation, NWB
II	4–8	Partial WB, gentle ROM
III	8–12	Strengthening, proprioception
IV	> 12	Sport, return to activity

Early malreduction – revision ORIF of syndesmosis

Malreduced syndesmosis (CT: fibula posterior in incisura) at 2 wk
remove screws, re-reduce, revise
✗ Wrong options: PT · observe · add anterior screw · loosen screws

Malrotated fibula + syndesmotic malreduction at 3 wk
revision ORIF fibula + syndesmosis
✗ Wrong options: WB · deltoid repair

Widened syndesmosis + shortened fibula at 4 mo
revision ORIF with open syndesmosis reduction
✗ Wrong options: just add screw · PT · bracing

Persistent MCS widening + ossicle after hardware removal
revision ORIF + explore syndesmosis + medial ankle (interposed scar / deltoid / tib post)

Athlete, persistent syndesmotic widening at 4 mo, fibula anatomic
revision ORIF syndesmosis
✗ Wrong options: screw removal · fibular osteotomy · deep deltoid repair (not feasible)

Malunion – fibular length / rotation & corrective osteotomy

Typical malunion	distal fibula **shortened + externally rotated** → talar tilt, instability, PTOA
Prevention	**restore fibular length + rotation** ✗ Wrong options: deltoid repair · quadricortical screw · two screws
Correction	**fibular lengthening (± rotational, oblique) osteotomy + syndesmosis reduction** · even with **early arthritis** · secondary fusion needed in only 10–15% ✗ Wrong options: shortening osteotomy · syndesmotic fixation only · fusion
Joint-preserving vs sacrificing	decided by **degree of tibiotalar arthritis** ✗ Wrong options: deformity · date · gender · WC
Chronic Maisonneuve (12 wk) with shortened fibula, no OA	**fibular osteotomy + syndesmosis ORIF**
Old malunion + MM nonunion	**corrective osteotomy of fibula + MM (graft) ± syndesmosis reconstruction** ✗ Wrong options: steroid / HA injection · fusion
Shortened fibula + deltoid instability, no OA	**medial arthrotomy + debridement + fibular correction** ✗ Wrong options: arthrodesis
Instability markers	asymmetric clear spaces · **talar tilt > 2 mm** · talar subluxation · abnormal TCA
Persistent pain 1 yr after shortened fixation	**fibular malunion** ✗ Wrong options: nonunion · "screw instead of dynamic" · deltoid

Delayed union / nonunion

Pain at fibula site 9 mo after nonop	**CT** (detects nonunion missed on X-ray) ✗ Wrong options: immobilise · sympathetic blocks · PT
Symptomatic MM nonunion, large fragment	**ORIF + autograft** · malleolar nonunion 5–30% ✗ Wrong options: excision (only small fragments) · percutaneous · PT

Outcomes & post-traumatic arthritis

Most important factor for outcome	**anatomic reduction** ✗ Wrong options: PT · early WB · early ROM · Ca/vit D
PM step-off ≥ 2 mm post-op	**independent risk for arthritis** (fragment size not clearly linked)
Overall	~90% success; final function at **~2 yr**; 90% mild / no pain at 1 yr
Good outcome predictors	**age < 40**, male, **ASA 1–2**, no diabetes
Adverse	older age, osteoporosis, **diabetes**, PVD, female, ↑ ASA, **smoking**, **alcohol**, ↓ education, **medial malleolar fracture**
Age	**no effect on outcomes / complications of unstable fractures** (EXCEPT / LEAST answer) ✗ Wrong options: osteoporosis · function · smoking · diabetes · skin

– – Active space – –

Fibular stress fracture

Who	athletes, **military recruits**; female, training change, older age
X-ray	negative or **periosteal reaction along outer distal fibular shaft** ✗ Wrong options: avulsion = sprain · syndesmotic HO = old high sprain · calcified lesion = enchondroma · anterior spur = impingement
Treatment	immobilise + rest → heals 6–12 wk, return ~9 wk

Paediatric transitional fractures (TO)

Fracture (TO)	Salter-Harris	Key fact
Tillaux	**SH III** of distal tibia	**AITFL avulses anterolateral epiphysis (Chaput)**; age **12–14**; forced **lateral rotation of foot / medial rotation of leg on fixed foot** (SER); ORIF if **> 2 mm**; ~5% of paediatric ankle fractures
Triplane	**SH IV**	Age **10–17**; **sagittal** epiphysis (AP view) + **coronal** posterior metaphysis (lateral view) + **axial** physis; lateral commonest (AITFL, weakest lateral physis); medial rare (adduction)

Both occur during **partial physeal closure** (medial closes first)

X vs Y

	SER	PER	PAB	SAD
Fibula	Short oblique **AI → PS** at syndesmosis	Spiral **AS → PI** above syndesmosis	**Transverse comminuted** above	**Low transverse**
Medial	Transverse MM / deltoid (last)	Transverse MM / deltoid (first)	Transverse MM / deltoid (first)	**Vertical MM**
Special	Commonest	**Maisonneuve**	Butterfly; elderly	**Plafond impaction**, talar OCD

	Tillaux–Chaput	LeFort–Wagstaffe
AITFL avulses	**Tibia** (anterolateral)	**Fibula** (anterior tubercle)

	Bosworth	Log-splitter
Lesion	Fibula locked **behind posterolateral tibial ridge**	Talus driven into **syndesmosis**
Key	Irreducible → posterolateral open reduction	**PTOA 54–84%**

How it's asked

"Vertical MM + fibular tip fracture — mechanism?"	→ **supination–adduction** (TO)
"PAB fibular pattern?"	→ **comminuted, at / above syndesmosis**
"Fracture from abduction of foot?"	→ **PAB image** ✗ Wrong options: isolated diastasis = ER · medial subtalar = inversion

"Medial buttress plate parallel screws implies?"	→ **SAD**
"Pattern most likely with talar injury?"	→ **SAD**
"Unique feature to recognise in SAD?"	→ **anteromedial plafond marginal impaction**
"Plating technique for SAD medial malleolus?"	→ **antiglide**
"SAD bimalleolar treatment?"	→ **fibular plate + elevate impaction + MM antiglide plate**
"Mortise view taken in?"	→ **15° internal rotation** (TO)
"Talocrural angle second line?"	→ **between malleolar tips**
"Indicates anatomic reduction?"	→ **TCA 83°**
"Post-op finding needing CT?"	→ **tibiofibular overlap 2 mm on AP**
"Best test of deltoid competence?"	→ **ER stress radiograph**
"ORIF indication in isolated fibula fracture?"	→ **MCS widening on gravity stress**
"Most predictive of medial injury?"	→ **stress MCS 6 mm vs superior space 3 mm**
"Stable Weber B – treatment?"	→ **splint 2–4 wk → WBAT boot**
"Axial CT fragment = ?"	→ **Chaput tubercle – AITFL** (TO)
"Fragment attached to PITFL?"	→ **posterolateral (Volkmann)**
"Cotton test evaluates?"	→ **syndesmosis**
"Direction of greatest syndesmotic instability?"	→ **anterior–posterior**
"Most important for outcome after syndesmotic injury?"	→ **anatomic syndesmotic reduction**
"Lowest malreduction rate?"	→ **suture button**
"Clamp technique EXCEPT?"	→ **lateral tine posterior to ridge**
"Most accurate syndesmotic reduction check?"	→ **direct view of anterolateral joint**
"Iatrogenic fibular shortening first causes?"	→ **syndesmotic malreduction**
"Restores strongest syndesmotic portion?"	→ **reduce + fix posterior malleolus**
"PM fixation EXCEPT?"	→ **biomechanically inferior to syndesmotic screw**
"Fixation contributing most to syndesmotic stability (Weber C + PM)?"	→ **posterior malleolus fixation**
"Intact vs broken syndesmotic screw?"	→ **intact may be worse than loosened**
"Maisonneuve mechanism?"	→ **pronation–external rotation** (TO)
"Isolated posterior malleolus fracture, swollen leg – next?"	→ **tib-fib radiographs**
"Large PM (45%) – treatment?"	→ **ORIF with absolute stability**
"Posterolateral approach interval?"	→ **SPN (peroneals) ↔ tibial (FHL)**

– – Active space – –

"Spur sign + large PM?"	→ fibula ORIF + PM buttress via posterolateral approach
"Irreducible fracture-dislocation — block?"	→ posterolateral tibial ridge (Bosworth)
"Lost dorsiflexion after Bosworth fixation?"	→ posterior fixation of fibula + PM
"Lateral plate risk vs posterior plate?"	→ intra-articular screw
"Antiglide plate injures?"	→ peroneus brevis
"Peroneal pathology risk?"	→ low plate + prominent distal screw
"Most rigid fibular fixation?"	→ locking plate
"Plate doing neutralisation + buttress?"	→ under-contoured plate protecting lag screw
"Nerve crossing proximal lateral fibular dissection?"	→ SPN — two terminal branches
"Neuropathy is independent risk for complications of?"	→ ankle fractures
"Most important predictor of complications in diabetic?"	→ peripheral neuropathy
"Diabetic neuropathic ankle ORIF — next after plating?"	→ syndesmotic screws + NWB 8–12 wk
"Diabetic ORIF — weight bearing?"	→ protected WB in CAM at 10 wk
"Adjunct improving stability in neuropathic diabetic?"	→ retrograde hindfoot pin / nail
"Best candidate for primary TTC nail?"	→ complicated diabetes with neuropathy
"TTC fusion position?"	→ neutral DF, 5–10° ER, 5° valgus, 0–5 mm posterior talus
"Diabetic nonop, lost to follow-up 12 wk?"	→ fracture displacement
"Age effect on outcome / complications?"	→ none
"Factor affecting wound complications?"	→ smoking
"Braking time after right ankle ORIF?"	→ ~9 wk (6–9 wk)
"Taxi driver 8 wk after ORIF — braking normal when?"	→ one week from now (9 wk rule)
"Athlete with on-field fracture-dislocation?"	→ splint in situ (TO)
"Pulseless after fracture-dislocation reduced?"	→ re-evaluate pulses
"Blisters over incisions after failed reduction?"	→ spanning ex-fix (relative stability)
"Allgöwer-Donati benefit?"	→ ↑ incisional perfusion
"Clean type I open ankle — NOT true?"	→ VAC + delayed closure ↓ infection
"Log-splitter PTOA rate?"	→ 50–85%
"PM fixed with 2 mm step-off?"	→ step-off is risk factor for arthritis
"Ball sign indicates?"	→ malreduced (shortened) fibula (TO)

"Shortened fibular malunion — treatment?"	→ **lengthening osteotomy + syndesmosis reduction**
"Fusion vs osteotomy decided by?"	→ **degree of tibiotalar arthritis**
"Missed SAD impaction, anteromedial pain?"	→ **surgical correction of malunion**
"Pain 9 mo after nonop fibula?"	→ **CT for nonunion**
"Symptomatic MM nonunion?"	→ **ORIF + autograft**
"Military recruit lateral ankle pain 7 cm above joint?"	→ **periosteal reaction (stress fracture)**
"Tillaux fracture?"	→ **SH III distal tibia** (TO)
"Triplane fracture?"	→ **SH IV** (TO)
"Fleck sign at lateral malleolus?"	→ **superior peroneal retinaculum injury** (TO)
"Clinical sign that soft tissues are ready for ankle ORIF?"	→ **wrinkle sign (~10–14 days)** (TO)
"Most unstable Weber type?"	→ **Type C (PER / Maisonneuve)** (TO)

– – Active space – –

– – Active space – –

BRACHIAL PLEXUS INJURIES

1 Basics

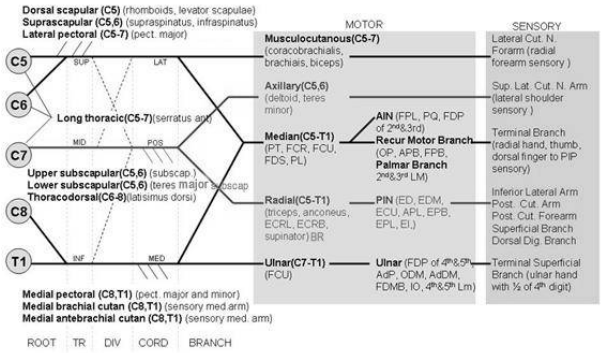
Epidemiology

Traumatic BPI pattern	Share
Complete (all roots, supraclavicular)	**75–80%** – commonest
C5–C6 upper trunk (Erb)	**20–25%**
C8–T1 (Klumpke)	**0.6–3%**

Cause **high-speed vehicle, mostly motorcycle (83%)**

Spectrum traumatic · obstetric (Erb, Klumpke) · burners / stingers · Parsonage-Turner

Plexus anatomy – cord / trunk innervation & lesion localisation



Brachial plexus motor and sensory innervation

Level	Branches
Roots	**Dorsal scapular (rhomboids, C5)** · **long thoracic (serratus anterior, C5–7)** · phrenic contribution (C3–5)
Upper trunk (C5–6)	**Suprascapular (SS, IS)** · nerve to subclavius
Lateral cord	**Musculocutaneous** · **lateral root of median** (FCR, PT) · lateral pectoral (clavicular pec major)
Medial cord	Medial pectoral · medial brachial + antebrachial cutaneous · **ulnar** · medial root of median
Posterior cord	**Upper + lower subscapular** (subscapularis, teres major) · **thoracodorsal (latissimus)** · **axillary (deltoid, teres minor)** · **radial** (triceps, BR, wrist + finger extensors)

Erb's point (TO) **C5 + C6 roots + suprascapular nerve + nerve to subclavius** converge
✗ Wrong options: **long thoracic** (does not meet there)

Upper trunk lesion (tackle / stinger) **axillary + musculocutaneous + suprascapular** deficits: lateral shoulder + radial forearm numbness, weak abduction, ER, elbow flexion, ↓ biceps reflex → **normal 2nd + 5th finger abduction** (ulnar, inferior trunk)
✗ Wrong options: C4 · middle trunk · lateral cord · axillary nerve

Posterior cord lesion spares **supraspinatus** (suprascapular, upper trunk) (EXCEPT answer) · clavicle fragment on posterior cord → radial numbness + weak abduction, IR, wrist extension

Subscapularis upper subscapular nerve consistently from **posterior cord** (lower subscapular variably from axillary)
✗ Wrong options: biceps = lateral cord · pec major = medial + lateral · rhomboid = C5 root · supraspinatus = upper trunk

Axillary nerve path (quadrilateral space syndrome) **C5–6 → upper trunk → posterior division → posterior cord**
✗ Wrong options: C5–7 / lateral cord = musculocutaneous · C5–6 / upper trunk = suprascapular · C5–7 roots = long thoracic · C8–T1 / medial cord = ulnar

Cord-level lesion (TO) **Tinel in infraclavicular fossa**; long thoracic, suprascapular, pectoral spared

Cord lesion (TO)	Deficit
Lateral	Musculocutaneous (biceps), lateral median (FCR, PT), lateral pectoral; **small anterolateral forearm sensory zone**; **GH subluxation**
Posterior	Subscapular, thoracodorsal, axillary, radial → can't IR, elevate, extend; sensory loss over deltoid
Medial	Combined ulnar + median (except FCR, PT) + medial arm / hand sensory loss

Can't flex fingers + no ulnar-side sensation (TO) **C8**

Root values (TO)

Root (TO)	Key motor	Key sensory
C5	Shoulder abduction, ER	Skin over deltoid
C5–C6	Shoulder abduction, **elbow flexion**, supination	–
C6	–	**Thumb + index**
C7	**Elbow + wrist extension** (not specific to one muscle)	**Middle finger**
C8–T1	**Finger flexion, hand intrinsics**	C8 ring + little; T1 medial forearm
T2	–	Medial arm
C5–T1	**Flail limb**	–

2 Mechanism

Caudally forced shoulder / head to opposite side **upper plexus** (high energy → all roots)

– – Active space – –

Forced abduction (clutching while falling)	**lower roots (C8–T1)**
Traction (TO)	arm pulled down + neck bent away · **compression** — lateral neck flexion · **direct blow at Erb's point**

3 Presentation

Horner syndrome	**ptosis, miosis, anhidrosis**; appears **~3 days** after injury → **C8 / T1 root avulsion** (sympathetic chain)
Severe pain in anaesthetic limb	**root avulsion**
Serratus anterior + rhomboids working	C5 injury more likely **postganglionic**
Pulses	check; arterial injury common with complete BPI

Upper vs lower vs total palsy

	Erb (C5–6)	Klumpke (C8–T1)	Total (C5–T1)
Cause	Head to opposite side + shoulder depression; difficult delivery; fall on shoulder	**Excess abduction** (falling, clutching); cervical rib, lung mets	High energy
Posture	**"Waiter's tip"**: **adducted + IR shoulder, extended elbow, pronated forearm**, flexed wrist	**Claw hand**: wrist extended, MCP hyperextended, IP flexed	**Flaccid arm**
Nerves	C5 → axillary, suprascapular, musculocutaneous · C6 → radial (BR, supinator)	Ulnar + median (all intrinsics)	All
Extras	C7 also in **50%** (TO)	**Preganglionic + Horner**	–
Prognosis	**Best**	**Poor**	**Worst**

Erb (TO)	weak deltoid, biceps, brachialis, SS, IS, serratus, rhomboids, levator scapulae, supinator; sensory loss lateral arm / forearm / radial hand

4 Diagnosis & Classification

Pre- vs post-ganglionic lesions

	Preganglionic (avulsion, proximal to DRG)	Postganglionic
Regeneration	**CNS – none → worst prognosis**	PNS – can regenerate
Horner	**Yes**	No
Scapular winging	**Medial winging** (serratus) · rhomboid palsy → lateral winging	No
Hemidiaphragm	**Elevated** (phrenic)	Normal
Tinel in neck	**Absent**	Present; **advancing Tinel** = regeneration
SNAP	**Normal / intact** (DRG cell body intact) despite anaesthesia	**Absent**
Motor NCV	Absent	Absent
EMG paraspinals	**Denervated**	**Normal**
Histamine test	**Normal triple response** (red, wheal, **flare**)	**Abnormal – no flare**
CT myelogram	**Pseudomeningocele** (> 95% predictive)	Roots intact
Wallerian degeneration of sensory fibres	**No**	Yes
Flail arm	**Yes**	**Yes** (both)

SNAP normal + insensate ulnar zone → preganglionic C8–T1 · SNAP normal + insensate median zone → preganglionic C5–C6

TO NOT-seen trap (TO)
preganglionic shows **absent motor but intact sensory conduction → "absent sensory conduction" is NOT seen** · if both absent → postganglionic

C8–T1 avulsion vs plexus injury
ptosis (Horner)
✗ Wrong options: weak radial pulse · SSSC double break · elevated hemidiaphragm = C5 · musculocutaneous deficit

Child, fall clutching ladder, Horner + hand weakness
root level (preganglionic)

Paraspinals normal on EMG
postganglionic
✗ Wrong options: avulsion · Horner · normal histamine

Postganglionic injury spares
rhomboid major (dorsal scapular from root)
✗ Wrong options: ECRL, biceps, deltoid, BR

Most suggestive of postganglionic
advancing Tinel in supraclavicular region
✗ Wrong options: rhomboid / serratus involved · pseudomeningocele · diaphragm palsy · Horner

– – Active space – –

Key radiology findings

Study	Finding / use
CXR (PA + lat)	**1st / 2nd rib fracture** → plexus injury · old rib fractures matter for **intercostal transfer** · inspiration / expiration → **paralysed diaphragm** (upper root)
C-spine AP + lateral	**Transverse process fracture** → root avulsion
Scapula / shoulder (AP + axillary)	**Scapulothoracic dissociation** → root avulsion + major vascular injury
Clavicle	Fracture may signal BPI
CT myelography	**Gold standard for root level**; avulsion → **meningocele**; do at **3–4 wk** (clot dissipates, meningocele forms)
MRI	Lesions **distal to roots**; **pseudomeningocele (T2)**, **empty root sleeves (T1)**, **cord shift away from midline (T1)**, neuromas

Preferred initial imaging (TO) **MRI brachial plexus** · CT myelography = gold standard for avulsion at 3–4 wk

Electrodiagnostics

Test	Fact
EMG	Fibrillations **10–14 d** proximal muscles, **3–6 wk** distal; test **rhomboids, serratus, cervical paraspinals** for pre vs post
NCV / SNAP	SNAP **preserved proximal to DRG** → preganglionic
NAP (intra-op)	Across lesion; detects reinnervation **months before EMG**; NAP negative = **neurapraxic** · NAP positive = **axonotmetic** (Orthobullets wording)
SSEP / MEP	More sensitive than EMG / NCV for root–cord continuity; at **4–6 wk** (Wallerian); stimulate **Erb's point**, record scalp

5 Management

Traumatic brachial plexus injury

- Sharp / open / iatrogenic / vascular / expanding haematoma → **explore < 1 wk**
- Near-total, high energy (avulsion likely) → **early surgery 3–6 wk**
- Partial upper plexus, low energy / GSW → **observe**
  - plateau at 3–6 mo → **delayed surgery (not > 6 mo)**

Operative criteria (timing)

Timing	Indication
Immediate (< 1 wk)	**Sharp penetrating** (not GSW), iatrogenic, open, progressive deficit, **vascular injury / expanding haematoma**
Early (3–6 wk)	**Near-total plexus, high energy** · preganglionic → explore + reconstruct **within 6 wk** (OB)
Delayed (3–6 mo)	**Partial upper plexus, low energy**, plateaued recovery · postganglionic: wait 6 wk–6 mo
Observe	Most; **GSW** without vascular injury → **3 months**

Motor end-plate deadline — irreversible loss **~1 yr** (OB) · **20–24 months** (another OB explanation) → nerve transfer > 20–24 mo relatively contraindicated → **free functioning muscle transfer** – read the options

Best clinical sign of regeneration — **advancing Tinel**

Techniques

Technique	Fact
Direct repair	Rare – acute sharp injuries only
Nerve graft	Common for **postganglionic traction**; upper + middle trunk best; donors **sural**, medial brachial cutaneous, MABC; **vascularised ulnar** (superior ulnar collateral a.) if C8–T1 avulsed
Neurotisation (nerve transfer)	Expendable motor nerve → important denervated muscle; **reduces high injury to low injury**, faster reinnervation
Extraplexal donors	**Spinal accessory (CN XI)**, **intercostals**, **contralateral C7**, hypoglossal (CN XII)
Intraplexal donors	Phrenic, median / ulnar fascicles, pectoral
Muscle / tendon transfer	**Isolated adult C8–T1** (too far to reinnervate intrinsics) · late presentation · **gracilis** = commonest free muscle

Nerve transfers (Oberlin, intercostal)

Oberlin / double fascicular transfer — **ulnar fascicle → biceps branch** + **median fascicle → brachialis branch** → elbow flexion

Indication — **acute (< 3–6 mo) C5–C6 avulsion / upper trunk** · restores **musculocutaneous (lateral cord)** function
✗ Wrong options: C8–T1 avulsion (ulnar dead) · 22 mo old palsy · medial + posterior cord GSW · C6 ASIA A

Other C5–6 options — **SA → suprascapular** · SA or intercostal → musculocutaneous

Pan-plexus avulsion (C5–T1, pseudomeningoceles) — ulnar / median useless → **intercostal → musculocutaneous**: donor is **expendable + functional**
✗ Wrong options: near end plate · synergistic · near sensory target · pure motor (intercostals carry sensation)

Contralateral C7 (TO) — harvested in total palsy with minimal donor deficit

Somsak transfer (TO) — **nerve to long head of triceps → axillary nerve** → shoulder abduction

– – Active space – –

Pedicled muscle transfers for elbow flexion (TO) — **latissimus dorsi** · **pectoralis major** · free functional **gracilis**

Reconstruction priorities

Priority	Nerve
1 Elbow flexion	**Musculocutaneous**
2 Shoulder stability / abduction	**Suprascapular**
3 Brachial–thoracic pinch	Pectoral
4 C6–C7 sensation	Lateral cord
5 Wrist extension / finger flexion	Lateral + posterior cords
6 Wrist flexion / finger extension	–
7 Intrinsics	–

Least important (TO) — **elbow extension**
✗ Wrong options: shoulder abduction · elbow flexion · medial-hand sensation

Medial-hand sensation (TO) — restore to prevent ulcers, burns, self-mutilation → median fascicles → ulnar sensory / dorsal ulnar branch; LABC → dorsal ulnar via 15 cm sural graft

Obstetric brachial plexus palsy

Erb — **commonest obstetric palsy** (1:1000) + **best prognosis → shoulder IR + forearm pronation** (adducted, elbow extended, wrist flexed)

Good prognosis — **biceps return before 3 months** + **full passive shoulder ROM**
✗ Wrong options: breech (preganglionic) · no clavicle fracture · Horner · absent Moro

Poor signs — **breech** (preganglionic), **Horner**, **absent Moro**

Deafferentation pain after plexus injury (TO)

Stem (TO) — severe chronic whole-limb pain weeks after BPI → options: baclofen pump, duroplasty, dorsal nerve ablation, **DREZ (dorsal root entry zone) procedure** – **answer key not provided by the source**

6 Complications

Prognosis — recovery up to **3 yr**; regeneration **1 mm/day**

Better — **infraclavicular > supraclavicular** · **upper plexus** (hand preserved) · postganglionic

Worst — **root avulsion (preganglionic)**: not repairable → arthrodesis, tendon transfer

X vs Y

	Oberlin transfer	Intercostal → MC
Use	**Upper trunk (C5–6)**, ulnar + median working	**Pan-plexus avulsion**
Donor	Intraplexal, synergistic, near target	Extraplexal, expendable, distant

	NAP negative	NAP positive
Lesion	Neurapraxic (Orthobullets)	Axonotmetic / regenerating

How it's asked

"Nerve that does NOT meet at Erb's point?"	→ **long thoracic** (TO)
"Can't flex fingers, ulnar-side numbness — root?"	→ **C8** (TO)
"Infraclavicular Tinel, anterolateral forearm numbness, GH subluxation?"	→ **lateral cord** (TO)
"Posterior cord transection spares?"	→ **supraspinatus**
"Superior trunk injury — what stays normal?"	→ **2nd + 5th finger abduction**
"Tackle: lateral shoulder + radial forearm numbness, weak abduction / ER / elbow flexion?"	→ **upper trunk**
"Muscle innervated by posterior cord?"	→ **subscapularis**
"Quadrilateral space nerve pathway?"	→ **C5–6, upper trunk, posterior division, posterior cord**
"Clavicle fragment, radial numbness, weak abduction + IR + wrist extension?"	→ **posterior cord**
"NOT seen in preganglionic palsy?"	→ **absent sensory conduction** (TO)
"Supports preganglionic injury?"	→ **intact SNAPs** ✗ Wrong options: abnormal histamine · flail arm · intact roots on myelogram · normal paraspinals
"Normal paraspinal EMG means?"	→ **postganglionic**
"Postganglionic — muscle NOT paralysed?"	→ **rhomboid major**
"Suggests C8–T1 root avulsion?"	→ **eyelid droop (Horner)**
"Most suggestive of postganglionic?"	→ **advancing Tinel**
"Gold standard for root level?"	→ **CT myelography at 3–4 wk**
"Ulnar → biceps + median → brachialis transfer indication?"	→ **C5–6 avulsion 2 months ago**
"Ulnar / median fascicle transfers treat injury to?"	→ **lateral cord**
"Intercostal → MC in pan-plexus avulsion — principle?"	→ **expendable, functional donor**
"Least important to reconstruct?"	→ **elbow extension** (TO)

– – Active space – –

"Top reconstruction priority?"	→ **elbow flexion (musculocutaneous)**
"Commonest obstetric palsy — posture?"	→ **shoulder IR + forearm pronation**
"Obstetric palsy — good prognosis?"	→ **biceps return before 3 mo + full passive shoulder ROM**
"GSW plexus injury without vascular injury?"	→ **observe 3 months**
"Sharp penetrating plexus injury?"	→ **explore < 1 week**
"Isolated adult C8–T1?"	→ **tendon / muscle transfer**

CALCANEUS FRACTURES

– Trauma · 6 / 67 –

1 Basics

Epidemiology

Commonest tarsal fracture	~65% of tarsal fractures (OB)
Intra-articular	**60–75%** (Orthobullets) · **75%** (OB) · tuberosity fractures **1–3%** · avulsion + tongue-type **< 15%**
Open	**17%** (15–20%) → no ↑ infection, ↑ wound complications
Tuberosity fractures	peak in **women, 7th decade**

Associated injuries

Association	Rate
Calcaneocuboid joint extension	**63%**
Vertebral fracture	**10%**
Contralateral calcaneus	**10%**
Peroneal tendon instability (SPR tear)	**~20%** – **commonest acute concurrent pathology**
Foot compartment syndrome	**10%** (Orthobullets) · **3–10%** (TO)

Fracture anatomy – constant fragment, sustentaculum & FHL

Facets	**posterior = largest, main weight-bearing** · **middle on sustentaculum (anteromedial)** · anterior often confluent with middle · **sinus tarsi** = calcaneal interosseous sulcus + talar sulcus
Constant (superomedial / sustentacular) fragment	**sustentaculum tali**; stays attached to talus by **interosseous + medial talocalcaneal ligaments** (+ deltoid); lateral fragments + tuberosity are reduced **to it** ✗ Wrong options: held by Achilles · by navicular · displaces superolateral
Sustentaculum intact (TO)	in **~60–70%** → reconstruction begins from it · **if fractured → medial approach** required (TO)
FHL	wraps **inferiorly around sustentaculum** in a groove · **medial to posterior facet, inferior to middle facet**; origin **inferior ⅔ posterior fibula + interosseous membrane**

FHL problem	Cause	Sign
Iatrogenic	**Lateral → medial drill / screws too long** into constant fragment	**Fixed flexed hallux** post-op
Stenosis / tear	**Sustentaculum tali fracture**	**Pain / popping with great-toe flexion**
Entrapment	**Marked posterior facet displacement**	**Limited hallux motion**

– – Active space – –

✗ FHL traps	beak / tuberosity (Achilles) · lateral wall · FDL runs **superior** to sustentaculum (lesser-toe flexion) · plantar paraesthesia = medial plantar nerve · spring ligament
Bifurcate ligament	dorsal **anterior process → cuboid + navicular**; resists **plantarflexion–inversion**; EDB also originates there

Blood supply

Extensile lateral flap	**lateral calcaneal branch of peroneal artery** (± lateral malleolar + lateral tarsal) ✗ Wrong options: branch of ATA · lateral malleolar a.
Heel pad	**medial calcaneal branch of posterior tibial artery** ✗ Wrong options: lateral calcaneal · sinus tarsi a. · tarsal canal a. (those feed talus)

2 Mechanism

Intra-articular	**axial load** (fall from height, MVC)
Primary fracture line	**oblique shear → superomedial (constant) + superolateral (posterior facet)** fragments · secondary line decides **tongue vs joint depression**
Primary line cause (TO)	**always lateral**, from **lateral process of talus** driven into calcaneus → medial sustentacular + anterolateral fragments (TO)
Tuberosity (avulsion)	osteoporosis + violent **triceps surae** contraction with forced dorsiflexion / knee extended · tight gastroc · neuropathy
Anterior process	**inversion + plantarflexion → bifurcate ligament avulsion**
Stress	↑ activity with relative energy deficiency; runners, **military recruits**, female athlete triad, obesity, **after hip / knee arthroplasty**

3 Presentation

Heel	swollen, **shortened, widened**, apparent varus
Posterior heel skin	**tenting, blanching, ecchymosis** with tuberosity / tongue fractures → **urgent fixation**
Fracture blisters	must epithelialise before surgery · **wrinkle sign (Langer lines)** = skin ready
Check	compartments, nerves, pulses (severe PVD may preclude surgery), Achilles continuity, plantarflexion strength
Plantar ecchymosis (TO)	think **midfoot injury** · **full foot views** to exclude missed **Lisfranc** · **spine + pelvis films** (associated injuries **10–21%**) (TO)
Foot compartments (TO)	**4: medial, lateral, superficial central, deep central** (TO – verify: usual teaching 9 compartments) – read the options



Posterior heel skin compromise over a displaced tuberosity fragment

4 Diagnosis & Classification

Key radiology findings

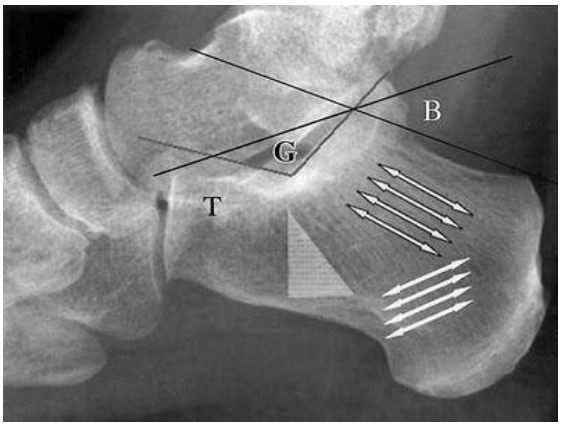
View / study	Use
AP, lateral, oblique foot	Standard
Broden	**Posterior facet**; ankle neutral, **~45° IR**, beam **40, 30, 20, 10° cephalad**; intra-op reduction check
Harris (axial)	Tuberosity **widening, shortening, varus**; max dorsiflexion, beam 45°
AP ankle	**Lateral wall extrusion → fibular impingement**
CT (2–3 mm)	**Gold standard** · **30° semicoronal** = posterior + middle facets (Sanders) · **axial** = **CC joint** · **sagittal** = tuberosity · axial CT shows peroneal dislocation
MRI	**Only for stress fracture** with normal X-ray: **T2 oedema first**, later **T1 low-signal fracture line**

Sign / angle	Normal	Fracture
Böhler angle	**20–40°**	**↓** (posterior facet collapse)
Crucial angle of Gissane	**120–145°**	**↑**
Double-density sign	–	**Lateral portion of posterior facet** depressed / rotated ~90° into body, separated from sustentaculum → subtalar incongruity
Calcaneal shortening, varus tuberosity	–	Present

Böhler angle line anterior process top → posterior facet top **vs** line along superior tuberosity (lateral view)

Gissane line along lateral posterior facet **vs** line anterior to calcaneal beak

– – Active space – –



Böhler angle and crucial angle of Gissane

Böhler normal **20–40°** (Orthobullets) · **25–40°** (TO) – read the options

CT slices **2–3 mm** (Orthobullets) · **≤ 2 mm** (TO), **30° semicoronal** – otherwise Sanders classification unreliable (TO)

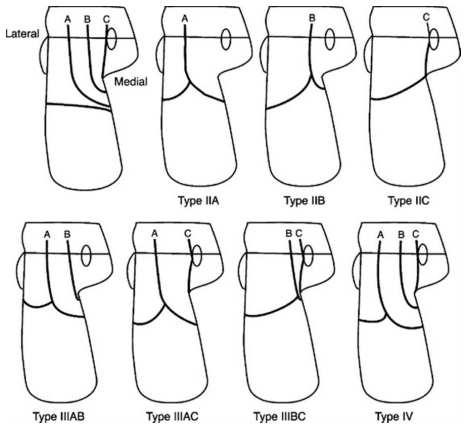
Essex-Lopresti classification

Type	Secondary line
Tongue-type	**Axial plane beneath facet, exiting posteriorly** – posterior facet stays attached to tuberosity → Achilles pulls fragment up
Joint depression	**Behind posterior facet**

Sanders classification (coronal CT, widest posterior facet)

Type	Posterior facet	Treatment
I	**Non-displaced** (any number of lines)	**Nonop**
II	**1 line → 2 fragments**	**ORIF**
III	**2 lines → 3 fragments**	**ORIF**
IV	**> 3 lines, comminuted (≥ 4 fragments)**	**ORIF + primary subtalar arthrodesis**

Sanders III vs II **6.5× PTOA, 4× subtalar fusion**



Sanders classification

Beavis classification (tuberosity)

Type	Fracture
1	**Sleeve** – small cortical shell avulsed
2	**Beak** – oblique line from superior posterior facet running posteriorly
3	**Infrabursal** – from mid-tuberosity

Anterior process fracture types

Type	Description
I	Non-displaced avulsion
II	Displaced, **no** CC joint
III	Displaced, **involves CC joint**

Extra-articular (25%) anterior process (bifurcate), sustentaculum, tuberosity (Achilles avulsion)

5 Management

Calcaneus fracture

- Tongue-type / tuberosity + skin threatened → splint 30° PF → **urgent fixation**
- Sanders I, small extra-articular, comorbid → **NWB cast 10–12 wk**
- Sanders II–III → **ORIF at 10–14 d (wrinkle sign)**
- Sanders IV → **ORIF + primary subtalar fusion**

Operative criteria

Nonoperative (NWB cast 10–12 wk)	Operative
Small extra-articular < 1 cm, Achilles intact, < 2 mm	**Displaced tongue-type > 1 cm**, **threatened skin**, open
Sanders I	**Sanders II + III**: facet **> 2–3 mm**, flattened Böhler, varus tuberosity
Near-normal Böhler (20–40°)	**Large extra-articular, > 10° varus / valgus**
Anterior process < 25% CC joint	**Anterior process > 25% CC joint**
Smoker, diabetes, PVD (avoid wound complications)	**Displaced sustentaculum**
Elderly low-demand tuberosity < 1 cm, skin OK	**Sanders IV → primary subtalar arthrodesis**

Stress fracture **NWB short-leg cast 6 wk** (well-padded heel) / CAM boot + NWB

Cast technique (10–12 wk) mild equinus, **posterior heel window**, **weekly cast changes** (skin complications)

Timing of ORIF **10–14 days** (swelling + blisters resolve, wrinkle sign) · no benefit to early surgery · **except displaced tuberosity / tongue with skin compromise → urgent**

– – Active space – –

Goals (TO)	**posterior facet congruity** · restore **height, width, length, Böhler angle** · **correct hindfoot varus** (TO) ✗ Wrong options: correct **talar declination** (that's talar neck)
Nonop indications (TO)	extra-articular undisplaced · Sanders I · **severe PVD** · **uncontrolled diabetes** · elderly minimally ambulatory · **extensive open wounds** · **life-threatening polytrauma** (TO)
Nonop protocol (TO)	BK slab → cast / boot after swelling · **ROM at 4 wk** · **NWB 6–12 wk** (TO) · NWB cast 10–12 wk (Orthobullets) – read the options
Operative share (TO)	**70–80%** of calcaneal fractures need surgery (TO)
Talar inclination	"correct talar declination" = **not** an aim (TO question) · **restore talar inclination; horizontal talus post-op = inadequate reduction** (TO lecture) – read the options

Tongue-type / tuberosity fracture – urgent fixation

Why urgent	Achilles pulls fragment **proximal + posterior → posterior skin necrosis** (> 20%) ✗ Wrong options: subtalar arthrosis · nonunion · AVN · compartment syndrome · better reduction
Splint	**30° plantarflexion** (max PF) to relax skin
Fixation	**immediate**, minimally invasive / percutaneous **lag screws posterosuperior tuberosity → inferior + distal** (perpendicular to fracture) · ORIF bone-to-bone if large · ± **gastrocnemius recession (Strayer)**
✗ Tuberosity traps	cast in equinus · **excise fragment + reattach Achilles** (large fragment, tension lost) · **percutaneous K-wires** (too weak vs triceps) · **extensile lateral approach** with swelling · delayed ORIF · primary arthrodesis (beak is extra-articular)
Posterior approach (tuberosity)	prone, midline; lag screws / tension band / Krackow through tunnels; restricted WB 6 wk + 6 wk progression

Approaches & techniques

Technique	Key facts
Closed reduction + percutaneous pinning	**Minimally displaced tongue-type**, severe PVD / soft-tissue compromise; Essex-Lopresti manoeuvre; pins out 8–10 wk; ± distracting ex-fix (tuberosity, cuboid, tibia)
Extensile lateral (L-shaped)	Vertical limb between fibula + Achilles, horizontal toward 5th MT base; full-thickness flap (**lateral calcaneal a.**); **high wound complications**; sural nerve + peroneals in flap; **bone graft adds nothing**
Sinus tarsi	**Tip of fibula → 4th MT base**, 2–4 cm; EDB up, peroneals back; sees **posterior facet, anterolateral fragment, lateral wall**; **↓ wound complications, ↓ infection, ↓ op time, better Maryland foot scores**; same Böhler restoration; **↑ radiation**; **sural nerve at greatest risk** (identify inferiorly)
Arthroscopic-assisted	Anterolateral + posterolateral portals; ↓ dissection; ↑ swelling, demanding
Primary subtalar arthrodesis	**Sanders IV**; restore height + fuse; avoids second surgery

Post-op (sinus tarsi) NWB 6–8 wk; ankle ROM from 2 wk

Sinus tarsi timing (TO)	can be done **day 1** · after **7–10 days** fragments become sticky → reduction difficult · best for **Sanders II + III** (TO)
Medial approach (TO)	**sustentacular involvement** · combined medial + lateral for complex fractures (TO)
Reduction sequence (TO)	posterior facet → medial sustentacular alignment → varus / valgus → height → tuberosity → CC joint (TO)
Lateral wall osteotomy (TO)	remove lateral wall → elevate depressed facet from **neutral triangle** → fix (TO)
Extensile flap handling (TO)	thick full-thickness flap · **no toothed forceps** · minimal retractors · **Allgöwer-Donati** closure (TO)
Primary subtalar fusion (TO)	Sanders IV not reconstructable · **chronic malunion with arthritis** · **poor bone stock** (TO)

Operative vs nonoperative – prognostic factors

Better outcome with ORIF	Worse outcome with ORIF
Female	**Male** (men do worse with surgery)
Age < 29–30	**Age > 50** (op ≈ nonop)
No workers' compensation	**Workers' comp**, litigation
Sedentary / light job	**Heavy manual labour**
Sanders II (less comminution)	Higher Sanders, **Böhler < 0°**
Böhler **0–14°** (moderately flat, OB) · **> 15°** (another OB)	**Smoking**, obesity, open, **bilateral**, polytrauma, vasculopathy
–	**Previous calcaneus fracture** (no benefit, EXCEPT answer)

Most likely eventual subtalar fusion	**male + WC + heavy labour + Böhler < 0°**
Highest PTOA risk	**Sanders III treated with ORIF** (articular damage drives arthritis, not sex / job)
Best predictor of PTOA	**initial energy / fracture severity (Sanders)** > reduction quality
Positive prognostic factor (Sanders II, 2 fragments)	**Sanders classification** ✗ Wrong options: sex · age 40 · labourer job · approach
Only undisputed op vs nonop difference	**ORIF ↑ wound complications + reoperation** (nonop → ↓ wound complications) · function, satisfaction, walking, shoe fit **equal at 5 yr**
Surgery reduces	**PTOA / subtalar fusion ~6× (5.5–6×)** ✗ Wrong options: faster RTW · ↑ subtalar ROM · ↓ nonunion (nonunion is rare)
Sanders II, young healthy non-smoker	**ORIF** ✗ Wrong options: cast · fusion · percutaneous (tongue only) · ex-fix
68-yo healthy, blisters	**delayed ORIF** (age > 50 no longer absolute contraindication; comorbidities matter more)

– – Active space – –

Open calcaneus fractures (TO)

Medial wound < 4 cm, closeable, stable (TO)	**serial I&D → standard ORIF** when swelling settles (Sanders III) ✗ Wrong options: percutaneous · ex-fix
Lateral wounds / medial > 4 cm / not closeable / unstable after 7–10 d (TO)	**open reduction through wound + percutaneous fixation**
Grade 1–2 medial open, clean at 3 wk	**infection ≈ closed fracture** with extensile ORIF; IV antibiotics 2–3 d ✗ Wrong options: 3–5× infection · ex-fix only

6 Complications

Complication	Fact
Wound complications	**10–25%** (Orthobullets) · **10–35%** (OB) · **delayed wound healing / dehiscence = commonest** (up to 25%) · deep infection 1–4% closed
Subtalar arthritis	↑ with nonop; **commonest complication of nonop**
Sural neuroma	Extensile lateral / sinus tarsi
FHL injury	Long lateral→medial screws at sustentaculum
Compartment syndrome	**10% → claw toes**
Peroneal tendinitis / impingement	Lateral wall blowout; **typical after nonop**
Malunion	Loss of height, widening, varus, impingement
Overall	**~40% complication rate**

Wound complications, approach & sural nerve

Risk factors (TO)	**smoking** · **diabetes**, **open fracture**, single-layer closure, higher Sanders, long op, ↑ EBL, many people in OR, delay to surgery, ↑ BMI, **less surgeon experience** ✗ Wrong options: RA · steroids · alcohol
Not wound risk factors (OB)	workers' comp, allograft, contralateral fracture, male sex
Prevention (TO)	tourniquet, no smoking, full-thickness flap, **drain**, **two-layer closure ends → middle**, sutures **2–3 wk**, no ROM during that time, keep hardware away from incision corner
Superficial dehiscence after sinus tarsi in smoker	**stop PT, immobilise, local wound care** (± oral antibiotics) · deep / purulent → serial debridement ✗ Wrong options: I&D · hardware removal
Sural neuroma (burning lateral pain, 7 cm above heel)	**excise + bury sural nerve in muscle or vein** ✗ Wrong options: SPN neuroplasty · subtalar fusion · hardware removal at 6 mo
Nerve by approach	sinus tarsi / extensile = **sural** · direct lateral fibula + anterolateral ankle = **SPN** · posteromedial = tibial nerve + PTA

Associated injuries – peroneal tendon dislocation

SPR tear + peroneal instability	**~20%**, from **lateral wall blowout** (joint depression); ↑ with higher Sanders + fracture-dislocation
Fleck / cortical avulsion off distal fibula on CT	**SPR disrupted → anterior peroneal subluxation** confirmed intra-op ✗ Wrong options: intact SPR · FHL entrapment · skin necrosis
CT	limited sensitivity → **test SPR intra-operatively**
TO stem (TO)	calcaneum X-ray, "which movement is weak?" (plantarflexion / dorsiflexion / inversion / eversion) – **answer key not provided by the source**

Compartment syndrome – claw toes

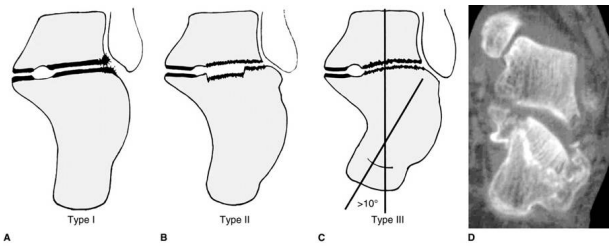
Strongest association (TO)	**Sanders IV** ✗ Wrong options: soft-tissue compromise · delay · crush
Lesser-toe clawing at 3 mo	**missed foot compartment syndrome** (intrinsic contracture) ✗ Wrong options: FHL tethering · sural · medial plantar neuropathy · tib post

Malunion sequelae & subtalar distraction bone-block arthrodesis

Malunion triad	**↓ height (talus dorsiflexed, talar declination < 20°)**, **↑ width (lateral wall exostosis)**, **varus**
Malunion pain sources (TO)	subfibular impingement · lateral wall blowout · loss of height → **horizontal talus** · heel varus · **facet incongruity = least common source of pain** (TO)
Fast union (TO)	calcaneus unites quickly → **after 3 wk treat as malunited** (TO)

Sequel	Cause
Anterior tibiotalar impingement / ↓ dorsiflexion	**Loss of posterior facet height** → dorsiflexed talus (**talar declination < 20°**)
Subfibular / peroneal impingement, peroneal tendinitis	**Lateral wall blowout + widening** (✗ Wrong options: hindfoot valgus (uncommon))
Shoe-wear problems	↑ width
Subtalar arthritis	Articular incongruity (most cases; often radiographic only)
Varus hindfoot	Tuberosity pulled into varus

Nonop DIACF frequently	**peroneal tendinitis** ✗ Wrong options: return to normal · narrowing · lengthening · plantarflexed talus



Calcaneal malunion classification

– – Active space – –

Malunion type (Stephens–Sanders)	Features	Treatment
I	Lateral exostosis, **no** subtalar arthritis	**Lateral wall resection (exostectomy)**
II	Exostosis **+ subtalar arthritis**	**Lateral wall resection + subtalar fusion**
III	+ **varus malunion**	Wall resection + subtalar fusion **± valgus osteotomy** (controversial)

Distraction bone-block subtalar arthrodesis	indicated by **loss of height with anterior ankle impingement / ↓ dorsiflexion** (+ subtalar arthritis); restores height (~8 mm), talocalcaneal angle (~9°), **talar declination** (~11°); corrects varus, TN subluxation
Bone block not for	**full dorsiflexion, no tibiotalar impingement** (EXCEPT) · **peroneal impingement alone** (needs **lateral wall exostectomy**) · isolated sinus tarsi pain (→ **in situ** fusion) · + tibiotalar arthritis (→ triple / larger fusion)
Ankle + lateral pain, subfibular impingement + arthritis	**distraction subtalar arthrodesis + lateral wall ostectomy**
Early post-traumatic subtalar arthritis, ankle pain-free	**Arizona / gauntlet brace** (limits inversion–eversion) before surgery ✗ Wrong options: ASO lace-up · UCBL · medial wedge
Subtalar fusion after prior ORIF	better function + fewer wound problems than after nonop malunion (Radnay)

Stress & insufficiency fractures

Clinical	**heel squeeze (medial–lateral compression) tender**; pain all day, worse with WB; walls of calcaneus
X-ray	**sclerotic line at 2–3 wk** (dense condensation in elderly insufficiency, e.g. after TKA)
MRI	most sensitive: **marrow oedema (T2 / STIR) + linear low-signal fracture line on T1** ✗ Wrong options: plantar fascia thickening · Achilles insertion signal · OCD · serpiginous AVN line
Next step if suspected	**restricted WB + MRI** ✗ Wrong options: heel-pad steroid (fat-pad atrophy) · PRP · nerve release · Graston · EMG
Treatment	**boot / cast + NWB**; 4–6 months cast excessive ✗ Wrong options: fascia stretching · eccentric Achilles · CRPP · activity as tolerated

Anterior process fracture – bifurcate ligament

Mechanism	**inversion + plantarflexion → bifurcate (Y) ligament avulsion** ✗ Wrong options: CFL (inversion + dorsiflexion) · ATFL (talus) · tibiocalcaneal (deltoid)

Often missed as "ankle sprain"; best seen on **oblique foot X-ray or CT**; prone to nonunion

Treatment	acute non-displaced → immobilise NWB · **> 25% CC joint → ORIF**

X vs Y

	Tongue-type	Joint depression
Secondary line	Exits **posteriorly** through tuberosity	**Behind** posterior facet
Facet	Attached to tuberosity	Depressed separately
Urgency	**Skin necrosis → urgent**	Delay 10–14 d
Option	**Percutaneous** reduction / screws	ORIF

	Extensile lateral	Sinus tarsi
Wounds / infection	**Higher**	**Lower**
Op time	Longer	**Shorter**
Radiation	Less	**More**
Böhler restoration	Same	Same

	In situ subtalar fusion	Distraction bone-block fusion
Indication	Isolated subtalar arthritis	**+ loss of height, anterior ankle impingement**

How it's asked

"Displaced tongue-type – feared sequel?"	→ **posterior skin necrosis**
"Immediate fixation of tuberosity avulsion prevents?"	→ **skin necrosis**
"Splint position before tongue-type fixation?"	→ **30° plantarflexion**
"Tuberosity fracture with blanching skin?"	→ **immediate percutaneous reduction + screws**
"Displaced tuberosity, threatened skin – definitive?"	→ **urgent ORIF** ✗ Wrong options: cast · excision · K-wires
"FHL at risk with lateral→medial drill in?"	→ **intra-articular calcaneus fracture**
"Superomedial fragment – true?"	→ **FHL wraps inferiorly around it**
"Fixed flexed hallux after ORIF?"	→ **screws too long into constant fragment**
"Pain / popping with great-toe flexion months later – initial injury?"	→ **sustentaculum tali fracture**
"Constant fragment is?"	→ **sustentaculum tali**
"Double density on lateral?"	→ **lateral portion of posterior facet**
"Tendon entrapped by posterior facet – origin?"	→ **inferior ⅔ posterior fibula (FHL)**
"NOT an aim of calcaneus management?"	→ **correct talar declination** (TO)
"Superior ORIF outcome predictors EXCEPT?"	→ **male sex**
"ORIF better than nonop EXCEPT?"	→ **previous calcaneus fracture**

– – Active space – –

"Most likely eventual subtalar fusion?"	→ **male, WC, heavy labour, Böhler < 0°**
"Highest PTOA risk?"	→ **male, Sanders III, ORIF**
"Surgical advantage over nonop?"	→ **↓ post-traumatic arthritis**
"Only undisputed difference op vs nonop?"	→ **nonop ↓ wound complications**
"Long-term difference ORIF vs nonop?"	→ **ORIF ↑ additional surgery**
"Favourable predictor after ORIF?"	→ **age < 30**
"Sanders III – counselling?"	→ **initial energy / severity best correlates with PTOA**
"Positive prognostic factor in Sanders II?"	→ **Sanders classification**
"Infection risk factor after lateral approach?"	→ **smoking** (TO)
"Sanders III with 3 cm closeable medial wound?"	→ **serial I&D → ORIF** (TO)
"Wound complication risk factor?"	→ **open injury**
"Commonest complication after extensile ORIF?"	→ **delayed wound healing / dehiscence**
"Grade 1–2 medial open, ORIF via lateral approach – infection?"	→ **same as closed**
"Burning lateral pain, injection helps – treatment?"	→ **excise + bury sural nerve**
"Superficial dehiscence after sinus tarsi in smoker?"	→ **stop PT, immobilise, local wound care**
"Approach riskiest for sural nerve?"	→ **sinus tarsi**
"Sinus tarsi vs extensile?"	→ **↓ operative time**
"Wound complications correlate with surgeon experience – fracture?"	→ **calcaneus**
"Extensile flap artery?"	→ **lateral calcaneal branch of peroneal**
"Heel pad supply?"	→ **medial calcaneal branch of PTA**
"Commonest acute concurrent pathology?"	→ **SPR tear / peroneal instability**
"Fibular fleck + lateral wall blowout?"	→ **SPR disrupted, peroneals sublux anteriorly**
"Compartment syndrome most associated with?"	→ **Sanders IV** (TO)
"Lesser-toe clawing at 3 months?"	→ **missed foot compartment syndrome**
"Bone-block distraction fusion improves?"	→ **anterior ankle impingement**
"Sanders IV nonop, block to dorsiflexion, height loss?"	→ **distraction bone-block subtalar arthrodesis**
"Bone-block indications EXCEPT?"	→ **full dorsiflexion without impingement**
"Bone block alone won't fix?"	→ **peroneal impingement** (add lateral wall exostectomy)

"Subtalar arthritis + subfibular impingement + ↓ dorsiflexion?"	→ **distraction fusion + lateral wall ostectomy**
"Factor deciding distraction fusion?"	→ **anterior impingement with ↓ dorsiflexion**
"Commonest complication of nonop DIACF?"	→ **subtalar arthritis**
"NOT seen after malunion?"	→ **hindfoot valgus with subfibular impingement**
"Failure to restore posterior facet height causes?"	→ **tibiotalar impingement**
"Talar declination < 20° causes?"	→ **limited dorsiflexion**
"Early subtalar arthritis, injection helps?"	→ **Arizona / gauntlet brace**
"Runner, + squeeze test — next?"	→ **MRI** (restricted WB)
"Recruit, heel pain after march — treatment?"	→ **boot + NWB**
"Elderly heel pain after starting to walk post-TKA?"	→ **calcaneal insufficiency fracture**
"Expected MRI in calcaneal stress fracture?"	→ **marrow oedema + T1 linear fracture line**
"Lateral foot pain, anterior process avulsion?"	→ **bifurcate ligament (inversion + plantarflexion)**
"Primary fracture line produced by?"	→ **lateral process of talus (always lateral)** (TO)
"Sinus tarsi approach — latest practical timing?"	→ **within 7–10 days** (fragments become sticky) (TO)

– – Active space – –

CAPITELLUM FRACTURES

– Trauma · 7 / 67 –

1 Basics

Epidemiology

Frequency	**1% of elbow fractures · 6% of distal humerus fractures**
Associated	**radial head and / or LUCL injury in up to 60%**

Anatomy

Radiocapitellar joint	key static stabiliser: **longitudinal + valgus stability**; capitellar excision → coronal instability if medial structures are torn
LUCL relationship	integral to posterolateral complex → **don't destabilise LUCL** during fixation
Blood supply	enters **posterolateral** → preserve during exposure

2 Mechanism

Mechanism	low-energy **FOOSH**; **axial compression through radial head with elbow semi-flexed** → **shear** of anterior capitellum

3 Presentation

Symptoms	elbow pain, swelling, ± wrist pain
Exam	**mechanical block to flexion / extension and / or rotation**; instability; NV exam

4 Diagnosis & Classification

Key radiology findings

Best view	**lateral elbow** (+ AP, radiocapitellar view)
Double-arc sign	**pathognomonic** on lateral: **subchondral bone of displaced capitellum + lateral trochlear ridge → coronal shear fracture extending into trochlea** · also called "double bubble" ✗ Wrong options: sagittal lateral epicondyle · coronal medial condyle · olecranon · trans-olecranon dislocation
CT	low threshold; delineates anatomy, trochlear involvement, posterior comminution

Bryan & Morrey classification (McKee modification)

Type	Description	Treatment
I – Hahn-Steinthal	**Large osseous capitellar fragment**, may involve trochlea	Displaced > 2 mm → **ORIF**
II – Kocher-Lorenz	**Articular cartilage shear** with **little subchondral bone**	Displaced → **excision**
III – Broberg-Morrey	**Comminuted / multifragmentary**	Displaced → **excision** (or TEA elderly)
IV – McKee	**Coronal shear including capitellum + trochlea**	**ORIF**



Type I (Hahn-Steinthal) capitellum fracture

5 Management

- **Capitellum fracture**
  - Non-displaced (< 2 mm) type I / II → **posterior splint < 3 wk**
  - Displaced type I or IV → **ORIF, headless screws**
    - posterior comminution / medial injury → **posterior ± olecranon osteotomy**
  - Displaced type II / III → **fragment excision**
  - Elderly, unreconstructable + medial column instability → **TEA**

Treatment choice – ORIF vs arthroplasty

Indication	Treatment
Non-displaced I + II (< 2 mm)	**Posterior splint < 3 wk**
Displaced type I (> 2 mm), type IV	**ORIF**
Isolated type I with good bone (experienced arthroscopist)	**Arthroscopic-assisted ORIF** (70° scope; proximal anterolateral portal trocar reduces; flex past 90° to lock; screws P→A)
Displaced type II, III	**Fragment excision**
Elderly, unreconstructable + medial column instability	**TEA**

– – Active space – –

Healthy 33-yo coronal shear (double bubble)	**ORIF** ✗ Wrong options: excision (large fragment excision → arthritis / instability with MCL tear) · sling · capitellar replacement
Displaced capitellum, no dislocation (diabetic 41-yo)	**ORIF** ✗ Wrong options: MRI for ligaments (unlikely injured without dislocation) · cast · splint + motion · radial head replacement
88-yo comminuted low distal humerus (capitellum + trochlea), poor bone	**TEA** · TEA = choice for comminuted DHF > 70 yr; lifting limit **10–15 lb**; interposition arthroplasty for young who can't comply ✗ Wrong options: ORIF · nonop · excise capitellum + fix trochlea

Surgical approach & fixation

Approach	Use	Detail
Lateral column / lateral extensile (Kaplan or Kocher)	**Isolated capitellum + capitellum into lateral trochlear ridge (type I, IV)**	**Supine**; incision over lateral epicondyle → 2 cm past radial head; strip **ECRL + distal BR** off lateral column (wrist-extensor origin), elevate brachialis off anterior capsule; triceps posterior / BR + ECRL anterior; distal ECU–anconeus (Kocher)
Kaplan interval	–	**EDC ↔ ECRB → PIN at risk → keep forearm pronated**
Posterior ± olecranon osteotomy	**Capitellum + trochlea with posterior comminution, medial injuries, distal humerus / olecranon fractures**	**Lateral decubitus**; radial + ulnar flaps

Isolated capitellar shear, no trochlear / posterior comminution	**supine + Kaplan (lateral column)** ✗ Wrong options: prone / lateral + osteotomy · triceps elevation · medial epicondylectomy
Capitellum + small piece of lateral trochlea	**lateral column approach** ✗ Wrong options: Bryan-Morrey · Hotchkiss (medial) · posterior ± osteotomy (too morbid)
PIN injury from Kaplan approach	**can't extend thumb** ✗ Wrong options: ulnar = medial approach · AIN = can't flex thumb / index IP · brachial artery = anterior · EIP
Fixation	**headless compression screws** (buried beneath cartilage — avoid radial-head damage) **A→P** or **P→A**; mini-fragment screws P→A; capitellar / mini plates for proximal extension
Headless screw most indicated	**capitellar coronal shear (double arc)** ✗ Wrong options: lateral condyle (pins / headed screw) · medial epicondyle (headed ± washer) · olecranon (plate / TBW) · supracondylar (plate / pins)
Lateral approach best for	**complete coronal shear capitellum** ✗ Wrong options: extra-articular DHF (posterior) · proximal radial shaft (Henry / Thompson) · undisplaced radial head (nonop) · trans-olecranon (posterior)
Reduction manoeuvre (TO)	**extension + supination + varus** · headless (Herbert) screws **posterior → anterior** (spares cartilage)

6 Complications

Complication	Fact
Elbow contracture / stiffness	**Most common**; **commonest reason for reoperation** (reoperation up to **48%**)
Nonunion	**1–11%** after ORIF
Heterotopic ossification	**4%** after ORIF
AVN of capitellum	Less common
Ulnar nerve injury, tardy ulnar palsy	–
Olecranon osteotomy nonunion	–
Instability, PTOA, cubital valgus, infection	–

Prognosis functional ROM but **residual stiffness** common; ORIF results generally favourable

Worst outcome predictor **posterior comminution** (needs osteotomy / TEA)
✗ Wrong options: single fragment > 2 cm · coronal shear 1 cm · 5 mm post-reduction displacement · lateral trochlear extension

X vs Y

	Type I Hahn-Steinthal	Type II Kocher-Lorenz	Type IV McKee
Fragment	Large bony capitellum	Cartilage shell	Capitellum + trochlea
Sign	–	–	**Double arc**
Displaced →	ORIF	Excision	ORIF

	Lateral column (Kaplan)	Posterior + olecranon osteotomy
Pattern	Isolated / lateral trochlear ridge	Posterior comminution, both columns, medial injury
Position	**Supine**	**Lateral decubitus**
Risk	**PIN** (pronate forearm)	Osteotomy nonunion, ulnar nerve

How it's asked

"Double-arc sign means?"	**→ coronal shear fracture of capitellum (into trochlea)**
"Bicycle fall, double arc on lateral – describe?"	**→ capitellum fracture extending into trochlea**
"Young adult with displaced capitellar shear?"	**→ ORIF**
"Elderly, comminuted distal articular fracture, poor bone?"	**→ TEA**
"Lateral approach elevating extensors off lateral column best for?"	**→ complete coronal shear capitellum fracture**

– – Active space – –

"Ideal position + approach for isolated capitellar shear?"	→ **supine, Kaplan**
"Capitellum + small lateral trochlea, headless screws — exposure?"	→ **lateral column approach**
"Complication of the approach?"	→ **can't extend thumb (PIN)**
"Headless screw most indicated for?"	→ **capitellar coronal shear**
"Most common complication after fixation?"	→ **elbow flexion contracture / stiffness**
"Most common complication requiring reoperation?"	→ **elbow contracture**
"Finding predicting worst outcome?"	→ **posterior comminution**
"Non-displaced type I or II?"	→ **posterior splint < 3 wk**
"Displaced type II (cartilage shell)?"	→ **fragment excision**

CLAVICLE FRACTURES - DISTAL

– Trauma · 8 / 67 –

1 Basics

Epidemiology

Clavicle fractures	**2.6–4%** of adult fractures · **distal third = 10–25%** of clavicle fractures
Who	older / osteoporotic adults; less common in children (intact periosteal sleeve)
Associated (rare)	floating shoulder · **scapulothoracic dissociation** (if widely distracted) · rib fracture · PTX · NV injury

Neer classification & coracoclavicular ligament anatomy

Stabiliser	Function	Insertion from distal clavicle
AC ligament (sup, inf, ant, post)	**Anterior–posterior** stability; **superior strongest**, then posterior (posterosuperior most important)	**~6 mm**
Trapezoid (lateral)	Vertical	**2–3 cm** (3 cm, OB); **trapezoid tubercle – anterolateral**
Conoid (medial)	**Most important for vertical stability**	**4–4.5 cm** (4.5 cm, OB); **conoid tubercle – posteromedial**
Dynamic	Deltoid + trapezius	–

✗ Conoid traps	at 2–3 cm · anterolateral to trapezoid · trapezoid as main vertical restraint
Displacement depends on	**location** (extra- vs intra-articular), **pattern** (simple vs comminuted), **CC ligament integrity**, **age**
CC ligaments	conoid + trapezoid = primary resistance to **superior displacement of medial fragment**

2 Mechanism

Mechanism	**direct compressive blow / fall onto point of shoulder** (as for midshaft)

3 Presentation

Exam	anterior shoulder pain, swelling, AC deformity, **skin tenting** (impending open)
Suprascapular nerve	at risk → **weak ER with arm adducted**

4 Diagnosis & Classification

Key radiology findings

View	Use
Upright AP of both shoulders	Displacement vs normal side
Axillary lateral	AP displacement
Zanca – 15° cephalic tilt	Superior / inferior displacement; ± **5–10 lb weight** in hand
CT (3D)	Displacement, shortening, comminution, articular extension, nonunion

Superior displacement of medial clavicle relative to coracoid — **CC (conoid) disruption**

Neer classification

Type	Fracture	CC ligaments	Stability	Treatment
I	**Extra-articular, lateral** to CC	**Intact**	**Stable**, minimal displacement	Nonop
IIA	**Medial** to CC	**Both intact** (on distal fragment)	**Unstable**; **nonunion up to 56%** nonop	**ORIF**
IIB	(1) **Between** CC – **conoid torn, trapezoid intact** · (2) **lateral** to CC – **both torn** (proposed "IIC")	Conoid torn	**Unstable**; **nonunion 30–45%**	**ORIF**
III	**Intra-articular** into AC joint, lateral to CC	Intact	Stable → later **AC arthritis**	Nonop
IV	**Physeal** (skeletally immature); clavicle pops out of periosteal sleeve superiorly	Intact	Stable	Nonop
V	**Comminuted**; inferior fragment stays on CC	Intact	Usually **unstable**	**ORIF**

Stable — **I, III, IV · Unstable – IIA, IIB, V**

Conoid injury on X-ray — **lateral fracture with superiorly displaced medial fragment**
✗ Wrong options: conoid still on butterfly fragment · medial to CC · undisplaced



Neer type IIB – fracture between conoid and trapezoid

Alternative classification

Type	Definition	Treatment
A	Non-displaced, CC intact (A1 extra-, A2 intra-articular)	Nonop
B	Displaced, CC intact (B1 extra-articular, B2 comminuted)	Nonop vs op
C	**Displaced, CC torn** (C1 extra-, C2 intra-articular)	**Op**

Differential

Mimics clavicle shaft fracture · paediatric medial / distal physeal injury · **AC separation**

5 Management

Distal clavicle fracture

- Stable (Neer I, III, IV), minimally displaced, children → **sling, ROM 2–4 wk**
- Unstable (IIA, IIB, V) / displaced in athlete → **ORIF**
  - tiny distal fragment → **hook plate or CC reconstruction**

Treatment indications – sling vs ORIF / CC reconstruction

Absolute	Relative
Open / impending open (skin tenting)	**Unstable pattern (IIA, IIB, V)**
Subclavian artery / vein injury	Brachial plexus injury (66% recover spontaneously)
Floating shoulder (+ scapular neck > 10 mm displaced)	Closed head injury, seizures, polytrauma
Symptomatic nonunion	–

Minimally displaced, 56-yo golfer **sling → gentle ROM**
✗ Wrong options: IM pin · bone stimulator · ORIF · distal clavicle resection

Displaced, professional motocross athlete **ORIF**
✗ Wrong options: sling · Kenny-Howard (skin breakdown) · smooth K-wires (migration) · Weaver-Dunn (unnecessary with good bone)

Displaced, wants to avoid second operation **small-fragment plate ± CC reconstruction**
✗ Wrong options: distal clavicle resection (instability) · transacromial wires (removal / migration) · CC screw (removal) · hook plate (removal for acromial osteolysis)

Sling immobilisation

Technique sling or figure-of-8 (**no difference** in function / cosmesis); **no reduction attempt**; ROM at **2–4 wk**; strengthening **6–10 wk**

Outcomes nonunion **~15%** (Neer II up to 56%); poorer cosmesis

– – Active space – –

Open reduction internal fixation

Construct	Key facts
Locking / precontoured / LC-DCP plate	Needs **large enough distal fragment** for multiple locking screws; **≥ 3–4 bicortical screws medially**
Hook plate	When **distal fragment too small**; hook **posterior to AC joint, far lateral**; correct depth; **needs removal (2nd surgery)** – acromial osteolysis
Others	AC-spanning fixation, transacromial wires, TBW, IM screw, **CC repair / reconstruction**, modified Weaver-Dunn (excisable fragment + CA ligament)

ORIF advantages ↑ union, faster union, less overhead pain, ↓ symptomatic malunion, better cosmesis

ORIF disadvantages further procedures, symptomatic hardware, infection

Rehab sling 7–10 d → active motion; strengthen ~6 wk; sport ~3 mo; hardware removal after 3 mo

6 Complications

Nonoperative	Operative
Nonunion ~15% – comminution, displacement, Z deformity, **female, older, smoker**, distal > middle third → asymptomatic: leave · symptomatic: **ORIF + graft**	**Hardware prominence** (~30% request removal; superior plates worst)
AC arthritis – Neer I + III → distal clavicle resection	**Hardware removal** – commonest with **hook plates**
–	**NV injury 3%** (superior plates → subclavian penetration), **subclavian thrombosis**
–	Nonunion **1–5%** · infection ~4.8% · mechanical failure ~1.4% · PTX
–	**Adhesive capsulitis 4%** needing surgery

X vs Y

	Neer IIA	Neer IIB
Fracture vs CC	**Medial** to CC	**Between** (or lateral to) CC
Conoid	**Intact**	**Torn**
Trapezoid	Intact	Intact (or torn)
Nonunion nonop	**Up to 56%**	**30–45%**

	Neer IIB distal clavicle fracture	AC separation (type III)
CC injury	**Conoid only**	**Conoid + trapezoid**

	Conoid	Trapezoid
Distance	**4–4.5 cm**	**2–3 cm**
Tubercle	**Posteromedial**	Anterolateral
Role	**Main vertical stabiliser**	Vertical

How it's asked

"X-ray indicating conoid injury?"	**→ lateral fracture with superiorly displaced medial fragment**
"Conoid ligament – true?"	**→ 4–4.5 cm medial, posteromedial to trapezoid tubercle, main vertical stabiliser**
"Neer IIB?"	**→ between CC ligaments, conoid torn, trapezoid intact**
"Fracture with ligament violation → vertical instability?"	**→ Neer IIB**
"Wants to avoid a second operation?"	**→ small-fragment plate ± CC reconstruction**
"Minimally displaced, golfer?"	**→ sling + gentle ROM**
"Displaced in professional athlete?"	**→ ORIF**
"RELATIVE indication for surgery?"	**→ unstable fracture pattern** ✗ Wrong options: tenting · subclavian vein injury · symptomatic nonunion · floating shoulder = absolute
"Hook plate downside?"	**→ removal needed (acromial osteolysis)**
"Nonunion risk in IIA nonop?"	**→ up to 56%**
"AC arthritis after Neer I / III?"	**→ distal clavicle resection**
"Nerve at risk, weak ER in adduction?"	**→ suprascapular**

– – Active space – –

CLAVICLE FRACTURES - MIDSHAFT

– Trauma · 9 / 67 –

1 Basics

Epidemiology

Incidence	**1 / 1000 / yr** · **2.6–4%** of adult fractures
Location	**middle third = 75–80%** (80–85%, OB) of clavicle fractures
Who	young active **males < 30**

Anatomy

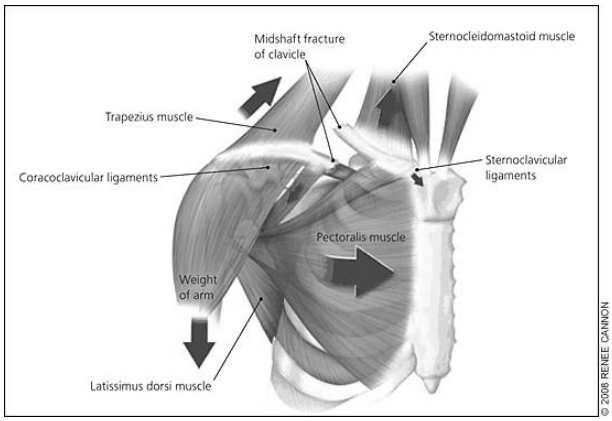
Shape	S-shaped; flat laterally, tubular centrally, prismatic medially
Weakest point	**junction of outer + middle thirds**: thinnest, transition in curvature + cross-section, **only part without muscle / ligament reinforcement**
SC joint stabilisers	posterior capsular ligament (key), anterior SC ligament, costoclavicular ligament, intra-articular disc · **AC joint** – CC + AC ligaments
CC insertions	trapezoid **3 cm**, conoid **4.5 cm** from distal end
Subclavian vessels	behind + below clavicle at **medial–middle junction**; **vein closest**, anterior to artery + plexus
Supraclavicular nerves	vertical cutaneous nerves over clavicle → injured by plating
Unique clavicle facts (TO)	**only long bone lying horizontally** · **first bone to ossify, last to complete** · **only long bone ossifying by intramembranous ossification**
Why middle third (TO)	junction of medial convex + lateral concave curves · thinnest, subcutaneous · **nutrient artery entry** weakens cortex · mechanism: **direct fall on shoulder (commonest)** > FOOSH

Deforming forces

Fragment	Pulled	By
Medial	**Posterosuperior**	**Sternocleidomastoid**
Lateral	**Inferomedial**	**Pectoralis major + weight of arm** (± deltoid)

Trapezius (TO)	elevates lateral end / acromion · distal fragment also **medial + anteriorly rotated** (TO)
Shortening	pec major pulls medially; shortening ↓ deltoid lever arm
Open fracture	medial fragment **buttonholes through platysma**

– – Active space – –



Weight of the arm pulls the lateral fragment down

Associated injuries

Medical	**pneumothorax**, **closed head injury**
Orthopaedic	**rib fracture = commonest associated fracture (TO)** · floating shoulder · **scapulothoracic dissociation** (wide distraction, widened scapula–spine interval, plexus / vascular injury) · NV injury
Brachial plexus injury (TO)	commonest with **medial-third** fractures
Open clavicle fracture	**closed head injury 65%**, pulmonary 75%, spine 35% ✗ Wrong options: STD · calcaneus · pelvis · open tibia
Ipsilateral 1st–4th rib fractures	↑ odds of **late displacement > 100%** of nonop fracture (SSSC instability) ✗ Wrong options: plexus palsy · contralateral scapula · deltoid atony · head injury
Weak pulses after fall (arteriogram)	**axillary artery** (3rd part, distal to subscapular) — collaterals keep hand perfused

2 Mechanism

Fall onto lateral shoulder	**85%** (Orthobullets) · **87%** (TO) · direct blow 7% · **FOOSH only 6% (TO)**

3 Presentation

History	pop / crack, anterior shoulder pain
Exam	tenderness, crepitus, deformity, **skin tenting** (impending open) · check subclavian vessels + plexus

4 Diagnosis & Classification

Key radiology findings

View / study	Use
Upright AP clavicle	Supine **under-estimates** displacement
15° cephalic tilt (Zanca)	Removes scapular overlap
Shoulder series	Proximal humerus, scapula injuries
Upright CXR	**Shortening vs contralateral**; PTX
CT (± contrast)	Comminution, shortening, nonunion; **vascular injury**, medial fracture, **SC dislocation**

Findings medial fragment up, lateral fragment down, shortening

Measures **shortening > 2 cm** → ↓ strength + endurance · **> 100% displacement** → nonunion risk

Classification

Displacement	Treatment
< 100% (non-displaced)	Nonop
> 100% (displaced, no cortical contact)	Op considered

AO type	Subtypes	Treatment
A – simple	A1 spiral, A2 oblique, A3 transverse	Nonop vs op
B – wedge	B1 spiral, B2 bending, B3 fragmented	Nonop vs op
C – complex	C1 complex spiral, **C2 segmental**, C3 irregular	Op

Allman medial / diaphyseal / lateral

Allman (TO)	Site	Frequency
I	Middle third	**80–85%**
II	Lateral third	**15%**
III	Medial third	**<5%**

AO location (TO) clavicle = **15**: **15A medial** · **15B middle** · **15C lateral**

Differential

Mimic	Clue
Distal-third fracture	Older, osteoporotic; ↑ CC distance
SC dislocation	High energy; dysphagia, stridor, asymmetric pulses → **serendipity view / CT**
AC injury	Pain + prominence over AC; Zanca / axillary

5 Management

Midshaft clavicle fracture

- < 100% displaced, < 2 cm short, closed, NV intact → **sling (≡ figure-of-8)**
- Open / tenting / subclavian injury / floating shoulder → **ORIF (absolute)**
- > 100% displaced, > 2 cm short, comminuted / Z, segmental, polytrauma → **ORIF (relative)**

Operative vs nonoperative – indications

Absolute	Relative / fracture-specific
Open	**Shortening > 2 cm** (> 15 mm, OB)
Displaced with skin tenting	**Displacement > 100% / > 2 cm (TO)**
Subclavian artery / vein injury	**Increasing comminution (> 3 fragments) (TO)**
Floating shoulder	**Segmental**
–	Scapular malposition / winging, obvious deformity (TO)
–	**Bilateral**, **polytrauma**, head injury, seizures, brachial plexus injury (66% recover)

Nonop (TO) sling, **no reduction attempt**, early pendulums

Comminuted displaced clavicle + bilateral plateau fractures **ORIF plate**
✗ Wrong options: figure-of-8 · pins · sling

TO true statement (TO) **increasing comminution is an indication for surgery**
✗ Wrong options: fixation rarely required · figure-of-8 treatment of choice · > 2 cm shortening remodels

Operative vs nonoperative – outcomes

Outcome	ORIF vs nonop
Nonunion	**↓** (op 1–5% / ~2–2.5%; union > 94%) vs nonop **~15%**
Symptomatic malunion	**↓**
Time to union	**16.4 vs 28.4 wk** (faster)
Constant + DASH	**Better at all time points to 1 yr** (COTS)
Strength / endurance	Nonop ↓ to ~**80%** of other side
Shoulder ROM	**Same**
Cosmesis / satisfaction	Better with ORIF
Cost	**↑ with ORIF**
Secondary surgery	↑ with ORIF (hardware removal commonest)

Pooled recent literature ORIF **↓ nonunion + symptomatic malunion**; **no clinically significant differences** in strength, ROM, return to activity, or unplanned surgery (contrasts with COTS 2007) — read the options

– – Active space – –

Primary goal of ORIF	**↓ nonunion** ✗ Wrong options: faster pain-free union · no cosmetic deformity · elite strength · fewer complications
Nonop, 1 yr outcome	**↓ strength + endurance** ✗ Wrong options: ↓ motion · instability
3 cm shortening – similar between op + nonop	**shoulder ROM** ✗ Wrong options: nonunion · strength · endurance
Lowest nonunion + malunion (2.6 cm short)	**ORIF with plate** ✗ Wrong options: pins · ex-fix · sling

Sling immobilisation

Indications	< 2 cm shortening / displacement, < 1 cm SSSC displacement, closed, NV intact, low demand
Sling ≡ figure-of-8	**no difference** in healing time, rate, alignment, function, cosmesis; **figure-of-8 → more pain, shortening, ↓ compliance** ✗ Wrong options: figure-of-8 advantages

No reduction attempt · passive ROM 2 wk · strengthening 6 wk · sport **4–6 mo**

Plating technique

Plate	Pros	Cons
Superior	**Higher load to failure**, stronger with inferior comminution, less deltoid detachment	**Prominent**, **↑ subclavian / neurovascular risk**
Anteroinferior	**Less prominent**, **longer screws** (wider front-to-back laterally), ↓ subclavian risk	**Detach part of deltoid origin**
Dual mini-fragment (2.7 sup + 2.4 ant)	**↓ hardware removal (0–3.7%)**; equivalent (not superior) biomechanics	Cost; union + reduction same

Single 3.5 mm superior vs dual	**single has more hardware removal / reoperation** ✗ Wrong options: dual 10× stronger · dual more reoperation · union time differs
Hardware removal risk	**low BMI**, **female** ✗ Wrong options: age · male · dual plates
Plates	LC-DCP 3.5 mm, reconstruction 3.5 mm, 2.0–2.7 mm for dual
Approach extras (TO)	superior = workhorse · anteroinferior = safer but demanding · protect the **three supraclavicular nerve branches** · **TEN = cosmetic but rotationally less stable** · lateral fragment: lateral locking plate or **hook plate under posterior acromion** · CC screw augments construct
Rehab	sling 7–10 d → active; strengthen ~6 wk; sport ~3 mo

Intramedullary fixation

Pros	small incision, less stripping, **spares supraclavicular nerves**, **best for simple patterns**
Cons	**can't lock → no length / rotation control**, biomechanically inferior, migration, irritation, removal at ~6 mo
Contraindicated	**substantial comminution, segmental**

TO EXCEPT trap (TO)	"**IM nailing gives better union than plates in comminuted fractures**" = false
Implant	nail ≈ **30–40% of midshaft diameter**; cannulated screws, TENS, Hagie pin

Nonunion management

Symptomatic atrophic nonunion	**ORIF compression plate + bone graft** ✗ Wrong options: stimulator · figure-of-8 · CRPP · IM nail
Graft	**local autograft ≈ iliac crest union rates** (✓ true)
After failed plating	biology (soft tissue, approaches, infection) vs mechanics (malreduction, weak / short plate, wrong stiffness) → dual mini-fragment revision
Commonest reoperation after ORIF	**implant removal** ✗ Wrong options: infection

6 Complications

Nonunion risk factors

Factor	Note
Displacement > 100%	**Strongest predictor** (Jørgensen)
Comminution / Z-deformity	✓
Shortening > 2 cm (> 15–20 mm)	✓
Female	✓ (highest in options)
Advanced age	✓
Smoking	✓
Not risk factors	Sling vs figure-of-8, early ROM, male, dominant arm, young age, simple pattern, > 50% displacement, 1 cm shortening, diaphyseal location

Nonunion rates	nonop displaced **~15%** (OB, Orthobullets) · **10–15%** · historic **1%** · 7–15% vs op 2% · overall 5–6% · **0.1–13% with 85% of nonunions in middle third (TO)** – read the options
Highest nonunion by location	**middle third** (TO, most nonunions) · **lateral third** has highest *rate* vs diaphysis (OB)
6-wk predictors	**fracture-site motion, no callus, DASH > 40** → 0 factors: 3% · 2–3 factors: **60% nonunion**
Elderly comminuted displaced	highest nonop nonunion risk = **comminution + displacement + age**

– – Active space – –

Nonop displaced, shortened – commonest complication

nonunion
✗ Wrong options: skin necrosis · CRPS 2% · SC / AC arthrosis 4–6%

Nonoperative	Operative
Nonunion ~15% → symptomatic: **plate + graft**	**Supraclavicular nerve numbness – most common** (83% at 2 wk, ~50% at 1 yr)
Malunion ~20%: shortening > 3 cm, angulation > 30°, translation > 1 cm → overhead fatigue, **thoracic outlet syndrome**, strap problems → **osteotomy + graft**	**Hardware prominence** – **8–30%** request removal (superior plates)
↓ strength + endurance (> 2 cm short; Lazarides > 18 mm men, > 14 mm women)	Subclavian injury / thrombosis (superior plates)
Poor cosmesis	Nonunion 1–5%, infection ~4.8% (drug use, diabetes, prior surgery), failure ~1.4%, PTX, adhesive capsulitis 4%

X vs Y

	Superior plate	Anteroinferior plate
Strength	**Higher load to failure**	–
Prominence	**More**	Less
Screws	Shorter	**Longer**
Subclavian risk	**Higher**	Lower
Deltoid	Less detachment	**Detach some deltoid**

	Plate	IM device
Comminuted / segmental	**Yes**	**No**
Length / rotation control	Yes	**No (unlocked)**
Supraclavicular nerves	At risk	Spared

How it's asked

"Nonunion risk factors with nonop?"	**→ displacement + comminution**
"Highest nonunion rate factor?"	**→ female gender**
"73-yo highest nonunion risk?"	**→ comminution + displacement + advanced age**
"Predisposes young male to nonunion?"	**→ fracture displacement**
"NOT a nonunion risk factor?"	**→ sling vs figure-of-8**
"Best estimate of nonunion, displaced, nonop?"	**→ 15%**
"Risk factor in mountain biker?"	**→ Z-type pattern**

"FALSE except (TO)?"	→ **highest incidence of nonunion in middle third** ✗ Wrong options: FOOSH commonest · plexus with middle third · humerus commonest associated
"True about clavicle fracture (TO)?"	→ **increasing comminution indicates surgery**
"Comminuted displaced + bilateral plateau fractures?"	→ **ORIF plate**
"Nonop at 1 yr?"	→ **↓ strength + endurance**
"Figure-of-8 vs sling?"	→ **no advantage**
"2.6 cm shortening – lowest nonunion + malunion?"	→ **plate ORIF**
"Op vs nonop – true?"	→ **nonop ↑ symptomatic malunion / nonunion**
"COTS – true?"	→ **better Constant + DASH at all time points**
"ORIF outcome at 1 yr?"	→ **↑ functional scores**
"ORIF vs sling – nonunion + cost?"	→ **↓ nonunion, ↑ cost**
"Similar between op + nonop?"	→ **shoulder ROM**
"ORIF statement most appropriate (22-yo, > 100%)?"	→ **improves functional outcome** ✗ Wrong options: union > 95% nonop
"Primary goal of ORIF?"	→ **reduce nonunion**
"Pooled literature – plating gives?"	→ **↓ nonunion**
"FALSE about surgical fixation (TO)?"	→ **IM nail better union in comminuted**
"Anteroinferior plate disadvantage?"	→ **detach deltoid origin**
"Risk for reoperation after dual plating in thin runner?"	→ **low BMI**
"Single 3.5 superior vs dual mini?"	→ **single → more hardware removal**
"Painful nonunion at 9 months?"	→ **compression plating** (+ graft)
"Nonunion repair graft?"	→ **local ≈ iliac crest**
"Commonest association with open clavicle fracture?"	→ **closed head injury**
"Predicts late displacement of nonop fracture?"	→ **ipsilateral 1st–4th rib fractures**
"Arteriogram after clavicle fracture, weak pulses – vessel?"	→ **axillary artery**
"Most common complication of plating?"	→ **supraclavicular nerve numbness**

– – Active space – –

CORONOID FRACTURES

– Trauma · 10 / 67 –

1 Basics

Epidemiology

Frequency	**10–15% of elbow injuries**
Meaning	essentially **pathognomonic of an elbow instability episode**; isolated fractures less common than thought; usually with dislocation → recurrent instability

Anatomy & biomechanics

Coronoid	**anterior buttress of greater sigmoid notch** → resists **posterior subluxation**; primary resistor of subluxation / dislocation
Tip	**intra-articular**; **nothing inserts on it** · **anterior capsule 6 mm** distal to tip ✗ Wrong options: avulsion
Anteromedial facet	**varus stability**; critical weight-bearing ulnohumeral surface
Sublime tubercle	**anterior bundle of MUCL**, **18 mm** distal to tip → basal fractures amplify instability
LUCL insertion	**crista supinatoris** of proximal ulna
MUCL origin	anteroinferior **medial epicondyle**

2 Mechanism

Shear	distal humerus driven against coronoid during **severe varus stress or posterior subluxation**

Mechanism	Fracture	Associated	Instability
Varus + posteromedial rotation (pronation, axial load)	**Anteromedial facet**	**LCL avulsed from humerus**; **radial head + MCL spared**	**Varus PMRI**
Valgus + supination (posterolateral)	**Small tip (transverse)**	**Radial head fracture + LCL**, posterolateral dislocation	**PLRI / terrible triad**
Olecranon fracture-dislocation	**Large basal** fragment	Trans-olecranon / trans-ulnar basal	–

Anteromedial facet sequence	**LCL fails first**, then medial coronoid compressed against / under medial trochlea
Isolated coronoid fracture (no radial head fracture)	think **varus PMRI** ✗ Wrong options: PLRI · valgus anterolateral · posterior · anterior
Trans-ulnar basal coronoid fracture-dislocation	basal coronoid displaced from olecranon + metaphysis → **worse ROM, PROMs, more complications + reoperation**

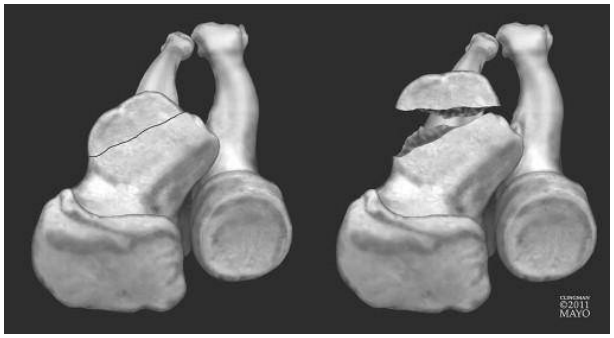
3 Presentation

Symptoms	pain, swelling, deformity; forearm / wrist pain → associated injury
PMRI clue	**crepitus / "crispy rice cereal" + pain moving the elbow with arm abducted** (varus stress); history of dislocation, clunk, spontaneous reduction; LCL tenderness
Gravity-assisted varus stress test	arm **abducted 90°**, neutral rotation, flex + extend → pain, grinding, instability → **most sensitive / specific for PMRI** ✗ Wrong options: lateral pivot shift · milking · chair rise · posterior drawer (PLRI / valgus tests)
Exam	varus / valgus stress (difficult but key), document ROM + crepitus, NV

4 Diagnosis & Classification

Key radiology findings

Radiographs	AP + lateral; overlap makes reading hard
PMRI signs	**lateral joint widening, medial joint collapse** on AP · **crescent sign** on lateral (depressed anteromedial facet)
CT	**next step for anteromedial facet fracture** (size, comminution, articular extent) ✗ Wrong options: EUA · traction views · immobilise 3 wk · immediate terrible-triad surgery

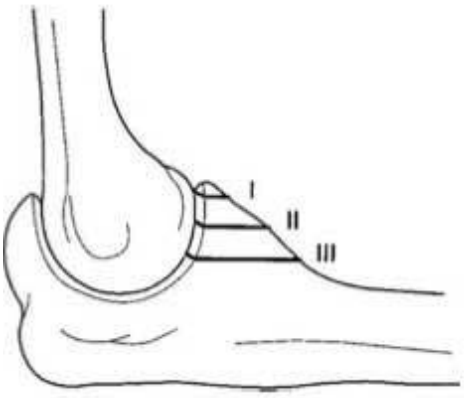


Anteromedial facet fracture on 3D CT

Regan & Morrey classification (height)

Type	Fragment
I	**Tip**
II	**≤ 50%** of height
III	**> 50%** of height

– – Active space – –



Regan and Morrey classification

O'Driscoll classification (location)

Type	Location	Association (TO)
I	Tip	**Terrible triad** (commonest)
II	Anteromedial facet	**Varus PMRI**
III	Base	**Olecranon / proximal ulna fracture-dislocation**

Anteromedial subtype (TO)	Involves
1	Anteromedial **rim** only
2	Rim + **tip**
3	Rim + **sublime tubercle** (± tip) → also **valgus instability** (MCL insertion)

5 Management

Coronoid fracture

- Minimally displaced + stable elbow (any type) → **brief splint, early ROM**
- Anteromedial facet (PMRI) → **medial approach ORIF + LCL repair**
  - still unstable → **hinged ex-fix**
- Terrible triad / olecranon fracture-dislocation → **fix coronoid (lateral / posterior)**

Operative criteria

Indication	Approach / fixation
Minimally displaced, stable elbow	**Nonop** – brief immobilisation, early ROM
Type I–III with persistent instability	**Medial ORIF**
PMRI (anteromedial facet) – most need surgery; residual subluxation / crepitus = dynamic incongruity	**Medial buttress plate / screws + LCL repair** (± sublime tubercle / MCL fix)
Olecranon fracture-dislocation, terrible triad	**Posterior approach**
Large fragments, poor bone, revisions	**Hinged external fixator**

Anteromedial facet – varus PMRI treatment

Medial approach	between **two heads of FCU**, or more anteriorly through **flexor-pronator split**
Fixation & medial approaches (TO)	tip: **suture anchor / transosseous** · anteromedial facet: buttress plate via medial approach · **Hotchkiss over-the-top** · **FCU split** (exposes ulnar nerve, also for MCL repair) · **Taylor & Scham**
Fixation by type	**type I tip → cerclage / No. 5 suture through ulnar drill holes** · **II / III → retrograde cannulated screws or plate** · **anteromedial facet → buttress plate / pins + lateral ligament repair**
Still unstable after coronoid ORIF	repair **LUCL at crista supinatoris** · LUCL injured in **83% (15/18)** ✗ Wrong options: MUCL origin · radial tuberosity = biceps · sublime tubercle (MCL rarely torn) · coronoid tip = capsule
Splinting	**LUCL torn (+ pivot shift) → forearm neutral** · **LUCL intact → full supination** (tightens medial side)
Untreated / nonunion	**varus laxity** persists → severe progressive arthritis ✗ Wrong options: valgus laxity (MUCL intact) · milking · rotation block (radial head fine) · ulnar paraesthesia
Rehab	thermoplastic splint 90°, forearm neutral; **block terminal 30° extension 2–4 wk**; **avoid shoulder abduction 4–6 wk** (varus moment); early active motion (dynamic compression of joint)

Terrible triad – coronoid tip & LCL

Terrible triad	**posterior / posterolateral dislocation + radial head fracture + coronoid (tip, transverse) fracture**; valgus + supination; MCL fails last if at all
Coronoid tip fracture associated with	**radial head fracture + posterolateral ulnohumeral dislocation** ✗ Wrong options: PMRI · capitellum · trans-olecranon · MCL rupture
Terrible triad, 50% coronoid, radial head replaced + LCL repaired → redislocates	**coronoid not fixed (from lateral approach)** · fix via **ulnar bone-tunnel sutures or dorsal → volar screws** ✗ Wrong options: immobilisation length / position (flexion + pronation stabilises lateral side) · medial buttress plate · MCL repair

Olecranon fracture-dislocation

Posterior approach	mobilise olecranon fragment to reach coronoid → **fix coronoid first**, then ulna with dorsal plate

6 Complications

Recurrent instability (esp. medial) · **stiffness** · **PTOA** · **HO** · **early failure** = unrecognised / unrepaired instability

Prognosis	high complication + reoperation rates

– – Active space – –

X vs Y

	Anteromedial facet (PMRI)	Tip (terrible triad, PLRI)
Force	**Varus + posteromedial**	**Valgus + supination**
Radial head	**Intact**	**Fractured**
Ligament	**LCL (humeral avulsion)**	LCL ± MCL
Test	**Gravity-assisted varus stress**	Lateral pivot shift, chair rise, posterior drawer
Approach	**Medial** (buttress) + LCL repair	Lateral (± posterior)

	Regan-Morrey	O'Driscoll
Basis	**Height** (tip, ≤ 50%, > 50%)	**Location** (tip, anteromedial, base)
Strength	Simple	**Recognises anteromedial facet**

How it's asked

"Dislocation, subtle coronoid, intact radial head, CT anteromedial – association?"	→ **varus PMRI** (TO)
"Anteromedial facet fracture, intact radial head – exam finding?"	→ **varus PMRI**
"Varus + posteromedial mechanism causes?"	→ **anteromedial coronoid fracture**
"Posteromedial instability 2 months after dislocation – fracture?"	→ **anteromedial facet**
"Isolated coronoid fracture – instability?"	→ **varus posteromedial**
"Mechanism for anteromedial facet?"	→ **varus + posteromedial rotation**
"Provocative test likely positive?"	→ **gravity-assisted varus stress**
"Valgus stress + milking positive, pivot shift negative, splinted supinated?"	→ **PMRI with intact LUCL**
"Injury most leading to PMRI?"	→ **anteromedial facet fracture**
"PMRI – fracture needing bony + soft-tissue fixation?"	→ **anteromedial facet**
"Nonunion of anteromedial facet – persistent sign?"	→ **varus laxity**
"Coronoid type in terrible triad?"	→ **tip** (TO)
"Terrible triad redislocates after RHA + LCL repair?"	→ **coronoid not fixed (lateral approach)**
"Coronoid tip on CT after valgus-supination fall – associated?"	→ **radial head fracture + posterolateral dislocation**
"Unstable after anteromedial facet ORIF – next insertion to address?"	→ **crista supinatoris (LUCL)**
"Anteromedial facet on X-ray – next step?"	→ **CT**

DISTAL FEMUR FRACTURES

– Trauma · 11 / 67 –

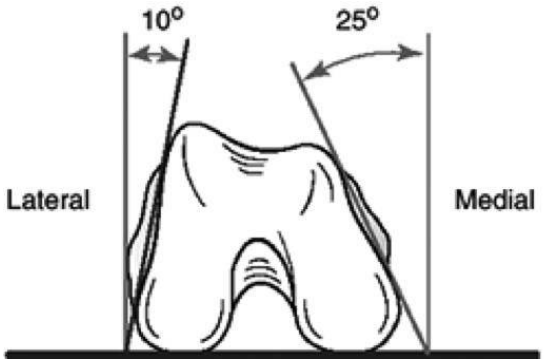
1 Basics

Epidemiology

Frequency	**3–6% of femur fractures** · < 1% of all fractures
Bimodal	**young males, high energy** (displaced) · **elderly osteopenic females, low-energy fall**
Open	**5–10%** of supracondylar fractures
Definition (TO)	fracture within the **transcondylar width** (square drawn on medial + lateral epicondyles)

Anatomy

Anatomic axis	**6–11° valgus**; medial condyle extends further distal
Trapezoidal cross-section	**lateral cortex slopes ~10°**, **medial cortex ~25°** → straight AP hides long medial screws → check with **AP in 30° internal rotation**
Condyles	posterior halves lie **behind** posterior shaft cortex
Adductor hiatus (TO)	femoral vessels pass posteriorly **10–12 cm above knee** → respect zone in **medial plating**
Posterior spike of distal fragment (TO)	may injure **popliteal artery**



Trapezoidal distal femur

Deforming forces

Muscle	Deformity
Quadriceps + hamstrings	**Shortening**
Adductor magnus	**Varus or valgus** (depends on comminution vs adductor tubercle)
Gastrocnemius	**Extension / apex-posterior** (recurvatum) + condylar rotation with intercondylar split

– – Active space – –

2 Mechanism

Young	high energy, displaced · **Elderly** – fall from standing, osteoporotic, less displaced

3 Presentation

Exam	distal thigh tenderness, swelling, varus / valgus deformity, effusion if intra-articular, open wounds
Vascular	**popliteal artery** at risk with displacement; femoral vessels tethered at **adductor hiatus**

Vascular assessment (ABI)

ABI	Meaning
> 0.9	**NPV 99%** (Orthobullets, OB) · **100%** (one OB explanation) → serial exams
< 0.9	**95% sensitive, 97% specific** for major arterial injury → **angiography / CTA**
0.9–1.3	Normal

Pulse returns after traction but "feels different"	**ABI** next ✗ Wrong options: serial exams (inaccurate) · angiography first (time / dye risk) · duplex (operator / availability) · compartment pressures
GSW, ABI 0.81	**CTA**
Angiography also for hard signs	pulselessness, expanding haematoma, massive bleeding

4 Diagnosis & Classification

Key radiology findings

Study	Use
AP + lateral	Standard
Traction views	Characterise (painful)
Whole femur, contralateral femur	Associated injuries; templating
CT	**Intra-articular extension**, **Hoffa**, notch osteochondral fragments; after ex-fix
Elderly	Look for **pre-existing knee OA**

Commonest intra-articular split	**sagittal**

Hoffa (coronal-plane) fractures

Fact	Value
Incidence in supra- / intercondylar fractures	**38%** (X-ray, Orthobullets) · **~40%** (CT, OB)
Condyle	**Lateral 80%**
Missed on X-ray	**up to 31%** (~30%)
Missed on X-ray (TO)	**30–40%**
Best seen	**Lateral view / CT** (✗ Wrong options: AP)
Association	**Open fractures ~3× more likely**; high energy (✗ Wrong options: ground-level fall)
OTA	**33B3**
Fixation	**Anterior → posterior lag screws across lateral condyle** (interfragmentary)
Implant to avoid	**95° blade plate / DCS contraindicated**

CT after motorcycle supracondylar fracture — likely finding

coronal fracture of lateral femoral condyle
✗ Wrong options: tibial eminence · sagittal MFC · Schatzker I · axial MFC



Hoffa fracture on lateral radiograph

AO / OTA 33 classification

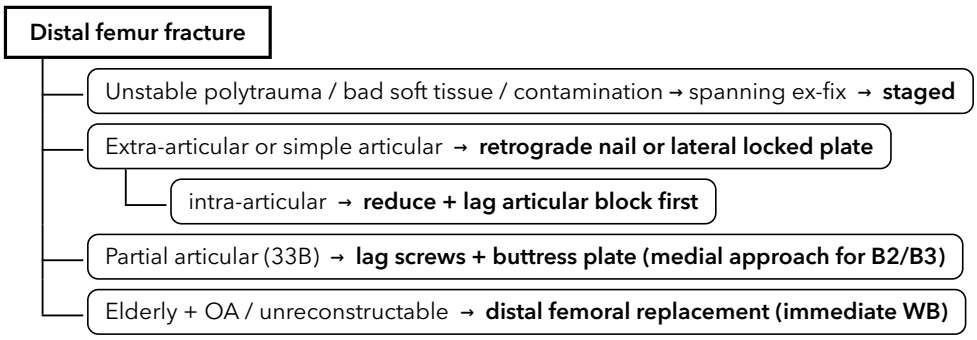
Type	Definition
33A	**Extra-articular**
33B	**Partial articular** — part of joint still attached to shaft; **B3 = coronal (Hoffa)**
33C	**Complete articular** — articular block separated from shaft
Subgroups (TO)	**A1** simple · A2 wedge · A3 comminuted · **B1 lateral condyle** · **B2 medial condyle** · **B3 Hoffa** · C1 simple · **C2 metaphyseal comminution** · **C3 articular comminution**

– – Active space – –



AO/OTA classification of distal femur fractures

5 Management



Treatment options

Option	Indications	Key facts
Hinged knee brace (rare)	Stable non-displaced, non-ambulatory, prohibitive risk	6–8 wk; ROM 3–4 wk; stiffness, wounds
Ex-fix	Temporising (polytrauma, soft tissue, contamination); definitive for severe open / comminuted	Definitive union **92–100% at 4–6 mo**; keep pins out of plate zone; narrow anterior safe zone
ORIF	Displaced, intra-articular, **osteoporotic periprosthetic**, nonunion	**Fixed-angle for metaphyseal comminution** (non-fixed → **varus collapse**); nonunion up to 18%
Retrograde IMN	**Extra-articular**, **simple intra-articular**, periprosthetic with **open-box** femoral component	**≥ 4 cm intact distal femur** (traditional); more symmetric callus, ↓ malunion, ↑ satisfaction
TKA / DFR	**Pre-existing OA**, **unreconstructable**, **loose TKA**, low demand	**Immediate WB**, better ambulation, ↓ nonunion; same complications + reoperations; ↓ longevity

Up to 20% wheelchair-bound at 1 yr after internal fixation (elderly)

Locked plating principles & implant choice

Locking plates — screw–plate interface (angular stability), no bone–plate contact needed → **comminuted metaphyseal / diaphyseal**, **short segments**, **osteoporotic periprosthetic**
✗ Wrong options: simple diaphyseal · displaced articular unless lagged first

Lag first, then lock — locked screws can't reduce or compress → **reduce + lag articular block, then attach to shaft**

Osteoporotic 68-yo, metaphyseal comminution + sagittal split — **lag screws then locking plate**
✗ Wrong options: non-locking plate · retrograde nail · ex-fix

Lateral locking plate NOT for — **33B partial articular (isolated medial condyle) → medial approach, lag + buttress plate**

Implant	Fact
Locking compression plate	**Best control of coronal-plane + small distal segments**; multiplanar screws for Hoffa; may be **too stiff → nonunion / failure**
95° angled blade plate	Precise insertion needed; **greater torsional strength** than LCP but **more subsidence, lower load to failure**; not C3 or Hoffa; poor in osteoporosis
Dynamic condylar screw	Sagittal alignment forgiving (rotates on barrel); **removes most bone**
Dual plating (lateral + medial)	**Greatest axial + torsional stiffness** – best for **medial comminution in osteoporosis**
Nail-plate combination	↓ **reoperation** vs lateral plate alone (LLP nonunion ~20%); same infection; allows immediate WB

Plate failure risk — **short working length**, plate-screw density **> 0.5**, short plates, non-locked diaphyseal screws, **stainless steel may be worse than titanium**, improper bridging
✗ Wrong options: early ROM · bridging · density < 0.5

Percutaneous submuscular precontoured locking plate — ↓ infection + nonunion, preserves perfusion, **but ↑ malreduction (rotational)**

Intercondylar screw length — **AP in 30° internal rotation**
✗ Wrong options: ER view · full extension AP

Implant facts (TO) — **blade plate** technically demanding, **obsolete** · **DCS** = **12.5 mm** lag screw, bone-consuming, needs **≥ 4 cm intact distal fragment** · **distal femoral locking plate = gold standard** (combi holes)

Plating tips (TO) — **distal-most screw parallel to joint surface** · **anterior screws short** (trapezoid) · **non-locking before locking** · correct apex-posterior angulation with **knee flexion + bolster** · K-wire / Steinmann joysticks

Far cortical locking (TO) — ↓ construct stiffness → ↑ callus

Ex-fix (TO) — DCO, Gustilo III, blisters / swelling, **vascular repair** · pins **away from future incisions**

– – Active space – –

Approaches

Approach	Use
Lateral	All types; arthrotomy for joint
Swashbuckler (modified anterior)	Extra-articular / lateral; submuscular plate
Lateral parapatellar	**Complex articular extension**, Hoffa; evert patella
Anterolateral (Swashbuckler) (TO)	Combines lateral plating access + parapatellar articular view; lateral parapatellar = best articular view (TO)
Medial parapatellar	Complex **medial condyle**
Medial	**B2, B3**; medial plate augment in C3
Posterior (prone, between gastrocnemius heads)	Very posterior Hoffa

Retrograde nail technique

Start	**superior margin of Blumensaat line (lateral)**, **centre of notch (AP)**; knee flexion **≥ 70°**
Start (TO)	AP: **just anterior + medial to PCL attachment** · lateral: **apex of Blumensaat line**
Approach	patellar tendon split or medial parapatellar (extend 2–8 cm if articular)
Rules	fix articular fragments first (lag screws out of nail path); **long nail to lesser trochanter** (short nails rarely; vascular risk)
Blocking (Poller) screws	narrow the metaphysis; place in **proximal part of distal segment** on **concave side**: **varus → AP screw medial to nail** · **recurvatum → ML screw anterior to nail** · valgus → lateral · procurvatum → posterior
Gastrocnemius	recurvatum + posterior translation; wide metaphysis → varus / valgus

Elderly & periprosthetic

Retrograde nail contraindicated	**distal femoral replacement** (long stem) ✗ Wrong options: PS knee (technically harder) · CR · spiral · **flexion** contracture helps; **extension** contracture blocks access
Periprosthetic nail	box needs **posterior start → recurvatum** risk
Elderly intra-articular + pre-existing OA (cane user)	**DFR** ✗ Wrong options: nail-plate · dual plates · lateral plate · stemmed TKA
Lateral locked plate, extra-articular elderly	**WBAT** immediately (no ↑ failure / reoperation) ✗ Wrong options: NWB · TDWB · partial
Plate vs nail, elderly extra-articular + immediate WB	**no difference in function**; no shorter op / stay

6 Complications

Complication	Fact
Knee pain / stiffness	Early ROM, PT; **nail → anterior knee pain**
Symptomatic hardware	**Lateral plate → IT band** pain with flexion / extension · **long medial screws**
Malunion	After plating: **rotation, recurvatum, coronal**; **"golf-club" deformity = plate too posterior distally → medial translation of articular block**; percutaneous plate → **rotational**; **malalignment commoner with nails**; acceptable < 5° any plane
Nonunion	**Up to 19%** (metaphyseal; articular part heals) → **revision ORIF + autograft** ± change fixation
Infection	Diabetics with foot ulcers → debride, antibiotics
Implant failure	Up to 9%
Loss of fixation	**Varus collapse (commonest)** – toggling non-fixed-angle screws; proximal screw failure with short plates / non-locked diaphysis

Oligotrophic supracondylar nonunion

open reduction + plating + autograft
✗ Wrong options: nail + BMP · ex-fix + BMP · percutaneous plate + BMP · calcium phosphate

External fixator pin placement

Anterior femoral half-pin safe zone

narrow: **femoral nerve branches** up to **58 ± 36 mm (0–136)** below lesser trochanter (174 ± 43 mm from ASIS) · **suprapatellar pouch** reaches **46 ± 13 mm (20–74)** above patella (up to 7.4 cm) → avoid **intra-articular pins (infection)**

X vs Y

	Lateral locked plate	Retrograde nail	DFR
Best for	Comminuted metaphyseal, periprosthetic, articular (after lagging)	Extra-articular, simple articular, open-box TKA	Elderly + OA, unreconstructable, loose TKA
Problem	Varus collapse, rotation malunion, IT band, nonunion ~20%	Malalignment, knee pain	Longevity, mechanical failure
WB	WBAT OK	WBAT OK	**Immediate**

	Blade plate	DCS	LCP
Insertion	Precise 3-plane	Forgiving sagittal	Multiplanar
Bone removed	–	**Most**	–
Coronal / Hoffa control	No	No	**Yes**

– – Active space – –

How it's asked

"Pre-op CT likely shows?"	→ **coronal (Hoffa) fracture of lateral condyle**
"Commonest Hoffa — screw orientation?"	→ **A→P across lateral condyle**
"Hoffa — true?"	→ **commoner in open fractures**
"Implant controlling coronal fractures best?"	→ **locking plate**
"Most appropriate use of locking plates?"	→ **osteoporotic periprosthetic distal femur**
"Lateral locking plate NOT appropriate for?"	→ **partial articular (33B) medial condyle**
"Osteoporotic comminuted metaphyseal + sagittal split?"	→ **lag screws then locking plate**
"Check intercondylar screw length?"	→ **AP with leg 30° internally rotated**
"Submuscular locking plate ↑ risk of?"	→ **malreduction (rotational)**
"Lateral plate failure risk factor?"	→ **short working length**
"Varus + recurvatum during nailing — blocking screws?"	→ **AP medial to nail + ML anterior to nail in distal segment**
"Highest axial stability, medial comminution, osteoporosis?"	→ **lateral locked + medial plate**
"Nail-plate vs lateral plate?"	→ **↓ reoperation**
"Contraindication to retrograde nail around TKA?"	→ **distal femoral replacement**
"WB after lateral locked plate, 73-yo?"	→ **WBAT**
"Elderly intra-articular + OA?"	→ **DFR**
"Plate + immediate WB vs nail in elderly?"	→ **no difference in function**
"Oligotrophic nonunion?"	→ **plate + autograft**
"Pulse back after traction but different?"	→ **ABI**
"ABI requiring CTA?"	→ **0.81**
"Safe anterior ex-fix pin site?"	→ **mid-femur, clear of femoral nerve branches + suprapatellar pouch**
"DCS needs minimum intact distal fragment?"	→ **4 cm** (TO)
"Distal-most screw in distal femoral plate?"	→ **parallel to articular surface** (TO)

DISTAL HUMERUS FRACTURES

– Trauma · 12 / 67 –

1 Basics

Epidemiology

Frequency	**2% of all fractures** · **30% of elbow fractures** · rising
Who	young males (high energy) · older females (low-energy falls)
Commonest pattern	**distal intercondylar (bicolumnar)**
Associated	elbow dislocation, terrible triad, floating elbow, **Volkmann contracture** (missed forearm compartment syndrome)
AO definition (TO)	fracture centred within a **square** whose side = distance between the epicondyles · bimodal: **young males 11–20 y** (high energy) · **elderly females >70 y** (osteoporotic)

Anatomy

Articular orientation	**6° valgus, 5° ER, 30° flexion** (anterior)
Trochlea ↔ sigmoid notch (flex / ext) · **capitellum** ↔ radial head (rotation)	
Origins	**medial epicondyle = flexor-pronators** (PT, FCR, PL, FDS, FCU) · **lateral = extensors** (anconeus, ECRL, ECRB, EDC, EDM, ECU)
MCL anterior bundle	medial epicondyle → **sublime tubercle**; primary valgus restraint **30–120°**; tight in pronation
LCL	lateral epicondyle → **crista supinatoris**; vs PLRI; tight in supination
Ulnar nerve	subcutaneous in cubital tunnel behind medial condyle

Radial nerve landmarks (posterior approach)

Landmark	Distance
Spiral groove	**15 cm** proximal to distal articular surface
From medial olecranon edge	**10 cm**
From lateral olecranon edge	**7.5 cm**
Proximal to triceps aponeurosis	**3.9 cm (≈ 2 finger-breadths)**
Pierces lateral intermuscular septum	**~14 cm** above lateral epicondyle (OB)

Course	between **BR + brachialis** above elbow → divides into **PIN + superficial radial** at radial head
PABCN	radial sensory branch **superficial to lateral intermuscular septum → posterior distal arm + forearm sensation**; trace it proximally to find radial nerve ✗ Wrong options: LABCN = musculocutaneous · MABCN = medial cord · SBRN = dorsoradial hand

– – Active space – –

2 Mechanism

Elbow flexed < 90° + axial load	transcolumnar fracture · direct posterior blow → olecranon ± distal humerus
Elbow flexed > 90°	**intercondylar** fracture

3 Presentation

Exam	open wounds (**posterior**), gross instability (**avoid ROM**), radial / ulnar / median function, **brachial artery** (↓ pulse → non-invasive studies → vascular surgery), **forearm compartment syndrome**

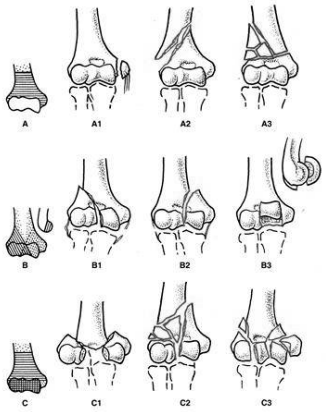
4 Diagnosis & Classification

Key radiology findings

View	Use
AP (≈ 40° flexion) + lateral	Standard
Humerus, forearm, **wrist**	Associated injuries
Oblique	Trochlea continuity with medial epicondyle → implant choice
Traction radiographs	Comminuted, telescoped / flexed fragments → approach, **ORIF vs TEA**, 3rd plate, tricortical graft (✗ Wrong options: oblique · flexion / extension films)
CT	Planning; shear fractures of capitellum + trochlea
MRI	Not indicated acutely

OTA classification

Type	Definition
A	**Extra-articular** – supracondylar (80% extension type), epicondyle
B	**Partial articular** – single column, coronal shear, epicondyle with articular extension
C	**Complete articular** – both columns, no joint contiguous with shaft



OTA classification of distal humerus fractures

Milch classification (single column – 5%, lateral > medial)

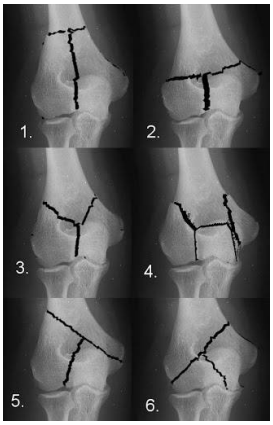
Type	Lateral trochlear ridge	Treatment
I	**Intact** (stable)	Non-displaced → **cast** · displaced → **CRPP**
II	**Fractured** through ridge (unstable)	**ORIF**

Casting **supination for lateral condyle**, **pronation for medial condyle**

Jupiter classification (bicolumnar)

Pattern	Feature
High-T	Transverse at / above olecranon fossa
Low-T	Just above trochlea (**common**)
Y	Oblique through both columns + vertical distal line
H	**Trochlea free fragment → AVN risk**
Medial / lateral lambda	Proximal line medial / lateral
Multiplane T	+ coronal fracture

5 articular fragments capitellum / lateral trochlea, lateral epicondyle, posterolateral epicondyle, posterior trochlea, medial trochlea / epicondyle



Jupiter classification of bicolumnar fractures

5 Management

Distal humerus fracture

- Non-displaced Milch I → cast · displaced Milch I → **CRPP**
- Supracondylar / bicolumnar / Milch II, adult → **ORIF, dual plates**
  - complex articular / coronal split → **olecranon osteotomy**
- Elderly, comminuted, osteoporotic / RA, low demand → **semiconstrained TEA**
- Unfit / can't comply → **"bag of bones"**

– – Active space – –

Exposure – approaches

Approach	Use / facts
Triceps-splitting (Campbell)	Midline split to olecranon; open fractures → use the wound; exposes **35%** of joint
Triceps-sparing (paratricipital, Alonso-Llames)	**Extra-articular + simple articular split**; medial + lateral windows; convertible to osteotomy
Olecranon osteotomy	**Complex intra-articular, coronal split**; **most exposure (57%)**
Triceps-reflecting (Bryan-Morrey)	Medial → lateral sleeve; transosseous repair, protect 4–6 wk; **46%** exposure; good medial, poor lateral view
TRAP (O'Driscoll)	Triceps + anconeus V-flap; repair via drill holes; no early mobilisation
Lateral muscle interval	Lift ECRB + part ECRL off supracondylar ridge

Young, complex articular (motorcycle) — **olecranon osteotomy + parallel plating** (✗ Wrong options: Bryan-Morrey · TRAP) · or posterior extensile + **chevron** osteotomy (✗ Wrong options: Henry · Kocher · medial column · CRPP)

TO approach notes — triceps tongue (**Van Gorder**) less used · TRAP preserves anconeus supply · if **TEA planned → lateral para-olecranon (modified paratricipital), avoid olecranon osteotomy** (TO)

Olecranon osteotomy

Technique — **chevron, apex distal**, ~**2 cm from tip**, through **bare area**; fix with **TBW, screws, plate**, or **intramedullary cancellous screw angled slightly medially** (proximal ulna varus bow)

TEA after osteotomy — **not contraindicated**; triceps-sparing preferred if TEA possible, but **all posterior approaches** (paratricipital, split, tongue, osteotomy) allow TEA

Complication	Rate
Symptomatic hardware – **most common**	**6–30%** (Orthobullets) · **8–33%** (OB) · removal up to **70%** in 90/90 + osteotomy series
Osteotomy nonunion	**0–9%** (Orthobullets) · **0–2%** (OB); healing 95–100%
Fixation failure	~5%
AIN injury (K-wires through anterior ulnar cortex)	**Can't flex thumb IP** – check in recovery (✗ Wrong options: wrist flexion · thumb sensation (AIN has none) · dorsal web · index abduction)

Osteotomy + TBW vs other fixation — **↑ AIN injury**; **not** ↑ ulnar neuritis

Fixation constructs – parallel vs orthogonal plating

O'Driscoll principles — **every distal screw through a plate**; **screws as long as possible**, engaging many articular fragments; **interdigitate** screws from both plates; **compress both columns** at supracondylar level

Sequence provisional K-wires → articular fragments with **countersunk / headless screws** (use **positional, not lag, screws across trochlea** to avoid narrowing) → plates to shaft

Construct	Fact
Parallel (medial + lateral, 180°)	**Greatest rigidity** – **medial + lateral LC-DCP** stiffest
Orthogonal (90-90)	**Posterolateral + medial**; clinical outcomes **= parallel**
1/3 tubular plates	**Too weak** → failure + nonunion
Single plate (Y, medial)	Weaker than any dual construct
Locking plates	Poor bone, comminution

50-yo displaced intra-articular **dual column plates**
✗ Wrong options: TEA · brace · cast · K-wires

Adult bicolumnar (any age except frail) dual plating (90° or 180°)

Post-op splint **70° flexion** → ROM by 7–10 d · osteotomy: active / active-assisted, **no active extension against gravity / resistance for 6 wk** · strengthening 6 wk, full 3 mo

Reduction tips (TO) **no screws in olecranon fossa** (impingement) · min 2 plates, occasionally triple · immobilise **< 3 wk**

Ulnar nerve dysfunction

Transposition at ORIF **no benefit**; **↑ ulnar neuritis** (33% vs 9%) **(TO: no effect or increased)**
✗ Wrong options: decreased

In situ release vs transposition **in situ → less ulnar neuritis**; PROMs equal

Transpose only if nerve **on stretch or touching implants**, subluxing, recurrent

Pre-existing cubital tunnel symptoms + H-type fracture **parallel plating (all screws through plate) + in situ cubital tunnel release**
✗ Wrong options: orthogonal + submuscular · observe neuropathy · lag screws outside plate + subcutaneous

Persistent ulnar symptoms after in situ release, no stretch / contact **observe + anti-claw splint**
✗ Wrong options: EMG acutely · revision transposition · Guyon release

Post-op ulnar palsy mostly **traction** during ORIF

Elderly / osteoporotic – TEA vs ORIF

TEA indication **elderly (> 65–70), low demand, osteoporotic, comminuted bicolumnar / distal articular fractures not amenable to stable fixation**; **RA**, **chronic steroids**; must accept **lifting limits**

Implant **semiconstrained (linked)** – collaterals need not heal; radial head not needed
✗ Wrong options: unconstrained in RA

– – Active space – –

Lifting limit **10 lb** (Orthobullets) · **5 lb** (OB) · **5–10 lb lifelong** (OB) — read the options

ORIF vs TEA (elderly)	Finding
Reoperation	**ORIF higher (27% vs 12%)** (OB) · similar frequency (OB) · some studies higher with TEA (OB) — read the options
Early PROMs / short-term function	**TEA better**
ROM	TEA greater (Orthobullets, OB) · similar (OB)
Complications	TEA fewer (OB) · TEA more frequent (OB)
Operative time	**TEA shorter**
Nonunion	**Not a TEA complication**
Ulnar nerve dysfunction	Both

TEA facts **> 75% survival at 10 yr**; ROM **26–125°**; commonest complication **loosening / bushing wear (10–15%)**; **men** revised more; **salvage TEA after failed ORIF ≈ primary TEA**; complications: loosening, poly wear, periprosthetic fracture

Failed ORIF in 83-yo (screws cut out at 5 wk) **elbow arthroplasty**
✗ Wrong options: revision ORIF · bag of bones · arthrodesis

Not TEA candidates young labourer open T · transolecranon fracture-dislocation · coronal shear · nonunion in 50-yo

"Bag of bones" brief immobilisation then ROM for unfit / non-compliant elderly → painless pseudarthrosis, fair function, high nonunion
✗ Wrong options: **immediate** ROM

Extra-articular distal-third shaft – functional bracing

Brace vs surgery **similar small loss of elbow motion** (6° vs 3°, NS); 100% union; malalignment without functional limits; shoulder motion within 10°

6 Complications

Complication	Fact
Stiffness	**Most common** (3–42%); mean arc 90–106°; static-progressive splint
HO	**8%** (Orthobullets) · ~4% (OB); no routine indomethacin (↑ nonunion); risk: **head injury**, floating elbow, type A + B, delayed fixation
Nonunion	**0–11%** (2–10%, OB); stripping, open, comminution, **low transcondylar**; → revision ORIF + graft
Malunion	Cubitus valgus (lateral column) / varus (medial column)
AIN injury	Olecranon osteotomy
Ulnar nerve	**10–38%** (Orthobullets) · **0–12%** (OB)
Wound	Up to 16%
Infection	0–14%
PTOA	–

Stiffness, contracture release & HO

Most common complication after ORIF	**↓ ROM / stiffness** → subsequent **open release of capsule + posterior bundle of MUCL** ✗ Wrong options: revision for malunion / nonunion · hardware removal · HO removal alone
Limited flexion (30–90° or 10–80°)	**open release of posterior bundle of MCL + posterior capsule + excise osteophytes / HO** · consider ulnar transposition if flexion < 90° ✗ Wrong options: release **anterior** bundle → valgus instability · complete MCL + LCL release · anterior arthroscopic release (that limits extension) · dome osteotomy · indomethacin / radiation (prophylaxis only)
Expected after ORIF	**~25% loss of flexion strength**; full ROM rare; most regain ~75% motion + strength; goal **30–130°**
Unsatisfactory	up to 25% (low fracture lines, metaphyseal fragmentation, articular comminution, poor bone)

Radial nerve & branches

Neurapraxia on NCV / EMG at 6 wk	**normal amplitude + velocity in superficial radial sensory nerve** (no Wallerian degeneration) ✗ Wrong options: LABCN · FCU · EDC fibrillations = worse injury · triceps MUPs
Recovery order	**proximal → distal**: **BR first … EIP last**
Laceration 2 finger-breadths above triceps aponeurosis	radial nerve → **can't extend index finger** ✗ Wrong options: thumb adduction = ulnar · small DIP = ulnar · thumb IP = AIN · pronation = median

X vs Y

	Parallel plating	**Orthogonal (90-90)**
Plates	Medial + lateral	Medial + posterolateral
Rigidity	**Higher**	Lower
Clinical outcome	Same	Same

	ORIF	**TEA (elderly)**
Who	Adults, good bone	Elderly, osteoporotic, comminuted, RA
Nonunion risk	Yes	**None**
Early function	Lower	**Better**
Restriction	–	**5–10 lb lifelong**

	In situ ulnar release	**Anterior transposition**
Post-op ulnar neuritis	**Lower**	**Higher / no benefit** (TO)

How it's asked

"85-yo osteoporotic comminuted, low-level ambulator?" → **TEA**

– – Active space – –

"92-yo comminuted, non-dominant?"	→ TEA
"Most appropriate TEA candidate?"	→ 86-yo comminuted bicolumnar
"71-yo on steroids, comminuted intra-articular?"	→ TEA
"ORIF vs TEA in 79-yo?"	→ ORIF ↑ reoperation
"TEA in RA patient – true?"	→ lifelong 5–10 lb lifting limit
"TEA vs ORIF – true?"	→ TEA better early PROMs
"Treatment with loosening risk – true?"	→ no risk of nonunion
"TEA – true?"	→ > 75% survival at 10 yr
"RA + steroids, comminuted – best?"	→ semiconstrained TEA
"Failed ORIF at 5 wk in 83-yo?"	→ elbow arthroplasty
"TEA vs ORIF in 86-yo – true?"	→ TEA better short-term function
"Approaches allowing TEA if ORIF abandoned?"	→ all posterior approaches
"50-yo intra-articular displaced?"	→ dual column plates
"Stiffest construct?"	→ parallel medial + lateral LC-DCP
"H-type + chronic ulnar symptoms?"	→ parallel plating + in situ release
"Osteotomy + TBW vs other – true?"	→ ↑ AIN injury
"AIN deficit after osteotomy – exam?"	→ can't flex thumb IP
"Best visualisation + early ROM, young complex?"	→ olecranon osteotomy + parallel plating
"Osteotomy characteristics?"	→ chevron apex distal, 2 cm from tip, screw angled medially
"Most common complication with 90/90 + osteotomy?"	→ pain from olecranon fixation
"Most common complication of osteotomy exposure?"	→ symptomatic hardware
"Intra-articular fracture – approach?"	→ posterior extensile + chevron osteotomy
"Most common complication after ORIF?"	→ ↓ elbow ROM
"Arc 30–90° with posteromedial HO – improve flexion?"	→ HO excision + release posterior band of UCL
"Arc 10–80° after nonunion repair?"	→ release posterior bundle MCL + osteophytes
"Expected outcome after ORIF?"	→ ~25% loss of flexion strength
"Subsequent surgery for commonest complication?"	→ capsule + posterior MUCL release
"Neurapraxia finding at 6 wk?"	→ normal superficial radial sensory conduction
"Radial palsy after posterior approach – last to recover?"	→ EIP
"Nerve 2 finger-breadths above triceps aponeurosis cut?"	→ can't extend index finger

"Branch superficial to lateral septum, 14 cm above epicondyle?"	→ **PABCN — posterior arm + forearm sensation**
"Anterior transposition effect?"	→ **no effect or ↑ ulnar neuritis** (TO)
"Ulnar symptoms persist after in situ release, no stretch?"	→ **observe + anti-claw splint**
"Extra radiographs that realign fragments?"	→ **traction views**
"Bracing extra-articular distal-third shaft vs surgery?"	→ **similar loss of elbow motion**

– – Active space – –

– – Active space – –

DISTAL RADIAL ULNAR JOINT (DRUJ) INJURIES

– Trauma · 13 / 67 –

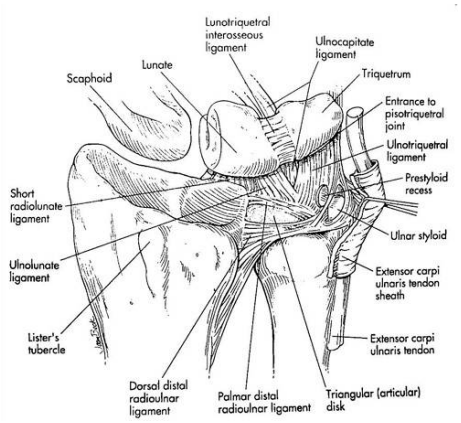
1 Basics

Overview

Why it matters	DRUJ injury with distal radius fractures is a common, **under-appreciated cause of pain + lost rotation** → assess it **independently** after fixing the radius
Associated	**ulnar styloid / distal ulna fractures**, **TFCC tears**, **ulnar impaction**, **Essex-Lopresti**, **Galeazzi**

Anatomy

Joint	ulnar head in **sigmoid notch** (shallow); motion = rotation + translation
Most stable	**supination**
Primary stabilisers	**volar + dorsal radioulnar ligaments** + **TFCC** (inserts at **fovea** at ulnar styloid base)
Secondary / back-up primary	**distal oblique bundle (DOB) of interosseous membrane** → can keep forearm stable when TFCC torn ✗ Wrong options: annular ligament = PRUJ · central band (not DRUJ) · meniscal homologue · proximal oblique cord
TFCC components	central articular disc · meniscal homologue · volar + dorsal radioulnar ligaments · ulnolunate + ulnotriquetral origins · **ECU sheath floor**
Ligament roles	**pronation: volar ligaments block dorsal subluxation** · **supination: dorsal ligaments block volar subluxation**



TFCC anatomy

2 Mechanism

TFCC tear	wrist **extension + forearm pronation**

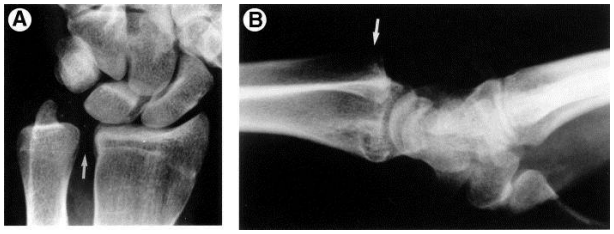
3 Presentation

Acute dislocation	pain + instability (often with open distal radius fractures)
Post-traumatic arthritis	dorsal wrist pain, ↓ **pronosupination**, snapping, crepitus, ↓ grip; pain on rotating with ulna compressed against radius

4 Diagnosis & Classification

Key radiology findings

Study	Finding
AP	**DRUJ widening**
Lateral	**Dorsal displacement** of ulnar head; instability = subluxed **full width** of sigmoid notch in neutral
Dynamic CT (neutral, full supination, full pronation, vs other side)	**> 50% translation = abnormal**; subtle chronic instability
MRI	**TFCC** tears



DRUJ instability on radiographs

Palmer TFCC classification

Type	Description
I	**Traumatic**
II – degenerative (ulnocarpal impaction)	**IIA** thinning · **IIB** + lunate / ulnar chondromalacia · **IIC** + perforation · **IID** + LT ligament tear · **IIE** + ulnocarpal + DRUJ arthritis

5 Management

DRUJ after distal radius / radial shaft fixation

- Stable → **no extra treatment (ulnar styloid ignored)**
- Unstable, ligamentous → reduce + splint (dorsal → supination, volar → **pronation) 4 wk**
  - highly unstable → **pin 0.062 K-wires ± ligament repair**
- Unstable + displaced styloid base / sigmoid notch / TFCC avulsion → **ORIF / repair**
- Irreducible after anatomic radius → ECU interposed → **open removal**

– – Active space – –

Ligamentous instability

Closed reduction + immobilise 4 wk	**dorsal instability → supination** · **volar instability → pronation**
Highly unstable	**pin DRUJ with 0.062-inch K-wires** ± radioulnar ligament repair
ECU interposition	blocks reduction

DRUJ instability after radius fixation – ulnar fixation & ECU

Incongruent DRUJ after anatomic radial shaft fixation (Galeazzi)	**interposed ECU** → extricate via dorsal approach ✗ Wrong options: revise plating · IM nail · pronator quadratus · FCU
Indication to fix distal ulna fracture	**intra-operative DRUJ instability** ✗ Wrong options: angulation, comminution, location, size alone
Galeazzi	distal-third radius fracture + DRUJ injury → **radial ORIF + DRUJ pinning** often; nonop (splint in supination) rarely

Ulnar styloid fractures

Meaning	marker of **high initial displacement**; **base fractures → TFCC rupture + instability**
Nonunion without instability	**does not worsen outcome** (least likely to affect function) ✗ Wrong options: DRUJ instability · step-off > 3 mm · radial shortening · workers' comp (all worsen)
Operative	**displaced base fracture + instability**, sigmoid notch fracture, Galeazzi, TFCC avulsion with unstable DRUJ → ORIF or excision **preserving TFCC ulnar attachment**
Non-displaced, proximal	cast

TFCC tears

Acute traumatic	immobilisation + NSAIDs first → arthroscopic / open debridement or repair if persistent
Degenerative (type II)	debride TFCC + treat impaction: **ulnar shortening osteotomy** (no DRUJ arthrosis) or **wafer resection**

Ulnar impaction syndrome

Cause	radial shortening → **positive ulnar variance** → **lunate chondromalacia + degenerative TFCC tear**
Surgery	TFCC debridement, radial osteotomy, **ulnar shortening**, **wafer (preserve TFCC attachment)**

Forearm stabilisers – Essex-Lopresti

Essex-Lopresti (longitudinal radioulnar dissociation)	**radial head fracture + interosseous membrane tear + DRUJ disruption** → **proximal radial migration** → DRUJ pathology + ulnocarpal abutment
Acute	fix / replace radial head (restore radiocapitellar contact + length) + **pin DRUJ 6 wk in neutral**; failed pinning / missed → radial head replacement

Chronic	restore radial length, level DRUJ, **reconstruct central band of IOM**
Ipsilateral radial head / neck + Galeazzi (functional Essex-Lopresti)	treatment causing chronic instability = **radial head excision** ✗ Wrong options: temporary ex-fix · radial head ORIF · arthroplasty · DR ORIF + cross-pinning

6 Complications

DRUJ arthrosis – salvage

Procedure	Key fact
Darrach (distal ulna excision)	**Low-demand elderly**; risk **unstable painful ulnar stump**
Hemiresection-interposition (Bowers)	Preserves **TFCC ulnar insertion**; soft tissue prevents impingement
Sauvé-Kapandji	**DRUJ fusion + proximal ulnar neck pseudarthrosis**; younger patients; keeps rotation
Ulnar head prosthesis	–
One-bone forearm	**Ultimate salvage**; **eliminates rotation**

Prevention **anatomic reduction of the distal radius** usually reduces the DRUJ



Sauvé-Kapandji procedure

X vs Y

	Darrach	Sauvé-Kapandji
Patient	Elderly, low demand	Younger
What	Distal ulna excised	DRUJ fused + proximal pseudarthrosis
Risk	**Unstable ulnar stump**	Painful proximal stump

	Dorsal DRUJ instability	Volar DRUJ instability
Stable position	**Supination**	**Pronation**

– – Active space – –

How it's asked

"DRUJ incongruent after anatomic radius fixation?"	→ **reduce interposed ECU**
"Indication for distal ulna fixation?"	→ **intra-operative DRUJ instability**
"Least likely to affect outcome after DR ORIF?"	→ **ulnar styloid nonunion**
"Treatment causing chronic instability in radial head + Galeazzi?"	→ **radial head excision**
"Gross forearm stability after radius fixation despite DRUJ injury – structure?"	→ **distal oblique bundle**
"Dynamic CT translation abnormal?"	→ **> 50% vs contralateral**
"Dorsal DRUJ dislocation – immobilise in?"	→ **supination**
"Essex-Lopresti – DRUJ?"	→ **pin 6 wk in neutral**
"Ultimate salvage for DRUJ?"	→ **one-bone forearm**

DISTAL RADIUS FRACTURES

– Trauma · 14 / 67 –

1 Basics

Epidemiology

Commonest orthopaedic injury	**17.5% of adult fractures**; lifetime risk **15% women, 2% men**; **F:M 2–3:1**
Bimodal	young high energy (MVC) · elderly low energy (FOOSH) = **fragility fracture → DEXA**
Pattern	**65% extra-articular**, 10% partial articular, 25% complete (Orthobullets) · **~50% intra-articular** (OB)
Predictor of later fractures	osteoporosis, women > 50
Associated soft-tissue injury (70%)	**TFCC 40%**, **SL 30%**, **LT 15%** · DRUJ injury · radial styloid fracture (higher energy)

Abbreviations

Tendons / nerves	EPL = extensor pollicis longus · EIP = extensor indicis proprius · FPL = flexor pollicis longus · EDC = extensor digitorum communis · PIN = posterior interosseous nerve · CTR = open carpal tunnel release

Anatomy

Load radius carries **80%** of axial load

Column	Contents	Roles
Radial	Styloid (**volar** structure) + scaphoid fossa	BR insertion, **long radiolunate** + **radioscaphocapitate** (blocks ulnar carpal translation); buttress; ulnar-deviation load
Intermediate	**Lunate fossa**	Load transfer; **short radiolunate** origin (radiocarpal stability)
Ulnar	TFCC + distal ulna	DRUJ stability, rotation

Three-column concept (Rikli & Regazzoni, TO)	lateral: radial styloid + scaphoid fossa · **intermediate: lunate facet + sigmoid notch** · medial: ulnar head + TFCC · **intermediate column injury = major predictor of instability**
Treatment principles "ARMS" (TO)	**A**rticular congruity · **R**adial alignment + length · **M**otion · **S**tability by preserving anatomical length
Lister tubercle	dorsal, ~1 cm above joint; **EPL just ulnar** → long volar screws rupture EPL
Watershed line	**most volar prominent rim** (lunate portion highest; short radiolunate) → **keep volar plate proximal**
Volar-ulnar ("critical") corner / volar lunate facet	radiolunate ligaments → loss → **volar carpal subluxation**

– – Active space – –

2 Mechanism

FOOSH (elderly) · MVC (young)

Chauffeur (Hutchinson) (TO)	**radial styloid** fracture from **scaphoid impacting styloid** ✗ Wrong options: ulnar styloid

3 Presentation

Exam	pain, swelling, **dinner-fork deformity** if displaced; median nerve check

4 Diagnosis & Classification

Key radiology findings

Measure	Normal	Acceptable (post-reduction)
Radial height (PA)	**13 mm** (Orthobullets) · **11 mm** (TO)	**< 5 mm shortening**
Radial inclination (PA)	**23°** (22° TO text)	**< 5° change**
Articular step-off	Congruous	**< 2 mm**
Volar tilt (lateral)	**11°**	**Dorsal angulation < 5° or within 20° of contralateral** (Orthobullets, TO) · **dorsal tilt < 10°** in healthy < 65 (OB)
Ulnar variance	Slightly negative	–

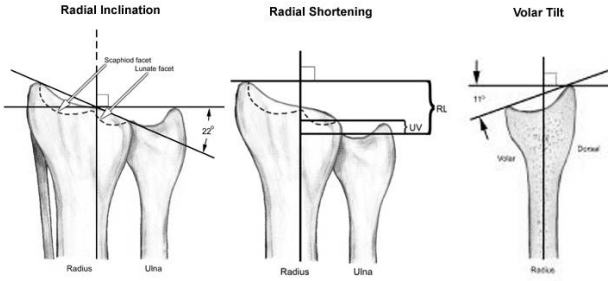
Extra parameter (TO)	Normal / meaning
Teardrop angle	**~70°** · ↑ = volar marginal fragment · **↓ = dorsally angulated fracture**
Dorsal vs volar rim (PA)	Dorsal rim projects **3–5 mm distal** to volar rim
AP / lunate diameter ratio	**~1** · **> 1.3 = AP widening → unstable**
Ulnar variance	Perpendicular method; normal < 1 mm difference

TO EXCEPT trap (TO)	"**dorsal tilt up to 7°**" is **not** acceptable
Radial length (TO)	radial styloid ~1 cm distal to ulnar head
CT	articular involvement, planning · **MRI** – TFCC, SL (DISI), LT (VISI)

Classifications & eponyms

Systems **Fernandez** (mechanism) · **Frykman** (radiocarpal / DRUJ ± ulnar styloid) · **Melone** (4 intra-articular types by displacement) · **AO** (comprehensive, cumbersome)

Eponym	Description
Colles	**Dorsally angulated, extra-articular**
Smith	**Volarly angulated, extra-articular**
Volar Barton	Radiocarpal fracture-dislocation with **volar lip** fragment
Dorsal Barton	Radiocarpal fracture-dislocation with **dorsal lip** fragment
Chauffeur (Hutchinson)	**Radial styloid** (transverse / oblique)
Die-punch	**Depressed lunate fossa**



Normal radiographic relationships of the distal radius

Colles complications (TO) TFCC tear, DRUJ instability, CRPS, stiffness, **malunion (dinner fork)**, **EPL tear**, nonunion
✗ Wrong options: **gunstock deformity** (supracondylar humerus)

Smith – Thomas type (TO)	Pattern
I	Extra-articular, volar angulation
II	**Intra-articular, volar displacement (= reverse / volar Barton)**
III	Extra-articular, volar displacement, more oblique line

Fernandez (mechanism, TO)	Pattern	Example
I Bending	Metaphyseal bending	Colles, Smith I
II Shear	Articular rim	**Barton, chauffeur**, volar / dorsal rim
III Compression	Subchondral collapse	**Die-punch**
IV Avulsion	Ligament avulsion	**Radiocarpal fracture-dislocation** → always ORIF + ligament repair
V Combined high energy	Multifragmentary	Combined dorsal + volar approach + bridge plate + bone graft

– – Active space – –

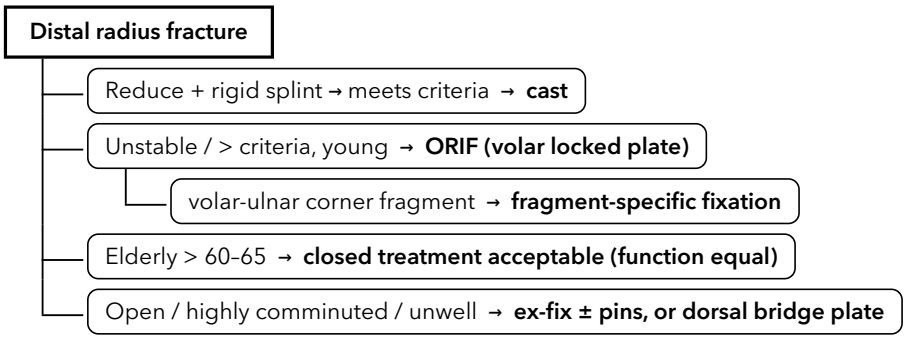
AO (TO)	Pattern
A	Extra-articular
B	**Partial articular** (part still attached to metaphysis)
C	**Complete articular** with metaphyseal detachment

Mayo (TO) extra-articular (non-displaced / displaced) · intra-articular (reducible / irreducible / unstable) · joint involved: I radioscaphoid · II radiolunate · III radioscapholunate

Frykman (TO) extra- vs intra-articular ± ulnar styloid · **limited prognostic value**

Chauffeur mechanism (TO) forced **dorsiflexion + radial deviation**

5 Management



Instability & indications for fixation

LaFontaine instability predictors **radial shortening (most predictive)**, then **dorsal comminution**; dorsal comminution > 50%, palmar / intra-articular comminution, **dorsal angulation > 20°**, **initial displacement > 1 cm**, **shortening > 5 mm**, severe osteoporosis, associated ulnar fracture; **≥ 3 criteria → ↑ loss of reduction**

TO instability list (any one) dorsal angulation > 20° · dorsal comminution > 50% · intra-articular · ulnar fracture · **age > 60** · **AP widening (ratio > 1.3)** · **carpal malalignment (capitate axis not intersecting radius axis)** · die-punch = treat as unstable

Redisplacement after closed reduction **severity of initial displacement** (esp. shortening), age, metaphyseal comminution
✗ Wrong options: TFCC tear · open · radial head · time to reduction

Best stability predictor **pre-reduction radiographs**

Repeat closed reduction	**< 50%** satisfactory

ORIF indications
Dorsal angulation > 5° / > 20° vs contralateral
Intra-articular displacement > 2 mm
Radial shortening > 5 mm
Volar or dorsal comminution · associated ulnar fracture (not styloid) · severe osteoporosis
Barton (articular margin) fractures · **Smith** (comminuted) · **die-punch**
Progressive loss of tilt / length in cast

Young, large dorsal articular fragment	**ORIF** ✗ Wrong options: casts · ex-fix · CRPP
Young high-energy metaphyseal comminution	**ORIF of radius** (locking resists shortening + dorsal angulation) ✗ Wrong options: cast · ex-fix + pins · styloid repair (only if DRUJ unstable / base displaced)

Closed reduction & casting

Technique	anaesthesia, traction + volar / dorsal pressure; below-elbow cast / splint (**rigid / sugar tong**)
Avoid Cotton-Loder position	**extreme flexion + ulnar deviation → CTS**
Within criteria (65-yo)	**rigid splint + clinic follow-up** ✗ Wrong options: admit · ORIF · soft splint
75-yo dorsally angulated extra-articular – initial treatment	**closed reduction + splint**
Complications	**acute CTS**, **EPL rupture**

Elderly / osteoporotic – closed vs operative

Meta-analyses	**no functional difference** closed vs operative in **> 65** (Orthobullets) / **≥ 60** (OB); **radiographic** results better with surgery
Osteoporotic, failed closed treatment	**peri-articular locked plate**
Volar comminution in 65-yo – contraindicated	**Kapandji intrafocal pinning alone** ✗ Wrong options: cast · pinning + ex-fix · dorsal bridge plate · volar plate
Dorsal bridge plate	ligamentotaxis for comminuted fractures; **polytrauma / elderly needing walker → immediate platform (front-wheel) walker** WB (~22% BW) · must be removed ✗ Wrong options: crutches ~50% BW · NWB wheelchair · no device

CRPP

Indication	extra-articular with **stable volar cortex**; 82–90% good
Techniques	**Kapandji intrafocal** (pins lever fracture) · Rayhack (arthroscopic)

– – Active space – –

Complications	**radial sensory nerve injury**, **pin-tract infection**

Volar plating & volar (Henry) approach

Volar plating preferred over dorsal; locked plates support subchondral bone

Classic Henry	between **BR and radial artery** (radial to artery) · **modified (FCR) Henry** – **ulnar (medial) to radial artery**, through FCR sheath (TO)
Distal third (TO)	FCR medial + radial artery lateral → **sweep FPL ulnarward with a finger, protecting palmar cutaneous branch → PQ incised along its RADIAL border** and elevated with periosteum
TO FALSE statement (TO)	"**PQ elevated from ulnar side**" ✗ Wrong options: true: FPL swept ulnarly, modified Henry medial to artery, Thompson more cosmetic for middle / distal-third shaft
Proximal third (TO)	**full supination** moves PIN away; incise supinator medially · **middle third** – pronate, partially detach pronator teres
Palmar cutaneous branch of median nerve	leaves median **4–6 cm** above wrist crease, between median nerve + FCR → **thenar base hyperaesthesia** from retraction ✗ Wrong options: CRPS · Wartenberg · CTS · C7
Bone void	**calcium phosphate** incorporates by ~1 yr ✗ Wrong options: calcium sulfate (resorbs 4–12 wk) · TCP (osteolysis) · OP-1 · freeze-dried allograft (not synthetic)

Volar lunate facet / volar-ulnar corner

Volar-ulnar corner fragment	→ **volar carpal translation** ✗ Wrong options: die-punch · radial styloid (ulnar translation) · extra-articular
Early failure 2 wk after volar plate	**lunate facet fragment not supported → fragment-specific fixation** (small plates, TBW, K-wires) ✗ Wrong options: non-locking plate · volar tilt · styloid column · 3 shaft screws
Region risking radiocarpal instability if missed	**volar lunate facet**

Intra-operative fluoroscopic views

View	Shows
Tilted lateral / radial inclination view (22–23° elevated)	**Radiocarpal (intra-articular) screw penetration**
Dorsal tangential (skyline) – wrist max flexed, beam ~15° volar to radial axis	**Dorsal cortex screw penetration**
PA anatomic tilt (~11°)	Articular congruity
Extended wrist tangential ("sigmoid notch")	DRUJ penetration
Standard lateral	Plate vs watershed

TO answer (TO)	**dorsal tangential + radial incline view**
Avoid extensor injury	**pegs** for articular block, **capture ~75%** of volar–dorsal width

External fixation

Indications — open, highly comminuted, unwell · **cannot reliably restore 10° palmar tilt alone** → add pins / plate

Spanning (small articular fragments) vs **non-spanning** (large fragments)

Ligamentotaxis — traction restores length + inclination; **volar translation of lunate restores volar tilt** (multiplanar, Agee)

Rules — open placement of radial pins (**superficial radial nerve** at "bare area" ~4 finger-breadths above styloid, dorsal to BR – **commonest nerve injured**); **avoid over-distraction (< 5 mm)**, flexion, ulnar deviation; **≤ 8 wk**; aggressive finger OT

Pin care — **daily showers + dry dressings**; chlorhexidine / peroxide **no better**; complication rate ~20%

Ex-fix + K-wires vs volar plate — **both give satisfactory results**; pin / wound complications **higher with ex-fix**
✗ Wrong options: tip styloid needs fixation · plate proximal to watershed ↑ flexor injury · pillar pain

DRUJ instability & ulnar styloid fractures

Stable DRUJ after radius fixation + ulnar styloid fracture — **no adverse effect without fixation**

Shuck test positive after fixation — **radioulnar ligaments of TFCC**
✗ Wrong options: thumb UCL · ulnolunate / ulnotriquetral

Shuck test — elbow 90°, translate ulna sagittally in neutral, pronation, supination

SL disruption seen intra-op — **separate dorsal incision → repair dorsal SL ligament + K-wires** · results better than isolated SL injuries
✗ Wrong options: immobilise · pins alone · volar extension · arthroscopic

DRUJ with DRF (TO)	Lesion	Treatment
I Stable	Sigmoid notch intact; styloid tip # or stable ulnar neck #	Cast, no DRUJ surgery
II Unstable	**TFCC foveal tear or styloid base #**; shuck +	TFCC repair / styloid fixation · **above-elbow cast in supination** ± radioulnar K-wire
III Sigmoid notch	Intra-articular palmar / dorsal lip fragments	**Anatomical reduction + fixation**; long arm cast

Rehabilitation & outcomes

Formal therapy vs home programme — **no long-term difference** (Wakefield-McQueen); **better DASH + ROM early (6 wk)** in women > 40 – improves early, not late PROMs

DASH variability (nonop) — **depression**
✗ Wrong options: neuroticism · occupation · handedness

Poor outcome — workers' comp, ↓ SES, ↓ education, ↓ BMD

Good outcome — **articular reduction accuracy**, restored anatomy, early finger + wrist motion

– – Active space – –

6 Complications

Complication	Fact
Median neuropathy / CTS	**Commonest neurological**; **1–12% low energy, 30% high energy** (Orthobullets) · acute CTS **5.4–8.6%** (OB)
EPL rupture	Nondisplaced / nonop **0.3–5% (3–5%)**; volar plate with **proud dorsal screw → EIP transfer**
FPL rupture	**Volar plate distal to watershed**
Radiocarpal arthrosis	2–30%; **90% of young adults with step-off > 1–2 mm**
Malunion / nonunion	Radial shortening → worst function
ECU / EDM entrapment, ulnar neuropathy	DRUJ injury
Compartment syndrome	–
CRPS	Vitamin C prophylaxis (AAOS) – later analyses show no benefit

Acute carpal tunnel syndrome

Urgent release indications	**progressive paraesthesia**, **weak thumb opposition**, paraesthesia not improving after reduction **> 24–48 h**
Worsening numbness after reduction / CRPP / ORIF	**urgent carpal tunnel release** (+ ORIF / pinning) ✗ Wrong options: bivalve + 1 wk · hand compartment pressures · NCS · EMG · plexus exploration · repeat closed reduction
Pre-op mild paraesthesia, 2PD 6 → > 10 mm day after volar plate	**immediate CTR**
Child SH II, worsening after bivalving	**CTR + pinning**
TO nuance (TO)	**immediately after reduction in palmar flexion / ulnar deviation → ORIF + CTR (TO)** · **delayed CTS in cast → release cast + neutral wrist first (TO)**; CTR if persists · no prophylactic CTR
Urgent-surgery sign on rounds	**can't oppose thumb** ✗ Wrong options: can't extend thumb IP (EPL) · initial displacement · humeral fracture · dorsal comminution
Mechanism	**nerve compression** ✗ Wrong options: laceration · arterial · RSD

EPL rupture after nondisplaced fracture & EIP transfer

Classic	**non- / minimally displaced** fracture in cast → sudden painless loss of thumb extension at **7 wk–4 mo** ✗ Wrong options: PIN palsy · intersection · malunion · EPB
Cause	**hypovascular zone in 3rd compartment**, attrition, callus / ischaemia ✗ Wrong options: displaced fragments
Most likely complication of nonop nondisplaced DRF	**EPL rupture** ✗ Wrong options: CTS · FPL · abutment · arthritis

Treatment	**EIP → EPL transfer** ± palmaris graft; pre-rupture → 3rd compartment release ✗ Wrong options: primary repair (attritional) · EPB (too little excursion) · ECRL · splinting
EIP	**4th compartment** with EDC + **PIN** · EPL innervated by **PIN**
Dorsal screw ulnar to Lister tubercle	irritates **EDC** (4th) ✗ Wrong options: ECRL / ECRB (radial to Lister) · ECU · EPB

FPL rupture – volar plate past watershed

Risk	plate **volar / distal to watershed (Soong) line**, prominent screws
Soong grade	**0** dorsal to critical line · **1** volar to line, proximal to rim · **2** on / distal to volar rim
Flexor injury	up to **12%** of volar plates; **FPL 57%**, FDP index next; mean **~9 mo**
Plate distal to watershed, asymptomatic at 2 wk	**plan removal after union** to protect FPL
FPL	origin volar radius + IOM → **base of thumb distal phalanx** → rupture = **can't flex thumb IP**

CRPS – vitamin C

Dose	**500 mg daily × 50 days** ✗ Wrong options: 200 mg · 1500 mg (no better) · alendronate
Evidence	AAOS 2010 / 2013 moderate support; subsequent analyses **no benefit** – read the options

Malunion

Nascent malunion at 2 mo in young drummer	**acute corrective osteotomy** (easier, less structural graft) ✗ Wrong options: wrist fusion · USO / Darrach alone
Dorsal angulation > 10°	stiffens IOM, ↓ rotation → opening-wedge osteotomy + graft

X vs Y

	EPL rupture	**FPL rupture**
Setting	**Nondisplaced, casted** · proud dorsal screw	**Volar plate distal to watershed**
Deficit	Can't extend thumb IP / "thumbs up"	**Can't flex thumb IP**
Fix	**EIP transfer**	Repair / graft / transfer + plate removal

	Tilted lateral (22–23°)	**Skyline**
Checks	**Joint penetration**	**Dorsal cortex penetration**

	Classic Henry	**Modified (FCR) Henry**
Interval	**BR ↔ radial artery**	**FCR ↔ radial artery**

– – Active space – –

How it's asked

"3 months after minimally displaced DRF, can't extend thumb — next?"	→ **EIP → EPL transfer** (TO)
"Spontaneous EPL rupture most associated with?"	→ **nondisplaced distal radius fracture**
"Nerve supplying the ruptured tendon?"	→ **PIN**
"Commonly transferred tendon's compartment?"	→ **4th**
"Donor tendon description?"	→ **in same compartment as PIN**
"Risk in nonop nondisplaced DRF?"	→ **EPL rupture**
"Tendon at risk 12 months after volar plating (plate prominent)?"	→ **FPL**
"Distal plate, grip weakness at 6 months?"	→ **can't flex thumb IP**
"Plate past watershed at 2 wk?"	→ **remove after union to protect FPL**
"Tendon at risk — origin / insertion?"	→ **volar radius + IOM → base of thumb distal phalanx**
"Prominent screw ulnar to Lister?"	→ **EDC**
"Pain + paraesthesia in lateral digits after casting comminuted fracture?"	→ **release cast, neutral position** (TO)
"Numbness right after reduction in flexion / ulnar deviation?"	→ **ORIF + CTR** (TO)
"Worsening median signs post-op day 1?"	→ **immediate CTR**
"Worsening numbness 2 h after reduction?"	→ **emergent ORIF + CTR**
"Indication for urgent surgery on rounds?"	→ **inability to oppose thumb**
"Vitamin C prevents?"	→ **CRPS**
"Vitamin C dose?"	→ **500 mg daily × 50 days**
"Early failure — cause?"	→ **unsupported volar lunate facet**
"Fracture causing volar carpal translation?"	→ **volar-ulnar corner fragment**
"Radiocarpal screw penetration view?"	→ **radial inclination (22–23° tilted lateral)**
"Dorsal + articular screw check (TO)?"	→ **dorsal tangential + radial incline views**
"Stable DRUJ + ulnar styloid fracture untreated?"	→ **no adverse effect**
"Shuck test positive — structure?"	→ **radioulnar ligaments of TFCC**
"SL disruption on intra-op film?"	→ **dorsal incision repair + K-wires**
"Osteoporotic failed closed treatment — plate?"	→ **peri-articular locked**
"Within criteria, 65-yo?"	→ **rigid splint + clinic**
"Elderly ORIF vs closed — meta-analysis?"	→ **no functional difference**
"Volar comminution — contraindicated?"	→ **Kapandji pinning alone**

"Dorsal bridge plate — weight bearing?"	→ **platform walker immediately**
"Acceptability EXCEPT (TO)?"	→ **dorsal tilt up to 7°**
"Redisplacement predictor?"	→ **severity of initial displacement**
"Ex-fix — volar translation of lunate?"	→ **restores volar tilt**
"Nerve most injured placing ex-fix?"	→ **superficial radial**
"Ex-fix vs volar plate?"	→ **both satisfactory**
"Pin-site care?"	→ **no special solution better than soap + water**
"Formal therapy vs home exercise?"	→ **no long-term difference** (early benefit only)
"DASH variability after nonop?"	→ **depression**
"Thenar base hyperaesthesia after Henry approach?"	→ **palmar cutaneous branch injury**
"Void filler incorporated at 1 yr?"	→ **calcium phosphate**
"Henry approach — FALSE (TO)?"	→ **PQ elevated from ulnar side**
"NOT a Colles complication (TO)?"	→ **gunstock deformity**
"NOT true about chauffeur fracture (TO)?"	→ **ulnar styloid fracture**
"Young drummer, nascent malunion at 2 mo?"	→ **corrective osteotomy**

– – Active space – –

– – Active space – –

DISTAL RADIUS MALUNION

1 Basics

Epidemiology

Most common complication of distal radius fractures

Rates **17% overall · 11% after operative · 24% after definitive casting**

Risk factors **post-reduction malalignment**, **osteoporosis**

Anatomy – three columns

Distal radius carries **80% of wrist axial load**

Column	Contents	Role
Radial	Radial styloid, **scaphoid fossa**	Buttress vs radiocarpal translation; load in ulnar deviation; anchors **radioscaphocapitate** (vs ulnar translation)
Intermediate	**Lunate fossa**	Main load transfer carpus → forearm
Ulnar	TFCC + distal ulna	DRUJ stability

Biomechanics

Deformity	Effect
↓ volar tilt (dorsal tilt)	**Limits flexion + supination**
↑ volar tilt	**Limits extension + pronation**
↓ radial inclination	↓ flexor mechanical advantage → **hand weakness**
Radial shortening	**↑ positive ulnar variance**; load shifts to ulnocarpal joint; **largest change in DRUJ kinematics**
Volar tilt change	DRUJ incongruity → IOM tightens → ↓ rotation + flexion–extension arc

Volar-angulated malunion (OB stem) **↓ supination + extension**, **dorsal subluxation of radius in supination** (OB) vs ↑ volar tilt limits extension + pronation (Orthobullets) – read the options

Dorsal malunion carpal instability (1) **adaptive midcarpal DISI** · (2) **dorsal radiocarpal subluxation** with maintained midcarpal alignment

2 Mechanism

How initial displacement · **secondary loss of reduction in cast** · failed fixation

Pattern	Note
Dorsally angulated (apex volar)	**Most common**; DISI-like collapse
Volarly angulated (apex dorsal)	–
Radial shortening	DRUJ kinematics
Intra-articular	With **initial intra-articular displacement > 2 mm**

3 Presentation

Symptoms pain (localise radial vs ulnar side → guides ulnar procedures), **grip weakness**, ↓ motion with **pronation / supination worst**

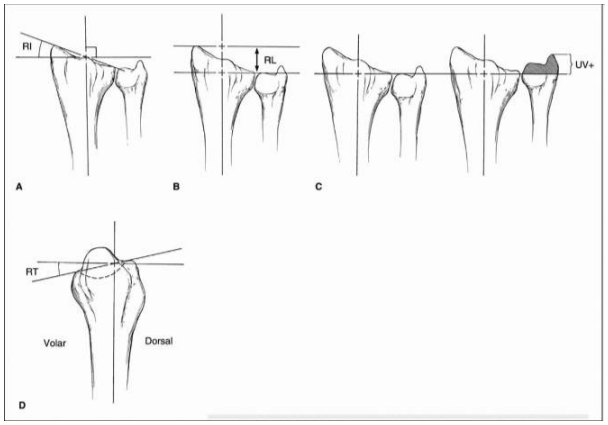
Exam deformity, tenderness map, ROM all planes, **bilateral grip dynamometry**, **DRUJ shuck test** (pain with instability), Allen test, CTS, CRPS

Late DRUJ instability in adult (longshoreman) from untreated **childhood distal radius fracture** malunion
✗ Wrong options: supracondylar (cubitus varus / valgus) · medial epicondyle · both-bone forearm (DRUJ preserved) · elbow dislocation

4 Diagnosis & Classification

Key radiology findings

Measure	Normal
Radial height (PA)	**8–14 mm**
Radial inclination (PA)	**21–25°**
Articular step-off	Congruous
Volar tilt (lateral)	**7–15°**
Ulnar variance	Neutral (< 1 mm change)



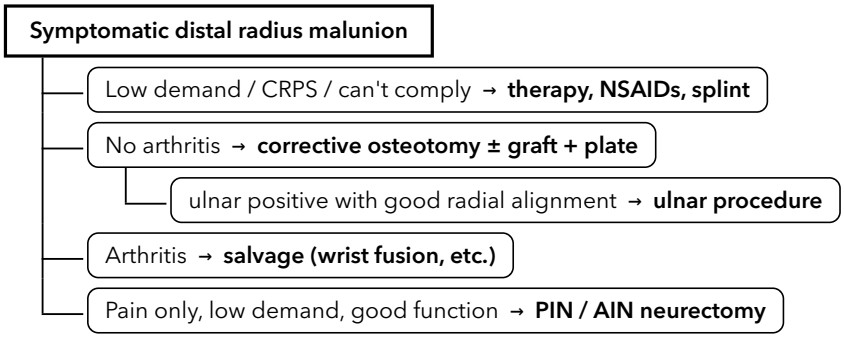
Radiographic parameters of the distal radius

Views **bilateral** PA, lateral, oblique · **anatomic (tilted) lateral** for lunate facet

– – Active space – –

CT	**intra-articular malunion**, axial rotation, **DRUJ instability (> 50% translation vs other side)**
MRI	TFCC

5 Management



Corrective osteotomy for symptomatic malunion

Indications	symptomatic after failed casting or failed fixation with **mechanical symptoms, ↓ grip, ↓ ROM, instability on stress**; **Fernandez**: active patient, extra-articular **> 25–30° angulation**, **no degenerative change**
Ballet dancer, dorsal angulation, joint space preserved	**corrective osteotomy** · arthritis → salvage ✗ Wrong options: wrist arthrodesis · PRC · SLAC procedures · interposition
Nonop outcomes	**age 18–65 → more adverse outcomes** when untreated

Approach	Pros	Cons
Dorsal (2nd–4th compartments)	Easy osteotomy; **tension-side** fixation; for dorsal subluxation	**Extensor tendon** irritation
Volar (extended FCR)	Align volar cortex → stability, **may avoid graft**; for volar subluxation; volar malunions	**Flexor tendon** irritation

Osteotomy	Fact
Opening wedge	**Needs bone graft**
Closing wedge	Bone contact + compression; **↑ ulnar variance → often needs ulnar shortening**
Intra-articular	Arthroscopy-aided; outcomes ≈ extra-articular osteotomy; **push-pull screw** restores height

Graft	**structural bicortical ICBG** → union ~7 wk, better supination · **cancellous distal radius ≈ structural ICBG** · allograft, DBM, calcium phosphate options
Complication rate	extra-articular correction **~50%**; **higher with distraction osteotomy**

Denervation

PIN ± AIN neurectomy	debilitating pain + infection, low demand, preserved function; dorsal incision proximal to DRUJ; PIN on radial floor of **4th compartment**; AIN after dividing IOM (1 cm excised)

Ulnar-sided procedures

Indications **ulnar-positive with preserved radial alignment**, DRUJ instability, TFCC tear, styloid nonunion

Procedure	Rule
Ulnar shortening osteotomy (Milch)	**Isolated radial shortening**; **not with DRUJ arthrosis**
Wafer (Feldon)	**Intact TFCC**; angular deformities
Hemiresection-interposition (Bowers)	**Intact TFCC**; DRUJ arthrosis / instability
Sauvé-Kapandji	**Younger**, incongruity + degeneration; keeps rotation
Darrach	Salvage; don't over-resect
Ulnar head / DRUJ arthroplasty	Salvage
One-bone forearm	**Ultimate salvage**; no rotation

TFCC central perforations usually settle; irreparable + unstable DRUJ → **Adams reconstruction (palmaris graft)** or BR wrap; confirm reparability arthroscopically

Styloid nonunion small + stable → **excise** · large + unstable → **repair** (headless screw, TBW, or excise + foveal TFCC repair)

6 Complications

Complication	Fact
Tendon irritation / rupture	**20%** of complications; **EPL commonest**; rupture ~**10 wk**; risk: **distraction osteotomy**, dorsal callus, volar plate, dorsal osteophyte, prominent hardware
Others	Nerve injury, CRPS, nonunion, infection, PTOA

X vs Y

	↓ volar tilt (dorsal malunion)	↑ volar tilt (volar malunion)
Lost motion	**Flexion + supination**	**Extension + pronation** (Orthobullets)
Carpus	DISI / dorsal subluxation	–

	Opening wedge	Closing wedge
Graft	**Needed**	Not needed
Ulnar variance	Restores length	**↑ positive → ulnar shortening**

How it's asked

"Dancer 11 mo after casting, dorsal angulation, no arthritis?"	→ **distal radius corrective osteotomy**
"Adult with DRUJ instability from childhood injury?"	→ **distal radius fracture (malunion)**
"Commonest malunion direction?"	→ **dorsal angulation (apex volar)**

– – Active space – –

"Deformity with biggest DRUJ kinematic change?"	→ radial shortening
"↓ radial inclination causes?"	→ grip weakness
"Ulnar shortening contraindicated with?"	→ DRUJ arthrosis
"Commonest tendon complication?"	→ EPL rupture (~10 wk)

DOMESTIC AND ELDER ABUSE

– Trauma · 16 / 67 –

1 Basics

IPV epidemiology

Fact	Value
Women experiencing domestic violence	**~25%** (Orthobullets) · lifetime **~30% / 33% (PRAISE)** (OB)
Women in ER with injuries due to DV	**up to 35%**
Past-year IPV in women at fracture clinics	**1 in 6** · **32%** within 1 yr before orthopaedic visit
Fracture-clinic visits directly from IPV	**1 in 50**
Ever asked about IPV in health care	Only **14%**
Men who are victims	15%
Children abused in DV homes	**~50%**
Surgeon-perceived vs actual prevalence	**~10% vs up to 40% → 4× under-estimated**

DV = commonest cause of non-fatal injury to women

Elder abuse epidemiology

Scale ~2 million Americans / yr; **1 in 25** over 65; widely **under-reported**; forms: physical, mental, neglect, financial, abandonment

2 Mechanism

IPV risk factors

Risk factor	Note
Female	–
Age 19–29 (Orthobullets) · **16–49** (OB) · 20–29 (OB)	Young
Pregnancy	**Highest risk**; 40–60% of battered women abused in pregnancy
Short-term relationship	–
Cohabitation	–
Low income / SES (< $10,000 / yr)	(✗ Wrong options: "equal in all incomes")
Alcohol / drug dependence	–
≥ 1 prior child	Higher prevalence with children
Not risk factors	**Prior divorce**, no children, older age, long relationship, living apart

– – Active space – –

Elder abuse risk factors

Victim	**↑ age**, **functional disability**, **cognitive impairment / dementia**, poor physical + emotional health, disruptive / aggressive behaviour, social isolation, poverty, race, prior violence toward abuser; regional child-abuse rates
Caretaker	**substance abuse**, mental illness, **financial dependence on elder**, burden perception, shared living, stress
Gender	**not** a risk factor for elder abuse

3 Presentation

Red flags (injury pattern)

Injuries	**inconsistent with history**, **long delay to treatment**, **repeat injuries**, **multiple injuries of varying ages**, multiple providers / EDs, unusual patterns
Most common IPV manifestation	**head, neck, face injuries** (unwitnessed)
Elder abuse	**long-bone + rib fractures**, bruises **> 5 cm on face, neck, back**, burns, lacerations, poor hygiene, behaviour change

Behaviour

Victim	changed affect, withdrawn, seeks partner's approval, **repeated ED visits**, wants to stay in hospital, **missed work / ↓ productivity**, chronic headaches, sleep disturbance, high-risk behaviour, GI / psychosomatic / pelvic symptoms
Abuser	won't leave, overly attentive, hostile, **answers for the patient**
Repeated ED visits, vague pain, frightened, ecchymoses, negative imaging	**IPV** ✗ Wrong options: malingering · CRPS · anxiety · fibromyalgia

Barriers to disclosure

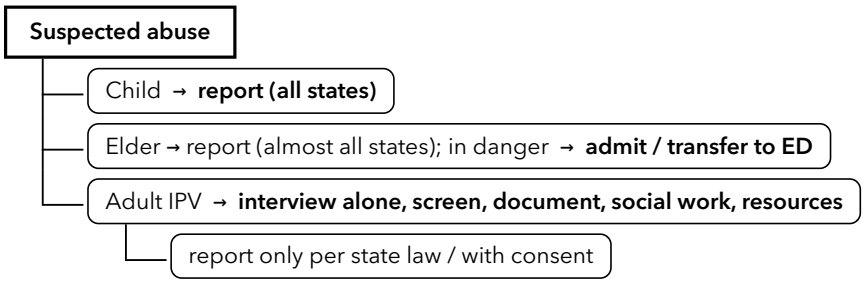
Barriers	**fear of retaliation**, shame, embarrassment, **reluctance with male physicians**, custody fears, downplaying
Emotional abuse	**harder to detect** than physical

4 Diagnosis & Classification

No skeletal survey in competent adults (or pregnancy)

Documentation	exact quotes, abuser's name, body maps / drawings, **photographs with consent**, demeanour, opinion if injuries don't fit story

5 Management



Screening & management

First step in suspected IPV	**speak with patient alone, privately, non-judgementally** → resources ✗ Wrong options: confront partner · treat + discharge · mandatory psych referral · call police first
Pregnant, inconsistent history, bruises	after treating fracture → **social work consult** ✗ Wrong options: arthroscopy · MRI · skeletal survey · urgent ORIF
Shoulder dislocation + positive pregnancy test	**ask privately about injury + home safety** ✗ Wrong options: skeletal survey · nutritionist · anorexia admission · MRI

Treat injuries promptly regardless of social situation
✗ Wrong options: "resume care once social situation assessed"

Interdisciplinary collaboration → better outcomes

Reporting obligations

Victim	Duty
Child	**Mandatory in all states**
Elder	**Mandatory in almost all states**; **hospitalise if in immediate danger**
Adult IPV	**Not standardised**; many states need consent; physician **has no authority to provide protection** → encourage victim to report

Federal law does NOT mandate photographs (EXCEPT answer) – only with consent

AAOS roles	know state laws, be aware you may be first physician, keep accurate records, transfer endangered elder to ED + notify, advocate legislation

X vs Y

	IPV	Elder abuse
Gender risk	**Female**	**Not a risk factor**
Reporting	Varies by state	Mandatory (almost all)
Classic injury	Head / neck / face	Long bones, ribs, large bruises

How it's asked

"DV management EXCEPT?"	→ **federal law mandates photographs**

– – Active space – –

"Orthopaedist's role EXCEPT?"	→ report all adult IPV in all states
"IPV – true?"	→ interdisciplinary collaboration helps
"Dislocation + pregnancy, inconsistent story?"	→ question privately about home safety
"AAOS statement EXCEPT?"	→ resume care once social situation assessed
"Pregnant, bruises, delay – next step?"	→ social worker consult
"Withdrawn, partner interrupts – first step?"	→ speak with patient alone
"IPV risk factors EXCEPT?"	→ prior divorce
"4th ED visit, vague pain, bruises?"	→ IPV
"IPV – true?"	→ 4× more prevalent than surgeons perceive
"Risk factor for IPV?"	→ cohabitation
"IPV – true (facial bruises)?"	→ head + neck injuries commonest
"IPV – true (partner answers)?"	→ highest prevalence age 16–49
"Elder abuse risk factors EXCEPT?"	→ gender

– – Active space – –

ELBOW DISLOCATION

– Trauma · 17 / 67 –

1 Basics

Epidemiology

Rank	**2nd commonest major joint dislocation** (after shoulder); **commonest dislocation in children**
Share	**10–25%** of elbow injuries · **posterolateral 80%** · peak **age 10–20**
Simple vs complex	**simple (no fracture) 50–60%** · complex 40–50%
Associated	shoulder / wrist injuries **10–15%** · radial head fracture up to 10%

Anatomy – stabilisers

Type	Structures
Primary static	**Ulnohumeral joint**, **anterior bundle MCL** (key valgus), **LCL complex incl. LUCL** (primary varus + posterolateral rotatory)
Secondary static	**Radiocapitellar joint**, **capsule**, **common flexor + extensor origins**
Dynamic	**Anconeus, brachialis, triceps** (+ biceps) – compression

LCL (TO)	radial collateral, annular, **lateral ulnar collateral** · **MCL (TO)** – **anterior bundle (valgus)**, posterior bundle (secondary valgus + ulnar rotation) · extensors resist varus, flexors resist valgus

2 Mechanism

Posterolateral dislocation	**axial load + forearm supination / ER + valgus posterolateral force** (O'Driscoll) · varus posteromedial mechanism also reported
Circle of Horii (lateral → medial)	**LCL fails first** → capsule → **MCL last** (energy dependent); some recent work suggests failure may start medially (TO)
LUCL failure mode	**avulsion from humeral (lateral epicondylar) origin** · McKee: proximal avulsion 32, midsubstance 18, bony humeral 5, ulnar 3 / 62 ✗ Wrong options: ulnar insertion · midsubstance (less common) · bony avulsion
MRI of torn LCL	avulsed off lateral epicondyle (MCL + flexor origin intact)

3 Presentation

Exam	pain, swelling, deformity; **skin (open?)**; **compartments**; **wrist + shoulder**; NV
Commonest nerve injury (TO)	**ulnar nerve** (stretch) · median rare (with brachial artery) · brachial artery (rare; open)

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

AP + lateral	congruency esp. post-reduction · **obliques** – periarticular fractures · **CT** – complex patterns, fractures, loose bodies
"Drop sign" (TO)	**↑ static ulnohumeral distance** on post-reduction lateral → residual posterolateral subluxation

Classification

Basis	Types
Anatomic (olecranon vs humerus)	**Posterolateral (commonest)** · posteromedial
Simple	No fracture
Complex	With fracture: **terrible triad** (dislocation + LUCL + radial head + coronoid tip) · **varus PMRI** (LCL + anteromedial coronoid; radial head spared)



Simple posterolateral elbow dislocation

5 Management

Elbow dislocation

- Simple, stable after reduction → splint 5–10 d → **early active ROM**
  - residual subluxation (drop sign) → **active ROM pronated, avoid varus**
- Unstable (needs > 50–60° flexion) / irreducible → **open, repair LCL ± MCL**
- Complex → **fix coronoid / radial head / olecranon + LCL ± MCL**
- Chronic → **open reduction, release, hinged ex-fix**

Simple dislocation – reduction & early motion

Reduction	analgesia + relaxation → **in-line traction** (coronal) + **forearm supination** (coronoid under trochlea) + **flex elbow with pressure on olecranon** → "clunk"
Parvin's reduction (TO)	flex **30°** → traction–countertraction → correct mediolateral shift → **supinate + push olecranon anteriorly while flexing** · irreducible / unstable → open (posterior) ± ligament repair

Post-reduction test	unstable in **extension** + **valgus**; test valgus with **forearm pronated** (locks lateral side)
Splint	posterior splint **≥ 90° flexion**, rotation of maximum stability: **LCL torn → pronation** · **MCL torn → supination** (O'Driscoll: reduce in supination, immobilise in pronation) · neutral acceptable
Both LCL + MCL repaired (TO)	immobilise in **neutral** rotation
Duration	**5–10 days** (5–7 d, OB) → **early active ROM** through stable arc; hinged brace with extension block **3–4 wk**; light duty at 2 wk; full extension by **6–8 wk** · avoid varus / valgus loading until 12 wk (TO)
Stable after reduction – start motion	**~1 week** ✗ Wrong options: immediately · 3 wk · 6 wk · after LCL repair
Best outcome predictor	**mobilisation within 5–10 days** (> 10 days → arthrofibrosis, pain) · immobilisation **> 3 wk → poor final ROM**
Avoid	**immobilisation 4 wk** ✗ Wrong options: vs active motion, 90° neutral splint, extension-limited early rehab, light duty (all OK)
Simple dislocation, can't move elbow	**closed reduction, splint, early ACTIVE ROM** ✗ Wrong options: passive ROM · hinged ex-fix · acute LCL repair · open reduction
Residual posterolateral subluxation (drop sign) (TO)	**splint 1 wk → active motion with forearm PRONATED through full range**; elbow at side (no varus / shoulder abduction); supinate only at ≥ 90° flexion ✗ Wrong options: LCL reconstruction · splint 1 month
Recurrent instability after simple dislocation	**rare (< 1–2%)**

Operative criteria

Indication	Treatment
Irreducible (soft tissue / osteochondral fragment)	Open reduction
Persistent instability – needs **> 50–60° flexion** to stay reduced (Orthobullets, OB) · **redislocates at < 30° flexion** (TO)	**Repair LCL ± MCL**
Complex (coronoid, radial head, olecranon)	ORIF / arthroplasty + LCL ± MCL
Open, vascular injury, recurrent	Surgery
Throwers	± MCL repair
Chronic dislocation	**Open reduction + capsular release + dynamic hinged ex-fix**
Still unstable after all repairs	**Hinged external fixator**

Highly unstable after closed reduction	**open reduction + LCL repair ± MCL repair** ✗ Wrong options: 6 wk splint · ex-fix alone · internal joint stabiliser alone · reconstruction + 6 wk immobilisation
Essential lesion in persistent instability	**LUCL** ✗ Wrong options: MCL (repaired only if still unstable) · coracohumeral · capsule

– – Active space – –

Structure needing repair	**LUCL** ✗ Wrong options: radial collateral · MCL bands · accessory collateral

Approaches

Kocher (ECU–anconeus)	LCL, extensor origin, coronoid, capitellum, radial head; incise **anterior to radial head midline** to spare posterior LCL fibres; careful retractors → **PIN**
Medial	MCL, flexor-pronator, comminuted coronoid; **protect ulnar nerve**

Complex dislocation – fixation

Coronoid	tip fractures rarely need ORIF (proximal to brachialis); lateral or medial approach
Radial head ORIF	**safe zone ≈ 90° non-articulating arc** (between Lister tubercle + radial styloid lines) (Orthobullets, OB) · **~110°** in original dissections · **69%** of head articulates with ulna
Radial head arthroplasty	if unreconstructable; **anatomic size / length** → restores varus + ER stability **only once LCL repaired** · **excision → varus / ER instability** without LCL
Dislocation + comminuted radial head (55-yo)	**open reduction, LCL repair, radial head ORIF or metallic replacement** ✗ Wrong options: cast 4 wk · brace + immediate ROM · silastic implant (contraindicated) · excision

Complex dislocation – coronoid & PMRI

Varus PMRI	**LCL rupture (humeral) + anteromedial coronoid facet fracture + ulnohumeral subluxation / dislocation**; radial head usually spared
Dislocation + anteromedial coronoid fracture on AP	highest risk **varus PMRI** ✗ Wrong options: AIN / PIN palsy · PLRI · chair-rise instability
Fixation preventing PMRI	**anteromedial coronoid facet** · failure to fix / malunion → poor function + arthrosis (Doornberg & Ring) ✗ Wrong options: radial head · coronoid tip · olecranon tip · medial capitellum
Transolecranon fracture-dislocation (TO)	**anterior dislocation with intact PRUJ**, mainly **bony**, usually **no ligament injury**, coronoid fracture common → **NOT a Monteggia variant** (EXCEPT) · type I simple / II complex · restore ulnar length, alignment, rotation + **sigmoid notch**; missed coronoid = failure

6 Complications

Complication	Fact
Loss of terminal extension	**Commonest** (simple + complex) → early active ROM; static progressive splint at 6–8 wk
Stiffness / contracture	**Immobilisation > 3 wk**
HO	May need excision
Varus PMRI	LCL + anteromedial facet
Nerve	**Ulnar** (stretch) · **AIN palsy** (weak pinch, can't flex thumb IP – misread as FPL rupture)
Brachial artery	Rare, open dislocations
Compartment syndrome, articular damage, recurrent instability	–

Post-traumatic stiffness – release & HO excision

Functional arc	**30–130°**
Arc 30–95° (flexion-limited) + HO	**excise HO + release posterior bundle of MCL + ulnar nerve decompression** (flexion < 100° → decompress) ✗ Wrong options: supracondylar osteotomy + anterior MCL release · anterior capsule release (helps extension) · **LCL release → instability**
MCL release order	**posterior bundle → transverse → (posterior half of anterior bundle)**
Extension loss 80° with mature anterolateral HO, no ulnar symptoms	**open resection + capsulectomy via lateral approach** (access anterior + posterior) ✗ Wrong options: arthroscopic (risky with tight capsule) · anterior approach (no posterior access) · indomethacin + radiation (too late once visible) · manipulation (contraindicated)
HO maturation	fluffy → sharp margins over **3–6 months**; labs + bone scan not prognostic; prophylaxis (NSAID, **700 cGy**) ineffective once visible

X vs Y

	Terrible triad (PLRI)	Varus PMRI
Force	Valgus, supination, axial	Varus, posteromedial, axial
Radial head	**Fractured**	**Spared**
Coronoid	**Tip**	**Anteromedial facet**
Ligament	LCL (± MCL last)	LCL humeral avulsion

	LCL torn	MCL torn
Most stable forearm position	**Pronation**	**Supination**

	Transolecranon fracture-dislocation	Posterior Monteggia
PRUJ	**Intact**	Disrupted
Ligaments	Usually intact	LCL injured in up to ⅔

– – Active space – –

How it's asked

"Commonest sequel of simple + complex dislocations?"	→ **loss of terminal extension**
"Rehab protocol to AVOID?"	→ **immobilisation 4 wk then PT**
"Simple dislocation — next step?"	→ **closed reduction, splint, early active ROM**
"Stable after reduction — when to start motion?"	→ **1 week**
"Best outcome strategy?"	→ **mobilise within 5–10 days**
"Post-reduction drop sign — management?"	→ **splint 1 wk, then active motion forearm pronated** (TO)
"LUCL failure mode?"	→ **avulsion from humeral origin**
"Essential lesion in recurrent instability?"	→ **LUCL**
"Torn structure on coronal MRI?"	→ **LCL (avulsed from epicondyle)**
"Highly unstable, won't stay reduced?"	→ **open reduction + LCL repair ± MCL repair**
"Structure most needing repair?"	→ **LUCL**
"Transolecranon fracture-dislocation EXCEPT (TO)?"	→ **Monteggia variant**
"Dislocation + anteromedial facet — risk?"	→ **varus PMRI**
"Dislocation + comminuted radial head?"	→ **open reduction, LCL repair, radial head ORIF / metal replacement**
"Long jumper, MCL attachment fracture — pattern?"	→ **LCL disruption + anteromedial coronoid facet + dislocation**
"Fixation preventing PMRI?"	→ **anteromedial coronoid facet**
"Stiff 30–95° with HO, ulnar symptoms?"	→ **HO excision + posterior MCL bundle release + ulnar decompression**
"Extension loss with mature anterior HO?"	→ **open resection + capsulectomy via lateral approach**
"Weak pinch, can't flex thumb IP after dislocation?"	→ **AIN palsy**
"Radial head articulating with ulna?"	→ **69%** (safe zone ~90°)
"Commonest nerve injured (TO)?"	→ **ulnar**

EVALUATION, RESUSCITATION & DCO

– Trauma · 18 / 67 –

1 Basics

Trimodal death after trauma

Peak	Share	Cause
Minutes	**50%**	Massive bleeding, neurological injury
Hours	**30%**	Shock, hypoxia, neurological injury
Days–weeks	**20%**	**MOF, infection**

Golden hour ~60% of preventable deaths

Trimodal timing – TO lecture (TO) immediate **< 1 h** · early **1–4 h** · late **> 24 h** (MOF, sepsis) → ATLS ↓ immediate + early deaths; late deaths remain

Trauma triad of death (TO) **coagulopathy + hypothermia + acidosis**

Airbags (head-on) ↓ head injury, facial fractures, thoraco-abdominal injury (splenic rupture, flail chest), extraction · **pelvic ring injuries NOT reduced**

Psychological **PTSD ~50%** after orthopaedic trauma (**females 4×**; LE / pelvic injuries 2× UE) · depression 33%

PTSD risk factor **high ISS** (> 15 = polytrauma)
✗ Wrong options: social history · timing of surgery · tobacco · rehab duration

Female sex is linked to **PTSD**
✗ Wrong options: nonunion · malunion · infection · fat embolism

Genes & lab markers

IL-6 **most reliable marker of systemic inflammatory response**; correlates with **injury severity, complications, mortality** · **> 500 pg/dL → DCO**
✗ Wrong options: IL-8, IL-10, CRP, TNF-α (too variable)

Elevated IL-6 ↑ **mortality** · also sensitive for **PJI**
✗ Wrong options: osteomyelitis, rhabdo, compartment syndrome

Lactate normal **< 2–2.5 mmol/L** (0.6–1.7 in one OB explanation); **most sensitive** marker of occult hypoperfusion; **last to normalise**; clearance within **12–24 h** predicts survival

Base deficit normal **−2 to +2**; global hypoperfusion; **most predictive of mortality** (incl. normotensive elderly); confounded by saline (hyperchloraemia), renal failure, DKA, CO_2 retention

Gastric mucosal pH normal **> 7.3 (7.30–7.35)**; regional endpoint

– – Active space – –

SIRS criteria (≥ 2)

Parameter	Threshold
Temperature	**> 38 °C or < 36 °C**
Heart rate	**> 90 / min**
Respiration	**RR > 20 or $PaCO_2$ < 32 mm Hg**
WBC	**> 12,000, < 4000, or > 10% bands**

Lactate is NOT a SIRS criterion

Adults vs children — **adults: robust systemic response, lungs fail first (day 2–3)** · children: dampened systemic, robust local, all organs simultaneously

2 Mechanism

Two-hit theory — **1st hit** (hypoxia, hypotension, soft tissue, fractures) + **2nd hit** (surgery, ischaemia-reperfusion, compartment syndrome, infection) → SIRS → ARDS / MOF

Compensated shock — normal vitals with ongoing hypoperfusion (**up to 85%**) → **"primed" immune system → exaggerated SIRS** to a 2nd hit
✗ Wrong options: "no risk" · ↑ nonunion / infection after staged fixation

Lethal triad (TO) — **hypothermia + acidosis + coagulopathy**
✗ Wrong options: haemorrhage · hypoxia · hypotension

3 Presentation

Primary / secondary / tertiary survey

Primary (ABCDE) — treat greatest threat first; **A** incl. C-spine · **B** · **C** incl. haemorrhage control (pelvis!) · **D** · **E**

Pregnancy — **left lateral decubitus** (or right hip wedge) → relieves aortocaval compression (supine ↓ CO 25%)

Tertiary survey — when mental status recovers → ↓ missed injuries

Missed fractures — commonest cause = **extremity never imaged**
✗ Wrong options: X-ray read wrongly · MRI / CT not done

Haemorrhagic shock classes (ATLS)

Class	Blood loss	HR	BP	Pulse pressure	Urine	Mental state	Treatment
I	**< 15% (< 750 mL)**	Normal	Normal	Normal / widened	> 30 mL/h	Anxious	Fluid
II	**15–30% (750–1500)**	**> 100**	**Normal** (↑ diastolic)	**Narrowed**	20–30 mL/h	Confused, irritable	Fluid
III	**30–40% (1500–2000)**	**> 120**	**↓ (first hypotension)**	Narrow	5–15 mL/h	Lethargic	**Fluid + blood**
IV	**> 40% (> 2000)**	**> 140**	↓↓	**Very narrow**	Negligible	Coma	Fluid + blood

Blood volume adult **~5 L (4.7–5 L, ~7% body weight)** · child 2–10 yr **75–80 mL/kg**

First tachycardia + narrow pulse pressure **~25% loss (class II)**

2 L loss in 70 kg **class IV**: ↓ SBP + narrow pulse pressure
✗ Wrong options: widened = class I · normal SBP with ↑ diastolic = class II

HR 135, BP 85/58, low urine **30–40% (class III)**

Each thigh holds **~20%** of blood volume; isolated femur fracture can cause **class II** shock

Non-haemorrhagic shock

	Hypovolaemic	Septic	Neurogenic
SVR	**↑**	**↓ (key discriminator)**	↓ (vasodilation)
Cardiac output	↓	**↑** (early; ↓ end-stage)	**↓**
PCWP	↓	↓	–
CVP	↓	↓	–
Mixed venous O_2	↓	**↑**	–
HR	**Tachycardia**	Tachycardia	**Bradycardia**
Skin	Cool, clammy	–	**Warm, dry**

Haemorrhagic vs neurogenic **tachycardia** favours haemorrhagic
✗ Wrong options: bradycardia · ↓ CO (both) · vasodilation · warm skin

Septic shock **↓ SVR** · mortality 25–50%
✗ Wrong options: ↑ SVR (EXCEPT answer)

Swan-Ganz (TO) differentiates shock types: **PAWP ↑ in cardiogenic**, ↓ in hypovolaemic / distributive (TO stem: cardiogenic shock after spinal injury)

- - Active space - -

4 Diagnosis & Classification

Key radiology findings

Study	Use
AP chest	Mediastinal widening, PTX
Lateral C-spine	Must see C7–T1; largely replaced by **CT**
AP pelvis	Ring, acetabulum, proximal femur; **CT after preliminary pelvic stabilisation**
CT C-spine, chest, abdo, pelvis	Rule out life-threatening injuries
Image any limb with pain, crepitus, ecchymosis, deformity	Avoid missed fractures

Resuscitation endpoints

Endpoint	Adequate
Lactate	**< 2 mmol/L** (best, TO) · **< 2.5** (Orthobullets)
Base deficit	**−2 to +2**
Gastric mucosal pH	**> 7.3**
Urine output	**≥ 0.5–1 mL/kg/h** (~30 mL/h)
Pulse pressure	> 15 mm Hg

Resuscitation complete — **oxygen debt repaid, aerobic metabolism restored in ALL tissue beds**
✗ Wrong options: MAP > 90 · pulse pressure · urine output · maximised O_2 delivery

Normal BP, HR, urine after fluids — resuscitation **cannot be determined** without lactate / BD / pHi

Occult hypoperfusion — **lactate 4.5–5 mmol/L**
✗ Wrong options: HR 80 · SBP 120 · BD −1.8 · urine 40 mL/h

Successfully resuscitated — **lactate 1.9**
✗ Wrong options: urine 0.25 mL/kg/h · BD 5.5 · gastric pH 6.5 · pulse pressure 15

Alternative to lactate / BD — **gastric intramucosal pH 7.4**

Normal lactate + BE despite abnormal vitals — **adequate tissue perfusion**
✗ Wrong options: arterial line best · normal Hct = no shock (Hct lags)

Variable best indicating readiness for definitive fixation — **base deficit / lactate**
✗ Wrong options: pulse pressure · HR · Hct · urine · SBP · platelets

Most predictive of mortality — **base deficit (e.g. BE −12)**
✗ Wrong options: $ETCO_2$ · Hct · HR · BP

Patient condition (Pape) – 3 of 4 criteria

– – Active space – –

Criterion	Stable	Borderline	Unstable	In extremis
SBP (mm Hg)	≥ 100	80–100	**< 90**	**≤ 70**
Blood units / 2 h	0–2	2–8	5–15	> 15
Lactate	Normal	≈ 2.5	**> 2.5**	Severe acidosis
Base deficit	Normal	–	–	> 6–8
ATLS class	I	II–III	III–IV	IV
Platelets	> 110,000	90–110,000	**70–90,000**	**< 70,000**
Fibrinogen (g/L)	> 1	≈ 1	< 1	DIC
Temperature (°C)	–	**33–35**	**30–32**	≤ 30
PaO_2/FiO_2	350–400	300–350	200–300	< 200
Chest AIS	1–2	≥ 2 (2–3 rib fractures)	≥ 3	≥ 3 (unstable chest)
Pelvic (AO)	A	B / C	C	C (crush)

OB unstable thresholds **SBP < 90**, **lactate > 2.5**, **platelets < 90,000**, fibrinogen < 1, **temp < 35 °C**, pulmonary contusion, severe liver / spleen, pelvic ring disruption · lactate **> 6 alone** → DCO

TO DCO criteria (TO) **pH < 7.24**, **lactate > 2.5**, **temp < 35 °C**, **platelets < 90,000**, op time > 90 min, > 10 U PRBC · **SBP < 90 is NOT among these (TO)**

5 Management

Polytrauma with long-bone / pelvic fractures

- Unstable / extremis / ongoing acidosis / head or chest injury → **DCO**
  - ex-fix long bones + pelvis, splint UE, I&D open → **definitive later**
- Stable, resuscitated → **early definitive fixation (EAC, < 36 h)**
- Borderline → **DCO if any parameter fails**

Fluids & transfusion

Resuscitate isotonic crystalloid → **O-negative / type-specific / cross-matched blood**

– – Active space – –

Massive transfusion **1 : 1 : 1 (PRBC : platelets : plasma)** + **IV calcium** (≥ 4 U)

Transfusion risk	Rate
HBV — highest viral risk	1 in 205,000 (Orthobullets)
HCV	1 in 1.8 million (Orthobullets) · 1 in 103,000 (OB)
HIV	1 in 1.9 million (Orthobullets) · 1 in 500,000 (OB)
Clerical error → haemolytic reaction — most prevalent adverse event	**1 in 12,000–50,000**
Anaphylaxis	1 in 150,000
Bacterial sepsis	1 in 1 million

Damage control vs early total care

ETC (1980s) fix everything at once ("too sick not to operate") → **2nd hit** in unstable, head, chest injuries

DCO (2000) temporary stabilisation (ex-fix, traction, splints, I&D) → definitive once physiology improves

First vs second hit (TO) first hit = injury → SIRS · second hit = **extensive early surgery, transfusion, hypoxia, pain** → ARDS / MODS

Phase (TO)	Response	Mediators
Early	**SIRS** (hyperinflammation)	**TNF-α, IL-1, IL-6**
Late	**CARS** (immunoparalysis → sepsis)	**IL-4, IL-10, IL-13**

DCO phases (TO) 1 life-saving (haemorrhage, airway, chest tube, binder) → 2 temporary stabilisation **≤ 48 h** (splints, ex-fix, damage-control laparotomy) → 3 ICU monitoring (MODS / ARDS) → 4 definitive fixation

Early fixation criteria – TO (TO) **pH > 7.25**, **BD > −5.5**, **lactate < 4.0**, normotensive · **not met within 8 h → DCO**

EAC (2013) definitive fixation of **spine, pelvis, acetabulum, femur within 36 h** once **any ONE** of: **lactate < 4.0 mmol/L**, **pH ≥ 7.25**, **base excess ≥ −5.5 (BD ≤ 5.5)** → ↓ delay, ↓ complications (ARDS, MOF, death), ↓ LOS, ↑ revenue; main delay = surgeon decision

Operative criteria – DCO indications

DCO indication	Threshold
ISS	**> 40** without chest trauma · **> 20** with chest trauma
GCS	**≤ 8** · **head injury AIS ≥ 3**
Bilateral femur fractures	Yes
Pulmonary contusion on radiograph	Yes (✗ Wrong options: on CT only · rib fractures alone)
Hypothermia	**< 35 °C**
Severe pelvic / abdominal injury + haemorrhagic shock	Yes (e.g. grade IV liver + SBP 85)
IL-6	**> 500 pg/dL**
Persistent acidosis	Lactate, BD, pH out of range

Not DCO triggers HR alone, SBP 90 alone, unilateral femur fracture, lactate 2.3–2.5, Hb < 9, rib fractures + PTX (borderline), pubic rami fracture (stable), temp 35.5 (borderline)

Head injury **intra-operative hypotension ↑ mortality** → avoid long surgery; SDH needing CPP monitoring → **ex-fix femur**; maintain **CPP > 70 mm Hg** (TO)
✗ Wrong options: reamed / unreamed nail · traction · beads

Inflammatory window & timing

Acute inflammatory window **days 2–5** → avoid elective definitive surgery; operate only for life / limb threats: **unstable pelvis**, compartment syndrome, vascular injury, unreduced dislocations, amputations, unstable spine, cauda equina, open fractures

Definitive conversion **pelvis 7–10 d** · **femur ex-fix → nail within 3 wk** · **tibia 7–10 d**

Ex-fix → nail at 7 d **equal union + infection rates**
✗ Wrong options: higher infection / nonunion / ARDS / mortality

Day (TO)	Physiological state	Surgery
1–2	Hyperinflammatory	**Life-saving only**
2–4	Borderline / resuscitating	Second look, temporary fixation
5–10	**Window of opportunity**	**Definitive fixation**
10–20	**Immunosuppression**	**Avoid surgery**
> 21	Recovery	Secondary reconstruction

ATLS 10th edition updates (TO)

Fluids (TO) initial **1 L warm isotonic crystalloid** (reduced from 2 L) → then **blood products early** · children **20 mL/kg** crystalloid once → 10–20 mL/kg blood

Massive transfusion (TO) **> 10 U PRBC / 24 h or > 4 U / 1 h** · **1:1:1** · TEG / ROTEM guided

– – Active space – –

TXA – CRASH-2 (TO) **1 g IV over 10 min + 1 g over 8 h**, **within 3 h** of injury

Urine output target (TO)	mL/kg/h
Adult / teen	**≥ 0.5**
Child (> 1 yr)	**1–1.5**
Infant	**1–2**

Tension pneumothorax (TO) adult needle at **4th / 5th ICS, anterior to mid-axillary line** (new) · child: **mid-clavicular** (2nd ICS)

Airway (TO) **needle cricothyroidotomy contraindicated < 12 yr** (subglottic stenosis) → tracheostomy definitive · suspected **tracheobronchial injury** (haemoptysis + surgical emphysema) → **fibre-optic intubation / bronchoscopy** · burns / inhalation → **pre-emptive intubation**

GCS updates (TO) record **best motor** · **"NT" items score 0** · **GCS-P** = GCS − pupil score (both non-reactive = 2; range **1–15**) · GCS-PACT adds age + CT

C-spine imaging rules (TO) **NEXUS**: no midline tenderness, not intoxicated, alert, no focal deficit, no distracting injury → no imaging · **Canadian C-spine**: high risk (**age ≥ 65**, dangerous mechanism, paraesthesia) → image; low risk + **45° active rotation** → no imaging

Log-roll (TO) ideal **5-person** team (minimum 4)

Pregnancy (TO) **15–30° left lateral tilt** · fetal Doppler > 10 wk, continuous CTG > 20–24 wk · indicated imaging should not be withheld

Limb haemorrhage ladder (TO) direct pressure → pressure dressing → proximal pressure → **tourniquet** · pneumatic tourniquet **UL ~250 mmHg · LL ~400 mmHg**

Urethral injury (TO) perineal / scrotal ecchymosis, blood at meatus → **retrograde urethrogram before Foley** (prostate exam no longer relied on)

Paediatric access (TO) **intra-osseous** (proximal tibia / distal femur) if IV fails

Mass vs multiple casualty (TO) multiple (capacity adequate) → treat life-threatening first · mass → **greatest chance of survival with least resource use**

DCO scenarios

– – Active space – –

Scenario	Answer
Unstable (MAP 51, BD 10, SDH) femur + open plateau	**I&D + ex-fix femur + ex-fix plateau**
BD 10, bilateral femurs, open tibia, forearm	**Ex-fix both femurs + tibia, I&D, splint forearm** (✗ Wrong options: ORIF forearm · nails)
Bilateral chest tubes, lactate 7, open femur + tibia	**I&D + bilateral ex-fix** (✗ Wrong options: nails · splints alone)
Lactate 10 unchanged after laparotomy, contusions	**Ex-fix tibia + femur** (✗ Wrong options: nails · slab + traction)
Lactate 8 → 7, femur + pelvis	**Ex-fix femur + pelvis** (✗ Wrong options: ORIF pelvis · splint femur)
High lactate, BD, lung injury – femur	**Ex-fix** (✗ Wrong options: nail · plate · traction (supine harmful))
BD 10, oliguria, open foot, femur, forearm, head injury	**I&D, ex-fix ankle + foot + femur, splint forearm**
Rising lactate 3 → 5, core temp 34 °C	**Ex-fix femur** (✗ Wrong options: reamed nail · traction · plate · MAST)
pH 7.21, BD −11, pelvis + distal femur + humerus	**Ex-fix pelvis + femur, splint humerus**
pH 7.21, BE −6.5	**DCO** (not EAC)
Vertical shear pelvis, lactate 9, laparotomy	**Skeletal traction** (✗ Wrong options: binder (worsens) · anterior ex-fix · ORIF · SI screws now)
SFA transection, unresuscitated, guarded prognosis	**Ex-fix femur + temporary arterial shunt** (+ fasciotomy) → definitive repair 24–48 h
Stable, lactate 1.7, femur + ankle + humerus	**IMN femur, splint ankle + humerus**
Stable, open ankle, femur + tibia	**IMN femur + tibia, I&D + ex-fix ankle**
Isolated floating knee, stable (ISS 27)	**IMN tibia + femur** (retrograde femur + antegrade tibia, one setup)

Upper-limb fractures in DCO	**splint**
Inadequately resuscitated for definitive care	**BE −6.5** ✗ Wrong options: lactate 2.0 · IL-6 300 · pH 7.28
Supports early definitive care	**BD 5.5** ✗ Wrong options: IL-6 550 · pH 7.21 · lactate 4.4 · BE −5.8
Femur fixation decision in polytrauma (TO)	**lactate < 4.0** · **best resuscitation marker (TO)** – **lactate < 2 mmol/L** ✗ Wrong options: BP · > 3 long bones · BE 5.5 as written

Pelvic haemorrhage

Order	**reduce pelvic volume (sheet, binder, ex-fix, traction)** → stable → **CT chest / abdo / pelvis** · unstable → laparotomy and / or **angiography + embolisation** (venous bleeding commoner than arterial)

– – Active space – –

Stabilised after sheet	**CT** next ✗ Wrong options: swap to binder · ER ex-fix · embolisation · ORIF
APC II – avoid (TO)	**log-rolling** (disturbs retroperitoneal clot) (✓ binder, ✓ perineal exam, ✓ early contrast CT)
Pelvic packing	**subumbilical midline incision** (6–8 cm up from symphysis) → split rectus fascia → **3 lap pads each side** below brim, SI joint → retropubic ✗ Wrong options: ASIS–symphysis · ASIS–ASIS · pararectus

Triage, transfer & battlefield

Mass casualty triage	immediate, delayed, minor, expectant
ATLS transfer to trauma centre	open pelvic fracture, major crush, multiple proximal long-bone fractures, solid-organ injury managed expectantly · **isolated clean open forearm fracture → may stay**
Battlefield mangled leg bleeding (TO)	**tourniquet first** ✗ Wrong options: pressure dressing · haemostatic dressing · evacuate first
Vascular reconstruction limit (TO)	**6 h warm ischaemia** (TO) · ideally **< 8 h** from injury (OB)

Head injury (TO)

GCS (TO)	Eye	Verbal	Motor
6	–	–	Obeys
5	–	Oriented	**Localises**
4	**Spontaneous**	Confused	Withdraws
3	To voice	**Inappropriate words**	Flexion
2	To pain	Incomprehensible	Extension
1	None	None	None

Localises, eyes open spontaneously, words make no sense (TO) **E4 V3 M5**

	Extradural (TO)	Acute subdural (TO)
Space	**Inner table ↔ dura**	**Under dura**
CT	**Biconvex**	**Crescentic**
Source	**Middle meningeal artery (85% arterial)**, pterion, temporoparietal fracture	Bridging veins / cortical laceration
Classic	**Lucid interval** (10–25%), contralateral hemiparesis, ipsilateral dilated pupil	More brain injury; worse prognosis
Surgery	**> 30 cm³** regardless of GCS; nonop if < 30 cm³ + < 15 mm + MLS < 5 mm + GCS > 8 + no deficit; **GCS < 9 + anisocoria → immediate**	**Thickness > 10 mm or MLS > 5 mm → evacuate regardless of GCS**; else if GCS ↓ ≥ 2, pupils abnormal, ICP > 20

Head CT read (TO)	scalp → **bone window** (fracture) → extra-axial blood → intra-axial lesions → oedema (lost sulci, Sylvian fissures, basal cisterns) · SAH = feathery in sulci/cisterns · **focal SAH → CT angiogram for aneurysm, secure before orthopaedic surgery** · fresh blood bright, **isodense at ~5–6 days** · central bleed (basal ganglia, thalamus, dentate) = **hypertensive**, not traumatic · **retained metallic fragment → no MRI**
SDH age on CT (TO)	acute 1–3 d **hyperdense** · subacute 4 d–3 wk **isodense** · chronic > 3 wk **hypodense** · anticoagulation ↑ ASDH 7× men, 26× women
Uncal herniation (TO)	**CN III palsy** (ipsilateral fixed dilated pupil, down + out)
Kernohan notch (TO)	midline shift pushes **contralateral cerebral peduncle** onto tentorium → **ipsilateral hemiplegia** (false localising)
CSF rhinorrhoea after facial trauma (TO)	**observe ~2 wk** (70% stop in 1 wk) · **β2-transferrin** confirms · **CT cisternography** localises · surgery if > 2 wk, delayed / spontaneous, recurrent meningitis; anosmia 78% ✗ Wrong options: immediate dural repair · acetazolamide first · cautery
Tension pneumocephalus (TO)	**Mount Fuji sign**; trauma commonest; burr holes if symptomatic
Carotico-cavernous fistula (TO)	**pulsatile proptosis, chemosis, bruit, ↓ vision**, **CN VI palsy** commonest diplopia; direct type A (high flow, traumatic 0.2%) → embolisation
Stems without answer key (TO)	elderly on antiplatelets with confusion + gait disturbance (options EDH, chronic SDH, SAH, infarct) · 14-month-old enlarging parietal mass after injury at 8 mo (options growing skull fracture, hygroma, NCC, cephalhaematoma) · CT image identification — **answer key not provided by the source**

Chest injuries (TO)

Flail chest (TO)	**≥ 2–3 adjacent ribs broken in ≥ 2 places → paradoxical movement**
Stove-in chest (TO)	flail segment **collapses inward**; **blunt** trauma ✗ Wrong options: penetrating (NOT true)

6 Complications

ARDS + MOF	2nd-hit during inflammatory window; ETC in unstable patients
Fat embolism	immediate nailing in unresuscitated chest-injured patients
Hypothermia < 35 °C	↑ mortality

X vs Y

	ETC	DCO	EAC
Era	1980s	2000	2013
Principle	Fix all at once	Stabilise → definitive later	Definitive < 36 h when resuscitated
Trigger	–	Unstable / extremis	Lactate < 4, pH ≥ 7.25, BE ≥ −5.5 (any one)

– – Active space – –

	Class II	Class III
Loss	15–30%	30–40%
BP	**Normal**	**↓**
HR	> 100	> 120
Blood needed	No	**Yes**

How it's asked

"Residual hypoperfusion shown by?"	**→ lactate 4.5**
"Successfully resuscitated?"	**→ lactic acid 1.9**
"Normal in compensated shock EXCEPT?"	**→ serum lactate**
"Occult hypoperfusion despite normal vitals?"	**→ lactate 5**
"Best variable for early definitive fixation decision?"	**→ base deficit**
"Alternative to lactate / BD?"	**→ gastric intramucosal pH 7.4**
"Normal lactate + BE but abnormal vitals?"	**→ adequate tissue perfusion**
"Resuscitation status with normal BP / HR / urine?"	**→ cannot be determined**
"Compensated shock risk?"	**→ primed immune system → SIRS**
"Resuscitation complete when?"	**→ aerobic metabolism restored in all tissue beds**
"Philosophy using lactate < 4 / pH ≥ 7.25 / BE ≥ −5.5?"	**→ early appropriate care**
"Parameter supporting early definitive care?"	**→ BD 5.5**
"Inadequately resuscitated?"	**→ BE −6.5**
"↑ Mortality in head-injured polytrauma?"	**→ intra-operative hypotension**
"Most predictive of mortality?"	**→ base excess −12**
"Elderly polytrauma – mortality predictor?"	**→ base deficit**
"Tachycardia + narrow pulse pressure first at?"	**→ 25%**
"HR 135, BP 85/58 – loss?"	**→ 30–40%**
"Class II → III on the floor?"	**→ presented class II, now class III**
"2 L acute loss – unique sign?"	**→ ↓ SBP + narrow pulse pressure**
"Haemorrhagic vs neurogenic?"	**→ tachycardia**
"End-stage septic shock EXCEPT?"	**→ ↑ SVR**
"Septic shock finding?"	**→ ↓ SVR**
"Swan-Ganz used in (TO)?"	**→ cardiogenic shock after spinal injury**
"NOT in lethal triad (TO)?"	**→ haemorrhage / hypoxia**
"Lethal triad (TO)?"	**→ hypothermia, acidosis, coagulopathy**

"Does NOT favour DCO (TO)?"	→ SBP < 90
"Isolated floating knee, ISS 27?"	→ IMN tibia + femur
"Unstable, SDH, femur + open plateau?"	→ ex-fix both
"BD 10, multiple long bones?"	→ ex-fix femurs + tibia, splint forearm
"Supports ex-fix over nail?"	→ grade IV liver + SBP 85
"Stable, femur + tibia + open ankle?"	→ IMN femur + tibia, ex-fix ankle
"Vertical shear, lactate 9?"	→ skeletal traction
"DCO indications EXCEPT?"	→ rib fractures
"Suggested DCO indication?"	→ bilateral pulmonary contusions on CXR
"Bilateral chest tubes, lactate 7, open femur + tibia?"	→ I&D + bilateral ex-fix
"SDH + open femur after splenectomy?"	→ ex-fix
"Conventional indicator supporting DCO?"	→ platelets 20,000
"Lactate 10 after laparotomy, contusions?"	→ ex-fix tibia + femur
"Lactate 8 → 7, pelvis + femur?"	→ ex-fix femur + pelvis
"High lactate + BD + lung injury – femur?"	→ external fixation
"BD 10, oliguria, open foot, femur, forearm?"	→ ex-fix ankle / foot / femur, splint forearm
"Lactate rising, 34 °C?"	→ ex-fix
"Ex-fix → nail at 7 d?"	→ equal union + infection
"pH 7.21, BD −11?"	→ ex-fix pelvis + femur, splint humerus
"Borderline – DCO indicator?"	→ ISS > 40 without chest injury
"pH 7.21, BE −6.5 – approach?"	→ damage control
"Stable, lactate 1.7?"	→ IMN femur, splint ankle + humerus
"SFA transection, unresuscitated?"	→ ex-fix + arterial shunt
"IL-6 associated with?"	→ ↑ mortality
"IL-6 reliable measure of?"	→ injury severity
"Reliable systemic inflammatory marker?"	→ IL-6
"Children vs adults?"	→ adults: robust response, lungs first
"Meets SIRS?"	→ RR 24 + HR 94
"Highest viral transfusion risk?"	→ hepatitis B
"Massive transfusion ratio?"	→ 1:1:1
"Most prevalent transfusion adverse event?"	→ clerical error
"Avoid in APC II (TO)?"	→ log-rolling
"Stable after pelvic sheet – next?"	→ CT chest / abdomen / pelvis

– – Active space – –

"Pelvic packing incision?"	→ **subumbilical midline**
"Missed wrist fracture after coma?"	→ **wrist never imaged**
"Pregnant trauma positioning?"	→ **left lateral decubitus**
"Managed without transfer?"	→ **isolated open forearm fracture**
"Airbags reduce all EXCEPT?"	→ **pelvic ring injuries**
"Female sex linked to?"	→ **PTSD**
"PTSD risk factor?"	→ **high ISS**
"GCS: eyes open, nonsense words, localises (TO)?"	→ **E4 V3 M5**
"Clear nasal discharge after facial trauma (TO)?"	→ **wait and watch**
"Right SDH, right CN III + right hemiplegia (TO)?"	→ **left cerebral peduncle (Kernohan)**
"Peaked frontal poles with air (TO)?"	→ **tension pneumocephalus**
"SDH – true (TO)?"	→ **MLS > 5 mm → evacuate regardless of GCS**
"Biconvex hyperdensity (TO)?"	→ **extradural haematoma**
"Pulsatile proptosis 2 wk after head injury (TO)?"	→ **carotico-cavernous fistula**
"SDH + SAH – likely CN deficit (TO)?"	→ **CN III**
"Battlefield mangled limb bleeding (TO)?"	→ **tourniquet**
"Max delay for vascular repair (TO)?"	→ **6 h warm ischaemia**
"NOT true of stove-in chest (TO)?"	→ **penetrating injury**
"TXA dosing per CRASH-2?"	→ **1 g over 10 min + 1 g over 8 h, within 3 h** (TO)
"Adult needle decompression site (ATLS 10)?"	→ **4th / 5th ICS anterior to mid-axillary line** (TO)
"Window of opportunity for definitive fixation?"	→ **days 5–10** (TO)
"Focal SAH in a trauma patient before orthopaedic surgery – next?"	→ **CT angiogram to exclude aneurysm** (TO)

FEMORAL HEAD FRACTURES

– Trauma · 19 / 67 –

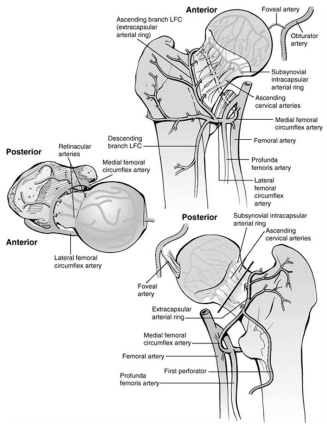
1 Basics

Epidemiology

Rare	**12% of hip dislocations**; rising (more MVAs, better resuscitation)
With posterior dislocation	**5–15%** (Orthobullets) · **~15%** (TO) · **95% of head fractures accompany a dislocation (TO)**

Blood supply

Vessel	Fact
MFCA (from profunda)	**Main supply to weight-bearing head**; runs **posteromedially** → endangered by posterior approach
Artery of ligamentum teres (obturator or MFCA)	**10–15%**; perifoveal area



Femoral head blood supply

2 Mechanism

Mechanism	impaction, avulsion, shear; **dashboard injury** (knee on dash, hip flexed + adducted + IR), falls, sport, industrial
Posterior dislocation	head **shears against posterior acetabular rim** → fracture (usually **anteromedial / anteroinferior** fragment)
Anterior dislocation	**impaction / indentation** fractures of head
Associated	femoral neck fracture, acetabular (posterior wall) fracture, **sciatic neurapraxia**, AVN, **ipsilateral knee ligament injury**

3 Presentation

Posterior dislocation	**flexed, adducted, IR**, shortened (little rotation with large wall fractures)
Anterior dislocation	flexed, **abducted, ER**

– – Active space – –

Check **sciatic nerve**, ipsilateral knee stability

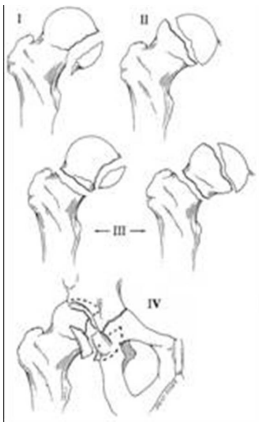
4 Diagnosis & Classification

Key radiology findings

Study	Use
AP pelvis + hip series (pre + post reduction)	–
Judet	Acetabular fracture
Inlet / outlet	Pelvic ring
Post-reduction CT	**Loose bodies / incarcerated fragments**, fragment size, plane, comminution, impaction, acetabulum

Pipkin classification

Type	Definition	Key fact
I	**Below fovea** (infrafoveal), **non-weight-bearing**, small	Nonop or **excision** (excision > ORIF)
II	**Above fovea** (suprafoveal), **weight-bearing**, larger; ligamentum stays on inferior fragment → may block reduction	ORIF if displaced (ORIF > excision)
III	I or II **+ femoral neck fracture**	**Highest AVN →** catastrophic ORIF failure
IV	I or II **+ acetabular fracture (usually posterior wall)**	Treat by acetabular injury



Pipkin classification

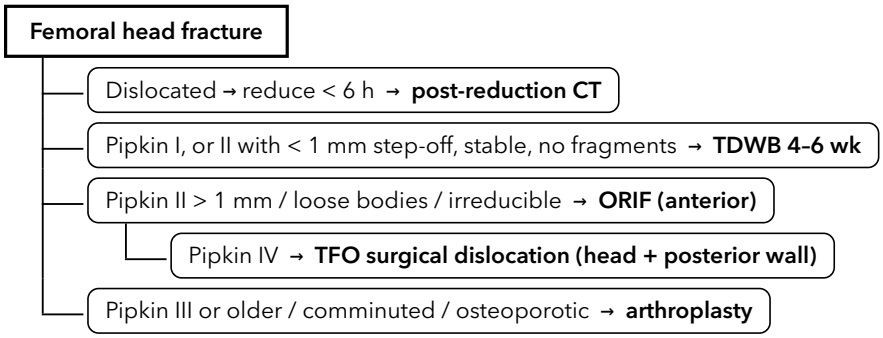
Chiron classification (TO)

Type (TO)	Extent
1	Osteochondral injury
2	¼ of head
3	⅓ of head
4	½ of head
5	**Cartilage impaction of weight-bearing dome** (superior collapse)

Subtypes (TO) **A** pure head · **B** + acetabulum · **C** + neck · **Brumback** – more prognostic than Pipkin (TO)

– – Active space – –

5 Management



Operative criteria

Nonoperative	Operative
Pipkin I	**Pipkin II with > 1 mm step-off**
Pipkin II with < 1 mm step-off (Orthobullets, OB) · **< 2 mm** (TO)	**Loose bodies / incarcerated fragments**
No interposed fragments	**Pipkin III, IV**
Stable, concentric hip	Irreducible fracture-dislocation, polytrauma
–	**Older patients**, comminution, osteoporosis → **arthroplasty**

1 mm, not 2 mm femoral head cut-off differs from other articular fractures (OB)

Strongest nonop indication on post-reduction CT **infrafoveal (Pipkin I) with congruent joint**
✗ Wrong options: suprafoveal displaced · neck fracture · displaced acetabulum · incarcerated posterior wall fragment

TO nonop indication (TO) **Pipkin II with 1 mm displacement**
✗ Wrong options: Pipkin I with 3 mm · Pipkin III · nonconcentric reduction

Displaced Pipkin II on CT (TO) **ORIF with Herbert (headless) screw**
✗ Wrong options: nonop 6 wk

Hip reduction

Timing **within 6 h** → AVN **5–40%**, ↑ with delay

Technique full sedation + relaxation; in-line traction, slight adduction, counter-pressure on pelvis; **no forceful manoeuvres → post-reduction CT**

Nonop regimen **TDWB 4–6 wk**, avoid **adduction + IR**; serial X-rays

– – Active space – –

Surgical approach

Approach	Facts
Smith-Petersen (anterior)	**Internervous: superior gluteal ↔ femoral**; best view of **anteromedial** fragment; shorter, less blood loss, better reduction; **spares MFCA → lower AVN** (KL **3.2× AVN**, TO); **↑ functionally significant HO**
Watson-Jones (anterolateral)	TFL ↔ gluteus medius (both superior gluteal → intermuscular)
Surgical hip dislocation + trochanteric flip osteotomy (TFO)	**Whole head + acetabulum through one approach** (head + posterior wall); **preserves MFCA**; ↓ AVN + **↓ PTOA**; **↑ HO**; same function; **lateral position**
Kocher-Langenbeck ± digastric (trochanteric) osteotomy	Posterior wall (Pipkin IV); gluteus max split (no internervous plane); **plain KL → worse outcomes, more AVN**

Exposure	**periacetabular capsulotomy** to protect head supply
Pipkin IV – preserving blood supply	**greater trochanteric osteotomy (surgical dislocation)** ✗ Wrong options: strip gluteals off crest = extended iliofemoral · ligate LFCA ascending branches · detach piriformis (not needed) · supine (lateral position)
TFO primary advantage	**address all injury sites through one approach** ✗ Wrong options: minimally invasive · minimal stripping · better union
Irreducible posterior fracture-dislocation – TFO true?	**lower PTOA** ✗ Wrong options: more AVN · less HO · incomplete view · worse function
Polytrauma, posterior dislocation + head fracture – definitive approach	**Smith-Petersen** ✗ Wrong options: Southern / posterior (MFCA) · Stoppa · KL · iliofemoral
Pipkin IV with small posterior wall	wall may be left; suprafoveal head via anterior approach

Fixation & outcomes

Fixation	**≥ 2 lag screws 2.7 / 3.5 mm (mini-fragment) countersunk**, headless screws, bioabsorbable pins ✗ Wrong options: 3.0 mm cannulated screws with washers → poor results
Pipkin II ORIF – true?	**mini-fragment screws suffice** ✗ Wrong options: HO uncommon (~15–20%) · KL optimal · excision better · 2 mm cut-off
Post-op	immediate ROM, **delay WB 6–8 wk**, quads + abductor strengthening; X-ray at 6 mo for AVN / OA
Outcome	mirrors associated injury (dislocation, neck fracture)
Pipkin III ORIF in 51-yo – 6-mo outcome	**fixation failure + AVN → THA** ✗ Wrong options: posterior HO revision · pain-free · core decompression · advanced OA at 6 mo

Arthroplasty

Indications	**older patients** with Pipkin I–IV displaced, osteoporotic, comminuted

Facts	immediate WB; **higher dislocation risk** than THA for OA; hemiarthroplasty if no acetabular fracture; **posterior approach best for posterior wall**
Arthroscopy	loose body removal

6 Complications

Complication	Rate / fact
HO	**6–64%**; **anterior approach > posterior**; ~15–20% after fixation; **radiation** prophylaxis (esp. head injury)
AVN	**0–23%**; ↑ with **delayed reduction**; anterior approach **not** ↑ AVN
Sciatic neurapraxia	**10–23%**; **peroneal division**; **60–70% recover**
DJD	**8–75%** (incongruity, cartilage damage)
↓ Internal rotation	Often not disabling

X vs Y

	Smith-Petersen	Kocher-Langenbeck	TFO surgical dislocation
View	Anteromedial head	Posterior wall	**Whole head + acetabulum**
AVN	**Lower**	**3.2× higher** (TO)	Lower
HO	**Higher**	Lower	**Higher**
Best for	Pipkin I / II	PW only	**Pipkin IV**

	Pipkin I	Pipkin II
Location	**Infrafoveal**	**Suprafoveal (weight-bearing)**
Displaced →	**Excise**	**ORIF**

How it's asked

"Nonop indication (TO)?"	**→ Pipkin II with 1 mm displacement**
"CT: displaced suprafoveal fragment (TO)?"	**→ Pipkin II – ORIF with Herbert screw**
"Which image is Pipkin II?"	**→ suprafoveal fragment without neck / acetabulum fracture**
"Strongest indication for nonop?"	**→ Pipkin I, congruent**
"Preserve blood supply in Pipkin IV?"	**→ greater trochanteric osteotomy**
"TFO primary advantage?"	**→ all injuries through one approach**
"Best definitive approach after posterior dislocation + head fracture?"	**→ Smith-Petersen**
"TFO – true?"	**→ lower post-traumatic OA**
"Pipkin II ORIF – true?"	**→ mini-fragment screws sufficient**

– – Active space – –

"Pipkin III ORIF at 6 months?"	**→ failure + AVN → THA**
"Reduce dislocation within?"	**→ 6 hours**
"Sciatic palsy – division?"	**→ peroneal; 60–70% recover**

FEMORAL NECK FRACTURE NONUNION

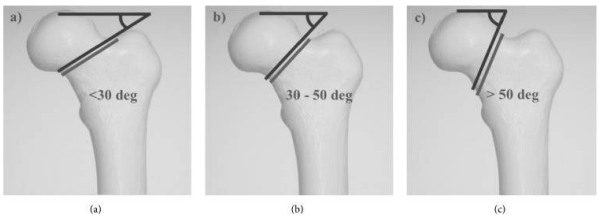
– Trauma · 20 / 67 –

1 Basics

Epidemiology

Rate **10–30% after femoral neck ORIF**

Risk factor	Note
Vertical fracture (Pauwels III)	Shear > compression
Significant initial displacement, **posterior comminution**	–
Malreduction, esp. varus	(✗ Wrong options: valgus)
Older age	–
Delay to fixation > 24 h	–



Pauwels classification

Why the neck fails to heal

Intracapsular **no periosteum / extraosseous supply**, **bathed in synovial fluid**, **intraosseous healing only (no callus)** → anatomic reduction essential

Anatomy

Neck-shaft angle **130 ± 7°** · **anteversion 10 ± 7°** · tensile + compressive trabeculae

Blood supply extracapsular ring → **ascending cervical (retinacular) arteries**; **MFCA (lateral epiphyseal) = main adult supply**; LFCA anterior ring; gluteal small; **ligamentum teres insignificant**

Load single-leg stance force **parallel to neck**, **4× body weight**

Associated **AVN**, malunion, hardware failure, **septic nonunion**, low BMD

3 Presentation

Symptoms persistent groin / buttock pain, pain on WB + hip extension, referred to medial thigh / knee

Exam **↓ internal rotation**, antalgic gait, **pain with log-roll**, pain on straight-leg raise; check old incision, sciatic + femoral nerves

– – Active space – –

Nonunion signal **persistent pain + no radiographic healing at 3 months** → consider revision

4 Diagnosis & Classification

Key radiology findings

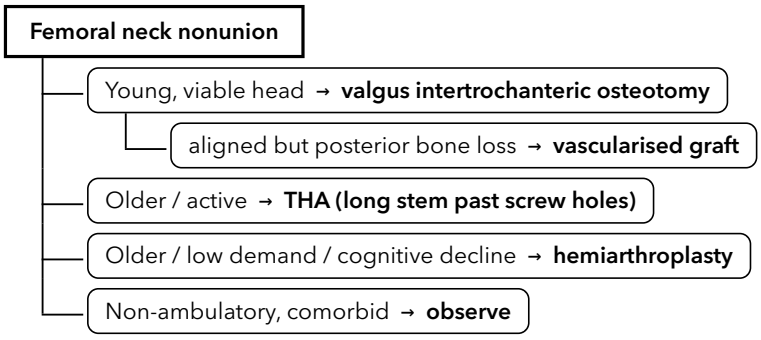
Study	Use
AP pelvis + hip, **cross-table lateral**, full femur	Lucent line, **settling**, **hardware backing out**, head perforation, **> 10 mm change in position**, collapse, calcar comminution, **varus**
CT	**Most definitive diagnosis**; planning
MRI	**Head AVN**
Bone scan	Uptake at nonunion; helps exclude AVN

Infection work-up **CBC, ESR, CRP** → aspirate if raised → intra-op frozen section → **staged** if infected

Classification (descriptive)

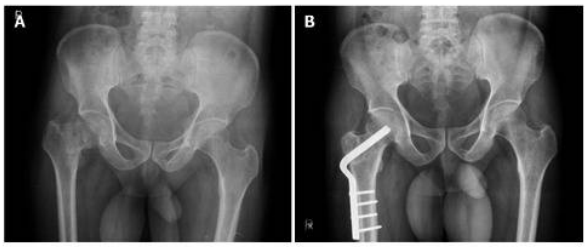
Type	Description
1	Inadequate fixation / non-anatomic reduction
2	Loss of fixation with displacement
3	Fibrous nonunion, no displacement, intact fixation

5 Management



Treatment options

Option	Indication	Fact
Observation	Non-ambulatory, comorbid	Poor function
Revision ORIF	**Not recommended**	Bone loss, LLD, high failure
Valgus intertrochanteric osteotomy	**Young**, viable head, varus collapse, nonunion > 3 mo	**Commonest salvage**; turns **vertical → ~30° (horizontal)** line → **compression**; restores length + abductor lever; **80–90% union**; not with bone loss, incongruity, **age > 65 (relative)**
Vascularised graft (quadratus femoris pedicle, free fibula)	Aligned nonunion with **posterior bone loss**, failed fixation	Up to 90% union; not if malalignment is the cause
Hemiarthroplasty	**Low demand, comorbid, cognitive decline**	**Lower dislocation than THA**; **cemented > uncemented**; bipolar ≈ unipolar; risk of acetabular erosion
THA	**Older, active, cognitively intact**; acetabular degeneration	**~90% 10-yr, 75% 20-yr survival**
Arthrodesis	Young, non-viable head, heavy labourer	**20–35° flexion, 0–5° adduction, 5–10° ER**; arthroplasty now favoured



Valgus intertrochanteric osteotomy

26-yo nonunion after screws at 3 mo — **valgus intertrochanteric osteotomy**
✗ Wrong options: graft + revision ORIF · hemiarthroplasty · DHS without correction · core decompression

76-yo community ambulator — **THA gives best long-term outcome**
✗ Wrong options: valgus malreduction causes failure (varus does) · revision ORIF + inverted triangle · valgus osteotomy · hemi ↑ dislocation (hemi is lower)

Healed fracture but AVN collapse + screw cut-out into acetabulum (62-yo) — **THA with long cementless stem** to bypass screw-hole stress risers
✗ Wrong options: primary tapered stem · hardware removal · valgus osteotomy · IM revision

Technique points

Valgus osteotomy — lateral approach; **blade plate or sliding hip screw**; risks: malrotation, **over-medialisation of shaft → ↓ offset + abductor weakness** (longer blade avoids)

– – Active space – –

Conversion THA	**longer stems** past screw holes (intra-op fracture risk), gentle cup impaction ± screws, **dual-mobility / large heads** vs dislocation; loosening ↑ if **age < 65 + BMI > 30**
Dual mobility vs fixed bearing (< 55 yr)	**lower dislocation** (same HHS, loosening, infection, op time)
Failed hemi → THA	worse than primary THA

6 Complications

Complication	Fact
AVN	**Up to 40%** with nonunion; ~15% symptomatic; risk: displacement, poor reduction → arthroplasty
Acetabular cartilage wear	→ THA
Trendelenburg gait	Lost offset / abductor damage
Prosthetic dislocation	**10–20% in salvage THA** vs < 10% primary; abductor insufficiency, posterior approach

X vs Y

	Valgus IT osteotomy	THA
Patient	**Young**, viable head	**Older**, active
Mechanism	Shear → compression	Replace joint
Result	80–90% union	~90% 10-yr survival

How it's asked

"26-yo, screws, 3-mo nonunion?"	→ **valgus intertrochanteric osteotomy**
"76-yo nonunion – true?"	→ **THA gives best long-term outcome**
"Healed but AVN collapse + screw cut-out?"	→ **THA with long cementless stem**
"Dual mobility vs fixed bearing?"	→ **lower dislocation**
"Most definitive imaging for nonunion?"	→ **CT**
"Malreduction most linked to failure?"	→ **varus**

FEMORAL NECK FRACTURES

– Trauma · 21 / 67 –

1 Basics

Epidemiology

Rising	ageing population · **women > men**, Caucasians > African Americans · **USA highest incidence**
Cost	**most expensive fracture per person**
Associated	**ipsilateral femoral shaft fracture: 6–9% of shaft fractures** (~10%, OB) → fix **neck first**
Burden (TO)	**> 300,000** hip fractures/yr in USA · **> 6 million worldwide by 2050** · **55–59% of hip fractures are intracapsular**
Risk factors (TO)	secondary: **HIV**, **primary hyperparathyroidism**, chronic liver disease, spinal injury · drugs: **PPIs**, anti-epileptics, **aromatase inhibitors**, inhaled corticosteroids, **thiazolidinediones**, GnRH inhibitors · modifiable: low weight, smoking, alcohol, inactivity

Healing biology

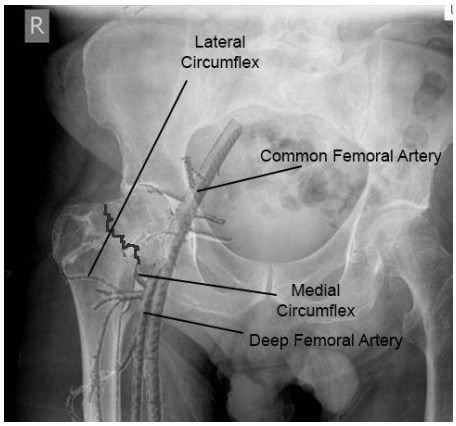
Intracapsular	**bathed in synovial fluid, no periosteum, limited callus** → poor healing; anatomic reduction essential
"Unsolved fracture" (TO)	MFCA dependence · **no cambium layer → endosteal healing only** · **angiogenesis-inhibiting factors in synovial fluid** · revascularisation by creeping substitution + vascular ingrowth

Anatomy

Neck-shaft angle 130 ± 7° · anteversion 10 ± 7°

Head supply	**MFCA (lateral epiphyseal artery) = main** · LFCA anterior / inferior head · inferior gluteal · ligamentum teres insignificant
Retinacular arteries (TO)	superior + anterior often torn · **inferior retinacular artery usually preserved** in displaced fractures
Calcar femorale (TO)	dense posteromedial bone plate → stability
Angles – TO lecture (TO)	neck-shaft **130° men, 128° women** · anteversion **15–25°** (TO) vs 10 ± 7° (Orthobullets) – read the options
Displacement	disrupts retinacular vessels + **intracapsular haematoma (tamponade, effect debated)**
Lowest intracapsular pressure	**flexion + abduction + ER** (patients lie this way) · highest = **extension + IR** ✗ Wrong options: flexion + IR · extension + ER

– – Active space – –



Blood supply to the femoral head

Genes & lab markers

FRAX (WHO)	uses **femoral neck BMD (DEXA)**, BMI, current smoking, parental hip fracture, prior fracture < 50, etc. → **spine T-score NOT used** · ~50% of fragility fractures occur without DEXA-defined osteoporosis

2 Mechanism

Young	high energy (vertical, Pauwels III) · **Elderly** – low-energy falls
Cyclic load	single-leg stance compression ~4× body weight → superior-neck tension fractures
Mechanism (TO)	axial load in flexion + abduction · sideways fall on greater trochanter · **external rotation** tensions anterior capsule + iliofemoral ligament → **posterior comminution**

3 Presentation

Impacted / stress	mild groin or medial thigh / knee pain, no deformity, pain at ROM extremes, **pain on percussion of greater trochanter**
Displaced	leg **shortened, externally rotated, abducted**; pain with log-roll
Intra- vs extracapsular (TO)	intracapsular: **ER < 45°**, shortening **< 1 inch**, minimal bruising · extracapsular: **marked ER** (near 90°), more bruising
X-ray signs (TO)	break in medial cortex · **prominent lesser trochanter** (ER) · overriding greater trochanter · broken trabeculae + **Shenton's line**

4 Diagnosis & Classification

Key radiology findings

Study	Use
AP pelvis, cross-table lateral, full femur	Standard
Traction-internal rotation AP (15° IR)	**Best to define fracture type** (accounts for anteversion)
AP in abduction + IR (TO) / 15–20° IR	**Best position for impacted neck fracture (TO)**
Contralateral hip	Templating
CT	Displacement, comminution
MRI	**Occult fracture – imaging of choice** (< 24 h); **not** reliable for head viability
Thin-slice MDCT (TO)	**Second-line after X-ray for occult fracture (TO)** – sensitivity + specificity 100% with experienced reader
Bone scan	Occult fracture **after 72 h** if MRI contraindicated
Duplex	**DVT if delayed presentation (> 2 days)**
Bone scan / PET-CT (TO)	**Early prediction of AVN**
MRI (TO)	100% sensitivity + specificity for occult fracture; dynamic MRI → head vascularity

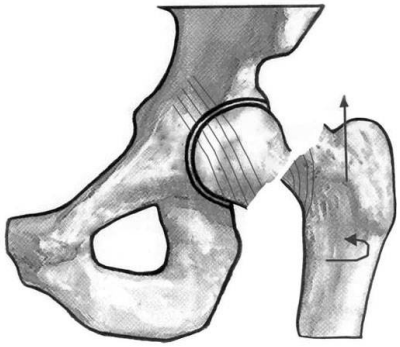
Occult fracture next step
MRI hip + pelvis (Orthobullets, OB) · **thin-slice MDCT** (TO) – read the options
✗ Wrong options: stress views · bone scan at 7 d · repeat CT at 48 h · WBAT

Delayed presentation 3 days
bilateral lower-limb duplex before surgery
✗ Wrong options: inlet / outlet · CT head without head signs · CT / MRI for displaced fracture

Garden classification

Type	Description	Group
I	**Incomplete, valgus impacted**	Non-displaced
II	**Complete, non-displaced**	Non-displaced
III	**Complete, partially displaced**	Displaced
IV	**Complete, fully displaced**	Displaced

– – Active space – –



Garden IV femoral neck fracture

Pauwels classification (angle from horizontal)

Type	Angle	Note
I	**< 30°**	Compression
II	**30–50°**	–
III	**> 50°**	**Shear; most unstable; highest nonunion + AVN**

AO/OTA (TO) **31B1 subcapital · 31B2 transcervical · 31B3 basicervical**

Femoral neck stress fractures (TO)

Type (TO)	X-ray	Treatment
Tension (type A)	**Superolateral** lucent crack → displaces	**Always internal fixation**
Compression (type B)	**Inferomedial** sclerosis / callus	**Nonop if < 50% neck width**; **fix if > 50%**

Stress fracture complications (TO) varus malunion · displacement · **AVN up to 42%**

5 Management

- **Femoral neck fracture**
  - Young (< 50), displaced → **urgent anatomic reduction (open if needed) + fixation**
    - vertical (Pauwels III) / basicervical → **sliding hip screw ± anti-rotation screw**
  - Elderly non-displaced (Garden I–II) → **cannulated screws (inverted triangle)**
  - Elderly displaced, active, independent / OA → **THA**
  - Elderly displaced, low demand, dementia, Parkinson, > 80 → **cemented hemiarthroplasty**

Treatment options

Option	Indication
Observation	Non-ambulatory, minimal pain, prohibitive risk
Closed reduction + cannulated screws	**Non-displaced**; Garden I–II in elderly; displaced in young if reduction anatomic
ORIF	**Displaced in young / physiologically young (< 50)** (female → ↑ reoperation)
Sliding hip screw	**Basicervical**, **vertical (Pauwels III) in young** ± anti-rotation screw
Hemiarthroplasty	Debilitated elderly, metabolic bone disease, low demand
THA	**Older active**, **pre-existing OA / inflammatory arthritis**, Garden III–IV < 85 yr

Young patients – urgent ORIF, reduction & osteonecrosis

Displaced in young — **ORIF**
✗ Wrong options: THA (salvage only) · closed parallel screws ± Pauwels screw · cephalomedullary nail

Most important for healing + outcome — **quality of reduction**
✗ Wrong options: surgery < 12 h · open reduction itself · capsular decompression · approach · comorbidities · implant

Unacceptable closed reduction — **> 5 mm displacement in any view** or **> 10° varus / valgus / version → open reduction + SHS**
✗ Wrong options: intra-op CT · repeated closed attempts (↑ AVN) · accept 7 mm · acute THA

Most prevalent complication after ORIF in young — **osteonecrosis** (> nonunion) · AVN **15–30% in < 60 with displaced fracture**

Highest AVN risk — **45-yo woman, displaced** (young + displaced)

Approaches — **Smith-Petersen** (sartorius–TFL, protect LFCN) · **Watson-Jones** (TFL–gluteus medius; capsulotomy **anterior to lesser trochanter** to spare MFCA)

Approach extras (TO) — **Hueter modification**: incision over TFL → ↓ LFCN injury; Smith-Petersen needs a 2nd incision for screws · **Watson-Jones**: reduction + fixation through one approach, **best for basicervical**

Closed reduction (TO)	Manoeuvre
Whitman (extension)	Traction, abduction + **internal rotation** (to 45°), full extension
Leadbetter (flexion)	Hip **flexed 90°**, slight adduction + IR, gradual extension
Flynn (atraumatic)	Flexion 90°, **lateral traction** to disimpact, IR + extension

Reduction attempts — **2 only, then open** (TO, Shetty) · avoid > 3–4 (TO, Upadhyay) — read the options

Check reduction (TO) — **Garden alignment index 160° AP / 180° lateral** (medial trabeculae vs medial femoral cortex) · **Lowell lines** = smooth S / reverse-S curves

Varus (TO) — **even 1° varus unacceptable** → ↑ nonunion

Pre-op (TO) — skin traction: **no benefit** for reduction or pain, ↑ sores · **capsulotomy**: ↓ pressure, **no proven ↓ AVN** · ideal fixation in young **< 24 h**

– – Active space – –

Reduction	joysticks (A→P K-wires), evacuate haematoma; **length-stable / compression** to avoid neck shortening (shortening → ↓ SF-36, QoL)
Ipsilateral neck + shaft	**lag screws for neck first, then retrograde femoral nail** (✗ Wrong options: shaft first · plate shaft · proximal plate) · in unresuscitated patient (lactate 6, BE −8) → **delay until resuscitated** (EAC), but still early (✗ Wrong options: ex-fix + 1 wk · immediately after IR · single implant proven better · THA)

Fixation constructs

Cannulated screws	**3 screws, inverted triangle**, **inferior screw posterior + along calcar**, max spread; **start at / above lesser trochanter** (below → subtrochanteric stress riser); avoid multiple lateral perforations; 4th screw for posterior comminution (unproven) ✗ Wrong options: central
Union by screw–cortex contact (Lindequist)	posterior + inferior contact **89%** · one site **59%** · none **0%**
Screw rules (TO)	inferior + posterior screws **within 3 mm of cortex** · tips **5–10 mm below subchondral bone** · **4 screws for calcar comminution** · **DHS + derotation screw for basicervical** / posterior comminution · **Moore / Knowles pins in children**
Pauwels III (TO)	simple screws fail → **primary valgus osteotomy** converts shear to compression
Newer implants (TO)	**medial buttress plate** (anti-varus) · **biplane double-supported screw fixation (BDSF)** (osteoporotic) · **Femoral Neck System** (small footprint + anti-rotation screw) · **Targon FN** (limits telescoping / shortening) · Tronzo screw
Head viability on MRI, young (TO)	viable → **abduction / McMurray osteotomy**, **Meyer muscle-pedicle graft** (also Bakshi), FVFG · non-viable → hemi / THA
Undisplaced but arthroplasty preferred (TO)	**pre-existing OA**, **CRF**, **RA**, steroid-induced osteoporosis
Conversion to THA after fixation (TO)	10% at 2 yr · 20% at 12 yr · **65% at 14 yr** (TO — verify: steep late rise; likely cohort-specific figure)
Improve success	**screws abutting posterior + inferior neck cortex** ✗ Wrong options: titanium · start below LT · DHS for transcervical · alignment guide for parallelism
Vertical (Pauwels III)	**SHS + side plate** mechanically optimal (± anti-rotation screw) ✗ Wrong options: 2–4 parallel screws (cantilever)
FAITH (SHS vs cannulated screws)	overall **no difference in reoperation / union**; **SHS ↑ AVN (9% vs 5%, NS)**; **SHS ↓ reoperation in displaced, basicervical, current smokers**
55-yo FNF fixation – true?	**SHS ↑ AVN vs screws** ✗ Wrong options: cut-out with TAD < 25 mm (> 25 mm) · fixation ↓ reoperation vs arthroplasty · timing > reduction for AVN · screws ↑ reoperation overall

Active elderly – THA vs hemiarthroplasty

Outcome	THA vs hemiarthroplasty
Function / outcome scores	**Better with THA** (more so long-term ≥ 4 yr)
Pain relief	Better, more predictable
Reoperation	**Lower** (hemi → acetabular erosion)
Dislocation	**Higher (~10%; 5× [Orthobullets] · 7× [Orthobullets complications, OB])**; ~5% vs 2% (OB)
Peri-operative complications, blood loss, time	**Higher**
Mortality, infection, LOS, readmission, periprosthetic fracture	**Similar**

THA best for — **active, independent community ambulator** (golf instructor, pickle-ball player), **pre-existing OA**, < 75–80 yr

Hemi best for — **> 80**, low demand / household ambulator, **dementia, Parkinson's**, hepatic encephalopathy, can't follow precautions, **dialysis / multiple comorbidities** (THA no better short-term; hemi more cost-effective)

85-yo Parkinson's, household (TO) — **hemiarthroplasty** (bipolar; if THA → dual mobility)
✗ Wrong options: THR · DHS · bed rest

65-yo active Garden IV (TO) — options AMP, bipolar, cemented, uncemented hemi — **answer key not provided by the source**

THA vs internal fixation (elderly displaced) — **lower reoperation**, less pain, better function; mortality similar; ADL better short-term only
✗ Wrong options: shorter OR · less blood · less infection · shorter stay

ORIF in dialysis patient — **higher complication rate than hemi** (2 / 13 united); mortality 45% at ~2 yr, same between groups

Hemiarthroplasty – cement, dislocation & conversion

Cemented > uncemented — **lower aseptic revision + fewer intra- / post-op periprosthetic fractures**, **better QoL at 4 mo**, better short / medium-term mobility · collared stems ↓ periprosthetic fracture
✗ Wrong options: in-hospital mortality difference · shorter OR (cement longer)

Unipolar ≈ bipolar outcomes · bipolar for instability risk (Parkinson, dementia)

Dislocation risk — **posterior approach** (✗ Wrong options: anterior, anterolateral, bipolar, unipolar) · **posterior wall insufficiency**, labral injury, poor capsular repair, oversized head (✗ Wrong options: 20° anteversion · ↑ offset · +1.5 cm length (tighten) · cementless stem (fracture not dislocation))

Anterolateral approach — ↑ abductor weakness

Failed bipolar with acetabular wear + superior migration — **convert to THA — add acetabular component only** (femoral stem well fixed)
✗ Wrong options: remove wires · walking aid · PT · revise stem

Elderly displaced insufficiency fracture (walker user) — **hemiarthroplasty**
✗ Wrong options: CMN · DHS (for extracapsular) · ORIF · resurfacing (FNF absolute contraindication)

– – Active space – –

Failed screw fixation in 82-yo, comminuted at revision	**remove hardware + arthroplasty**
Internal fixation vs hemi at 3 mo–1 yr	**higher failure with fixation** (most within 2 yr) · **46% vs 8% failure at 2 yr** (> 70 yr) · mortality same

Perioperative care

Timing	**as soon as medically optimised, ideally < 24–48 h**; **delay > 4 days → mortality × 2.25** (TO); **regional anaesthesia ↓ DVT, PE, pneumonia, transfusion** (TO); **< 24 h of admission ↓ 30-d, 90-d, 1-yr mortality**; **delay > 4 days ↑ 1-yr mortality**; delay > 48 h acceptable **only to optimise multiple comorbidities**
> 3 comorbidities	**pre-op medical optimisation** matters more than speed
Delays surgery without changing treatment	**pre-op echocardiogram** ✗ Wrong options: fascia iliaca block · geriatric admission · Hb < 10 · aspirin
DOAC (rivaroxaban 12 h ago)	**proceed to surgery** ✗ Wrong options: andexanet (life-threatening bleeding only) · FFP (ineffective) · delay 24 h · PCC
Anaesthesia	spinal vs general: **no difference** in delirium, mortality, ambulation at 60 d (Orthobullets) · **general ↑ delirium + mortality** (OB) — read the options
↓ Mortality + complications	**surgery < 24 h**, **multidisciplinary / ortho-geriatric co-management**, neuraxial anaesthesia, **TXA**, **early WBAT + mobilisation**, opioid-sparing analgesia ✗ Wrong options: delayed WB · echo · general anaesthesia · PCA · traction · SNF discharge
Post-op delirium (~25%)	**↑ length of stay**, nursing home placement · risk: age, dementia, pre-injury nursing home ✗ Wrong options: ICU · in-hospital death

Mortality & prognosis

Fact	Value
1-yr mortality	**~25–30%** (14–36%; 20% in 85-yo)
In-hospital mortality	**~6%**
CRF / dialysis	**~45% at 2 yr — highest-risk pre-op factor**
Loss of independence	**~50%** at 1 yr

Strongest determinant of survival	**pre-injury mobility**
Poor pre-injury cognition	↑ mortality after **hip fracture**
Poor outcome predictors (TO)	older age, not at home, poverty, unmarried, ↓ pre-fracture function, **cognitive impairment**, comorbidities, **high ASA**, ↓ grip, depression, ↓ albumin / folate, visual impairment, **extracapsular** type, delay > 48 h, **NWB post-op**, long LOS · BMI unrelated ✗ Wrong options: **intracapsular fracture** (NOT a poor predictor)

Loss of independence after fixation	**age > 80, ASA > 1, prior walking aid, smoking**, implant placement quality · **pre-op ambulation without aid → independent at 1 yr**

Timed Up and Go (TUG)

Normal	**< 12 s** → predicts **walking without aid at 2 yr** · **> 26 s** → persistent walking aid ✗ Wrong options: predicts fracture, life expectancy, independent living, falls (**UPST predicts falls**, ASA predicts mortality)

Neglected fracture (TO)

AIIMS box technique (TO)	**neglected femoral neck fracture** with large neck defect: box from GT base, head anterior wall + **quadratus femoris pedicle graft** posteriorly, cancellous autograft + screws

6 Complications

Complication	Fact
Osteonecrosis	**10–45%**; ↑ displacement, **non-anatomic reduction**, **SHS (FAITH)**; can occur in undisplaced; time to reduction not proven → young > 50% head: FVFG vs THA · old: arthroplasty
Nonunion	**5–30%**; displaced; **varus malreduction** most linked to failure → **valgus IT osteotomy** (young), FVFG, arthroplasty
Dislocation	THA ~10% (5–7× hemi); hemi: posterior approach, labral / wall injury, oversized head
Failure	Fixation ≫ arthroplasty, mostly first 2 yr
Loss of independence	See above

Nonunion / malunion – valgus intertrochanteric osteotomy

Concept	turns **vertical shear** line into **horizontal compression**; corrects varus; works even with **non-collapsed AVN**
Nonunion causes (TO)	**posterior comminution** · initial displacement · imperfect reduction · non-compressive fixation
Osteotomy options (TO)	**McMurray displacement** · Schanz angulation · **Pauwels (Müller) reorientation = best accepted** · template on **normal side**, wedge **~25°**
AVN rates (TO)	**5–8% undisplaced** · **20–45% displaced** · causes: inadequate / delayed reduction, displacement, **associated dislocation** · young → osteotomy or arthrodesis; elderly → arthroplasty
Valgus osteotomy indication	**young nonunion with viable head** ✗ Wrong options: collapsed AVN → arthroplasty · retroversion / cross-over → PAO · high dysplasia → subtroch osteotomy + THA · metastases → cemented arthroplasty
48-yo runner, varus collapse at 6 mo	**valgus IT osteotomy + blade plate** ✗ Wrong options: hemi · in-situ DHS · core decompression · revision screws

– – Active space – –

22-yo vertical nonunion at 9 mo	**valgus IT osteotomy + 130° blade plate** ✗ Wrong options: convert to SHS · THA · perpendicular 7.0 screw · varus osteotomy
Wedge + plate maths	**side-plate angle = guide-pin angle + wedge angle** → pin 130° + **20° wedge → 150° plate** ✗ Wrong options: 110° / 95° = varus · 140° = lateral gap · 35° + 165° (too short, non-standard)

X vs Y

	THA	Hemiarthroplasty
Patient	Active, independent, OA	> 80, low demand, dementia, PD
Function	**Better**	Lower
Dislocation	**Higher**	Lower
Reoperation	Lower	Higher (acetabular wear)

	Cemented stem	Uncemented stem
Periprosthetic fracture	**Lower**	Higher
Aseptic revision	**Lower**	Higher
QoL at 4 mo	**Better**	Worse
OR time	Longer	Shorter

	Cannulated screws	Sliding hip screw
Best for	Non-displaced transcervical	**Basicervical, Pauwels III**
FAITH	Less AVN	↓ reoperation in smokers / displaced / basicervical

How it's asked

"Lowest intracapsular pressure position?"	**→ flexion + abduction + ER**
"Young – most likely complication at 2 yr?"	**→ AVN**
"Most prevalent complication after urgent ORIF in young?"	**→ osteonecrosis**
"Failed closed reduction in 46-yo cyclist?"	**→ open reduction + SHS**
"Highest AVN risk?"	**→ 45-yo woman, displaced**
"Displaced FNF in 32-yo?"	**→ ORIF**
"Most important predictor of healing / outcome?"	**→ quality of reduction**
"Ipsilateral neck + shaft, unresuscitated?"	**→ delay for resuscitation**
"Optimal cannulated screw configuration?"	**→ inverted triangle, inferior screw posterior along calcar**
"Improve fixation success?"	**→ screws abutting posterior + inferior cortex**

"Vertical (Pauwels III) — best construct?"	→ **SHS + side plate**
"SHS vs screws in smoker?"	→ **lower reoperation**
"FAITH — true?"	→ **SHS ↑ AVN**
"Ipsilateral non-displaced neck + shaft?"	→ **lag neck first, then retrograde nail**
"Indication for valgus IT osteotomy?"	→ **young FNF nonunion**
"Varus collapse at 6 mo in 48-yo?"	→ **valgus osteotomy + blade plate**
"40° fracture line, pin at 130° — correction?"	→ **20° wedge + 150° plate**
"22-yo vertical nonunion?"	→ **valgus osteotomy + 130° blade plate**
"70-yo golfer displaced?"	→ **THA**
"Hemi vs THA — hemi has?"	→ **lower dislocation**
"Active 72-yo — best long-term?"	→ **THA**
"THA vs IF — long term?"	→ **THA less reoperation**
"THA most appropriate for?"	→ **66-yo golf instructor**
"Lowest reoperation + best scores in active 65-yo?"	→ **THA**
"Greatest painless function, lowest reoperation (60-yo independent)?"	→ **THA**
"HA vs THA — true?"	→ **THA ↑ peri-op complications**
"Best function at 7–10 yr (pickle-ball)?"	→ **THA**
"THA over IF advantage?"	→ **lower reoperation**
"THA vs hemi in active 75-yo?"	→ **better outcome scores**
"THA vs hemi in 68-yo?"	→ **improved function**
"Dialysis, wheelchair, RA, COPD — counsel?"	→ **THA not better than HA**
"THA vs hemi (68-yo men)?"	→ **THA ↑ dislocation**
"85-yo PD household (TO)?"	→ **hemiarthroplasty**
"87-yo walker + aide + dementia?"	→ **hemiarthroplasty**
"70-yo PD — best option + rationale?"	→ **hemi to lower dislocation**
"Dialysis — ORIF vs hemi?"	→ **ORIF higher complications**
"Bipolar 5 yr, acetabular wear?"	→ **add acetabular component (THA)**
"Hemi dislocation risk?"	→ **posterior approach**
"Improved short / medium mobility?"	→ **cemented hemiarthroplasty**
"Elderly insufficiency fracture displaced?"	→ **hemiarthroplasty**
"Cemented vs uncemented — true?"	→ **lower revision + periprosthetic fracture**
"Hemi dislocation factor (posterior approach)?"	→ **posterior wall insufficiency**

– – Active space – –

"90-yo — best option + outcome?"	→ cemented hemi, better QoL at 4 mo
"Cemented vs cementless — biggest difference?"	→ less periprosthetic fracture
"Failed screws, comminuted at revision (82-yo)?"	→ remove hardware + arthroplasty
"IF vs hemi to 1 yr?"	→ higher failure with IF
"Minimally displaced, IF — true?"	→ pre-op unaided walking → independence
"NOT poor-outcome predictor (TO)?"	→ intracapsular fracture
"Delay > 4 days?"	→ ↑ 1-yr mortality
"Poor cognition ↑ mortality in?"	→ hip fracture
"Dialysis — 2-yr mortality?"	→ 45%
"20% mortality + 50% dependence at 1 yr?"	→ femoral neck fracture
"In-hospital + 1-yr mortality?"	→ 6% + 30%
"Highest-risk pre-op factor?"	→ chronic renal failure
"Best outcome in comorbid 87-yo?"	→ pre-op medical optimisation
"Presented 3 days late — pre-op test?"	→ duplex both legs
"67-yo — true?"	→ delay > 48 h OK to optimise comorbidities
"Associated with surgical delay?"	→ pre-op echocardiogram
"↓ morbidity?"	→ early WBAT + mobility
"↓ mortality + complications?"	→ multidisciplinary team
"Post-op delirium risk?"	→ ↑ length of stay
"↓ mortality + complications?"	→ surgery < 24 h of admission
"Surgery < 24 h affects?"	→ mortality
"Rivaroxaban 12 h ago?"	→ proceed to surgery
"TUG 11 s predicts?"	→ walking without aid at 2 yr
"↑ TUG time predicts?"	→ need for walking aid
"Impacted neck — best X-ray position (TO)?"	→ abduction + internal rotation
"Second-line after pelvic X-ray (TO)?"	→ thin-slice MDCT
"Negative X-ray, pain on log-roll?"	→ MRI hip + pelvis
"15-yo — view to classify?"	→ traction AP with 15° IR
"NOT in FRAX?"	→ spine T-score
"AIIMS box technique (TO)?"	→ neglected fracture neck of femur
"Garden alignment index for acceptable reduction?"	→ 160° AP, 180° lateral (TO)
"Tension-side femoral neck stress fracture?"	→ internal fixation (TO)
"Artery usually preserved in displaced neck fracture?"	→ inferior retinacular (TO)

FEMORAL SHAFT FRACTURES

– Trauma · 22 / 67 –

1 Basics

Epidemiology

Incidence	**37.1 / 100,000 / yr**
Who	high energy in young (MVC) · low energy in elderly · gunshot
Blood loss (TO)	**1–1.5 L** into thigh · associated **fat embolism**
Nonop traction options (TO)	skin traction · **Russell (balanced) traction** · **boot and bar (derotation bar)**
Retrograde nail indications (TO)	**bilateral shaft fractures**, polytrauma, **floating knee**, ipsilateral **acetabulum / patella** fracture · entry **anterior + medial to PCL attachment / apex of Blumensaat**, nail up to lesser trochanter
Reaming (TO)	larger nail, better stability, **autograft deposition** vs thermal necrosis + fat embolism

Associated injuries

Association	Fact
Ipsilateral femoral neck fracture	**2–6%** (Orthobullets) · ~5% · 1–9% / 2.5–9% / 0–15% (OB) · 7.5% (Tornetta); **basicervical, vertical, non-displaced**; **missed 19–31% (~30%)**
Bilateral femur fractures	↑ **mortality** (5.6% vs 1.5%), initial hypotension, head + pelvic injuries (✗ Wrong options: **not** ↑ rib fractures); pulmonary complications
Ipsilateral tibia (floating knee), acetabulum	–
Pulmonary injury	Early fixation → ARDS ~2% if under-resuscitated
Head injury	Intra-op hypotension → ↓ cerebral perfusion

Blood loss **closed femur 1000–1500 mL** · closed tibia 500–1000 mL · **open ≈ double** · **bilateral closed femur 2000–3000 mL**

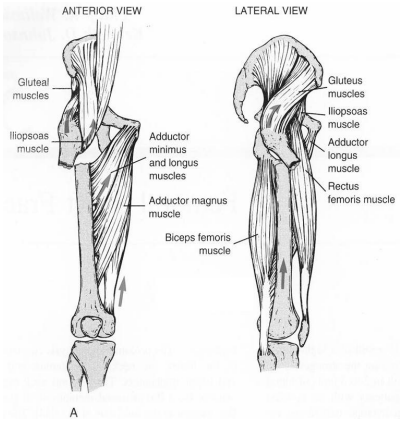
Anatomy

Femur	largest, strongest; **anterior bow** (radius of curvature **120 ± 36 cm**); **linea aspera** = posterior compressive strut
Compartments	anterior (sartorius, quads) · posterior (hamstrings) · adductor (gracilis, adductors)
Blood supply after fracture	**periosteal + soft-tissue supply predominates**; reaming disrupts endosteal but stimulates periosteal

– – Active space – –

Deforming forces

Fragment	Deformity	Muscle
Proximal	**Abducted**	Gluteus medius + minimus
Proximal	**Flexed** (+ ER)	**Iliopsoas** (+ short rotators)
Distal	**Varus**	Adductors
Distal	**Extension**	Gastrocnemius



Deforming forces on femoral shaft fractures

2 Mechanism

Force	Pattern
Pure bending	**Transverse**
Rotation	Spiral
Uneven bending	Oblique
4-point bending	Segmental
High-speed crush / torsion	Comminuted

3 Presentation

Exam tense swollen thigh, shortened, tender; hip exam unreliable (neck fracture easily missed); **document distal NV status**

Adequate resuscitation HR < 100, SBP > 100, DBP > 70, temp > 35 °C, urine 0.5–1 mL/kg/h, **lactate < 2.5**, **BD −2 to +2**, **IL-6 < 500 pg/dL**, gastric pH > 7.3

Compensated shock clues normotensive, afebrile tachycardia, cool limbs, narrow pulse pressure, weak pulses, slow refill

4 Diagnosis & Classification

Key radiology findings

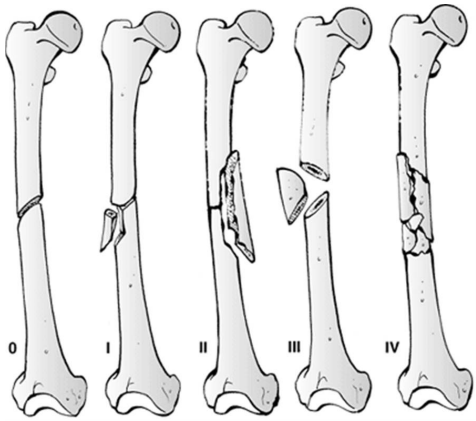
Study	Use
AP + lateral whole femur, ipsilateral **hip** + **knee**	Standard
AP hip in ~10° internal rotation	Neck in profile (**✗** Wrong options: ER view)
Fine-cut (2 mm) CT of hip	**Best pre-op study for occult neck fracture**; **CT capsular sign** = **> 1 mm more capsular distension** than uninjured side
Intra-operative fluoroscopy of neck	**Still needed even after negative 2 mm CT**
Post-op hip X-ray in OR + follow-up **AP hip in 15–20° IR**	**New groin pain after nailing → hip X-ray first**

Neck protocol 10° IR AP hip + 2 mm CT + intra-op fluoro + post-op film
✗ Wrong options: bone scan · traction views · MRI routinely · frog-leg (too painful)

Septic nonunion labs **CRP most sensitive for occult infection**, ESR, WBC

Winquist classification

Type	Comminution
0	None
I	Insignificant
II	**> 50% cortical contact**
III	**< 50% cortical contact**
IV	**Segmental** – no contact between main fragments



Winquist classification

– – Active space – –

AO / OTA 32

Type	Subtypes
A – simple	A1 spiral · A2 oblique > 30° · A3 transverse < 30°
B – wedge	B1 spiral · B2 bending · B3 fragmented
C – complex	C1 spiral · C2 segmental · C3 irregular

5 Management

Femoral shaft fracture

- Resuscitated → **reamed, statically locked IMN < 24 h (WBAT)**
  - antegrade = gold standard
  - retrograde: neck fracture, floating knee, acetabulum, bilateral, obese, pregnant, polytrauma
- Unstable / head injury with ICP problems / vascular / severe open → ex-fix → **IMN within 2–3 wk**
- Ipsilateral neck fracture → **fix NECK first (open if displaced), then retrograde nail / plate**

Treatment options

Option	Indication	Fact
Spica / long leg cast	Children (< 5 yr spica), unfit adults	–
Antegrade reamed IMN	**Gold standard**	**98–99% union**, infection ~2%; **< 24 h** → ↓ ARDS, VTE, LOS, cost
Retrograde IMN	**Ipsilateral neck fracture**, **floating knee**, **acetabular fracture**, **bilateral femurs**, **polytrauma**, **morbid obesity**, **pregnancy (1st trimester)**, infra-isthmal	Not with skeletal immaturity, knee sepsis, knee soft-tissue injury
Ex-fix → IMN within 2–3 wk	Unstable polytrauma, vascular injury, severe open	Union + infection = acute nailing; **infection ↑ if ex-fix > 28 days**; ↓ ARDS / fat embolism; knee stiffness
Plate	Neck fracture needing screws, distal meta-diaphyseal fracture, no canal access	**↑ infection, nonunion, hardware failure**

Immediate (retrograde / antegrade) nailing — safe for **GSW** femur fractures

WB after statically locked appropriately sized nail — **immediately** (even comminuted)

Antegrade vs retrograde nailing

	Antegrade	Retrograde
Union	Equal	Equal
Pain	**Hip pain**, abductor weakness, **abductor HO**	**Knee pain**, **symptomatic distal locking screws**
Setup	Fracture table / lateral	**Supine, all limbs accessible, compare rotation + length**
Bilateral femurs	Reposition	**Shorter op — no re-prep**

Retrograde preferred because — **less hip pain**
✗ Wrong options: union · infection · time · less knee pain

Long-term weakness after antegrade nail — **hip abduction + knee extension** (quads + abductors weaker)

Stable, floating knee + open ankle — **retrograde femur nail + tibial nail + ankle I&D + ORIF**
✗ Wrong options: ex-fix

Femur + ipsilateral open tibia, stable — **reamed retrograde nail, static locking**
✗ Wrong options: unlocked · unreamed · dynamic locking (length unstable)

Bilateral femurs, resuscitated (lactate 1.2) — **reamed retrograde nailing of both**
✗ Wrong options: traction · ex-fix · plates · unreamed antegrade

Antegrade technique & entry points

Positioning — fracture table (perineal post, well-leg lithotomy) or **sloppy lateral**

Entry	Pros	Cons
Piriformis fossa	**Colinear with shaft**	Hard in obese; **most damage to piriformis tendon, abductors, MFCA** (→ AVN in children)
Trochanteric	Less abductor injury, easier	Not colinear → **trochanteric-design nail** required; **too lateral → varus**

Trochanteric start — **depends on trochanter position relative to shaft axis** + nail geometry (tip or just medial)
✗ Wrong options: "always at the tip"

Varus causes — **trochanteric start too lateral** · **straight (piriformis) nail through trochanteric portal** · wrong-side trochanteric nail · (trochanteric nail through piriformis → **valgus**)

Too ANTERIOR start (≥ 6 mm) — ↑ **hoop stress → iatrogenic proximal femur fracture ("bursting")** (esp. piriformis nail); **over-ream ≥ 0.5 mm** to reduce · 3 mm anterior improves **cephalic screw position** in CMN

Too POSTERIOR start + straight nail (larger radius of curvature) + short / bowed femur — **anterior cortical perforation** of distal femur / anterior thigh pain → prevent: **more anterior start**, **smaller radius of curvature (more curved) nail**, **anterior** blocking screws, **smaller nail**, notched tip
✗ Wrong options: larger nail · posterior blocking screw · straighter nail

Insertion — nail inserted **rotated 90° internally** then rotated back (uses bow, avoids medial comminution)

– – Active space – –

Piriformis nail	single sagittal bow (either side) · **trochanteric nail** – double bow, side-specific (twist on insertion); 3-point fit

Reamed vs unreamed nailing

Reamed	**↑ union**, ↓ time to union, **no ↑ pulmonary complications**, **no ↑ infection in open**; unreamed → **4.5× nonunion** (Canadian RCT)
Why reaming helps	**reaming debris ≈ iliac crest autograft** (osteogenic cells) · stimulates periosteal flow · **larger nail (stiffness ∝ r^4)** · more isthmic contact · fluted nails ↓ marrow pressure (greatest at nail insertion)
Unreamed	consider only with **bilateral pulmonary injury**; **biggest nonunion risk factor** among options ✗ Wrong options: open reduction · cerclage · retrograde · staged ex-fix
Nail diameter	**10 mm ≈ larger nails** for union ✗ Wrong options: bigger canal needs bigger nail

Retrograde technique

Start	**centre of notch (AP)** + **extension of Blumensaat line (lateral)** (posterior → cruciate injury); medial condylar start needs curved nail (else valgus) · knee flexed **30–50°**
Steps	reduce before reaming (F-tool, bumps, Schanz joysticks); ream 1–1.5 mm over nail; **nail ~1 cm deep to cartilage**; guidewire **proximal to lesser trochanter**; **long nail to LT**
Proximal interlocks	safe **above lesser trochanter**; **A→P screws below LT → femoral nerve branches + deep femoral artery** at risk
Periprosthetic (CR TKA)	start **more posterior → hyperextension (recurvatum)**; arthrotomy needed; PS box may need burr; backup = locking plates (multiple distal points)

Ipsilateral femoral neck + shaft

Priority	**neck first** (AVN / nonunion) → **open reduction + screws** if displaced → **reamed retrograde nail** (or plate) for shaft
Two implants	**lower malreduction** than single cephalomedullary nail ✗ Wrong options: single CMN (**highest malreduction**; no faster neck healing, less pain)
Missed neck found after antegrade nail	screws **anterior to nail** (difficult)

Polytrauma & head injury

Head injury	early stabilisation **does not worsen neurological outcome** if **hypotension + hypoxia avoided**; **intra-op hypotension → lower GCS at discharge**
Severe head injury with ↑ ICP (bifrontal contusion, ICP 30)	**ex-fix femur** ✗ Wrong options: due to L4 burst, open tibia, LC1, rib fractures
SAH + femur – adverse outcome from?	**intra-operative hypotension** (traction if ICP labile)
GCS 13, normal CT head, stable	**early fixation, no neuro risk** ✗ Wrong options: ↑ pulmonary risk · delay · DCO · chest injury always contraindicates

Unilateral contusion + chest tube, now stable (BD −2)	**reamed IMN**
Shock, bilateral contusions, BD −10	**ex-fix** ✗ Wrong options: plating · reamed / unreamed nails
Borderline after splenectomy, bilateral open femurs + pelvis	**bilateral femoral ex-fix + anterior pelvic ex-fix**
BD −6, lactate 3.5, contusions, hip reduced	**bilateral temporising ex-fix**

Distal interlocking

Perfect circles	hole widened **proximal–distal → adjust abduction / adduction** · widened **anterior–posterior → adjust internal / external rotation**
Computer navigation	**↓ fluoroscopy time** ✗ Wrong options: accuracy, time, image quality, length
Single distal screw failure	**4-point bending → screw shaft fractures inside the nail** ✗ Wrong options: pull-out · head shear

6 Complications

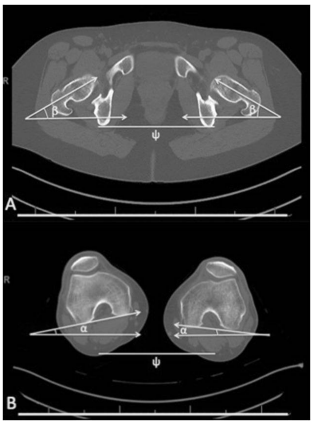
Complication	Fact
HO	**25%**, rarely significant (abductors after antegrade)
Pudendal nerve	**10%** with traction on fracture table
Femoral artery / nerve	Retrograde proximal locking screws
Malrotation	**Proximal fractures 30%**, distal 10%; fracture table, **comminution**, night surgery
Delayed union	Dynamise ± graft
Nonunion	**< 10%**; NSAIDs, **smoking**, **indomethacin**
Infection	< 1% → remove nail, ream, ex-fix if unhealed
Weakness	Quads + abductors
Iatrogenic fracture	Anterior start ≥ 6 mm, under-reaming
MAD	Lengthening → **lateral** MAD · shortening → **medial** MAD
Anterior cortical perforation	Radius mismatch; posterior start + straight nail
Compartment syndrome	**Ballistic > blunt**

Rotational malalignment

Most associated factor	**fracture comminution** (TO, OB) · also night-time surgery ✗ Wrong options: entry point · level · surgeon experience · closed technique
Fracture table	**↑ internal malrotation** (vs manual traction supine)

– – Active space – –

Direction by level	**proximal fracture → INTERNAL malrotation** (proximal fragment ER by iliopsoas, rotators, abductors) · **distal fracture → EXTERNAL malrotation** (gastrocnemius / plantaris ER distal fragment) · antegrade vs retrograde and entry type don't matter
Lesser trochanter larger on injured side (ER proximal fragment)	**internal rotation malalignment**
Iatrogenic internal malrotation	**distal segment internally rotated relative to proximal** (↑ anteversion)
CT measurement (Jeanmart)	angle between **posterior condylar tangent** and **femoral neck axis**; rotation = β − α; compare sides
Thresholds	**< 10° = normal variation** · **≤ 15° well tolerated** · **> 15° + symptomatic → derotation** ✗ Wrong options: dynamisation
Acceptable differences	**9° IR + 14° ER** ✗ Wrong options: 18°
CT: ER malalignment on exam	**more ER, less IR** of hip (tibia as pointer)
Worked example	right neck IR13 / condyles ER17, left neck ER13 / condyles ER3 → right distal ER30 vs left IR10 → **rotate right distal fragment INTERNALLY 40°**
Correction	intra-op: remove distal locks + derotate · after union: **osteotomy**



CT measurement of femoral version (Jeanmart)

Nonunion

Definition	**no healing by 9 months** or **no progress over 3 months**; broken distal locks = race lost

Nonunion	Treatment
Aseptic, after reamed nail (smoker)	**Reamed antegrade exchange nailing** (✗ Wrong options: graft in situ · stimulator · biopsy)
Hypertrophic, **undersized nail**	**Reamed exchange nailing**
Hypertrophic, sub-isthmal	**Nail retention + plate augmentation + graft** (highest union + immediate WB)
Failed exchange nailing (union only ~53%)	**Bone graft + compression plating**

Exchange nailing works by	stiffer larger nail, better isthmic fit, reaming graft · **dynamic locking** + grafting ↑ union · **smoking** = biggest culprit when it fails
Indomethacin HO prophylaxis (25 mg TDS × 6 wk)	↑ **long-bone nonunion** (inhibits IGF-1); radiation preferred

Skeletal traction (TO)

Upper tibial pin (sliding) traction (TO)	**counter-traction by raising foot end of bed** (~1 inch per lb) with BB splint · **Thomas splint = fixed traction** (ring in groin) ✗ Wrong options: couch · none · muscle spasm

X vs Y

	Piriformis entry	Trochanteric entry
Alignment with shaft	**Colinear**	Lateral → nail-specific design
Soft tissue	Piriformis, abductors, MFCA	Gluteus medius
Error	Too anterior → bursting	Too lateral → **varus**

	Proximal shaft fracture	Distal shaft fracture
Malrotation tendency	**Internal**	**External**

	Exchange nailing	Plate augmentation
Best for	Isthmal, undersized nail	**Sub-isthmal hypertrophic**
WB	Immediate	Immediate (nail kept)

How it's asked

"Neck fractures diagnosed late without fine-cut CT?"	→ **~30%**
"Associated with ~5% of shaft fractures?"	→ **ipsilateral neck fracture**
"Vertical basicervical neck line usually with?"	→ **ipsilateral shaft fracture**
"Highest malreduction for neck + shaft?"	→ **antegrade cephalomedullary nail**
"Best algorithm (displaced neck + shaft)?"	→ **ORIF neck, then reamed retrograde nail**
"Next step before antegrade nail?"	→ **hip radiograph / CT**
"2 mm CT negative – what else?"	→ **intra-op fluoroscopy of neck**
"Groin pain 2 wk after nailing?"	→ **hip radiograph**
"Best study for occult neck fracture?"	→ **2 mm CT**
"6% concomitant fracture – image?"	→ **ipsilateral hip**
"CT sign of occult neck fracture?"	→ **> 1 mm capsular distension**
"Most piriformis tendon injury?"	→ **piriformis-entry antegrade nail**
"Larger radius of curvature nail risks?"	→ **anterior perforation of distal femur**

– – Active space – –

"Varus after subtroch nailing related to?"	→ piriformis nail through trochanteric portal
"Start 3 mm anterior with CMN benefit?"	→ better head screw placement
"Start too anterior → ?"	→ iatrogenic proximal fracture
"Trochanteric start A (lateral) → ?"	→ varus
"Periprosthetic CR TKA retrograde nail – true?"	→ posterior start → hyperextension
"Proximal femur burst after nailing – error?"	→ too anterior piriformis start
"Proper trochanteric start?"	→ depends on trochanter vs shaft axis
"Too lateral trochanteric start?"	→ varus
"Anterior perforation in bisphosphonate patient – prevention?"	→ more anterior start
"Distal anterior thigh pain after long nail?"	→ posterior start, piriformis nail, short patient
"Lateral guidewire for trochanteric CMN?"	→ varus
"Retrograde ↑ vs antegrade?"	→ symptomatic distal locking screws
"Why retrograde?"	→ less hip pain
"Stable polytrauma: femur, tibia, open ankle?"	→ retrograde femur + tibial nail + ankle ORIF
"Muscle deficits after antegrade nail?"	→ hip abduction + knee extension
"Antegrade ↑ vs retrograde?"	→ hip pain
"Femur + ipsilateral open tibia?"	→ reamed retrograde, static locking
"Bilateral femurs, resuscitated?"	→ reamed retrograde nailing both
"Malrotation most associated with (TO)?"	→ fracture comminution
"Manual traction vs fracture table?"	→ fewer internal malrotations
"Proximal vs distal fracture malrotation?"	→ proximal internal, distal external
"CT: right ER17 + IR13 vs left ER3 + ER13?"	→ rotate right distal fragment internally 40°
"Injured left vs right on CT?"	→ external malrotation 13°
"Larger lesser trochanter on injured side?"	→ internal rotation malalignment
"8° ER on CT scanogram?"	→ observe
"Anteversion 44° vs 23°?"	→ internal rotation malalignment 21°
"CT ER malalignment – exam?"	→ less IR than ER
"Neutral vs 6° normal side?"	→ no further procedure
"30° IR after retrograde nailing – cause?"	→ distal segment internally rotated
"Acceptable rotation differences?"	→ 9° IR + 14° ER
"Fracture table causes?"	→ internal malrotation

"Comminution associated with?"	→ femoral malrotation
"Reamed vs unreamed (distal shaft) — higher?"	→ union
"Reaming debris like?"	→ iliac crest autograft
"Reamed vs non-reamed — higher?"	→ union
"Nail diameter — true?"	→ 10 mm = larger nails
"Biggest nonunion risk among options?"	→ unreamed nail
"Indomethacin for HO — true?"	→ ↑ femoral nonunion
"Persistent oligotrophic nonunion after exchange nail?"	→ smoking
"Failed exchange nail?"	→ bone graft + compression plating
"Aseptic nonunion with broken distal locks, smoker?"	→ reamed antegrade exchange nailing
"Hypertrophic nonunion, undersized nail?"	→ reamed exchange nailing
"Sub-isthmal hypertrophic, immediate WB?"	→ nail retention + plate augmentation + graft
"Head injury + intra-op hypotension?"	→ lower GCS at discharge
"Bilateral open femurs, borderline, after splenectomy?"	→ bilateral ex-fix + pelvic ex-fix
"NOT ↑ with bilateral femurs?"	→ rib fractures
"Injury dictating ex-fix (ICP 30)?"	→ bifrontal contusion
"BD −6, contusions, hip reduced?"	→ bilateral temporising ex-fix
"SAH + femur — worst factor?"	→ intra-op hypotension
"Resuscitated after chest tube?"	→ reamed IMN
"Shock + bilateral contusions?"	→ ex-fix
"GCS 13, normal head CT?"	→ early fixation doesn't worsen neuro outcome
"Perfect circles — proximal-distal widening?"	→ abduct / adduct leg
"Navigation advantage?"	→ less fluoroscopy
"Retrograde nail — screws endangering femoral nerve / deep femoral artery?"	→ A→P below lesser trochanter
"Single distal screw failure?"	→ shaft fractures inside the nail
"Bilateral closed femur blood loss?"	→ 2000–3000 mL
"Pure bending →?"	→ transverse
"GSW femur ↑ risk of?"	→ compartment syndrome
"WB after locked 12 mm nail?"	→ immediately
"UTP traction counter-traction (TO)?"	→ raise foot end of bed

– – Active space – –

FLAIL CHEST

– Trauma · 23 / 67 –

1 Basics

Definition & epidemiology

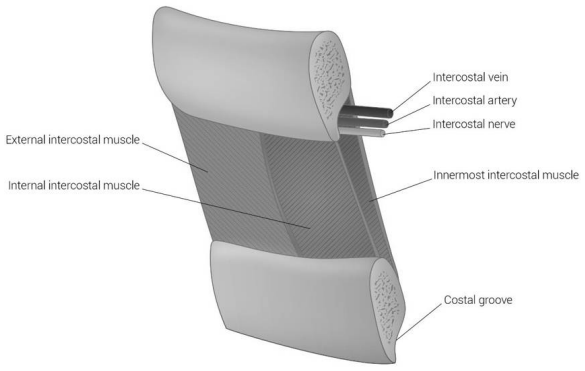
Flail chest	**segmental fractures of ≥ 3 consecutive ribs** → hemothorax / pneumothorax
Incidence	~300,000 / yr; **~10% of blunt thoracic trauma** (OB); more ribs → ↑ morbidity + mortality
Who	bimodal: young trauma · elderly osteopenic
Commonest ribs	**7–10**
Mortality	**up to 33%** with flail chest · **2.5% with surgical stabilisation**

Associated injuries

Ribs	Association
1–3	**Mediastinal injury (aorta, brachial plexus)**, worse mortality
1–4 (upper)	**Progressive displacement of ipsilateral clavicle fracture**
9–12	**Intra-abdominal injury – liver + spleen**
Any	Hemopneumothorax, pulmonary contusion, scapula + clavicle fractures

Anatomy

Ribs	12 pairs; **11–12 = floating** (no costal cartilage to sternum)
Intercostal neurovascular bundle	**posteroinferior**, in **costal groove** → avoid when plating



Intercostal neurovascular bundle

2 Mechanism

Mechanism	direct blunt / penetrating; fracture at impact or **rib angle (posterolateral bend) = weakest point**; pathological / metastatic

Pathophysiology negative-pressure ventilation → **paradoxical motion** → pain → ↓ respiratory effort → **pneumonia, ARDS**, intubation

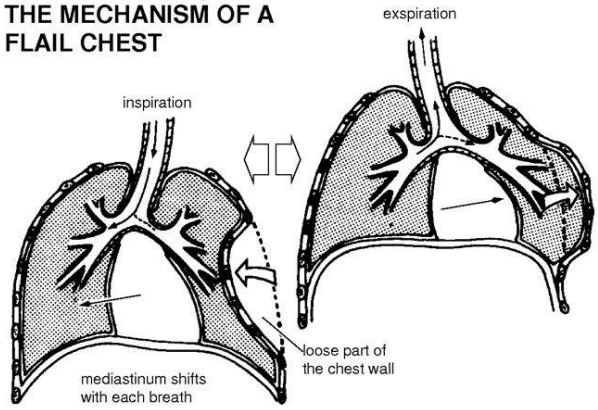
– – Active space – –

3 Presentation

Symptoms pain, chest-wall bruising, respiratory difficulty

Paradoxical respiration segment **sinks in on inspiration**, **expands on expiration**

Palpation bony / soft-tissue crepitus; visible deformity



Flail chest and paradoxical respiration

4 Diagnosis & Classification

Key radiology findings

Study	Fact
Standard AP / lateral CXR	**Low sensitivity**; may show hemothorax
Dedicated rib series	**More sensitive** (oblique views, higher energy)
CT	**Best** for displacement, segmental injury, **need for surgery** + thoracic / abdominal injuries

5 Management

Rib fractures

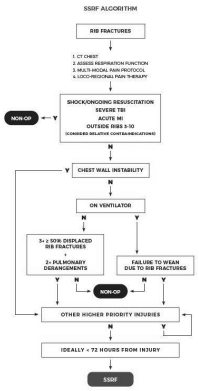
- No respiratory compromise, no flail → **O2, incentive spirometry, multimodal analgesia**
- Flail (≥ 3 consecutive segmental) / failure to wean / open / intractable pain → **ORIF**
  - haemodynamically unstable / spine or pelvis first → **delay**

– – Active space – –

Operative criteria

Indication (ORIF)	Contraindication
Flail segment (≥ 3 consecutive segmental ribs)	**Haemodynamic instability**
Failure to wean from ventilator	**Spinal / pelvic fractures needing fixation first**
Open rib fractures	–
Displaced fractures with **intractable pain**	–
Symptomatic nonunion	–

Guidelines Chest Wall Injury Society SSRF algorithm



SSRF rib fracture algorithm

Technique

Approach	Fractures
Lateral thoracotomy (lateral decubitus)	Anterolateral + posterolateral
Posterior paramedian	Very posterior, near **costovertebral junction**
Inframammary (supine)	Anterior + **costochondral dislocations**

Implants plate + screw constructs · intramedullary splints

Operative rib fixation — outcomes

Surgery vs nonop **↓ ICU stay** · **↓ total hospital stay** · **↓ pneumonia** · ↓ ventilation days, ↓ tracheostomy, ↓ deformity, ↓ mortality, ↓ cost

Long-term respiratory function **equivalent**
✗ Wrong options: superior

6 Complications

– – Active space – –

Complication	Fact
Nonunion	Chest pain **> 3 months** → nonunion work-up before fixation
Intercostal neuralgia	Bundle **posteroinferior** to rib
Periscapular weakness	–
Pneumonia	–
Restrictive pulmonary function	–

X vs Y

	Ribs 1–3	Ribs 9–12
Associated injury	**Aorta, brachial plexus** (mediastinal)	**Liver, spleen**

How it's asked

"Flail ORIF vs nonop?"	→ **shorter ICU stay** ✗ Wrong options: longer / equal hospital stay · superior long-term lung function · equal pneumonia
"Definition of flail chest?"	→ **≥ 3 consecutive ribs fractured segmentally**
"Lower rib fractures associated with?"	→ **liver + spleen injury**
"Upper rib fractures + clavicle?"	→ **progressive clavicle displacement**
"Best imaging for surgical decision?"	→ **CT**

– – Active space – –

FOOT COMPARTMENT SYNDROME

– Trauma · 24 / 67 –

1 Basics

Epidemiology & causes

Pathophysiology trauma → bleeding + oedema → ↑ interstitial pressure → vascular occlusion → **myoneural ischaemia**

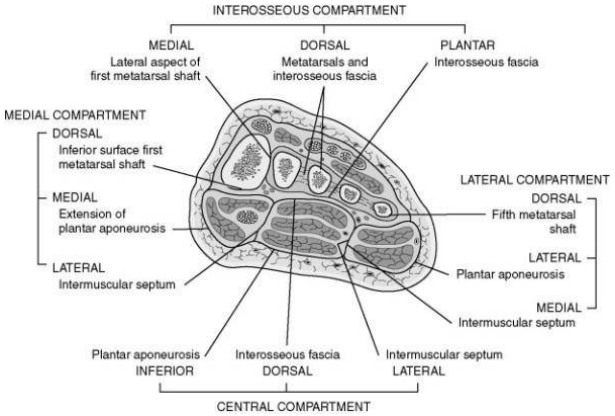
Causes **fractures 69%** (**5–17% of calcaneus fractures**), **crush**, severe ankle sprain, GSW, tight casts / dressings, burns, post-ischaemic swelling, bleeding disorders, vascular injury

Foot compartment anatomy

Number **9** (Orthobullets, TO; controversial) · **8** (OB: lateral, medial, central, 4 interosseous)

Compartment	Contents
Medial	**Abductor hallucis, flexor hallucis brevis**
Lateral	**Abductor + flexor digiti minimi brevis**
Interosseous (× 4)	**Interossei – each in its own compartment**
Central – **superficial**	**Flexor digitorum brevis**
Central – middle	**Quadratus plantae**
Central – **deep (calcaneal / adductor)**	**Adductor hallucis** + posterior tibial NV bundle

Trap (TO) "**adductor compartment = adductor hallucis + interossei**" = **wrong**; interossei form **4 separate** compartments



The 9 compartments of the foot

3 Presentation

Symptom **pain out of proportion**

Exam pain on **passive toe dorsiflexion (MTPJ)** (stretches intrinsics; may be absent in foot) · tense swollen foot · ↓ 2-point discrimination

Pulses **present pulses do not exclude**; absent DP may be expected with severe foot trauma

4 Diagnosis & Classification

Key radiology findings

Radiographs rule out fracture only; **diagnosis is clinical ± pressures**

Compartment pressure measurement & thresholds

Indications polytrauma, obtunded / unreliable, **children**, equivocal exam (not needed when clinically unequivocal → straight to theatre)

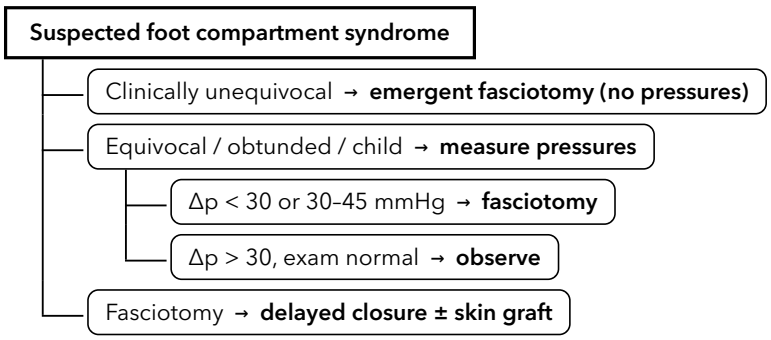
Compartment	Needle entry
Medial	**Base of 1st MT**, 2 cm into abductor hallucis
Central	**Base of 1st MT**, directed **lateral + plantar through abductor hallucis** (deeper than medial)
Interosseous	**2nd, 3rd, 4th web spaces**, plantar 2 cm through extensor fascia
Lateral	**Midshaft 5th MT**, 1 cm medial + plantar

Diagnostic criteria

Fasciotomy if	Observe if
Δp (diastolic – compartment) < 30 mmHg	**Δp > 30 mmHg**
Absolute 30–45 mmHg (historic, controversial)	Exam not consistent
Clinical diagnosis	–

Intra-op anaesthesia lowers diastolic → **compare to pre-op diastolic**

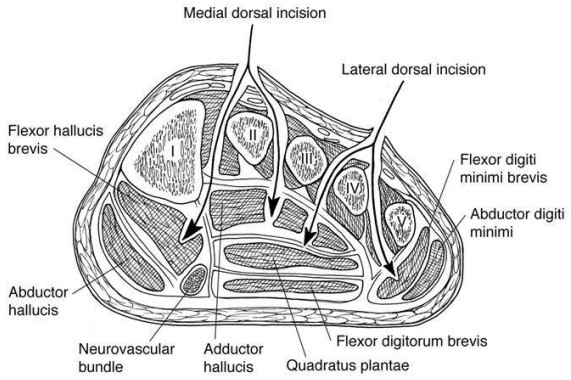
5 Management



– – Active space – –

Fasciotomy technique

Approach	Releases	Notes
Dual dorsal (gold standard) – medial to 2nd MT	1st + 2nd interosseous, **medial**, **deep central**	Access to **Chopart, Lisfranc, TMT**; no calcaneus fixation
Dual dorsal – **lateral to 4th MT**	3rd + 4th interosseous, **lateral**, superficial + middle central	–
+ Medial incision	**Calcaneal compartment**	–
Single medial (along abductor hallucis in arch)	**All 9**	**Technically demanding**



Dual dorsal fasciotomy incisions

6 Complications

Complication	Fact
Claw toes (fixed flexion)	Intrinsic contracture → **release FDB + FDL at digits**
Chronic pain + hypersensitivity	Difficult to manage

X vs Y

	Dorsal incisions	Medial incision
Access	All compartments + **midfoot fractures**	**All 9** / calcaneal
Downside	No calcaneus access	Technically challenging

How it's asked

"Central compartment needle path + abnormal values?"	→ **lateral + plantar through abductor hallucis under 1st MT base; 30–45 mmHg or Δp < 30** ✗ Wrong options: Δp > 30
"Foot compartments – EXCEPT (TO)?"	→ **adductor compartment = adductor hallucis + interossei** (interossei are separate)
"Superficial central compartment (TO)?"	→ **flexor digitorum brevis**

"Late complication?" → **claw toes**

– – Active space – –

– – Active space – –

GALEAZZI FRACTURES

– Trauma · 25 / 67 –

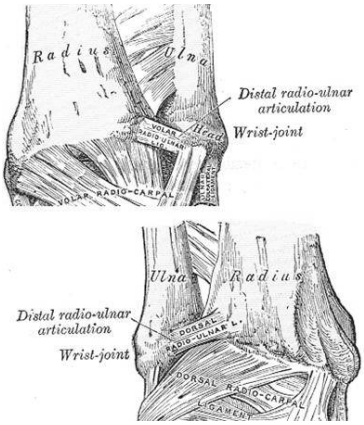
1 Basics

Definition & epidemiology

Galeazzi	**distal ⅓ radial shaft fracture + DRUJ injury**
DRUJ instability by fracture level	radial fracture **< 7.5 cm from articular surface → 55% unstable** · **> 7.5 cm → 6%** · closer = more unstable

Anatomy

Sigmoid notch	shallow concavity on ulnar border of distal radius
Primary DRUJ stabilisers	**volar + dorsal radioulnar ligaments**
DRUJ most stable in	**supination**



Distal radioulnar articulation

Associated DRUJ ligament injury

Most commonly injured	**dorsal radioulnar ligaments** ✗ Wrong options: volar long radiolunate · RSC · ligament of Testut · SL

2 Mechanism

Mechanism	**FOOSH with forearm in pronation** · direct blow to dorsolateral wrist

3 Presentation

Exam	point tenderness; test **pronation / supination for instability**; DRUJ stress → wrist / midline forearm pain

4 Diagnosis & Classification

Key radiology findings

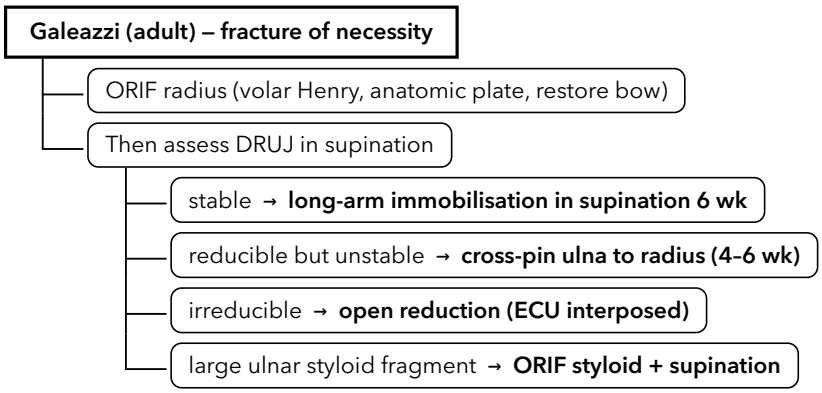
Views AP + lateral **forearm, elbow and wrist**

Sign of DRUJ injury	View
Widened DRUJ	**AP**
Dorsal or volar ulnar displacement	**Lateral**
Radial shortening ≥ 5 mm	AP
Ulnar styloid fracture	AP

OTA classification (radius / ulna 22)

Code	Description
22-A2.3	Simple radius fracture + DRUJ dislocation
22-A3.3	Simple both-bone (distal radius zone) + DRUJ dislocation
22-B2.3	Wedge radius + DRUJ dislocation
22-B3.3	Wedge both-bone + DRUJ dislocation

5 Management



TO classification **Walsh / Rettig & Raskin type I: within 7.5 cm of articular surface** · type II > 7.5 cm · apex volar (volar ulnar dislocation) vs apex dorsal (dorsal ulnar dislocation) (TO)

DRUJ fixation (TO) reduced + unstable → **2 parallel K-wires 1 cm proximal to the joint** · irreducible → open (**ECU / EDM interposition**)

Immobilisation position **supination** (Orthobullets) · **dorsal Galeazzi in supination, volar Galeazzi in pronation** (TO) – read the options

Synonyms (TO) fracture of necessity · **Piedmont** · **reverse Monteggia** · DRUJ sign: **ulnar variance > 5 mm** (TO)

Treatment – radius ORIF then DRUJ assessment

Indication for surgery **all adults**; **acute fixation far superior to late reconstruction**

Radius **volar (Henry) approach**, **anatomic plating**, **restore radial bow**

– – Active space – –

DRUJ assessment	only **after rigid radius fixation**; gross ulnar motion in supination = unstable; **radius fixation alone does not guarantee DRUJ stability**
Pinning	**1–2 × 1.2 / 1.6 mm K-wires transversely, proximal to sigmoid notch → above-elbow cast in supination × 6 wk** ✗ Wrong options: pinning in **pronation**
Approach for DRUJ	**dorsal capsulotomy**

Irreducible DRUJ – ECU interposition

Block to reduction after anatomic radius	**ECU (extensor carpi ulnaris) tendon** ✗ Wrong options: median nerve · FCR · pronator quadratus · FCU

6 Complications

Complication	Fact
Compartment syndrome	High-energy crush, open, vascular injury, coagulopathy; **pain on passive stretch most sensitive**
NV injury	Uncommon except **type III open**
Refracture after plate removal	Early removal, **large 4.5 mm plates**, comminution, persistent lucency → **remove ≥ 18 months**
Nonunion / malunion	–
DRUJ subluxation	Gravity, **pronator quadratus**, **brachioradialis**

X vs Y

	Galeazzi	Monteggia	Nightstick
Fracture	**Distal ⅓ radius**	**Proximal ulna**	Isolated ulnar shaft
Joint	**DRUJ**	**Radial head / PRUJ**	None
Fix	Plate radius → assess DRUJ	Plate ulna → radial head reduces	–

	Rolando	Smith
Fracture	Comminuted intra-articular **1st MC base**	Distal radius, **volar angulation**

How it's asked

"Plate radius + evaluate DRUJ?"	**→ Galeazzi** ✗ Wrong options: nightstick · Monteggia · Rolando · Smith
"Definitive management?"	**→ ORIF radius → assess DRUJ → percutaneous pin if unstable** ✗ Wrong options: closed reduction + cast · PRUJ · palmaris graft (chronic only)

"High-speed MVA, Galeazzi in adult?"	→ **plate radius, closed reduction + evaluate DRUJ** ✗ Wrong options: open TFCC repair · fix both bones · pin in pronation
"Irreducible DRUJ after radius reduced?"	→ **ECU**
"Ligament most commonly injured?"	→ **dorsal radioulnar ligaments**
"DRUJ instability if fracture < 7.5 cm from joint?"	→ **55%**

– – Active space – –

– – Active space – –

GERIATRIC PATIENT TRAUMA

– Trauma · 26 / 67 –

1 Basics

Definition & epidemiology

Elderly	commonly **age > 65**; **physiological age / frailty** more clinically relevant
Growth	**> 85-year group fastest growing** (30%, 2000–2010)
Falls	**30% of people > 65 fall ≥ 1×/yr · 2.1 million fragility fractures / yr**

3 Presentation

History	mechanism: mechanical, **syncope**, **elder abuse** · **accurate medication list** (polypharmacy, **anticoagulants**) · baseline ambulation + mental status

4 Diagnosis & Classification

Key radiology findings

Radiographs	**joint above + below** (other injuries, adjacent arthroplasty / implants)

Pre-operative work-up

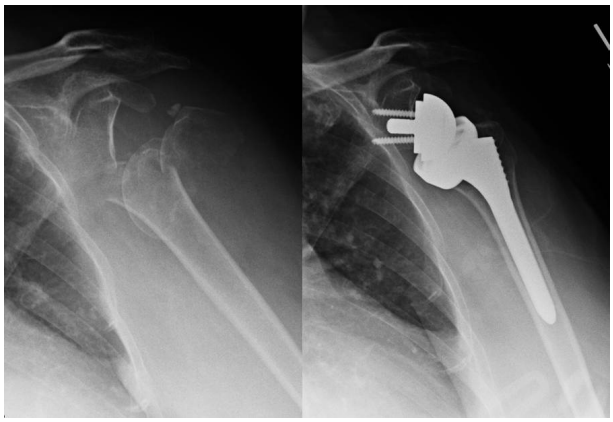
ED	**ECG, CXR, urinalysis, labs**
Optimise	coagulopathy, **heart failure**, sepsis · prevent: **avoid over-sedation + anticholinergics**, ulcer prophylaxis
Cardiac risk	**Revised Cardiac Risk Index (RCRI)** or **NSQIP calculator** → guides work-up + timing

Situation	Next step
Low risk, normal ECG, no CHF signs	**Proceed with surgery** (general anaesthesia safe)
Signs / symptoms of **CHF**	**BNP, echocardiogram**
Elevated MACE risk + very poor functional capacity	**Pharmacological stress test**

5 Management

Treatment principles

Nonop	comorbidities, low function, family wishes → pain control, ease of transfer
Operative goals	**early mobilisation**, return to pre-injury function, pain control
Acute arthroplasty	**comminuted distal humerus (TEA)**, **proximal humerus (reverse TSA)**, **femoral neck**, acetabulum, intra-articular distal femur
Adjuncts	**fibular strut allograft (proximal humerus)**, multiple implants, **locking plates**



Reverse TSA for proximal humerus fracture

Perioperative optimisation — regional block, co-management, timing

↓ Narcotic use while awaiting optimisation	**pre-op fascia iliaca block** ✗ Wrong options: PCA · skeletal traction · pain service consult alone · prophylactic haloperidol
Hip fracture < 48 h	↓ complications, pain, LOS, **mortality**
Ortho-geriatric co-management — proven	**↓ mortality**, **↓ complications**, **↓ time to surgery**, **↑ mobility at 4 months**, ↓ cost; LOS conflicting
Co-management — NOT proven	**↓ need for post-discharge rehab facility**

6 Complications

Mortality predictors

Factor	1-yr mortality
Mechanical ventilation at admission	**78.7% — highest**
ISS > 25	73.2%
Head injury	51.1%
Hip fracture	~30% (in-hospital **6%**)
Open fracture	≈ hip fracture

Spinal cord injury	**higher mortality** than in young

Delirium

Incidence	**13–50%** post-op
Risk factors	dementia, polypharmacy, **narcotics**, dehydration, restraints, **Foley catheters**, frequent awakenings
Treatment	**co-management**, reorientation, minimise narcotics (**fascia iliaca block**), early mobilisation, early catheter removal
Medical complications	pneumonia, **PE / DVT**, **pressure ulcers**, MI, stroke

– – Active space – –

How it's asked

"Reduce narcotics pre-op in hip fracture?"	→ **fascia iliaca block**
"Ortho-geriatric co-management — EXCEPT?"	→ **decreased need for rehab facility**
"Low-risk patient, normal ECG — next?"	→ **proceed with surgical fixation** ✗ Wrong options: BNP · echo · stress test · neuraxial only
"Highest 1-yr mortality predictor?"	→ **need for mechanical ventilation** ✗ Wrong options: head injury · ISS > 25 · hip fracture · open fracture

– – Active space – –

GUN SHOT WOUNDS

– Trauma · 27 / 67 –

1 Basics

Epidemiology

GSW	**2nd leading cause of death in US youth** (after MVA) · buttock GSW ≈ **8%** of extremity GSWs

Ballistics – kinetic energy transfer

KE = ½ m v²	**velocity matters more than mass** (m at 2v > 2m at v); injury rises **exponentially** with velocity
Damage	missile passage + **secondary shock wave** + **cavitation**; fractures possible without direct impact
Max energy transfer	**penetrating (no exit)** > perforating (exits) · **yaw 90° (sideways)** > nose-on; yaw ↓ with longer travel
Thresholds (TO)	**50 m/s penetrates skin** · **65 m/s fractures bone** · larger calibre → slower (TO)
Contamination (TO)	**bullets are not sterile** (heat + barrel friction insufficient); skin flora + clothing; **civilian GSW infection risk is LOW**

Associated conditions

Lead intoxication (plumbism)	**intra-articular** missile; **lead soluble in synovial fluid**; blood lead **peaks 4–6 months (TO)** → neurotoxicity, **anaemia**, emesis, **abdominal colic**; local **lead synovitis → chondrolysis, hypertrophic arthritis**
GSW to hip / acetabulum	**bowel perforation > vascular > urogenital**

4 Diagnosis & Classification

Velocity classification

Velocity	Muzzle speed	Weapon	Gustilo equivalent
Low	**< 350 m/s** (**< 1,200 ft/s**)	**Handguns (except magnums)**	**I or II** (by wound size)
Intermediate	**350–650 m/s** (Orthobullets) · 350–600 m/s (TO); 1,200–2,000 ft/s	**Shotguns**, magnum handguns	Variable – mostly III
High	**> 600 m/s** (**> 2,000 ft/s**)	**Military + hunting rifles**	**III regardless of wound size**

Low-velocity cut-off	< 350 m/s or **< 1,200 ft/s** (Orthobullets, TO) · **< 2,000 ft/s** (OB) – read the options
Shotgun wounding	**shot pattern, load (pellet size), distance**; **< 21 ft (close range) ≈ high velocity**; **wadding** → contamination

– – Active space – –

Low-velocity GSW + fracture (OB) **< 8 h = Gustilo I · > 8 h = Gustilo II**

Key radiology findings

Study	Use
Radiographs	Bone involvement, pattern
CT	**Intra-articular missile**; **hollow viscus injury** communicating with fracture (pelvis / spine)
CT abdomen	**Hip / pelvic GSW → exclude bowel perforation (commonest associated injury)** (✗ Wrong options: bladder US · MRI · CTA · CT acetabulum first)
Sigmoidoscopy	**Blood on DRE** or path **crossing midline at / below pelvic brim**
CT with intra-articular air	**~100% sensitive + specific for traumatic arthrotomy** (more than saline load test; less cost-effective)

Vascular injury assessment & infection risk

Finding	Action
Hard signs – pulsatile bleeding, **expanding haematoma**, thrill / bruit, pulse deficit, ischaemia	**Operative exploration**
Soft signs – haematoma, bleeding at scene, unexplained hypotension	Further assessment
ABI < 0.9 (e.g. 0.85)	**Vascular surgery consult** ± CTA
Symmetric pulses	**Do not exclude** injury → **ABI first** (✗ Wrong options: CTA as screen)

ABI **highly sensitive + specific**, quick, non-invasive · CTA expensive, finds irrelevant occult findings · duplex user-dependent

Wound location alone **not** an indication for exploration

Vascular injury + fracture **↑ infection** → antibiotics · **retained fragments do not ↑ infection**
✗ Wrong options: ↑ malunion · hardware failure · hip disarticulation

5 Management

Gunshot wound

- Low velocity, no fracture / stable fracture → **local wound care, tetanus ± oral antibiotic**
- Low velocity, unstable fracture → **treat as CLOSED fracture (nail / ORIF)**
- High velocity / close-range shotgun → **open-fracture protocol: debride + ex-fix**
- Transabdominal → **laparotomy first, then hip / spine decisions**

Low-velocity wounds without fracture – local wound care

Treatment **irrigation + local wound care in ED**, tetanus, ± **short oral antibiotic** (cephalosporin) (TO, OB)
✗ Wrong options: OR debridement · IV × 7 d · primary closure · remove all fragments

True statement (TO)	**low-velocity GSW → local debridement, oral antibiotics, tetanus** ✗ Wrong options: infection risk high · all grade III · formal debridement always
Low-velocity arm wound, entry + exit, no fracture (TO)	**local wound care + oral antibiotics** ✗ Wrong options: surgical debridement + IV
Primary closure	**contraindicated**
Antibiotics	controversial; give if contaminated · **single long-acting IM cephalosporin = 48 h IV (TO, Geissler)** · **oral = IV** for prophylaxis (TO)
Subcutaneous bones (ulna, tibia) (TO)	exposed fragments → **treat as open fracture**
Retained fragments (TO)	benign; remove if **palm, sole, periarticular, intra-articular**

Fracture management after GSW (nail, external fixation, debridement)

Low-velocity femoral shaft	**reamed antegrade or retrograde IMN** (direction + timing don't matter); union + infection = closed fractures ✗ Wrong options: ex-fix · formal I&D · IV antibiotics 6 wk
IMN (TO)	**statically locked** for low / mid-velocity femur + tibia (incl. subtroch, supracondylar)
High-velocity / shotgun / vascular injury (TO)	**ex-fix → IMN ~2 weeks** when wound healed
Shotgun humerus with radial nerve weakness	**I&D + ex-fix** ✗ Wrong options: acute fibular osteoseptocutaneous flap · nerve repair + retrograde nail · CT + ORIF · coaptation splint
Shotgun wadding (TO)	**must be removed** (organic) · **lead shot – leave** (removal damages tissue)
Humerus + dense radial nerve palsy	**closed IMN relatively contraindicated** → plate + explore, or open the fracture to protect nerve ✗ Wrong options: age < 25 · GSW · athlete · comminution

GSW to hand / foot

Antibiotics	gross contamination, **joint penetration**, contamination unclear
Surgery	**articular involvement**, **unstable fracture**, **≥ 8 h presentation**, **tendon involvement**, **superficial fragments in palm / sole**
Foot: unstable 1st MT base + cuneiform, plantar fragments	**surgical debridement + ex-fix / ORIF + IV antibiotics** ✗ Wrong options: irrigation + cast ± antibiotics
Type I stable, < 8 h	tetanus, surface irrigation, cast / hard-sole shoe · **type II unstable → ex-fix + repeat debridement**

Transabdominal / bowel-contaminated GSW – antibiotics & debridement

Peritonism + positive FAST	**laparotomy first** ✗ Wrong options: DPL · CT · saline load · arthroscopic bullet removal

Laparotomy precedes arthrotomy for transabdominal trajectories

– – Active space – –

Spine bullet + bowel injury, neuro intact	**broad-spectrum IV antibiotics × 7 days** ✗ Wrong options: gram-negative only · oral · observation · decompression
Spine bullet + bowel (cecum), weakness, missile not in canal	**broad-spectrum IV antibiotics 7–14 days** ✗ Wrong options: fragment removal · oral 3–5 d · **methylprednisolone — contraindicated in GSW**
Bullets through alimentary canal → SCI	**removal not required**
Spine surgery indications	**instability** · **deficit matching compression by missile** · **lead in contact with CSF** · retained canal fragment + **incomplete** deficit (relative)
Transabdominal hip GSW (blood on DRE)	**hip debridement + ≥ 24 h antibiotics (gram +, gram —, enteric)** ✗ Wrong options: bullet removal unless intra-articular · single dose · omit gram +
Extra-articular, stable, no rectal blood (low velocity, bullet in ilium)	**empiric antibiotics + observation** ✗ Wrong options: laparotomy · bullet removal · tract debridement · sigmoidoscopy
Acetabular GSW — septic arthritis risk	**associated Letournel pattern** + **bowel injury** = independent risks → **least risk: anterior wall, no bowel injury**
Pelvic / acetabular GSW complications	septic arthritis, **enterocutaneous / enteroacetabular / vesicoacetabular fistula**, infected nonunion, iliac vessel injury



Bullet embedded in L3 vertebral body

Intra-articular bullets & traumatic arthrotomy

Retained intra-articular bullet	**arthrotomy / removal** → lead synovitis, plumbism, **third-body wear**, sepsis
Transabdominal intra-articular GSW	contaminated → **arthrotomy** (after laparotomy)
Transarticular, no retained bullet / IA pathology	**IV antibiotics + local care, no arthrotomy** (OB)
Low-velocity knee GSW, no fracture (TO)	**arthroscopic debridement + IV antibiotics** (TO) · contrast OB: no IA pathology → antibiotics alone — **read the options** ✗ Wrong options: open debridement · local care + oral
Arthroscopy (TO)	↓ morbidity, faster recovery; locate bullet → mini-arthrotomy
Antibiotics for IA fractures (TO)	**1–2 days** if bullet through clean skin / clothing · **1–2 weeks broad-spectrum** if through **lung, bowel, contaminated skin**

Small traumatic arthrotomy (< 5 cm), minimal contamination	**no surgery**: bedside irrigation, tetanus, antibiotics ✗ Wrong options: saline load always needed · CT air not sensitive · retained lead needs no surgery · cost equal (surgery ~$10,000 more)



CT of intra-articular ankle (pilon) GSW

6 Complications

Complication	Fact
Compartment syndrome	Muscle oedema with higher-velocity wounds; ballistic femur > blunt
Plumbism / lead arthropathy	Intra-articular retained bullet
Infection	↑ with **vascular injury**, **bowel transit**, high velocity, associated acetabular pattern
Septic arthritis	Transabdominal → hip / acetabulum

X vs Y

	Low velocity	High velocity / shotgun close range
Treat as	**Closed fracture**	**Open fracture (Gustilo III)**
Debridement	Superficial / local	**Aggressive, repeated**
Antibiotics	Gram + / oral, short	**Broad spectrum**
Femur	**Immediate reamed IMN**	**Ex-fix → IMN ~2 wk**

	Bullet passed through bowel → spine	Bullet passed through bowel → hip joint
Treatment	**IV broad-spectrum 7–14 d; leave bullet**	**Debride hip** + ≥ 24 h antibiotics

How it's asked

"Greatest kinetic energy transfer?"	**→ penetrating, mass m, velocity 2v, yaw 90°**
"True about firearm wounds (TO)?"	**→ low velocity → local debridement, oral antibiotics, tetanus**

– – Active space – –

"Handgun arm wound, no fracture (TO)?"	→ local wound care + oral antibiotics
"Forearm GSW 1,000 ft/s, no fracture?"	→ irrigation + local wound care in ED
"Spine bullet, cecum injury, weakness?"	→ IV broad-spectrum antibiotics 7–14 days
"L2 bullet, bowel laceration, intact?"	→ IV broad-spectrum 7 days
"Bullet in ilium, no rectal blood, stable?"	→ empiric antibiotics + observation
"Hip GSW – image for commonest associated injury?"	→ CT abdomen (bowel perforation)
"Buttock GSW, blood on DRE, hip involved?"	→ hip debridement + 24 h gram +/–/ enteric cover
"Flank GSW, peritonism, FAST positive?"	→ laparotomy
"Acetabular GSW least likely septic?"	→ anterior wall, no bowel injury
"Low-velocity femoral shaft GSW?"	→ reamed antegrade or retrograde IMN
"Shotgun humerus?"	→ I&D + external fixation
"Foot GSW, unstable fracture, plantar fragments?"	→ debridement + fixation + IV antibiotics
"Relative contraindication to closed humeral nail?"	→ radial nerve palsy
"Low-velocity femur GSW + vascular injury ↑?"	→ infection
"Thigh GSW, ABI 0.85?"	→ vascular surgery consult
"Low-velocity knee GSW, no fracture (TO)?"	→ arthroscopic debridement + IV antibiotic
"Bullet passed through hip into iliac wing, no IA pathology?"	→ IV antibiotics without arthrotomy
"Traumatic arthrotomy – true?"	→ small minimally contaminated wounds need no surgery

GUSTILO CLASSIFICATION

– Trauma · 28 / 67 –

4 Diagnosis & Classification

Gustilo-Anderson classification

	I	II	IIIA	IIIB	IIIC
Energy	Low	Moderate	High	High	High
Wound	**≤ 1 cm**	**1–10 cm**	Usually > 10 cm	Usually > 10 cm	Usually > 10 cm
Soft tissue	Minimal	Moderate	Extensive	Extensive	Extensive
Contamination	Clean	Moderate	Extensive	Extensive	Extensive
Comminution	Minimal	Moderate	Severe	Severe	Severe
Periosteal stripping	No	No	Yes	Yes	Yes
Coverage	Local	Local	**Local (incl. skin graft / primary closure)**	**Rotational or free flap**	Typically flap
Vascular	Normal	Normal	Normal	Normal	**Arterial injury requiring repair**



Gustilo IIIB – periosteal stripping, needs flap

Grading rules

Most accurate grading	**intra-operatively** after debridement (definitive)
Automatically ≥ IIIA	**farm injuries** · **open femur** (energy needed) · **segmental fractures** · **high-velocity GSW**
IIIA vs IIIB	decided by **coverage**: **skin graft / primary closure = IIIA** · **flap (gastrocnemius, free) = IIIB** – **not wound size**
IIIC	**vascular repair needed**, regardless of soft tissue; ABI 0.6 without repair = **not IIIC**
Not part of Gustilo	**fasciotomy need**, closed fractures

- - Active space - -

5 Management

Antibiotics by grade

Grade	Antibiotic
I + II	**1st-generation cephalosporin (cefazolin)**
III (A, B, C)	**Cefazolin + aminoglycoside (gentamicin)** – gram-negative cover; **controversial**
Farm injury / bowel contamination	**+ Penicillin** (**Clostridium**, anaerobes)
Fresh or salt water	**Fluoroquinolone (ciprofloxacin)**; also for cephalosporin / clindamycin allergy
Salt water	**Doxycycline + 3rd / 4th-gen cephalosporin (ceftazidime)**

Timing

Start	**ASAP**; **infection ↑ if delayed > 3 h** from injury
Duration	**24–72 h after I&D**
Tetanus booster	if **none in last 5 years**

X vs Y

	IIIA	IIIB	IIIC
Key feature	Coverage possible locally	**Flap needed**	**Vascular repair needed**
Example	8 cm avulsion **skin-grafted**; open femur closed primarily	**Gastrocnemius flap** for proximal tibia	**Pale foot after reduction**

How it's asked

"Segmental tibia, 8 cm avulsion, skin graft?"	→ **IIIA** ✗ Wrong options: IIIB (no flap)
"Which matches IIIB?"	→ **proximal tibia with soft-tissue loss needing gastrocnemius flap** ✗ Wrong options: pale foot = IIIC · 8 cm closed primarily = IIIA · fasciotomy not graded · closed pilon
"Open femur, 3 cm wound, ABI 0.6, closed primarily?"	→ **IIIA**
"Most accurate grading?"	→ **intra-operative**
"Farm injury – add?"	→ **penicillin**

HAND & FOREARM COMPARTMENT SYNDROME

– Trauma · 29 / 67 –

1 Basics

Causes

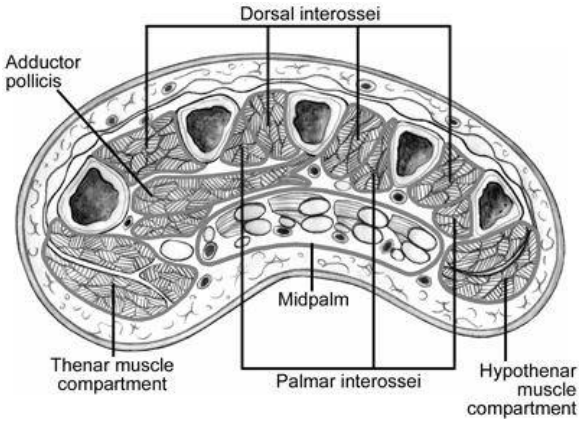
Pathophysiology	trauma → bleeding + oedema → ↑ interstitial pressure → vascular occlusion → **myoneural ischaemia**; muscle damage from **3–4 h warm ischaemia** ("time is tissue")
Fractures (commonest)	**distal radius (adults)** · **supracondylar humerus (children)**; **floating elbow = highest paediatric risk**
Other	**crush**, contusion, GSW, tight casts / dressings, **IV / intra-arterial extravasation**, **burns (circumferential eschar)**, post-ischaemic swelling, bleeding disorders, arterial injury, **drug use (found down)**
Obtunded patients	**79% obtunded**, **68% from IV / intra-arterial extravasation** (Ouellette)

Anatomy

Upper extremity commonest site	**forearm, volar compartment**

Region	Compartments
Arm	**2** – anterior, posterior
Forearm	**3** – **volar (most often)**, dorsal, **mobile wad (BR, ECRL, ECRB; rarely)** (Orthobullets, OB) · **4** – dorsal, **superficial volar, deep volar**, mobile wad (TO)
Hand	**10** – **thenar, hypothenar, adductor pollicis, 4 dorsal interossei, 3 volar interossei**

VIC – commonest muscle + nerve (TO)	**FDP** + **median nerve**



Compartments of the hand

– – Active space – –

3 Presentation

Diagnosis & fasciotomy

Finding	Significance
Pain out of proportion	**First symptom**; absent with nerve damage; hard in polytrauma, sedated, children
Pain on passive finger stretch (MCP / phalangeal extension)	**Most sensitive** / early
Tense hand in intrinsic MINUS position	**Most consistent** clinical finding (✗ Wrong options: intrinsic plus)
Paraesthesia / hypoaesthesia	Nerve ischaemia
Paralysis	**Late** – full recovery rare
Absent pulses	**Late** – amputation usually inevitable
Cool, pale hand	Late; suggests **arterial injury**
Loss of vibration sense	Not specific

Earliest feature of VIC (TO) — **pain on passive stretch of middle + ring fingers** (FDP of these fingers first)
✗ Wrong options: skin changes · thumb stretch · intrinsic minus

Forearm CS + numb thumb / index — **acute carpal tunnel** → release CTR with fasciotomy

4 Diagnosis & Classification

Key radiology findings

Radiographs — rule out fracture only · **no advanced imaging, CTA or Doppler** – delays surgery

Diagnostic criteria

Fasciotomy	Observe
Clinical diagnosis (no pressures needed)	Exam not consistent
Δp < 30 mmHg (diastolic – compartment)	**Δp > 30**
Absolute **> 30 mmHg**	–

Pressures indicated — obtunded, intubated, **after a block**, polytrauma, equivocal

Intra-op Δp < 30 — check **pre-op diastolic**; follow post-op (anaesthesia lowers diastolic)

Normal hand compartment pressure

Normal hand — **0–15 mmHg** (e.g. **10 mmHg**) · **> 15 mmHg pathological** (some) · finger pressures **not measured** · volar–dorsal interosseous fascia incompetent

5 Management

Suspected hand / forearm compartment syndrome

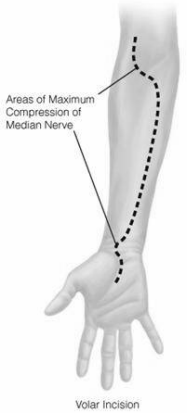
- Clinical diagnosis → **emergent fasciotomy of all involved compartments**
  - median nerve symptoms → **add carpal tunnel release**
- Obtunded / equivocal → pressures → Δp < 30 → **fasciotomy**
- Never: analgesia + wait, nerve block, overnight observation, CTA

Crush hand, pain on passive MCP / wrist motion, normal NV	**surgery** ✗ Wrong options: imaging · Doppler · admit + observe · analgesia
Peripheral nerve block	**contraindicated** when CS suspected (masks symptoms)

Forearm fasciotomy

Incision	Releases	Interval
Volar – radial to FCU at wrist → **medial epicondyle**, extend distally	**Volar compartment + carpal tunnel** + lacertus fibrosus	**FCU ulnar, FDS radial** → open deep fascia
Dorsal – 2 cm distal to lateral epicondyle → mid-wrist	**Mobile wad + dorsal**	**EDC / ECRB**

Post-op **leave open** (VAC / wet-to-dry) → **re-debride 48–72 h** → delayed closure or **STSG**



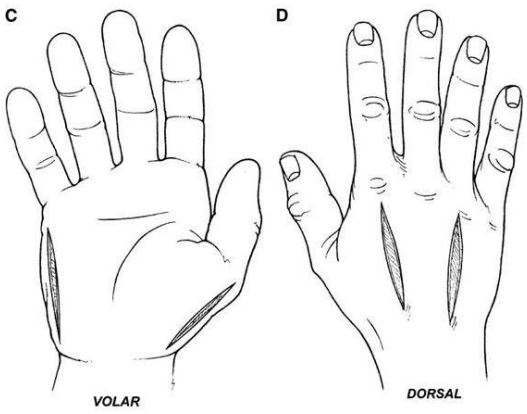
Volar forearm fasciotomy incision

Hand fasciotomy

Incision	Releases
2 dorsal – over 2nd + 4th MCs	**Dorsal + volar interossei**, **adductor pollicis** (1st volar interosseous + adductor via ulnar side of 2nd MC)
Radial side of 1st MC	**Thenar**
Ulnar side of 5th MC	**Hypothenar**

All 10 compartments released; wounds open → primary closure or STSG

– – Active space – –



Hand fasciotomy incisions

6 Complications

Volkmann ischaemic contracture

Cause missed CS; **brachial artery obstruction (tight plaster, tourniquet)** → muscle necrosis → fibrosis

Posture **elbow flexion, forearm pronation, wrist flexion, thumb adduction (+ flexion), MCP extension, IP flexion = intrinsic-minus (claw) hand**

Progression (TO) FDP (middle + ring) → **FPL** → FDS, FCR, FCU → extensors (severe); median + ulnar neuropathy add to claw

Tsuge stage	Muscles	Treatment
Mild	**Finger flexors**	**Dynamic splinting, tendon lengthening**
Moderate	**Wrist + finger flexors**	Excise necrotic muscle, **median + ulnar neurolysis**, **BR → FPL, ECRL → FDP** transfers, **flexor slide**
Severe	**Flexors + extensors**	As moderate ± **free functioning muscle transfer**

Tsuge – TO wording (TO) mild = **partial FDP (1–2 slips)** · moderate = **all flexors + nerve involvement** · severe = **flexors + extensors + skin / joint**

Seddon's ellipsoid (TO) **central** part of flexor mass (**FDP + FPL**) most ischaemic → necrosis → fibrosis

Volkmann sign / fixed-length phenomenon (TO) fingers **cannot be passively extended with wrist extended**; extend when wrist flexed · **ring + middle** fingers commonest

VIC treatment (TO) **turnbuckle (dynamic) splint** + physio (mild) · muscle–tendon lengthening / Z-plasty (mild) · **Max Page muscle slide** = flexor–pronator origin released from **medial epicondyle** + slid distally · **arthrodesis / tenodesis / capsulodesis** for severe rigid deformity

Prognosis loss of function, VIC, neurological deficit, infection, amputation

Extravasation injury mimic

Purple glove syndrome **IV phenytoin extravasation**: oedema, **violaceous** discoloration, pain; resolves with **stopping IV + elevation + monitoring**

Mechanism	**vesicant leaking into extravascular space** (phenytoin pH ≈ 12, precipitates) ✗ Wrong options: axillary DVT · necrotising fasciitis · dermatomyositis · flexor tenosynovitis
Irritant vs vesicant	irritant = inflammation only · **vesicant = direct toxicity** (slough, blisters, necrosis, ± CS)
Agents	phenytoin, **TPN**, vasopressors, arginine, chemotherapy, contrast
Management	stop infusion, mark border, elevate, immobilise, heat / ice, antidotes, irrigation, debridement ± fasciotomy

X vs Y

	Early	Late
Findings	**Pain out of proportion, passive stretch pain**, swelling	**Paralysis, absent pulses, pallor**

	Compartment syndrome	Purple glove (phenytoin)
Treatment	**Fasciotomy**	**Stop IV, elevate, observe**

How it's asked

"Earliest feature of VIC (TO)?"	→ **pain on passive stretch of middle + ring fingers**
"Crush hand, passive pain, no fracture — next?"	→ **surgical intervention**
"Burn eschar — early CS finding?"	→ **pain on phalangeal extension** ✗ Wrong options: absent pulses · Δp 38 · intrinsic plus · paralysis
"Most supports hand CS?"	→ **swelling + pain out of proportion** ✗ Wrong options: vibration loss · finger pressure 15 · intrinsic plus · cool pale hand
"Forearm crush, thumb / index numbness, passive pain?"	→ **emergent forearm fasciotomy + carpal tunnel release**
"Normal hand compartment pressure?"	→ **10 mmHg**
"Muscle-sliding operation for VIC?"	→ **Max Page (flexor-pronator slide)** (TO)
"Fingers extend only with wrist flexed?"	→ **Volkmann sign (VIC)** (TO)
"Swollen violaceous hand on IV phenytoin?"	→ **extravasation of a vesicant**

– – Active space – –

HIP DISLOCATION

– Trauma · 30 / 67 –

1 Basics

Epidemiology

Who	young, **high-energy** trauma (MVA **dashboard**); rare but **high rate of associated injuries** (**95%**)
Direction	**posterior 90%** · anterior **~10%** (Orthobullets) · **10–15%** (TO)
Stability	bony geometry + **labrum, capsule, ligamentum teres**, muscles
Anatomy (TO)	head = **⅔ of a sphere** · acetabulum covers ~**40%**, **labrum adds 10%** · inverted-U articular surface · **iliofemoral (Y) ligament strongest in body** (AIIS → intertrochanteric line)
Capsule (TO)	anterior **on intertrochanteric line** · posterior **~1–1.5 cm (½ inch) proximal to intertrochanteric crest**
Head supply (TO)	**MFCA → posterosuperior retinacular vessels (most crucial)** · ligamentum teres artery significant only in children · dislocation **kinks MFCA** → urgent reduction
Sciatic nerve variants (TO)	exits **below piriformis in 80–90%** · pierces / passes above in 10–20%

Associated injuries (knee, aorta)

Dislocation	Associated injury
Posterior	**Posterior wall fracture (commonest; up to 70%, TO)**, femoral head fracture (**5–15%**, TO · 10–15%, OB) – anterior head fragment, **sciatic nerve (peroneal division)**, **ipsilateral knee (up to 25%)**, AVN
Anterior	**Femoral head impaction / chondral fracture – commonest (TO)** (superolateral, **Hill-Sachs-like**); acetabulum, neck, GT, shaft; **AIIS avulsion** (superior type); femoral vessels + **femoral nerve** (less common)

Ipsilateral knee on MRI	**93% abnormal**: **effusion 37%**, **bone bruise 33%**, **meniscal tear 30%** → most likely concomitant = **ipsilateral meniscus tear** ✗ Wrong options: contralateral ACL · SDH · ankle · lumbar burst
Thoracic aortic rupture	associated with **posterior hip dislocation** (deceleration); **bilateral pulse deficit → proximal large-vessel injury** → CXR mediastinum, thoracic surgery

2 Mechanism

Hip position	Result
Flexed + adducted (+ IR), axial load via knee (**dashboard**)	**Posterior — pure** (more flexion / adduction → simple)
Flexed + **abducted**	**Posterior fracture-dislocation**
Abduction + ER + extension	**Anterior superior (pubic)**
Abduction + ER + flexion	**Anterior inferior (obturator)**

Pure vs fracture-dislocation (TO) **↑ flexion + IR / adduction → pure** · ↓ flexion / adduction → **posterior wall fracture-dislocation** · **low femoral anteversion → pure posterior**; high anteversion → fracture-dislocation (TO)



Dashboard injury mechanism

3 Presentation

Type	Limb attitude
Posterior	**Flexion, adduction, internal rotation**, shortened
Anterior pubic (superior)	**Extension + external rotation**
Anterior obturator (inferior)	**Flexion, abduction, external rotation**
Irreducible variant	**Neutral, mildly shortened** — not overtly flexed / IR, limb locked

Exam ATLS; **sciatic nerve (10–20%)**; knee stability; **CXR for aortic injury**

Narath sign (TO) **feeble / absent femoral pulse** in posterior dislocation (no counter-surface behind artery)

Why foot drop (TO) peroneal fibres **peripheral / superficial** + phylogenetically newer → more stretch-sensitive

Anterior pubic type (TO) compresses **femoral artery → risk of gangrene**; head palpable in groin, limb may be **lengthened**

Windswept dislocation (TO) one hip anterior, other posterior · atraumatic → **cerebral palsy** · adult → high-velocity trauma

FNB attitude question (TO) answer key not provided by the source; map: **FADIR → posterior**, **FABER → anterior obturator**, **extension + abduction (ER) → anterior pubic**

4 Diagnosis & Classification

Classification

Simple pure dislocation · **Complex** – + acetabulum or proximal femur fracture

Thompson-Epstein (posterior) (TO)	Description
1	± Minor fracture
2	**Large single posterior rim** fragment
3	**Comminuted rim** ± major fragment
4	**Acetabular floor** fracture
5	**Femoral head** fracture

Epstein (anterior) (TO)	Description
Type 1 – superior	**Pubic, subspinous**
Type 2 – inferior	**Obturator, thyroid, perineal**
Subtypes	**A** simple · **B** head fracture · **C** acetabular fracture

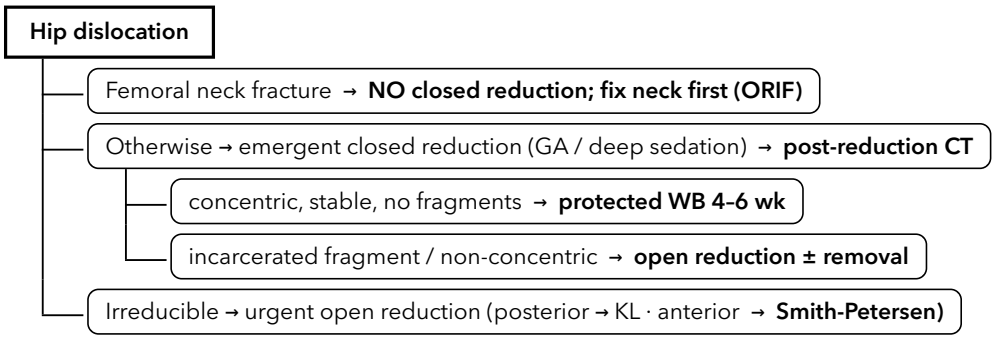
AO/OTA hip dislocation (TO) **30D10 anterior · 30D11 posterior · 30D30 obturator**

Key radiology findings

Finding	Meaning
AP + **cross-table lateral**	**Anterior vs posterior**; **check femoral neck before reduction**
Shenton's line broken	**All dislocations** (inferior neck ↔ superior obturator foramen)
Head smaller than other side, over roof, **lesser trochanter hidden (IR)**	**Posterior**
Head larger, **medial / inferior** to acetabulum	**Anterior**
Cranial head close to ilium + ER (prominent LT) + IA fragments	**Irreducible variant** (✗ Wrong options: absent fragments · hidden LT · Shenton's (all dislocations))
Post-reduction CT – mandatory in ALL	Head fractures, **loose bodies / incarcerated fragments** (even if concentric on X-ray), acetabular fractures
Post-reduction AP, inlet / outlet, **Judet**	–
MRI	Controversial, not routine (labrum, cartilage, vascularity)
Post-reduction 5 views (TO)	**AP + inlet + outlet + 2 Judets**
MRI – early AVN predictor (TO)	**Obturator externus / piriformis injury**; AVN visible at **6–8 wk**

Advanced imaging before reduction **no** → reduce first

5 Management



Reduction – urgency & technique (including children)

Timing — **within 6 h** (OB, TO) · **within 12 h** (Orthobullets, OB) – read the options; **> 12 h → 5.6× AVN odds (TO)**; ↑ adverse events

Hip before wrist — reduce hip first; **CT after reduction**
✗ Wrong options: CT / MRI first

Technique — supine, **traction in line with deformity**, adequate sedation + relaxation, assess stability

Manoeuvre (TO)	Dislocation	Key steps
Allis	Posterior	Inline traction, **knee 90°**, assistant stabilises pelvis, extend + ER
Bigelow	Posterior	Traction, knee 90°, **extension, abduction, ER**
East Baltimore lift	Posterior	Hip + knee 90°, physician + assistant lift under popliteal fossa
Allis leg extension	Anterior	Inline traction; **pubic → hyperextension**
Reverse Bigelow	Anterior	Traction in line → **adduct, IR, extend**
Stimson (TO)	Posterior	**Prone**, gravity-assisted
Rochester (TO)	Posterior	Single assistant; needs **both hips flexed** → not for bilateral
PGI technique (TO)	Posterior	**Single surgeon, most atraumatic**: flexion → abduction → ER
Walker modification of Allis (TO)	**Anterior**	Preferred for anterior

Mnemonic SABER (TO) — Stimson · Allis · Bigelow · East Baltimore · Rochester · Bigelow carries **higher fracture risk**

Child (10-yr, posterior) — **closed reduction under GA with fluoroscopy** → avoids **physeal separation (epiphyseal displacement)**, confirms concentricity
✗ Wrong options: ED sedation · no fluoroscopy · open first

After reduction (TO) — **WBAT**, ROM / strengthening, **jog 6–8 wk**, **running / sport 3–4 months**; follow **18 months–2 yr** for AVN

Post-reduction (TO) — stability test at **90° flexion** · **no role for traction** after concentric reduction · ROM day 1

Simple dislocation (Orthobullets) — **protected WB 4–6 wk**

– – Active space – –

Post-reduction CT

Next step after closed reduction (any)	**CT hip + pelvis** ✗ Wrong options: skeletal traction · EUA fluoroscopy · spica · TDWB · observation
Skeletal traction	if acetabular fracture / loose body / unstable

Operative criteria

Indication	Treatment
Irreducible dislocation	**Urgent open reduction**
Failure of 2 closed attempts (TO) · **iatrogenic sciatic palsy** (TO) · instability after reduction (TO)	Open reduction (posterior pure → **Ganz surgical dislocation** (TO); PW → KL)
Incarcerated fragment on CT, **non-concentric** reduction	Open reduction / fragment removal
Delayed presentation	Open reduction
Associated femoral neck fracture	**Stabilise neck before reduction**
Acetabular / femoral head fractures	ORIF
Arthroscopy	No established indication (loose bodies, labrum)

Anterior / obturator & irreducible dislocations — recognition & open approach

Approach	**posterior → Kocher-Langenbeck** · **anterior → Smith-Petersen** (or Watson-Jones)
Irreducible femoral head fracture-dislocation	head **buttonholes through labrum-bone interval**; **Smith-Petersen**: ↓ op time, ↓ blood loss, better view, **spares head supply** (anterior capsule contributes little)
Smith-Petersen interval	**sartorius (femoral) ↔ TFL (superior gluteal)**; deep: rectus femoris ↔ gluteus medius
Other intervals	Watson-Jones: TFL ↔ gluteus medius (both **superior gluteal**) · Hardinge: splits gluteus medius + vastus lateralis (no internervous plane) · **posterior (Moore / Southern)**: splits gluteus maximus
Posterior approach risk	**lateral epiphyseal artery within 15 mm of external rotator insertions**
Irreducible without PW fracture (Mehta)	**urgent ORIF without closed attempts** → avoid iatrogenic neck fracture

Associated femoral head fracture — management

Infrafoveal head fracture, stable hip, 8 mm displacement (TO)	**ORIF via anterior approach** ✗ Wrong options: traction · excision · posterior ORIF
Approach rule (TO)	**Pipkin I / II → anterior** · **Pipkin IV → posterior**
Nonop Pipkin I / II (TO)	**< 2 mm displacement**, stable hip, **no interposed fragment**

6 Complications

Complication	Rate / fact
Post-traumatic arthritis	**Up to 20% simple** (commonest long-term, TO); markedly ↑ complex; **anterior 17%** (TO)
OA – TO lecture (TO)	**16% isolated** dislocations · **88% with acetabular fracture**
AVN	**5–40%** (Orthobullets) · **2–10% if reduced < 6 h** (TO); ↑ with time to reduction; **anterior 4%** (TO)
Sciatic nerve (peroneal division)	**8–20%** (Orthobullets) · **10–15%** (TO); ↑ with delayed reduction
HO	**2.8–9%** (TO); repeated reduction attempts
Recurrent dislocation	**< 2%**

X vs Y

	Posterior	Anterior
Frequency	**90%**	~10%
Attitude	**Flexion, adduction, IR**	**ER** + abduction (obturator: flexion · pubic: extension)
X-ray head size	**Smaller**	**Larger**
Associated	**PW fracture**, sciatic nerve, knee	**Head impaction**, femoral NV
AVN	Higher	**4%**
Open approach	**Kocher-Langenbeck**	**Smith-Petersen**

	Anterior pubic (superior)	Anterior obturator (inferior)
Hip at injury	**Extension**	**Flexion**
Attitude	**Extension + ER**	**Flexion, abduction, ER**
X-ray	Head superior	**Head inferior**

How it's asked

"After closed reduction – next?"	→ **CT hip + pelvis**
"Normal exam after reduction – next?"	→ **CT**
"Commonest association with anterior dislocation (TO)?"	→ **femoral head (impaction) fracture**
"Most likely concomitant injury (dashboard)?"	→ **ipsilateral knee meniscus tear**
"Injury associated with thoracic aortic rupture?"	→ **posterior hip dislocation**
"Flexed, abducted, ER limb?"	→ **anterior obturator dislocation**

– – Active space – –

"Irreducible fracture-dislocation — Smith-Petersen interval?"	→ **sartorius (femoral) + TFL (superior gluteal)**
"X-ray warning of irreducible dislocation?"	→ **cranial head + externally rotated femur**
"Return to running after dislocation (TO)?"	→ **WBAT, jog by 8 wk, run by 3 months**
"10-yr-old posterior dislocation?"	→ **closed reduction under GA with fluoroscopy**
"Hip + wrist injury — next?"	→ **closed reduction of hip**
"Infrafoveal fracture, 8 mm, stable (TO)?"	→ **ORIF via anterior approach**
"Most atraumatic single-surgeon reduction?"	→ **PGI technique** (TO)
"Earliest MRI predictor of AVN after dislocation?"	→ **obturator externus / piriformis injury** (TO)

HUMERAL SHAFT FRACTURES

– Trauma · 31 / 67 –

1 Basics

Epidemiology

Incidence **3–5% of all fractures** · **20% of humeral fractures** · 7–11.3 / 100,000

Age bimodal; **60% > 50 yr** · < 50: 70% men · > 50: 70% women

Location proximal ⅓ 30% · **middle ⅓ 60% (commonest)** · distal ⅓ 10%

Mechanism **ground-level fall 60%** · MVA ~30% · pathological 4.3% · open 3%

Risk factors previous fracture, **smoking (men)**, age, osteoporosis

Level	Mechanism
Proximal ⅓	FOOSH, elderly (impaction at surgical neck)
Middle ⅓ **transverse**	**Direct blow**
Middle ⅓ **spiral**	FOOSH / **torsion (throwing, arm wrestling)**
Distal ⅓	Fall on flexed elbow

Associated **floating elbow** (+ radius / ulna; commoner in children) · ipsilateral shoulder dislocation (**posterior** > anterior)

Surgical anatomy & approaches

Canal ends **2–3 cm proximal to olecranon fossa**

Deforming forces fracture **proximal to pectoralis major → deltoid abducts proximal fragment**

Radial nerve course exits axilla posterior to brachial artery → **triangular interval** → between **medial + long heads** of triceps → **spiral groove** (**medial to long + lateral heads, 2 cm proximal to deep head**) → pierces **lateral intermuscular septum ~10 cm proximal to radiocapitellar joint** (**20 cm** above medial, **14 cm** above lateral epicondyle) → anterior **~7.5 cm** from articular surface (**safe zone – never posterior within 7.5 cm**)

Radial nerve in contact with bone **~6.5 cm** of diaphysis in spiral groove

Radial nerve – surgical (TO) identify on posterior approach **10–15 cm above lateral epicondyle** · MIPO anterior: **forearm supinated** to protect radial nerve · palsy rate **3–30%**, usually neuropraxia

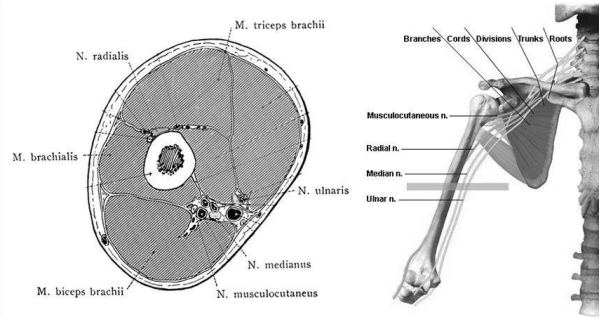
Ulnar nerve enters posterior compartment at **arcade of Struthers**, ~8 cm above medial epicondyle

Axillary nerve **4–7 cm below acromion** (limits proximal posterior exposure)

Posterior antebrachial / inferior lateral brachial cutaneous nerve branch of **radial**; trace proximally to find radial nerve (**Gerwin modification**, paratricipital)

– – Active space – –

Compartments anterior: biceps, brachialis, coracobrachialis, brachial vessels, **musculocutaneous, median, ulnar** · posterior: triceps, **radial nerve**



Radial nerve course in the arm

3 Presentation

Exam shortening + **varus**; **document radial nerve pre- and post-reduction** (wrist + thumb IP extension, dorsal hand sensation)

Radial nerve palsy "**A-OK**" + finger crossing intact, ↓ wrist / finger extension, dorsal hand numbness → **Holstein-Lewis**

4 Diagnosis & Classification

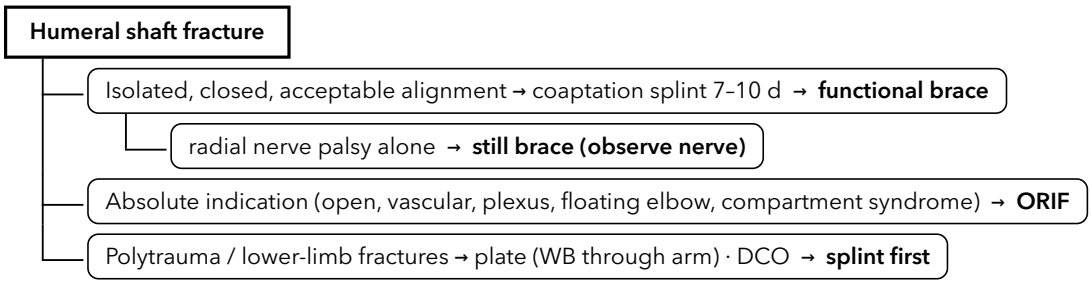
Key radiology findings

Study	Use
AP + lateral humerus, joint above + below	Standard
Transthoracic lateral	**Sagittal (apex anterior) deformity**; rotate patient, not distal fragment
Traction views	Shortened / extension (not routine)
CT	Intra-articular extension
CTA	Vascular injury
EMG / NCS	Nerve palsy follow-up – **not acutely**

Classification

System	Categories
Descriptive	Proximal / middle / distal ⅓ · spiral, transverse, comminuted
OTA 12	**A simple · B wedge · C complex**
Garnavos	Location **P / M / D** (+ j = joint) · morphology **S** simple (T transverse / oblique, S spiral) · **I** intermediate (1–2 butterflies) · **C** complex (≥ 3 butterflies / comminution)
Holstein-Lewis	**Spiral distal ⅓** → radial nerve palsy **7–22%** (Orthobullets) · **22%** (OB); entrapment risk; **7.5% of shaft fractures**

5 Management



Functional bracing – indications, acceptable alignment & contraindications

Acceptable alignment (adult)	Limit
Anterior (sagittal) angulation	**< 20°**
Varus / valgus	**< 30°**
Rotation	**< 30°** (Orthobullets, TO) · **< 15°** (OB)
Shortening	**< 3 cm**

TO trap (TO)	"**angulation < 30°**" is **NOT** acceptable (must be < 20°) · **shortening < 3 cm, rotation < 30°, angulation < 20°** (TO)
Child (TO)	**20–30° varus**, **20° apex-anterior**, **15° IR**, **≤ 2 cm shortening**
Infant / toddler	**coaptation splint + bandage arm to thorax** ✗ Wrong options: hanging cast · flexible nails · ex-fix · plate
Technique	coaptation splint / hanging cast **7–10 days → Sarmiento brace** (2.5 cm below axilla → 2.5 cm above condyles); **no sling** (gravity reduction); **valgus mould** vs varus; weekly X-ray × 3 wk, then every 3–4 wk
Hanging cast	**avoid in transverse fractures** (distraction)
At 9 days, acceptable	**functional brace 6–8 wk** ✗ Wrong options: continue splint (stiffness) · ORIF
Contraindications (absolute)	severe soft-tissue injury / bone loss · **vascular injury needing repair** · **brachial plexus injury** (no muscle tone → nonunion)
NOT contraindications	**radial nerve palsy (closed)** · comminution · 1 cm shortening · 20° varus
Outcomes	union **93.5%** (Sarmiento < 2% nonunion); time **10.7 wk**; malunion 12%; **proximal ⅓ oblique / spiral nonunion 29% (Rutgers-Ring)** · 54% proximal ⅓ (Orthobullets)
Varus malunion	**commonest deformity** (esp. distal / transverse) → **no effect on function** ✗ Wrong options: nonunion · pain · satisfaction · nerve recovery
Nonop nonunion	higher than all surgical techniques (contemporary literature); 33% nonunion in one series

– – Active space – –

Operative indications (floating elbow, polytrauma, plexus injury) & weight bearing

Absolute	Relative
Open fracture	**Bilateral humerus**
Vascular injury requiring repair	**Polytrauma / lower-limb fracture** (crutch WB)
Brachial plexus injury	**Pathological** (life expectancy > 6 mo)
Floating elbow (ipsilateral forearm)	Burns / soft tissue precluding brace
Compartment syndrome	**Distraction**, **segmental**, **short oblique / transverse (OTA A)**, intra-articular, long oblique proximal
Periprosthetic at stem tip	**Obesity**, large breasts
Inability to maintain reduction	Radial nerve palsy (relative only)
Progressive nerve deficit after manipulation	**Young athlete wanting quick return**

NOT an operative indication (TO) — **closed fracture with radial nerve palsy**
✗ Wrong options: nonunion · floating elbow · pathological with secondaries

Strongest indication — **ipsilateral both-bone forearm fracture**
✗ Wrong options: radial nerve palsy · long oblique · comminution · Holstein-Lewis

Flaccid limb, C5–T1 avulsions — **ORIF with plate**
✗ Wrong options: splint · brace · ex-fix · sling

Floating elbow (olecranon + humerus) — **ORIF both + early ROM**
✗ Wrong options: brace humerus · splint 4 wk

Polytrauma (humerus + femur) — **plate humerus + IMN femur** → crutch WB

WB after plate / IMN — **full WB**; **no effect on union**
✗ Wrong options: partial (can't use crutches)

Plateau + humerus ORIF + contralateral femur IMN — **NWB plateau side leg**, **WBAT arm + nailed femur**

DCO, multiple GSWs + abdominal injury — **closed reduction + splint in ED**
✗ Wrong options: I&D · ex-fix in ED · ORIF · IMN

15-yr quarterback, torsional spiral, wants early return — **surgical fixation**
✗ Wrong options: splint / brace · EMG acutely

Ex-fix indications — high-energy comminution, open, defect, floating elbow, **unstable polytrauma**, **vascular injury**; 80% good, **pin-track infection 12%**

Special situation (TO)	Preferred
Polytrauma / DCO · open 3B/3C	**External fixator**
Paraplegic (weight-bearing on arms)	**Plating**
Pathological / segmental	**IM nail**

TEA — comminuted distal fractures, **low-demand > 65**, osteoporosis / **RA**

Plate vs intramedullary nail

	Plate (ORIF)	IMN
Union	**Equal** (90–92%)	**Equal**
Time to union	11.9 wk	**10 wk**
Op time	88 min	**61 min (faster)**
Shoulder pain / impingement / stiffness	**Lower**	**Higher (16–37%)** – rotator cuff (supraspinatus)
Functional shoulder score (ASES) at 1 yr	**Equal**	**Equal**
Radial nerve palsy	Equal	Equal
Infection	5.4%	**1.2%** – **lower** (Orthobullets, OB) · no difference (OB) – read the options
Reoperation	7.6%	11.6% (NS) · **↑ revision** (McCormack, OB)
Total complications	Lower	**Higher**
Hardware complications	Lower	**Higher**
Malrotation	Lower	**Higher → ↓ ROM**

IMN – EXCEPT "**increased shoulder ROM**" is false

Antegrade IMN – true **higher shoulder impingement**
✗ Wrong options: ↑ ROM · less radial nerve palsy · less nonunion

IMN relative indications **pathological**, segmental, osteoporotic, poor skin, polytrauma

MIPO anterolateral incisions + extraperiosteal tunnel; **lacks nerve visualisation → ↑ radial nerve risk** (Orthobullets) · **lower transient palsy than ORIF** (OB) – read the options

Plate technique

Implants **3.5 / 4.5 mm (narrow or broad) DCP / LC-DCP** (LC-DCP: easier contour, ↓ stress shielding, periosteum spared)

Simple pattern **lag + compression** (absolute stability) · **comminuted** – **bridge** mode

Bone defect **≤ 3 cm → shorten · > 3 cm → graft**

Nerve transposition during ORIF **no benefit**

– – Active space – –

Approaches

Approach	Use	Key facts
Anterior (brachialis split)	Middle ⅓	**Intramuscular internervous plane** – **lateral brachialis (radial) / medial brachialis (musculocutaneous)** in ~80%
Anterolateral (extends deltopectoral)	Proximal–middle ⅓	Radial nerve **between brachialis + brachioradialis** distally; cephalic vein, anterior circumflex
Posterior triceps-split	Distal–middle ⅓	**55%** exposed (no nerve mobilisation), **76%** with; no internervous plane; avoids ulnar nerve dissection
Paratricipital (triceps-sparing) / medial reflection of lateral + medial heads	Distal	**~94% exposure (greatest)**; lateral window → radial nerve; medial window → mobilise ulnar nerve
Lateral	Distal ⅔	**Highest iatrogenic radial palsy (20%)**
Medial	Brachial artery, median + ulnar nerves	Rarely for fractures

Iatrogenic radial palsy **lateral 20%** > **posterior 11%**; only **approach** predicts risk

All posterior exposures limited proximally by **axillary nerve**



Anterolateral plating of the humeral shaft

IMN technique & interlock nerve risk

Antegrade supine; split deltoid (axillary nerve); entry **through supraspinatus as medial as possible** at apex, near **musculotendinous junction** (avascular insertion), **≤ 1 cm**

Retrograde prone / lateral; posterior supracondylar cortex

Don't ream across the fracture radial nerve

– – Active space – –

Interlock	Nerve at risk
Distal anterior → posterior	**Musculocutaneous → lateral antebrachial cutaneous** (numb radial volar forearm)
Distal lateral → medial	**Radial** (over-penetration → **ulnar**)
Anteromedial → posterolateral oblique	Median
Proximal (antegrade)	**Axillary** + circumflex vessels

External fixation pin safe zones

Distal pins **no safe zone → mini-open** to bone (radial nerve)
✗ Wrong options: percutaneous arc · direct posterior (triceps, supine patient)

Proximal pins anterolateral, mini-open (**axillary nerve**, biceps tendon); **most distal pin just above olecranon fossa**

6 Complications

Radial nerve palsy — natural history, recovery order, EMG timing, exploration & tendon transfer

Incidence **12.3% (8–15%)** (Orthobullets) · **11–11.8%** (OB); **distal ⅓ 22%**; **10–20% secondary** (develop during treatment, TO)

Fracture type **transverse 21.2% > spiral 19.8%** > oblique 8.4% > comminuted 6.8% · level: distal > middle > proximal

Injury type **closed → neurapraxia** · **low-velocity GSW → axonotmesis** · **open (high energy) → neurotmesis** (laceration) · complete palsy 50–68% (TO)

Spontaneous recovery **~70% by 7 weeks** (OB, TO > 70%) · **77.2%** (Orthobullets) · **85–90% within 3 months** · full at **~6 months** · 100% of low-energy by 4 months (TO) · **88.6% overall** · iatrogenic 93.9%

At exploration **in continuity 62.7%** · lacerated 26.8% · incarcerated 10.5% · transected only **12%** (Pollock, TO)

Recovery order (proximal → distal) **brachioradialis (radial styloid) → ECRL (2nd MC base)** → supinator → ECRB (3rd MC) → EDC → ECU → EDQ → APL → **EPL** → EPB → **EIP (last)**

First to recover crossing the wrist **ECRL** (inserts **base of 2nd MC**) → **wrist extension in radial deviation** first

– – Active space – –

Last **EIP → index MCP hyperextension**

Situation	Management
Closed, primary palsy	**Brace + observe** (✗ Wrong options: explore · IMN + explore)
Low-velocity GSW + palsy	**Splint + observe**, short antibiotics
Open fracture + palsy	**I&D, ORIF + explore nerve**
High-energy open + brachial artery	**Neurotmesis** likely
Secondary palsy after closed reduction	**Early exploration** (controversial)
Holstein-Lewis	**Controversial** ORIF + explore

Early exploration (TO) **open**, vascular injury, **secondary palsy after reduction**, **high-velocity GSW**, sharp / penetrating, severe soft tissue

Complete RNP after shaft fracture (TO) **hanging cast + explore if no recovery at 3 months**
✗ Wrong options: explore + plate · brace + explore at 3 wk · nail + observe

EMG / NCS timing	Source
~2 months (baseline)	Orthobullets
1–2 months → repeat ~4 months; **no exploration between 1–4 months**	OB
3–4 / 3–5 months	OB
No recovery on EMG at **6 wk → observe**, repeat at 12 wk	OB

EMG interpretation **fasciculations / reinnervation = recovery → observe** · **fibrillations = denervation → explore**

Exploration timing **4–6 months** (Orthobullets, OB) · **3–4 months** if fracture healed (TO) · **3–6 months**, before 6 (TO) · **6 months** (OB) · **no advantage waiting > 7 months** (TO) – read the options

Healed at 4 months, no recovery **EMG / NCS**
✗ Wrong options: observe · explore immediately · MRI · CT

Pre-op palsy, nerve protected at ORIF, 2 months **NCS / EMG**
✗ Wrong options: tendon transfer · re-explore

Exploration recovery **early (< 3 wk) ~90%** · **late (≥ 8 wk) ~68%**

Tendon transfers (≥ 1 year) **wrist: PT → ECRB** · **fingers: FCR (or FCU) → EDC** · **thumb: PL → EPL** (FCU loses ulnar deviation power)

Abbreviations PT pronator teres · ECRL / ECRB extensor carpi radialis longus / brevis · EIP extensor indicis proprius · EMG electromyography · NCS nerve conduction studies

Predicting nonunion during bracing

Motion + no callus at 6 weeks **PPV 90–100%**, **specificity 99–100%**, sensitivity 82%, NPV 97%

Too early **4 weeks without callus → continue brace, review in 4 wk**
✗ Wrong options: LIPUS · ORIF · IMN

6 wk persistent motion (proximal ⅓) **surgery now**

Gap predicts mobility	**each mm of gap ↑ mobility risk 40%**; **gap, smoking, female sex → 8 mm gap at 8 wk → ORIF plate** ✗ Wrong options: stimulator · pinning
RUSHU	score per cortex at 6 wk (1 absent · 2 non-bridging · 3 bridging); **≥ 8 → 86% NPV** · < 8 → 65% PPV
Nonunion risk by pattern (nonop)	**OTA A (15–29%) > B (4%) > C (0%)**
Nonunion treatment	**plate + autograft > exchange nailing** · atrophic: debride + compression plate + graft (ICBG, **RIA**) or exchange nail (**ream ≥ 1 mm**) · hypertrophic: exchange nail / locked plate
Malunion	varus common, rarely matters; **transverse** patterns

X vs Y

	Anterior approach	**Triceps-split**	**Paratricipital**
Plane	**Brachialis split: radial / musculocutaneous (internervous)**	**No internervous plane** (all radial)	Not intramuscular
Exposure	Middle ⅓	**55%**	**94%**

	Neurapraxia	**Axonotmesis**	**Neurotmesis**
Typical cause	**Closed fracture**	**Low-velocity GSW**	**Open high-energy**
Recovery	Spontaneous	Regenerates (Wallerian)	**Needs repair**

How it's asked

"Maximum acceptable deformity (TO)?"	**→ shortening < 3 cm, rotation < 30°, angulation < 20°**
"Acceptable – EXCEPT (TO)?"	**→ angulation < 30°**
"NOT a contraindication to bracing?"	**→ closed mid-shaft fracture with radial nerve palsy**
"9 days, normal exam, acceptable?"	**→ functional brace 6–8 wk**
"Contraindication to functional brace?"	**→ brachial plexus injury**
"Closed fracture, normal neuro?"	**→ functional bracing**
"18-month-old humeral shaft?"	**→ coaptation splint + bandage to thorax**
"Thin 23-yr-old, transverse?"	**→ coaptation splint → fracture brace** ✗ Wrong options: hanging cast
"Arm wrestling + wrist drop?"	**→ splint → functional brace**
"4 weeks, no callus, in brace?"	**→ continue brace, review 4 wk**
"25° varus persists – effect?"	**→ no effect on function**
"15° coronal, 18° sagittal, isolated?"	**→ functional brace**
"Distal shaft in Sarmiento – likely deformity?"	**→ varus**

– – Active space – –

"NOT an operative indication (TO)?"	→ closed fracture with radial nerve palsy
"Strongest surgical indication?"	→ ipsilateral both-bone forearm fracture
"Weight bearing: plateau + humerus ORIF + femur IMN?"	→ NWB plateau leg, WBAT arm + other leg
"DCO with multiple GSWs?"	→ closed reduction + splint in ED
"Absolute indication?"	→ floating elbow
"Flaccid limb, C5–T1 avulsion?"	→ plate fixation
"Polytrauma humerus + femur?"	→ plate humerus, IMN femur
"Floating elbow?"	→ ORIF both + early motion
"6 weeks persistent motion?"	→ surgery now
"Humerus with tibia nail + ankle – plan?"	→ plate + full WB
"Teen quarterback wants early return?"	→ surgical fixation
"Plate vs nail – plate has?"	→ lower shoulder pain
"IMN vs plate – EXCEPT?"	→ increased shoulder ROM
"IMN vs ORIF – accurate?"	→ higher total complication rate
"Antegrade nail – true?"	→ higher shoulder impingement
"IMN vs ORIF at 1 year?"	→ equal functional shoulder outcomes
"Contemporary IMN vs ORIF?"	→ IMN lower infection
"Complete RNP – plan (TO)?"	→ hanging cast, explore if no recovery at 3 months
"4 months, healed, palsy unchanged?"	→ EMG + NCS
"Last to recover?"	→ index MCP hyperextension (EIP)
"Loss of ECRL→EPL, open fracture + brachial artery?"	→ radial nerve neurotmesis
"Low-velocity GSW + palsy – true?"	→ splint + observe
"6 weeks, EMG no recovery?"	→ observation
"Open comminuted fracture + palsy?"	→ ORIF + radial nerve exploration
"First muscle to recover – insertion?"	→ radial styloid (brachioradialis)
"Last to return (figure)?"	→ EIP
"Highest radial nerve palsy risk pattern?"	→ distal ⅓ spiral (Holstein-Lewis)
"Palsy persists 2 months after ORIF?"	→ NCS / EMG
"12 months, no wrist extension?"	→ PT → ECRB
"First to recover after post-op palsy?"	→ brachioradialis / ECRL
"A-OK intact, wrist drop, dorsal numbness?"	→ Holstein-Lewis fracture
"Recovery by 7 weeks?"	→ 70%

"First muscle crossing the wrist to recover?"	→ **inserts on base of 2nd MC (ECRL)**
"Distal A-P interlock risks?"	→ **musculocutaneous nerve**
"Radial nerve enters anterior compartment?"	→ **10 cm proximal to radiocapitellar joint**
"Posterior antebrachial cutaneous traces to?"	→ **radial nerve**
"Greatest exposure?"	→ **posterior with elevation / medial reflection of lateral + medial heads**
"Posterior approach — nerve at risk?"	→ **radial**
"A-P distal interlock injury — symptom nerve?"	→ **lateral antebrachial cutaneous**
"Anterior approach deep interval?"	→ **musculocutaneous + radial**
"Intramuscular internervous approach?"	→ **anterior (brachialis split)**
"Gerwin modification — landmark?"	→ **inferior lateral brachial cutaneous nerve**
"Ex-fix distal pins?"	→ **no safe zone — mini-open**
"When does no callus + motion become concerning?"	→ **6 weeks**
"8 mm gap at 8 weeks?"	→ **ORIF with plate**
"Specificity of motion at 6 weeks?"	→ **99%**

– – Active space – –

HUMERAL SHAFT NONUNION

1 Basics

Epidemiology

Definition failure to heal by **6 months** (OB) · bone-stimulator criteria: **> 9 months**, or **no progress over 3 months**

Primary nonunion **nonop 2–33%** (Orthobullets, OB) · 2–10% (OB) · **surgery 5–10%** · ~10% either (OB)

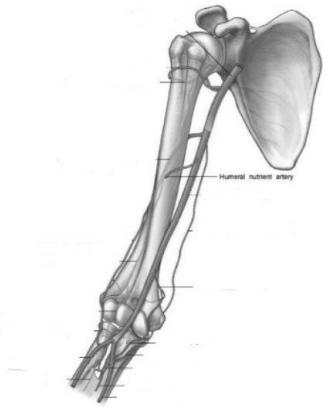
Location **proximal ⅓** highest (pectoralis major + deltoid deforming forces; **biceps tendon interposition**)

Risk factor	Type
Vitamin D deficiency (commonest metabolic), osteoporosis	Biological
Infection (incl. **P. acnes**)	Biological
Smoking, obesity, malnutrition, **NSAIDs**, alcohol, noncompliance	Patient
Open fracture, segmental	Injury
Unstable pattern, inadequate fixation	Mechanical
Shoulder / elbow stiffness (motion → fracture site), GH arthritis	Mechanical

Not predictive radial nerve palsy, no callus at **3 weeks**, hand dominance

Anatomy

Nutrient artery **medial** aspect of mid–distal ⅓ diaphysis



Nutrient artery of the humerus

Genes & lab markers

Metabolic work-up – vitamin D deficiency **most common** correctable abnormality → correct before surgery
✗ Wrong options: hypogonadism · thyroid · PTH · calcium · iron · magnesium

Abnormality (nonunions, OB)	Attributed share
Calcium deficiency	35%
Thyroid disorder	24%
Central hypogonadism	15%
Parathyroid disorder	11%

Labs **CRP, ESR, CBC** (infection) · **total protein, albumin** · **vitamin D, TSH, PTH**

3 Presentation

Symptoms pain with use; **gross motion** at fracture

Exam brace fit, skin, atrophy, angulation; **radial nerve function**

4 Diagnosis & Classification

Key radiology findings

Study	Use
AP + lateral humerus, shoulder, elbow	No consolidation, **hypertrophic callus**, **pseudarthrosis**, broken hardware
CT	**Extent of bridging**, planning; **not needed if X-ray already diagnostic**

Predictors of nonunion

Most predictive **persistent fracture mobility at 6 weeks** (extremely sensitive + specific)

Infection work-up

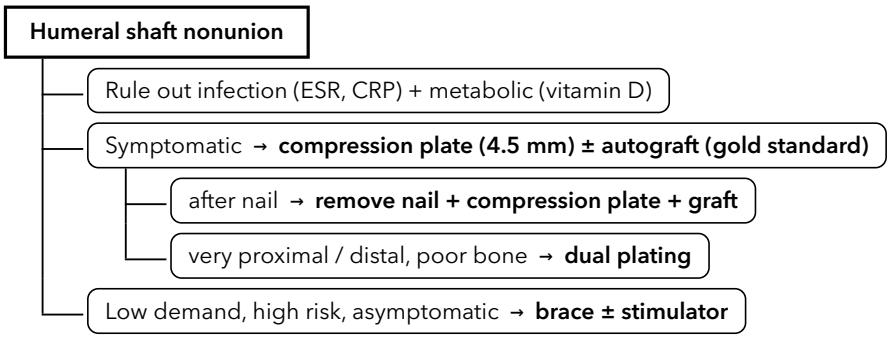
Before any revision **ESR + CRP (labs)** · **3–5 cultures** for atrophic · septic → **two-stage**
✗ Wrong options: CT · brace · revision without infection work-up

Septic nonunion with bone loss resect + **antibiotic spacer**

Types **hypertrophic, oligotrophic, atrophic** (atrophic ↑ with bone loss, open fracture, infection)

– – Active space – –

5 Management



Treatment options

Option	Indication	Fact
Brace ± **bone stimulator**	**Low demand, high risk, asymptomatic**	Stimulator not if **pseudarthrosis, gap > 5 mm, poor blood supply**
Compression plating + bone graft	**Symptomatic nonunion – gold standard**	**Superior to IMN**; ~100% union; **ICBG donor morbidity 44%**
Dual plating (orthogonal)	**Very proximal / distal**, poor metaphyseal bone, micromotion after one plate	**92–100% union at 16 wk**
Cortical strut allo- / autograft (intramedullary or medial + lateral plate)	**Severe osteopenia**, bone loss, recalcitrant	**95–100%** union
Free fibula	**Recalcitrant atrophic**	–
Induced membrane / fibula / cancellous allograft	**Bone loss**	–
BMP	**Limited role** – no proven benefit	–

Prognosis **83–100%** union after surgery · plating ± graft or IMN + graft **> 85%**

Compression plating with bone graft (after nail or brace)

Technique anterior / anterolateral / **posterior (radial nerve neurolysis)**; debride fibrous tissue + bone ends; **maximal cortical contact**; **4.5 mm compression plate**

Graft **ICBG autograft if atrophic** · DBM, **RIA**, local callus · **graft in hypertrophic – controversial**

After failed IMN **nail removal + plate + graft**: **9/9 union vs exchange nailing 4/10 (McKee)**
✗ Wrong options: exchange nailing · dynamisation · MUA · nail removal + brace

Atrophic after nail, smoker **nail removal + compression plate + bone graft**
✗ Wrong options: augmentative plate (**hypertrophic only**) · plate without graft · smoking cessation alone

After failed ORIF (screw pull-out), normal ESR / CRP **compression plating + autograft**
✗ Wrong options: spacer · ex-fix · brace · retain implants

Delayed union in brace at 16 wk (atrophic)	**ORIF + autograft** ✗ Wrong options: stimulator + sling · IMN (nonunion + cuff damage) · locked plate + allograft
Transverse, 6 months, gross motion	**ORIF + autologous graft** ✗ Wrong options: ultrasound · bisphosphonates · nails

6 Complications

Postoperative nerve deficit with nonunion

Commonest **radial nerve** palsy after nonunion surgery — **iatrogenic 7–19%**

Procedure / pathology (OB)	Typical nerve palsy
Humeral nonunion fixation	**Radial**
Reverse TSA	Axillary
Hook of hamate fracture	Ulnar
Medial-entry supracondylar pins	Ulnar
Lunate dislocation	Median (acute CTS)

X vs Y

	Hypertrophic	Atrophic
Problem	**Stability**	**Biology**
Treatment	Compression plate (± augment plate); graft controversial	**Compression plate + ICBG**; resect necrotic bone ± shorten

	Plate + graft	Exchange nail
Union after failed nail	**9/9**	4/10
Preferred	**Yes**	No (cuff damage)

How it's asked

"Painful nonunion 13 months after IMN?"	→ **nail removal + plate fixation**
"Hypertrophic nonunion after nail, 9 months?"	→ **nail removal, ORIF + bone graft**
"Transverse, 6 months, gross motion?"	→ **ORIF + autologous bone graft**
"16 weeks in brace, gross motion?"	→ **open plating + autograft**
"Atrophic after nail, smoker?"	→ **nail removal + compression plate + bone graft**
"Nonunion after ORIF, normal markers?"	→ **compression plating + autograft**
"Metabolic cause of nonunion?"	→ **vitamin D deficiency**
"Broken screw 8 months after nonunion surgery — next?"	→ **laboratory evaluation (ESR, CRP)**
"Most predictive of nonunion?"	→ **persistent mobility at 6 weeks**

– – Active space – –

"Nerve palsy after humeral nonunion surgery?" → **radial nerve**

INTERTROCHANTERIC FRACTURES

– Trauma · 33 / 67 –

1 Basics

Epidemiology

Share	**~50% of hip fractures** · 150,000 / yr (US) · F 500 vs M 200 / 100,000 / yr
Who	average **~80 yr** (older than neck fractures) · **F:M 2:1 to 8:1**
Risk factors	osteoporosis, age, comorbidities, ADL dependency, recurrent falls (dementia, Parkinson's, vision, drugs)
Cumming's four factors (TO)	faller **orientation** at impact · **soft-tissue padding** over hip · adequacy of **protective responses** · **residual bone strength**

Osteoporosis – sentinel fractures & bisphosphonates

Sentinel fracture	**proximal humerus fracture → ↑ hip fracture risk (for 1 yr)** ✗ Wrong options: sacral · radius · fibula · distal humerus
Early bisphosphonate after fragility hip fracture	**greatest BMD gain in first year**; **no ↑ nonunion, delayed union or peri-implant fracture**; oral ≈ IV

Anatomy

Neck-shaft	**130 ± 7°** · anteversion **10 ± 7°**
Calcar femorale	dense vertical wall from **posteromedial shaft → posterior neck**; its involvement decides stability

Calcar = Adams arch (TO) · capsule: anterior at **intertrochanteric line**, posterior ends **~1.5 inch proximal to crest** → IT fracture **extracapsular**

Femoral radius of curvature	**114–120 cm**; **smaller (more bowed)** in **elderly, Asian, short stature**

Fragment	Deformity	Muscles
Proximal	**Flexion**	**Iliopsoas**, sartorius, rectus, pectineus
Proximal	**External rotation**	Short external rotators
Proximal	**Abduction**	**Gluteus medius + minimus**, maximus, TFL
Distal	**Adduction + shortening → varus**	Adductors, gracilis

Blood supply	rich (**trochanteric anastomosis**: MFCA + LFCA ascending branches, superior + inferior gluteal) → **nonunion < 2%**
Trabeculae	**Ward's triangle** = lowest BMD, bounded by primary compressive, principal tensile, secondary compressive trabeculae

3 Presentation

Exam	**shortened, externally rotated**, GT tenderness, pain on log-roll, **no straight-leg raise**

– – Active space – –

History pre-injury function (predicts outcome), antecedent pain (OA / pathological), **anticoagulants**, ASA

4 Diagnosis & Classification

Key radiology findings

Study	Use / fact
AP pelvis, AP hip, **cross-table lateral**, **full-length femur**	Pattern; **posterior comminution** (lateral); subtroch extension, lesions, nail length, **bow**, canal
Traction internal-rotation view	**Better classification**, assess reduction
MRI	**Occult fracture (best; T1 100% sensitive, T2 84%)** · **isolated GT fracture → MRI for IT extension** (✗ Wrong options: CT · traction view · fix first)
CT (1–2 mm)	**Second line** if no MRI: sens 86%, spec 98%
Bone scan	Rare; **false negative up to 72 h**; sens up to 98%

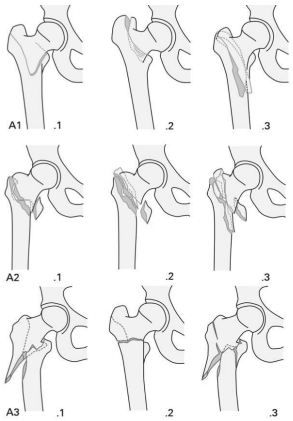
MRI findings marrow oedema on **STIR / fat-suppressed T2**; low-signal **T1 line**

Stability classification

	Stable	Unstable
Definition	**Intact posteromedial cortex** → resists medial compression	Collapses into **varus** / shaft **medialises**
Examples	2-part; small LT fragment without medial comminution	**Large / comminuted posteromedial (LT) fragment** · **reverse obliquity (transtrochanteric)** · **subtrochanteric extension** · **lateral wall < 20.5 mm**

Lateral wall thickness measured **3 cm below innominate tubercle at 135°** to fracture line; **< 20.5 mm → postop lateral wall fracture → CMN, not SHS**; 25 mm = **not** a risk

AO / OTA 31A	Description
A1	Simple 2-part pertrochanteric, **intact lateral cortex**
A2	Separate **posteromedial fragment**, **intact lateral cortex**
A3	Through **lateral + medial cortex** (reverse oblique / transverse)



AO/OTA 31A classification

Evans	Description
Stable	Posteromedial cortex intact / minimal comminution
Unstable	Greater posteromedial comminution — **convertible to stable with medial cortical opposition**
Reverse obliquity	Unstable — **adductors medialise shaft**

Boyd & Griffin (TO)	Pattern
I	Stable two-part
II	Unstable, comminuted
III	**Reverse oblique**
IV	IT + **subtrochanteric extension**

Jensen modification of Evans (TO) type 3 **GT** fragment · type 4 **LT** fragment · type 5 **both** · **type R = reverse oblique**

AO 2018 revision (TO) **31A1 = simple, lateral wall > 20.5 mm · 31A2 = multifragmentary, lateral wall < 20.5 mm · 31A3 = reverse oblique**

CT (TO) lateral wall thickness, posteromedial comminution, reverse obliquity, subtroch extension



Reverse obliquity intertrochanteric fracture

– – Active space – –

Reduction quality – cortical support

Medial cortical support (TO)	Proximal fragment
Positive – most acceptable	**Medial** to shaft cortex
Neutral	Anatomical
Negative	**Lateral** – collapses

5 Management

Intertrochanteric fracture

- Stable (intact posteromedial + lateral wall ≥ 20.5 mm) → **SHS or CMN (equal outcomes)**
- Unstable / reverse obliquity / subtroch extension / thin lateral wall → **CMN**
  - subtroch extension ≤ 3–4 cm → short CMN · more distal → **long CMN**
- Comminuted + severe OA / bone won't hold → **calcar-replacing arthroplasty**

Treatment options

Option	Indication	Fact
Nonop (NWB, bed → chair)	**Non-ambulatory**, too high risk, skin breakdown, incomplete fracture	**1-yr mortality 84.4%**; pneumonia, UTI, sores, DVT
SHS (most common ORIF)	**Stable**	**Cheapest**, **spares abductors**, dynamic compression; **2-hole side plate = 4-hole**
CMN	**Stable or unstable – AAOS strong for unstable, reverse oblique, subtroch extension**	Load-sharing, **shorter lever arm**, **intramedullary buttress limits medialisation**, ↓ / = blood loss; **violates abductors**; dearer; **↑ use over last decade**
Proximal femoral locking plate	Young + unstable (rare)	Surgeon-dependent; **no early WB**
95° blade plate / DCS	Rare	Acceptable for reverse oblique
Arthroplasty (long-stem, calcar-replacing)	Salvage, severe comminution, **pre-existing OA**, very osteoporotic	Fix GT; more blood loss, time, cost

Stable vs unstable patterns – implant choice (SHS vs cephalomedullary nail, reverse obliquity)

Stable	**SHS = CMN outcomes**; **reduction quality > implant choice**
Reverse obliquity	**NOT SHS (56% failure)** → shear, **shaft medialisation + cut-out** → **CMN** (trochanteric or piriformis), **blade plate, DCS, locking plate, ex-fix** all acceptable
SHS advantages over CMN	**none** for DVT, blood loss, nonunion, biomechanics ✗ Wrong options: all

CMN preferred because — **stability for the fracture pattern**
✗ Wrong options: shorter LOS · less abductor damage · easier THA conversion · cheaper

Most cost-effective (stable) — **SHS**
✗ Wrong options: cannulated screws · hemi

Spare abductors — **SHS**
✗ Wrong options: lateral position · lag screw vs blade · omitting distal lock

Short vs long CMN — equal function, reoperation, peri-implant fracture, cut-out, mortality · **long: ↑ OR time, ↑ blood loss, ↑ postop anaemia + transfusion**, radiation, bow mismatch · **short tolerates 3–4 cm subtroch extension**

Distal locking screw — **protective against peri-implant fracture** (omitting ↑ refracture)

SHS contraindicated — reverse oblique, **lateral wall comminution / < 20.5 mm**, **subtrochanteric fractures**

Richard (DHS) screw (TO)	Value
Thread / core diameter	**14 mm / 8 mm**
Thread length	22 mm
Barrel	**25 mm (short) / 38 mm (long)**
Barrel rule	Lag screw **> 85 mm → long barrel** · < 85 mm → short barrel

DHS + thin lateral wall (TO) — add **trochanteric stabilisation plate (TSP)** + short barrel + 5-hole side plate · lag screw **centre-centre**

Pattern (TO)	Preferred implant
Stable, good lateral wall	**DHS**
Lateral wall incompetent	**PFN** or **DHS + TSP**
Reverse oblique	**PFN** (blade plate acceptable)
Posteromedial comminution	PFN or valgus osteotomy
Subtroch extension	**Long PFN + cerclage**
Young, narrow canal	Proximal femoral locking plate
Elderly + arthritis	Arthroplasty (fix GT)

Medoff plate (TO) — **biaxial dynamisation**; A3 fractures

Cement-augmented PFN (TO) — fenestrated screws, severe osteoporosis

Implant history (TO) — Jewett nail → sliding nail → DHS → condylocephalic (Ender) → Gamma → **PFN (current gold standard)**

Isolated GT / LT fracture (TO) — usually nonop · **isolated LT avulsion in adult → suspect tumour** → fix / work-up

Tip-apex distance & cut-out

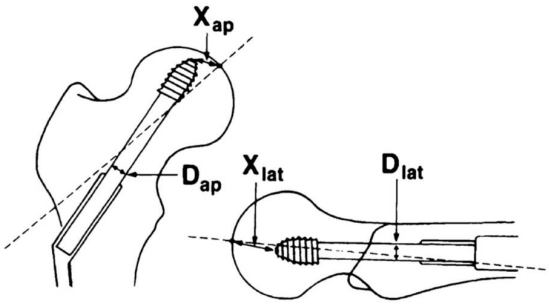
TAD — **sum of tip → apex distances on AP + lateral**, corrected for magnification

Goal — **TAD < 25 mm** · **> 45 mm → 60% failure** · 27 / 32 / 40 mm → ↑ **cut-out**

Parker ratio (TO) — lag-screw position along head: AP **0 inferior → 100 superior** · lateral **0 posterior → 100 anterior**

– – Active space – –

Garden alignment index – IT (TO)	AP **160–180°** · lateral **180°**
Strongest predictor of cut-out	**TAD** ✗ Wrong options: age · stability · reduction · plate angle
Lag screw position	**centre–centre (TO)**; **low-central AP + central lateral** acceptable; **never peripheral / posterior** (head rotates → superior cut-out) ✗ Wrong options: posterior on lateral
Lag screw vs helical blade	controversial · blade compacts bone, ↑ cut-out resistance (biomechanical), but **much higher "cut-through"**



$$TAD = \left(X_{ap} \times \frac{D_{true}}{D_{ap}}\right) + \left(X_{lat} \times \frac{D_{true}}{D_{lat}}\right)$$

Tip-apex distance

Failure	Definition
Cut-out	**Anterocranial** perforation **with varus collapse**
Cut-through	**Central** perforation **without collapse** – **helical blade**

Lateral wall integrity & sliding hip screw failure

Intra-op lateral wall fracture	**↑ collapse / sliding** ✗ Wrong options: nail length · WB status
Predictor of postop lateral wall fracture with SHS	**lateral wall thickness** ✗ Wrong options: reverse obliquity · T-score · calcar comminution
Highest SHS failure	**wall 18.5 mm + TAD 28.5 mm**
DHS failure – EXCEPT	**2-hole side plate** ✗ Wrong options: reverse oblique · TAD 27 / 32 · lateral wall comminution
Left-sided unstable fracture + DHS	**anterior spike** (screw torque) → **derotation wire** first
SHS problems in unstable fractures	excessive collapse, **limb shortening**, shaft medialisation

Anterior cortical perforation & nail technique

Cause	**radius-of-curvature mismatch (nail straighter / larger radius than bowed femur)** + **posterior start on GT** ✗ Wrong options: anterior start · large implant · lateral position
Prevent	lateral view of distal femur while seating · avoid posterior start
Too-lateral start (> 3 mm)	**varus** → collapse + cut-out

Guidewire stuck in nail (no fragment)	**remove nail, check guidewire bend** ✗ Wrong options: insert lag wire · re-ream · smaller nail · tighten jig
PFN-A2 vs PFN-A (TO)	**all**: **less lateral cortex pressure (flat lateral surface)**, **smaller proximal diameter (16.5 vs 17.0 mm)**, **smaller blade**; higher bend point, **5° ML bend** – for **Asian femora**
PFN types (TO)	conventional PFN (**2 screws**: lag + anti-rotation) → **Z / reverse Z** · **PFN-A** (single **helical blade**, unreamed, 2003)

6 Complications

Complication	Fact
Cut-out / implant failure	**Commonest after CMN**; **4–20%**, within **4 months** (Orthobullets) · 3 months (OB)
Nonunion / malunion	**< 2%**; diagnosis: pain + lucency **4–7 months**; rule out infection
Peri-implant fracture	**1–3% at 1 yr**; short = long; short → **at nail tip**, long → around rod
Anterior perforation	Long CMN, bow mismatch, posterior start
Anaemia	**> 30% transfused**; threshold **≤ 8 g/dL** if asymptomatic; **TXA (strong)**

Failed fixation – conversion to THA

Z effect (TO)	**lag screw migrates laterally**, varus collapse, **derotation screw perforates head**
Reverse Z effect (TO)	**derotation screw migrates laterally + lag screw migrates medially** (TO case answer)
Z effect predisposition (TO)	**two-screw PFN** + **posteromedial comminution**
Cause	unstable calcar, **deficient lateral buttress**, no medial support; head vs neck compressive-strength mismatch

Failure	Treatment
Elderly / joint damaged by cut-out	**THA** (✗ Wrong options: NWB · valgus osteotomy · revision ORIF · resection)
Malunion + hardware into acetabulum	**THA with calcar-replacing stem bypassing hardware** (✗ Wrong options: bipolar (acetabular erosion))
Active 72-yr, cut-out + acetabular erosion	**Conversion THA** (✗ Wrong options: exchange CMN · osteotomy · proximal femoral replacement · hemi)
Young / intact head	**Valgus osteotomy ± graft** / revision ORIF

Mortality, surgical timing & perioperative care

1-yr mortality	**15–30%** (Orthobullets) · 20–30% (OB) · ~10% at 1 mo, 20% at 4 mo, **30% at 1 yr**; **nonop 84.4%**
↑ Mortality	**male (25–30% vs 20%)**, **IT > neck fracture**, **age > 85**, **≥ 2 comorbidities**, **ASA III–IV**, delay **> 48 h** (> 2 days, Orthobullets · ≥ 3 days, OB)
Recognised predictor	**ASA class** ✗ Wrong options: WB status · comminution · device · anaesthetic

– – Active space – –

Not associated with ↑ mortality	**surgery within 48 h**, **spinal vs general (REGAIN)**, nail length, interdisciplinary care (↓)
Delay > 48 h	**↑ 30-day mortality**, **↑ pneumonia** ✗ Wrong options: OR time · transfusion · infection
Timing	**24–48 h** (AAOS moderate) · **co-management** (AAOS strong)
Dementia patient	**geriatric co-management, avoid centrally-acting drugs (meperidine)**, rapid optimisation, **surgery < 48 h** ✗ Wrong options: neurology consult · routine echo · nephrology for normal CrCl (low muscle mass masks renal impairment)
Routine echo	**delays surgery** without changing outcome (unless valve disease, arrhythmia, worsening CHF, CAD) ✗ Wrong options: hospitalist · fascia iliaca block · Foley · fluids cause delay
DAPT (aspirin + clopidogrel)	**continue, proceed when optimised** ✗ Wrong options: vitamin K (warfarin) · idarucizumab (dabigatran) · FFP · stop + wait 48 h
Function at 1 yr	**41%** regain status · 40% more aids · 12% household · 8% non-functional · **rule of thirds**: ⅓ regain, ⅓ lose a level, ⅓ die
Post-op	**WBAT** ↓ complications

X vs Y

	SHS	CMN
Stable fractures	**Equal**	**Equal**
Unstable / reverse oblique / subtroch	No	**Yes**
Cost	**Lower**	Higher
Abductors	**Spared**	Violated
Lever arm	Longer (extramedullary)	**Shorter**
THA conversion	**Easier**	Harder

	Short CMN	Long CMN
Blood loss / transfusion	Lower	**Higher (anaemia)**
Peri-implant fracture	Equal (at tip)	Equal (around rod)
Anterior perforation	–	**Risk**

How it's asked

"NOT appropriate for reverse obliquity?"	**→ sliding hip screw**
"55-yr MVA, reverse obliquity?"	**→ cephalomedullary nail**
"Adequate fixation EXCEPT (reverse oblique)?"	**→ dynamic hip screw**
"IMN vs SHS – true?"	**→ IMN use has increased in last 10 years**
"SHS advantage over CMN?"	**→ none of the above**

"Most cost-effective implant?"	→ **sliding hip screw**
"Stable fracture — true?"	→ **no difference between SHS and IMN**
"Why CMN over SHS?"	→ **improved stability for fracture pattern**
"SHS appropriate for?"	→ **stable IT + femoral neck stress fracture**
"2 cm subtroch extension, cardiac patient?"	→ **short CMN**
"Reduce abductor injury?"	→ **sliding hip screw**
"Long vs short nail?"	→ **↑ postoperative anaemia**
"Optimal lag screw position (TO)?"	→ **central AP + central lateral**
"Strongest predictor of cut-out?"	→ **tip-apex distance**
"TAD 40 mm, superior screw?"	→ **cut-out**
"How is TAD measured?"	→ **sum of tip–apex on AP + lateral**
"Commonest CMN complication?"	→ **implant cut-out**
"Failure commoner with helical blade?"	→ **central perforation without collapse (cut-through)**
"↑ collapse / sliding?"	→ **intra-op lateral wall fracture**
"Minimum side-plate holes (stable 2-part)?"	→ **two**
"Left unstable fracture + DHS deformity?"	→ **anterior spike**
"DHS failure — EXCEPT?"	→ **2-hole side plate**
"Predictor of postop lateral wall fracture?"	→ **lateral wall thickness**
"Lateral wall 25 mm risk?"	→ **none of the above**
"Highest SHS failure combination?"	→ **wall 18.5 mm + TAD 28.5 mm**
"SHS inadvisable if?"	→ **lateral wall 20 mm**
"PFN-A2 vs PFN-A (TO)?"	→ **all of the above**
"Anterior perforation from antegrade nailing?"	→ **radius-of-curvature mismatch**
"Anterior perforation risk factors?"	→ **mismatch + posterior start**
"Guidewire won't come out?"	→ **remove nail, assess wire bend**
"Nail radius > femur?"	→ **anterior cortical perforation**
"PFN failure — screws migrating opposite (TO)?"	→ **reverse Z effect**
"Lateral wall thickness defining unstable IT fracture?"	→ **< 20.5 mm** (TO)
"DHS lag screw 90 mm — barrel?"	→ **long (38 mm)** (TO)
"Failed SHS, 82-yr?"	→ **THA**
"Malunion + hardware into acetabulum?"	→ **THA with calcar replacement bypassing hardware**

– – Active space – –

"Cut-out + acetabular erosion, active 72-yr?"	→ **conversion THA**
"Predictor of mortality?"	→ **ASA classification**
"↑ 1-yr mortality EXCEPT?"	→ **fixation within 48 hours**
"Dementia + IT fracture — plan?"	→ **geriatric co-management, avoid central drugs, surgery < 48 h**
"Delay > 48 h causes?"	→ **↑ 30-day mortality**
"Aspirin + clopidogrel?"	→ **no change, proceed when optimised**
"What delays surgery?"	→ **pre-op echocardiogram**
"Negatively affects mortality?"	→ **delay beyond 48 hours**
"Isolated GT fracture — next?"	→ **MRI**
"Fracture most increasing later fracture risk?"	→ **hip fracture**
"Early bisphosphonate — true?"	→ **greatest BMD gain within first year**
"Most acceptable cortical support reduction (TO)?"	→ **positive**

ISOLATED ULNAR SHAFT FRACTURE

– Trauma · 34 / 67 –

1 Basics

Epidemiology

Rare	**< 1% of upper-limb fractures** · men 10–20 yr, women > 60 yr

Associated conditions

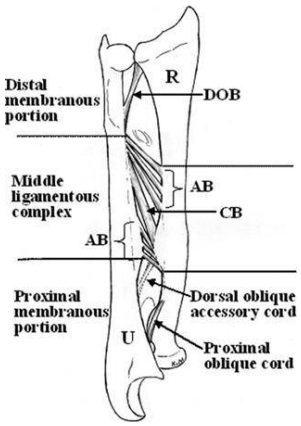
Elbow / DRUJ	**Monteggia**, **Essex-Lopresti** → image wrist + elbow
Compartment syndrome	up to **15%**

Atypical (bisphosphonate-associated) fracture

Pattern	**isolated transverse proximal ⅓ ulna**, **thickened posterior cortex**, minimal comminution, prodromal ache → **alendronate** ✗ Wrong options: teriparatide · semaglutide · levothyroxine · methotrexate
Location	**20–45% of ulnar length from olecranon tip** · ASBMR major / minor criteria (as for atypical subtrochanteric)
Outcome	**delayed / nonunion ~50%** → resect diseased bone + plate

Anatomy

Forearm axis	**radial head (proximal) → ulnar fovea (distal)**; radius rotates around ulna; ulnar malangulation → ↓ rotation
Interosseous membrane	**5 ligaments**; **central band = key part to reconstruct** · accessory band, distal oblique bundle, proximal oblique cord, dorsal oblique accessory cord



Interosseous membrane

2 Mechanism

Direct blow ("nightstick") · indirect: MVA, fall from height (axial load)

– – Active space – –

3 Presentation

Exam deformity, open wounds, **tense compartments**, **pain on passive finger stretch**; radial + ulnar pulses; median, radial, ulnar nerves

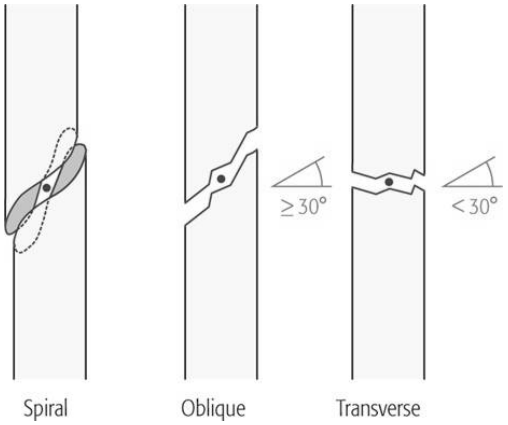
4 Diagnosis & Classification

Key radiology findings

Views **AP + lateral forearm** ± obliques + **ipsilateral wrist + elbow**

Radial head aligned with capitellum on every view excludes Monteggia

OTA (ulna shaft)	Subtypes
A – simple	A1 spiral · A2 oblique · A3 transverse
B – wedge	B2 intact · B3 fragmentary
C – multifragmentary	C2 intact segmental · C3 fragmentary



OTA type A ulnar shaft fractures

5 Management

Isolated ulnar shaft fracture

- Distal ⅔, < 50% displacement + < 10° angulation → **brace / Muenster cast / soft dressing**
- Displaced distal ⅔ or any proximal ⅓ → **ORIF (3.5 mm DCP, 6 cortices each side)**
- Gustilo I–IIIA open → primary ORIF · IIIB → **ex-fix**
- Bone loss / nonunion → **autograft, vascularised fibula, Masquelet**

Operative criteria

Nonoperative	Operative
Distal ⅔ (nightstick)	**Displaced distal ⅔**
< 50% displacement	**Proximal ⅓**
< 10° angulation	**Open** (I–IIIA primary ORIF; IIIB ex-fix)
e.g. **20% translation, < 5° angulation**	**Any adult radial shaft or both-bone fracture** (even undisplaced)

Nonoperative management — outcomes & acceptable displacement

Minimally displaced — **equivalent clinical outcomes** to surgery; **slower union**; union **> 92%**
✗ Wrong options: long-arm cast needed · 20° rotation loss · wrist / elbow stiffness

> 10° angulation or > 50% displacement — **loss of forearm rotation** if treated nonop

Options — **functional brace**, **Muenster cast** (interosseous mould), **soft compression + immediate motion** – all similar; **above- vs below-elbow = no difference**; review at ~1 wk; ≥ 6 wk



Muenster cast

Operative technique

ORIF — **subcutaneous approach**; **3.5 mm DCP**; compression mode; **6 cortices** above + below; locking in osteoporosis; bridge if comminuted; 2.0 / 2.7 lag screws
✗ Wrong options: **4.5 mm — refracture after removal**

Post-op — **early ROM**; **NWB ~6 wk** (secondary displacement)

Ex-fix — **Gustilo IIIB**; distal pins in **2nd + 3rd MC**, **≤ 4 mm** pins

IMN — **poor soft tissue**; through **posterior olecranon**; union ≈ ORIF; **7% nonunion** forearm nailing (Street)

Grafting — open fracture with bone loss (delayed) / nonunion; **autograft is critical** (ICBG, proximal tibia, **RIA**); **vascularised fibula — large defects, ↓ infection**

Masquelet (induced membrane) — stage 1: I&D + **cement spacer** + temporary fixation → stage 2: graft into membrane at **4–6 wk**

– – Active space – –

6 Complications

Complication	Fact
Infection	**3%** after ORIF; open fractures
Compartment syndrome	**Up to 15%**; crush, open, **low-velocity GSW**, vascular, **DIC**
Nonunion	**< 5%** compression plate · **up to 12%** bridge plating of comminution
Malunion	**Radial bow restoration ∝ function**
NV injury	**PIN with Monteggia**, type III open → **observe 3 months**, explore if no return
Refracture	**Up to 10%** with early removal; **remove ≥ 1 yr** (ulna, Orthobullets) · ≥ 18 months (Galeazzi note); 4.5 mm plates, comminution, lucency → **brace 6 wk + protect 3 months** after removal

Nonunion

Type	Treatment
Atrophic	**ORIF (3.5 mm plate) + autogenous cancellous graft** (✗ Wrong options: dynamic splint · graft alone · pinning · ultrasound)
Infected / atrophic with defect	**Masquelet**
Hypertrophic	**Debridement + compression plating**

X vs Y

	Isolated ulna (nightstick)	Radius shaft / both-bone (adult)
Minimal displacement	**Nonop OK**	**ORIF always**
Risk if nonop	Little	Displacement, malunion, ↓ rotation

	Nightstick	Atypical (bisphosphonate)
Site	Distal ⅔	**Proximal ⅓**
Pattern	Direct blow	**Transverse + thick posterior cortex**, prodrome
Union	> 92%	**~50% delayed / nonunion**

How it's asked

"Nonop vs operative – true?"	**→ equivalent clinical outcomes**
"Which forearm fracture can be nonop in adult?"	**→ isolated mid-shaft ulna, 20% translation, < 5° angulation**
"Atrophic nonunion after nonop at 1 year?"	**→ ORIF + autogenous bone graft**
"Proximal ⅓ transverse ulna + thick posterior cortex?"	**→ alendronate**

KNEE DISLOCATION

– Trauma · 35 / 67 –

1 Basics

Epidemiology

Incidence	**0.02%** of orthopaedic injuries; under-reported – **~50% self-reduce** before ED
Who	**M:F 4:1** · **morbid obesity → "ultra-low-energy" dislocation** with ADLs + ↑ neurovascular injury
Mechanism	high energy (MVC, **dashboard → axial load on flexed knee**, fall) · low energy (sport, walking) · **hyperextension → anterior** · **posterior force on proximal tibia → posterior**
Associated	**vascular**, **common peroneal nerve (~25%)**, **fractures 60%**, patellar tendon rupture, avulsions, displaced menisci

Anatomy

Ligaments at risk	ACL, PCL, MCL, LCL, PLC (bony stability minimal)
Popliteal artery tethered	**proximally at adductor hiatus** (fibrous tunnel) + **distally at soleus arch**
Geniculate collaterals	can give **palpable pulses despite complete popliteal occlusion**

KD classification & cruciate injury pattern

Kennedy (tibial direction)	Share	Key fact
Anterior	**30–50% – most common**	**Hyperextension**; vascular injury = **intimal tear (traction)**
Posterior	30–40% – 2nd	**Highest vascular injury → complete popliteal artery tear**
Lateral	13%	Varus / valgus
Medial	3%	Varus / valgus
Rotational (posterolateral)	4%	**Usually irreducible** (buttonholed MFC)

Direction frequency (TO)	anterior **40%** · posterior **33%** · lateral **18%** · medial **4%** · rotatory **5%** – read the options vs table above

– – Active space – –

Anterior dislocation (TO) hyperextension **> 30°** → posterior capsule → ACL → PCL → **popliteal artery** · vascular injury = **stretch** (anterior) vs **contusion / transection** (posterior)

Time of presentation (TO)	Duration
Acute	**< 3 weeks**
Subacute	3 weeks – 3 months
Chronic	**> 3 months**

Schenck	Ligaments torn
KD I	Multiligament with **ACL or PCL** intact
KD II	**ACL + PCL** only
KD III	ACL + PCL + **PMC (IIIM)** or **PLC / LCL (IIIL)** – **IIIL → highest peroneal nerve risk**
KD IV	ACL + PCL + PMC + PLC – **highest vascular injury (5–15%)**
KD V	Multiligament + **periarticular fracture**

Schenck – TO frequencies (TO) **commonest = KD IIIM** · **peroneal palsy = KD IIIL** · Rocher modifiers **C = arterial**, **N = nerve** (e.g. **KD IIIMC**)

Kennedy (direction) (TO) not useful for treatment · **posterior → extensor mechanism injury** · **anterior → ACL rupture**

KD IV cruciate pattern (TO) **ACL midsubstance + PCL femoral avulsion** · ACL: midsubstance 45% > femoral 34% > tibial 21% · PCL: **femoral avulsion 76%** > midsubstance 17% > tibial 7%



Buttonholing of the medial femoral condyle

3 Presentation

Diagnosis **instability on exam** (X-ray may be normal); **recurvatum** in extension
✗ Wrong options: procurvatum

Dimple / pucker sign **MFC buttonholed through medial capsule → irreducible posterolateral** dislocation → **no repeated closed reduction (skin necrosis)**

Exam before + after reduction DP + PT pulses vs other side; peroneal + tibial nerves; **serial exams mandatory**

Clinical tests for combined injuries (dial, ERR, collaterals)

Test (TO)	Positive means
Dial test ↑ ER (> 10–15°) at 30° only	**PLC**
Dial test at 30° AND 90°	**PLC + PCL**
Anterior drawer with foot in EXTERNAL rotation	**ACL + MCL** (anteromedial rotatory instability)
External rotation recurvatum test	**ACL + PLC** (recurvatum, varus, anterior tibial translation; sensitivity 7.5%)
Varus / valgus stress at 30°	**Isolated collateral (LCL / MCL)** – **best position**; > 10 mm + soft end-point = complete
Varus stress opens **in full extension**	**LCL + PLC + PCL**
Valgus stress opens **in full extension**	**MCL + POL / posteromedial capsule + PCL**

4 Diagnosis & Classification

Key radiology findings

Study	Finding / use
Pre- + post-reduction AP / lateral	**Asymmetric joint space**, **Segond (lateral tibial avulsion)**, osteochondral defects
CT	Fractures on post-reduction films (eminence, tubercle, plateau)
MRI	**After reduction, before hardware** – ligaments, menisci, planning
Arcuate sign (TO)	**Fibular head avulsion → PLC / LCL** injury
Stress views (TO)	Varus → LCL · valgus → MCL



Knee dislocation radiograph

Avulsion signs (TO) **Segond → ACL · reverse Segond → PCL / PLC · arcuate sign → PLC**

Vascular injury – incidence, ABI, CT angiography & exploration

Incidence **5–15% all** (Orthobullets) · 10–40% / 18–45% / ~16% / ~25% (OB) · **40–50% anterior or posterior** · **low-velocity anterior ~40%** (OB) – read the options

– – Active space – –

ABI = ankle systolic ÷ brachial systolic (normal 0.9–1.1) – **< 0.9 → concern** (e.g. **0.8**) · **< 0.85** (TO); ABI < 0.9: **sens 87%, spec 97%**; **> 0.9 → NPV 100%**

After reduction

- Pulses absent / cool limb → **emergent vascular exploration (no imaging delay)**
- Pulses present, ABI < 0.9 → CT angiography → **vascular surgery if injury**
- Pulses present, ABI > 0.9 → **serial exams (2–4 hourly, 48 h)**
- Pulses absent pre-reduction, return after (TO) → **CT angiography (intimal tear)**

Present pulses	**do not exclude** popliteal injury (collaterals) · **pulse oximetry** doesn't exclude either
Ischaemia > 8 h	**amputation up to 86%** (Orthobullets) · > 6–8 h **~85%** (OB)
Vascular incidence (TO)	**16–64%** · highest in **posterior (44%)**
Pulses present, ABI > 0.9 (TO)	observe **24 h** (TO) · serial exams 48 h (above) – read the options
Warm ischaemia > 6 h (TO)	**mandatory fasciotomy**
Bilateral, ABI 0.78 left / 0.96 right	**CTA left only, observe right**
ABI 0.0 after posterior dislocation	**complete popliteal artery tear**
Ischaemic after reduction, BMI 52	**emergent vascular exploration** · on-table angiogram ✗ Wrong options: ex-fix first · CTA · IR stenting
Surgery for	flow-limiting intimal flap, absent pulses, ischaemia, haemorrhage, expanding haematoma · **non-flow-limiting flap → observe** (anticoagulation doesn't help)

5 Management

Reduction & initial stabilisation (irreducible posterolateral dislocation, spanning ex-fix)

Orthopaedic emergency	**reduce first, then reassess pulses**

Direction	Reduction
Anterior	Traction + anterior translation of **femur**
Posterior	Traction, extension, anterior translation of **tibia**
Medial / lateral	Traction + translation
Rotatory	Traction + rotate opposite to deformity

Splint	**20–30° flexion**
Splint (TO)	long leg splint **in extension** with towel bump under calf · vs 20–30° flexion (above) – read the options · persistent instability → **hinged knee-spanning ex-fix**
Reduction trick (TO)	**exaggerate the deformity, then reverse** · posterolateral + dimple → **open reduction only**

Pre-patellar abrasion (TO)	dashboard injury → **check hip**
Stener-like lesion (knee) (TO)	**distal MCL avulsion** with **pes anserinus interposed** → will not heal → **surgical repair**
Immobilisation options (TO)	ex-fix (instability, poor skin, open, vascular repair): **femoral pins anterolateral, tibial anteromedial**, clear of future tunnels · cast if stable · **transarticular Steinmann pin obsolete** (breakage, infection)
Absent pulses, peroneal palsy (wrestler)	**closed reduction in ED + reassess** ✗ Wrong options: acute reconstruction · MRI · arteriogram first
Symmetric pulses, ACL + PCL + PLC on MRI	**emergent reduction + EUA, repeat NV exam, admit overnight, delayed reconstruction** ✗ Wrong options: discharge · angiogram routinely · immediate reconstruction
Irreducible posterolateral + pucker sign	**reduction in OR → open reduction via anteromedial approach** (medial parapatellar) + **spanning ex-fix**; pulses still absent → **popliteal exploration** ✗ Wrong options: femoral distractor closed · traction pin · posterior approach
After vascular repair + fasciotomy, unstable knee	**spanning ex-fix** ✗ Wrong options: cylinder cast · acute ligament surgery · arthroscopy

Operative criteria

Open reduction	Spanning ex-fix
Irreducible / posterolateral	**Vascular repair** (protect it)
Open fracture-dislocation	Open fracture-dislocation
Obesity (hard closed)	**Compartment syndrome**
Vascular injury	**Obesity (BMI > 50)**, polytrauma

Ex-fix position	**20–30° flexion**, reduced in AP + sagittal planes
Nonop definitive	rare; **worse outcomes** (stiffness + instability)

Timing of ligament reconstruction

Acute (< 3 wk)	**↑ residual instability** + **↑ stiffness recalcitrant to treatment** ✗ Wrong options: less instability · best outcomes
Staged	more good / excellent outcomes (older data; now ≈ equal); **no algorithm proven superior**

Early treatment improves outcomes (Orthobullets treatment section) vs staged better (technique section) – **read the options**

Arthroscopic	risky if large capsular tear (**fluid extravasation → compartment syndrome**); **PLC + PMC need open** reconstruction
Hinged ex-fix	adjunct, ↓ re-rupture
Timing – TO (TO)	ideal **10–21 days** once swelling settles · bony avulsion → screws / anchors · soft-tissue avulsion → anchors / spiked washer · midsubstance → reconstruct

– – Active space – –

Graft choice (TO) hamstrings, BPTB, quadriceps, peroneus longus · **avoid hamstrings if MCL injured**

Combined ligament reconstruction — timing, sequence & graft fixation order

ACL + PCL (± collaterals) (TO) **simultaneous reconstruction at 2–4 weeks**

Fixation order (TO) **PCL femoral → ACL femoral → PCL tibial → ACL tibial**

Tension angles (TO) **PCL at 70–90°** (anterolateral bundle 90°, restore step-off) **first → LCL at 20–30°** + slight valgus → rest of PLC at 60°, neutral rotation → **ACL at 30°** → posteromedial corner last

Early surgery criteria (TO) stable reduction, **ROM > 90°**, healthy skin → repair / reconstruction **within 3 weeks** · polytrauma, poor skin, vascular injury → delayed / staged

Combination (TO, Mohindra)	Acute < 3 wk	Chronic > 3 months
Cruciate + MCL / POL	Cruciate reconstruction + MCL / POL repair	Cruciates first, delayed MCL reconstruction if unstable
ACL + PCL	One-stage ACL + PCL	Two-stage: **PCL first → ACL after 3 months**
ACL + PCL + MCL or PLC	One-stage ACL + PCL + collateral	Stage 1 PCL + PLC / MCL → stage 2 ACL (3 months)

Tensioning sequence (TO, Mohindra) **PCL at 90°** (anterior drawer) → PLC in slight varus + IR → **ACL in full extension** · vs ACL at 30°, PLC neutral rotation (above) — read the options

MLKI rehab (TO)	Key rule
0–6 wk	PCL brace, NWB, **prone ROM only (no supine knee bending)**
6–8 wk	Partial WB, ROM > 90°, closed chain only, avoid ER
8–12 wk	Full WB, squats < 70°, brace
> 12 wk	Open chain, proprioception
RTS	**9 months**

PLC before ACL (TO) ACL tension in PLC-deficient knee **↑ external tibial rotation**

ACL + grade 1 MCL (TO) **hinged brace (locked 30°) ~2 wk, then ACL reconstruction**
✗ Wrong options: MCL repair · brace 1 month · conservative

MCL grade with ACL (TO)	Plan
1	ACL surgery at **2 wk**, no motion restriction
2	ACL within 2 wk; **7 days** brace locked at 30°
3	ACL at **3 wk**; 10–14 days locked at 30°

Acute MCL repair (TO) intra-articular entrapment, large bony avulsion, plateau fracture, complete tibial-side avulsion (athlete), AMRI, valgus laxity at 0° in valgus knee · **MCL heals nonop > 90%**

ACL + PLC (LCL, popliteus, biceps), PCL intact (TO) **ACL + PLC reconstruction** within 3 wk
✗ Wrong options: PLC **repair** (higher failure) · delayed ACL at 6 wk

PLC repair only for (TO) **avulsions off fibular head** (FCL, PFL, biceps) or popliteus, capsular / meniscocapsular

PLC static stabilisers (TO) **FCL (LCL), popliteus tendon, popliteofibular ligament**; up to **16%** of knee ligament injuries

Fanelli PLC (TO)	Findings
A	↑ ER 10° – PFL, popliteus
B	+ slight varus (5–10 mm) – + FCL
C	+ severe varus (> 10 mm) – + capsule, cruciate

Dial + at 30° and 90° (TO) **arthroscopic PCL + LaPrade PLC reconstruction** · **LaPrade = anatomical (LCL, popliteus, PFL)** · Larson = LCL + PFL only

O'Donoghue unhappy triad

Triad (TO) **ACL + MCL + medial meniscus** (lateral force, fixed foot) · arthroscopically **lateral meniscus** tears commoner with acute ACL
✗ Wrong options: PCL · LCL

6 Complications

Complication	Fact
Stiffness (arthrofibrosis)	**Most common (38%)**; delayed mobilisation → early motion, LOA, MUA
Laxity / instability	**37%**; redislocation rare
Vascular	Emergent repair + **prophylactic fasciotomy**
Peroneal nerve	**25%** (16–40%)

Peroneal nerve injury – examination & EMG

Commonest nerve **common peroneal (25%)** > tibial (subclinical tibial in ~half of peroneal cases); **most associated structure** after posterior dislocation
✗ Wrong options: popliteal artery 16%

Pattern **KD IIIL**, anterior + posterolateral dislocations; ↑ with **vascular injury**, male, **↑ BMI**, **fibular head fracture**

Exam **weak ankle dorsiflexion**, EHL (great toe extension), **eversion (peroneus brevis)**, numb **dorsum + 1st web**
✗ Wrong options: plantar foot · plantarflexion · sural · saphenous

Level **common peroneal at fibular neck**
✗ Wrong options: deep / superficial branches alone

Recovery **~50% partial** (Orthobullets) · complete 30–50% · **partial palsy ~85% vs complete ~35%** · < ⅓ regain dorsiflexion (OB)

Peroneal palsy (TO) **10–40% (avg 15%)**, most with **posterolateral** dislocation · traction > transection · only **50%** regain full function · **superficial peroneal recovers first** · persistent foot drop **50%**

EMG / NCS **unreliable before 4–6 wk**; at **3 months with weak TA firing → NCS / EMG**
✗ Wrong options: PT alone · neurolysis · PTT transfer yet

Partial palsy wait **6–12 months**; **AFO** vs equinus · **complete neurotmesis on early EMG → early repair / transfer**

– – Active space – –

Foot drop — tendon transfer (tibialis posterior)

Transfer	**tibialis posterior: medial navicular → dorsal lateral cuneiform through interosseous membrane** (gold standard) ✗ Wrong options: FDL · PL · PB (peroneal-innervated) · FHL→EHL
Bridle procedure	PTT + **tibialis anterior + peroneus longus** anastomosis · tenodesis to TA → weaker
1 yr foot drop, nerve in continuity	**PTT transfer** ✗ Wrong options: re-neurolysis · cable graft · EHL transfer · ankle fusion
Contraindication	**rigid deformity / can't passively dorsiflex past neutral** · **weak PTT / S1 radiculopathy** also limits ✗ Wrong options: TA 0/5 · prior nerve surgery · L4 radiculopathy · corrected gastrocnemius contracture

Septic arthritis after reconstruction

4 wk after ACL, fever, ESR 105, CRP 44 (TO)	**arthroscopic debridement + graft retention + IV antibiotics** ✗ Wrong options: graft removal first

Gächter (TO)	Findings	Approach
1	Turbid effusion, hyperaemic synovium	Arthroscopic
2	**Purulent**, fibrin, hypertrophic synovium	Arthroscopic
3	Adhesions, necrotic synovium / cartilage	Arthrotomy
4	Cartilage necrosis, **bone erosion**	Arthrotomy

X vs Y

	Anterior	Posterior
Frequency	**Most common**	2nd
Mechanism	**Hyperextension**	Dashboard
Artery	**Intimal tear**	**Complete tear**

	Dial + 30° only	Dial + 30° and 90°
Lesion	**PLC**	**PLC + PCL**

How it's asked

"Pulses return after reduction (TO)?"	**→ CT angiography**
"Vascular injury rate (low-velocity anterior)?"	**→ 40%**
"Irreducible, pucker sign, pulses lost?"	**→ open reduction anteromedial + spanning ex-fix → explore artery**
"Bilateral, ABI 0.78 / 0.96?"	**→ CTA left, monitor right**
"Reduced, ABI 0.95?"	**→ serial examination**
"Ischaemic after reduction, BMI 52?"	**→ emergent vascular exploration**

"Absent pulses + peroneal palsy — initial?"	→ closed reduction in ED, reassess
"How is ABI calculated + threshold?"	→ ankle systolic / brachial systolic; 0.8 = concern
"ABI 0.8 after reduction?"	→ CT angiography
"ABI 0.0 posterior dislocation?"	→ complete popliteal artery tear
"Can't dorsiflex after multiligament injury?"	→ common peroneal nerve
"Nerve injury level (figure)?"	→ common peroneal nerve
"Additional exam finding with ACL + PCL + varus laxity?"	→ difficulty dorsiflexing ankle
"Highest rate of associated injury?"	→ peroneal nerve
"3 months, weak TA firing?"	→ NCS / EMG
"KDIIIL deficit?"	→ weak ankle dorsiflexion
"Muscle affected if nerve injured?"	→ EHL
"Structure affected after anterior dislocation?"	→ peroneus brevis (5th MT base)
"Most common nerve finding?"	→ weak ankle dorsiflexion
"Dynamic transfer for foot drop?"	→ medial navicular → dorsal lateral cuneiform (PTT)
"Foot drop 1 yr, nerve in continuity?"	→ tibialis posterior transfer through IOM
"Contraindication to tendon transfer?"	→ rigid deformity, can't dorsiflex past neutral
"Anterior dislocation, symmetric pulses?"	→ reduce + EUA, observe overnight, delayed reconstruction
"After vascular repair, unstable knee?"	→ spanning external fixation
"Irreducible, ABI 0.92?"	→ reduction in the operating room
"Failed ED reduction — clinical image?"	→ skin pucker (dimple)
"Acute (< 3 wk) reconstruction — true?"	→ more stiffness recalcitrant to treatment
"ACL + PCL graft fixation sequence (TO)?"	→ PCL femoral, ACL femoral, PCL tibial, ACL tibial
"ACL + grade 1 MCL (TO)?"	→ brace 2 wk then ACL reconstruction
"ACL + PCL timing (TO)?"	→ simultaneous at 2–4 weeks
"ACL + LCL / popliteus / biceps, PCL intact (TO)?"	→ ACL + PLC reconstruction
"Dial + at 30° and 90° — treatment (TO)?"	→ arthroscopic PCL + LaPrade reconstruction
"Confirms PCL + PLC (TO)?"	→ dial test + at 30° and 90°
"Test for ACL + MCL (TO)?"	→ anterior drawer with foot externally rotated

– – Active space – –

"Position to test collaterals (TO)?"	→ **30° flexion**
"ER 15° more at 30°, more at 90° (TO)?"	→ **PCL + PLC**
"Positive ER recurvatum test (TO)?"	→ **ACL + PLC**
"Infection 4 wk after ACL (TO)?"	→ **arthroscopic debridement, graft retention, IV antibiotics**
"Unhappy triad (TO)?"	→ **MCL + medial meniscus + ACL**
"Unhappy triad EXCEPT (TO)?"	→ **PCL**
"KD IV cruciate pattern (TO)?"	→ **ACL midsubstance + PCL femoral avulsion**
"Commonest Schenck type?"	→ **KD IIIM** (TO)
"Arcuate sign indicates?"	→ **PLC / LCL injury** (TO)
"Irreducible knee dislocation with dimple sign (TO)?"	→ **posterolateral — open reduction**
"Surgical tensioning sequence in multiligament reconstruction (TO)?"	→ **PCL at 90°, then PLC, then ACL**

LEG COMPARTMENT SYNDROME

– Trauma · 36 / 67 –

1 Basics

Epidemiology & causes

Commonest site	**leg** · **anterior compartment** most often (TO)
Fractures	**69%** (Orthobullets) · **75%** (TO) · **tibial shaft = commonest cause** (TO) · **proximal tibia** commonest overall (other TO) · **plateau 12% vs shaft 3%**, **Schatzker VI** high risk (TO)
Other causes	crush, contusion, GSW, **tight casts / dressings**, IV extravasation, burns, post-ischaemic swelling, bleeding disorders / anticoagulants, arterial injury, osteotomy (**Blount's**), **lithotomy position**, snake bite, exercise

Risk factor	Fact
Young age	**Highest 12–19 yr**, then 20–29 (more muscle, less room)
Diaphyseal fracture	More muscle than tendon
Open, **segmental / comminuted**	Higher energy (Orthobullets, OB) · open **not** associated (one OB item) – read the options
Higher BMI (children)	**Most significant paediatric factor** in OB
Paediatric	**Older (> 14 yr)**, male, MVC / motorbike, **ipsilateral fibula fracture**, IMN · **< 12 yr protective**
Polytrauma (TO)	↑ lactate / base deficit, transfusion, coagulopathy (warfarin)

Not associated (OB)	male sex, BMI ≥ 30 (adult), open fracture, nicotine
Anterior compartment vulnerability (TO)	**type I (slow-twitch, oxidative) fibres** → most sensitive to hypoxia

Pathophysiology – venous outflow obstruction

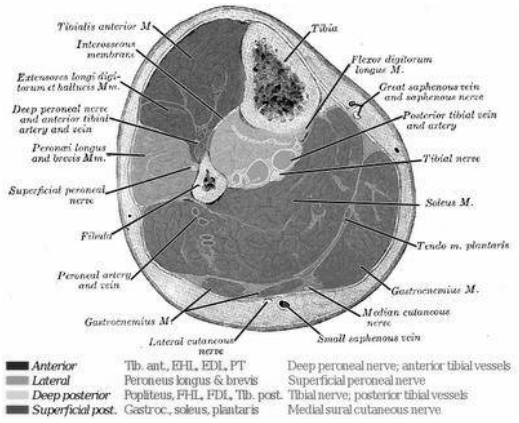
Cascade	trauma → bleeding + oedema → ↑ interstitial pressure → **↓ venous outflow relative to arterial inflow** → myoneural ischaemia
Fasciotomy effect	**↑ venous outflow** ✗ Wrong options: ↓ arterial inflow · ↑ interstitial pressure · ↓ perfusion · ↓ bleeding (may ↑ by losing tamponade)
Irreversible damage	within **6 h**
Causes by mechanism (TO)	**↓ compartment size** (tight cast / bandage, circumferential burn eschar) · **↑ compartment content** (haematoma, reperfusion / crush oedema, snake bite, bleeding disorder)
Eaton & Green ischaemia–oedema cycle (TO)	arterial spasm → ischaemia → ↑ capillary permeability → oedema → ↑ intramuscular pressure → more ischaemia
Critical closing pressure theory (TO)	**Laplace**: transmural pressure = wall tension ÷ radius → interstitial pressure > vessel wall tension → **capillary collapse**

– – Active space – –

AV gradient theory (TO) flow ∝ **(Pa − Pv) ÷ R** → ↑ interstitial → ↑ venous pressure → ↓ AV gradient → ischaemia

Compartment anatomy & nerve deficits

Compartment	Muscles	Nerve / vessel	Deficit
Anterior	TA, EHL, EDL, peroneus tertius	**Deep peroneal**, anterior tibial artery	**1st web space** numb, ↓ dorsiflexion
Lateral	Peroneus longus + brevis	**Superficial peroneal** (only nerve)	**Dorsum of foot – hallux, 3rd + 4th toes**
Deep posterior	**TP, FDL, FHL, popliteus**	**Tibial nerve**, posterior tibial artery	**Plantar foot numb + pain on passive toe EXTENSION**
Superficial posterior	Gastrocnemius, soleus, plantaris	Sural	Plantarflexion



Leg compartments

3 Presentation

Clinical signs & risk factors (analgesia requirement, children, young age)

Earliest / most important (TO)	**pain out of proportion + pain on passive stretch** ✗ Wrong options: tight compartment · ↑ pressure · pulse loss + paraesthesia
Most sensitive before ischaemia	**pain on passive stretch**
Late	**paralysis** (rare full recovery), **absent pulses** (amputation usually inevitable), pallor · **pulses and capillary refill intact in ACS** (TO)
Early sensory (TO)	↓ 2-point discrimination / vibration
Children	**"3 A's": anxiety, agitation, increasing analgesic requirement · increasing analgesia = most sensitive · duration before fasciotomy = most important outcome variable · mechanism is NOT the best predictor** ✗ Wrong options: 5 P's
Pain unreliable	polytrauma, **sedated (impossible)**, children, nerve injury
After 5-h tibial nailing: pain on passive toe flexion, pulse present (TO)	**operative decompression** ✗ Wrong options: measure pressure · Doppler · dextran

4 Diagnosis & Classification

Key radiology findings

Radiographs rule out fracture only · **CTA / Doppler / ABI / duplex not diagnostic** (duplex = DVT)

Diagnosis – pressure measurement & delta-P thresholds

Diagnostic criteria

Fasciotomy	Observe
Δp (diastolic – compartment) ≤ 30 mmHg (TO: **< 30 for > 2 h** if minimal signs)	**Δp > 30** + exam not consistent
Clinical diagnosis (no pressures needed)	–
Absolute **> 30–45 mmHg** (historic; **43% unnecessary fasciotomies** – McQueen)	–

Normal compartment pressure **< 10 mmHg** (TO)

Most accurate **Δp < 30**
✗ Wrong options: palpation · paraesthesia · absolute 25

Intra-op anaesthesia ↓ diastolic → **use PRE-op diastolic** · e.g. **intra-op 29 with pre-op DBP 58 → concerning**
✗ Wrong options: MAP

Intra-op 28 mmHg, BP 84/57 (pre-op 132/84) **re-evaluate in recovery**
✗ Wrong options: fasciotomy · pressors

31–35 mmHg, BP 115/78, pain unchanged with stretch **support BP + serial exams**
✗ Wrong options: fasciotomy · MRA · EMG

Hypotensive polytrauma (80/50), pressures 21–25 **Δp < 30 → immediate fasciotomy**

Alert, stable pain, DBP 98, pressure 35 (Δp 63) **reduce + splint, serial exams**
✗ Wrong options: fasciotomy · analgesia alone

Technique point	Fact
Site	**Within 5 cm of fracture** (highest pressure; also centre of muscle belly)
Interobserver reliability	**Poor**
Anterior	1 cm lateral to tibial crest
Deep posterior	Just posterior to medial tibial border → toward fibula
Lateral	Just anterior to posterior fibular border
Superficial posterior	Mid-calf

– – Active space – –

Device (TO)	Fact
Whitesides needle	**Mercury manometer + syringe + 3-way stopcock** · simplest ward method
Wick / slit catheter	Fluid-displacement pressure; continuous
Stryker monitor	Hand-held electronic, real-time

Pressure measurement indicated (TO)	unconscious / polytrauma · **tourniquet palsy / peripheral nerve injury** (pain unreliable)
Obtunded / intubated	**measure pressures** · AAOS CPG: **repeated / continuous** monitoring ✗ Wrong options: palpation · myoglobinuria (lactate / troponin better but inferior)
Continuous monitoring (slit catheter, anterior)	**most sensitive + specific in sedated**: sens 94%, spec 98%, PPV 93%, NPV 99%
Child post-op: tense calf, cool foot, ↓ pulses, little pain	**measure compartment pressures** ✗ Wrong options: vascular consult · ABI · duplex
Child after Blount osteotomies, sedated, passive toe pain	**remove cast + measure pressures** ✗ Wrong options: elevate · more analgesia · muscle relaxant
Children	measure **under sedation** if suspected · **haemophilia → Factor VIII before** measuring

5 Management

Suspected leg compartment syndrome

- Cast / dressing, no clear CS → **bivalve + spread, split padding, ankle neutral–30° PF**
- Clinically certain → **emergent 4-compartment fasciotomy (+ stabilise fracture)**
- Equivocal / obtunded → pressures → Δp ≤ 30 → **fasciotomy**
- Missed (late, established) → **no fasciotomy**

Cast-related – ankle position & bivalving

Safest ankle position in cast (TO)	**neutral to 30° plantarflexion** · anterior lowest at neutral, deep posterior lowest in PF · resting PF ~37° ✗ Wrong options: dorsiflexion
Cast + increasing pain + toe numbness	**bivalve + spread cast, remove padding** ✗ Wrong options: elevate · morphine · ice · observe
Already bivalved, ankle dorsiflexed, recurrent numbness	**reposition ankle into slight plantarflexion** ✗ Wrong options: fasciotomy · re-reduction
Impending CS (TO)	**limb at heart level** (elevation ↓ arterial inflow)

Emergent fasciotomy with fracture stabilisation

Tibial shaft fracture + CS at 6 h (TO)	**fasciotomy then reamed IMN in same sitting** · reamed = unreamed ICP ✗ Wrong options: unreamed · delay nail · ex-fix

Sequence (TO)	**fasciotomy first**, then stabilise; **cast contraindicated** with ACS; avoid excess traction, thigh-bar pressure, high elevation
Severe pain despite narcotics, passive stretch pain	**immediate 4-compartment fasciotomy + ex-fix** ✗ Wrong options: 2-compartment · fix first
Teen, knee injury, firm compartments, pressures 33–38 (BP 100/50)	**emergent 4-compartment fasciotomy + ORIF**

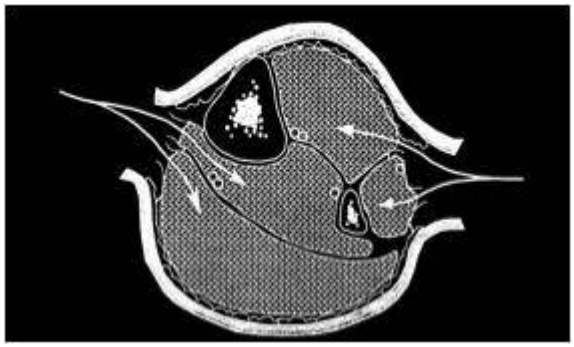
Fasciotomy technique (incisions, compartments, SPN risk)

Technique	Detail
Dual incision – 2 × 15–18 cm, **8 cm skin bridge**	**Anterolateral**: protect **SPN**; anterior fasciotomy **1 cm in front of** septum, lateral **1 cm behind** · **Posteromedial**: protect **saphenous vein + nerve**; superficial posterior; **detach soleal bridge from tibia → deep posterior**
Single lateral (perifibular)	Fibular head → ankle; anterior + lateral, then superficial posterior; **deep posterior via interosseous membrane behind fibula**

Soleus elevated from tibia	to reach **deep posterior** compartment
Medial incision does NOT release	**EHL (anterior)** ✗ Wrong options: TP, FDL, popliteus, soleus
Single lateral – nerve at risk proximally	**superficial peroneal** (OB) · **common peroneal** (Orthobullets) – read the options
Closure	**delayed primary / STSG at 3–7 days** (Orthobullets) · 7–10 days (TO)
Principles (TO)	extensile incisions, release all involved compartments, protect vital structures, debride
Contraindication	**missed (established) compartment syndrome**
Timing (TO)	ideal **within 12 h** · **> 24 h usually contraindicated** (infection of necrotic muscle)
Limited / subcutaneous fasciotomy (TO)	**no role in acute** compartment syndrome

Region (TO)	Fasciotomy
Leg	**Two-incision (Mubarak)** – anterolateral + posteromedial → all 4
Thigh	**Lateral** incision (anterior + posterior) ± **medial** for adductors
Arm	Anterolateral; anterior compartment first
Forearm	Volar **lazy-S** (curved laterally → mobile wad) ± dorsal
Foot	**2 dorsal (over 2nd + 4th MT)** ± medial

– – Active space – –



Dual-incision fasciotomy

Hyperbaric oxygen

Mechanism	**↑ oxygen diffusion gradient** (100% O_2 at ↑ pressure)
Indications	CO poisoning, **gas gangrene**, crush, compartment syndrome, **necrotising fasciitis**, chronic osteomyelitis, grafts / flaps, burns

6 Complications

Late sequelae – contracture & tendon transfers

Missed compartment	Deformity	Treatment
Anterior + deep posterior	Weak TA + TP → **PL overpowers → cavovarus** (flexible)	**Peroneus longus → dorsolateral midfoot** + Achilles lengthening
Anterior + lateral	**Foot drop**, flexible equinovarus (TP 5/5)	**PTT through IOM to TA + peroneus tertius (above ankle)** + anterior debridement + **TAL** (✗ Wrong options: PL transfer (weak) · fusion)
Deep posterior	**Clawing all toes**, equinocavovarus, no heel rise; equinus **not relieved by knee flexion**	**Posterior capsule release + TAL + excise scarred muscle / tendon**

Transfer rule	transferred muscle **loses one grade** → needs **5/5**
Equinus relieved by knee flexion	gastrocnemius slide · not relieved → **Achilles lengthening**

X vs Y

	Lateral compartment	Anterior compartment	Deep posterior
Nerve	**SPN**	**DPN**	**Tibial**
Sensory loss	**Dorsum (hallux, 3rd, 4th)**	**1st web**	**Plantar foot**
Stretch pain	Inversion	Plantarflexion of toes / ankle	**Toe extension**

How it's asked

"Decompression threshold (TO)?"	**→ Δp < 30 mmHg for > 2 h**
"Most accurate diagnosis?"	**→ diastolic – compartment < 30**

"Intra-op 28 mmHg, hypotensive?"	→ repeat evaluation in recovery
"Post-op 31–35 mmHg, BP 115/78?"	→ continued monitoring + serial exams
"Concerning measurement?"	→ intra-op 29 with pre-op diastolic 58
"Hypotensive polytrauma, pressures 21–25?"	→ immediate fasciotomy
"8-yr-old tense calf, cool foot post-op?"	→ measure compartment pressures
"Pressure measurement — true?"	→ poor interobserver reliability
"Sedated on pressors, swollen calf?"	→ measure pressures
"Most sensitive + specific in sedated?"	→ continuous pressure monitoring
"Alert, DBP 98, pressure 35?"	→ closed reduction + splint, serial exams
"Most important sign (TO)?"	→ pain out of proportion + pain on passive movement
"Pain after 5-h nailing, pulse present (TO)?"	→ operative decompression
"Most sensitive sign in child (supracondylar)?"	→ increasing analgesia requirement
"Simplest ward method of pressure measurement?"	→ Whitesides needle manometer (TO)
"Fasciotomy at 30 h after onset?"	→ usually contraindicated (> 24 h) (TO)
"Earliest sign in child?"	→ pain out of proportion
"Paediatric CS — EXCEPT?"	→ mechanism is the best predictor
"Blount osteotomy, passive toe pain?"	→ remove cast + measure pressures
"Factor most associated (18-yr-old)?"	→ age < 20
"Most significant paediatric risk factor?"	→ higher BMI
"Fasciotomy effect?"	→ ↑ venous outflow
"Lateral compartment pressure 47 — pathophysiology + deficit?"	→ ↓ venous outflow; numb dorsum (hallux, 3rd, 4th)
"Isolated lateral compartment — deficit?"	→ numb dorsum of foot (hallux, 3rd, 4th)
"Deep posterior compartment signs?"	→ plantar paraesthesia + pain on passive toe extension
"Perifibular approach — proximal risk?"	→ superficial peroneal nerve
"Soleus elevated to reach?"	→ deep posterior compartment
"Not released by medial incision?"	→ EHL
"Safest ankle position in cast (TO)?"	→ neutral to 30° plantarflexion
"Already bivalved, recurrent numbness?"	→ reposition ankle into slight plantarflexion
"Long-leg cast, increasing pain + tingling?"	→ bivalve + spread cast

– – Active space – –

"Tibial fracture + CS at 6 h (TO)?"	→ fasciotomy + reamed IMN same sitting
"Teen, pressures 33–38?"	→ emergent 4-compartment fasciotomy + ORIF
"Uncontrolled pain, passive stretch pain?"	→ immediate 4-compartment fasciotomy + ex-fix
"Anterior + deep posterior CS → cavovarus?"	→ peroneus longus to dorsolateral midfoot
"Foot drop 2 yr, PTT 5/5?"	→ PTT through IOM to TA + peroneus tertius, TAL
"Toe clawing + equinus after segmental tibia?"	→ posterior capsule release, TAL, excise scar
"↑ oxygen diffusion gradient?"	→ hyperbaric oxygen

MONTEGGIA FRACTURES

– Trauma · 37 / 67 –

1 Basics

Definition & epidemiology

Monteggia	**proximal ⅓ ulna fracture + radial head dislocation** (PRUJ disruption)
Who	**rare in adults** (high energy) · **children 4–10 yr** commoner, different protocol
Associated	olecranon fracture-dislocation, radial head (~5%), coronoid, LCL injury, **terrible triad**
Key ligament	**annular ligament**

4 Diagnosis & Classification

Bado classification – direction of radial head dislocation

Rule **radial head dislocates toward the apex of the ulnar fracture** (apex anterior → anterior)

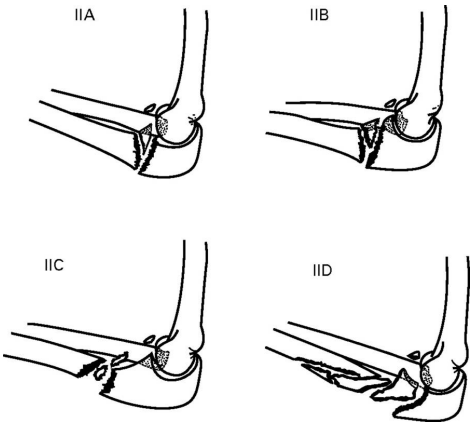
Bado	Share	Pattern
I	**60%**	Proximal / middle ulna, **anterior** dislocation – **commonest in children + young adults**
II	15%	**Posterior** dislocation – **70–80% of adult Monteggia**; **worst prognosis**
III	20%	Ulnar **metaphysis** (distal to coronoid), **lateral** dislocation – **most often irreducible (annular ligament)**
IV	5%	**Ulna + radius** fractures, dislocation any direction



Bado type I – anterior dislocation

Jupiter (type II)	Ulnar fracture level
IIA	**Coronoid** level
IIB	Metaphyseal-diaphyseal junction
IIC	Distal to coronoid (diaphysis)
IID	Extends into distal half

– – Active space – –



Jupiter classification of Bado II

Monteggia "variant" + **radial head / neck fracture** → separate approach; ↑ OA, instability, nonunion, ↓ ROM

Key radiology findings

Views AP + lateral **elbow, wrist, forearm**; **radiocapitellar line** congruent on every view

CT coronoid, olecranon, radial head involvement

3 Presentation

PIN palsy **most likely nerve injury**; **wrist extension in radial deviation** (ECRL spared), weak thumb + **MCP extension**, sensation intact

5 Management

Monteggia fracture

- Child, stable ulna, reduced RC joint → **closed reduction + cast (supination for Bado I / III)**
- Adult (almost all) → ORIF ulna (plate) → **radial head usually reduces**
  - radial head still out → check ulna (revise) → **then open reduction (annular ligament)**
- Transverse / short oblique, easily reduced → **closed reduction + IM fixation of ulna**

TO sequence ORIF ulna (plate) → check radial head reduction → persistent dislocation → **correct PUDA (~15°)** → still out → open reduction (annular ligament interposition) (TO)

Treatment options

Option	Indication
Closed reduction + cast	**Children**, stable transverse ulna; **supination for Bado I + III**; oblique / unstable → pinning
ORIF ulna + closed radial head reduction	**Adults**, open, **unstable (long oblique)**, comminuted
ORIF ulna + open radial head reduction	**Radial head won't reduce after correct ulnar ORIF**, complex injury, **variants**
Closed reduction + **IM ulnar fixation**	Transverse / short oblique, easily reduced

Technique lateral decubitus; posterior incision lateral to olecranon tip; **FCU / anconeus** proximally, **FCU / ECU** distally; **dorsal plating**; **dorsal cortex** to judge length; **distractor** if comminuted; fluoroscopy through full motion

Radial head open reduction **posterolateral (Kocher)**; separate lateral incision

Persistent radial head dislocation — ulnar malreduction / length

After ulnar fixation, radiocapitellar joint dislocated **revise ulnar ORIF**
✗ Wrong options: radial neck osteotomy first · radial head arthroplasty · splint in supination · shortening osteotomy

Commonest cause of persistent / recurrent dislocation **failure to restore ulnar length / anatomic alignment**
✗ Wrong options: annular ligament not repaired · missed radial head fracture · LCL · fixation rigidity

Late (6 mo) subluxation + stiffness **ulnar corrective osteotomy + open radial head reduction**

Irreducible radial head — annular ligament interposition

Non-concentric after reduction (child, posterolateral) **annular ligament interposed** · **Bado III** most often
✗ Wrong options: LUCL · MCL bands · anconeus

How common "often found interposed" (Orthobullets) · "rare complication" in adults (OB) — read the options

Radial head excision — MCL competence

Chronic malunion ulnar osteotomy + radial head reduction · or **radial head excision** (restores rotation)

Must be competent before excision **anterior bundle of MCL** (primary valgus stabiliser)
✗ Wrong options: posterior bundle (secondary) · brachialis · RCL / LUCL (varus)

– – Active space – –

6 Complications

Complication	Fact
PIN palsy	**Up to 10%** acute; blunt trauma from radial head → **observe 2–3 months** (usually resolves) → NCS if not
Radial head malunion / redislocation	**Ulnar malalignment**
Stiffness	**Diagnosis delayed > 2–3 wk** → complications ↑↑
Synostosis, nonunion, plate failure	Type II: **nonunion + plate failure**

Outcomes & prognostic factors

Bado II — **highest complication rate**, worst prognosis; nearly all lose some ROM
✗ Wrong options: full recovery · casting = surgery · nerve injury common · ulnohumeral instability

Worse DASH — **radial head / neck fracture**
✗ Wrong options: age · radius injury type · treatment method

X vs Y

	Monteggia	Galeazzi
Fracture	**Proximal ulna**	**Distal ⅓ radius**
Joint	**Radial head / PRUJ**	**DRUJ**
Key to reduction	**Ulnar length + alignment**	Radial ORIF then DRUJ
Nerve	**PIN**	–

How it's asked

"Apex-anterior proximal ulna — radial head direction?"	→ **anterior**
"Radial head out after ulnar fixation?"	→ **revision ORIF of ulna**
"Commonest cause of persistent dislocation?"	→ **failure to restore ulnar length**
"Pain + ↓ motion 6 months after ORIF?"	→ **inadequate ulnar reduction**
"Child, non-concentric after closed reduction?"	→ **annular ligament interposition**
"Bado II in 54-yr-old — counsel?"	→ **high complication rate**
"Worse functional scores?"	→ **radial head / neck fracture**
"Must be competent before radial head excision?"	→ **anterior bundle of MCL**

MOREL-LAVALLEE LESION

– Trauma · 38 / 67 –

1 Basics

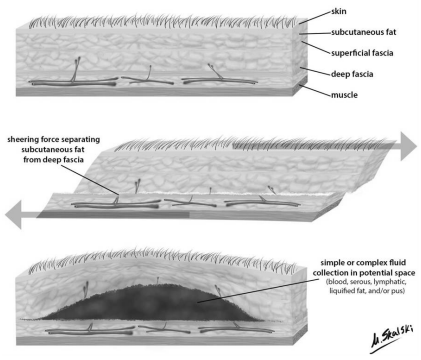
Definition & epidemiology

Morel-Lavallée lesion (MLL)	**closed degloving**: shear separates **dermis + subcutaneous fat from deep fascia**
Incidence	**up to 8% of acetabular fractures** (Orthobullets) · up to 10% (TO)
Site	**proximal lateral thigh (peritrochanteric) – commonest**; gluteal, lumbosacral, abdominal (hypermobile skin)
Risk	**high-energy shear**; **obese**; with **pelvic, acetabular, proximal femur** fractures

Anatomy – plane of fluid collection

Layers	epidermis → dermis → subcutaneous fat → **[MLL] → deep fascia** → muscle
Plane	**deep to subcutaneous fat, superficial to deep fascia** ✗ Wrong options: dermis (seroma) · subcutaneous (oedema) · intramuscular (haematoma) · deep to muscle

Morel-Lavallée mechanism



Morel-Lavallée anatomy

Pathophysiology (4 stages)

Stages	shear separation → **subdermal plexus** lymphatics + vessels bleed → replaced by **serosanguinous fluid** → **pseudocapsule** (chronic, untreated)
Bacterial colonisation	**up to 46%**

3 Presentation

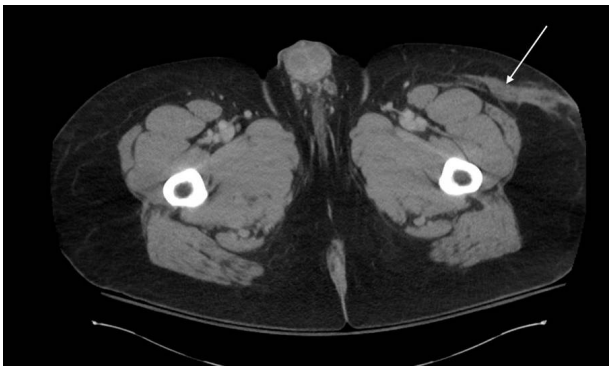
Missed	**up to 33%** at index evaluation (distracting injuries, habitus)
Exam	**ecchymosis, swelling, fluctuance, skin hypermobility** · discoloration may take **days**
Earliest indicator (TO)	**paraesthesia of overlying skin**
Chronic	dull pain, contour deformity, cosmesis

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

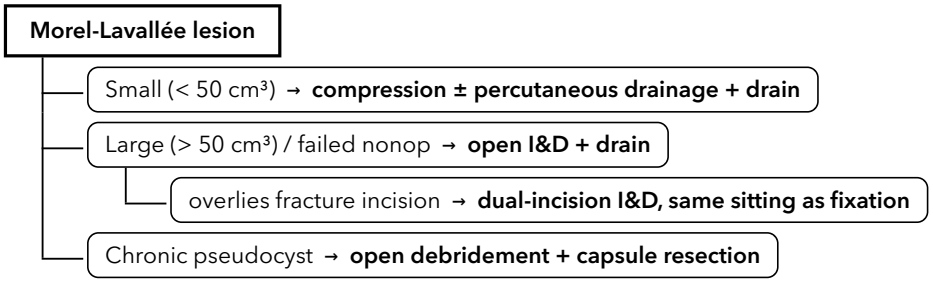
Study	Use / finding
Radiographs	Underlying fractures; soft-tissue swelling
CT	**Size → management**; fluid in fascial plane (soft-tissue window)
Ultrasound	**Hypoechoic** space superficial to fascia; localise for drainage
MRI	**Best extent** (avg **30 × 12 cm**); **acute: T1 low, T2 high** · **subacute: T1 + T2 high** with **capsule low on both**



CT soft-tissue window — Morel-Lavallée lesion of the thigh

Not to do for MLL MRI + core biopsy (sarcoma work-up), CT chest / abdomen / pelvis (metastasis work-up), admit for antibiotics (abscess)

5 Management



Operative criteria

Treatment	Indication	Outcome
Compression (ace wraps, bike shorts)	**< 50 cm³**	Needs compliance
Percutaneous drainage + drain (US-guided, 1–2 cm incision, irrigate)	**< 50 cm³**	**Inferior** to open (can't debride necrotic tissue); **needle aspiration → high recurrence**
Single-incision I&D	**> 50 cm³** or failed nonop; **not in fracture approach**	**75% resolution**
Dual-incision I&D	**Lesion overlies fracture approach** / found intra-op	Higher infection; serial I&D
Open debridement + capsule resection	**Chronic pseudocyst**	Often multiple surgeries

Chronic lesion – open debridement & drain

Large symptomatic lesion at 1 month — **open I&D + drain** via **limited approach** (protect compromised subdermal plexus)
✗ Wrong options: needle decompression · biopsy · antibiotics

Trochanteric fracture + MLL (TO) — **fix fracture + incision, open debridement at same time**
✗ Wrong options: delay fixation · observe · fix + ambulate only

Technique — scrub-brush / Cobb debridement · **≥ 15 Fr drain** exiting away, remove at **< 20–30 mL/day** · primary vs delayed closure controversial (VAC + repeat I&D at **48–72 h**)

Dual-incision — proximal + distal ends of lesion; **close proximal primarily**, **distal: VAC → repeat I&D 48–72 h**

6 Complications

Complication	Fact
Recurrence	**Most common – up to 56% nonop**, **15–20% after open** → repeat I&D + drain, VAC, **sclerotherapy (talc / doxycycline)**
Pseudocyst	Chronic untreated → **resect capsule**
Skin necrosis	Delay, repeated I&D → skin graft
SSI after pelvic / acetabular surgery	**MLL = independent risk factor**

Prognosis — better if **small, acute, away from incisions**

Differentiation from necrotising fasciitis (LRINEC)

LRINEC (TO) — **CRP, WBC, haemoglobin, sodium, creatinine, glucose**; **≥ 6 = high risk** · **not included: SGPT (ALT)**

– – Active space – –

X vs Y

	Morel-Lavallée	Haematoma / seroma
Plane	**Subcutaneous fat ↔ deep fascia**	Intramuscular / dermis
Skin	**Hypermobile**, fluctuant	Not hypermobile

How it's asked

"Plane of fluid collection?"	**→ deep to subcutaneous fat, superficial to deep fascia**
"Trochanteric fracture + thigh fullness (TO)?"	**→ fixation combined with incision + open debridement**
"Large fluctuant lesion 1 month after crush?"	**→ open I&D + drain**
"NOT in LRINEC (TO)?"	**→ SGPT**

NECROTIZING FASCIITIS

– Trauma · 39 / 67 –

1 Basics

Definition & risk factors

Necrotising fasciitis — life-threatening infection spreading rapidly **along fascial planes**; overlying cellulitis ± (early skin deceptively benign)

Immune suppression — **diabetes (commonest comorbidity, 71%)**, AIDS / HIV, cancer, **obesity**, alcohol

Bacterial entry — **IV drug use**, injections, insect bites, **skin abrasions (turf)**, abdominal / perineal surgery

Microbiology & empiric antibiotics (polymicrobial vs group A strep)

Type	Organisms	Setting
1 – polymicrobial (80–90%)	**4–5 species**: **non-group A strep**, **anaerobes (Clostridia)**, facultative anaerobes, **gram-negative enteric bacilli (Enterobacteriaceae)**; synergy	**Diabetics, cancer**, post-op abdominal / perineal
2 – monomicrobial (5%)	**Group A β-haemolytic Streptococcus** (commonest monomicrobial)	**Healthy**, extremities
3	**Vibrio vulnificus** (gram-negative rods)	**Marine** exposure
4	Fungal	–

Type 3 (TO) — **fungal: mucormycosis** in immunocompromised (COVID period) · Vibrio = type 3 (Orthobullets) · fungal = type 3 (TO) – read the options

Spread pattern (TO) — toxins spread **lengthwise along fascial planes** (few barriers) → microcirculation thrombosis → skin red → purple → blue-grey → bullae

Commonest overall — **polymicrobial** · **commonest single isolate** – **group A strep** (Fontes, Wong)
✗ Wrong options: GAS · MRSA · S. aureus

Polymicrobial culture most likely in — **diabetic with necrotising soft-tissue infection**
✗ Wrong options: chronic PJI (CoNS) · nail through shoe (Pseudomonas) · neonate hip (GBS) · acute PJI (S. aureus)

MRSA — **not** the commonest monomicrobial cause



Necrotising fasciitis – bullae and discoloration

– – Active space – –

3 Presentation

Early	rapidly progressing cellulitis / abscess, **minimal swelling**, no discoloration
Late	**severe pain (out of proportion)**, high fever, rigors, **tachycardia, hypotension**
Skin	**bullae (purple / haemorrhagic)**, ischaemic patches, cutaneous gangrene, induration, **subcutaneous emphysema (gas only 13%)**
Red flag	**worsening despite IV antibiotics** + systemic toxicity
Clinical clues (TO)	**non-pitting oedema beyond cellulitis** · **peau d'orange** beyond swelling · **hypo-/anaesthetic skin** · **tenderness beyond erythema** · anxiety, "impending doom"
Finger test (TO)	**finger / probe dissects easily along fascia beneath intact skin** = hallmark

4 Diagnosis & Classification

Key radiology findings

Imaging	**not required**; **CT / MRI delay surgery** (only show oedema / thickening) · **gas → urgent debridement**
Frozen section	1 × 1 × 1 cm, bedside or OR; confirms early cases; **never delay surgery** · histology: **fascial necrosis**, organisms in fascia, PMNs, **fibrinous thrombi + vessel wall necrosis**

Diagnosis – LRINEC score & laboratory markers

LRINEC component	Points
CRP ≥ 150 mg/L	**4 (largest)**
WBC 15–25 · **> 25 × 10^9/L**	1 · **2**
Hb 11–13.5 · **< 11 g/dL**	1 · 2
Na < 135 mmol/L	2
Creatinine > 141 µmol/L	2
Glucose > 10 mmol/L	1

Score max **13**; **≥ 6 → high risk**; original **PPV 92%** · later **sens 68%, spec 85%**

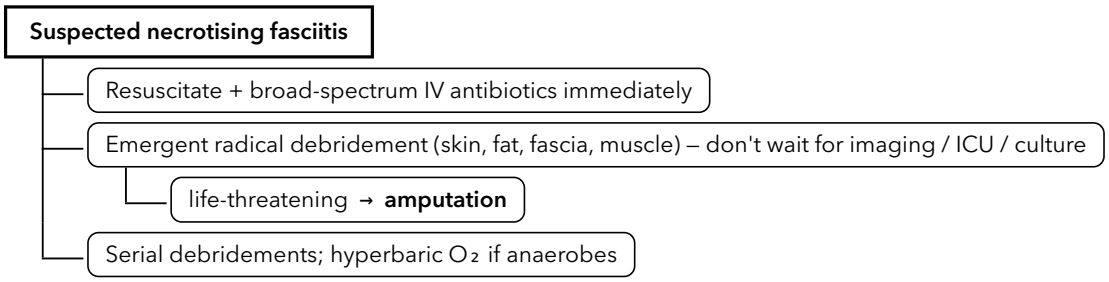
LRINEC score (TO)	Risk
≤ 5	Low (< 50%)
6–7	Intermediate (50–75%)
≥ 8	**High (> 75%)**

LRINEC 6–7	**high risk** (Orthobullets, ≥ 6) · **intermediate** (TO) – read the options
NOT in LRINEC	**ESR**, potassium, bicarbonate, calcium, **SGPT / ALT**
Missing value given CRP, WBC, glucose, Cr, Na	**haemoglobin**

Most valuable lab (with Na 130, CRP > 200)	**WBC > 25 → emergent debridement ± amputation** ✗ Wrong options: ESR · MRI

– – Active space – –

5 Management



Emergent surgical debridement

Worsening despite cefazolin, hypotension	**urgent aggressive debridement of skin, fat + fascia** ✗ Wrong options: triple antibiotics alone · ICU + ID consult · gentle closure · HBO / IVIG first
Diabetic, bullae, CRP > 200, WBC 25k, Na 127	**emergent debridement ± amputation** ✗ Wrong options: vascular exploration · ICU + escalate antibiotics · MRI · CT
IVDU / HIV hand, CRP 178, WBC 29k	**emergent debridement** ✗ Wrong options: mark margins + observe · vascular consult · oral antibiotics · next-day surgery
Operative findings	**liquefied fat**, **"dishwater" pus**, muscle necrosis, venous thrombosis
Fascia at surgery (TO)	**grey-green / liquefied fascia** with normal-looking patches · fingers pass easily along normally tough planes · muscle involvement = poor sign
Multiple debridements	often required · **multidisciplinary team**

Antibiotics

Stage	Regimen
Empiric	**Penicillin + clindamycin + metronidazole + aminoglycoside** (clindamycin ↓ amputation odds) · or **vancomycin + clindamycin + cefepime**
Empiric with CKD (avoid nephrotoxins)	**Linezolid + meropenem** (MRSA, GAS, gram-neg, anaerobes) (✗ Wrong options: vanc + gent · vanc alone · gent alone · micafungin)
Strep / Clostridium	**Penicillin G**
Polymicrobial	**Imipenem / doripenem / meropenem**
MRSA	+ **vancomycin or daptomycin**

Hyperbaric oxygen	adjunct, esp. **anaerobes / Clostridia** (↑ leukocyte O_2 killing, perfusion, antibiotic uptake) · HBO + IVIG only after surgery ✗ Wrong options: "contraindicated in Clostridia"

– – Active space – –

6 Complications

Mortality predictors — time to surgery, vasopressors

Mortality	**~30% (32%)**; **higher in older** patients; **amputation vs salvage — similar** ✗ Wrong options: 15%
Reduces mortality	**shorter time from admission to surgery** ✗ Wrong options: MRI findings · pressors · organism ID · location
Delay > 24 h	**↑ mortality (≈ 4×)** ✗ Wrong options: always trial antibiotics · single debridement always enough · HBO useless · single-discipline team
Strongest predictor of death	**vasopressor requirement outside anaesthesia** ✗ Wrong options: IVDU · cellulitis · ICU admission · debridement size (predicts final defect)
Mortality factors (TO)	**age > 50** · diabetes · **truncal involvement** · **bacteraemia at presentation** · ↑ creatinine · **hypoalbuminaemia** · delay in diagnosis/treatment
Upper-limb NF reaching shoulder / chest wall (TO)	**> 75% mortality**

Gas gangrene — clostridial toxins

C. perfringens myonecrosis toxin (TO)	**alpha toxin** (haemolysis, myonecrosis, gas gangrene) ✗ Wrong options: beta · epsilon · iota

X vs Y

	Type 1	Type 2
Organisms	**Polymicrobial**	**Group A strep**
Host	**Diabetic / immunosuppressed**	**Healthy**
Frequency	**80–90%**	5%

	Necrotising fasciitis	Cellulitis
Progression	Rapid despite antibiotics	Responds
Signs	**Bullae, pain out of proportion, shock**	Erythema
LRINEC	**≥ 6**	Low

How it's asked

"Most common single isolate?"	**→ group A streptococcus**
"Initial antibiotics, CKD + HIV + IVDU?"	**→ linezolid + meropenem**
"Polymicrobial culture most likely?"	**→ diabetic necrotising soft-tissue infection**
"Diabetic foot spreading despite IV antibiotics — organism?"	**→ polymicrobial**

"True statement?"	→ **gram-negative enteric bacilli common in polymicrobial NF**
"Teen with turf abrasion, bullae, shock – true?"	→ **polymicrobial > monomicrobial**
"Most valuable lab + next step?"	→ **WBC > 25; emergent debridement ± amputation**
"Missing value for LRINEC?"	→ **haemoglobin**
"Worsening on cefazolin, hypotensive?"	→ **aggressive debridement of skin, fat + fascia**
"Diabetic calcaneal ulcer, new bullae?"	→ **emergent debridement ± amputation**
"IVDU hand, CRP 178, WBC 29k?"	→ **emergent surgical debridement**
"Reduces mortality?"	→ **shorter time from admission to surgery**
"Strongest mortality predictor?"	→ **vasopressor requirement**
"True about treatment?"	→ **delay > 24 h ↑ mortality**
"Clostridial myonecrosis toxin (TO)?"	→ **alpha**

– – Active space – –

NEONATAL FOREARM COMPARTMENT SYNDROME

– Trauma · 40 / 67 –

1 Basics

Epidemiology

Rare case reports; largest series **24 cases / 20 yr**

When **first 24 h of life**, present **at birth**

Where forearm, wrist, hand; **unilateral**; R = L; **dorsum > volar**

Risk factor	Type
Hypercoagulable – polycythaemia	Intrinsic
Prematurity, multiple gestation	–
Oligohydramnios	Extrinsic
Maternal diabetes, abnormal lie, neonatal respiratory distress	–

Pathophysiology

Commonest **idiopathic**

Extrinsic (compression) fetal posture, **oligohydramnios**, **umbilical cord loops**, **amniotic bands**, birth trauma

Intrinsic (clotting) **hypercoagulable state** → arterial / venous thrombosis

3 Presentation

Pathognomonic **skin lesion in ALL**: **bullae**, erythema, ulcers, **eschar**, distal hand / digit **oedema**, fingertip gangrene

Nerves **radial + PIN > ulnar = median**; ↓ spontaneous movement

Usually missed → found only after complications



Neonatal forearm compartment syndrome

4 Diagnosis & Classification

Key radiology findings

Radiographs	skeletal changes **late**: **physeal distortion** (widening, flaring, premature closure, angulation), shortening, LLD
MRI	**late cases** with full-thickness necrosis + contracture (fasciotomy futile) → map necrosis, guide debridement
Labs	**CBC, ESR, CRP normal** (vs infection)
Compartment pressure – NOT measured	**no neonatal Δp standards**; **DBP at birth < 40 mmHg** → small rise impairs perfusion

Differential

Condition	Distinguishing feature
Cellulitis	Mother well, cultures negative
Necrotising fasciitis	Most easily confused; **skin + subcutaneous only → excision**, not fasciotomy; confirmed at surgery
Vascular injury	**Absent pulses / Doppler**; with brachial plexus lesions

5 Management

Neonatal forearm compartment syndrome

- Diagnosed < 24 h → **emergency fasciotomy (volar, dorsal, mobile wad + carpal tunnel) ± STSG**
- Hypercoagulable → **anticoagulants / thrombolytics**
- Late → **salvage (debride, neurolysis, release, lengthening, angular correction, tendon reconstruction)**

Best outcome	**fasciotomy within first 24 h · salvage inferior**
6-h-old, 34 wk, oligohydramnios, polycythaemia, bullae	**neonatal compartment syndrome → fasciotomy** ✗ Wrong options: cellulitis + antibiotics · NF debridement / HBO · needle manometry

6 Complications

Complication	Treatment
Ischaemic muscle contracture	Debridement + release
Fingertip gangrene	–
Physeal distortion	Lengthening, angular correction
Nerve dysfunction	Neurolysis

Prognosis **missed diagnosis = worst**; **early fasciotomy = best**

– – Active space – –

How it's asked

"Neonate with bullous forearm, oligohydramnios, polycythaemia?"	**→ neonatal compartment syndrome; fasciotomy**
"Why no needle manometry in neonates?"	**→ no Δp standards; DBP < 40 mmHg**

OLECRANON FRACTURES

– Trauma · 41 / 67 –

1 Basics

Epidemiology

Incidence	12 / 100,000 / yr · **~10% of upper-limb fractures** · mean age **~57**; **bimodal**
Mechanism	**direct blow → comminuted** · **indirect (FOOSH) → transverse / oblique**
Transolecranon fracture-dislocation	severe axial load; **distal humerus driven through olecranon** = **anterior** dislocation; **PRUJ intact** (vs Monteggia)

Anatomy

Greater sigmoid (semilunar) notch	olecranon + coronoid; articulates with trochlea (flexion-extension + stability)
Triceps	inserts posterior proximal ulna, blends with periosteum (**radial nerve, C7**) · **anconeus** – lateral olecranon (radial nerve)

3 Presentation

Pain	posterior elbow · **palpable gap** → displaced / comminuted
Can't extend elbow	**extensor (triceps) mechanism discontinuous**

4 Diagnosis & Classification

Key radiology findings

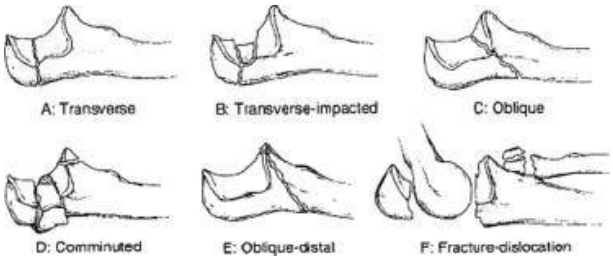
True lateral	**essential** for fracture pattern · radiocapitellar view → radial head / capitellum · CT → comminuted planning

Classifications

Colton	Description
Nondisplaced	Displacement **doesn't increase with flexion**
Avulsion	Displaced
Oblique / transverse	Displaced
Comminuted	Displaced
Fracture-dislocation	**Transolecranon**

– – Active space – –

Schatzker	Pattern
A	Simple transverse
B	Transverse **impacted**
C	**Oblique**
D	**Comminuted**
E	More distal, extra-articular
F	Fracture-dislocation



Schatzker classification

AO	Pattern
A	Extra-articular
B	Intra-articular
C	Intra-articular **radial head + olecranon**

Mayo comminution, displacement, fracture-dislocation

Mayo (TO)	Pattern	Treatment
1A / 1B	Undisplaced, non-comminuted / comminuted	Non-operative (slab / cast)
2A	Displaced, stable, non-comminuted	**TBW**
2B	Displaced, stable, comminuted	Plate
3A / 3B	Unstable ± comminution	Plate

Angles to restore (TO) **PUDA (proximal ulna dorsal angulation) ~15°** · olecranon–diaphyseal angle ~30° · proximal ulna varus 5–6°

5 Management

Olecranon fracture

- Nondisplaced, extensor intact → **cast 45–90°, motion at 1 wk**
- Displaced, transverse, no comminution → **tension band (or IM screw)**
- Comminuted / oblique past coronoid / Monteggia / fracture-dislocation → **plate**
- Elderly low-demand → **nonop or excision + triceps advancement (< 50% joint)**

Operative indications (TO) displacement **> 2 mm** · extensor mechanism disruption · articular incongruity · elbow instability · open

TBW (TO)	Mayo 2A · **contraindicated: comminution, unstable elbow, fracture distal to sigmoid notch** · K-wires toward **anteromedial cortex** · bury under triceps
IM screw (TO)	**6.5 mm cancellous + washer** ± TBW · start centre (AP), slightly lateral (ML)
Plate (TO)	pre-contoured locking = gold standard for unstable / comminuted · **"home run" screw** olecranon tip → anterior cortex · avoid corner screw (prominence)
Excision + triceps advancement	fragment **< 50% of joint** (Orthobullets) · **< 75%** with **coronoid intact** (TO) – read the options · triceps reattached **adjacent to articular surface** (Orthobullets) · **posteriorly** for lever arm (TO) – read the options

Operative criteria

Implant	Indication
Tension band / IM screw	**Transverse, no comminution**, through **midpoint of trochlear notch**, **proximal to coronoid**
Plate	**Comminution**, **oblique extending distal to coronoid**, **coronoid involvement**, **Monteggia**, **fracture-dislocation**, articular impaction, **open**
Excision + triceps advancement	**Elderly osteoporotic**, **< 50% of joint**, too comminuted / small, nonunion; **no instability**
Nonop cast	Nondisplaced + intact extensor; **displaced in low-demand elderly**

Tension band wiring – indications & biomechanics

Mechanism	**converts triceps distraction (tension) into articular compression** ✗ Wrong options: neutralisation · shear · bending
Best for	**transverse fracture through midpoint of trochlear notch** ✗ Wrong options: comminuted · coronoid · Monteggia · oblique distal
Technique	K-wires **engaging anterior cortex** (↓ migration) but **no over-penetration**; **18-gauge wire / thick suture** figure-of-8 through drill hole
Cons	**hardware removal 40–80%**; **no axial stability in comminution**; can **narrow sigmoid notch** if comminuted
Relative TBW indications	tip fractures, transverse, osteoporotic, displaced

Plate fixation – comminution, fracture-dislocation & open fractures

Comminution	**the** indication for plate over TBW ✗ Wrong options: displacement · intra-articular · obliquity alone
Comminuted, past coronoid (24-yr)	**plate + screws** ✗ Wrong options: IMN · TBW + IM screw · triceps advancement · TEA
Articular comminution + depressed fragment (62-yr)	**plate**
Bridge plating	**comminuted, non-reconstructable, young / active (45-yr)** ✗ Wrong options: osteotomy fixation · simple transverse · 91-yr osteoporotic (excise) · child apophysis

– – Active space – –

Transolecranon fracture-dislocation (open or closed)	**plate** → best function ✗ Wrong options: TBW / IM screw (need cortical contact) · excision (anterior instability) · cast
Technique	**dorsal (tension) side**; **lag screws for oblique**; ⅓ tubular too weak for comminution; triceps advanced over plate; **removal ~20%**

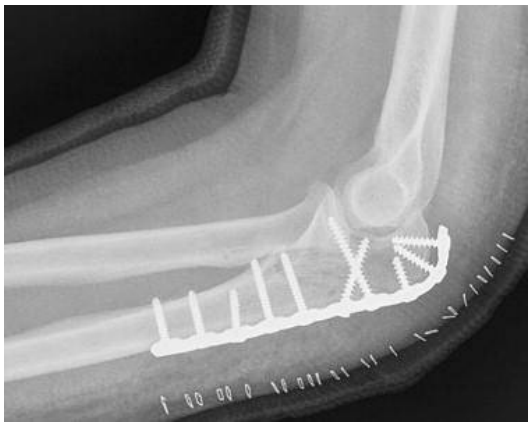


Plate and screw fixation

Elderly low-demand — nonoperative or excision with triceps advancement

82-yr nursing home, comminuted 25% joint, osteopenic	**cast in 90° flexion, short immobilisation then ROM** · Duckworth: good long-term results ✗ Wrong options: 45° × 8 wk (stiffness) · pinning · TBW · locking plate
79-yr RA + osteoporosis, comminuted proximal 30%	**fragment excision + triceps advancement** ✗ Wrong options: IM screw · TBW · TEA · lag + plate
Excision ideal	**85-yr, comminuted, < 50% of proximal joint** ✗ Wrong options: young · > 50% · coronoid-level oblique
Technique	triceps reattached **adjacent to articular surface** with nonabsorbable sutures through drill holes
Downside	**↓ extension strength**; **instability** if ligament injury missed

6 Complications

Complication	Fact
Symptomatic hardware	**Most common** (subcutaneous; poor seating)
Stiffness	**~50%**; rarely functional
HO	↑ with **head injury**
AIN injury	**K-wire over-penetration of anterior (volar) cortex** — can't flex thumb IP / index DIP → observe
↓ Forearm rotation	**K-wire through volar cortex** or too radial
PTOA, ulnar nerve symptoms, ↓ extension strength	—
Nonunion	**Rare**

Postoperative stiffness — therapy & splinting

Arc 40–90° at 2 months, rotation preserved	**formal therapy + static progressive splinting** first ✗ Wrong options: arthroscopic / open release · hardware removal · steroid + aggressive ROM
Failed nonop	limited open release, interposition, arthroscopy, **TEA if > 65**

X vs Y

	Tension band	Plate
Fracture	**Transverse, simple**	**Comminuted, oblique past coronoid, dislocation**
Principle	**Tension → compression**	Neutralisation / bridging
Removal	**40–80%**	~20%

	Transolecranon dislocation	Monteggia
PRUJ	**Intact**	**Disrupted**
Radial head	Moves with ulna	**Dislocated**

How it's asked

"Force at articular surface with tension band?"	**→ compression**
"Best indication for tension band?"	**→ transverse through midpoint of trochlear notch**
"Comminuted, distal past coronoid?"	**→ plate + screws**
"Articular comminution + depressed fragment?"	**→ ORIF with plate**
"Bridge plating most appropriate?"	**→ comminuted fracture in 45-yr-old**
"Transolecranon fracture-dislocation — stability?"	**→ plate fixation**
"Justifies plate choice?"	**→ comminution**
"Plate over tension band when?"	**→ comminuted**
"Open comminuted transolecranon — best function?"	**→ ORIF + plate**
"58-yr osteopenic, comminuted, marginal impaction?"	**→ ORIF + plate**
"Most common TBW complication?"	**→ symptomatic implants**
"Nerve at risk with anterior K-wire over-penetration?"	**→ AIN**
"Impaired forearm rotation after TBW?"	**→ K-wire through volar cortex**
"79-yr RA, comminuted proximal 30%?"	**→ fragment excision + triceps advancement**
"82-yr low-demand, comminuted 25%?"	**→ cast in 90° flexion**
"Excision + triceps reattachment ideal for?"	**→ 85-yr, comminuted, < 50% of joint**
"Stiff 40–90° at 2 months?"	**→ therapy + static progressive splinting**

– – Active space – –

OPEN FRACTURES MANAGEMENT

– Trauma · 42 / 67 –

1 Basics

Epidemiology

Incidence	**30.7 / 100,000 / yr** · average age **45**
Commonest sites	**tibia + finger phalanx** · tibia = commonest long-bone fracture, **up to ~24% open**
Associated injuries	**30%** · **compartment syndrome still possible** with an open wound
Wound near a fracture	treat as **open until proven otherwise**
Psychiatric history	**pre-existing depression = independent predictor of complications**; depression + substance abuse commonest in polytrauma

3 Presentation

Exam	wound size **may not reflect deep damage** · **ABI if vascular concern** (> 0.9 normal; **< 0.9 → vascular consult + angiogram**)
Traumatic arthrotomy	**saline load test** or **CT (more sensitive for knee)**

4 Diagnosis & Classification

Key radiology findings

Radiographs	joint above + below · **CT** – periarticular injury, **knee arthrotomy (intra-articular air)**

Classification & scoring (Gustilo, Ganga, AO soft tissue)

Gustilo (TO)	Wound	Soft tissue
I	**< 1 cm**, clean	Minimal
II	**1–10 cm**, moderate	**No periosteal stripping**
IIIA	Any size (**high energy**)	**Local coverage possible**
IIIB	–	**Periosteal stripping + exposed bone → flap needed**
IIIC	–	**Arterial injury needing repair**

6 × 4 cm dirty crush wound, distal tibia, exposed bone + periosteal stripping (TO)	**IIIB**
Automatically type III (TO)	**gunshot**, **farm injury**

Gustilo drawbacks (TO) many modifications, wide IIIB spectrum, wound-size based, **doesn't address salvage**, **poor interobserver reliability**

Ganga Hospital Open Injury Score (TO)	Points
Covering (skin + fascia)	1–5 (circumferential skin loss = 5)
Skeletal (bone + joint)	1–5 (**bone loss ≥ 4 cm = 5**)
Functional (musculotendinous + nerve)	1–5 (loss of ≥ 2 compartments / subtotal amputation = 5)
Comorbidities – 2 each	**Debridement > 12 h**, **age > 65**, sewage / farm contamination, **SBP < 90**, compartment syndrome / other major injury same limb, DM / cardiorespiratory, **polytrauma ISS > 25** / fat embolism

Ganga facts (TO) designed for **IIIB**; **scored AFTER debridement** (not in ED before); **max 29**; **better reliability than Gustilo**; **doesn't predict survival** of polytrauma

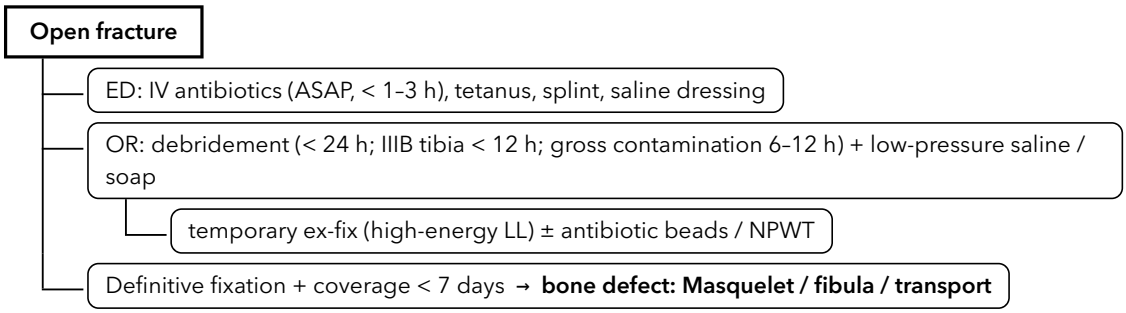
Ganga thresholds (TO) **≤ 14 → salvage** · **15–16 = grey zone** · **≥ 17 → amputation** · ("≥ 14 amputation threshold" in another TO item – read the options)

Ganga component → closure (TO) component score **1–2 → primary closure** · **3–4 → early / delayed flap** · **5 → staged reconstruction** · developed by **Rajasekaran** for **non-ischaemic** limbs (where MESS fails)

AO soft-tissue (TO) **extensive closed degloving = IC4** (IC closed vs IO open)

Tscherne closed / open soft-tissue grading

5 Management



Antibiotic prophylaxis – choice, timing & infection risk

First thing to do (grade III tibia) **IV antibiotics**
✗ Wrong options: bedside I&D · splint · CTA · OR first

Timing **as soon as possible**; **> 3 h → ↑ infection (4.7% → 7.4%)**; **within 1 h of arrival**; **> 66 min independent risk in type III tibia** (OB)

Most modifiable infection risk **delay in antibiotics**
✗ Wrong options: free vs rotational flap · flap at 4 days · nail vs plate · debridement at 15 h · personal transport · reamed nailing

Septic nonunion after 15-h transport delay **prolonged time to antibiotics**
✗ Wrong options: coverage < 7 days · unreamed nail · vitamin D · low-pressure irrigation

– – Active space – –

Multidisciplinary training **↓ time to antibiotics**

Wound	Regimen
Gustilo I / II (e.g. 3 cm wound)	**Cefazolin** (clindamycin / vancomycin if allergic; fluoroquinolone 2nd line)
Gustilo III	**Cefazolin + aminoglycoside** · or 3rd-gen cephalosporin alone · or vancomycin + cefepime
Farm / soil / bowel (e.g. hay loft, lawnmower)	**+ high-dose penicillin → Clostridium**
Fresh water	Fluoroquinolone or 3rd / 4th-gen cephalosporin
Salt water	Doxycycline + ceftazidime or fluoroquinolone

Clostridium **gram-positive, obligate anaerobic, spore-forming rod**, **catalase negative**; C. perfringens (gas gangrene), C. tetani, C. difficile, **C. botulinum (botulism)**

Duration **24 h after injury if closed primarily** · **24 h after final closure** (**48 h type III**) (Orthobullets) · **72 h or until closure** for grade III tibia (TO) · 24–72 h after I&D (Gustilo note) – read the options

Tetanus prophylaxis

Booster within 5 yr **no prophylaxis**
✗ Wrong options: vaccine · TIG

Grade II, no booster since high school **Td** + cefazolin within 3 h + I&D

Toxoid **0.5 mL** any age · **TIG** – < 5 yr 75 U · 5–10 yr 125 U · > 10 yr 250 U · **separate syringes + sites, IM**

Depends on vaccination history (3 doses), last dose, wound severity

Debridement as the key infection deterrent

Most important infection deterrent **surgical debridement of devitalised tissue**
✗ Wrong options: bacitracin · NPWT · 6-h window · pulse lavage

6-hour rule **not supported** · **delayed debridement doesn't ↑ infection if antibiotics given** (TO) · **< 24 h** (GOLIATH: type III); **IIIB tibia < 12 h**; gross contamination **6–12 h**

Technique extend wound longitudinally; **remove bone fragments without soft-tissue attachment**; debride **beyond** zone of injury; repeat every **24–48 h**

Exception to I&D **low-energy puncture open fracture in children**

ED remove gross debris, **don't remove bone**, saline-soaked dressing; **no aggressive / antiseptic irrigation in ED**

Most common source of infection (TO) **hospital-acquired**
✗ Wrong options: site of injury · soil · haematogenous

Infection risk factors (TO) **inadequate skeletal stabilisation**, organic contamination, poor debridement, dead space, debridement > 24 h, shock, ARDS, age > 65, DM, smoking, compartment syndrome, malnutrition

Irrigation — pressure & solution (FLOW)

Pressure	**low-pressure (bulb / gravity) = high-pressure** for **infection + union**; **cheaper**; pulse lavage damages bone, pushes bacteria deep, ↑ rebound contamination at 48 h ✗ Wrong options: nerve injury · wound problems
Solution	**saline (± castile soap)**; **soap < antibiotic solution for wound-healing problems** · **saline = betadine** · peroxide / iodine / alcohol harm osteoblasts + cartilage
Volume	**3 L per Gustilo grade** (**9 L for type III**)
After debridement of IIIB	**saline, low-pressure gravity flow**
Low-pressure parameters (TO)	14 psi / 550 pulses·min^{-1} vs high 70 psi / 1,050

Negative pressure wound therapy

Benefits	**↑ granulation tissue**, **↑ angiogenesis**, ↓ oedema, ↑ wound-bed flow, ↓ complex reconstruction, **outpatient** care ✗ Wrong options: **↓ angiogenesis** · ↓ bacterial count · ↓ hospital stay · fewer debridements
NPWT vs conventional (WOLLF)	**higher cost**; no difference in deep infection, amputation, LOS, disability at 1 yr
Not an indication (TO)	**ulcer with necrotic wound bed** ✗ Wrong options: necrotising fasciitis post-debridement · large diabetic ulcer · dehiscence
Open > 7 days	↑ infection

Local antibiotics

Beads / spacer	PMMA + **heat-stable vancomycin / tobramycin**; heavy contamination, large soft-tissue or bony defect

Temporising spanning external fixation

Polytrauma, open distal femur + proximal tibia	**I&D + knee-spanning ex-fix** ✗ Wrong options: traction · I&D alone · ORIF · nails
Child (10 yr), open tibia, gravel	**I&D + ex-fix** · flexible nails also an option ✗ Wrong options: reamed nail (physes) · K-wires · cast · submuscular plate
Definitive internal fixation vs ex-fix	**IF: faster union, better function, shorter stay**; **definitive ex-fix → poor function (LEAP)**

Soft-tissue coverage timing — flap within 7 days

Target	**< 7 days**; **infection odds +16% per day beyond day 7** · flap **> 7 days → ↑ amputation, osteomyelitis, pneumonia** (DVT same)
Older thresholds	≤ 72 h (early studies, OB218814) · **LEAP: no difference at 3 days** · closure > 5 days risk (one OB item)
Flap by tibial level	**proximal ⅓ → gastrocnemius** · **middle ⅓ → soleus** · **distal ⅓ → free flap (within 7 days)**

– – Active space – –

ORIF before / at / after fasciotomy closure	no difference

Bone defects – induced membrane (Masquelet) & vascularised fibula

Masquelet stage 1	**debridement + cement spacer + temporary fixation**
Stage 2 at 4–6 wk	**remove spacer, PRESERVE membrane, graft into cavity (around retained nail)** ✗ Wrong options: debride membrane · exchange nail · remove nail · spacer only
Membrane biology	BMP-2, **VEGF, IL-6, type I collagen** – **peak at 1 month (~4 wk)**, baseline ~6 months · defects **up to 25 cm**
7 cm tibial defect, refuses transport	**ORIF + vascularised fibular autograft** (osteogenic + inductive + conductive) ✗ Wrong options: DBM · calcium phosphate · cortical allograft (not osteogenic) · amputation
Other	distraction osteogenesis (transport)

Aseptic nonunion after nailing – exchange nailing

"Elephant foot", gaps on 4 cortices after unreamed nail	**hypertrophic nonunion = lack of stability** ✗ Wrong options: infection · blood supply (atrophic) · vitamin D · OI
Treatment	**reamed exchange nailing** (stiffness ∝ diameter⁴, isthmic fit, reaming graft) ✗ Wrong options: observe · antibiotic nail · new locking screws · unreamed exchange
Nonunion rates	**up to 17% closed, 40% open** tibia
SPRINT	reamed benefit **only in closed** tibia; open: reamed = unreamed

6 Complications

Complication	Fact
Fracture-related infection	**< 1% type I → 30% type III**
Osteomyelitis	**1.8–27%**; **tibia commonest**
Risk factors	**Blast mechanism**, acute amputation, **coverage > 7 days**, higher Gustilo, contamination, **tibial location**
Depression	**Commonest post-op issue** after open tibia (✗ Wrong options: arthritis · malunion)

Outcomes & infection risk factors (blast, definitive ex-fix, location)

↑ Infection	**blast mechanism** ✗ Wrong options: coverage 3–7 days · NPWT · debridement 6–24 h · grafting > 3 months
Poor function in type III tibia	**definitive external fixation** ✗ Wrong options: coverage < 3 days · grafting at 3 months · debridement 6–24 h · free flap

Ballistic fracture – highest infection	**tibial shaft** (subcutaneous) · **blunt > penetrating** infection ✗ Wrong options: radius, femur, metacarpal, humerus

Psychological sequelae – depression

Pre-existing depression	**independent predictor of complications**, ↑ readmission; psychiatric comorbidity **up to 40%**; ortho patients **less** likely to get psych follow-up
Predictor of depressive symptoms	**previous psychiatric diagnosis** (also unemployment) ✗ Wrong options: employment · implant · male · private insurance

Jehovah's Witness – bloodless resuscitation

Unstable intra-op, GCS 8, open pelvis	**plasma expanders** ✗ Wrong options: blood · cell saver · ask family · ethics delay
Prohibited	allogeneic blood, pre-donated autologous · **personal choice** – cell salvage, haemodilution, EPO, dialysis

X vs Y

	Low-pressure	High-pressure lavage
Infection / union	Equal	Equal
Cost	**Lower**	Higher
Tissue damage	Less	**Bone + soft tissue damage**

	Gustilo	Ganga
Timing	Intra-op	**After debridement**
Purpose	Grade + antibiotics	**Salvage vs amputation (IIIB)**
Reliability	Poor	**Better**



Type IIIA open tibia fracture

How it's asked

"Grade 3 tibia antibiotic duration (TO)?"	→ **72 hours**
"Lawnmower soil wound – penicillin organism true?"	→ **may cause botulism (Clostridium)**
"3 cm wound, exposed bone – antibiotic?"	→ **cefazolin**
"Factor increasing infection (10 h to hospital)?"	→ **delay in IV antibiotics**

– – Active space – –

"Farm injury, 0.5 cm wound?"	→ cephalosporin + aminoglycoside + penicillin
"Penicillin covers?"	→ Clostridium
"Septic nonunion — associated factor?"	→ prolonged time to antibiotics
"Independent infection risk (zip line)?"	→ timing of antibiotics
"Multidisciplinary training effect?"	→ ↓ time to antibiotics
"IIIB — factor most ↑ infection?"	→ no prompt antibiotics
"Grade II, no recent booster?"	→ I&D, cefazolin within 3 h, Td
"First thing for grade III tibia?"	→ IV antibiotics
"Booster 12 months ago?"	→ no tetanus prophylaxis
"Irrigation after debridement?"	→ saline, low-pressure gravity flow
"High vs low pressure — true?"	→ no difference in union
"Shotgun hand — accurate?"	→ saline = betadine
"True about debridement (TO)?"	→ delay doesn't ↑ infection with antibiotics
"Most important infection deterrent?"	→ surgical debridement of devitalised tissue
"Not an NPWT indication (TO)?"	→ ulcer with necrotic bed
"NPWT benefits EXCEPT?"	→ decreases angiogenesis
"NPWT vs damp-to-dry advantage?"	→ accelerated granulation tissue
"NPWT vs conventional — true?"	→ higher cost
"Flap > 7 days?"	→ ↑ amputation
"Distal ⅓ tibia coverage?"	→ free flap within 7 days
"Masquelet stage 2?"	→ remove spacer, preserve membrane, graft cavity
"Masquelet membrane — true?"	→ 1-month membrane has highest IL-6
"7 cm defect, no transport?"	→ ORIF + vascularised fibula
"Masquelet stage 1?"	→ debridement, cement spacer, temporary fixation
"True about compound fractures (TO)?"	→ inadequate stabilisation ↑ infection
"Commonest source of infection (TO)?"	→ hospital-acquired
"Poor function in type III tibia?"	→ definitive ex-fix
"↑ Infection risk (blast soldier)?"	→ blast mechanism
"Ballistic fracture with highest infection?"	→ tibial shaft
"10-yr-old open tibia with gravel?"	→ I&D + ex-fix
"Polytrauma, open distal femur + proximal tibia?"	→ I&D + spanning ex-fix

"Elephant-foot nonunion — cause?"	→ **lack of stability**
"Hypertrophic nonunion, broken locks?"	→ **reamed exchange nailing**
"Open tibia — highest post-op risk?"	→ **depression**
"Psychiatric comorbidity — true?"	→ **pre-existing depression predicts complications**
"Depression risk factor?"	→ **previous psychiatric diagnosis**
"Jehovah's Witness unstable?"	→ **plasma expanders**
"Ganga — false (TO)?"	→ **predicts polytrauma survival**
"6 × 4 cm crush, exposed bone (TO)?"	→ **IIIB**
"Closed extensive degloving AO (TO)?"	→ **IC4**
"Ganga — EXCEPT (TO)?"	→ **scored in ED before debridement**
"Ganga grey zone (TO)?"	→ **15–16**

– – Active space – –

OSTEOMYELITIS - ADULT

1 Basics

Epidemiology & sites

Host	Typical site
Dialysis	**Spine + ribs**; **sternoclavicular joint**
IV drug use	**Medial / lateral clavicle**; SC / AC joint (**Pseudomonas**)
Diabetes	Foot, decubitus ulcers
Adult haematogenous	**Vertebrae (commonest)**

Risk factors trauma / surgery, immunocompromise, IVDU, poor vascularity, DM, sickle cell, neuropathy, malnutrition

Mechanism

Route	Fact
Haematogenous	**Commonest in children**; **vertebrae in adults**; **S. aureus**
Contiguous	Surgery, trauma, wounds, **poor vascularity**; bacterial > mycobacterial / fungal
Direct inoculation	Penetrating, **open fracture**, surgical contamination

Organisms

Age / setting	Organism
Neonate (< 4 mo)	S. aureus, Enterobacter, **GAS + GBS**
4 mo–4 yr	S. aureus, GAS, **Kingella kingae**, Enterobacter
4 yr–adult	**S. aureus (80%)**, GAS, H. influenzae
Adult	**S. aureus** ± Enterobacter / Strep
Sickle cell	**Salmonella** (S. aureus still commonest)
IVDU (SC / AC joint), nail through rubber sole	**Pseudomonas**
HIV + cat scratch	**Bartonella**
Immunosuppressed, long IV lines, TPN	Fungal
Pott's disease	TB

Mycotic (infected) aneurysm (TO) **S. aureus** (+ Salmonella); "mycotic" ≠ fungal
✗ Wrong options: H. influenzae · Strep viridans · Pseudomonas

Biofilm & bacterial growth stages

Stage	Fact
Planktonic	**Free-floating** bacteria **attach to inert substrate** + apoptosis → biofilm matrix
Sessile	**Quorum sensing** → matures; **no-growth** state; virulence genes; **antibiotic resistant**
Dispersion	Releases planktonic bacteria

Biofilm — **extracellular polymeric substance / exopolysaccharide (glycocalyx)**; poor antibiotic penetration + ↓ metabolism · **PJI > 4 wk → explant**

Metaphyseal osteomyelitis & joint spread

Intracapsular metaphyses (TO) — **proximal radius**, **proximal humerus**, **proximal femur**, **distal tibia / fibula** → septic arthritis · **NOT head of ulna**

3 Presentation

Symptoms — pain; **fever (acute)**; sepsis signs

Draining sinus — **chronic**; **probe to bone → chronic osteomyelitis** (diabetic ulcer probing bone = osteomyelitis, not Charcot – improves with elevation)

4 Diagnosis & Classification

Key radiology findings

Study	Fact
Radiographs	Lag **~2 wk**; need **50% bone loss**; chronic: **sequestrum** (dead bone, nidus), **involucrum** (new bone around), **Brodie abscess**, sclerosis, periosteal reaction
CT	Necrotic bone, planning (hardware artifact)
MRI	**Best for early diagnosis + localisation**; **negative rules out**; **may overestimate extent**; T2 oedema
Penumbra sign (T1)	**Bright rim** (vascularised membrane) around dark abscess, dark sclerotic outer rim → **intraosseous abscess → surgery**
Tc bone scan	Normal X-ray + no MRI; **sensitive, not specific**; negative rules out
Gallium	Diabetic foot / no MRI; **cellulitis false positive**

– – Active space – –



Penumbra sign on T1 MRI

Genes & lab markers

Test	Fact
WBC	**Raised in only ⅓** of acute
ESR	Raised **~90%**; **fall = favourable**
CRP	**Most sensitive (97%)**; falls faster than ESR
Blood cultures	Often negative; guide haematogenous cases
Sinus tract culture	**Unreliable** (contaminants)
Bone culture	**Gold standard** for antibiotics (off antibiotics)

Histology — **acute: live osteocytes + neutrophils** · **chronic: empty lacunae, marrow fibrosis, lymphocytes + plasma cells** (plasma cell = chronic)

Most accurate diagnosis vs Ewing (TO) — **bone biopsy** · **"biopsy all infections, culture all tumours"**
✗ Wrong options: indium scan · MRI (next step) · LDH

Classification

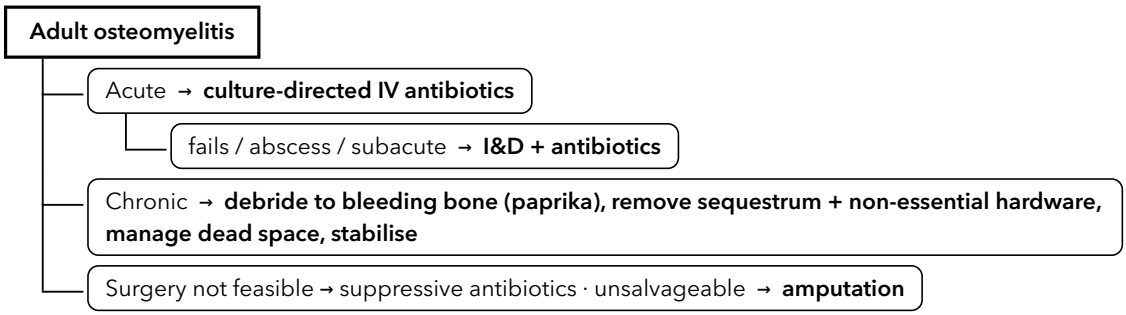
Timing — acute **< 2 wk** · subacute 1–several months · chronic > several months

Cierny-Mader stage (anatomic)	Extent
1	**Medullary**
2	**Superficial (cortex only)**
3	**Localised** – cortex + medulla, **stable**
4	**Diffuse** – full thickness, **unstable**

Host	Status
A	Normal
BL / BS	**Locally / systemically compromised** (smoking, venous stasis)
C	**Treatment worse than infection**

Cierny-Mader uses — **host status + extent of infected bone** · e.g. medullary, smoker + venous stasis → **1B**

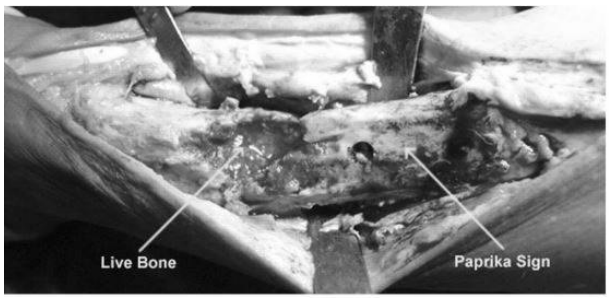
5 Management



Chronic post-traumatic osteomyelitis — debridement, hardware removal & local antibiotics

Principles	remove **all necrotic tissue + sequestrum**; debride to **punctate bleeding ("paprika sign")**; **remove non-essential hardware**; **stability required** (**ex-fix preferred**); dead-space management; culture-directed antibiotics; often **staged**
Dead space	vascularised bone graft, local / free flaps, **PMMA antibiotic beads**, VAC, Masquelet, Ilizarov
Draining sinus 18 months after GSW	**surgical debridement + culture + IV antibiotics** ✗ Wrong options: oral · needle biopsy + IV · tumour work-up
Infected nonunion with hardware	**I&D, remove hardware, over-ream canal, ex-fix + culture-directed antibiotics** ✗ Wrong options: suppression · AKA · NPWT without stability · retain hardware + acute graft
MRSA infected nonunion with nail + sinus	**debride, excise sinus (rule out Marjolin), exchange to antibiotic-cement nail + 6 wk IV** ✗ Wrong options: retain nail · larger nail · remove without stabilising · add ring fixator
Localised, healed tibia, low-virulence S. aureus	**curettage + absorbable antibiotic beads + IV 1–4 wk + oral 6 wk** ✗ Wrong options: segmental resection · amputation · free flap
Stable patient, multiple sinuses after forearm ORIF	**scheduled (non-urgent) I&D + hardware removal** ✗ Wrong options: suppression · sinus culture · MRI · urgent admission
Antibiotic nail	PMMA + **vancomycin + tobramycin**; **peak elution 24 h**; elutes **up to 4 months** · tobramycin concentration-dependent, vancomycin time-dependent
Prognosis better	**metaphyseal > diaphyseal** · worse: polymicrobial, **MRSA**, contralateral amputation, poor host
Medical failure	infection **> 3 months**, multidrug-resistant, polymicrobial ✗ Wrong options: CRP > 20 · gram-positive · long bone
Adjuncts	**suppressive antibiotics** (high relapse when stopped) · **hyperbaric O_2** (refractory)
Recurrence	**~30%** despite treatment

– – Active space – –



Paprika sign — debridement to bleeding bone

Negative pressure dressing mechanism

Benefits	removes inhibitory interstitial fluid, protein + electrolytes; ↓ slime / anaerobic colonisation; arteriolar dilation → granulation; **↓ capillary afterload → ↑ inflow**; draws edges in; **VEGF release** · **EXCEPT "↑ capillary afterload"**

Vertebral osteomyelitis & discitis

Dialysis, back pain, ↑ ESR, MRI: disc T2 bright + T1 dark, endplate enhancement	**osteomyelitis + discitis** ✗ Wrong options: lymphoma · renal osteodystrophy · metastasis (disc spared) · osteoporotic fracture
MSSA bacteraemia, L3–4, neuro intact	**IV antibiotics + lumbar orthosis** ✗ Wrong options: biopsy (organism known) · repeat MRI · surgery

CRMO

CRMO / SAPHO (TO)	**associated with skin lesions** (palmoplantar pustulosis, pustular psoriasis); **no organism**; autoimmune; **antibiotics no role**

Variant (TO)	Who	Site
Sternocostoclavicular hyperostosis	**Men 40–50**	Sternum, medial clavicles, spine
Subacute recurrent multifocal	**Children + adolescents**	Metaphyses (distal femur, tibia), medial clavicle

6 Complications

Marjolin ulcer (squamous cell carcinoma)

Marjolin ulcer	**SCC (~90%)** in a **chronic draining sinus**; **~1%** of chronic osteomyelitis; tibia > femur; growing tissue + **foul smell**
Precursor	**chronic osteomyelitis** ✗ Wrong options: adamantinoma · OFD · EG · cortical desmoid
Treatment	**wide resection** (Orthobullets) · **amputation** usually (OB) — read the options
Other	persistence, sepsis, amputation

X vs Y

	Sequestrum	Involucrum
What	**Dead bone — nidus**	**New bone around it**
Treatment	**Remove**	—

	Acute	Chronic
Histology	**Neutrophils**, live osteocytes	**Plasma cells**, empty lacunae, fibrosis
Treatment	Antibiotics ± drainage	**Debridement** essential

How it's asked

"Better prognosis in chronic tibial osteomyelitis?"	**→ metaphyseal infection**
"Medullary, smoker + venous stasis?"	**→ stage 1B**
"Cierny-Mader considers?"	**→ host status + extent of bone**
"Draining sinus after GSW?"	**→ debridement, culture, IV antibiotics**
"Infected nonunion with hardware, 2 yr?"	**→ I&D, remove hardware, over-ream, ex-fix, culture-directed antibiotics**
"MRSA sinus after nail, 4 yr?"	**→ debride, excise sinus, antibiotic cement nail, 6 wk IV**
"Localised chronic tibial OM 42 yr later?"	**→ curettage + absorbable beads + short IV + oral**
"Arrow on CT (dead bone)?"	**→ sequestrum**
"Forearm sinuses after ORIF, stable?"	**→ scheduled I&D + hardware removal**
"Tissue growing from 35-yr sinus?"	**→ squamous cell carcinoma**
"Precursor of SCC over tibia?"	**→ chronic osteomyelitis**
"Peak elution of antibiotic nail?"	**→ 24 hours**
"Free bacteria attaching to substrate?"	**→ planktonic**
"Dialysis patient with disc destruction?"	**→ osteomyelitis + discitis**
"MSSA discitis, neuro intact?"	**→ lumbar orthosis + IV antibiotics**
"Most accurate test (TO)?"	**→ bone biopsy**
"Dialysis + SC joint swelling?"	**→ chronic osteomyelitis**
"Diabetic ulcer probing to bone?"	**→ osteomyelitis**
"Needs surgery?"	**→ penumbra sign on MRI**
"Post ex-fix wrist infection image?"	**→ Brodie's abscess**
"Cell suggesting chronic OM?"	**→ plasma cell**
"NPWT benefits EXCEPT?"	**→ ↑ capillary afterload**
"OM → septic arthritis EXCEPT (TO)?"	**→ head of ulna**

– – Active space – –

"Mycotic aneurysm organism (TO)?"	**→ Staphylococcus aureus**
"CRMO – true (TO)?"	**→ associated with skin lesions**

OTHER TRAUMA TOPICS

– Trauma · 44 / 67 –

1 Basics

Transfer to Level 1 trauma centre

Associated with transfer	**multiple medical comorbidities** ✗ Wrong options: female · private insurance · Caucasian · ISS < 10
Unnecessary transfers associated with	**African-American race, comorbidities, Medicaid, male sex** · severe polytrauma → better outcomes at Level 1

4 Diagnosis & Classification

Associated Lisfranc injury

Cuboid compression ("nutcracker") fracture	twisting, midfoot swelling + ecchymosis → **look for subtle Lisfranc injury** ✗ Wrong options: peroneal tear · lateral process talus · talar neck · deltoid

5 Management

Common peroneal nerve palsy – tendon transfer

Lacerated CPN, 1 yr, no anterior / lateral function, posterior compartments 5/5	**split posterior tibial tendon transfer** ✗ Wrong options: split anterior tibial (denervated) · gastrocsoleus recession (no contracture) · subtalar fusion (doesn't fix drop) · FHL transfer

Nonunion – compression plating

Hypertrophic forearm nonunion (1 yr)	**compression plating alone** (stability problem, biology intact) ✗ Wrong options: bracing · ICBG (atrophic) · reamed IMN (not used in forearm) · stimulator (atrophic adjunct)

Masquelet induced-membrane technique

Stage 1	debride, **antibiotic cement spacer**, provisional fixation (ex-fix or internal) → 6 wk culture-specific antibiotics if infected, repeat stage 1 as needed

Stage 2 at 4–6 wk (peak **BMP, VEGF, IL-6**) – incise membrane, remove spacer, **autograft (RIA or ICBG – equivalent)**, close membrane, definitive fixation

True	**induced membrane prevents graft resorption** ✗ Wrong options: large articular defects (use allograft / prosthesis) · ICBG > RIA · stage 2 at 8 wk · antibiotics disrupt membrane
Uses	tibia (original, 2000), femur, humerus, **forearm** (e.g. GSW defects)

– – Active space – –

X vs Y

	Hypertrophic nonunion	Atrophic nonunion
Problem	**Instability**	**Biology**
Treatment	**Compression plate**	Plate + **graft** ± stimulator

How it's asked

"Cuboid nutcracker – problematic associated injury?"	**→ Lisfranc injury**
"CPN laceration, PTT 5/5?"	**→ split posterior tibial tendon transfer**
"Hypertrophic forearm nonunion?"	**→ compression plating**
"Masquelet – true?"	**→ membrane prevents graft resorption**
"Factor associated with Level 1 transfer?"	**→ multiple medical comorbidities**

PATELLA FRACTURE

– Trauma · 45 / 67 –

1 Basics

Epidemiology

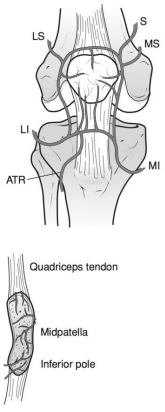
Incidence	**1% of skeletal injuries** · **6–9% open** · M:F 2:1 · age 20–50
Associated	femoral neck fracture, **posterior wall acetabulum**, knee dislocation (dashboard)

Anatomy

Patella	**largest sesamoid**; cartilage on **superior ¾** (up to 1 cm), **inferior ¼ non-articular**; **lateral facet larger**
Function	↑ extensor power **30–50%** (moves line of pull anterior)
MPFL	medial epicondyle ↔ adductor tubercle → upper ⅔ medial patella; restrains lateral translation **0–30°**
Quadriceps tendon layers	superficial **rectus femoris** · middle **VM + VL** · deep **vastus intermedius**
Retinaculum	fascia lata, VM, VL → **repair with fixation**

Patellar blood supply

Ring	genicular arteries (descending / supreme, superior + inferior medial + lateral, anterior tibial recurrent) → **mid-patellar + polar systems**
Dominant supply	**distal (inferior) pole**, **inferomedial** quadrant → its excision **most ↓ perfusion** ✗ Wrong options: superior pole · superolateral (least)



Blood supply of the patella

– – Active space – –

2 Mechanism

Mechanism	Pattern	Retinaculum
Direct blow (dashboard, fall)	**Comminuted**, chondral damage	**May be intact**
Indirect eccentric quad contraction (flexing against contracted quads)	**Transverse / inferior pole avulsion**	**Torn (typical)**
Child **8–10 yr**	**Sleeve fracture**	High suspicion

3 Presentation

Exam palpable defect, haemarthrosis; **can't straight-leg raise → extensor mechanism + retinaculum disrupted** (aspirate + lidocaine if pain limits)

Saline load rule out joint communication (open)

4 Diagnosis & Classification

Key radiology findings

View / study	Fact
Lateral (30° flexion)	**Best for transverse**; step-off
Axial (sunrise / Merchant)	**Best for vertical**
Patella alta (Insall-Salvati > 1.0)	**Patellar tendon** disruption
Patella baja (< 1.0)	**Quadriceps tendon** disruption
CT	**Distal pole comminution (sagittal)**, stress fracture, nonunion; **changes plan in 50%**
MRI	Not typically needed

Classification

Pattern	Note
Nondisplaced	–
Displaced	**Step-off > 2–3 mm** or **gap > 1–4 mm**
Transverse · pole / sleeve · **vertical** · marginal · osteochondral · comminuted (stellate)	–

AO / OTA 34 **A extra-articular · B partial articular · C complete articular**

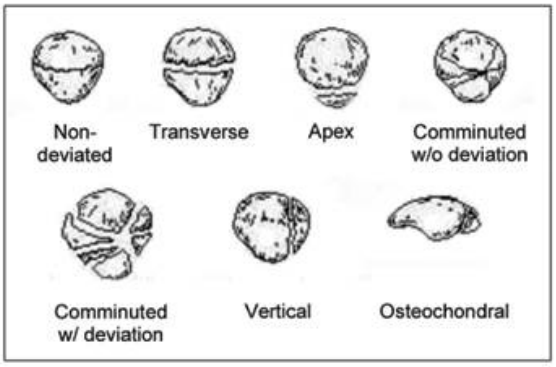


***Figure 1** - Classification of patellar fractures*

Patella fracture patterns

Bipartite patella

Bipartite	**failure of fusion of ossification centres**; **2–3%**; **superolateral**; **bilateral 50%**; **smooth corticated borders**; SLR intact ✗ Wrong options: failed apoptosis (discoid meniscus) · eccentric contraction · direct blow

5 Management

Patella fracture

- SLR intact + step-off < 2–3 mm + gap < 3 mm (or vertical) → **brace / cast in extension, WBAT**
- SLR lost / step-off > 2–3 mm / gap > 3 mm / open / osteochondral → **ORIF (keep patella)**
  - comminuted extra-articular inferior pole < 40–50% → **partial patellectomy + tendon advancement**
- Unsalvageable comminution / infection / tumour → **total patellectomy (+ VMO advancement)**

Operative criteria

Nonop	ORIF
Intact extensor mechanism (SLR)	**Extensor failure (no SLR)**
Step-off < 2–3 mm, gap < 3 mm	**Step-off > 2–3 mm**, **gap > 3 mm** (Orthobullets) · gap > 4 mm, step > 2 mm (TO)
Vertical patterns	**Open**, loose bodies, **osteochondral**, **sleeve (child)**
Major comorbidity	–

Most important factor for surgery in minimally displaced (TO)	**ability to extend the knee** ✗ Wrong options: > 2 mm · age · WB
Nonop	immobiliser / brace in extension, **WBAT**; ROM 1–2 wk; resistance at 6 wk; **> 95% good** · brace ~10° flexion avoids baja (TO)

– – Active space – –

Fixation choice — tension band, screws & partial patellectomy

Tension band — **converts anterior tensile force → articular compression**; needs compressible fracture + intact opposite cortex

Construct	Fact
K-wires (0.062) + 18-gauge wire	**Highest migration (K-wires bent only proximally)**, re-operation
K-wires + **suture**	**75% tensile strength** of wire, similar clinically, **less removal**
4.0 mm cannulated screws + wire	**Stronger than K-wires**; **less symptomatic hardware**
Plate (mini-fragment / mesh)	**Biomechanically superior**; comminution, distal pole, osteoporosis; less gapping
Cerclage	Alone or augment

Plate construct associated with — **improved biomechanical stability**
✗ Wrong options: less knee pain · less infection

Technique — midline incision; minimal stripping; retinacular repair; tension wire **superiorly** (more cover); tension in **2 places**; avoid over-tension + prominent screw tips

Post-op — WBAT in hinged brace, **flexion limited to 30° × 4 wk**

Union — **> 95%** any technique (nonunion higher if open)

Partial patellectomy — **comminuted extra-articular inferior pole < 40%** of height (Orthobullets) · **< 50%** (OB) when ORIF impossible · **advance tendon close to articular surface** (prevents tilt) · hyperextension, transosseous tunnels / anchors ± cerclage to tubercle
✗ Wrong options: vertical · bipartite · stellate · chronic quad rupture

Inferior pole, no SLR — **ORIF or excision + patellar tendon advancement** (ORIF preferred; ✗ Wrong options: cast · brace · allograft · cerclage alone)

Total patellectomy — rare (unsalvageable, infection, tumour); **> 50% loss of extensor strength**; **VMO advancement** improves

Osteochondral fracture after patellar dislocation (vertical, TO) — **internal fixation**
✗ Wrong options: arthroscopy · cast · patellectomy

6 Complications

Symptomatic hardware & anterior knee pain

Most common — **symptomatic hardware (up to 50%)**; thin, open fracture, **K-wire TBW → remove after union**

Removal rate — **~18%** elect removal (OB) · **≥ 50% after TBW** (OB) — read the options

Painful kneeling at 1 yr, no OA — **hardware removal**
✗ Wrong options: PF arthritis treatment · MUA · nonunion surgery · steroid

First 6 months — **anterior knee pain** + quadriceps deficit = main impairment
✗ Wrong options: osteonecrosis · refracture

After cannulated screw TBW **implant removal ≥ 50%**; flexion + quad strength reduced, extensor lag may persist, pain worse than norms

Complication	Fact
PF osteoarthritis	**Up to 50%** → TKA
Osteonecrosis	**Up to 25%**, **proximal pole**, asymptomatic, **revascularises by 2 yr** (Orthobullets) · < 5%, inferior pole (OB)
Loss of reduction	0–22%; age, osteoporosis
Nonunion	**< 1–5%**; open
Infection	0–5% (open **10–11%**)
Weakness / extensor lag	Patellectomy, poor retinacular repair
Stiffness	Long immobilisation

Poor prognosis comminution, open, **smoking**

X vs Y

	Patella alta	Patella baja
Insall-Salvati	**> 1.0**	**< 1.0**
Tendon torn	**Patellar**	**Quadriceps**

	Bipartite	Fracture
Edges	**Smooth, corticated**	Sharp
Site	**Superolateral**	Any
SLR	Intact	Often lost

How it's asked

"Deciding factor for surgery (TO)?"	**→ ability to extend the knee**
"Partial patellectomy indication?"	**→ severely comminuted inferior pole**
"Inferior pole, no SLR – best?"	**→ ORIF or excision + tendon advancement**
"Highest hardware migration?"	**→ K-wire TBW bent proximally only**
"Plate construct associated with?"	**→ improved biomechanical stability**
"Painful kneeling 1 yr after TBW?"	**→ remove hardware**
"Most common complication after fixation?"	**→ symptomatic implant**
"Main impairment at 6 months?"	**→ anterior knee pain**
"Cannulated screw TBW – true?"	**→ implant removal ≥ 50%**
"Dominant arterial entry?"	**→ distal pole**
"Excision causing most ischaemia?"	**→ inferomedial quadrant**

– – Active space – –

"Superolateral smooth fragment, SLR intact?"	**→ failed fusion of ossification centres**
"Vertical osteochondral fracture after dislocation (TO)?"	**→ internal fixation**

PELVIC RING FRACTURES

– Trauma · 46 / 67 –

1 Basics

Associated injuries

Orthopaedic	chest up to **63%** · long bones **50%** · spine **25%**
Non-orthopaedic	head + abdomen **40%** · urogenital (**sexual dysfunction up to 50%**) · **> 50%** have other significant injuries
Children (open triradiate)	iliac wing weaker than ligaments → **rami + iliac wing** fractures, rarely surgical

Anatomy

Ring sacrum + 2 innominates; **displacement needs disruption in 2 places**

Ligament	Resists
Symphyseal	External rotation
Sacrospinous	**External rotation**
Sacrotuberous	**Shear + flexion** (and ER)
Posterior SI complex	**Strongest in body**; **more important than anterior**
Anterior SI	ER after floor + anterior fail
Interosseous SI	AP translation
Posterior SI	**Cephalad-caudad displacement**
Iliolumbar	Rotation; augments posterior SI

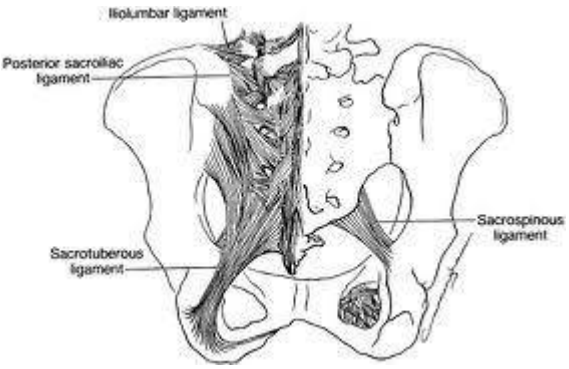
Anterior structures	~**40%** of pelvic stability (OB)
Three stabilising elements (TO)	**ring** · **keystone sacrum** · **suspension bridge** (posterior ligaments)
Load path (TO)	standing: sacrum → SI joints → **sciatic buttress** → acetabulum → femur · sitting: posterior ring → **ischium**
Ligament roles – TO wording (TO)	**sacrotuberous resists vertical shear** · **sacrospinous resists ER** · anterior SI resists ER · **posterior SI = primary stabiliser** · **iliolumbar (L5 TP → ilium)** important in LC
Vessels	internal iliac dives posteriorly at SI joint → **posterior division → superior gluteal** · anterior → obturator · **posterior venous plexus = 80–90% of bleeding**
Nerves	**lumbosacral trunk crosses ala + SI joint**; **L5 root over sacral ala ~2 cm medial to SI joint**

Corona mortis

Anastomosis (TO)	**external iliac (inferior epigastric / accessory obturator) ↔ internal iliac (obturator)** · **NOT internal pudendal**

– – Active space – –

Location behind superior ramus, **3–7 cm** (TO) · mean **6.2 cm** (Orthobullets) from symphysis → risk in **ilioinguinal / modified Stoppa**



Posterior sacroiliac ligaments

3 Presentation

Exam	gentle iliac crest rotation **once** (low sensitivity) · ER limb, LLD · **perineal / scrotal / labial / flank haematoma**, **Morel-Lavallée**
Neuro	**L5 + S1 commonest**; rectal tone; **10–15%** neuro injury
Mandatory	**vaginal + rectal exam** (occult open fracture) · **gross haematuria** commonest GU sign (M 21% vs F 8%)

4 Diagnosis & Classification

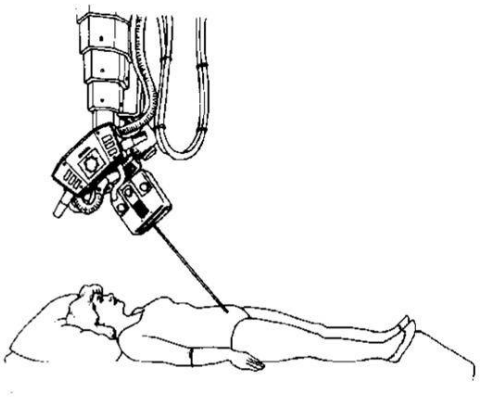
Key radiology findings

View	Beam	Shows
AP	–	ATLS; asymmetry, rotation
Inlet	**~40° caudad** (25–45°); **S1 overlaps S2**	**AP translation**, **internal / external rotation**, **SI widening**, **ala impaction**
Outlet	**~40° cephalad** (up to 60°); **symphysis over S2**	**Vertical translation**, flexion / extension, **sacral foramina + fracture location**
Lateral sacrum	–	**Transverse sacral / U-type → spinopelvic dissociation** (kyphosis)
Single-leg stance (flamingo)	Alternating	**Chronic instability – vertical translation of pubis**
CT	Routine	Posterior ring, comminution, foramina, **sacral dysmorphism**

Instability signs	**> 5 mm posterior SI displacement**, **posterior sacral gap**, **avulsions (ischial spine / tuberosity, sacrum, L5 TP)** · **lateral sacral avulsion = instability (APC)**
Avulsion → ligament (TO)	**L5 transverse process → iliolumbar** (high-energy LC) · **ischial spine → sacrospinous** · **lateral sacral avulsion → sacrotuberous**
AP pelvis (TO)	diagnoses **~90%** · outlet view: **sacral foramina appear round**, best for **vertical displacement** · inlet: AP displacement, rotation, sacral fractures, diastasis

– – Active space – –

Exam caution (TO)	**do not repeat pelvic compression / springing tests** (clot dislodgement) · check perineal sensation, femoral pulses, blood at meatus, vaginal / rectal exam
Isolated rami fractures – next view (TO)	answer key not provided by the source; inlet / outlet views complete the pelvic series
Labs	**Hb, lactate, base excess**



Inlet view technique

Occult instability – stress exam, single-leg stance & lateral views

Binder + early CT	**underestimate severity → fluoroscopic stress exam under anaesthesia (EUA)** ✗ Wrong options: remove binder in unstable patient · thinner CT cuts · Judet
Binder on, CT "normal" but retroperitoneal haematoma	**keep binder; EUA in OR**
APC-I (< 2.5 cm), EUA + CT normal	**protected WB + pain control**

Classifications

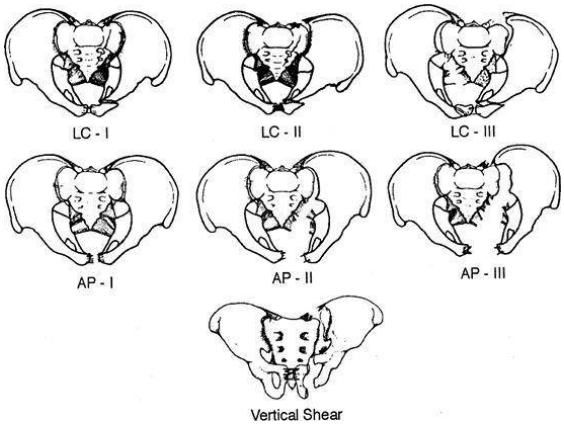
Tile	Description
A – stable	A1 avulsion / iliac wing · A2 stable ring · **A3 transverse sacral (Denis III)**
B – rotationally unstable, vertically stable	**B1 open book (ER)** · **B2 LC (IR)** (B2-1 ipsilateral rami, **B2-2 contralateral = bucket handle**) · B3 bilateral
C – rotationally + vertically unstable	C1 unilateral (iliac / SI / sacral) · C2 one B + one C · C3 bilateral C
Tile subtypes (TO)	A2 iliac wing / anterior arch (rami) · A3 transverse sacral / coccygeal · **C1.1 iliac · C1.2 SI dislocation · C1.3 sacral**

– – Active space – –

Young-Burgess	Injury
APC I	Symphysis **< 2.5 cm**
APC II	Symphysis **> 2.5 cm**, **anterior SI open**, **sacrospinous + sacrotuberous torn**, **posterior SI INTACT**
APC III	**Anterior + posterior SI torn (SI dislocation)**
LC I	**Oblique / transverse ramus + ipsilateral anterior sacral ala compression**
LC II	Rami + **ipsilateral posterior ilium crescent fracture-dislocation**
LC III	Ipsilateral LC + **contralateral APC (windswept)** – rollover, pedestrian
Vertical shear	Posterior + superior force; **highest hypovolaemic shock (63%)**, mortality up to 25%
Combined mechanism (TO)	LC / VS or APC / VS vectors (ejection)

Tilt fracture (displaced superior ramus, women) – **lateral compression**; vaginal exam for open fracture

Sacrospinous + sacrotuberous torn — **APC II**
✗ Wrong options: APC I · LC II (protected by IR)



Young-Burgess classification

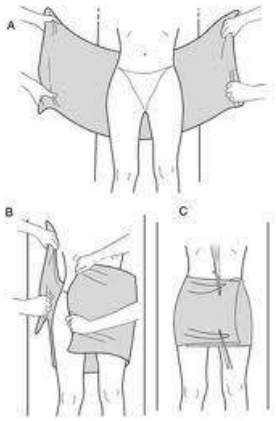
5 Management

Unstable pelvis + haemodynamic instability
- Binder / sheet over greater trochanters + 1:1:1 transfusion
- Positive FAST / laparotomy → **anterior ex-fix (± pelvic packing)**
- Still unstable → **angiography + embolisation (or packing – institutional)**
  - still bleeding → **repeat angiogram (new source 68%)**
- Resuscitated (lactate ≤ 4, pH ≥ 7.25, BE ±5.5) → **definitive fixation**

Haemorrhage control – pelvic binder / sheet & placement

First step in unstable patient — **pelvic binder / sheet**
✗ Wrong options: C-spine film · ex-fix in OR · angiography · packing · IS screws · inlet / outlet first · CT first

Centre over	**greater trochanters** · + IR of legs, tape ankles ✗ Wrong options: iliac crest / ASIS / umbilicus (ineffective, blocks abdomen)
Force	**180 ± 50 N** to stabilise symphysis diastasis (Bottlang)
Controls	**retroperitoneal venous bleeding** (holds **up to 4 L**)
Cautions	skin necrosis (brief "binder holiday" 1–2 h after 24 h) · hypothetical LC over-rotation / bladder injury (no clinical evidence) · **bladder injury is NOT a contraindication**
Iliac wing fracture + superior migration	**skeletal traction** ✗ Wrong options: binder (worsens) · ex-fix
Vertical shear	**skeletal traction + binder**



Pelvic sheet application over the greater trochanters

Haemorrhage — angiography, embolisation, packing & arterial sources

Sources	**venous 80% (Orthobullets) · 90% (OB)**, cancellous bone · **arterial 10–20%** · intra-abdominal up to 40%

Pattern	Artery
APC (posterior ring)	**Superior gluteal**
LC (anterior ring)	**Internal pudendal**, **obturator**

Commonest embolised	**internal iliac** + branches > superior gluteal > obturator > internal pudendal
Laparotomy needed	**ex-fix** (stabilises clot, aids laparotomy) ✗ Wrong options: internal fixation (contamination)
Ex-fix benefit	**maintains stable clot** ✗ Wrong options: rigid fixation (long lever arm)
Packing	**ex-fix first**, then **6–8 cm subumbilical midline**, 3 lap pads each side **below brim** (SI joint → anterior → retropubic)
Class III shock despite binder + 1:1:1, FAST negative	**emergent angiography**
Persistent bleeding after SGA embolisation	repeat angiogram → **new source (68%)** > re-bleed (18%) > both (14%)

– – Active space – –

Haemorrhage – TO (TO)	**80–90% venous** (retroperitoneal plexus) + bone · arterial ~10% · **CTA contrast extravasation (blush) predicts need for embolisation** · **elderly more likely to need angioembolisation**
Pre-peritoneal packing (TO)	persistent instability · **works only with pelvic stabilisation** (ex-fix first) · retropubic tamponade
REBOA / resuscitative thoracotomy (TO)	**only in extremis**, temporary proximal aortic control to buy time
Binder removal (TO)	within **16–24 h** once physiology stable → replace with ex-fix if still unstable
Pelvic C-clamp (TO)	**posterior ring / vertical shear** → rapid closure of SI diastasis · anterior frames: iliac crest vs **supra-acetabular (better bone)**
Pudendal bleed after splenectomy, lactate > 5	**IR embolisation** before definitive fixation
CTA	**NPV 98–100%** for arterial bleeding
Embolisation complications	**gluteal necrosis, impotence**, ↑ infection (**bilateral nonselective internal iliac**)

Massive transfusion ratio

PRBC : FFP : platelets = 1 : 1 : 1 (↓ mortality); coagulopathy hardest to correct

External fixator pins — iliac crest vs supra-acetabular

	Supra-acetabular (AIIS)	Iliac crest
Purchase	**Superior bone (biomechanically better)**	Thin pillar (gluteal pillar)
Fluoroscopy	**Needed**: **obturator outlet** (entry, teardrop), **iliac oblique** (above sciatic notch), **obturator inlet** (between tables)	**Less reliance** – fast
Future incisions	**Doesn't interfere** with ilioinguinal / lateral window	Interferes
Risk	**LFCN**, hip capsule (**start ≥ 2 cm above joint**)	–
Pin infection	Equal	Equal

Neither gives definitive posterior stability

Ex-fix indications	ER patterns (**APC, VS**), ongoing bleeding, **before laparotomy** · not for ilium / acetabular fracture

Anterior fixation nerve complications (LFCN, femoral nerve)

Supra-acetabular pin / INFIX	**LFCN (lateral thigh numbness) — commonest**
INFIX too deep	**femoral nerve palsy** (weak knee extension, **medial thigh + leg numbness**) · also HO, infection

Iliosacral & transsacral screws — fluoroscopy, L5 root injury & sacral dysmorphism

– – Active space – –

View	Guides
Outlet	**Superior-inferior**
Inlet	**Anterior-posterior**
Lateral sacral	**Iliac cortical density** (anterior safe limit) → avoids **L5**

Errant (anterior) screw	**L5 root → weak great toe extension** ✗ Wrong options: knee extension L4 · plantarflexion S1
AP breach (TO)	answer key not provided by the source; **inlet** shows anterior-posterior position (Orthobullets)
Transiliac screw start	posterior iliac crest at S1 level, behind sacrum (lateral view)
Sacral dysmorphism (20–40%)	**up-sloping ala**, **irregular / elliptical foramina**, **residual S1 disc**, **mamillary processes**, **tongue-and-groove SI** · **S1 corridor smaller + oblique**, **S2 corridor LARGER** → transiliac-transsacral at S2 · NOT unique: "S1 narrower than S2" · not a contraindication
Anterior sacral retractors (Stoppa / lateral window)	**L4 + L5** roots

Fixation strategy & failure (vertical sacral fracture, symphyseal plates, sequence)

Injury	Definitive treatment
APC I	Nonop, protected WB
APC II	Symphyseal plate / ex-fix **± posterior**
APC III	**Multi-hole symphyseal plate + SI screws** / posterior plate
LC I	Mostly nonop (**WBAT incomplete**; protected if complete / comminuted)
LC II	**ORIF ilium (crescent)**
LC III / VS	Posterior plate / SI screws

APC III stable after embolisation	**multi-hole non-locking symphyseal plate + IS screw** ✗ Wrong options: 2-hole · ex-fix definitive · locked plate + sacral plate
Multi-hole vs 2-hole	**lower failure + malunion**; reoperation equal · **locked = non-locked** stability
APC II anterior + posterior	**↓ anterior plate failure + malunion** (no better outcome)
Most stable for VS	**IS screw + anterior internal fixation**
Highest loss of reduction	**vertical sacral fracture** ✗ Wrong options: anterior fixation type · screw length · number · age
Anterior approach to SI — relative contraindications	**comminuted sacral fracture** (L5 at risk), morbid obesity, iliac crest ex-fix, colostomy
Ipsilateral pelvic ring + acetabulum	**fix ring (SI joint) first**
Subacute malunion (6 wk), flexed / shortened / ER	**anterior ORIF + posterior fixation**

– – Active space – –

S2 alar-iliac vs iliac screws (TO)	alar: deeper start (15 mm, less prominence), no connector, **MORE SI joint violation** (incorrect: "less violation")
Timing (TO)	definitive fixation best at **24–48 h**; otherwise **delay ≥ 7 days**
Reduction tools (TO)	symphysis: **Weber clamp on pectineal eminence** · **Jungbluth clamp** (screw-held) · ramus: 6-hole reconstruction plate · anterior ring reduction realigns sacrum indirectly
Approaches (TO)	anterior ring: **Pfannenstiel**, **modified Stoppa** · posterior: anterior iliac window for SI joint, posterior for sacral / crescent fractures
Iliosacral screws (TO)	**S1 + S2 only**, cannulated, fully threaded for stability
Crescent fracture (TO)	iliac wing fracture + SI joint involvement → **ORIF plate + SI screws**
Sacral U / H / Y fractures (TO)	**lumbopelvic fixation** (highest vertical stability) · **trans-iliac plate** for sacral comminution
Biomechanics (TO)	more fixation points → more stability · **anterior plate > ex-fix for rotation** · **neither controls vertical shear** → posterior fixation essential

Operative criteria

Operative	Nonoperative
Symphysis > 2.5 cm	**APC I < 2.5 cm**, intact posterior ring
SI displacement > 1 cm	**LC I** (< 1 cm posterior displacement)
–	**LC I < 1.5 cm displacement** (TO) – read the options
Sacral displacement > 1 cm	Isolated rami
Rotated / displaced hemipelvis	**Peripartum diastasis < 4 cm** (bed rest + binder)
Open fracture	Children with open triradiate (most)
Peripartum **> 4–6 cm** / chronic pain	–

Open pelvic fracture	aggressive debridement; **diverting colostomy** (perineal / rectal)

Stable injuries – nonoperative management (LC1, children, peripartum, elderly)

Peripartum diastasis (2 wk post-delivery)	**protected WB + binder + observation** ✗ Wrong options: ex-fix · plate · symphysiotomy
LC I + contralateral pilon	**WBAT pelvis side, NWB pilon side**
Child 7 yr, parasymphyseal, posterior stable	**WBAT** · surgery if acetabular / triradiate displacement **> 2 mm**; ex-fix if ring **> 2 cm**
Elderly LC I (incomplete sacral buckle)	**bilateral WBAT** ✗ Wrong options: TDWB · screws
AP I in cowgirl	**nonop + gradual WBAT** ✗ Wrong options: binder long-term · ORIF
LC incomplete sacral, 56-yr	**immediate WBAT** (screws if pain prevents mobilising or complete / comminuted)
LC1 surgery	**↓ short-term pain** (3–12 months) but **↑ narcotics post-op**, not complication-free

Rehab after fixation	weight-of-limb on posterior injury side, full WB other side; bilateral posterior → bed-to-chair

6 Complications

Urological injury, sexual dysfunction & childbirth

GU injury	**12–20%** (males 21%) · ~15% (OB)
Commonest urological	**posterior urethral tear** > bladder rupture (mortality 22–34%)
Urethral injury signs	**blood at meatus**, **high-riding mobile prostate**, haematuria, can't void → **retrograde urethrogram (RUG)** ✗ Wrong options: cystogram · urinalysis · CT · DPL
Site (TO)	**membranous urethra** (classic); commonest adult site distal to membranous; children more proximal
Management	**suprapubic catheter** (relative contraindication to symphyseal plating) · repair rupture before / with fixation
Long-term GU	**urethral stricture (commonest)**, impotence, incontinence, infection
APC / saddle-horn (horse) injury	**sexual dysfunction = commonest long-term sequela** (18 / 20); **males up to 90%**, **dyspareunia 60%** females · **35–56%** overall
Women	**dyspareunia + caesarean section** ↑ · dyspareunia risk: diastasis, bladder disruption, symphyseal plating, residual > 5 mm

Mortality predictors & causes of death

Mortality	**1–15% closed** · **up to 50% open** · isolated pelvic ring **< 15%** · shock: **25–52%**

Pattern	Mortality / cause
APC III	**Highest mortality (~37%)**, blood loss, **transfusion** · APC overall 26%
LC	~13%; **closed head injury = commonest cause of death**
VS	25%
CMI	17.1%
Acetabulum	~1.5%

Best predictor	**shock on presentation** · **transfusion in first 24 h** · also **age > 60**, SBP < 90, ISS / RTS, **> 4 units**, APC III
APC deaths	**pelvic + visceral injury / haemorrhage**

– – Active space – –

Postoperative infection risk (obesity, embolisation)

Strongest predictor — **obesity (↑ BMI)** · also DM, long op, long ICU, **transfusion volume**, GU / abdominal injury, **Morel-Lavallée**, open fracture, **bilateral internal iliac embolisation**
✗ Wrong options: angiography alone · venous injury · ex-fix · early fixation · head injury (HO)

Other	Fact
DVT / PE	DVT **~60%**, PE ~27%, fatal 2% → LMWH; **IVC filter if head injury**
Chronic instability	Rare → flamingo views
L5 injury	Anterior SI screw
Poor outcome	**SI incongruity > 1 cm**, **LLD > 2 cm**, malunion, nonunion, neuro / urethral injury

X vs Y

	APC II	APC III
Posterior SI ligaments	**Intact**	**Torn**
Stability	Rotationally unstable	**Rotationally + vertically unstable**

	Inlet	Outlet
Beam	**Caudad**	**Cephalad**
Shows	**AP translation, rotation, SI widening**	**Vertical translation**, foramina
IS screw guide	**Anterior-posterior**	**Superior-inferior**

How it's asked

"Tile B1 (TO)?"	→ **open book**
"APC II – not disrupted?"	→ **posterior SI ligaments**
"APC II – true description?"	→ **symphysis diastasis, anterior SI + sacrotuberous torn, posterior SI intact**
"LC II radiograph?"	→ **crescent fracture on impact side**
"Sacrospinous + sacrotuberous torn?"	→ **APC II**
"LC III image?"	→ **windswept pelvis**
"Lateral sacral avulsion indicates?"	→ **pelvic instability**
"Tilt fracture mechanism?"	→ **lateral compression**
"Hypotensive rollover, widened symphysis?"	→ **pelvic binder**
"Binder centred over?"	→ **greater trochanters**
"Binder controls?"	→ **retroperitoneal venous bleeding**
"L5 transverse process fracture implies?"	→ **iliolumbar ligament injury (high-energy LC)** (TO)

"Iliosacral screws placed in?"	→ **S1 and S2 only** (TO)
"Laparotomy needed — orthopaedic step?"	→ **external fixation**
"Ex-fix greatest benefit?"	→ **maintains stable clot**
"Iliac wing fracture + superior migration after splenectomy?"	→ **skeletal traction**
"Pelvic packing technique?"	→ **ex-fix then subumbilical lap pads**
"Class III shock despite binder?"	→ **emergent angiography**
"Repeat angiogram after SGA embolisation?"	→ **new arterial source**
"Persistent bleed, pudendal haematoma?"	→ **IR embolisation**
"APC arterial bleeder?"	→ **superior gluteal artery**
"Commonest embolised vessel?"	→ **internal iliac artery**
"Supra-acetabular pin advantage?"	→ **less interference with pelvic incisions**
"Pins in position B advantage?"	→ **superior bone purchase**
"AIIS pin start point view?"	→ **obturator outlet**
"Iliac crest pin advantage?"	→ **less fluoroscopy**
"Nerve at risk with supra-acetabular pins?"	→ **LFCN**
"INFIX nerve injury symptom?"	→ **lateral thigh numbness**
"INFIX, weak knee extension at 1 month?"	→ **medial thigh + leg numbness (femoral nerve)**
"Lateral sacral view protects?"	→ **L5 nerve root**
"Stoppa sacral retractor risks?"	→ **L5 nerve root**
"Lateral sacral view used for?"	→ **percutaneous IS screws**
"IS screw complication?"	→ **weak great toe extension**
"Outlet view guides?"	→ **superior-inferior orientation**
"Transiliac screw start?"	→ **posterior iliac crest at S1 behind sacrum**
"Dysmorphic sacrum — true?"	→ **S2 safe zone larger**
"NOT unique to dysmorphism?"	→ **S1 corridor narrower than S2**
"S2AI vs iliac screws — incorrect (TO)?"	→ **less SI joint violation**
"Fixation failure predictor?"	→ **vertical sacral fracture**
"Most stable construct?"	→ **IS screw + anterior internal fixation**
"Contraindication to anterior SI plating?"	→ **sacral fracture**
"Ring + transverse acetabulum — fix first?"	→ **SI joint**
"Highest loss of reduction?"	→ **vertical sacral fracture**

– – Active space – –

"APC III definitive?"	→ multi-hole non-locking plate + IS screw
"Subacute malunion 6 wk?"	→ anterior ORIF + posterior fixation
"APC II anterior + posterior vs anterior only?"	→ less anterior plate failure
"Anterior fixation — true?"	→ multi-hole lower failure than 2-hole
"Transfusions predict?"	→ mortality
"APC III highest rate of?"	→ transfusion
"LC death cause?"	→ closed head injury
"Most predictive of mortality?"	→ transfusions in first 24 h
"Best predictor of mortality?"	→ shock on presentation
"Isolated pelvic ring mortality?"	→ < 15%
"Highest mortality pattern?"	→ APC III
"Closed head injury death — pattern?"	→ LC II
"Urethral injury site (TO)?"	→ membranous urethra
"Commonest urological injury?"	→ posterior urethral tear
"Saddle horn → erectile dysfunction?"	→ APC injury
"Rule out urethral injury?"	→ retrograde urethrogram
"Female after displaced pelvic fracture?"	→ dyspareunia + caesarean section
"Horse rider groin pain — long-term?"	→ sexual dysfunction
"APC II plated female — long-term risk?"	→ dyspareunia
"Most frequent APC II complication?"	→ sexual dysfunction
"APC I after EUA?"	→ protected WB + pain control
"Alternating single-leg stance views?"	→ pelvic ring instability
"Lateral pelvic view identifies?"	→ spinopelvic dissociation
"Binder, CT negative, retroperitoneal haematoma?"	→ keep binder, EUA in OR
"Binder + early CT effect?"	→ underestimates; EUA
"Flamingo views assess?"	→ vertical translation of pubis (chronic)
"Outlet view shows?"	→ vertical translation
"Postpartum symphysis pain?"	→ protected WB, binder, observation
"LC I + right pilon WB?"	→ WBAT left, NWB right
"7-yr parasymphyseal, stable posterior?"	→ nonop WBAT
"72-yr LC I?"	→ bilateral WBAT
"Cowgirl AP I?"	→ nonop + gradual WBAT
"56-yr LC incomplete sacral?"	→ immediate WBAT

"LC1 surgery – true?"	→ **↓ short-term pain**
"Strongest infection predictor?"	→ **obesity**
"BMI 37 – true?"	→ **higher infection risk**
"↑ infection after angiography?"	→ **bilateral internal iliac embolisation**
"Massive transfusion?"	→ **PRBC, FFP, platelets 1:1:1**
"Corona mortis EXCEPT (TO)?"	→ **internal pudendal artery**

– – Active space – –

– – Active space – –

PROXIMAL HUMERUS FRACTURE NONUNION AND MALUNION

– Trauma · 47 / 67 –

1 Basics

Epidemiology & risk factors

Proximal humerus **4–5% of all fractures**

Nonunion risk	Factor
Fracture	**2-part surgical neck**, **head split**, **displaced tuberosity**
Patient	**Osteoporosis**, CKD, alcohol / steroids, smoking

Malunion origin initial displacement, **secondary displacement**, fixation failure

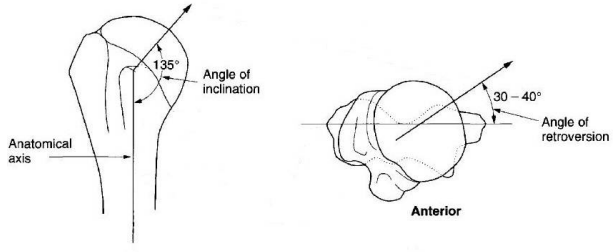
Malunion	Typical position
Head	Varus / valgus, **impacted > 1 cm**, articular incongruity
Greater tuberosity	**Posterior, superior, externally rotated**
Lesser tuberosity	**Medial**

Associated cuff tear, **osteonecrosis**, stiffness, PTOA, instability, **subacromial impingement**

Anatomy

Head spheroidal 90%, ~**43 mm**; **30° retroversion** (to transepicondylar axis); **neck-shaft 130–140°**

Greater tuberosity medial edge **~10 mm lateral** to canal axis · superior edge **~6 mm below** head top



Normal proximal humeral angles

3 Presentation

Symptoms pain, weakness, atrophy

Cuff **GT malunion → weak abduction + ER** · **LT malunion → weak IR** · head malunion → apprehension

Check **axillary nerve**

4 Diagnosis & Classification

Key radiology findings

Views true AP, scapular Y, **axillary** ± apical oblique, Velpeau, West Point

Diagnostic criteria

Malunion	Threshold
Head	**> 45° in any plane**, symptomatic incongruity, **neck-shaft < 120° or > 150°**
Tuberosity	**> 1 cm** from native position

CT bone stock, **glenoid version + stock**, articular injury · **MRI** – cuff / labrum, **deltoid atrophy (axillary nerve)**, LHB, **osteonecrosis**

Labs CBC, ESR, CRP, cultures (infection) · EMG if nerve concern

Classifications

Beredjiklian	Description
I	**Tuberosity malposition (> 1 cm)**
II	**Articular incongruity** (extension, OA)
III	**Articular surface malalignment (> 45°)**

Boileau (fracture sequelae)	Description	Arthroplasty
I	**Head necrosis / impaction**	Anatomic possible (tuberosities preserved)
II	**Chronic dislocation / fracture-dislocation**	Anatomic possible
III	**Surgical neck nonunion**	**Reverse**
IV	**Severe tuberosity malunion**	**Reverse**

5 Management

Proximal humerus nonunion / malunion

- Low demand / painless / can't rehab → **NSAIDs, physio, injection**
- Good bone stock, articular surface + head blood supply intact → **ORIF ± osteotomy ± graft**
- Poor bone stock, incongruity, AVN, chronic dislocation, resorbed tuberosities, cuff tear → **arthroplasty (reverse preferred)**

Malunion – osteotomy vs reverse shoulder arthroplasty

Head deformity minor: **tuberoplasty ± acromioplasty, capsular release, bursectomy** · severe: **varus / valgus ± rotational osteotomy + lateral plate** (young / active; ± strut allograft)

– – Active space – –

Greater tuberosity	**< 1.5 cm → arthroscopic subacromial decompression ± cuff repair · > 1.5 cm → tuberosity osteotomy**
24-yr, 45° varus after ORIF, stiff at 9 months	**revision ORIF + osteotomy** ✗ Wrong options: MUA · resurfacing · hemi · RSA
Reverse vs hemiarthroplasty for sequelae	**greater functional improvement**, **lower complication + reoperation** (less cuff dependence) ✗ Wrong options: more complications · weaker
68-yr active, surgical neck nonunion, osteoporotic	**reverse TSA** (Boileau III) ✗ Wrong options: anatomic TSA · hemi (inferior) · locking plate
Nonop outcome	impacted varus / valgus → good–excellent, **~90%** of normal function
Surgery for malunion	**higher complications** than acute fracture surgery

Nonunion — ORIF with bone graft vs arthroplasty

69-yr, painful nonunion after ORIF	**shoulder arthroplasty** ✗ Wrong options: revision ORIF · valgus osteotomy · fusion · head resection
72-yr, 2-part nonunion, GOOD bone stock, no OA / AVN, tuberosities fine	**ORIF (locked plate) ± bone graft** ✗ Wrong options: CRPP · hemi · graft alone · RSA + latissimus (latissimus gives **ER**, not IR)
Arthroplasty instead when	**marked osteopenia, AVN, resorbed tuberosities, pre-existing cuff tear**
45-yr nonunion after IM nail	**compression plating + bone graft** · union **> 90%** ✗ Wrong options: exchange nail · hemi · RSA (too young) · add screws

6 Complications

After surgery	persistent pain + weakness, stiffness, loss of fixation, infection, bleeding

X vs Y

	ORIF ± osteotomy	Reverse TSA
Bone stock	**Adequate**	Poor
Head	Intact blood supply, congruent	AVN, incongruity, chronic dislocation
Patient	Young / active	Older; Boileau III–IV

How it's asked

"24-yr, 45° varus malunion?"	→ **revision ORIF + osteotomy**
"Reverse vs hemi for malunion?"	→ **greater functional improvement**
"68-yr osteoporotic surgical neck nonunion?"	→ **reverse TSA**
"69-yr painful nonunion after ORIF?"	→ **shoulder arthroplasty**
"72-yr nonunion, good bone stock?"	→ **ORIF ± bone graft**

"45-yr nonunion after IM nail?" → **compression plating + bone graft**

– – Active space – –

– – Active space – –

PROXIMAL HUMERUS FRACTURES

– Trauma · 48 / 67 –

1 Basics

Epidemiology

Incidence	**4–6% of all fractures · 3rd commonest non-vertebral fracture > 65 yr · F:M 2:1**
Commonest pattern	**2-part surgical neck** · ~80–85% minimally displaced
Risk factors	osteoporosis, diabetes, **epilepsy**, female · complexity ↑ with age
Mechanism	elderly low-energy fall (90% in > 60) · young high-energy
Associated	**axillary nerve (commonest)**; arterial injury **5–6%** (older, surgical neck, subcoracoid dislocation; collaterals mask pulses)

Anatomy

Anatomic neck = old physis · **surgical neck** fractures more often · neck-shaft **135°**

Deforming force	Effect
Pectoralis major	Shaft **anterior + medial**
Deltoid	Proximal fragment lateral (abducts – axillary nerve, **posterior cord**)
SS, IS, TM	**GT superior + posterior (ER)**
Subscapularis	**IR articular segment / LT**

Ligaments	SGHL: inferior translation at 0° abduction · **MGHL: AP at ~45°** · **IGHL: AP at 90°**
Head geometry (TO)	articular arc **160°** · radius of curvature **~25 mm** (23–29) · **head–tuberosity distance (HTD) 8–10 mm** · head height **18 mm** (¾ of radius) · NSA **135°** (valgus tolerated, **varus poorly tolerated**) · retroversion **20° (10–40°)** · bicipital groove **~7 mm anterior to medullary axis** = retroversion landmark
Mechanisms (TO)	glenoid compression of head · bending at surgical neck · cuff tension → tuberosity avulsion · bimodal: young males high-energy, older females osteoporotic · **3rd commonest non-vertebral osteoporotic fracture**

Humeral head vascularity & ischaemia predictors (Hertel criteria)

Main blood supply (TO)	**posterior humeral circumflex artery (PCHA) – 64%** (Hettrich) ✗ Wrong options: ACHA (old view) · axillary · profunda
ACHA	**anterolateral branch → arcuate artery** (main GT supply); 1990 study: supplies whole head except posteroinferior GT + head · **disrupted in ~80%** of fractures yet AVN infrequent → PCHA dominant
PCHA	through **quadrilateral space** with axillary nerve

First site of osteonecrosis (TO) **superomedial** head (poorly vascularised)

Hertel predictor	Fact
Calcar (posteromedial metaphyseal extension) < 8 mm	**Most predictive**
Disrupted medial hinge (> 2 mm)	2nd
Anatomic neck fracture	Independent predictor
Angulation > 45° (> 30° in one OB item)	3rd
Head split	Least of these
Displacement > 10 mm, complexity, medial metaphyseal comminution	–

Order of prediction **calcar < 8 mm > medial hinge > angulation > 45° > head split**

NOT predictors (TO) **valgus displacement**, **surgical neck fracture**

Long calcar (> 8 mm) **protective** · Hertel PPV **97%** (TO)

Ischaemia ≠ AVN acute ischaemia **doesn't predict** later necrosis (Bastian & Hertel)

Highest AVN risk **4-part fracture-dislocation** · 4-part head split **21–75%** · valgus-impacted 4-part low (posteromedial periosteum intact) · 3-part valgus impacted 5–10%



Long vs short calcar

Campochiaro 2015 (TO) Hertel criteria **less predictive than quality of reduction** for AVN · anatomical neck fracture **90–100%** AVN

3 Presentation

Exam ecchymosis chest / arm / forearm · **deltoid + lateral shoulder sensation (axillary)** · chest wall

Lesser tuberosity fracture **posterior dislocation until proven otherwise** (seizure, **electrocution** – subscapularis avulsion)

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

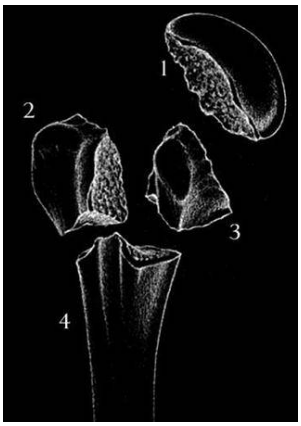
Study	Fact
Trauma series: true AP (Grashey), scapular Y, axillary	**Axillary view mandatory** (→ Velpeau if painful → CT)
Apical oblique, Velpeau, West Point	Additional
Combined cortical thickness (medial + lateral)	**> 4 mm → good screw purchase**; **< 4 mm precludes fixation (TO)** → sling / transosseous suture / hemi
Pseudosubluxation	Inferior head (haemarthrosis + atony)
AP in **IR**	LT in profile; **lightbulb sign** (posterior dislocation)
CT	Head / GT position, comminution, **head split**
MRI	Rarely; cuff tear

Classification

Neer — 4 segments: **GT, LT, articular surface, shaft**; part = **> 1 cm displacement or > 45° angulation**

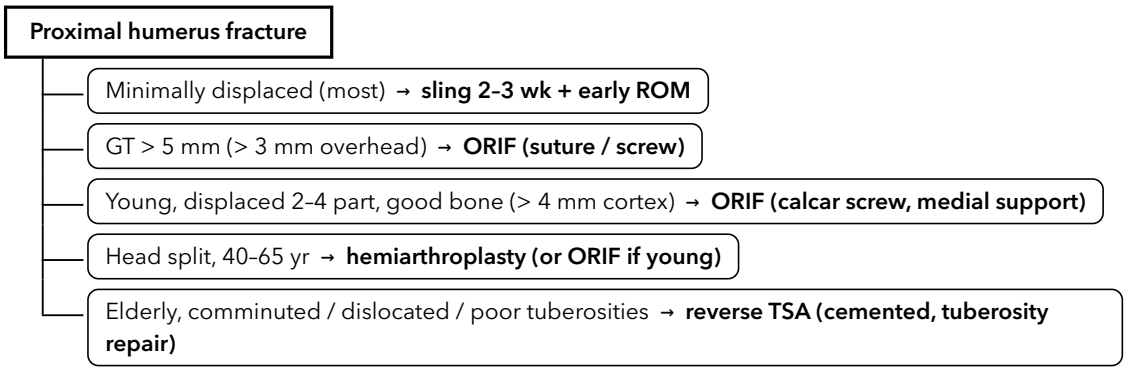
AO / OTA — location, surgical neck status, dislocation

Codman (TO) — 4 parts: head · GT · LT · shaft (proximal to pec major insertion) · AO/OTA **11-A extra-articular unifocal · 11-B extra-articular bifocal · 11-C intra-articular** (TO)



Neer four-part classification

5 Management



Abbreviations ORIF open reduction and internal fixation · CRPP closed reduction percutaneous pinning · RSA reverse shoulder arthroplasty · GT / LT greater / lesser tuberosity · MRI magnetic resonance imaging

Nonoperative management & natural history

Indications minimally displaced, **GT < 5 mm**, poor surgical candidates; consider age, bone, dominance, health

Technique **sling 2–3 wk + immediate PT / pendulum + elbow ROM within 1–2 wk** · early ROM **doesn't displace**
✗ Wrong options: immobiliser 6 wk · hanging cast 6–8 wk

Minimally displaced surgical neck **sling + recheck X-ray at 1 wk**

One-part fracture **early structured PT**; union **> 90%**; at 1 yr **forward flexion restored, IR + ER slightly ↓, no functional deficit**; most return to work

Impacted varus fractures (Court-Brown) **age = main outcome factor** · union 93% at 1 yr, **smoking ↑ nonunion 5.5×**
✗ Wrong options: angulation · PT · dominance · bone

Varus posteromedial vs valgus impacted nonop (TO) **varus posteromedial better** (TO)

Surgical indications **head split**, irreducible dislocation, GT > 5 mm, young, **+ humeral shaft fracture**
✗ Wrong options: age > 70 · comorbidity

10 months, can't elevate, motion at fracture **nonunion** (surgical neck commonest)

PROFHER (JAMA 2015) (TO) no long-term difference surgery vs sling in selected displaced fractures

Neer rehab (TO)	Activity
0–3 wk	Pendulums, avoid ER
3–9 wk	Active-assisted elevation after radiographic healing
>9 wk	Active + strengthening

– – Active space – –

Greater tuberosity fractures — displacement thresholds & fixation

Threshold	**> 5 mm (general)**, **> 3 mm overhead / active** (OB, Orthobullets) · **> 1 cm** or > 20% head overlap; elderly tolerate **2 cm / 40%** (TO) — read the options
Incorrect statement (TO)	"**GT fixation required for ≥ 5 mm in active young**" (TO uses > 1 cm)
> 5 mm residual	**impingement → loss of abduction + ER**, **altered cuff mechanics** (✗ Wrong options: nonunion · AVN (rich supply)) (✗ Wrong options: IR loss · instability)
Normal GT	**5–8 mm below** top of head (≈ 6 mm → nonop sling)
Fixation	**heavy non-absorbable suture figure-of-8 / tension band (osteoporotic)** · **isolated screws young + good bone** · avoid prominent hardware
Associations	**anterior dislocation**, **longitudinal cuff tear**; **subscapularis NOT** a GT deforming force
Healed GT, pain + weak overhead, full passive ROM at 5 months	**MRI (cuff tear)** ✗ Wrong options: more PT · MUA (passive full) · arthroscopy first · TSA
Valgus-impacted 4-part, young laborer	**ORIF**
3-part, cortex 4.2 mm, active 54-yr	**ORIF** ✗ Wrong options: sling · hemi (< 4 mm) · RSA
Seizure fracture-dislocation (young)	**ORIF** after reduction (watch for occult neck extension before closed reduction)
25-yr 4-part	**ORIF** (retain native head; arthroplasty for elderly / failed fixation)
Spot AP after trauma	**axillary view** next
GT / LT fixation by size (TO)	GT >5 mm: fragment **>2.5 cm → screw**, **<2.5 cm → suture anchor** · LT: <2.5 cm → suture/anchor, >2.5 cm → ORIF

CRPP

Indication	**2-part surgical neck**; 3-part / valgus-impacted 4-part with good bone, intact calcar · **highest complication rate**
Technique	threaded pins, don't cross cartilage, **ER the arm**, enter 2 cm below head

Pin	At risk
Lateral	**Anterior branch of axillary nerve**
Anterior	**LHB tendon**, **musculocutaneous nerve** (~17 mm), **cephalic vein** (~11 mm)
GT pin through medial cortex	**PCHA**

Modified Resch (TO)	**2 lateral-to-medial + 1 anterior-to-posterior** K-wires · contraindicated: calcar deficiency, LT fracture, metaphyseal comminution

Locked plating – screw penetration, varus collapse & calcar support

Commonest complication	**screw penetration / cut-out**: **14%** (Orthobullets, Südkamp) · **16–30%**, 23% (OB) ✗ Wrong options: infection · AVN · axillary nerve · HO
Mechanism	inadequate head fixation → **varus collapse → penetration** · 40% of complications technical
Prevent varus collapse	**inferomedial calcar screw(s)** · **anatomic medial cortical apposition** · **fibular strut** (↓ head height loss + varus) ✗ Wrong options: cortical + locking mix · tension band · CRPP · IMN
Loss-of-reduction risks (TO)	age, osteoporosis, smoking, **varus**, **medial comminution**, no cuff sutures
Plate placement	**lateral to bicipital groove + pectoralis tendon** (spare **ascending branch of ACHA**); locking plate more elastic than blade plate
Rehab	guided 3-phase program; **stiffness at 4 months with healed fracture → scar tissue**
ORIF in > 60 yr	**failure > 30%**, complications > 40%; **fracture type predicts failure**; fibular strut doesn't beat RSA
PHILOS position (TO)	**5–8 mm below GT tip**, **2–4 mm posterior to bicipital groove** · **4–6 head screws incl. 2 inferomedial calcar** · ≥3 shaft screws · **Gavaskar 2016: ≥7 locking head screws** · 10 suture holes for cuff · fill defects with calcium phosphate cement · **torus concept** = head on a "three-legged stool"
Bone quality (TO)	**deltoid tuberosity index** correlates with BMD (CACT < 4 mm = poor bone, as above)
Beach chair (TO)	BP cuff on dependent leg misreads cerebral perfusion

Surgical approaches – axillary nerve at risk

Axillary nerve	**~5–7 cm below acromion** (7 cm); **deltoid-split / extended anterolateral acromial approach → risk** (vs deltopectoral)

Intramedullary nailing

Indications	**2-part surgical neck** (± shaft extension), **pathological**, 3-part GT (young), proximal humerus + shaft · not varus posteromedial with both tuberosities, head split, 4-part
Facts	superior deltoid split; straight nail through articular cartilage; **cuff pain**, migration · **distal lateral → medial screw: radial nerve**; **A→P: musculocutaneous**
Nail (TO)	2-part surgical neck / 3-part GT · **8–12 mm** · 3 cm anterolateral incision, split cuff along fibres · straight nails enter via cuff split · **MultiLoc** (2nd generation) · complication: **cuff dysfunction + shoulder pain**

Hemiarthroplasty – height, tuberosity position & indications

Indication	**40–65 yr** with **head split** / complex fracture-dislocation likely to fail ORIF (**convertible stem**) · head depression > 40% (TO)
Best predictor of success	**anatomic tuberosity healing** ✗ Wrong options: 25° retroversion · < 24 h surgery · labral repair; age > 75 worse

– – Active space – –

Poor outcome (TO)	**GT migration** ✗ Wrong options: anteversion (**retroversion > 40°** worse) · no glenoid resurfacing
Height	**top of head 5.6 cm above superior pectoralis major tendon** (height + **retroversion** reference)
GT position	**~8 mm below top of prosthetic head** (head-to-tuberosity distance 7–8 mm); malposition → **8× ER torque**
Head too low / below GT	**impingement, ↓ elevation + abduction**
Can't elevate 6 months after hemi	**GT malunion** · LT nonunion → weak lift-off
Post-op caution	**passive ER** (+ active IR) stresses **LT / subscapularis**
Best hemi indication	**46-yr 4-part with comminuted head split** ✗ Wrong options: GT dislocation (ORIF) · 3-part intact calcar (ORIF) · OA (TSA) · cuff arthropathy (RSA)
59-yr carpenter – hemi over ORIF because	**head-split pattern**

Reverse shoulder arthroplasty vs hemiarthroplasty / ORIF in the elderly

Indications	**low-demand elderly**, **non-reconstructible tuberosities**, poor bone, **fracture-dislocation**, > 70 yr comminuted · **60° initial varus** (> 20°) → arthroplasty
70-yr comminuted metaphysis	**cemented RSA + tuberosity repair** ✗ Wrong options: uncemented (metaphyseal loss) · no tuberosity repair
Biomechanics	**COR medial (+ inferior) to native** (lateralised designs still medial) → **deltoid longer lever**; less tuberosity-dependent
↑ ER after RSA	**GT healing** (teres minor) · larger glenosphere, lateralised, **135° NSA**, more retroversion help ✗ Wrong options: inferior tilt · small glenosphere · medialisation · 155° NSA
GT not repaired / resorbed	**clinically significant loss of ER**
Contraindication	**non-functioning deltoid (upper trunk / C5 / axillary palsy)** ✗ Wrong options: irreparable cuff (that contraindicates hemi / TSA)
RSA vs HA	RSA: **higher complication rate** (OB) but **lower revision**; HA: **more IR**, more resorption issues, **greater motion loss**, unpredictable pain · equal complications (OB) – read the options
Octogenarian fracture-dislocation	**RSA > plate + hemi** outcomes
Primary vs salvage RSA	primary: **more ER + elevation**, better PROs · salvage after failed ORIF: worse function, ↑ instability, infection, complications · **RSA after failed nonop > after failed ORIF**
Chronic fracture-dislocation + shaft (81-yr)	**CT → long-stem RSA + cable / plate**
Elderly 4-part, independent	**arthroplasty (hemi or RSA)** · 73-yr 4-part → **RSA** · anatomic TSA not used for fractures

RSA complications scapular notching (↓ by 135° NSA, inferior tilt, inferior offset), dislocation, **glenoid loosening (commonest revision cause)**, **acromial stress fracture**, infection

6 Complications

Complication	Fact
Screw cut-out / penetration	**Commonest after locking plate**
AVN	Better tolerated than LL; not related to fixation type
Axillary nerve	**Up to 58% (EMG)**; suprascapular 48%
Malunion	**Varus apex-anterior** or GT → **RSA** (not TSA)
Nonunion	**2-part surgical neck**; **age + smoking**
Missed posterior dislocation	**LT fracture**
LHB injury	Anterior CRPP pin
Adhesive capsulitis, PTOA, infection	–

Related anatomy (long head of biceps, deltoid innervation)

LHB **supraglenoid tubercle**; **musculocutaneous**; **supination**; landmark in comminuted fractures

Deltoid **axillary nerve (C5–6) ← posterior cord**

X vs Y

	Hemiarthroplasty	Reverse TSA
Patient	**40–65**, head split	**Elderly**, poor tuberosities
Depends on	**Tuberosity healing**	**Deltoid**
Revision	Higher	Lower

	ACHA	PCHA
Role	Old "main"; arcuate → GT	**Main head supply (64%)**

How it's asked

"Main humeral head blood supply (TO)?"	**→ PCHA**
"Best predictor of ischaemia (TO)?"	**→ calcar length**
"NOT a predictor (TO)?"	**→ valgus displacement**
"Hertel EXCEPT (TO)?"	**→ surgical neck fracture**
"Initial AVN site (TO)?"	**→ superomedial**
"1990 study – ACHA supplies?"	**→ whole head except posteroinferior GT + head**
"Lowest ischaemia association?"	**→ long calcar segment**

– – Active space – –

"Highest AVN risk?"	→ 4-part fracture-dislocation
"Order of predictors?"	→ calcar > medial hinge > angulation > head split
"Most predictive (choose)?"	→ calcar 4–6 mm / disrupted medial hinge
"Incorrect statement (TO)?"	→ GT fixation for ≥ 5 mm in young
"Valgus-impacted 4-part, laborer?"	→ ORIF
"Posteriorly displaced GT 1 cm — true?"	→ ORIF is treatment of choice
"3-part, cortex 4.2 mm?"	→ ORIF
"5 mm residual GT?"	→ altered rotator cuff mechanics
"Closed treatment — altered cuff mechanics?"	→ GT displacement 6 mm
"GT fracture-dislocation, can't move shoulder?"	→ ORIF
"72-yr, GT 7 mm?"	→ ORIF
"GT 1 cm — motion lost?"	→ abduction
"Nonop displaced GT complication?"	→ impingement + loss of abduction
"Healed GT, painful overhead?"	→ MRI
"GT 6 mm below head top?"	→ sling + protected ROM
"Seizure fracture-dislocation, young?"	→ ORIF
"25-yr 4-part?"	→ ORIF
"Spot AP only?"	→ axillary view
"Electrocution LT fracture — ask about?"	→ posterior dislocation
"Cortical thickness precluding fixation (TO)?"	→ 4 mm
"Commonest locking plate complication?"	→ screw cut-out / penetration
"Prevent varus collapse?"	→ inferomedial calcar screw
"Stiff at 4 months, healed?"	→ scar tissue
"Calcar screw quadrant?"	→ inferomedial
"Poor calcar support →?"	→ varus collapse
"Lower fixation failure?"	→ anatomic medial cortical apposition
"Lateral CRPP pin risks?"	→ anterior branch of axillary nerve
"Anterolateral acromial approach risks?"	→ axillary nerve
"7 cm from acromion?"	→ axillary nerve
"Indication for surgery (figure)?"	→ head-split fracture
"10-month pain, can't elevate?"	→ nonunion
"Minimally displaced surgical neck?"	→ sling, recheck at 1 wk
"One-part fracture?"	→ early structured PT

"Outcome predictor (varus impacted)?"	→ age
"78-yr acceptable alignment?"	→ sling + pendulums at 1–2 wk
"1-yr outcome of 1-part?"	→ ↓ ROM without functional deficit
"RSA with GT removed?"	→ loss of ER
"Poor hemi outcome (TO)?"	→ GT migration
"Can't elevate 6 months after hemi?"	→ GT malunion
"Head below GT after hemi?"	→ ↓ elevation + abduction
"Pectoralis major landmark restores?"	→ height + retroversion
"Head top above pectoralis tendon?"	→ 5.6 cm
"Failed hemi at 2 yr?"	→ tuberosity malposition
"Most important for hemi success?"	→ anatomic tuberosity healing
"Avoid after hemi (LT)?"	→ passive external rotation
"GT position in hemi?"	→ 8 mm below head top
"Hemi over ORIF because?"	→ head split
"Best hemi candidate?"	→ 46-yr 4-part comminuted head split
"70-yr comminuted, minimise complications?"	→ cemented RSA + tuberosity repair
"Strongest arthroplasty indication?"	→ 60° initial varus
"82-yr displaced 4-part?"	→ hemi or RSA
"76-yr fracture-dislocation, nerve deficit?"	→ arthroplasty
"RSA vs HA — true?"	→ higher complication rate
"73-yr 4-part?"	→ reverse shoulder arthroplasty
"↑ ER after RSA?"	→ GT healing
"ORIF vs RSA in 75-yr — true?"	→ fracture type predicts ORIF failure
"Hemi vs RSA outcome?"	→ greater loss of motion with hemi
"Lateralised RSA — true?"	→ COR medial to native
"Maintain ER after RSA?"	→ fix + heal GT
"RSA after failed nonop vs failed ORIF?"	→ better after failed nonop
"Primary vs salvage RSA?"	→ primary has more ER
"81-yr chronic fracture-dislocation + shaft?"	→ CT + long-stem RSA with cable / plate
"RSA contraindication?"	→ upper trunk plexopathy
"87-yr smoker fracture-dislocation?"	→ RSA superior to plate + hemi
"LHB nerve, function, origin?"	→ musculocutaneous, supination, supraglenoid tubercle
"Deltoid nerve cord?"	→ posterior

– – Active space – –

PROXIMAL THIRD TIBIA FRACTURE

1 Basics

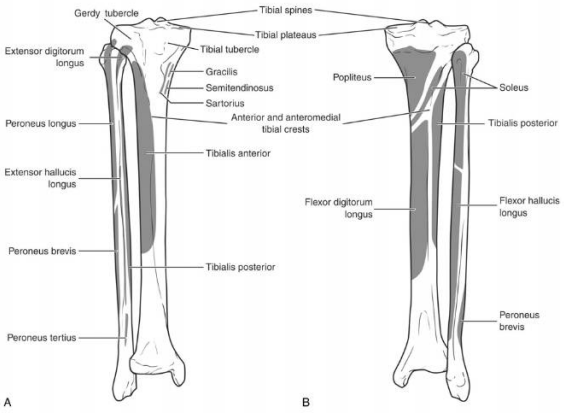
Epidemiology

Incidence	**5–11% of tibial shaft fractures** (~10%)
Mechanism	low energy: **torsion → spiral oblique** (indirect) · high energy: direct
Associated	**compartment syndrome**, soft-tissue injury (**muscle injury severity → amputation need**)

Characteristic deformity – valgus & procurvatum and its causes

Force	Effect
Patellar tendon	**Proximal fragment extends → apex anterior (procurvatum)**
Gastrocnemius	**Distal fragment flexes**
Pes anserinus	Proximal fragment **varus** (per note)
Anterior compartment	**Valgus**

Classic malunion after IMN	**valgus + procurvatum (apex anterior)** + anterior translation of proximal fragment ✗ Wrong options: varus · recurvatum · rotation
Causes	wide metaphysis (no canal fit), **eccentric / medial start**, deforming muscles, **knee hyperflexion**, nail **Herzog curve** ("wedge effect")
Malalignment rate	**20–60%** (Orthobullets) · **58%** proximal ⅓ (Freedman; vs 7% middle, 8% distal) · **up to 84%** (OB)
Other fractures' typical malunions	humeral shaft: varus, shortening, IR · distal femur: varus, shortening, recurvatum · distal ⅓ tibia + intact fibula: varus + rotation · subtroch: **flexion, abduction, ER**



Deforming forces on proximal tibia fractures

3 Presentation

Exam contusions, blisters, open wounds, **compartments** (passive toe stretch ± pressures) · DPN, SPN, sural, tibial, saphenous; DP + PT pulses

4 Diagnosis & Classification

Key radiology findings

Views AP + lateral tibia + **knee + ankle** · findings: **proximal fragment extended, apex anterior, varus**; **distal fragment flexed**

CT **intra-articular extension** (differential: plateau fracture, knee dislocation)

AO 42	Pattern
A	Simple
B	Wedge
C	Comminuted

5 Management

Proximal ⅓ tibia fracture

- Closed low energy, acceptable alignment → long leg cast (10–20° flexion) → **brace at 4 wk**
- Soft-tissue compromise / polytrauma → **ex-fix (convert to IMN 7–21 d)**
- ≥ 5–6 cm proximal bone for 2 locking screws → **IMN + anti-deformity tricks**
- Very proximal / inadequate nail fixation → **percutaneous locking plate**

Operative criteria

Acceptable for cast	Limit
Varus / valgus	**< 5°**
AP angulation	**< 10°**
Cortical apposition	**> 50%**
Shortening	**< 1 cm** (hardest to control)
Rotation	**< 10°**

Option	Indication / fact
Ex-fix	Soft-tissue compromise, polytrauma; circular frame for very proximal; **more malalignment than IMN**
IMN	**≥ 5–6 cm proximal bone (2 locking screws)**; static locking proximal + distal; ≥ 2 proximal screws, **oblique** screws help
Percutaneous locking plate	Extreme proximal, transverse / oblique, good soft tissue; **lateral safer**; **lateral plate + medial comminution → varus collapse**; **↑ infection vs IMN in open**

– – Active space – –

Nailing technique – lateral start point, semi-extended / suprapatellar nailing, unicortical plate

Start point	**just medial to lateral tibial spine (AP)** + **proximal, anterior to articular margin (lateral)** → **lateral start prevents valgus** ✗ Wrong options: medial start → valgus · distal + medial → anterior cortex cut-out
Approach	**lateral parapatellar** (needs mobile patella) helps lateral start · medial parapatellar → valgus
Semi-extended / suprapatellar nailing	neutralises extensor pull → **prevents procurvatum** · **suprapatellar does NOT affect valgus (coronal)** ✗ Wrong options: hyperflexion (worsens) · radiolucent triangle
Suprapatellar vs infrapatellar	infrapatellar → **more anterior knee pain**; ROM, union, PF joint same
Unicortical plate	short ⅓-tubular plate anterior / anteromedial / posteromedial, 2 unicortical screws each side → holds reduction (prevents procurvatum + anterior translation)
Universal / femoral distractor	Schanz pins medial, parallel to joint (can double as blocking pins)
Prevent apex anterior – all EXCEPT	**anterior blocking screw** (posterior blocking screw, posterior starting hole, semi-extended interlocking, anteriorly directed nail all help)
Preserve reduction – EXCEPT	**countersinking the nail** (loses aperture fixation) ✗ Wrong options: oblique proximal screws · blocking screws · lateral eminence start · unicortical plate
Valgus prevention – NOT shown for	**suprapatellar portal**
Nail choice	**Herzog curve proximal to fracture** (distal curve → anterior translation)
Locking	**static**; no acute dynamisation

Blocking (Poller) screws – position & function

Rule place in the **concavity** of the deformity, in the **proximal (metaphyseal) fragment** → narrows canal, redirects nail

Screw	Prevents
Posterior to nail, proximal fragment (coronal screw)	**Procurvatum** (counters patellar tendon + gastrocnemius)
Lateral to nail, proximal fragment (sagittal screw)	**Valgus**
Anterior	**Worsens** procurvatum
Medial	**Worsens** valgus

Benefits obtain + maintain reduction, ↑ **construct stiffness** (if left) · Ricci: < 5° valgus with screws vs > 6° without · union ~5.4 months (Krettek)
✗ Wrong options: ↓ nail breakage · replace interlocks · larger nail



Coronal blocking screw posterior in the proximal fragment

Minimally invasive plating – superficial peroneal nerve risk

Long lateral plate, percutaneous distal screws	**superficial peroneal nerve** (± deep peroneal) → **small open incision for distal holes** ✗ Wrong options: common peroneal · saphenous (medial) · peroneal artery · tendons

6 Complications

Complication	Fact
Anterior knee pain	**> 30% after IMN**; **nail removal helps ~50%**
Malunion	**Valgus + procurvatum (20–60%)** → ↑ arthrosis; revision IMN / osteotomy
Nonunion	**Rule out infection**; dynamise if axially stable

X vs Y

	Posterior blocking screw	Lateral blocking screw
Prevents	**Procurvatum**	**Valgus**
Plane	Coronal screw (sagittal deformity)	Sagittal screw (coronal deformity)

	Lateral start	Medial start
Effect	**Prevents valgus**	**Causes valgus**

How it's asked

"IMN deformities?"	**→ apex anterior + valgus**
"Ideal entry point?"	**→ medial to lateral spine, anterior to articular margin**
"Fracture most associated with procurvatum + valgus?"	**→ proximal tibia**
"Most common malalignment?"	**→ valgus + procurvatum**
"Counter patellar tendon pull?"	**→ posterior blocking screw in proximal fragment**

– – Active space – –

"Blocking screw advantage?"	→ ↑ construct stiffness
"Blocking screw positions?"	→ posterior + lateral in proximal segment
"Avoid procurvatum?"	→ posterior blocking screw / semi-extended nailing
"Start point + blocking screw combo?"	→ lateral start + posterior screw
"Screws at marked sites prevent?"	→ valgus + procurvatum
"Herzog curve + screws?"	→ lateral + posterior screws; Herzog proximal to fracture
"Avoid apex-anterior complication?"	→ blocking screws
"Prevent apex anterior EXCEPT?"	→ anterior blocking screw
"Avoid characteristic deformity?"	→ lateral starting point
"NOT shown to prevent valgus?"	→ suprapatellar portal
"Prevent procurvatum + anterior translation?"	→ anterior unicortical plate
"Facilitate reduction (choose)?"	→ suprapatellar nailing
"Preserve reduction EXCEPT?"	→ countersinking the nail
"Infrapatellar vs suprapatellar?"	→ more anterior knee pain
"MIPO lateral plate nerve risk?"	→ superficial peroneal nerve

RADIAL HEAD FRACTURES

– Trauma · 50 / 67 –

1 Basics

Epidemiology

Incidence	1–4% of adult fractures · **20–30% of elbow fractures (commonest)** · mean age ~45 (85% 30–60) · **F:M 3:2**
Mechanism	**FOOSH with elbow extended + forearm pronated**
Associated (30%)	**LCL injury (commonest, up to 80% on MRI)**, MCL, **Essex-Lopresti**, coronoid, olecranon, **Monteggia**, **terrible triad**, **scaphoid**

Anatomy

Head	**15° offset** from neck · **anterolateral third lacks subchondral bone** (fractures here) · 40° concave dish
Safe zone (non-articular)	**90–110° arc between radial styloid + Lister's tubercle** (straight lateral in neutral) · ~25% of head doesn't articulate with ulna
Radiocapitellar joint	**60% of elbow load**

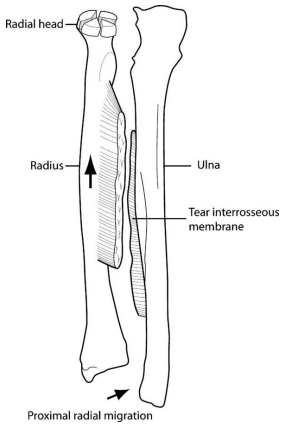
Ligament	Role
LUCL (supinator crest)	**Varus + ER (hypersupination)**; deficiency → **PLRI**
RCL	Into annular ligament
Annular	Holds head to lesser sigmoid notch; interposition → irreducible
MCL anterior bundle	**Primary valgus stabiliser** (radial head 2nd)

Radial head roles	**secondary valgus restraint** (key if MCL deficient) · **longitudinal stability** (with IOM + DRUJ)

Longitudinal instability (Essex-Lopresti) – radial head as buttress, excision contraindicated

Essex-Lopresti	**radial head fracture + IOM disruption + DRUJ injury** → proximal radial migration, ulnocarpal impaction ✗ Wrong options: distal radius fracture
Single best answer (TO)	**interosseous membrane injury**; IOM gives **> 70% axial stability** (TO)
Primary buttress against proximal migration	**radial head** (then IOM) (OB) – read the options · TFCC (triangular fibrocartilage complex) also maintains length
Excision in Essex-Lopresti (TO)	if ever, delay **3 weeks** (IOM heals)
Clues	**wrist / DRUJ tenderness**, ballottement +, ulnar styloid base fracture, **radius pull test > 3 mm** (intra-op), DRUJ translation > 50% vs other side
Late presentation	only **20% satisfactory** vs 80% if acute

– – Active space – –



Essex-Lopresti lesion

Load share (TO)	**~75–80%** of wrist axial load crosses the radiocarpal joint → IOM transfers it to the ulna · secondary stabilisers = **IOM central band** + TFCC / DRUJ
IOM (TO)	**5 bands**; **central band** most important, runs at **~20–25°** (angle used for reconstruction, PL or gracilis graft – rarely done)
Recognise early (TO)	radial head + forearm tenderness, DRUJ tenderness / instability → **X-ray elbow AND wrist**; **ultrasound** can show IOM tear · missed → painful loss of pronation / supination
Pull test < 3 mm (TO)	excision alone acceptable · **≥ 3 mm → radial head replacement**
DRUJ after head fixation / replacement (TO)	unstable → **above-elbow slab in supination**, **DRUJ pinning**, or **TFCC repair**
Neglected Essex-Lopresti (TO)	salvage **single-bone forearm** (proximal ulna to distal radius, DCP) → pain relief, no rotation · position **slightly more pronated than mid-prone**

3 Presentation

Exam	lateral tenderness; **mechanical block to motion** (normal flexion 0–150°, pronation 75°, supination 85°) → **aspirate haematoma + intra-articular lidocaine** · PIN + ulnar nerve · pivot shift / posterolateral drawer (LUCL), valgus stress (MCL) · **palpate wrist + IOM**
Pain limits exam at 1 wk	**intra-articular lidocaine + re-examine** ✗ Wrong options: CT · MRI · splint 2–3 wk · EUA

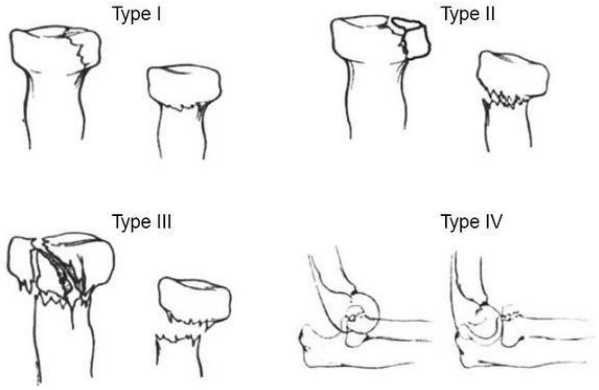
4 Diagnosis & Classification

Key radiology findings

Study	Fact
AP + lateral elbow, **forearm + wrist**	–
Radiocapitellar (Greenspan)	Oblique lateral, **45° cephalad** → head without coronoid overlap
Fat pads	Haemarthrosis; **posterior fat pad more sensitive**
CT	Comminution, fracture-dislocations (3D planning)
MRI	Ligaments; rarely changes management
Dynamic CT	Subtle DRUJ instability

Mason classification (Hotchkiss / Broberg-Morrey)

Type	Description
I	**< 2 mm**, **no block**
II	**> 2 mm** / angulated, **possible block**
III	**Comminuted**, displaced, **block**
IV	**+ elbow dislocation**



Mason classification

OTA 2R1 A extra-articular · B partial articular · C complete articular · high interobserver variability

5 Management

Radial head fracture

- No block (Mason I / II) → **sling 2–7 d + early ROM**
- Block / displaced, ≤ 3 fragments, young → **ORIF (headless screws ± plate in safe zone)**
- > 3 fragments / dislocation / Essex-Lopresti / unstable → **radial head arthroplasty (+ LCL repair)**
- Low-demand, no instability → **fragment excision (< 25–30%) or resection (never if unstable)**

Nonoperative vs ORIF — mechanical block & lidocaine test

Mason I (violinist, no block) **sling ~2 days → active motion** · good **85–95%** · Mason II without block: early failure up to **12%**
✗ Wrong options: cast ≥ 1 wk (worse function)

Displaced partial articular + block (20–30° / 45–65° arcs) **ORIF** via **ECU / anconeus (Kocher)** with plate or screws
✗ Wrong options: excision > 30% · medial approach · arthroplasty (older / ≥ 3 fragments)

Mason II on CT, young **ORIF**
✗ Wrong options: cast · excision (< 25%) · CRPP (none in adults) · replacement

Correct pairing **38-yr Mason II, 40° arc → ORIF**
✗ Wrong options: 4-part young → arthroplasty · resection in young Mason III · arthroplasty for minimally comminuted 3-part · excision with ulnar styloid / DRUJ injury

– – Active space – –

Operative criteria

Treatment	Indication
Nonop	Mason I; Mason II **without block**
ORIF	**Mason II with block**; Mason III / IV if feasible (**≤ 3 fragments**); open; complex elbow injury
Arthroplasty	**> 3 fragments** (≥ 3, OB), fracture-dislocation (> 30% articular), **Essex-Lopresti**, terrible triad, plastic deformity, malunion / nonunion
Fragment excision	Low demand, no instability, fragment **< 25%** (Orthobullets) · **< 30%** (OB)
Resection	Sedentary, **delayed isolated** fracture, salvage · not with **IOM injury (pull > 3 mm)**, coronoid fracture, MCL deficiency

ORIF outcomes **> 3 fragments → > 50% unsatisfactory** · isolated vs complex similar at 4 yr · fixation fails more than arthroplasty if elbow unstable

Surgical approach & safe zone (PIN, forearm rotation, LUCL)

Approach	Interval	Pros / cons
Kocher	**ECU (PIN) / anconeus (radial)**	**Less PIN risk**, extensile · **LUCL risk** if capsulotomy too posterior (stay anterior to crista supinatoris)
Kaplan	**EDC (PIN) / ECRB (radial)**	**Protects LUCL**, anterior fragment access · **more PIN risk**
EDC split	Through EDC	Anterior half, LCL spared
Posterior	Lateral flap	With olecranon / Monteggia

TO approach intervals written as **Kocher = ECRB–anconeus** and **Kaplan = ECRL–ECRB** (TO elbow-injuries lecture – verify: standard Kocher = **anconeus–ECU**, Kaplan = **ECRB–EDC**, as in the table above) · avoid deep retraction near annular ligament (PIN) · isolated low-energy comminution → excision (TO)

Approach **forearm pronated** (moves **PIN** anterior) · PIN crosses radius in supinator **~4 cm distal** to head

Plate application **forearm NEUTRAL → plate straight lateral** in safe zone · TO answer: "**just posterior to mid-axial line with forearm in pronation**" – read the options
✗ Wrong options: pronation / supination

Limits **within 90° arc (styloid–Lister's)** · **not distal to bicipital tuberosity (PIN)** · countersink articular implants

Fixation headless / 2.0–2.4 mm screws (simple partial articular; **better ROM than plates**) · 1.5–2.0 mm plates for head + neck · plate removal common

Least LUCL risk for arthroplasty **Kaplan (EDC / ECRB)**
✗ Wrong options: Kocher (anconeus / ECU)

Head-neck fracture past bicipital tuberosity + PIN palsy (GSW) **Kaplan + ORIF in safe zone** (neurolysis possible)

Radial head arthroplasty – indications & sizing (overstuffing)

4-fragment + dislocation	**arthroplasty** ✗ Wrong options: excision · ORIF · splint · hinged ex-fix
Comminuted + dislocation	**arthroplasty + LCL repair** ✗ Wrong options: ORIF + LCL · excision
Osteoporotic 65-yr comminuted	**metallic anatomic arthroplasty** ✗ Wrong options: excision (wrist pain, instability, migration)
Best indication	**Mason III + IOM injury**
Implants	**loose-stem (spacer)**, press-fit, **bipolar** (may destabilise), **monoblock vs modular**, pyrocarbon (investigational) · **silicone abandoned** (fracture, synovitis)
Overstuffing	**> 2.5 mm** alters kinematics → pain, stiffness, **capitellar erosion** · **detect: visual widening of LATERAL ulnohumeral joint → remove + downsize** ✗ Wrong options: medial on AP (only after 6 mm)
Height	proximal implant at **proximal lesser sigmoid notch**; **dish at level of lateral coronoid facet** · TO coronoid question: answer key not provided by the source

Terrible triad & complex instability

Terrible triad	**elbow dislocation + radial head + coronoid fractures**
Treatment	**coronoid ORIF + radial head ORIF / arthroplasty + LUCL repair** → MCL repair ± hinged ex-fix if still unstable ✗ Wrong options: **radial head excision – never** · transarticular screw · coronoid excision
Most common complication	**restricted motion (stiffness)** ✗ Wrong options: redislocation · PIN · infection
Distal humerus partial articular + radial head (23-yr)	**ORIF both** (headless screws)

6 Complications

Complication	Fact
Stiffness	3–20%; **supination loss commonest**; functional arc **30–130° + 50 / 50 rotation**
PIN injury	Distal to bicipital tuberosity, retraction → splint, EMG later, transfers
Secondary displacement	**< 5%** of nonop
Nonunion / malunion	Common, often asymptomatic
Radiocapitellar OA	Common radiographically; overstuffing
HO	CNS injury, burns → indomethacin 6 wk; excise when mature (~10% recur)
Resection sequelae	**Proximal migration, cubitus valgus, wrist pain, OA 73%**

– – Active space – –

Late sequelae — malunion & post-traumatic stiffness

Malunited comminuted head 4 yr, stable DRUJ + elbow	**radial head resection** ✗ Wrong options: ligament reconstruction · ORIF · debridement · TEA
Stiff 4 months after minimally displaced fracture	**static / dynamic progressive splinting** ✗ Wrong options: steroid · MUA (worsens) · home program · CPM
8 months, arc 50–85°, ulnar paraesthesia, healed	**anterior + posterior capsular release + ulnar nerve transposition** ✗ Wrong options: head resection · partial release · injection
60-yr, 1 yr after fracture, near-normal motion, intermittent pain (TO)	answer key not provided by the source

Paediatric radial neck — acceptable angulation

10-yr-old (TO)	**< 30° angulation + < 2 mm translation** (2–3 mm) → cast only; more → closed reduction (Israeli / Kaufman, Patterson, Neher-Torch), Metaizeau, joystick

X vs Y

	Kocher	Kaplan
Interval	**ECU / anconeus**	**EDC / ECRB**
Risk	**LUCL**	**PIN**

	ORIF	Arthroplasty
Fragments	**≤ 3**	**> 3**
Setting	Young, isolated	**Dislocation, Essex-Lopresti**

How it's asked

"2 fragments + block?"	**→ plate via ECU / anconeus**
"Violinist Mason I?"	**→ sling 2 days + active motion**
"Mason II, young?"	**→ ORIF**
"Block 70–100°, rotation 10 / 10?"	**→ ORIF**
"Pain limits exam?"	**→ intra-articular lidocaine + re-examine**
"Correct pairing?"	**→ ORIF for Mason II with 40° arc**
"4 fragments + dislocation?"	**→ radial head arthroplasty**
"Intra-op overstuffing?"	**→ remove + downsize implant**
"Comminuted + dislocation in 32-yr?"	**→ arthroplasty + LCL repair**

"65-yr osteoporotic comminuted?"	→ metallic radial head arthroplasty
"40% fragment in 4 pieces after dislocation?"	→ radial head replacement
"Detect thick implant?"	→ visual widening of lateral ulnohumeral joint
"Safe plate position (TO)?"	→ just posterior to mid-axial line, forearm pronated
"Pronation during Kocher protects?"	→ PIN
"Forearm position for plating?"	→ neutral
"Plate limited to?"	→ 90° arc safe zone
"Approach + fixation position?"	→ pronated for approach, neutral for plate
"Least LUCL injury?"	→ EDC / ECRB (Kaplan)
"GSW head-neck fracture + PIN palsy?"	→ Kaplan + ORIF in safe zone
"Terrible triad EXCEPT?"	→ radial head excision
"When is excision indicated (triad)?"	→ generally not indicated
"Distal humerus + radial head, 23-yr?"	→ ORIF both
"Terrible triad management?"	→ coronoid ORIF + radial head ORIF / arthroplasty + LUCL repair
"Commonest triad complication?"	→ restricted motion
"Essex-Lopresti involves (TO)?"	→ interosseous membrane injury
"Maintains radial length?"	→ TFCC
"Best arthroplasty indication?"	→ Mason III + IOM injury
"Radial head + positive ballottement?"	→ Essex-Lopresti
"Contraindicated with wrist tenderness?"	→ radial head excision
"Contraindication to excision?"	→ radius pull > 3 mm
"Ulnocarpal impingement after excision — missed?"	→ DRUJ + IOM injury
"Primary buttress against proximal migration?"	→ radial head
"Pull test 5 mm — contraindicated?"	→ radial head resection
"Malunited head, stable DRUJ, 4 yr?"	→ radial head resection
"Stiff after 10 wk therapy?"	→ progressive splinting
"Stiff 50–85° + ulnar symptoms?"	→ anterior + posterior release + ulnar transposition
"Child radial neck acceptable (TO)?"	→ < 30° angulation, < 2 mm translation

– – Active space – –

RADIUS AND ULNAR SHAFT FRACTURES

– Trauma · 51 / 67 –

1 Basics

Epidemiology

Who	men **10–20 yr**, women **> 60 yr**
Mechanism	**high energy** (direct blow, MVA, fall from height, sport) · low-energy both-bone → suspect wrist / elbow injury
Associated	**Galeazzi, Monteggia, Essex-Lopresti**, compartment syndrome

Anatomy

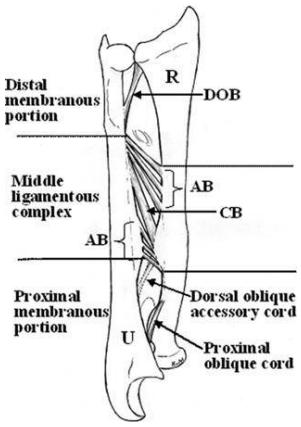
Rotation axis	**radial head → ulnar fovea**; radius rotates around ulna
Radial bow	complex; **max ~15 mm at 60%** of radial length · ulna slight bow in distal 75%
Nerves	median between PT heads then FDS / FDP · ulnar between FCU heads → FDP ring + small · **PIN around radial neck through supinator** → posterior IOM → 4th compartment
Vessels	brachial bifurcates **1 cm below elbow** · radial artery adherent to FCR

Interosseous membrane anatomy

IOM	oblique radius → ulna; load passes **distal radius → proximal ulna** · distal fibres tight in supination, central in neutral
IOM central band (TO)	runs **proximal radius → distal ulna** · torn in Essex-Lopresti
Displacement / cast position by level (TO)	proximal ⅓ → **supination** · middle ⅓ → neutral · distal ⅓ → pronation
Compartment syndrome (TO)	forearm = **2nd commonest site** after leg

IOM ligaments (5)	Note
Central band	**Key load-bearer; reconstruct this**
Accessory band, distal oblique bundle, proximal oblique cord, dorsal oblique accessory cord	–
NOT part of IOM	**Annular ligament**

Essex-Lopresti (TO)	**injury to interosseous membrane** (+ radial head fracture + DRUJ)



Interosseous membrane

3 Presentation

Exam deformity, open wounds, **tense compartments**, **passive finger stretch pain**; radial + ulnar pulses; median, radial, ulnar nerves; hand dominance + occupation

4 Diagnosis & Classification

Key radiology findings

Views AP + lateral forearm **+ wrist + elbow**; **radial head aligned with capitellum on all views**; contralateral for planning · CT rarely

OTA (radius / ulna shaft)	Subtypes
A simple	A1 spiral · A2 oblique · A3 transverse
B wedge	B2 intact · B3 fragmentary
C multifragmentary	C2 intact segmental · C3 fragmentary

Monteggia & Galeazzi variants (Bado, transolecranon, chronic)

Bado IV (TO) radius + ulna fractures with radial head dislocation (TO image: anterior angulation + anterior head)

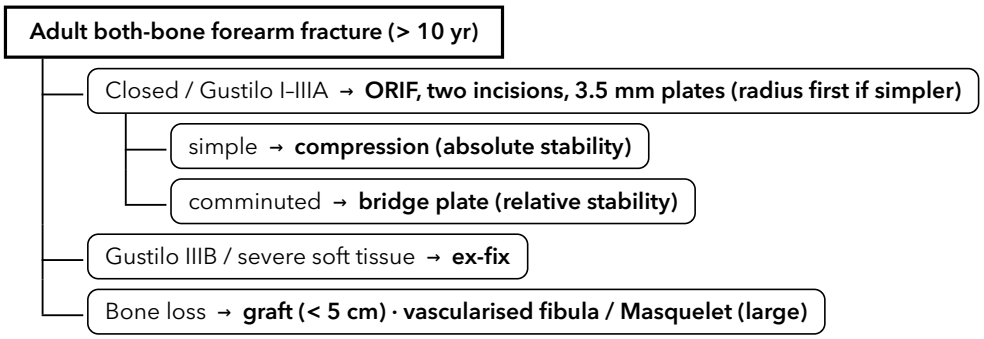
Transolecranon fracture-dislocation – EXCEPT (TO) "**a Monteggia variant**" – it is an **anterior elbow dislocation with intact PRUJ**, mostly **bony**, often **coronoid** fracture

Monteggia ORIF sequence (TO) ulna → coronoid → radial head (fix / replace) → LCL repair

Old malunited ulna + posterior radial head dislocation, 60-yr, painful (TO) answer key not provided by the source

– – Active space – –

5 Management



Treatment options

Option	Indication	Fact
Cast / brace (Muenster, above elbow)	**Rare** – non-displaced, non-surgical candidates	**> 50° rotation loss in 30%**, high nonunion
ORIF	**Nearly all ≥ 10 yr**; **Gustilo I–IIIA primary ORIF**	**> 95% union** compression · **88%** bridge plating
Ex-fix	**Gustilo IIIB**	Distal pins **2nd + 3rd MC**, **≤ 4 mm**
IMN	**Very poor soft tissue** only	No rotational / axial control, can't keep bow, **high nonunion** · **flexible nails contraindicated in length-unstable adult fractures**

Fixation principles & constructs

Comminuted closed both-bone (22-yr) — **ORIF without acute graft**
✗ Wrong options: ex-fix · flexible nails · graft · brace

Transverse + compression plates — **absolute stability → direct (primary) healing by internal remodelling** (intramembranous)

Comminuted (GSW) — **ORIF with relative stability (bridge plate)** → endochondral (secondary) healing

Greatest bending rigidity — **3.5 mm stainless steel DCP on both bones**
✗ Wrong options: titanium · ⅓ tubular

Technique — **3.5 mm DCP / LC-DCP** · **≥ 6 cortices** each side · **long plates** (torsion) · **contoured radius plate**, straight ulnar plate · locking in osteoporosis · 2.0 / 2.7 lag screws · dorsal = tension side (volar seats better)
✗ Wrong options: 4.5 mm (refracture)

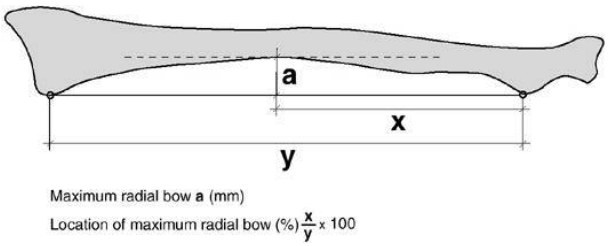
Post-op — early ROM; **NWB ~6 wk** (Orthobullets) · **WBAT if ≥ 8 cortices each side** (OB) – read the options · clinical healing ~3 months

Polytrauma (femur IMN, simple fibula ORIF, forearm ORIF, humerus ORIF) — **WBAT all extremities** (OB)

Restoration of radial bow & forearm rotation

Radial bow restoration — **most important for function → forearm rotation**
✗ Wrong options: wrist strength · elbow ROM · synostosis

Limited pronosupination after long straight radial plate	**loss of radial bow** ✗ Wrong options: nonunion · PIN · ECU at DRUJ · HO
Cause of poor pronation	**straight plate on radius** ✗ Wrong options: bridge plating · straight ulnar plate · dual incision
Malunion > 20°	limits rotation



Radial bow measurement

Surgical approaches (Henry, PIN, two-incision synostosis risk)

Two incisions	**↓ synostosis** (1 / 65 vs 5 / 12 single) · no wound, neuritis, malunion benefit

Approach	Use
Volar Henry	Distal + middle ⅓ radius
Dorsal Thompson	Proximal ⅓ radius
Subcutaneous ulnar	Ulna (dorsal ulnar cutaneous nerve at risk)

Proximal Henry	**radial artery medially**, **supinator laterally**, forearm **SUPINATED** (moves PIN dorsally)
Pronated during proximal Henry	**PIN injury → weak thumb (+ finger) extension** ✗ Wrong options: 1st web numbness (superficial radial) · opposition (median) · abduction (ulnar)
Ulnar approach (TO)	subcutaneous, **FCU–ECU** plane, protect dorsal cutaneous branch of ulnar n.
Intra-op rotation check (TO)	radius: **bicipital tuberosity medial vs radial styloid lateral (160–180°)** · ulna: coronoid vs styloid 180°
Nailing (TO)	length = olecranon → ulnar styloid (X): **ulnar nail X − 1 cm, radial nail X − 3 cm** · entry radius just lateral to Lister's, ulna slightly lateral to olecranon tip · indications: open, segmental, osteoporotic, polytrauma
Nonop (TO)	isolated stable ulnar fracture (**< 50% displacement, < 10° angulation**, nightstick)

Open fractures & bone loss (bone graft, vascularised fibula, GSW)

Clean 2 cm open wound	**I&D + ORIF + immediate closure**
1 cm segmental gap after debridement	**autograft** ✗ Wrong options: TCP, $CaPO_4$, $CaSO_4$, **BMP-3 — not osteoinductive**
Graft by defect size	**< 5 cm → cancellous autograft** · larger → **vascularised fibula** (↓ infection), allograft, transport, Masquelet
10 cm defect, strut allograft vs fibula	**allograft → more infection** (+ nonunion, fracture, delayed union)

– – Active space – –

GSW, large soft-tissue + bone loss	**ORIF + free fibular osteoseptocutaneous flap** ✗ Wrong options: strut / cancellous allograft (infection) · ex-fix definitive
GSW comminuted, 2 cm wound	**I&D + 3.5 mm LC-DCP bridge plate** ✗ Wrong options: 4.5 mm · compression · delay · ex-fix

6 Complications

Complication	Fact
Synostosis	**3–9%**; **single incision**, proximal ⅓, head injury, polytrauma, fragments / hardware in interosseous space
Infection	3%
Compartment syndrome	~1% (up to 15%); **superficial volar release may suffice** (connected)
Nonunion	**< 5%** compression · 12% bridge; IMN
Malunion	Radial bow
Nerve	**PIN (Henry proximal, Monteggia)**; median (modified Henry); dorsal ulnar cutaneous → observe 3 months
Refracture	**Up to 10%**

Synostosis – further risks (TO)	**same-level fractures** · comminution · **bone graft at the interosseous border**
Expected losses	grip **25%**, pronosupination strength **30%**, wrist flexion 16%, extension 37% strength; motion loss < 10°

Nonunion

Atrophic	**plate exchange + autogenous cancellous graft (ICBG) + compression plate** · smokers high risk ✗ Wrong options: Ilizarov · stimulators · ulnar shortening · IMN · bone scan if labs normal
Hypertrophic	↑ fixation (compression plate) · infected / atrophic → **Masquelet**

Refracture after plate removal

Greatest risk	**removal < 1 yr** · **diaphyseal** location (correlates most) · **4.5 mm plates** · comminution, displacement, lucency, immediate activity
NOT a risk	**3.5 mm DCP** ✗ Wrong options: locking screw removal · metaphyseal implants (lower)
Prevention	**functional brace 6 wk + protect 3 months**

Heterotopic ossification – timing of excision

Excision	**as early as 6 months** (4–6 months) with **radiation ± indomethacin** ✗ Wrong options: wait for ankylosis / maturity

Paediatric forearm — deformity limiting rotation

Most limiting (TO) **angulation** (10° → 20° rotation loss) > malrotation (10° → 10°) · **bayonet apposition doesn't limit**

Age (TO)	Sagittal	Coronal
< 9 yr	20°	15°
9–11 yr	15°	5°
11–13 yr	10°	0°

X vs Y

	Absolute stability	Relative stability
Construct	**Compression plate, lag screws**	**Bridge plate**, IMN, cast, ex-fix
Healing	**Primary (internal remodelling)**	**Secondary (callus, endochondral)**
Fracture	Simple transverse	Comminuted

How it's asked

"Restoring radial bow improves?"	**→ forearm rotation**
"Limited pronosupination after straight plate?"	**→ loss of radial bow**
"Pronation 15° one year later — cause?"	**→ straight plate on radius**
"Comminuted closed both-bone?"	**→ ORIF**
"Compression plating healing?"	**→ absolute stability, direct healing by remodelling**
"Comminuted GSW radius?"	**→ ORIF with relative stability**
"Greatest bending rigidity?"	**→ 3.5 mm stainless steel DCP on both**
"Two incisions advantage?"	**→ lower synostosis**
"Proximal Henry retraction + position?"	**→ artery medial, supinator lateral, supinated**
"Pronated during Henry → deficit?"	**→ weak thumb extension**
"Clean 2 cm open wound?"	**→ ORIF + immediate closure**
"1 cm segmental gap?"	**→ bone grafting**
"Strut allograft vs fibula?"	**→ higher infection**
"GSW with large bone + soft-tissue loss?"	**→ ORIF + free fibular flap**
"GSW comminuted both-bone?"	**→ I&D + 3.5 mm bridge plate**
"Atrophic radial nonunion?"	**→ plate exchange + autogenous graft**
"Atrophic ulnar nonunion, smoker?"	**→ ICBG + compression plating**
"Refracture risk EXCEPT?"	**→ 3.5 mm DCP**

– – Active space – –

"Greatest refracture risk?"	**→ removal < 1 yr**
"Pianist refracture — accurate?"	**→ diaphyseal removals higher risk**
"Essex-Lopresti (TO)?"	**→ interosseous membrane injury**
"IOM ligaments EXCEPT?"	**→ annular ligament**
"Earliest HO excision?"	**→ 6 months**
"Polytrauma WB status?"	**→ WBAT all extremities**
"Bado type (TO image)?"	**→ IV**
"Transolecranon EXCEPT (TO)?"	**→ Monteggia variant**
"Paediatric deformity most limiting rotation (TO)?"	**→ angulation**

SACRAL FRACTURES

– Trauma · 52 / 67 –

1 Basics

Epidemiology

Incidence	**30–45% of pelvic ring injuries**; bimodal (young high energy · elderly low-energy / insufficiency)
Neurological injury	**25%**; **missed in 75% of intact** and 50% with deficits

Anatomy

Sacrum	5 fused vertebrae; **4 foramina**; load from S1 → iliac wings → acetabulum
Nerves	**L5 root on top of sacral ala** · S1–S4 through foramina (**S1 + S2 most injured**) · **S2–S5: sphincter tone, bulbocavernosus reflex, perianal sensation**; **unilateral preservation enough for bowel + bladder**
Perianal numbness	**S2 root** ✗ Wrong options: L3–S1

2 Mechanism

MVA / fall from height (landing on feet → U-type) · repetitive stress / osteoporosis → insufficiency

3 Presentation

Exam	peripelvic pain; rotate iliac wings; **Morel-Lavallée** (lumbosacral degloving); **vaginal exam** (open); **rectal exam**, S2–S5 sensation, **anal wink, bulbocavernosus, cremasteric**; pulses ± ABI

4 Diagnosis & Classification

Key radiology findings

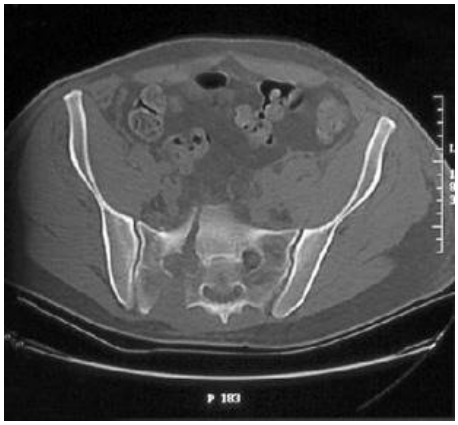
Study	Fact
Radiographs	**Show only 30%**
Lateral	Screening; poor quality
Inlet	Sacral canal, superior S1
Outlet	**True AP of sacrum**
Signs	**L4 / L5 TP fractures**, asymmetric foramina, anterior ring injury, **stepladder sign** (AP; overriding transverse fragments)
CT	**Study of choice** (coronal + sagittal) · **sagittal → lumbopelvic instability**
MRI	**Neural compromise** suspected

– – Active space – –

Denis zones & neurological injury

Denis	Location	Neuro deficit
1	**Lateral to foramina (ala)**	**5.9%** – lowest
2	**Through foramina**	**28.4%**
3	**Medial → spinal canal**	**~60% – highest** (bowel / bladder)

Classified by most medial extent

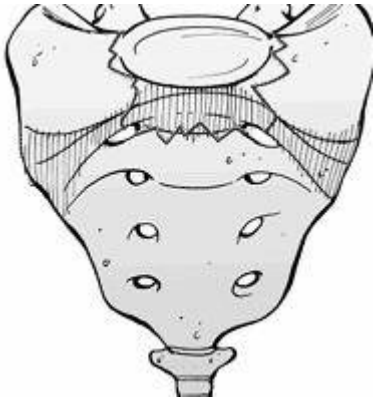


Denis zone 2 sacral fracture

Transverse, U-type & Roy-Camille

Transverse sacral fracture ↑ nerve dysfunction (Tile A3 / Denis zone III)

U-type (H / λ) **bilateral vertical (zone II) + transverse (S2) → spinopelvic dissociation**; axial load (fall onto feet); high neuro deficit, **cauda equina**



U-type sacral fracture

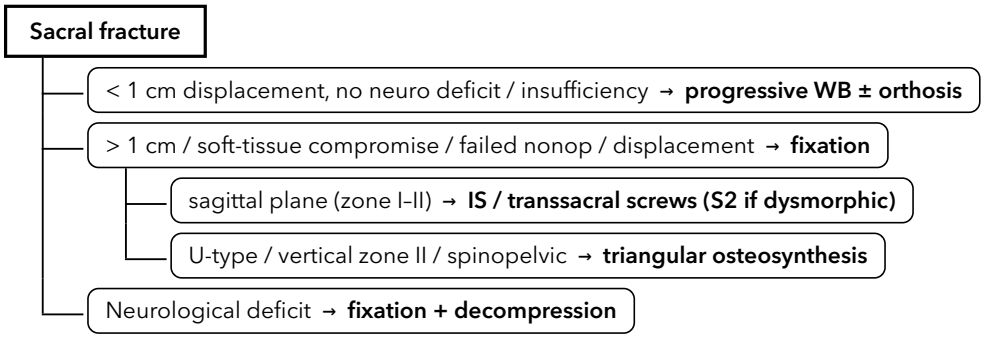
Roy-Camille (zone 3 transverse, sagittal) (TO)	Description
1	**Angulated, no translation**
2	**Angulated + incomplete translation**
3	**Complete translation + displacement**
4 (Strange-Vognsen)	**Comminuted (axial load)**

Isler (TO) L5–S1 facet: 1 lateral (stable) · 2 through facet · 3 medial (2 + 3 unstable)

AO (TO) A sacrococcygeal (stable) · B unilateral longitudinal · C spinopelvic instability

– – Active space – –

5 Management



Operative criteria

Nonoperative	Operative
< 1 cm displacement, no deficit	**Displacement > 1 cm**
Insufficiency fractures	Soft-tissue compromise, open
–	**Persistent pain / displacement after nonop**
–	**Neurological deficit → + decompression** (indirect traction or laminectomy / foraminotomy)
–	Posterior cortical displacement **> 4 mm**, + EUA, lumbopelvic dissociation (TO)

Spinopelvic dissociation (U-type) & triangular osteosynthesis

Most stable for unstable transforaminal / U-type
triangular osteosynthesis = lumbopelvic (L4 / L5 pedicle → iliac screws) + iliosacral screw
✗ Wrong options: ex-fix · IS alone · anterior plate + IS · transiliac bars · tension band

U-type + bladder loss (TO)
triangular osteosynthesis (allows laminectomy + early WB)
✗ Wrong options: bilateral S1 screws · laminectomy + plating · transsacral screws alone

Vertical zone II + dysmorphic sacrum (TO)
triangular osteosynthesis with S2 transsacral screw
✗ Wrong options: horizontal S1 screw (L5 injury) · posterior tension band (wounds, weak vs shear) · S1 TITS

Neuro prognosis predictor
initial fracture displacement
✗ Wrong options: time to surgery · smoking · BMI · steroids

Elderly bilateral ala fractures
U-type / lumbopelvic instability → fixation · frail: 6 wk NWB then progressive (TO)
✗ Wrong options: rami fractures alone · unilateral ala

– – Active space – –

Iliosacral / transsacral screws – fluoroscopy & dysmorphism

View	Shows
Lateral sacral	**Iliac cortical density (ICD)** – start **posterior to ICD** (normal sacrum) → protects **L5**
Inlet	Anterior limit; **anterior alar indentation** in dysmorphism (in-out-in → L5) (OB)
Outlet	Trajectory vs **S1 foramen**; **cephalad alar border** (TO: critical to avoid superior breach → L5)

L5 breach view	**outlet (TO)** · **inlet (OB)** – read the options
Start point (S1 transsacral)	**posterior to ICD, anterior to neural tunnels** (narrow safe zone); SI screws slightly more posterior
Trajectory	**sacral fracture → horizontal transverse screw** · **SI joint disruption → caudal-to-cranial, posterior-to-anterior** · avoid **over-compression** (nerve injury) → positional screw
Vessel at risk	**deep superior branch of superior gluteal artery** ✗ Wrong options: inferior gluteal
Dysmorphism	**ICD not colinear with alar slope** → misleading · **S1 corridor narrow + oblique**, **S2 corridor larger → S2 transsacral screw ("safe harbour")** (TO)
Dysmorphic signs (TO)	non-colinear SI joints (sacrum high), **acute alar slope**, **mamillary processes**, **tongue-in-groove SI**, residual S1–S2 disc / accessory foramina, crescent void
Screws	not in osteoporotic bone; can't remove fragments; malreduction possible
Posterior tension band plating	two posterior incisions; ± IS screws; **wound complications**

Sacral fragility fractures

Nondisplaced zone 1 insufficiency (79-yr)	**WBAT + gait PT** · fixation if > 1 cm, neuro deficit, or failed nonop ✗ Wrong options: EUA · NWB · lumbar MRI · fixation

6 Complications

Complication	Fact
VTE	Immobility
Iatrogenic nerve injury	Over-compression, malpositioned screws
Malreduction	**Vertical displacement**
Late	LE deficits, **urinary, rectal, sexual dysfunction**

Prognosis	**neurological deficit = most important outcome factor**; **displacement ↑ neuro risk**

X vs Y

	Zone 1	Zone 2	Zone 3
Site	Ala	Foramina	Canal
Deficit	5.9%	28.4%	**~60%**

	SI joint screw	Sacral fracture screw
Trajectory	**Caudal→cranial, posterior→anterior**	**Horizontal, transverse**

How it's asked

"Perianal numbness root?"	**→ S2**
"Highest nerve injury pattern?"	**→ Denis zone 3**
"Most neuro deficit?"	**→ zone 3 (central canal)**
"Predicts neuro dysfunction (U-type)?"	**→ fracture displacement**
"Bowel / bladder dysfunction – pattern?"	**→ zone III (medial to foramen)**
"Vertical zone II + dysmorphism (TO)?"	**→ triangular osteosynthesis with S2 transsacral screw**
"U-type + bladder loss (TO)?"	**→ triangular osteosynthesis**
"Most stable for transforaminal fracture?"	**→ triangular osteosynthesis**
"Fall onto feet, perineal numbness?"	**→ U-type spinopelvic dissociation**
"Most stable for U-type?"	**→ triangular osteosynthesis**
"View preventing cephalad breach (TO)?"	**→ outlet view**
"Constricted S1 corridor – alternative (TO)?"	**→ S2 transsacral screw**
"View showing anterior indentation (dysmorphic)?"	**→ inlet view**
"SI screw – true?"	**→ guide wire posterior to ICD is safe in normal sacra**
"S1 transsacral start point?"	**→ posterior to ICD, anterior to neural tunnels**
"79-yr nondisplaced zone 1 insufficiency?"	**→ WBAT + PT**
"Lumbopelvic instability in elderly?"	**→ bilateral sacral ala fractures**
"Roy-Camille type (TO)?"	**→ type 2 (angulated + incomplete translation)**

– – Active space – –

SCAPULA FRACTURES

– Trauma · 53 / 67 –

1 Basics

Epidemiology

Incidence	**< 1% of fractures**, 3–5% of shoulder girdle · age 25–50, M > F · mortality 2–5%
Location	**body / spine 45–50%** · glenoid 35% (neck 25%, fossa / rim 10%) · acromion 8% · coracoid 7%
Mechanism	**high energy 80–90%** (**MVC > 70%**) · FOOSH · dislocation: **anterior → anterior rim**, **posterior (seizure, shock) → posterior rim**

Associated injuries (rib / chest injury, ISS)

Associated injuries **80–95%** · **↑ Injury Severity Score**
✗ Wrong options: ↑ mortality · LOS · ICU

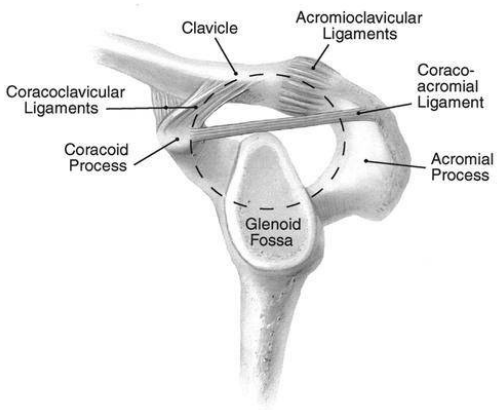
Injury	Rate
Chest injury (commonest)	**80%** – pneumo / haemothorax > 30%, **contusion > 40%**
Rib fractures (commonest fracture)	**52–53%**
Head injury	35–50%
Ipsilateral limb	50% (**clavicle 25%**)
Spine	26–30%
Pelvis / acetabulum	15%
Vascular (subclavian / axillary)	11%
Brachial plexus	5–13% (**75% resolve**)

CT chest consider for ribs + lung injury

Anatomy

Body	**18 muscles**; spine separates supra- / infraspinatus; **spinoglenoid notch** → infraspinatus only · scapular notch → supra + infraspinatus
Glenoid	pear-shaped, **1–5° retroversion**, **15° superior tilt**; labrum deepens **50%**
Acromion	pre- (tip), meso- (mid), meta- (base) acromion · **os acromiale** (unfused meso / meta) ≠ fracture
Coracoid	ossification centres open **to ~25 yr** (angle, tip); conjoint tendon, pectoralis minor; CC ligaments **~25 mm from tip**
SSSC (superior shoulder suspensory complex)	**distal clavicle, acromion, coracoid, glenoid + AC + CC ligaments · double disruption = unstable ("floating shoulder")**
Axillary nerve	inferior to glenoid · motion ⅓ scapulothoracic, ⅔ glenohumeral

– – Active space – –



Superior shoulder suspensory complex

3 Presentation

Often missed with distracting injuries · diffuse pain, dyspnoea, chest pain · passive ROM only · check **axillary, radial, median, ulnar**, pulses

4 Diagnosis & Classification

Key radiology findings

Study	Measurement
True AP, **Grashey**, **scapular Y**, axillary; **CXR**	Pneumothorax; **medial border–spine > 1 cm → scapulothoracic dissociation**
Glenopolar angle (Grashey)	Normal **30–45°**; **< 20–22° → poor / operative**
Lateral border offset	Medialisation
Angulation	**Scapular Y**
CT (3D)	Intra-articular, displacement > 1 cm, thoracic / spine injuries
MRI	SSSC ligament injury

Classifications

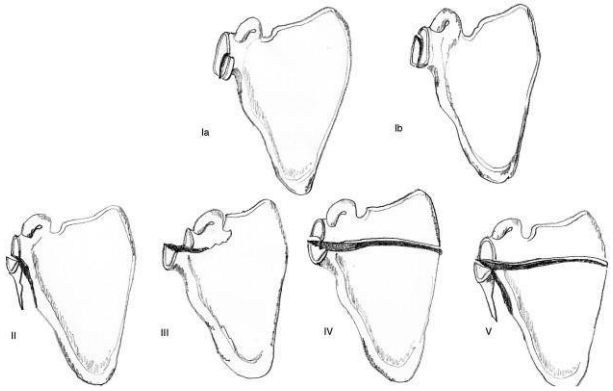
Ideberg (glenoid, Goss)	Fracture
Ia	**Anterior rim**
Ib	**Posterior rim**
II	Fossa → exits **inferiorly**
III	Fossa → exits **superiorly**
IV	Fossa → exits **medially** through body
Va / Vb / Vc	II + IV · III + IV · II + III + IV
VI	**Severe comminution**

– – Active space – –

AO-OTA glenoid (more reliable) 14F0 extra-articular (neck) · 14F1 simple intra-articular (rim, transverse) · 14F2 multifragmentary · 14B body extension

Kuhn (acromion)	Fracture
I	Non / minimally displaced
II	Displaced, subacromial space OK
III	**Displaced → narrows subacromial space**

Ogawa (coracoid)	Fracture
I	**Proximal to CC ligaments**
II	**Tip** (distal)



Ideberg classification

5 Management

Scapula fracture

- Most (90% minimally displaced) → **sling 2–3 wk + early motion (weekly X-ray × 3 wk)**
- Glenoid: step-off > 4 mm or > 20–25% rim with subluxation → **ORIF**
- Body / neck: medialisation > 20 mm, angulation > 40°, GPA < 20–22°, translation > 1 cm → **ORIF (Judet)**
- Floating shoulder (SSSC double disruption) → **fix (clavicle ± scapula)**

Operative indications — glenoid neck displacement, SSSC double disruption, scapular spine

– – Active space – –

Fracture	ORIF if
Body	**Medialisation > 20 mm**, **GPA < 20–22°**, **angulation > 40°** (> 45°, OB), medialisation > 15 mm + angulation > 35°
Neck	**Angulation > 40°**, **translation > 1 cm**, GPA < 20–22°, **SSSC double disruption**
Glenoid	**> 20–25% anterior / posterior rim + subluxation**; **step-off > 4 mm** (> 5 mm OB · > 3–4 mm OB)
Acromion	**Kuhn III**, painful nonunion, impingement, double disruption
Coracoid	> 1 cm, nonunion, Ogawa I into body, double disruption (✗ Wrong options: Ogawa II tip (nonop even > 1 cm))
Any	Open fracture

Neck: medial 3 mm, angulated 20° — **sling 2 wk + PT**
✗ Wrong options: ORIF

Displaced lateral scapular spine, weak abduction — **ORIF** (poor deltoid lever, impingement)
✗ Wrong options: stimulator · brace · acromioplasty · excision

Displaced extra-articular scapula + comminuted clavicle — **ORIF scapula + clavicle**
✗ Wrong options: sling · fix one only

Glenoid neck + strongest operative indication — **ipsilateral distal clavicle fracture (floating shoulder)**
✗ Wrong options: plexus injury (75% resolve) · LC1 pelvis · pneumothorax · 2 rib fractures

Double disruption with neck fracture — **AC ligament injury**
✗ Wrong options: labrum, supraspinatus, transverse scapular ligament, coracohumeral ligament

Nonop outcomes — union 6–8 wk; near-normal function; **may displace in first 3 wk** (rib fractures, SSSC injury) · poor if GPA < 20°

Floating shoulder

Definition (TO) — **ipsilateral clavicle + scapular neck fracture LATERAL to coracoid (anatomical neck)** · surgical neck (medial to coracoid) + clavicle + **CA + CC ligament disruption** also counts — TO explanation calls both correct

Surgical neck — medial to coracoid base (common) · **anatomical neck** — lateral (rare, unstable)

Floating-shoulder history (TO)	Contribution
1799 Vogt	Ipsilateral clavicle + scapula fracture → "unstable shoulder"
1975 Ganz & Noesberger	Unstable shoulder: scapular neck + clavicle fracture / AC dislocation + CC + CA ligament rupture
Herscovici (1980s–90s)	Coined **"floating shoulder"** = ipsilateral midshaft clavicle + scapular neck
Goss 1993	**Superior shoulder suspensory complex (SSSC)**
Goss 2006	Floating shoulder = **double disruption of SSSC**

– – Active space – –

SSSC (TO) osseoligamentous ring: **glenoid, coracoid, CC ligament, distal clavicle, AC joint, acromion** (+ CA ligament later) · **superior strut = middle third + distal clavicle** · **inferior strut = lateral scapular body–medial glenoid neck junction** · **pelvic-ring analogy**: one break → look for another

Fragments (TO) distal = **glenoid + coracoid** · proximal = acromion, spine, body · double disruption → drooping shoulder, malunion/nonunion, impingement, weakness; high plexus + axillary vessel association

Sequence (TO) **fix clavicle first** (ligamentotaxis realigns neck) → reassess → fix scapula only if GPA <20°, lateralisation, persistent displacement

Strong bone for fixation (TO) **glenoid neck · lateral scapular border · base of spine · coracoid** · 3.5 / 2.7 mm recon or LC-DCP; locking for comminuted/osteoporotic

Literature (TO)	Recommendation
Edwards (JBJS 2000)	Nonop if displacement **<5 mm**
Van Noort	Nonop if glenoid well aligned (no caudal displacement)
Goss	Fix both; nonop → **drooping shoulder**
Ada & Miller	Cuff dysfunction if both displaced
Romero / Romeo	Poor outcome if **GPA <20°** or displacement **>1 cm**

Glenoid inclination angle (TO) AP, glenoid vs medial-border perpendiculars = **caudal tilt**

Nonop thresholds (TO) displacement **< 5 mm**, GPA **> 30°** · operate at **100% translation**

Posterior (Judet) approach — internervous interval

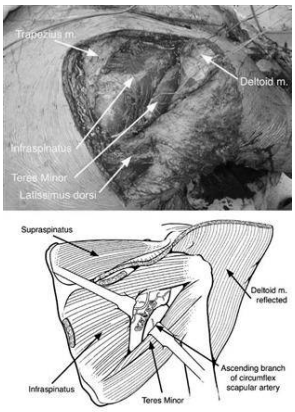
Modified Judet **infraspinatus (suprascapular n.) ↔ teres minor (axillary n.)**
✗ Wrong options: supra / infraspinatus · teres minor / major (axillary / lower subscapular) · deltopectoral

Incision along spine + down vertebral border ("L" / boomerang); detach posterior deltoid

At risk **posterior branch axillary nerve** (teres minor retracted inferiorly), **suprascapular nerve + artery** (infraspinatus superiorly), circumflex scapular + posterior humeral circumflex vessels

Other approaches straight posterior (isolated neck / lateral border) · **deltopectoral** (anterior glenoid, Ideberg II; axillary + musculocutaneous risk; subscapularis failure) · lateral mid-axial (inferior glenoid) · posterior over spine (acromion)

Fixation 2.7 / 3.5 mm plates (locking on thin border) · rim: screws / arthroscopic anchors · acromion: lag + recon plate / tension band · coracoid: 2.7 / 3.5 mm screws



Judet approach

Glenoid rim fracture – instability

Large anterior-inferior rim fracture after dislocation	untreated → **recurrent instability** (loss of concavity compression) → ORIF

6 Complications

Complication	Fact
PTOA	Residual step-off → arthroplasty
Malunion	Nonop, angulation / medialisation; glenoid → RSA
Recurrent instability	Young, large bone loss
Neurovascular	Scapulothoracic dissociation → **EMG at 3–6 wk**

Carpal fracture epidemiology

Commonest carpal fracture (TO)	**scaphoid** > triquetrum > trapezium

X vs Y

	Surgical neck of scapula	Anatomical neck
Site	**Medial** to coracoid	**Lateral** to coracoid
Frequency	Common	Rare
+ clavicle	Floating shoulder if CA + CC torn	**Floating shoulder**

How it's asked

"Scapula fracture association in polytrauma?"	→ **↑ Injury Severity Score**
"Most associated orthopaedic injury?"	→ **rib fracture**
"Most common associated fracture?"	→ **rib fracture**
"Most commonly associated injury?"	→ **chest injury**

– – Active space – –

"Neck 3 mm medial, 20° angulated?"	→ **sling 2 wk + PT**
"Displaced scapular spine, weak abduction?"	→ **ORIF**
"Scapula + comminuted clavicle?"	→ **ORIF both**
"Strongest operative indication with glenoid neck fracture?"	→ **ipsilateral distal clavicle fracture**
"Double disruption with neck fracture?"	→ **AC ligament injury**
"Judet interval?"	→ **infraspinatus / teres minor (suprascapular / axillary)**
"Large anterior rim fracture untreated?"	→ **recurrent instability**
"Floating shoulder definition (TO)?"	→ **clavicle + scapular neck lateral to coracoid**
"Commonest carpal fracture (TO)?"	→ **scaphoid**
"Floating shoulder (Goss 2006)?"	→ **double disruption of the SSSC** (TO)

SCAPULOTHORACIC DISSOCIATION

– Trauma · 54 / 67 –

1 Basics

Mechanism & associations

Mechanism	**lateral traction** on the shoulder girdle (high energy, motorcycle) · heart, chest wall, lung trauma
Orthopaedic	scapula, **clavicle** (widely displaced), **AC separation**, SC dislocation · **flail extremity 52%**
Vascular	**subclavian artery (commonest)** > axillary
Neurological	**up to 90% (> 90%)**; ipsilateral **brachial plexus**, often **complete** · **neuro > vascular** injuries

Anatomy

Scapulothoracic joint	sliding articulation over **ribs 2–7** · abduction **2:1 (GH 120° : ST 60°)**

3 Presentation

Exam	massive shoulder swelling + bruising · **↓ / absent pulses** (hallmark) · neurological deficits (**critical**)

4 Diagnosis & Classification

Key radiology findings

Study	Finding
AP chest	**Medial scapular border > 1 cm further from spinous process** than other side
Shoulder views	Displaced clavicle fracture, **AC widening**, SC dislocation
Chest CT (axial)	**Asymmetric spinous process–scapula distance**
Angiogram	Subclavian / axillary injury

– – Active space – –



Scapulothoracic dissociation — laterally displaced scapula

5 Management

Scapulothoracic dissociation

- No significant vascular injury, stable → **supportive care (collaterals)**
- Unstable + axillary injury → **high lateral thoracotomy + repair**
- Unstable + subclavian injury → **median sternotomy + repair**
- Clavicle / AC injury → **ORIF (avoid neurogenic nonunion)**
- Complete brachial plexus injury → **early forequarter amputation**

Flail extremity treatment	sling, tendon transfers, **forequarter amputation** · **~21% early above-elbow amputation**

6 Complications

Prognosis — neurovascular injury and flail extremity

Mortality	**10%** ✗ Wrong options: 30%
Best predictor of function	**neurological status / deficit** ✗ Wrong options: open wound size, contamination, debridement timing · pulses · age
Worst outcome	**complete brachial plexus injury** · vascular injuries usually repairable → don't worsen outcome
Most likely outcome with neuro injury	**flail extremity** (death also common)
Uniquely associated	**flail upper extremity** ✗ Wrong options: malunion · intubation · abdominal / C-spine injury
EMG	**3–6 wk** after injury

How it's asked

"Best predictor of functional outcome?"	→ **neurological deficit**

"Likely outcome with neuro injury?"	→ **flail extremity**
"Exam finding predicting outcome?"	→ **neurological compromise**
"Most important outcome predictor (open humerus too)?"	→ **neurovascular status**
"Uniquely associated?"	→ **flail upper extremity**
"Prognosis – true?"	→ **complete brachial plexus injury has the worst outcome**

– – Active space – –

SEPTIC ARTHRITIS - ADULT

1 Basics

Epidemiology

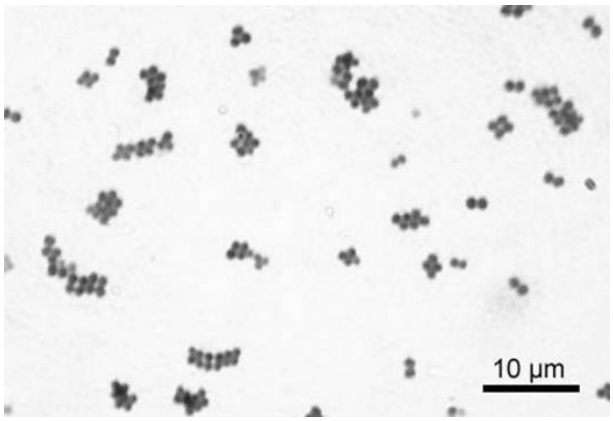
Joints (descending)	**knee (~50%)** > hip > shoulder > elbow > ankle > **sternoclavicular** (IV drug users)
Risk factors	**age > 80**, diabetes, RA, cirrhosis, HIV, crystal arthropathy, endocarditis / recent bacteraemia, **IV drug use**, recent joint surgery, prosthetic implant

Pathophysiology

Seeding	**bacteraemia** · **direct inoculation** (trauma, surgery) · **contiguous spread** (adjacent osteomyelitis)
Cartilage destruction	irreversible; **begins by 8 hours** · proteolytic enzymes from **PMNs**
S. aureus vs gonococcus – joint destruction	**S. aureus → greater WBC influx (> 50,000)** ✗ Wrong options: filamentous appendages (that is gonococcal adhesion; S. aureus uses MSCRAMMs) · IL-1 inhibitors / endotoxin (both organisms induce IL-1 + have endotoxin activity) · penicillin sensitivity

Organisms

Organism	Setting
Staphylococcus aureus	**Commonest (> 50%)** in all patients, **including IV drug users** · gram +ve cocci in **grape-like clusters**
Neisseria gonorrhoeae	**~20%** · commonest in **healthy sexually active adolescents / young adults** · **intracellular + extracellular gram −ve diplococci**
Gram −ve bacilli (E. coli, Proteus, Klebsiella, Enterobacter)	**10–20%** · neonates, IV drug users, elderly, immunocompromised diabetics
Streptococcus (Group A pyogenes commonest; Group B agalactiae)	Gram +ve cocci in **chains** · Group B → infants, elderly, diabetics
Pseudomonas aeruginosa	**IV drug abuse** (commonest SC organism in the **1980s**)
Propionibacterium acnes	**Shoulder surgery**
Salmonella / Strep. pneumoniae	**Sickle cell disease**
Bartonella henselae	**HIV**
Pasteurella multocida	**Dog / cat bite**
Eikenella corrodens	**Human bite**
Fungal / Candida	Immunocompromised



Staphylococcus aureus – gram-positive cocci in clusters

Sternoclavicular septic arthritis

Frequency — **~1% of septic joints** · IV drug users, **diabetes**, renal insufficiency / **haemodialysis**, immunocompromise, alcoholism, HIV

Organism — **S. aureus** – even in IV drug users · gonococcal, fungal, candida in HIV
✗ Wrong options: Pseudomonas (↑ incidence but not commonest) · Strep. pneumoniae · S. epidermidis · P. acnes · gonococcus

Presenting symptom — **chest wall tenderness**
✗ Wrong options: shoulder pain (much less often)

Imaging — **CT / MRI routinely before surgery → retrosternal abscess, chest wall phlegmon**, extent of osteomyelitis (may need general / cardiothoracic surgeon) · MRI better; CT if MRI unavailable

Treatment — **I&D + antibiotics**; **aspiration alone ineffective** · with **medial clavicular osteolysis (osteomyelitis) → resection of SC joint** (best outcome, lowest recurrence)
✗ Wrong options: antibiotics alone · repeat aspiration · hyperbaric O_2 · simple I&D (high recurrence)



Sternoclavicular septic arthritis

Septic olecranon / prepatellar bursitis

Organism — **S. aureus ~80% → gram +ve cocci in pairs and clusters** · mixed flora common
✗ Wrong options: chains (Strep) · branching bacilli (Nocardia / Actinomyces) · gram –ve diplococci (Neisseria) · thin rods (E. coli)

Route — **direct inoculation** (poor bursal blood supply) · middle-aged men

– – Active space – –

3 Presentation

Symptoms joint pain · **fever in only 60%** · may look toxic

Exam erythema, effusion, warmth, tenderness · **can't bear weight**, **can't tolerate PROM** · limb held in position of **maximum joint volume → hip in FABER**



Hip held in FABER

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Radiographs (AP + lateral)	Joint space widening / effusion, periarticular osteopenia
Ultrasound	Confirms effusion in **large joints (hip)**; guides aspiration
MRI	Effusion + adjacent **osteomyelitis**
CT	Intra-articular air (traumatic arthrotomy adjunct) · SC joint abscess



Lateral knee — septic effusion

Diagnostic criteria

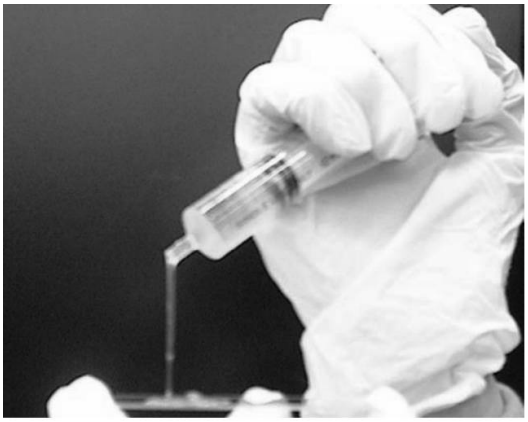
Test	Septic value
Serum WBC	**> 10,000** with left shift
ESR	**> 30 mm/h** · may be normal early; rises in 2 days, may rise 3–5 days after antibiotics, normal by 3–4 wk
CRP	**> 1 mg/dL** · **most helpful** · rises within hours, **normalises within 1 wk → best to monitor treatment**
Synovial WBC	**> 50,000** (native joint) = diagnostic · lower counts still infected if gram stain / culture +ve
Synovial PMN	High (e.g. 90%)
Gram stain	Identifies organism only **⅓ of the time**
Glucose	**< 60% of serum**
Viscosity	**Negative "string" sign** (low viscosity)

Aspiration — **gold standard**; send **cell count + differential, gram stain, culture, glucose, crystals**

Antibiotics within 24 h before arthrocentesis — **↓ synovial WBC → false negative**

Crystals + septic — coexist in **< 5%** · **crystals do NOT exclude infection**

Septic shock — hypotension needing **vasopressors to keep MAP ≥ 65 mmHg** + **lactate > 2 mmol/L** despite fluids → mortality **> 40%** · aspirate, then **immediate broad-spectrum IV antibiotics**



Negative string sign

Diagnosis — aspiration & coexisting crystals

Red, hot 1st MTP, ↑ ESR / CRP — **joint aspiration first** (septic vs gout)
✗ Wrong options: I&D before diagnosis · oral antibiotics · indomethacin before aspiration · MRI

IV drug user, ankle WBC 95,000 + negatively birefringent crystals — **urgent I&D** · gout / pseudogout does not remove need for surgery
✗ Wrong options: await culture · 24 h IV antibiotics then reassess · analgesia · ESR / CRP / urate

– – Active space – –

Diabetic bacteraemia → septic knee, septic shock	**S. aureus** ✗ Wrong options: gonococcus · gram −ve bacilli · Strep. pyogenes · Pseudomonas

Traumatic arthrotomy – saline load test

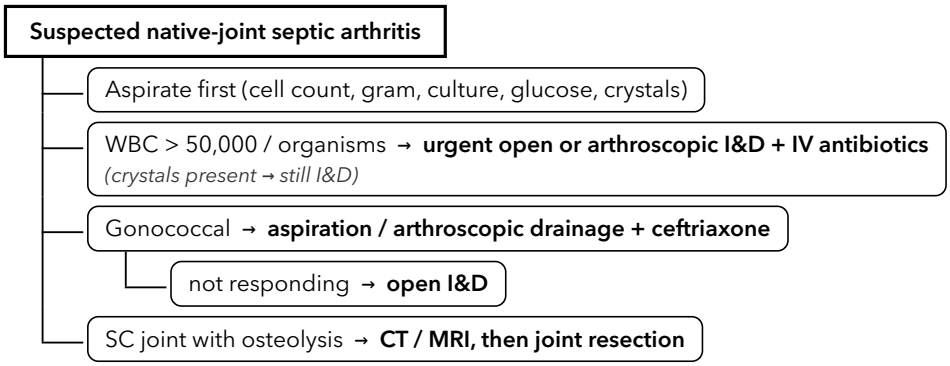
Saline into knee	Sensitivity (1 cm inferolateral wound)
100 mL	**< 75%** – insufficient
110 mL	**75%**
145 mL	90%
155 mL	**95%**
175 mL	**99%**

Technique	**18 G** needle, saline with **passive ROM** · **egress through wound = positive** (arthrotomy)
Traps	**methylene blue does NOT ↑ sensitivity** · **exam alone performs worse** than SLT · **CT → intra-articular air** (adjunct; ✗ Wrong options: MRI routinely)
100 mL, no egress, wound near patella	**inject at least another 55 mL** ✗ Wrong options: close + discharge · arthroscopy yet · MRI · secondary intention
Positive	surgical I&D (open / arthroscopic) to prevent septic arthritis

Differential

Crystal arthropathy (gout, pseudogout) · **cellulitis** · **bursitis** (prepatellar)

5 Management



Emergency	orthopaedic surgical emergency; damage within **8 hours**
Empirical antibiotics	**young healthy: S. aureus + N. gonorrhoeae** · **immunocompromised: S. aureus + Pseudomonas** → switch to culture-directed
Monitoring	serum WBC, ESR, **CRP**
Technique	open or arthroscopic · remove all pus, irrigate · **synovectomy** as needed · fluid + tissue for culture

Gonococcal arthritis

Antibiotic	**ceftriaxone** (3rd-gen cephalosporin; or cefotaxime / ceftizoxime) · **fluoroquinolones** if sensitive · **+ doxycycline** where chlamydia common ✗ Wrong options: penicillin · tetracycline (high resistance) · erythromycin · vancomycin
Forms	**arthritis-dermatitis syndrome (~60%)** · localised purulent arthritis (~40%) · bacteraemic infection; newborns + adolescents
Disseminated (polyarthralgia + trunk rash, gram −ve diplococci)	**serial aspiration + antibiotics** ✗ Wrong options: open I&D first (reserved for non-responders / severe) · I&D without antibiotics · colchicine · observation
Workup	screen for **HIV, syphilis, chlamydia** · response → fever, pain, leucocytosis resolve
Clue	young, sexually active, new vaginal discharge → take sexual history

Paediatric septic hip — Kocher criteria & urgent washout (TO)

Criterion (1 point each) (TO)	In original Kocher?
Fever > 38.5 °C	Yes
ESR > 40	Yes
WBC > 12,000/mm³	Yes
Refusal to bear weight	Yes
CRP	No — added later by Caird

Sensitivity order (TO)	**fever > CRP > ESR > refusal to bear weight > WBC** · **CRP most sensitive lab**
5-yr, fever 38.5 °C, ↑ CRP / WBC, effusion (TO)	**emergency aspiration + washout, then IV antibiotics** · **US = investigation of choice** for the painful child's hip · arthrotomy gives tissue for culture ✗ Wrong options: blood cultures + antibiotics · US-guided aspiration + antibiotics · antibiotics alone
4-yr, fever, ↑ ESR / CRP, hip collection on US (TO)	answer key not provided by the source

Metaphyseal osteomyelitis & joint spread (TO)

Intracapsular metaphyses (TO)	**proximal radius, proximal humerus, proximal femur, distal tibia + fibula** → osteomyelitis decompresses into joint → septic arthritis
EXCEPT (TO)	**head (distal end) of ulna**

Tuberculosis of joints (TO)

Rice / millet / melon seed bodies (TO)	**tuberculous synovitis / tenosynovitis** ✗ Wrong options: PVNS · RA · Madura foot
First radiographic sign of healing (TO)	**remineralisation of bone** (trabecular pattern, ↓ osteopenia) · spine: + sharpened disc margins, ↓ paravertebral shadow, fusion ✗ Wrong options: joint space reappears · sequestrum removal · bony ankylosis

– – Active space – –

TB hip (TO) **second commonest osteoarticular TB** · usually ends in **fibrous ankylosis** · advanced → **FADIR** usually; complete iliofemoral ligament resorption → **FABER**

FALSE statement (TO) "**Shanmugasundaram classification applies only to paediatric hip TB**" — it applies to **children and adults**

Shanmugasundaram (1983, radiological) (TO)	Feature
Normal	Synovial disease
Travelling / wandering acetabulum	Roof lesion → head migrates upward
Dislocating	Posterior dislocation / subluxation (lax capsule)
Perthes type	Fragmented, dense, flat epiphysis
Protrusio acetabuli	Floor lesion → medial migration
Atrophic	Marked joint narrowing
Mortar & pestle	Head / acetabulum destroyed → mismatch
Unclassified (Agarwal, 8th)	Triradiate, pseudarthrosis coxae, ankylosed

Other TB hip classifications (TO) **Babhulkar + Pande 2002** (clinico-radiological) · Tuli modification · **Moon 2012**: **stable** — Perthes, atrophic, protrusio · **unstable** — travelling acetabulum, mortar + pestle, dislocating

6 Complications

Complication	Fact
Arthritis	Cartilage loss from **8 h**
Fibrous ankylosis	–
Osteomyelitis	Contiguous spread

X vs Y

	S. aureus	N. gonorrhoeae
Share	**> 50%**	**~20%**
Gram	+ve cocci, clusters	−ve diplococci, intra + extracellular
Host	All, incl. IV drug users	Healthy sexually active young
Synovial WBC	**Higher → more destruction**	Lower
Treatment	**Urgent I&D** + IV antibiotics	**Aspiration / drainage + ceftriaxone**

	Septic arthritis	Gout
Synovial WBC	**> 50,000**	Usually lower
Crystals	Coexist < 5%	**Negatively birefringent** urate
Treatment	**I&D**	**Indomethacin**

How it's asked

"SC septic arthritis organism, IV drug user?"	→ **S. aureus** ✗ Wrong options: Pseudomonas
"SC septic arthritis, true?"	→ **CT / MRI routinely before surgery**
"SC infection with medial clavicle osteolysis?"	→ **resection of SC joint**
"Saline volume for 95%?"	→ **155 mL**
"175 mL saline detects?"	→ **99%**
"100 mL, no egress – next?"	→ **inject another 55 mL**
"Saline load test – true?"	→ **175 mL diagnoses 99%** ✗ Wrong options: methylene blue ↑ sensitivity
"Gonococcal antibiotic?"	→ **ceftriaxone**
"Young woman, vaginal discharge, septic knee – gram stain?"	→ **intra + extracellular gram −ve diplococci**
"Why S. aureus destroys more than gonococcus?"	→ **greater WBC influx**
"Disseminated gonococcal arthritis, first treatment?"	→ **serial aspiration + antibiotics**
"Red 1st MTP, ↑ CRP – next?"	→ **joint aspiration**
"WBC 95,000 + urate crystals?"	→ **urgent I&D**
"Septic knee in septic shock – organism?"	→ **S. aureus**
"Septic olecranon bursitis organism?"	→ **gram +ve cocci in pairs and clusters**
"Not in Kocher criteria (TO)?"	→ **CRP**
"Child, fever, ↑ CRP, hip effusion (TO)?"	→ **emergency aspiration + washout + IV antibiotics**
"Osteomyelitis NOT spreading to joint (TO)?"	→ **head of ulna**
"Melon seed bodies (TO)?"	→ **tubercular synovitis**
"First healing sign, TB foot (TO)?"	→ **remineralisation**
"False about TB hip (TO)?"	→ **Shanmugasundaram only paediatric**

– – Active space – –

SI DISLOCATION & CRESCENT FRACTURES

– Trauma · 56 / 67 –

1 Basics

Spectrum

Injury	Posterior SI ligaments	Stability
Incomplete SI dislocation	**Intact**	**Rotationally unstable**
Complete SI dislocation	**Ruptured**	**Rotationally + vertically unstable**
Crescent fracture (SI fracture-dislocation)	Posterior ilium stays attached to sacrum	Anterior ilium dislocates with **internal rotation**

Crescent fracture — **iliac wing fracture entering the SI joint**; posterior iliac "crescent" stays with sacrum = **Young-Burgess LC-II**

Anatomy

SI stabilisers — sacrospinous, sacrotuberous, anterior + **posterior SI ligaments**

L5 root — crosses sacral ala **~2 cm medial to SI joint**

Superior gluteal artery — crosses SI joint, exits via **greater sciatic notch**

2 Mechanism

Lateral compression, high energy · iliac wing fractures may be **open** or trap **bowel**

3 Presentation

Exam — haemodynamics, detailed neuro, abdominal distension, **rectal exam**, **blood at meatus**

4 Diagnosis & Classification

Key radiology findings

Views — AP pelvis + **inlet / outlet** · **CT** – posterior ring + sacrum

Classification — within **Young-Burgess** (crescent = **LC-II**) and **Tile**



Crescent fracture on AP pelvis

5 Management

Injury	Treatment
Vertical translation	**Immediate skeletal traction**
Incomplete SI + symphysis diastasis	**Anterior ring ORIF**
Complete SI dislocation	**Anterior + posterior fixation**
Crescent fracture	**ORIF of ilium** (restores posterior ligament stability)

Approaches anterior (lateral window, elevate iliacus to SI) · **posterior** (crescent fragment to intact ilium) · plates + **iliosacral lag screws**

Iliosacral screws – fluoroscopic views & L5 root injury

View	Shows
Inlet	**Anterior-posterior** position
Outlet	**Cephalad-caudad** position + **sacral neural foramina (best)**
Lateral sacral	Anterior alar cortex + nerve tunnel → **L5 safety**

In-out-in (too anterior) screw **L5 root → weak great toe extension / dorsiflexion (EHL)**
✗ Wrong options: ankle DF (L4 TA) · plantarflexion (S1) · knee extension · bowel / bladder

Positioning traction aids reduction; midline sacral bump

6 Complications

Complication	Fact
DVT	**35–50%**
Neurological injury	L5
Loss of reduction / fixation failure	–

Prognosis **accurate, stable SI joint reduction**

– – Active space – –

How it's asked

"Outlet view purpose during IS screws?"	**→ sacral neural foramina**
"In-out-in screw finding?"	**→ no great toe extension**
"Anterior alar penetration → weakness?"	**→ great toe dorsiflexion**
"Crescent fracture pattern?"	**→ LC-II**

STERNOCLAVICULAR DISLOCATION

– Trauma · 57 / 67 –

1 Basics

Types

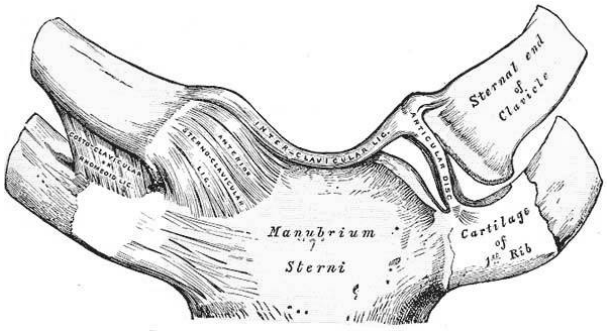
Traumatic	**anterior (commoner)** · **posterior → mediastinal structures at risk** · high energy (MVA, contact sport)
Atraumatic subluxation	younger, **generalised ligamentous laxity**; with **overhead elevation**, reduces on lowering arm
Medial clavicle physis	**first bone to ossify, last physis to close (20–25 yr)** · under 20–25 yr "dislocation" is usually a **physeal fracture** · closure ~25 yr (OB)

Stabilisers – posterior capsular ligament

SC joint **diarthrodial saddle joint**, **incongruous (~50% contact)**, **fibrocartilage**, intra-articular disc · stability is **ligamentous**

Structure	Function
Posterior capsular (SC) ligament	**Most important – primary restraint to anterior-posterior stability**
Anterior SC ligament	**Primary restraint to superior displacement** of medial clavicle
Costoclavicular (rhomboid) – anterior fasciculus	Resists **superior rotation + lateral displacement**
Costoclavicular – posterior fasciculus	Resists **inferior rotation + medial displacement**
Intra-articular disc ligament	Prevents **medial displacement**; secondary restraint to superior displacement

Prevents posterior dislocation **posterior SC joint capsule**
✗ Wrong options: first rib · disc ligament · costoclavicular · interclavicular



Sternoclavicular joint ligaments

3 Presentation

Anterior palpable bump; prominence ↑ with abduction + elevation

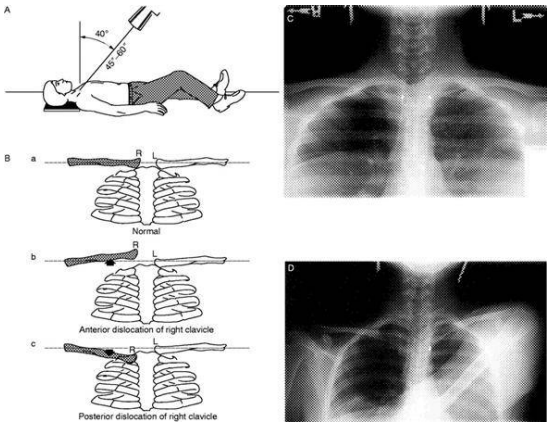
– – Active space – –

Posterior	**dyspnoea, dysphagia**, tachypnoea, **stridor worse supine** · **paraesthesia** in arm · **venous congestion (unilateral jugular engorgement)** / ↓ pulse · ↓ arm ROM · turning head to affected side relieves pain
Posterior complications	**early**: pneumothorax, great vessel laceration / erosion / occlusion, **oesophageal rupture**, **brachial plexus compression**, respiratory distress, myocardial conduction abnormality · **late**: **tracheo-oesophageal fistula**, stridor, dysphagia

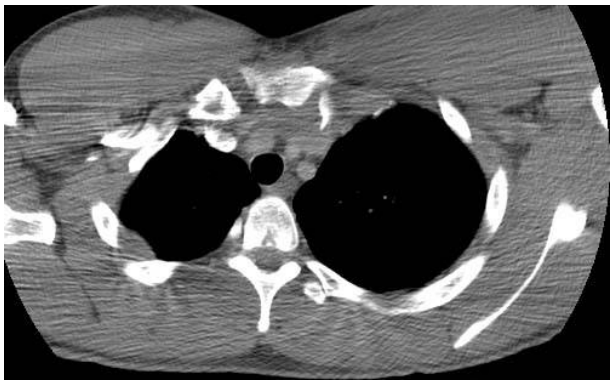
4 Diagnosis & Classification

Key radiology findings

Study	Finding
AP chest	Hard to see; often read as normal
Serendipity view (40° cephalic tilt)	**Anterior → clavicle above** contralateral · **posterior → clavicle below**
CT (axial)	**Study of choice** – direction, **mediastinal injury**, **distinguishes physeal fracture**
MRI	Adds nothing; delays treatment (avoid)

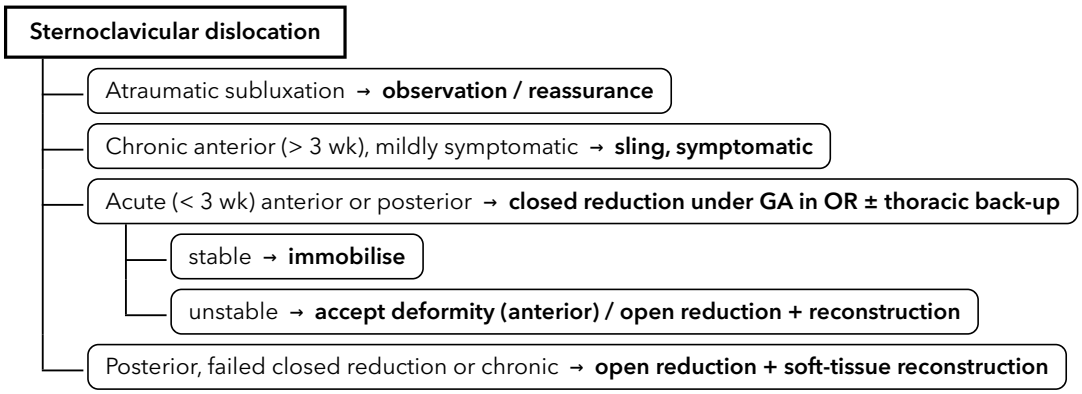


Serendipity view



Axial CT – posterior SC dislocation

5 Management



Operative criteria

Situation	Treatment
Atraumatic subluxation	**Reassurance**, symptomatic
Chronic anterior, minimally symptomatic (> 3 wk)	**Sling**, unrestricted activity by **3 months**
Acute anterior / posterior (< 3 wk)	**Closed reduction under GA ± thoracic back-up**
Acute posterior, failed closed reduction (dysphagia, dyspnoea, ↓ pulses)	**Open reduction + soft-tissue reconstruction**
Chronic anterior / posterior, **persistently symptomatic**	**Open reduction + reconstruction**
Chronic posterior (> 3 wk) with dysphagia	**Open reduction in OR + thoracic back-up** (✗ Wrong options: sling (avoid late mediastinal problems))

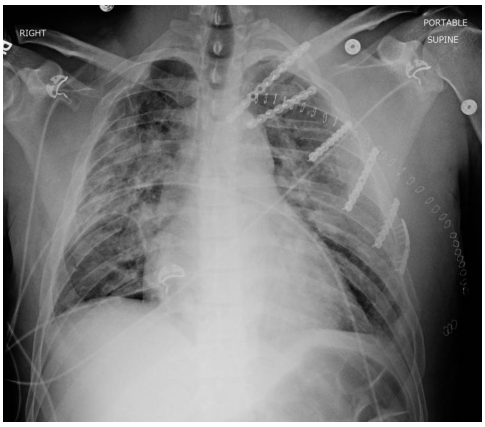
Posterior dislocation — reduction in OR with thoracic backup

Posterior + ↓ pulses (16-yr)	**reduction in OR with thoracic surgery back-up** ✗ Wrong options: sling · ED reduction · ORIF + pinning of physis
Closed reduction fails	in **up to 75%** → open reduction
Initial treatment	**closed reduction in OR with towel clip** (anterior translation) · usually stable ✗ Wrong options: observation · open first · ORIF · allograft reconstruction first
After closed reduction	**immobilisation 4 wk** (OB) · **Velpeau bandage 6 wk** (Orthobullets) — read the options · elbow exercises at 3 wk · sport at 3 months
Cardiothoracic stand-by	**reduction of posterior SC dislocation** · routine thoracic surgeon now **controversial** but still recommended ✗ Wrong options: midshaft clavicle ORIF · distal clavicle excision · SC joint I&D · Judet scapula
Never	**ED reduction** (mediastinal injury), **compression plating across SC joint**, **K-wires** (migration)
Closed reduction technique	supine, arm at table edge, **abduct + extend** with **axial traction**, direct posterior pressure over medial clavicle · **towel clip / fingers** pull medial clavicle

– – Active space – –

Open reduction & reconstruction

Figure-of-8 tendon graft	**2 drill holes in manubrium + 2 in medial clavicle**, **1 cm apart** · **gracilis or semitendinosus** · don't disrupt SCM sheath; clean capsulotomy for repair
Medial clavicle excision	outdated · resect **< 15 mm** to **preserve costoclavicular ligaments** (repair if injured) · Wirth + Rockwood: costoclavicular reconstruction ± medial head resection for unstable injuries



SC joint reconstruction

Fixation hazards — pin migration

Highest life-threatening complications	**percutaneous pins (K-wires / Steinmann)** → migrate to **heart, pulmonary artery, aorta** ✗ Wrong options: cannulated screws · cerclage · Balser plate · locking plate

Anterior dislocation — natural history

Anterior dislocation — true	**rarely symptomatic when left unreduced** · surgery mainly **cosmetic** ✗ Wrong options: stable after reduction (persistent instability common) · tracheal injury on reduction · fusion · acute medial clavicle excision

Atraumatic spontaneous subluxation

Diagnosis	no trauma, **lump at base of neck**, worse **overhead** · settles in **4–6 wk** nonoperatively ✗ Wrong options: unreduced posterior · hypoplasia · arthritis · hyperostosis

Beighton-Horan ≥ 5 / 9 = hypermobility · **treatment → observation**
✗ Wrong options: figure-of-8 brace · resection · reconstruction · arthrodesis

Type III SC instability (TO)	**non-structural, abnormal muscle patterning** (inappropriate **pectoralis major**, EMG) → **biofeedback physiotherapy** · surgery only if structural abnormality ✗ Wrong options: closed reduction + sling · open reduction + reconstruction · open reduction + sling

6 Complications

– – Active space – –

Complication	Fact
Cosmetic deformity	Anterior — usually well tolerated
Mediastinal injury	Posterior — great vessels, trachea, oesophagus
Pin migration	Heart, pulmonary artery, aorta → **avoid K-wires**

X vs Y

	Anterior SC dislocation	Posterior SC dislocation
Frequency	**Commoner**	Rarer
Risk	Cosmetic	**Mediastinal**
Serendipity	Clavicle **above**	Clavicle **below**
Acute	Closed reduction; accept if unstable	**Closed reduction in OR + thoracic back-up**
Chronic	**Observe** (if mild)	**Reduce** (open)

How it's asked

"Posterior SC + ↓ pulses?"	**→ reduction in OR with thoracic back-up**
"3-month-old posterior SC with dysphagia?"	**→ open reduction in OR with thoracic back-up**
"Unilateral JVD, posterior SC — next?"	**→ closed reduction in OR under GA, immobilise 4 wk**
"Footballer posterior SC on CT — next?"	**→ closed reduction in OR**
"Best initial treatment, posterior SC?"	**→ closed reduction with towel clip**
"Which surgery needs cardiothoracic stand-by?"	**→ posterior SC dislocation reduction**
"Structure preventing posterior dislocation?"	**→ posterior SC joint capsule**
"Primary A-P restraint?"	**→ posterior SC capsular ligament**
"Anterior SC dislocation — true?"	**→ rarely symptomatic when unreduced**
"Atraumatic subluxation, Beighton 6?"	**→ observation**
"No trauma, neck lump, overhead?"	**→ spontaneous SC subluxation**
"Type III SC instability (TO)?"	**→ biofeedback physiotherapy**
"Device with most life-threatening complications?"	**→ percutaneous pins**

– – Active space – –

SUBTALAR DISLOCATIONS

– Trauma · 58 / 67 –

1 Basics

Epidemiology

Incidence **1% of all dislocations** · < 1 / 100,000 / yr · young / middle-aged men

Definition dislocation of **both subtalar (talocalcaneal) + talonavicular** joints · + tibiotalar = **total talar dislocation (extrusion)**

High energy **25% open** (**lateral more often open**) · irreducible **~33%** (OB)

Anatomy

Talus **no muscle or tendon attachments** · inferior surface → posterior facet of calcaneus · head → navicular + sustentaculum tali · lateral process → posterior facet + fibula · posterior process: medial + lateral tubercles with **FHL groove**

Talar blood supply	Supplies
Artery of tarsal canal (posterior tibial)	**Most of body – main supply**
Calcaneal branches (posterior tibial)	Posterior talus
Anterior tibial artery	Head + neck
Artery of tarsal sinus (perforating peroneal)	Head + neck
Deltoid artery (deep deltoid)	Body · may be the only supply left after talar neck fracture

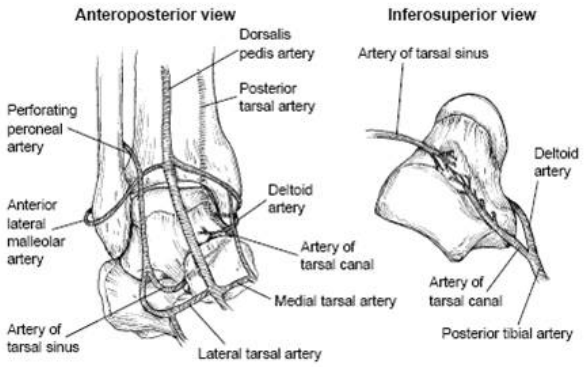


Figure 1 Blood supply to the talus.

Blood supply of the talus

2 Mechanism

Medial **inversion of plantarflexed foot**; talar neck pivots on **sustentaculum tali**

Lateral **eversion of plantarflexed foot**; **anterior process of calcaneus = fulcrum** for anterolateral talus

3 Presentation

Medial foot **locked in supination** = **"acquired clubfoot"**

Lateral foot **locked in pronation** = **"acquired flatfoot"**



Medial subtalar dislocation – foot locked in supination

4 Diagnosis & Classification

Key radiology findings

Study	Finding
AP + lateral foot	**Medial → talar head superior to navicular** (lateral view) · **lateral → talar head collinear / inferior to navicular**
CT after reduction	Associated fractures, **subtalar osteochondral debris** (residual subluxation)

Radiograph talonavicular dislocation is easily missed; it is part of every subtalar dislocation

Anatomic classification (direction of midfoot / forefoot)

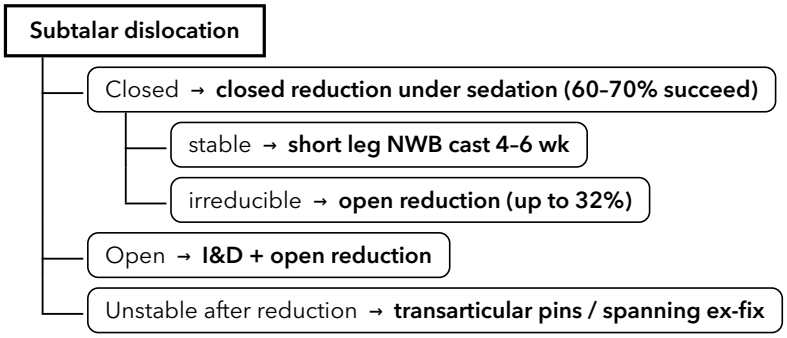
Type	Features	Image
Medial	**Commonest – 65–80%** (Orthobullets) · 65% / ~75% (OB) · **lateral malleolus buttress** · fractures: **dorsomedial talar head, posterior process of talus, navicular**	
Lateral	**More often open**, more associated fractures, **higher energy** · locked in pronation · fractures: **cuboid, anterior calcaneus, lateral process of talus, fibula**	
Anterior	Rare	–
Posterior	Rare	–
Total (talar extrusion)	Out of ankle + subtalar + talonavicular · **usually open**	–

– – Active space – –

Patterns & associated fractures (lateral more often open)

Associated fractures	**up to 44%** · **commonest: talus** · occult **lateral process of talus** ✗ Wrong options: fibula · cuboid · calcaneus · navicular
Medial vs lateral – true	**lateral more likely open** ✗ Wrong options: medial more associated fractures · lateral blocked by EDB · medial blocked by tibialis posterior · lateral more stable (medial more stable)
Isolated injury, no fractures – other joint dislocated	**talonavicular** ✗ Wrong options: tibiotalar, calcaneocuboid, MTP, TMT

5 Management



Closed reduction	knee flexion + ankle plantarflexion → **distraction** → hindfoot inversion / eversion (opposite to dislocation) · lateral: traction + hyperpronation, then supination (OB) · **post-reduction CT**

Irreducible dislocation – blocks (EDB medially, tibialis posterior laterally)

Dislocation	Blocks (lateral / medial structures)	Approach
Medial	**EDB** (talar head **buttonholes** through EDB), **peroneal tendons**, **talonavicular capsule**, deep peroneal NV bundle	**Sinus tarsi**
Lateral	**Tibialis posterior (commonest)**, **FHL**, **FDL**, posterior tibial NV bundle	**Medial** (between tibialis anterior + posterior) ± sinus tarsi for debris

Not blocks	tibialis anterior, EHL (too dorsal), plantar fascia, extensor retinaculum, CFL
Irreducible medial	**~10%** even under anaesthesia (OB)



Tibialis posterior interposed — lateral dislocation

Total talar dislocation

Failed closed reduction (closed pantalar, no fracture)	**percutaneous / open reduction + soft-tissue repair** + temporary pins / spanning ex-fix · open → urgent I&D + reduction (reimplant) ✗ Wrong options: primary talectomy + tibiocalcaneal fusion · talectomy + total talus replacement · primary tibiotalar / subtalar arthrodesis
Salvage later	talectomy, fusion, talus replacement for **osteonecrosis / arthritis**

6 Complications

Outcomes — arthritis & osteonecrosis

Complication	Fact
Post-traumatic arthritis	**Subtalar** commonest — radiographic **up to 89%**, **symptomatic 63%**
Stiffness	**Most common complication** (Orthobullets)
Talar osteonecrosis	**Total talar dislocation** — common; **without collapse at 12 months**
Infection	Open — poor muscle cover, poor vascularity

Most common complication	**stiffness** (Orthobullets) · **post-traumatic arthritis** (OB) — read the options; ✗ Wrong options: osteonecrosis of talus
Most symptomatic degenerated joint	**subtalar** ✗ Wrong options: tibiotalar · talonavicular · calcaneocuboid · Lisfranc
12 months after total talar extrusion	**talar osteonecrosis without collapse** ✗ Wrong options: pantalar arthritis · collapse yet · navicular / calcaneus ON
Poor prognosis	**high energy**, **lateral**, **open**, **concomitant subtalar fractures**

– – Active space – –

X vs Y

	Medial	Lateral
Frequency	**Commoner (65–80%)**	Less common
Mechanism	Inversion	Eversion
Foot	Supinated (acquired clubfoot)	Pronated (acquired flatfoot)
Talar head (lateral X-ray)	**Superior** to navicular	Collinear / inferior
Block	**EDB**, peroneals, TN capsule	**Tibialis posterior**, FHL, FDL
Open / fractures	Less	**More**
Outcome	Better, more stable	**Worse**

How it's asked

"Irreducible lateral subtalar – block?"	→ **tibialis posterior tendon**
"Irreducible medial subtalar – block?"	→ **extensor digitorum brevis**
"Most common associated fracture?"	→ **talus**
"Joint dislocated with subtalar?"	→ **talonavicular**
"Lateral vs medial – true?"	→ **lateral more likely open**
"Most symptomatic arthritis?"	→ **subtalar joint**
"Most common complication (OB)?"	→ **post-traumatic arthritis**
"12 months after extruded talus?"	→ **osteonecrosis without collapse**
"Closed pantalar dislocation, failed closed reduction?"	→ **open reduction + soft-tissue repair**

SUBTROCHANTERIC FRACTURES

– Trauma · 59 / 67 –

1 Basics

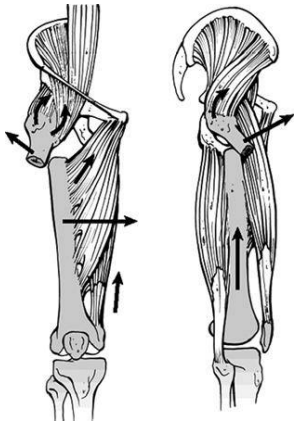
Epidemiology

Definition	proximal femur fracture **from the lesser trochanter to 5 cm distal** to it
Incidence	**7–34% of femur fractures** · bimodal: **young high energy (MVC)** · **elderly low energy (fall)**
Rule out	pathological / **atypical femur fracture** (bisphosphonate – esp. **alendronate**, **denosumab**)

Deforming forces & reduction manoeuvres

Fragment	Deformity	Muscle
Proximal	**Abduction**	**Gluteus medius + minimus** (greater trochanter)
Proximal	**Flexion**	**Iliopsoas** (lesser trochanter)
Proximal	**External rotation**	**Short external rotators** (piriformis, gemelli, obturator internus) ± iliopsoas, gluteus maximus
Distal	**Adduction + shortening**	**Adductors**

Typical proximal deformity	**flexion, abduction, external rotation** · lesser trochanter intact → **abduction + flexion** ✗ Wrong options: adduction · extension · internal rotation
Responsible muscles (proximal displacement)	**gluteus medius + iliopsoas** ✗ Wrong options: gluteus maximus + adductors · rectus · hamstrings
Reduction of proximal fragment	**adduction + extension + internal rotation** ✗ Wrong options: abduction · flexion
Bring distal to proximal	after traction, **flex + externally rotate the distal fragment** ✗ Wrong options: traction alone
Biomechanics	weight bearing → **compression medial cortex, tension lateral cortex**



Deforming forces in a subtrochanteric fracture

– – Active space – –

3 Presentation

History	long-term **bisphosphonate / denosumab** · **prodromal thigh / groin pain** before a low-energy fall
Exam	can't bear weight · **shortening + varus** deformity · flexed proximal fragment may threaten skin

4 Diagnosis & Classification

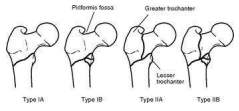
Key radiology findings

Views	AP + lateral hip, AP pelvis, **full-length femur incl. knee** · traction views optional (comminution)
Findings	proximal fragment **flexed + abducted** · distal **adducted + ER**
Bisphosphonate fracture	**lateral cortical thickening ("beaking", "dreaded black line")** · ↑ diaphyseal cortical thickness · **transverse / short oblique** · **medial spike** · **no comminution** · varus remodelling



Atypical femur fracture – lateral cortical thickening

Russell-Taylor classification

Type	Description	Image
I	**No extension into piriformis fossa** (historically → IM nail)	
II	**Extends into greater trochanter + piriformis fossa** (see on **lateral X-ray**) – historically fixed-angle plate; now nailable	

Seinsheimer (TO)	types **1–5** by number + pattern of fragments
Subtrochanteric (TO)	**strongest deforming forces** of proximal femur · reduce **before** nail entry; tools: blunt Steinmann pin, bent guide wire, reamer as lever, temporary K-wires · **never accept varus**, IR or retroversion

AO/OTA (32, subtrochanteric = .1)

Code	Pattern
32-A3.1	Simple, **transverse**
32-B3.1	Wedge, fragmented
32-C1.1	Complex, spiral

Diagnostic criteria – atypical femur fracture (ASBMR revised)

Major (4 of 5 needed)	Minor (optional)
No / minimal trauma (standing-height fall or less)	Generalised ↑ diaphyseal cortical thickness
Starts at **lateral cortex**, **substantially transverse** (may go oblique medially)	**Prodromal groin / thigh pain**
Noncomminuted	**Bilateral** fractures
Complete: both cortices ± **medial spike** · incomplete: lateral cortex only	**Delayed healing**
Localised periosteal / endosteal thickening of lateral cortex	–

Excluded femoral neck, intertrochanteric with spiral subtrochanteric extension, pathological (tumour), periprosthetic

5 Management

- **Subtrochanteric fracture**
  - Non-ambulatory, unfit for surgery → **observation + analgesia (limited role)**
  - Most → **cephalomedullary IM nail (reamed, statically locked)**
  - Fixed-angle plate → **femoral neck fracture, narrow canal, shaft deformity, preference**
    - 95° blade plate / condylar screw (not sliding hip screw)
  - Symptomatic impending AFF (lateral cortical stress line) → **prophylactic IM nail**

Implant choice – cephalomedullary nail vs plate

Implant	Fact
Cephalomedullary nail	**Load-sharing**, preserves vascularity, **stronger in unstable patterns** · **reamed + statically locked → highest union** (✗ Wrong options: unreamed (reaming ↑ vascularity + union without ↑ pulmonary / transfusion risk))
95° blade plate / condylar screw	**Load-bearing**; lateral approach (split / elevate vastus lateralis, **profunda perforators**) · **tension band** · devitalises fragments, weaker in comminution
135° sliding hip screw	**Contraindicated** → loss of fixation, **malunion**, distal fragment displaces **medially + proximally**, proximal rotates on single screw

CMN vs SHS CMN: **lower revision, less pain, better mobility, higher QoL + satisfaction** (EXCEPT; ✗ Wrong options: "higher pain")

– – Active space – –

Failed 135° plate (reverse obliquity nonunion) – would have prevented	**cephalomedullary nail** ✗ Wrong options: longer plate · autograft · strut allograft · derotation screw
Tension band principles	**eccentric loading + implant on tensile side + opposite cortex resists compression → subtrochanteric femur** ✗ Wrong options: comminuted tibia / humerus / olecranon distal to coronoid · valgus impacted proximal humerus
Anterior distal femoral cortex perforation	nail with **greater radius of curvature (straighter)** than femoral bow

Nailing technique

Position	Advantages
Lateral	**Easier reduction of distal to flexed proximal fragment**, easier entry (piriformis), vastus retraction
Supine	**Protects injured spine**, **access in polytrauma**, **easier fluoroscopy**, **easier to judge rotation**, shorter set-up

Lateral disadvantages	harder imaging, rotation hard to judge, impractical in polytrauma · supine: harder start point
Reduction	**reduce before + during nail passage** (nail can't reduce) · percutaneous aids / **open clamp** · evaluate proximal fragment **before guidewire**
Entry	piriformis nail may avoid malreduction from the **valgus proximal bend** of trochanteric nails



Clamp-assisted reduction

Atypical femoral fractures (bisphosphonate-associated)

Cause	**prolonged bisphosphonate (> 5 yr)** · low-energy fall · prodromal pain ✗ Wrong options: acute trauma · metastasis · fibrous cortical defect · osteomyelitis
Symptomatic lateral cortical thickening / "dreaded black line"	**IM nail** ✗ Wrong options: observe · physio · MRI · THA
Bilateral changes, one side painful	**nail the symptomatic side only** · **image the contralateral femur** in all ✗ Wrong options: both · bone scan · oncology

Progression of contralateral AFF
continued post-op bisphosphonate · longer use (> 5 yr), **Asian race**, **varus proximal femur → stop bisphosphonates**
✗ Wrong options: 2 yr · Caucasian · valgus · female sex

Fixation of AFF	Risk
IM nail	**↑ iatrogenic fracture** (brittle, thick cortices; canal–nail mismatch) · **↑ nonunion → revision**
Plate	**↑ hardware failure** (varus, intramembranous healing inhibited) · no ↑ iatrogenic fracture

Main complication
delayed healing + nonunion · **anatomic reduction = best predictor of union**
✗ Wrong options: LLD · metastasis · femoral head AVN · occult neck fracture

Teriparatide
anabolic; **↑ serum calcium via ↑ bone resorption / turnover** (intermittent) · for non-union despite vit D, calcium, stopping bisphosphonate
✗ Wrong options: vitamin D (gut uptake) · ATP analogues (non-nitrogen bisphosphonates) · mevalonate pathway (nitrogen bisphosphonates) · RANKL antibody (denosumab)

6 Complications

Nonunion, malunion & infection

Complication	Fact
Varus / procurvatum (flexion) malreduction	**Most frequent intra-op complication** of antegrade nailing — abductors + iliopsoas vs adductors; portal made with distal femur adducted
Nonunion	Lowest with **reamed statically locked IMN** · treat with plating (corrects varus)
AFF	Iatrogenic fracture, nonunion, hardware failure

Most common deformity after nailing — muscles
hip abductors + iliopsoas
✗ Wrong options: internal rotators · quadriceps · hamstrings

Atrophic nonunion + varus collapse after CMN
remove hardware, correct alignment, plate (or nail) + autograft
✗ Wrong options: dynamisation (fails with deformity) · Taylor frame (poorly tolerated) · unreamed exchange nail (same path) · plate around nail

Reamed exchange nailing
choice for femoral nonunion **without deformity**

Draining sinus, healed fracture, heterotopic bone (3 yr)
I&D + hardware removal + HO excision + culture-directed antibiotics
✗ Wrong options: disarticulation · suppression alone · radiation / NSAIDs

– – Active space – –



Varus malreduction

X vs Y

	Typical subtrochanteric	Atypical (bisphosphonate)
Patient	Young, high energy	Elderly, low energy, prodromal pain
Pattern	Comminuted, spiral	**Transverse, medial spike, no comminution**
Cortex	Normal	**Lateral thickening / beaking**
Nail risk	Varus malreduction	**Iatrogenic fracture, nonunion**

How it's asked

"Proximal fragment deformity?"	**→ flexion, abduction, external rotation**
"Muscles displacing proximal fragment?"	**→ gluteus medius + iliopsoas**
"Muscles causing common post-nail deformity?"	**→ hip abductors + iliopsoas**
"Reduction force on proximal fragment?"	**→ adduction + extension**
"Intra-op manoeuvre (proximal, metastatic hx)?"	**→ extension + internal rotation**
"Distal fragment manipulation after traction?"	**→ flexion + external rotation**
"Supine advantages EXCEPT?"	**→ easy reduction of distal to flexed proximal fragment**
"Best treatment?"	**→ cephalomedullary nail**
"Young, comminuted – maximise union?"	**→ reamed statically locked IMN**
"CMN vs SHS EXCEPT?"	**→ higher pain**
"Failed 135° plate – prevention?"	**→ cephalomedullary nail**
"Tension band construct suitable?"	**→ subtrochanteric femur**
"Anterior cortex perforation cause?"	**→ nail with greater radius of curvature**
"Painful lateral cortical thickening?"	**→ IM nail**
"Prodromal pain then curb-step fracture?"	**→ bisphosphonate**
"AFF nailing vs conventional – true?"	**→ ↑ iatrogenic fracture with IMN**

"One painful side, bilateral changes?"	→ **IM nail symptomatic side**
"Contralateral progression?"	→ **post-op bisphosphonate use**
"Primary AFF complication?"	→ **delayed healing + nonunion**
"Teriparatide mechanism?"	→ **↑ serum calcium through ↑ bone resorption**
"Atrophic nonunion with varus collapse?"	→ **remove nail, correct, plate + autograft**
"Draining sinus 3 yr after fixation?"	→ **I&D, remove hardware, excise HO, culture antibiotics**

– – Active space – –

TALAR NECK FRACTURES

– Trauma · 60 / 67 –

1 Basics

Epidemiology

Frequency — **commonest talus fracture (~50%)** · talus fractures 0.1–2.5% of all fractures (TO) · **ipsilateral foot injuries up to 89%** (CT)

Mechanism — high energy · **forced dorsiflexion + axial load**

Anatomy

Talus — **60–70% cartilage**, **no muscle / tendon attachments** · inferior → posterior facet of calcaneus · head → navicular + sustentaculum tali · posterior process: medial + lateral tubercles, **FHL groove**

Neck vs body — defined by the **lateral process** (articulates with **posterior facet of calcaneus**) · fracture **anterior → neck**, **posterior → body**

Aviator's fracture (TO) — **forced dorsiflexion** → talar neck impacts anterior tibia · body fracture = **axial load** (TO)

Blood supply – artery of the tarsal canal & deltoid branch

Source	Branch	Supplies
Posterior tibial	**Artery of tarsal canal – dominant**	**Most of body** (lateral ⅔ of body, OB) · forms anastomotic sling under neck with tarsal sinus artery
Posterior tibial	**Deltoid branch** (deep deltoid)	**Medial body · may be the only supply left in displaced fractures**
Anterior tibial (dorsalis pedis)	–	**Head + neck**
Perforating peroneal	**Artery of tarsal sinus**	Head + neck, lateral body

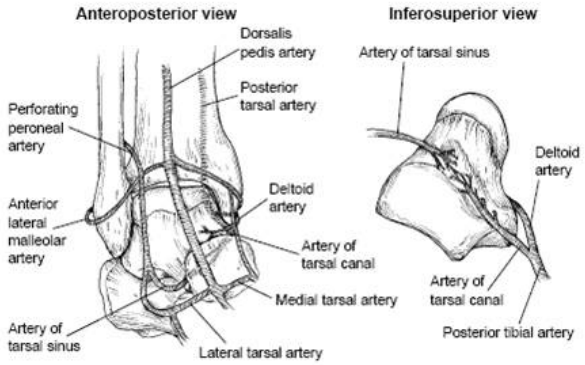
Deltoid artery (TO) — **most important**, supplies **~70% of talar body** · displaced neck / body fractures disrupt it · **do not remove small medial neck fragments attached to deltoid** (TO)

Extraosseous sources (TO) — **posterior tibial, anterior tibial, peroneal** · **artery of tarsal canal = mainly intraosseous** → "NOT extraosseous" (TO)

Retrograde supply — so displaced neck fractures → high AVN

Hawkins III – preserve — **deltoid ligament / deltoid branch of artery of tarsal canal**
✗ Wrong options: medial malleolus (**osteotomise** it) · dorsalis pedis · tarsal sinus · perforating peroneal · ATFL

Vessel whose injury ↑ osteonecrosis — **artery of the tarsal canal**
✗ Wrong options: dorsalis pedis · perforating peroneal · lateral tarsal · tarsal sinus



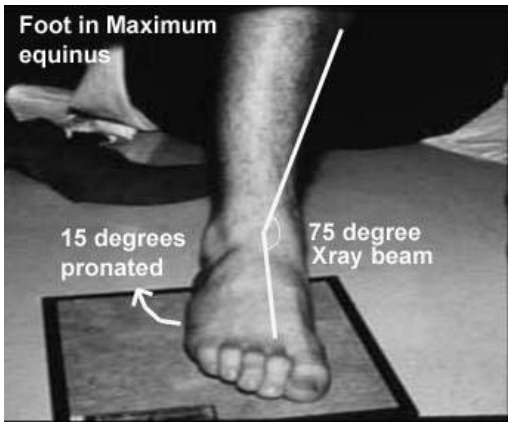
Blood supply of the talus

4 Diagnosis & Classification

Key radiology findings

Study	Fact
AP, lateral foot	Radiograph sensitivity for talar injury only **74%** (TO)
Canale view	**Best view for talar neck – maximum equinus, 15° pronated, beam 75° cephalad**
CT	**Displacement, comminution, congruity**; **gold standard (TO)**; confirms Hawkins I has no step-off
Mortise X-ray 6–8 wk	**Hawkins sign**

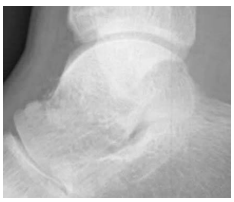
CT (TO) for **all suspected talar fractures** · detects **medial comminution** (present in nearly all neck fractures → varus malunion if compressed) · subtalar / tibiotalar involvement (TO)



Canale view

– – Active space – –

Hawkins classification & neck vs body distinction

Hawkins	Dislocation	AVN – Orthobullets	Image
I	Nondisplaced	0–13%	
II	Subtalar	20–50%	
III	Subtalar + tibiotalar	20–100%	
IV (Canale)	Subtalar + tibiotalar + talonavicular	70–100%	

AVN rates – other sources **TO: I 0–50%, II 20–50%, III 50–80%, IV 90–100%** · TO question: **Hawkins III → 70%** · OB (Hawkins): I 10%, II > 40%, III > 90%, IV 100% · II 15–40% · III 69–100% – read the options

Hawkins II with 20–50% risk pick the image with **subtalar dislocation only** · posterior process fractures → little AVN

5 Management

Talar neck fracture

- All → **emergent closed reduction in ER**
- Hawkins I (CT: no step-off) → **SLC 8–12 wk, NWB first 6 wk**
- Displaced (II–IV), extruded → **ORIF (may delay for soft tissues)**
  - soft tissue / patient unfit → **ex-fix until safe**
  - medial comminution → **medial bridge plate + positional screws**
- Open / extruded talus → **I&D + reimplant + ORIF ± ex-fix**

Operative criteria

Situation	Treatment
Hawkins I	Cast **8–12 wk** (NWB 6 wk) · TO: type I nonoperative · nondisplaced neck **still usually fixed** (TO) – cast for non-ambulatory / unfit
Hawkins II–IV	**ORIF** – even if closed reduction looks good (✗ Wrong options: definitive closed · pins · primary subtalar / TTC fusion)
Interposed fragment	**Open reduction before screws** (✗ Wrong options: percutaneous screws)
Talar extrusion	**Reimplant + ORIF**

Treatment – ORIF, fixation construct & timing

Approaches **two incisions** to see medial + lateral neck (comminution **dorsal + medial**)

Approach	Interval / key point
Anteromedial	**Tibialis anterior ↔ tibialis posterior** · preserve **deep deltoid** · **medial malleolar osteotomy** if needed
Anterolateral	Tibia–fibula proximally, **line of 4th ray** · elevate **EDB**, clear subtalar debris

Dual approach (TO) **standard of care**; avoids rotational errors · anteromedial: **tip of MM → 1st ray**, protect **saphenous vein** · anterolateral: **tip of fibula → 4th MT base**, split **EDB** (TO)

Medial malleolar osteotomy "if needed" for neck exposure (Orthobullets) · **only for body fractures, never for neck** (TO) – read the options · **short oblique**; **crack the posterior cortex** rather than cutting through (TO)

Avoid dissection superior + inferior to the neck (remaining blood supply)

Strongest construct (TO) **posterior-to-anterior parallel dual screws**
✗ Wrong options: anterior-to-posterior · crossed

Medial comminution (Hawkins II) **ORIF with medial bridge plate + lateral screw in non-lagging mode** · mini-fragment plates buttress against varus
✗ Wrong options: lag screws (shorten medial column → **varus**) · TTC fusion · closed + pins

Fixation by comminution (TO) non-comminuted: **2 compression screws** ⊥ **fracture** · comminuted (medial): **lateral screw compressive, medial screw positional** ± small medial buttress plate ± bone graft (TO)

Lateral screw with medial comminution **non-lagging** (Orthobullets) · **compressive** (TO) – read the options; both agree: **no compression across the medial comminution**

Timing **delayed fixation does NOT ↑ AVN** · **acute fixation → ↑ infection, skin necrosis, wound dehiscence** (77% vs 2–10%, TO) · open fractures → emergent debridement
✗ Wrong options: acute → ↓ AVN · ↓ subtalar OA · ↓ varus · ↓ dehiscence

Post-op **NWB 10–12 wk**

– – Active space – –



Talar neck ORIF

Extruded talus – reimplantation

Extruded body + neck fracture	**I&D + reduce native talar body + ORIF neck** ✗ Wrong options: cast · Steinmann pins + ex-fix definitive · allograft · remove + spacer
Extruded talus with debris	**bacitracin solution, irrigate, debride, reimplant**, ex-fix and / or ORIF ✗ Wrong options: talectomy + ex-fix · tibiocalcaneal fusion · tibiotalar fusion · Syme
Reimplantation – true	**safe and can succeed** · **low infection** with I&D + antibiotics · **AVN ~90%, most without collapse** · ~63% need no secondary procedure ✗ Wrong options: acute triple arthrodesis · BKA

6 Complications

Complication	Rate / fact
Subtalar arthritis	**Most common complication – ~50%** (50–100%, OB; 54% PTA)
Tibiotalar arthritis	**33%**
Osteonecrosis	**31% overall** · **2nd commonest**
Varus malunion	**25–30%** (Orthobullets) · **one-third** (TO)
Nonunion	10–12%
Symptomatic implants	~18%
Wound dehiscence	~3%

Order (OB)	**subtalar arthritis > tibiotalar arthritis > osteonecrosis > varus malunion**
Infection (TO)	up to **25% in open** fractures (TO)
Arthritis rates	subtalar **~50%**, tibiotalar **33%** (Orthobullets) · subtalar **up to 81%**, tibiotalar **~18%** (TO) – read the options
Commonest complication	**subtalar arthritis** (Orthobullets) · **varus malunion** (TO), from medial comminution + overcompression → osteotomy, subtalar fusion if advanced – read the options

Hawkins sign & osteonecrosis prognosis

Hawkins sign	**subchondral lucency of talar dome**, **mortise / AP at 6–8 wk → resorption = intact vascularity → AVN unlikely** → observe, progress weight bearing ✗ Wrong options: cement · fusion · arthroscopy
Hawkins sign positive	**low risk of osteonecrosis** ✗ Wrong options: sclerosis (bad) · diffuse osteopenia · medial malleolus fracture
Absent sign	doesn't prove AVN · up to **44% revascularise without collapse**
Not related to Hawkins sign	fixation construct; **reduction quality** affects AVN
Absent Hawkins sign (TO)	no subchondral lucency **after 12 wk → MRI** (TO)

Osteonecrosis risk factors (displacement, open fracture; not surgical timing)

Greatest impact	**extent of initial fracture displacement** ✗ Wrong options: fixation within 6–8 h · nicotine · PA screws · reduction quality
↑ AVN	displacement, **comminution**, **subtalar dislocation**, **extension into body**
Not risk factors	**delay > 24 h**, time to reduction > 12 h, surgery duration, polytrauma, staged ex-fix then ORIF
Open fracture	**independent AVN risk factor** (Orthobullets; OB – extruded talus) · **no significant difference** open vs closed (OB, Hawkins II) – read the options
Dual-incision approach	**associated with AVN** (Orthobullets; OB – vs lateral only) · **not linked** (OB, Hawkins II) – read the options

Post-traumatic arthritis – most common complication

Most likely long-term complication after anatomic ORIF	**tibiotalar and / or subtalar arthritis** ✗ Wrong options: osteonecrosis · nonunion · infection · forefoot supination
Hawkins II	**subtalar arthritis** ✗ Wrong options: AVN · tibiotalar · talonavicular (not involved)
Hawkins III – second commonest	**osteonecrosis** (subtalar arthritis first)

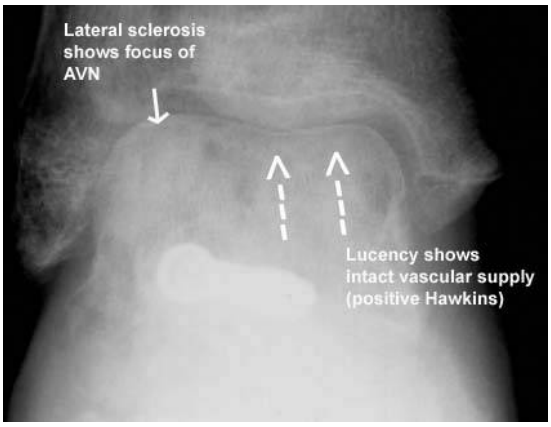
Varus malunion – effects & corrective osteotomy

Cause	**medial comminution** / non-anatomic reduction · Hawkins II treated closed
Effects	**↓ subtalar eversion** (TO + OB) · locked midfoot + hindfoot · **supination deformity** · **walks on lateral border** · later subtalar arthrosis ✗ Wrong options: planovalgus (pronation)
Treatment	**talar neck medial opening wedge osteotomy** (tricortical ICBG) ✗ Wrong options: TAA · lateral calcaneal closing wedge · calcaneal neck opening wedge · triple
Dorsal impingement + ↓ ankle dorsiflexion	**isolated dorsal malunion** ✗ Wrong options: varus · valgus · plantar
Hawkins sign 6 wk = poor prognosis (TO)	false; **varus malunion ↓ eversion** = true

– – Active space – –

Nonunion & salvage (revision ORIF, TTC fusion)

Nonunion, viable body, no arthritis	**revision ORIF + bone graft** (preserves motion) · vascularised graft if AVN ✗ Wrong options: talectomy · ankle / TTC / triple fusion
AVN + tibiotalar + subtalar arthritis	**tibiotalocalcaneal fusion** (hindfoot nail) ✗ Wrong options: TAA (poor talar stock) · isolated tibiotalar fusion · supramalleolar osteotomy · arthroscopic debridement
Isolated tibiotalar arthritis	tibiotalar fusion or TAA (needs talar bone stock)



Hawkins sign

X vs Y

	Hawkins sign present	Absent / sclerosis
Meaning	**Vascularity intact** (resorption)	Possible AVN
Prognosis	**Low AVN risk**	Risk – but 44% revascularise
Action	**Observe**, progress WB	Monitor

	Varus malunion	Dorsal malunion
Effect	↓ **subtalar eversion**, lateral border WB	**Dorsal impingement**, ↓ dorsiflexion
Fix	**Medial opening wedge osteotomy**	–

How it's asked

"NOT extraosseous supply (TO)?"	→ **artery of tarsal canal**
"Structure to keep intact in ORIF?"	→ **deltoid ligament**
"Two-incision Hawkins III – preserve?"	→ **deltoid branch of artery of tarsal canal**
"Vessel injury ↑ osteonecrosis?"	→ **artery of tarsal canal**
"Subchondral lucency at 8 wk – next?"	→ **observe; vascularity intact**
"Good prognostic radiographic sign?"	→ **talar dome subchondral lucency**
"Hawkins sign positive?"	→ **low osteonecrosis risk**
"Bony resorption at 8 wk means?"	→ **talar perfusion**

"Hawkins III AVN rate (TO)?"	→ **70%**
"Greatest impact on AVN?"	→ **initial displacement**
"Delayed fixation — true?"	→ **doesn't increase AVN**
"AVN after open extruded neck fracture — risk factor?"	→ **open fracture**
"Hawkins II AVN — correlated?"	→ **degree of initial displacement**
"Second commonest complication, Hawkins III?"	→ **osteonecrosis**
"Most common complication?"	→ **post-traumatic (subtalar) arthritis**
"Varus malunion causes (TO)?"	→ **↓ eversion**
"Varus malalignment ↓ which motion?"	→ **subtalar eversion**
"Supination deformity after percutaneous screws?"	→ **varus malunion**
"Reconstruct varus malunion?"	→ **talar neck medial opening wedge osteotomy**
"Dorsal impingement, ↓ dorsiflexion?"	→ **isolated dorsal malunion**
"Strongest construct (TO)?"	→ **posterior-to-anterior parallel screws**
"Medial comminution — fixation?"	→ **medial bridge plate + lateral non-lag screw**
"Hawkins II reduced closed — next?"	→ **ORIF**
"Acute definitive fixation — true?"	→ **higher infection**
"Extruded talar body?"	→ **reduce native body + ORIF**
"Reimplantation — true?"	→ **safe and can succeed**
"Talar neck nonunion, viable body?"	→ **revision ORIF + bone graft**
"AVN + ankle + subtalar arthritis?"	→ **TTC fusion**
"Neck vs body landmark?"	→ **lateral process (articulates with posterior facet)**
"Medial comminution in talar neck — medial screw mode?"	→ **positional (non-compressive)** (TO)
"Medial malleolar osteotomy is used for?"	→ **talar body fractures only, never neck** (TO)

TALUS FRACTURE (OTHER THAN NECK)

1 Basics

Epidemiology

Incidence **< 1% of all fractures** · talus = **2nd commonest tarsal fracture** (after calcaneus)

Site	Share of talus fractures
Neck	Commonest (~50%)
Body	**13–23%**
Lateral process	**10.4%**
Head	**Least common**

Mechanism body: high energy, hindfoot supinated or pronated · lateral process: see below

Anatomy

Talus **5 articular surfaces**, **~70% cartilage**, **no muscle / tendon attachments** → direct extraosseous blood supply (artery of tarsal canal main; deltoid artery to body)

Lateral process articulates with **posterior facet of calcaneus** + **lateral malleolus** (lateral margin of talofibular joint) · "lateral facet of calcaneus" (OB) – not medial facet

Posterior process **medial + lateral tubercles** separated by **FHL groove** · **os trigonum** in up to **50%** – don't call it a fracture

3 Presentation

Lateral process anterolateral ankle pain after snowboarding, **misdiagnosed as ankle sprain**

Posterolateral tubercle pain with **FHL flexion / extension**

4 Diagnosis & Classification

Key radiology findings

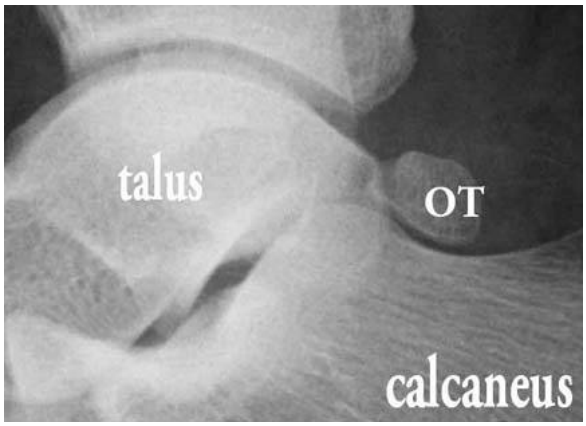
Study	Fact
AP + lateral	**Lateral process seen on AP**; X-rays may be **falsely negative**
Canale view	Talar neck (max equinus, 15° pronation, 75° cephalad)
CT	**Best for posterior, lateral, posteromedial process**; displacement, comminution, congruity; when suspicion high + X-rays negative
MRI	Confirms diagnosis when X-rays negative

Snowboarder, negative X-rays, can't bear weight – next

MRI (OB) · **CT** when suspicion high (Orthobullets) – read the options
✗ Wrong options: cast · bone scan · injection · repeat X-rays



Lateral process fracture on AP



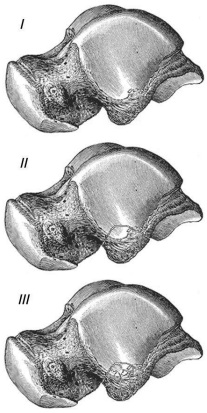
Os trigonum – not a fracture

– – Active space – –

Anatomic classification

Fracture	Type / description
Lateral process 1	**Extra-articular**
Lateral process 2	**Into subtalar + talofibular joint**
Lateral process 3	**Comminuted**
Posteromedial tubercle	Avulsion of **posterior talotibial / posterior deltoid** ligament · **CeDell fracture** (TO)
Posterolateral tubercle	Avulsion of **posterior talofibular ligament** · **Shepherd fracture** (TO)
Head	Least common
Body	Posterior to lateral process

Aviator's astragalus (TO) **talar neck** fracture (RAF pilots)



Lateral process classification

Lateral process fracture (snowboarder's fracture)

Snowboarder's ankle (TO) **lateral process of talus fracture** · **15× likelier** in snowboarders; **⅓ of their ankle fractures**
✗ Wrong options: ATFL avulsion · PTFL avulsion · medial process + deltoid

Mechanism **dorsiflexion + axial load + eversion + external rotation** (Orthobullets, TO) · **inversion + ER** (OB) – read the options

Subtalar arthritis **45%** regardless of treatment = **most common complication** · prognosis good with prompt treatment

Excising a 1 cm fragment **lateral talocalcaneal ligament incompetent** (✗ Wrong options: bifurcate · peroneal retinaculum · arcuate · PTFL) · **tolerated – no ankle / subtalar instability** (✗ Wrong options: 5 mm)

5 Management

Operative criteria

Fracture	Nonop (< 2 mm)	Operative (> 2 mm)	Excise
Lateral process	**SLC 6 wk**, moulded to **support longitudinal arch**	**ORIF / K-wires** (lateral approach)	**Comminuted**
Posterior process	SLC 6 wk	ORIF posteromedial (± posterior / medial malleolar osteotomy)	**Comminuted, nonunion**
Head	SLC 6 wk	ORIF	–
Body	SLC 6 wk	**ORIF – combined medial + lateral** ± malleolar osteotomy	–

Displaced, non-comminuted, intra-articular lateral process
ORIF · **4 mm displaced → ORIF, not cast**
✗ Wrong options: excision · brace + PT · hard-soled shoe · cast (that is < 2 mm)

Comminuted intra-articular, failed nonop
fragment excision

Lateral + posterior process (TO)
large fragments → fix · small → excise · posterior process: forced inversion in slight DF or PF (TO)

Approach	Use	Interval / risk
Lateral	Lateral process	Over **tarsal sinus**, reflect EDB distally
Posteromedial	Medial tubercle / whole posterior process displaced medially	**FDL ↔ NV bundle**
Posterolateral	Lateral tubercle	**Peroneals ↔ Achilles** · **sural nerve** · NV bundle **medial to FHL**

Talar body fracture – articular reduction

Improves long-term outcome
primary bone healing at the articular surface (anatomic reduction, **absolute stability**)
✗ Wrong options: surgery < 48 h · minimally invasive · early WB · relative stability

Open talus fracture – staged management

Diabetic (HbA1c 10.1), contaminated open neck fracture + body extrusion
urgent (< 24 h) I&D + ankle-spanning ex-fix, delayed definitive fixation
✗ Wrong options: discharge + delayed single stage · Steinmann pins definitive · acute medial plating · acute TTC nail

Timing
acute definitive fixation of open fractures → **up to 75% early complications** (dehiscence, skin necrosis, infection) vs **2–10% delayed** · definitive at **1–3 wk** when swelling settles

Talus cartilage
> 50% articular → limited vascular inflow · **quality of reduction = most important outcome predictor**

6 Complications

Complication	Fact
AVN	**Hawkins sign (lucency) = revascularisation** · **sclerosis without Hawkins sign = AVN**
Subtalar arthritis	**45%** after lateral process fracture · anatomic reduction ↓
Talonavicular arthritis	→ **TN arthrodesis**
Malunion	–
Symptomatic nonunion	Pain **up to 2 yr** · posterior process → excise

X vs Y

	Shepherd fracture	CeDell fracture
Site	**Lateral tubercle**, posterior process	**Medial tubercle**
Ligament	Posterior talofibular	Posterior talotibial / deltoid
Frequency	Commoner	Less frequent
Approach	Posterolateral	Posteromedial

	Lateral process fracture	Ankle sprain
Clue	Snowboarder, dorsiflexion + ER	Inversion
X-ray	**AP view**; may be negative	Normal
Next if negative	**MRI / CT**	–

How it's asked

"Snowboarder's ankle (TO)?"	→ **lateral process of talus fracture**
"Snowboarder, negative X-rays – next?"	→ **MRI ankle**
"Excising 1 cm fragment → incompetent?"	→ **lateral talocalcaneal ligament**
"Displaced non-comminuted lateral process – diagnosis + treatment?"	→ **lateral process fracture → ORIF**
"Displaced lateral process fracture – treatment?"	→ **operative fixation**
"Lateral process fracture – true?"	→ **subtalar arthritis commonest regardless of treatment**
"Talar body – improves outcome?"	→ **primary bone healing at articular surface**
"Open contaminated talus, diabetic?"	→ **I&D + spanning ex-fix, delayed fixation**
"Shepherd fracture (TO)?"	→ **lateral part of posterior process**

TERRIBLE TRIAD INJURY OF ELBOW

– Trauma · 62 / 67 –

1 Basics

Components of the triad

Triad (Hotchkiss 1996) **elbow dislocation (posterolateral)** + **radial head / neck fracture** + **coronoid fracture** · LCL injury almost always

NOT part of it (TO) **olecranon fracture** · **biceps tendon tear**
✗ Wrong options: LCL injury · coronoid · radial head (all part)

Mason-Johnston type IV (radial head fracture + dislocation) = terrible triad pattern (TO)

Coronoid fracture types in the triad (TO) mostly **tip**, less often **anteromedial facet** (O'Driscoll)



Terrible triad – lateral radiograph

Anatomy – elbow stabilisers

Primary (TO)	Secondary (TO)
Ulnohumeral joint (coronoid) – buttress vs varus + posterior dislocation	**Radial head** – valgus + PLRI
Anterior bundle of MCL	Flexor-pronator + extensor origins
LCL complex	Joint capsule

Radial head **secondary restraint to PLRI + valgus** (Orthobullets) · "**primary restraint to PLRI**" (OB) – read the options · neutral rotation: lateral non-articular margin between **radial styloid + Lister's tubercle**

Coronoid **anterior + varus buttress**; ~60% of anterior articulation (TO); **resists posterior subluxation beyond 30° flexion** · fragment often has **anterior capsule** attached

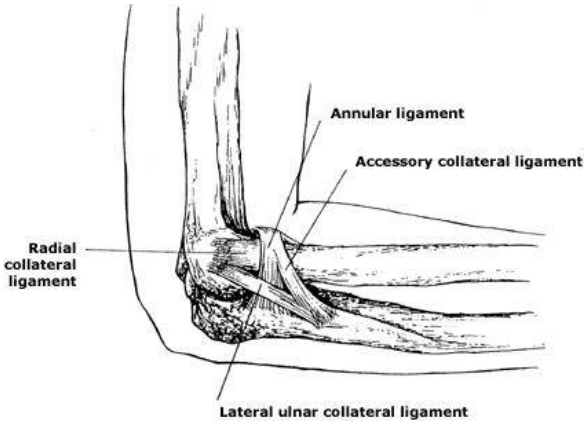
– – Active space – –

Anterior capsule insertion — **5.9 mm from tip** (TO) · inserts more distally – small tip fractures often lack capsule (OB) · coronoid tip fracture is a **shear**, not an avulsion (OB)

MCL bundle	Fact
Anterior	**Most important – valgus + PMRI restraint** · inserts **sublime tubercle (anteromedial facet of coronoid)**, **18.4 mm dorsal to tip** (✗ Wrong options: 3 mm distal to tip · radial tuberosity)
Posterior	Valgus instability in flexion
Transverse	–

LCL component (insert: supinator crest)	Fact
LUCL	**Most important; primary restraint to PLRI** · pushing up from chair
Radial collateral	Varus instability
Annular	Little stability
Accessory collateral	–

LCL injury — usually **avulsed from the lateral epicondyle (humeral origin)** ± extensor origin rupture



LCL complex

2 Mechanism

Mechanism & sequence of soft-tissue failure

Mechanism — fall on **extended arm, forearm supinated → valgus + axial + posterolateral rotatory** forces → **posterolateral dislocation** · ~60% of complex dislocations from **standing-height fall** (TO)

Sequence (Horii circle) — **LCL → anterior capsule (coronoid) → MCL** · **MCL torn last** (TO)
✗ Wrong options: MCL first

EXCEPT (TO) — "**LCL tear is uncommon due to valgus force**" – false; LCL fails first

Supination + axial + posterolateral + valgus — **coronoid + radial head fracture + dislocation**
✗ Wrong options: supracondylar · anterior olecranon fracture-dislocation (direct blow at 90°) · coronoid + olecranon + LCL (**varus**)

Pattern	Force	Fracture
PLRI (terrible triad)	**Valgus**, supination, axial	Radial head + coronoid **tip**
PMRI	**Varus**	**Anteromedial coronoid facet**, LCL; often no radial head fracture

PLRI far commoner than PMRI

3 Presentation

Symptoms — pain, **clicking + locking in extension** · varus / valgus instability · **check DRUJ (Essex-Lopresti)**

4 Diagnosis & Classification

Key radiology findings

Study	Fact
Pre- + post-reduction X-rays	**Concentric ulnohumeral + radiocapitellar joints** · **radial neck line through capitellum centre** on every view
Lateral	**Coronoid fracture**
Wrist / forearm	When indicated (Essex-Lopresti)
CT ± 3D	Coronoid detail, fracture lines

Coronoid anatomy & biomechanics (MCL insertion, Regan-Morrey, approach)

Regan-Morrey	Coronoid height	Fixation
I	**Tip** (< 10%)	**Not needed** – no stability conferred (cadaver)
II	**≤ 50%**	Usually not – fix MCL first if unstable
III	**> 50%**	**Fix**

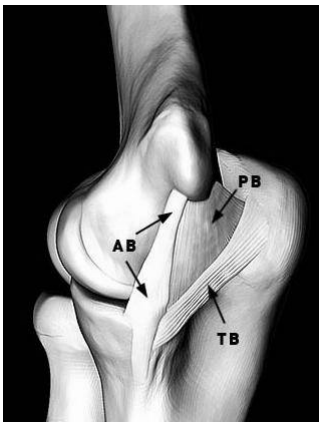
Biomechanics — **> 50% fractured → ↑ posterior axial displacement at all flexion angles incl. 30° · < 50% (I + II) → no significant change · anteromedial facet → PMRI**
✗ Wrong options: 25% → displacement

Fix the coronoid — when > 10% (OB) · > 30–50% base (OB) · **type III** (OB) · decided by **fragment size + stability after radial head** – read the options
✗ Wrong options: displacement > 5 mm · comminution · age

Repair even type I (TO) — **restores anterior bony buttress + anterior capsule insertion → prevents posterior subluxation** · OB: type I / II repair adds little
✗ Wrong options: brachialis attachment · HO prevention · LUCL isometric point

Isolated large coronoid fracture — **medial FCU-splitting approach + buttress plate** → FCU = **ulnar nerve** · triad → through lateral radial head defect

– – Active space – –



Anterior bundle of MCL

5 Management

Terrible triad (after reduction)

- Rare: concentric, small coronoid, radial head non-surgical, stable → **splint 90° 7–10 d, early ROM**
- Most → **sequential repair, deep to superficial, then retest**
  - 1 coronoid (fix / capsular suture) → **2 radial head ORIF or replace**
  - 3 LCL repair to humerus → **4 MCL repair if still unstable**
  - 5 hinged ex-fix if still unstable

Operative criteria

Structure	Fix / replace when
Radial head ORIF	**Non-comminuted, < 40% articular** · 2-part > 30% → ORIF
Radial head arthroplasty	**Comminuted > 3 fragments** (≥ 3 involving > 25%, OB) · > 40% articular · not age > 40, sagittal fracture, PIN palsy
Radial head excision	**Never** with instability / Essex-Lopresti · fragment excision only if < 25%, non-articulating, stable
Coronoid	Type III / large base (see above)
MCL	**Instability after LCL + fractures**, esp. **extension beyond 30°** · PMRI with type I / II coronoid

Standard fixation — radial head fixation / replacement, coronoid ORIF & LCL repair

Standard (comminuted head + coronoid)	**radial head arthroplasty + coronoid ORIF + LCL / LUCL repair** (McKee: 34 / 36 stable) ✗ Wrong options: closed + early ROM · radial head resection · hinged brace · static ex-fix 6 wk · MCL routinely · ulnar nerve transposition
LUCL	**repair to the humerus** (avulsed from origin) ✗ Wrong options: reconstruction acutely · repair to ulna · TEA
Radial head ORIF	1.5 / 2.0 / 2.4 mm countersunk screws · plate **posterolateral, safe zone 90–110°** (styloid → Lister's) · 2.0 plate on neck → little motion loss

Radial head implant	articulate **2 mm distal to coronoid tip** · **avoid overstuffing** · undersizing → valgus instability
Coronoid fixation	through radial head defect · sutures via 2 drill holes ("lasso"), anchors · **posterior-to-anterior lag screws** if large (anterior-to-posterior, TO) · basal → medial plate
LCL repair	anchors / transosseous at **centre of capitellar curvature**, elbow 90° · **MCL intact → forearm pronated** · **MCL injured → supinated** (avoid medial gapping)
Posterior skin incision	reaches medial + lateral, fewer cutaneous nerve injuries, cosmetic
Elbow stable after triad – single most important for rotatory stability	**LCL complex repair / reconstruction** ✗ Wrong options: radial head replacement / ORIF · capsular plication · MCL

Persistent instability after lateral repair – MCL repair

Still unstable after coronoid + radial head + LCL	**repair MCL** ✗ Wrong options: larger radial head (overstuffing) · resect head · cast 6–8 wk · reinforce LCL · Steinmann pin · ex-fix before MCL
Residual posterolateral instability after fracture fixation alone	**repair LCL complex** ✗ Wrong options: MCL first · cast / brace · ex-fix
Type I coronoid, unstable with valgus after head ORIF + LCL	**MCL repair** ✗ Wrong options: coronoid ORIF · convert to arthroplasty · splint
Posteromedial subluxation after head + LCL, tip fracture	**MCL via medial approach, protect MABCN (medial antebrachial cutaneous nerve)** ✗ Wrong options: coronoid through lateral · medial brachial cutaneous · musculocutaneous
Hinged ex-fix	**last step**, after MCL
Hinged ex-fix axis (TO)	**lateral epicondyle → medial trochlear sulcus** (centre of capitellum / trochlea) · Pugh–McKee sequence as above (TO)

Postoperative splinting position

Repaired	Splint / passive motion
LCL only (MCL intact)	**Flexion + pronation** – pronation tightens MCL sling
MCL	Supination
Both	**Flexion + neutral**

Rehab	**active ROM at 48 h** (dynamic stabilisers) · may block last 30° of extension early · prolonged immobilisation → poor results

6 Complications

Complications — stiffness

Complication	Fact
Post-traumatic stiffness	**Most common** complication · **commonest needing reoperation** · prevent with early ROM
Instability	Commoner after **type I / II coronoid** not fixed · "most commonly persistent instability" (OB) — read the options
Fixation failure	**Radial neck** ORIF (poor vascularity → AVN, nonunion)
HO	Common; doesn't always need surgery · prophylaxis with **head injury / revision**
PTOA	Chondral damage, residual instability
PIN injury	Lateral approach — stay proximal to annular ligament, pronate, release supinator posteriorly

Prognosis historically poor: **persistent instability, stiffness, arthrosis**; good with stable fixation + early motion

X vs Y

	PLRI (terrible triad)	PMRI
Force	**Valgus**, supination	**Varus**
Coronoid	**Tip**	**Anteromedial facet**
Radial head	**Fractured**	Usually intact
Key repair	**LCL**	Coronoid buttress (medial) + LCL

	LCL repair, MCL intact	MCL also injured
Forearm during LCL repair	**Pronation**	**Supination**
Splint	**Flexion + pronation**	Neutral (both repaired)

How it's asked

"Not part of terrible triad (TO)?"	**→ olecranon fracture**
"Not part of terrible triad, biceps option (TO)?"	**→ biceps tendon tear**
"Valgus + axial + posterolateral force (supinated)?"	**→ coronoid + radial head fracture + dislocation**
"Sequence of injury?"	**→ LCL → anterior capsule → MCL**
"Terrible triad EXCEPT (TO)?"	**→ LCL tear uncommon due to valgus**
"PLRI radiograph?"	**→ terrible triad (radial head + coronoid tip)**
"Coronoid > 50%, comminuted head?"	**→ arthroplasty + coronoid ORIF + LCL repair**

"Comminuted head + basal coronoid, 62-yr?"	→ head replacement + coronoid ORIF + LUCL ± MCL
"What decides ulna (coronoid) fixation?"	→ fragment size + stability after head replacement
"Restore stability in most cases?"	→ head fixation / replacement + coronoid fixation + LUCL repair
"Type II coronoid, nonreconstructable head?"	→ arthroplasty + coronoid ORIF + LCL repair
"Arthroplasty indication?"	→ > 3 fragments
"Unstable in valgus at 40° (comminuted head)?"	→ replacement + coronoid ORIF + LUCL repair
"Comminuted head in triad?"	→ radial head arthroplasty
"Where to repair LUCL?"	→ to the humerus
"Still unstable after full lateral repair?"	→ repair MCL
"Residual posterolateral instability < 35°?"	→ repair LCL complex
"Type I coronoid, valgus instability after head + LCL?"	→ MCL repair
"Unstable 0–20° after full repair?"	→ repair medial UCL
"Posteromedial subluxation — next step + nerve?"	→ MCL via medial approach, protect MABCN
"Most rotatory stability?"	→ LCL complex repair
"Late instability pushing from chair?"	→ LUCL
"Repair type I coronoid — rationale (TO)?"	→ anterior buttress + capsule insertion vs posterior subluxation
"Anterior MCL bundle inserts?"	→ anteromedial (sublime tubercle) coronoid
"Isolated coronoid — approach muscle nerve?"	→ ulnar (FCU split)
"Cadaver — > 50% coronoid?"	→ ↑ posterior axial displacement at 30°
"Coronoid needing fixation?"	→ Regan-Morrey type III
"Splint if MCL intact?"	→ flexion + pronation
"Commonest complication requiring reoperation?"	→ loss of motion
"Most common complication?"	→ post-traumatic stiffness

– – Active space – –

TIBIAL PLAFOND FRACTURES

– Trauma · 63 / 67 –

1 Basics

Epidemiology

Incidence	**5–10% of tibia fractures** (Orthobullets) · 5–7% (TO) · < 10% of lower-limb injuries · age 35–45, M > F
Associated	**fibula fracture 75% (commonest association, TO)** · ipsilateral limb injury 30% · **open 20%** · bilateral 5–10% · **syndesmotic injury ~3.4%**

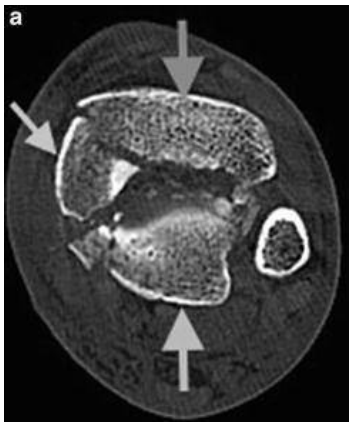
Fracture anatomy & patterns (Chaput fragment)

Mechanism	**high-energy axial load** (talus driven into plafond → **articular impaction + metaphyseal comminution**) · fall from height, MVC · low-energy rotation (skiing) less common
Posterior pilon	**rotation + axial impaction** (plantarflexed foot) → large posterior malleolus fragment, posteromedial comminution, **double-density sign** → ORIF
Fibula intact (TO)	**medial (varus) impaction pattern** more likely (TO)
Foot position (TO)	**forced plantarflexion → posterior plafond** · **forced dorsiflexion → anterior plafond** injury (TO)
Associated (TO)	talar body / neck, tibial shaft, plateau, calcaneus fractures · rule out spine, head, abdominal injury · open injuries → **Ganga score** for salvage decision (TO)
Pilon deforming forces (TO)	talus drives plafond into comminution · distal fragment **ER**, proximal fragment IR · fibula carries **20–25%** of load (TO)

Fragment (ligament attached)	Position
Medial malleolar	**Deltoid**
Chaput	**Anterolateral** – **AITFL**
Volkmann	**Posterolateral** – **PITFL**
Wagstaffe	**Fibular** (anterior tubercle) – other end of AITFL

3 fragments typical (intact ligaments) · **4 classic** on X-ray: medial, anterior (Chaput), lateral (Wagstaffe), posterior (Volkmann)

Syndesmosis	AITFL (Chaput → Wagstaffe), **PITFL – strongest** (Volkmann → lateral malleolus), interosseous membrane + ligament, inferior transverse ligament



Pilon fracture fragments

3 Presentation

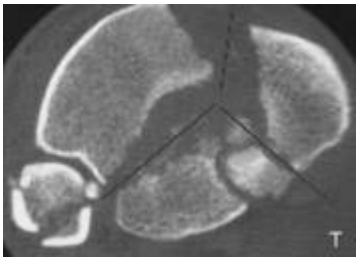
Exam swelling, **fracture blisters**, open wounds, chronic skin / vascular change · **DP + PT pulses** (ABI / CTA if needed) · nerves · **compartment syndrome**

4 Diagnosis & Classification

Key radiology findings

Study	Fact
X-ray	AP, lateral, mortise + **full-length tibia / fibula + foot** · lumbar films if indicated
CT	**After spanning ex-fix** (ligamentotaxis → better planning) (✗ Wrong options: before frame) · fine cuts ± 3D · **"Mercedes-Benz" sign** on axials
CT soft-tissue windows	**Tibialis posterior tendon entrapment** (posteromedial fragment)
MRI	Not used — poor bony detail

CT impact (TO) **changes the surgical plan in > 60%** · obtain only after spanning + length restored ("**span, scan, plan**") (TO)

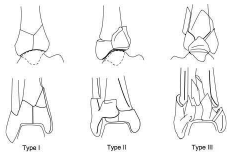


Mercedes-Benz sign on axial CT

– – Active space – –

Classifications

AO/OTA 43	Pattern	Subtypes (TO)
A	**Extra-articular**	1 simple · 2 wedge · 3 multifragmentary
B	**Partial articular**	1 split · 2 split-depression · 3 depression
C	**Complete articular** (articular dissociation from shaft)	1 simple · 2 simple articular + multifragmentary metaphysis · 3 multifragmentary

Rüedi-Allgöwer	Pattern	Image
I	**Nondisplaced** intra-articular	
II	**Displaced, no comminution** (incongruous joint)	–
III	**Displaced + comminuted / impacted**	

Leonetti + Tigani (TO) **only CT-based classification** · I non-displaced (Ia) / extra-articular (Ib) · **II** 2-part (S sagittal, F frontal) · **III** 3-part (S / F) · **IV** 4-part / comminuted ("die-punch") · AO + Rüedi-Allgöwer are radiograph-based, poor for planning / prognosis

5 Management

Pilon fracture

- Non-displaced / unfit / high skin risk → **long leg cast 6 wk**
- Length-unstable, high energy → splint → spanning ex-fix (± fibula) → CT → **delayed ORIF (10–14 d)**
  - open → **I&D + ex-fix**
- Simple articular extension → **percutaneous screws + IM nail**
- Non-reconstructable / elderly / labourer → **primary arthrodesis (rare)**

Staged management – spanning ex-fix then delayed ORIF

Sequence **splint → spanning ex-fix → CT → delayed ORIF**
✗ Wrong options: CT before ex-fix · acute ORIF · cast

Comminuted closed pilon – next **spanning external fixation**
✗ Wrong options: ORIF · percutaneous · cast · arthrodesis · CT / MRI first

Open pilon (medial wound) **I&D + external fixation** · open + tibial defect → **ex-fix of tibia** only; graft at staged ORIF
✗ Wrong options: immediate ORIF · I&D + splint · amputation

Decreases wound complications	**immediate spanning ex-fix ± immediate fibula fixation, delayed tibial reconstruction** (Sirkin) ✗ Wrong options: immediate tibia · splint + ORIF at 1 wk · Ilizarov + limited fixation
Obese smoker, swollen	**closed reduction + spanning ex-fix** ✗ Wrong options: splint → cast · ORIF · Blair fusion
3 wk, swelling settled	**ORIF tibia + fibula** ✗ Wrong options: fibula only · walking cast · dynamise · fusion in young
Two-stage ORIF vs LIFEF	**two-stage ORIF preferred** (LIFEF: ↑ superficial infection, healing problems) · acute ORIF infection up to **55%** · staged ~**3%** · definitive ex-fix: pin infection, malalignment, nonunion
Timing	**skin wrinkles, blisters epithelialised, ecchymosis resolving (10–14 d)** · ex-fix until then
Ex-fix	**2 tibial half-pins + 1 transcalcaneal pin** ('delta' / 'A' frame) · **1st MT pin prevents equinus** · trans-navicular if calcaneal fracture · pins out of zone of injury / incisions
Options (TO)	**single-stage ORIF rare** (minimal swelling only) · **staged = most common** (stage 1 fibula fixation for length / ligamentotaxis + spanning ex-fix → stage 2 plafond) · **ex-fix + limited internal fixation** for poor soft tissue / open · **primary arthrodesis for non-reconstructable C3** · **slab** acceptable if fibula intact + alignment acceptable (TO)
Post-op (TO)	early ROM · **WB after 8–12 wk** (TO)

Operative criteria

Option	Indication
Cast	Stable, no articular displacement · critically ill / non-ambulatory · diabetes, PVD, neuropathy
Spanning ex-fix	**Most length-unstable** fractures (temporary)
ORIF	**Definitive for most**; outcome = **articular reduction**
Ex-fix / circular frame alone	Bone / soft tissue precludes ORIF · high union, pin-tract infection
IM nail + percutaneous screws	**Simple intra-articular extension** · ↑ **valgus + recurvatum** vs plate
Primary arthrodesis	Non-reconstructable, select elderly, labourers

Simple articular extension, swollen	**percutaneous articular fixation + IM nail** ✗ Wrong options: cast · immediate ORIF · uniplanar ex-fix

Surgical technique & approaches

Rüedi-Allgöwer principles (TO)	**1 restore fibular length · 2 anatomic articular reduction · 3 cancellous autograft for defect · 4 medial column buttress** (now often anterolateral locking plate)
Order after fibula plated	**anatomic reduction + stabilisation of articular surface first** → then metaphysis (length, rotation, angulation — not anatomic) → plate to shaft · simple fibula: plate first · **comminuted fibula: after tibia** ✗ Wrong options: plate first
Alignment goals	**< 5–10° varus / valgus**, **< 5–10° pro- / recurvatum** · skin bridge **> 7 cm**

– – Active space – –

Skin bridge — **> 7 cm** (Orthobullets) · **minimum 5–7 cm** (TO) — read the options · **curvilinear incisions** across ankle, long incisions with thick flaps (TO)

Fragment-specific fixation (TO) — identify anterior, medial, posterior, Chaput fragments · **reduce posterior malleolus early → converts C into B pattern** · **low-profile buttress plates (1/3 tubular / recon)** for medial + anterior shear · **avoid bulky anatomical medial plates** · **elevate impacted fragments before compressing** · graft metaphyseal voids · check syndesmosis intra-op (TO)

Approach	Plane / fact
Anterolateral	**Deep ↔ superficial peroneal nerves** · in line with **4th ray** · valgus-impacted / intact fibula · **whole articular surface visible** · extensors retracted **medially** · **SPN → dorsal foot numbness (commonest nerve problem)** · DPN at risk deep
Anteromedial	**Deep peroneal ↔ tibial nerve** · **extensile access to central impaction**
Direct anterior	**No true internervous plane**; not extensile
Posterolateral	Fibula (separate incision for skin bridge) · **bone graft for distal tibial nonunion**

Fibular fixation — not essential · **↑ overall complications + hardware removal**; **no difference in reduction, alignment, satisfaction, arthrosis** (OB) · **TO: alignment DOES differ** (EXCEPT: "no difference in alignment" is false) — read the options

Post-op — ankle ROM at **2 wk** · NWB **6–12 wk**

Staging (TO) — stage 1 **delta-frame spanning ex-fix** (length + alignment, avoid over-reduction) → stage 2 **CT by column** → stage 3 definitive once **wrinkle sign** returns · **fibular fixation helps restore tibial length** (TO)

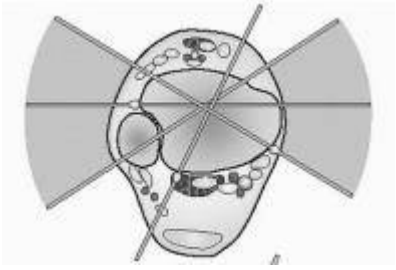
Approach (TO)	Fragment
Anterolateral	**Chaput** / anterolateral plafond
Medial	Medial malleolar fragment
Posterolateral	**Volkmann** fragment

Fixation (TO) — buttress plating · **raft screws** under articular surface · graft central defect

Fine-wire fixation vs ORIF

43C open, FWF vs ORIF — **FWF → higher complication rate** · same function, malunion, nonunion, PTOA · meta-analysis: frame vs ORIF — **no difference** in infection / complications (Orthobullets) — read the options

Circular frame — **distraction = reduction** · 2 AP half-pins proximally · **wires ≥ 1–2 cm from joint** (septic arthritis) · **60° safe arc** · HA-coated pins ↓ loosening



Circular frame wire safe zones

Primary arthrodesis

Position	**neutral DF, 5–10° ER, 5° hindfoot valgus, 5 mm posterior talar translation** · anterior plate · cast 8 wk, FWB 12 wk, CT at 3 months
TTC nail	sacrifices subtalar motion, ↑ transverse tarsal arthritis, immediate WB

6 Complications

Complication	Rate / fact
Wound slough / dehiscence	**9–30%** · wait 1–2 wk · free flap
Infection	**5–15%** · swelling at surgery, severity · **intra-wound vancomycin powder ↓ gram +ve deep SSI**
Malunion	6–14%
Nonunion	**5%**, **metaphyseal junction** · comminution, open, bone loss, **smoking, NSAIDs** → rule out infection, graft + rigid fixation
PTOA	**1–2 yr** · non-anatomic reduction · TAA / fusion
Stiffness	**Up to 33% at 3 yr** · severity, obesity, ASA ≥ 3
Tibialis posterior entrapment	Posteromedial fragment, **↓ inversion**, gap · **CT soft-tissue windows**

Malunion (TO)	**medial collapse** if impaction not addressed · **missed syndesmotic injury** (TO)

Complications — nonunion grafting, tendon entrapment, equinus

Distal tibial nonunion graft — approach	**posterolateral** (best soft-tissue cover) ✗ Wrong options: anterior · anterolateral · anteromedial · lateral
Inversion weakness + posteromedial gap	**tibialis posterior entrapment** ✗ Wrong options: deltoid · FHL (more posterior) · peroneus brevis · tibial nerve (plantar numbness)
Tendoachilles lengthening indicated	**DF −10° with knee flexed AND extended** (Silfverskiöld) ✗ Wrong options: only with knee extended (gastrocnemius) · subtalar 5° (arthritis)

Post-traumatic arthritis — chondrocyte death & IL-6

Initial chondrocyte death	**necrosis**, **superficial zone**, **at fracture margins** · **apoptosis = delayed**, also at margins ✗ Wrong options: deep zone · remote
Synovial fluid	**↑ IL-6** (+ IL-8, MMP-1, -2, -3, -9, -10) · not linked to NWB duration or compliance ✗ Wrong options: TGF-β · RANKL · IL-2 · cAMP

Outcomes & predictors (education level, prolonged recovery, syndesmotic injury)

Poor outcome / no return to work	**lower education**, comorbidities (≥ 2), **male**, **work-related**, low income, married, definitive ex-fix ✗ Wrong options: joint line · displacement · time to ORIF · open · varus collapse · marijuana · two plates · temporary ex-fix

– – Active space – –

Recovery	**improves for > 2 yr** · type C → SF-36 lower than pelvic fracture, AIDS, CAD
Syndesmotic injury	**~3.4%** · **↑ amputation**, deep infection, reoperation · more segmental fibula fractures · no ↑ nonunion

Return to driving – braking time

Brake travel time	**↑ until 6 wk after weight bearing begins** (long-bone + articular, right leg) ✗ Wrong options: 8 wk · normal at WB

X vs Y

	Anterolateral approach	Anteromedial approach
Plane	**Deep ↔ superficial peroneal**	**Deep peroneal ↔ tibial**
Best for	Valgus impaction, intact fibula	**Central impaction** (extensile)
Nerve risk	**SPN – dorsal foot numbness**	Saphenous

	Gastrocnemius contracture	Achilles contracture
DF knee flexed	Normal	**Limited**
DF knee extended	**Limited**	**Limited**
Surgery	Gastrocnemius recession	**TA lengthening**

How it's asked

"Comminuted closed pilon – next?"	**→ spanning ex-fix**
"Sign that soft tissues are ready for pilon ORIF?"	**→ wrinkle sign** (TO)
"Ex-fix for 3 wk, swelling settled?"	**→ ORIF tibia + fibula**
"Open pilon, medial wound?"	**→ I&D + ex-fix**
"Open pilon with defect – with I&D?"	**→ ex-fix of tibia**
"Correct sequence?"	**→ splint → ex-fix → CT → delayed ORIF**
"Decreases wound complications?"	**→ spanning ex-fix ± fibula fixation, delayed tibia**
"Obese smoker, swollen?"	**→ closed reduction + spanning ex-fix**
"CT timing?"	**→ after ex-fix**
"Optimise outcome, 43C, no wrinkles?"	**→ ex-fix then delayed ORIF**
"Chaput fragment ligament?"	**→ AITFL**
"Rotation + axial, plantarflexed?"	**→ posterior pilon**
"Pilon EXCEPT (TO)?"	**→ no alignment difference with fibular fixation**
"Next after plating fibula?"	**→ anatomic articular reduction**
"Simple articular extension, swollen?"	**→ percutaneous screws + IM nail**

"Fibular fixation — true?"	→ **higher complication rate**
"Anterolateral approach — true?"	→ **dorsal foot numbness commonest nerve complication**
"Central impaction — plane?"	→ **tibial ↔ deep peroneal**
"Poor outcome predictor?"	→ **lower education**
"Perceived outcome at 1 yr?"	→ **improves over > 2 yr**
"Syndesmotic injury — true?"	→ **higher amputation rate**
"Initial chondrocyte death?"	→ **necrosis, superficial zone, fracture margins**
"Synovial factor in PTOA?"	→ **IL-6**
"FWF vs ORIF in 43C?"	→ **FWF higher complications**
"Nonunion graft approach?"	→ **posterolateral**
"Inversion weakness, posteromedial gap?"	→ **tibialis posterior entrapment**
"TA lengthening indication?"	→ **DF −10° knee flexed + extended**
"Brake travel time?"	→ **↑ until 6 wk after WB**
"Pilon with intact fibula — impaction pattern?"	→ **medial (varus) impaction** (TO)
"Effect of early posterior malleolus reduction in pilon?"	→ **converts C-type into B-type pattern** (TO)

– – Active space – –

TIBIAL PLATEAU FRACTURES

– Trauma · 64 / 67 –

1 Basics

Epidemiology

Incidence **1–2% of all fractures** · 10.3 / 100,000 / yr · mean age 52 · **bimodal**: men 40s (high energy), women 70s (low energy)

Location **lateral 70–80%** · bicondylar 10–30% · medial 10–20%

Mechanism **valgus → lateral** · **varus → medial** · **axial → bicondylar** · high energy → medial, fracture-dislocation, soft-tissue injury · low energy → lateral

Lateral plateau share (TO) **55–70%** · **worst prognosis = medial plateau** (TO)

Force (TO)	Pattern	Age
Valgus / varus	**Split (cleavage)**	Young, good bone
Axial compression	**Depression**	Elderly osteoporotic
Combined	**Split-depression**	Any
Flexion + varus + IR	**Posteromedial fragment**	–

Causes (TO) bumper injury **52%** · fall from height 17%

Plateau anatomy (medial vs lateral)

	Medial plateau	Lateral plateau
Shape	**Concave, larger** (TO)	**Convex**, smaller
Height	**More distal**	**More proximal**
Load	**60% – primary weight-bearing**	40%
Bone	Stronger	Less dense
Meniscus	Less mobile (coronary ligaments)	**More mobile → submeniscal arthrotomy**

Alignment **posterior slope 6–10°** · **3° varus** to tibial mechanical axis

Slope / height (TO) posterior slope **3–10° (avg 7°)** (Tiwari) · sagittal **5°**, coronal **3°** (Shetty) · lateral plateau **2–3 mm higher**, cartilage **~4 mm lateral vs 2–3 mm medial**

Raft-screw tip (TO) screws parallel to lateral plateau → medial plateau lies **lower** → beware joint penetration

Kinematics 0–140° (walking 0–70°) · concave medial pivot + convex lateral → **femoral rollback**, screw-home (tibia ER in extension)

Insertions ACL between spines · PCL posterior sulcus · superficial MCL deep to pes · **LCL anterolateral fibular head** · ITB **Gerdy's tubercle** · pes anteromedial

Nerves **common peroneal around fibular neck (commonest nerve injury)** · tibial with popliteal artery

Associated meniscal & ligament injuries

Injury	Association
Lateral meniscus	**Commoner than medial** · **Schatzker II** · **> 10 mm depression + > 6 mm widening** (Orthobullets; OB) · **≥ 6 mm depression + 5–6 mm widening** (OB) – read the options
Lateral meniscus (TO)	**80%** in Schatzker II if **> 5 mm widening + > 6 mm depression** (TO)
Segond / reverse Segond (TO)	Lateral tibial avulsion → **ACL** · medial avulsion (deep MCL fibres) → reverse Segond
Medial meniscus	**Schatzker IV**
ACL	**Types IV + VI (25%)** (Orthobullets; OB) · **types V + VI** (TO; OB) – read the options
Neurovascular	**Schatzker IV fracture-dislocation**; hyperextension bicondylar
Compartment syndrome	High energy

Meniscal tear type — **peripheral / meniscocapsular detachment** (vascular zone, heals) · meniscus stays with femur as plateau widens
✗ Wrong options: horizontal cleavage · radial · root · bucket-handle medial

Rates — meniscal tear ~30% at surgery, **up to 91% on MRI** · ligament injury > 75% on MRI · LM injury > 75% of displaced Schatzker II

Lateral meniscal tear risk — **> 10 mm depression**, Schatzker II > I, **age < 48, male** (✗ Wrong options: > 50 · female · calcaneus fracture) · **6 mm widening** suggests LM tear (✗ Wrong options: 3 mm depression · spine fracture (ACL))

Management — **submeniscal arthrotomy dividing meniscotibial ligaments**, repair at fixation (longitudinal → repair · radial → partial resection)
✗ Wrong options: meniscofemoral · leave untreated

Repaired LM tear – 1-yr outcome — **similar function + QoL**
✗ Wrong options: ↑ TKA · ↓ ROM · ↑ OA grade · ↑ re-arthroscopy

ACL + PLC with plateau fracture — grade 3 Lachman + varus laxity 0° + 30° + ER asymmetry → **ACL + PLC reconstruction (LCL + popliteofibular)**, LM repair · missed PLC → ACL graft fails

3 Presentation

Exam — rule out **open** wound · **compartment syndrome** · varus / valgus stress in **full extension: laxity > 10° = instability** · pulses → **ABI if asymmetric; < 0.9 → arteriogram** · tibial + common peroneal nerves

4 Diagnosis & Classification

Key radiology findings

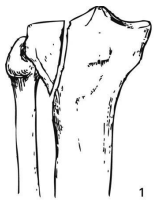
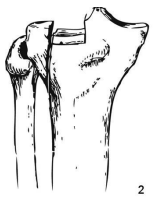
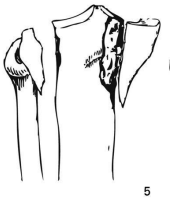
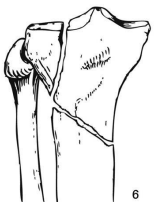
Study	Finding
AP, lateral, **oblique** (depression)	Depressed surface, **sclerotic band**, alignment · lateral: **posteromedial fracture lines**, slope
Plateau view	**10° caudal tilt** (matches posterior slope)
CT	Depression, comminution, **posterior coronal split** (axial + sagittal) · **after ex-fix** if staged
MRI	Meniscus + ligaments, occult fracture
Traction view (TO)	Restores geometry, ↓ overlap
Caudal tilt (TO)	**10–15°** (beam parallel to slope)

Lipohaemarthrosis — occult fracture

Lipohaemarthrosis — **intra-articular fracture**: **tibial plateau commonest** (TO) · **occult fracture** (OB)
✗ Wrong options: meniscus tear · MPFL · tendon rupture · ligament injury · arthritis

Layers — **fat (top), serum, RBCs (bottom)** · separation up to **3 h** · CT / MRI as sensitive as arthrocentesis

Classification (Schatzker, Moore, hyperextension varus)

Schatzker	Pattern	Note	Image
I	**Lateral split (wedge)**	**Young, strong subchondral bone**	
II	**Lateral split-depression**	**Commonest** · average age > 50 (TO) · **LM tear**	
III	**Lateral pure depression**	**Elderly osteoporotic**, uncommon	
IV	**Medial plateau**	**Fracture-dislocation, NV + ligament injury, MM tear**	
V	**Bicondylar**	**Spines continuous with shaft**	
VI	**Metaphyseal-diaphyseal dissociation**	Severe soft tissue	

FALSE (TO) "**type V = bicondylar wedges with metaphyseal-diaphyseal discontinuity**" – that is **VI**

Hohl + Moore (fracture-dislocations; 10% don't fit Schatzker)	Pattern
I	**Coronal split** (medial, displaces distally; doesn't cross eminence)
II	**Entire condyle** (≈ Schatzker IV)
III	Rim avulsion, lateral plateau
IV	Rim compression
V	Four-part

– – Active space – –

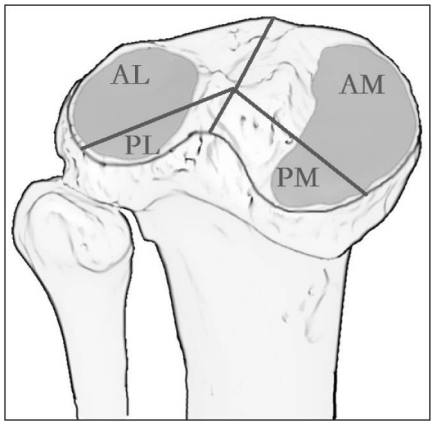
3-column concept — **medial, lateral, posterior** (CT) · captures **posterior** fractures Schatzker misses · guides approach

Luo 3-column (TO) — axial CT · **zero column = pure central depression (Schatzker III)** · Schatzker limitation = AP-based, misses posterior fragments

Column (TO)	Approach
Lateral / anterolateral	**Anterolateral** – incision 1 cm lateral to crest, **submeniscal arthrotomy**
Medial / posteromedial	**Posteromedial** (Lobenhoffer)
Posterolateral	**Frosch** – between **biceps femoris + lateral gastrocnemius**, protect **CPN**, plate **≤ 5 cm below joint line**
Combined (Kfuri–Schatzker)	Dual anterolateral + posteromedial

Hyperextension varus bicondylar (HEVBTP) — **loss / reversal of posterior slope** · posterior cortex tension failure, anterior compression, coronal varus · **popliteal artery 12%** (vs 1%), **peroneal nerve 16%**, compartment syndrome 12%, **PLC injury** · fix: **anteromedial buttress + graft anterior defect**
✗ Wrong options: posteromedial corner · ↓ peroneal risk · posterior tibial artery

Posterior column + varus — **poor function + sports limitation**
✗ Wrong options: low energy · highest meniscal injury · high satisfaction · Schatzker-classifiable



Three-column concept

5 Management

Tibial plateau fracture

- Minimally displaced, stable to varus / valgus → **hinged brace, NWB / PWB 8–12 wk, early ROM**
- Unicondylar lateral (I–III) → **lateral approach, elevate + graft, lateral buttress plate / rafts**
- Medial (IV) + ABI → **posteromedial / medial buttress plate**
- Bicondylar / VI, swollen or polytrauma + spanning ex-fix + CT → **dual-incision dual plating**
- > 65, osteoporotic, non-reconstructable → **acute TKA**

Operative criteria

Indication	Threshold
Articular depression / step-off	**> 5–10 mm** (Orthobullets) · **> 3 mm** step-off (OB) – read the options
Condylar widening	**> 5 mm**
Varus / valgus instability	**> 10°** (in extension)
Medial plateau fracture	All
Bicondylar	All
Step-off (TO)	**> 2 mm** – read the options
Open fracture, compartment / vascular injury (TO)	All

Decision for minimally displaced fracture — **physical exam (varus / valgus stability)** · may decide up to 2 wk
✗ Wrong options: MRI · age · compensation

Nonoperative — minimally displaced split / depression, stable, non-ambulatory, unfit · PTB cast / immobiliser / **hinged brace** · **early passive ROM**

Nonop duration (TO) — cast / hinged brace **4–8 wk** + early quadriceps exercises (Orthobullets: NWB / PWB 8–12 wk) – read the options

Span – scan – plan (TO) — spanning ex-fix → **CT after spanning** (reduced position) → definitive fixation once soft tissue settles

Schatzker (TO)	Fixation
I	Closed / percutaneous **6.5 mm cancellous lag screws + washer**; elderly → add buttress plate
II	Anterolateral approach, elevate + graft / calcium phosphate, **lateral buttress plate + raft screws**
III	Elevate via **metaphyseal window**, graft, **raft (3.5 mm subchondral) screws**
IV	**Medial buttress plate** + check popliteal artery
V / VI	**Dual plating ("post and lintel")**, 2 approaches 180° apart, **medial side fixed first**

Spanning external fixation – indications & effects

Indication — **axially / length-unstable** (**Schatzker VI with shortening**, bicondylar) or **knee subluxation** · significant swelling / **blisters**, polytrauma, open
✗ Wrong options: I–III · minimally displaced IV

Bicondylar VI + blisters next day — **knee-spanning ex-fix**
✗ Wrong options: splint / cast · ORIF

Polytrauma, open femur + plateau, head + abdominal injury — **ex-fix femur + plateau**, splint ankle + forearm
✗ Wrong options: acute nailing / ORIF

Technique — **two 5 mm half-pins in femur + two in tibia**, outside future incisions · **ligamentotaxis** · lock in **slight flexion** · **wires > 14 mm from joint** (capsule / septic arthritis)

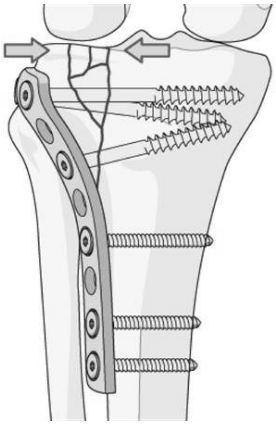
– – Active space – –

Expected	**transient ↑ compartment pressure** during placement · (✗ Wrong options: ↓ pulses · pulmonary injury · blocks wound access · ↑ deep infection / malunion) ✗ Wrong options: ↑ compartment syndrome
Downside	**arthrofibrosis ~15% → return to OR** (MUA, quadricepsplasty, lysis) · **each day in frame ↑ odds 10%** · **CPM ↓ risk up to 70%**
Posterior shear fracture, swollen, no skin wrinkles	**ex-fix + realign now**, delayed posteromedial ORIF ✗ Wrong options: immobiliser · fasciotomy without signs · prompt ORIF

Lateral plateau fixation — buttress principle & constructs

Partial articular (Schatzker II)	**lateral plate + metaphyseal graft** · rafting subchondral screws ✗ Wrong options: medial plate
Good bone	**undercontoured lateral nonlocking plate (buttress)** ± graft substitute · **osteopenic (63-yr, 15 mm depression) → still lateral nonlocking** ✗ Wrong options: locking (no buttress in pure locking mode, + cost) · ex-fix · screws alone · dual plates
Osteoporotic screw order	**3.5 mm nonlocking (lag) first → then 3.5 mm locking** (compress / narrow rim, then hold elevation) ✗ Wrong options: locking first · 6.5 mm screws
Buttress technique	**undercontoured plate + nonlocked screw near apex** of vertical fracture · resists axial load at **90° to deformity** · best for **simple partial articular / Schatzker I** (not depression, not metadiaphyseal)
Split-depression (skiing)	**ORIF with absolute stability** ✗ Wrong options: relative · IMN · ex-fix · cast

Plate function	Stability
Buttress	Resists axial / shear
Compression	Absolute
Neutralisation (with lag screw)	Absolute
Bridge	Relative



Buttress plate

Bone void fillers — calcium phosphate cement

Least subsidence + highest compressive strength	**calcium phosphate cement** · osteoconductive, porous, biodegradable ✗ Wrong options: ICBG · cancellous allograft · reamings · calcium sulfate (no structural strength) · tricalcium phosphate · hydroxyapatite · rhBMP-7
Depression technique	"open the book" or **cortical window + tamps** → fill void → rafts

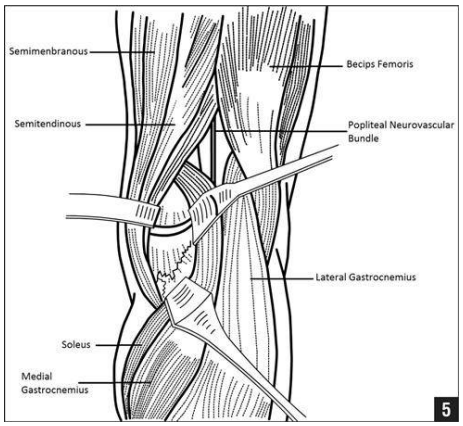
Bicondylar fractures — dual plating & posteromedial fragment

Posteromedial fragment	common; **lateral locking plate can't capture it → lateral + posteromedial plates via two incisions** ✗ Wrong options: single lateral / polyaxial plate · single anterior approach (infection) · hybrid frame · IMN
Bicondylar + blisters	**ex-fix → delayed dual-incision dual plating** ✗ Wrong options: lateral locked plate · single-incision dual plating · splint
Posterolateral fragment	must be **fixed separately** or fails ✗ Wrong options: single plate · rafting screws · cement
Hybrid ex-fix definitive (V / VI)	**unacceptably high malunion / nonunion** · shorter surgery, ↑ superficial infection, faster early return, shorter stay

Posteromedial approach — interval, positioning & buttress plating

Interval (Lobenhoffer)	**pes anserinus (sartorius, gracilis, semitendinosus) ↔ medial head of gastrocnemius** · semimembranosus ↔ medial gastrocnemius (OB) — read the options ✗ Wrong options: superficial MCL · POL · soleus · popliteus
Position	supine figure-4 or **prone** (extensile posterior / posterolateral shear) · elevate popliteus subperiosteally · knee **extension + valgus** aids reduction
At risk	**saphenous nerve + vein**, **posterior tibial recurrent artery** (distal / lateral)
Lobenhoffer steps (TO)	**prone** → dorsal incision → **medial gastrocnemius reflected laterally** → partial **soleus** detachment → buttress posteromedial fragment
Schatzker IV / medial condyle	**medial (posteromedial) buttress plate, absolute stability** ✗ Wrong options: lateral locking (shear failure) · medial bridge · lag screws alone (leg lever arm) · IMN · ex-fix
Posteromedial shear (fall on flexed knee)	**posteromedial approach + buttress plate** ✗ Wrong options: direct medial · lateral · anteromedial · hybrid
Posterior fragment only	**posterior buttress plate**

– – Active space – –



Posteromedial approach

Other approaches

Approach	Interval / fact
Anterolateral (commonest)	Hockey-stick over **Gerdy's**, elevate anterior compartment + ITB, **submeniscal arthrotomy**
Posterolateral	**Gastrocnemius-soleus ↔ peroneals** (TO; OB) · lateral gastrocnemius ↔ biceps, **CPN in plane** (OB)
Direct posterior	Medial ↔ lateral gastrocnemius – popliteal artery, small saphenous vein, CPN
Fibular neck osteotomy	Rare – NV risk
Midline	**Avoid** (stripping)

Elderly – acute arthroplasty

Primary TKA > 65–70 yr, osteoporotic — **earlier full weight bearing**
✗ Wrong options: ↑ implant failure (that's ORIF) · fewer complications than TKA for OA · worse than TKA after failed ORIF

Infection risk & fixation timing

Infection after ORIF — **male, smoking, pulmonary disease, bicondylar, operative time > 3 h** · vancomycin powder ↓ deep infection
✗ Wrong options: age · unicondylar · prior injury · mechanism

Fixation vs fasciotomy closure timing — **does not alter infection risk**

Articular cartilage healing – early motion

Best for surviving chondrocytes — **gentle compressive loading via early ROM**
✗ Wrong options: immobilisation · shear · distraction · glucosamine

6 Complications

Complication	Rate / fact
Post-traumatic arthritis	**25–35%** · meniscectomy, malalignment > 5°, instability
Conversion to TKA	**3–7% at 10+ yr** (Orthobullets) · **5–10% at 10 yr** (OB) · **~5× general population** (3–7×) – read the options
Compartment syndrome	**7–20%** · **young age**, high energy, **Schatzker IV** (Orthobullets) / **VI**, **V + VI** (OB), fibula fracture, plateau-shaft
Infection	**2–11%** (Orthobullets) · 6–12% (OB)
Nonunion / malunion	2–4% · **Schatzker VI**, comminution
Stiffness / arthrofibrosis	**10–25%** · **~15% after temporary ex-fix = commonest early complication** · MUA + lysis if < 90° flexion at 4 wk
Loss of reduction	5–30% · e.g. posteromedial coronal fragment missed by lateral plate
DVT	Nonop 9%, op 6%

Vascular injury – medial / Schatzker IV patterns & ABI

Schatzker IV — **fracture-dislocation variant**: femur follows the **medial** fragment · **popliteal artery + peroneal nerve stretch → ABI** · ABI < 0.9 → CTA / MRA
✗ Wrong options: compartment pressures · ex-fix before vascular check

Highest vascular risk — **Schatzker IV**
✗ Wrong options: II · V · tibial shaft · pilon · subtrochanteric

Schatzker IV – true — **medial meniscus + ACL injuries common**
✗ Wrong options: femur follows lateral plateau · lateral plate · bicondylar plating · vascular injury rare

Outcomes & arthritis – alignment, stability, age, conversion to TKA

Strongest predictor of long-term outcome — **restoration of joint stability**
✗ Wrong options: meniscal repair · surgery < 48 h · depression correction · diastasis

Most ↑ arthritis — **alteration of mechanical axis (> 5°)** · mild incongruity tolerated
✗ Wrong options: step-off · type · male · age

Age > 50 — **worse outcomes even in low-energy fractures**; only 35% satisfied · complication rates **don't differ** with age
✗ Wrong options: older → more infection · younger → more nonunion

Prognosis — 1-yr mortality 5% · return to work 70–90% · ROM 10–145°

X vs Y

	Schatzker II	Schatzker IV
Side	Lateral split-depression	**Medial**
Meniscus	**Lateral**	**Medial**
Danger	–	**Popliteal artery, peroneal nerve, ACL**
Fixation	**Lateral buttress + graft**	**Medial / posteromedial buttress**

– – Active space – –

	Lateral locking plate	Nonlocking buttress plate
Buttress effect	**None** (pure locking)	**Yes**
Use	Comminution, osteoporosis (after lag screws)	**Partial articular, good bone**

How it's asked

"Least subsidence filler?"	→ **calcium phosphate cement**
"Highest compressive strength filler?"	→ **calcium phosphate**
"Medial meniscus tear – which fracture?"	→ **Schatzker IV**
"Soft-tissue injury with lateral split-depression?"	→ **peripheral lateral meniscus tear**
"Radiographic sign of LM injury?"	→ **6 mm joint widening**
"Grade 3 Lachman + varus laxity + ER asymmetry – repair?"	→ **LCL (PLC)**
"Schatzker II – commonest associated injury?"	→ **lateral meniscal tear**
"Biggest LM tear risk factor?"	→ **> 10 mm depression**
"Meniscus tears – true?"	→ **lateral, Schatzker II, divide meniscotibial ligaments**
"ACL injured most in?"	→ **Schatzker IV / VI**
"Schatzker IV – true?"	→ **medial meniscus + ACL injuries**
"Repaired LM tear – 1-yr outcome?"	→ **similar function**
"Skiing – commonest injury pattern?"	→ **peripheral lateral meniscus tear**
"Schatzker FALSE (TO)?"	→ **type V with metaphyseal-diaphyseal discontinuity**
"69-yr pure lateral depression?"	→ **Schatzker III**
"Entire condyle fracture-dislocation?"	→ **Moore II**
"Hyperextension varus bicondylar – hallmark?"	→ **loss / reversal of tibial slope**
"Schatzker VI after ex-fix, good soft tissue?"	→ **medial + lateral plates, two approaches**
"Posteromedial fragment – why it matters?"	→ **may need dual plating**
"Most stable for bicondylar?"	→ **lateral + posteromedial plating**
"Bicondylar + blisters?"	→ **ex-fix → delayed dual-incision dual plating**
"Medial condylar split – construct?"	→ **posteromedial buttress plate, absolute stability**
"Posterior fragment only?"	→ **posterior buttress plate**
"Medial plateau fracture fixation?"	→ **medial buttress plate**
"Posteromedial interval?"	→ **pes anserinus ↔ medial head of gastrocnemius**

"Prone position fracture?"	→ posteromedial / posterolateral shear
"Buttress plating best for?"	→ Schatzker I / simple partial articular
"Schatzker II surgical plan?"	→ lateral plate + metaphyseal graft
"Schatzker II, 40-yr?"	→ lateral nonlocking plate ± graft substitute
"Osteoporotic screw sequence?"	→ nonlocking then locking 3.5 mm
"Buttress technique?"	→ undercontoured plate + nonlocked screw at apex
"Osteopenic Schatzker II, 15 mm depression?"	→ lateral nonlocking construct
"Split-depression treatment principle?"	→ ORIF with absolute stability
"Plate functioning as buttress?"	→ force at 90° to deformity axis
"Initial treatment, high-energy bicondylar?"	→ spanning ex-fix
"Polytrauma open plateau + femur?"	→ ex-fix femur + plateau
"Most benefit from temporary ex-fix?"	→ Schatzker VI with shortening
"Ex-fix sequela?"	→ transient compartment pressure rise
"Ex-fix duration ↑ risk of?"	→ return to OR (arthrofibrosis)
"Hybrid ex-fix definitive?"	→ high malunion / nonunion
"Vascular complications commonest?"	→ Schatzker IV
"ABI most indicated?"	→ Schatzker IV
"Schatzker IV + foot numbness after reduction?"	→ ABI
"Strongest long-term predictor?"	→ joint stability
"Most ↑ arthritis?"	→ mechanical axis alteration
"Age > 50 outcomes?"	→ worse in low-energy fractures
"Surgery decision, minimally displaced?"	→ physical exam
"Commonest complication within 6 months?"	→ arthrofibrosis
"Future TKA risk?"	→ ~5× general population
"Counselling – TKA at 10 yr?"	→ 5–10%
"Posterior column varus fracture?"	→ poor function + sports limitation
"Elderly acute TKA advantage?"	→ earlier full weight bearing
"Failed dual plating – missed?"	→ posterolateral fragment
"Fixation timing vs fasciotomy closure?"	→ doesn't alter infection risk
"Infection risk factor?"	→ operative time > 3 h
"Young age associated with?"	→ compartment syndrome
"Help reconstructed chondrocytes?"	→ gentle compression via early ROM

– – Active space – –

"Medial plateau shape + position?"	→ **concave, larger, more distal**
"Lipohaemarthrosis (TO)?"	→ **tibial plateau fracture**
"Lipohaemarthrosis means?"	→ **occult fracture**
"Posterolateral approach plane (TO)?"	→ **gastrocnemius-soleus ↔ peroneals**
"Zero column in 3-column classification?"	→ **pure depression (Schatzker III)** (TO)
"Dual plating principle for bicondylar fractures?"	→ **post and lintel** (TO)

TIBIAL SHAFT FRACTURES

– Trauma · 65 / 67 –

1 Basics

Epidemiology

Commonest long-bone fracture — **17% of lower-limb fractures** · 4% of Medicare fractures · M > F, bimodal · proximal ⅓ = 5–10% of shaft fractures · open ~20%

Mechanism	Pattern	Fibula	Soft tissue
Low energy, torsion	**Spiral**	**Different level**	Less · **posterior malleolus** with spiral distal ⅓
High energy, direct	**Wedge / short oblique**, comminuted	**Same level**	Severe (Tscherne II / III, open)

Amputation predictor — **severity of muscle / soft-tissue injury**

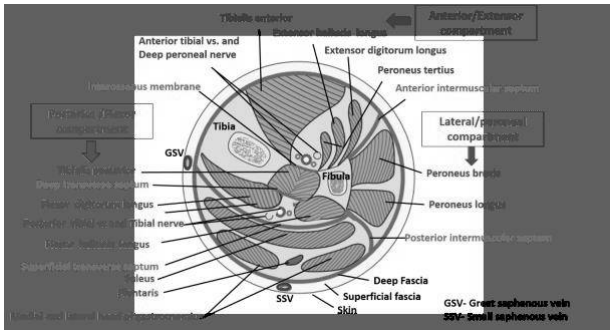
Anatomy

Shaft — triangular · **proximal canal centred laterally** (start point) · subcutaneous anteromedial crest · tubercle ~3 cm below joint · **tibia carries 80–85% of load**

Blood supply (TO) — **nutrient artery (from posterior tibial) enters posterolaterally** · periosteal branches from **anterior tibial** · fibula carries **up to 16%** of load (TO)

Deforming forces (TO) — proximal fragment **flexion + valgus** · distal fragment **varus + extension**

Compartment	Contents
Anterior	Tibialis anterior, EDL, EHL · deep peroneal n.
Lateral	Peroneus longus + brevis · superficial peroneal n.
Superficial posterior	Gastrocnemius, soleus, plantaris
Deep posterior	Popliteus, tibialis posterior, FDL, FHL · tibial n.

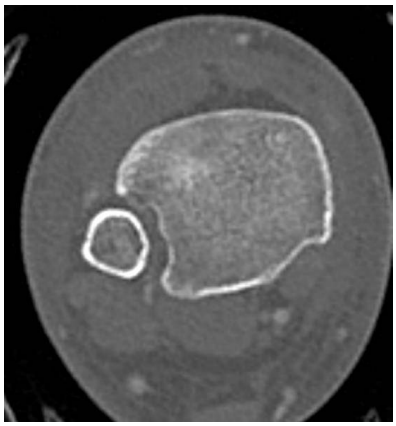


Compartments of the leg

– – Active space – –

Associated injuries (posterior malleolus in spiral distal-third fractures)

Spiral distal ⅓ (twisting)	**posterior malleolus (Volkmann) fracture** — 25–60% · **CT ankle** (or ankle X-rays) before nailing · nail can displace it → **AP lag screw first** · may affect syndesmosis ✗ Wrong options: MRI knee / hip · CT knee · calcaneal view · CT chest · arterial duplex
Not associated	vertical medial malleolus shear, Chaput, talus, Tillaux, knee dislocation · **PITFL intact** with posterior malleolus fracture
Compartment syndrome	**diaphyseal 8.1%** (proximal 1.6%, distal 1.4%) · **occurs in open fractures too** · check all 4
Floating knee	antegrade tibial nail + retrograde femoral nail



Posterior malleolus on axial CT

3 Presentation

Exam	deformity, blisters, wounds · **compartments** · DPN, SPN, sural, tibial, saphenous · DP + PT pulses (Doppler → CTA if absent)

4 Diagnosis & Classification

Key radiology findings

X-ray	full-length AP + lateral tibia · **knee + ankle** AP / lateral / oblique · repeat after splinting
CT	plateau / plafond extension · **distal ⅓ or spiral → exclude posterior malleolus** · nonunion

Classifications

OTA 42	Pattern
A	Simple
B	Wedge
C	Complex / comminuted

– – Active space – –

Tscherne (closed soft tissue)	Injury
0	Indirect, negligible
I	Superficial abrasion, simple fracture
II	Deep abrasion, contusion, **impending compartment syndrome**
III	Crush, degloving, muscle destruction, **compartment syndrome, vessel / nerve rupture**

Gustilo-Anderson	Wound / soft tissue
I	**< 1 cm, clean**, simple
II	**> 1 cm**, no extensive damage
IIIA	High energy / comminuted / segmental; usually > 5 cm; **coverage adequate** · **1 cm wound with high-energy comminution = IIIA**
IIIB	**Flap needed** (STSG doesn't count) · proximal ⅓ **gastrocnemius**, middle ⅓ **soleus**, **distal ⅓ free flap**
IIIC	**Vascular injury needing repair**
Always III	Barnyard, shotgun / high-velocity GSW, > 24 h to treatment

Exposed bone, dopplerable pulses, peroneal numbness

IIIB → spanning ex-fix, delayed fixation + coverage
✗ Wrong options: immediate plating (infection up to 44%)



Gustilo IIIB

5 Management

- **Tibial shaft fracture**
  - Closed, low energy, acceptable alignment → long leg cast → **PTB brace ~4 wk**
  - Most → **reamed, statically locked IM nail**
    - proximal ⅓ → **lateral start, semi-extended / suprapatellar, blocking screws**
    - distal ⅓ → **fibular plating, center-center guidewire**
  - Open → **antibiotics < 3 h, I&D < 12–24 h, nail or ex-fix**
  - Polytrauma / contaminated → ex-fix → **nail when resuscitated (ideally < 7 d)**

– – Active space – –

Nonoperative management – deformity & contraindications

Acceptable alignment (cast / functional brace)	Limit
Varus-valgus	**< 5°**
Anterior-posterior	**< 10°**
Cortical apposition	**> 50%**
Shortening	**< 1 cm**
Rotation	**< 10°**

Contraindication to brace	**shortening > 10 mm** (24 mm comminuted) · open, intra-articular, non-compliant ✗ Wrong options: same-level fibula · transverse · 9° sagittal · 7° rotation
Cast risk	**shortening hardest to control** (oblique / comminuted; mean 4 mm) · angulation + rotation held well · nonunion ~1% (Sarmiento 1.1%) · union ~18 wk
Intact fibula	**varus malunion (~25%)** – fibula = strut · relative contraindication to bracing
Nonop most common deformity	**varus** ✗ Wrong options: valgus + procurvatum (proximal ⅓ nailing)
IM nail vs cast EXCEPT	"**↓ compartment syndrome**" – nailing may ↑ it (11%, Finkemeier) · nail: faster union, ↓ malunion, ↓ shortening, quicker return to work

Operative criteria

Option	Indications
IM nail	Unacceptable closed alignment, soft tissue won't tolerate cast, **segmental, comminuted, floating knee, polytrauma, bilateral, morbid obesity** · union > 80% closed
Ex-fix	**Damage control**, contaminated / soft-tissue defect, metaphyseal
Plate	Proximal / distal with poor nail fixation, **adjacent implant (TKA)**
Amputation (relative)	**Severe soft tissue, warm ischaemia > 6 h, severe ipsilateral foot trauma**

Open fractures – classification, antibiotics & infection

Antibiotics	**within 3 h** · **cefazolin for 24 h after definitive surgery (I, II, IIIA)** · **+ aminoglycoside IIIB** (Orthobullets) / **type III** (OB) – read the options · + **penicillin** farm (Clostridium) · **tetanus**
Type III regimen	**1st-gen cephalosporin (cell wall) + aminoglycoside (30S, bactericidal)** ✗ Wrong options: 50S macrolide / clindamycin · tetracycline (bacteriostatic 30S) · fluoroquinolone alone (water contamination) · vancomycin alone
Most important to ↓ infection	**early antibiotic administration** ✗ Wrong options: debridement < 6 h · immediate internal fixation · antibiotic irrigation · high-pressure lavage · reamed nail · VAC · tetanus · 9 L irrigation
Second (after antibiotics, LEAP)	**time to transfer to definitive trauma centre** ✗ Wrong options: ex-fix · debridement < 6 h · coverage < 48 h

I&D	within **12–24 h** · remove all dead tissue + cortical bone · primary closure if clean, no tension · NPWT doesn't ↓ infection
Persistent infection with retained implants	strongest predictor **open fracture** ✗ Wrong options: smoking · Pseudomonas
BOA-BAPRAS protocol (TO)	IV antibiotics **< 3 h** → tetanus → primary debridement → temporary ex-fix → **definitive soft-tissue cover within 72 h**

Limb salvage vs amputation (LEAP)

Most important in salvage vs amputation	**severity of soft-tissue injury** ✗ Wrong options: plantar sensation · bone loss · education · contralateral injury
Insensate foot	**not** an indication · **55% normal plantar sensation at 2 yr** in both groups
LEAP	**amputation = salvage** at 2 + 7 yr (function, psychosocial, return to work) · **depression at 3 months → worse outcome** ✗ Wrong options: plantar sensation · nail · free vs rotational flap · number of debridements
METALS (military)	**amputation better function** than salvage · civilians: equal or slight benefit to amputation (OUTLET) · disability common either way

Soft-tissue coverage – flaps

Zone	Flap	Supply
Proximal ⅓	**Medial gastrocnemius** rotation (larger, no CPN risk)	**Medial sural artery** (popliteal branch)
Middle ⅓	**Soleus** rotation	–
Distal ⅓	**Free flap** (latissimus, rectus, gracilis) or reverse sural	–

Staged ORIF + free flap	**IIIB intra-articular distal tibia** ✗ Wrong options: midshaft · proximal Schatzker I · IIIC

Reamed vs unreamed nailing & reaming hazards

	Closed	Open
Reamed	**↓ time to union, ↓ reoperation** (SPRINT) · larger nail	**Equivalent** to unreamed
Unreamed	↑ grafting, dynamisation, exchange · ↑ **locking screw breakage**	Same union, infection, reoperation

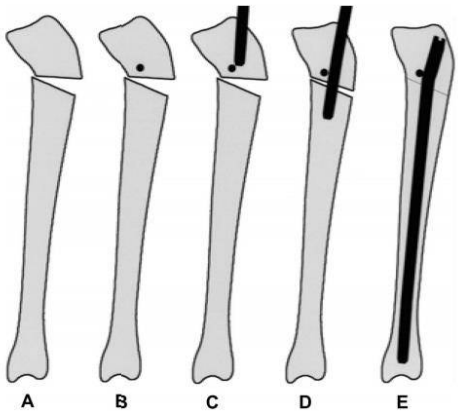
Unreamed in open	**no difference** · not an adverse factor · surgery maybe shorter · pulmonary benefit unproven ✗ Wrong options: ↑ nonunion · ↑ / ↓ infection
Adverse outcome (open tibia nail)	**high energy, stainless steel nail, gap > 1 cm, smoking** ✗ Wrong options: unreamed nail
Biology	**mineral apposition similar** reamed vs unreamed · **reamed → most new bone** · **limited reaming → lowest porosity** · **tight nails ↓ cortical reperfusion** · over-ream **1–1.5 mm**
Massive necrosis after reaming	**over-reaming** ✗ Wrong options: tourniquet · SLE · steroids · hyperflexion

– – Active space – –

Tourniquet during reaming	**no thermal necrosis** · with reamed femoral nailing in polytrauma → ↑ **pulmonary morbidity** (reperfusion + emboli) · ↑ post-op pain

Nail start point & malalignment (proximal & distal fractures)

Start point	**medial border of lateral tibial eminence** (AP) · just anterior to articular margin (lateral) · proximal ⅓ → **more lateral + proximal** (on lateral eminence) · **lateral knee X-ray** checks nail not proud
Proximal ⅓ deformity	**valgus + procurvatum (apex anterior)** – **patellar tendon extends proximal fragment**, **pes anserinus → valgus**, hamstrings / gastrocnemius flex distal · **medial start point → valgus** · malunion up to 50%
Prevent	**semi-extended / suprapatellar** (↓ procurvatum), **lateral start**, **blocking (Poller) screws in concavity: posterior in proximal fragment + lateral in metaphysis** (+25% stability), unicortical plate, lateral parapatellar
Nailing problems (TO)	**malalignment commonest** (varus / valgus) · anterior knee pain · delayed union · **varus = commonest malunion** (TO) · entry: patellar-tendon split (commonest), parapatellar, **suprapatellar** (better alignment, less pain, easier reduction)
Nonop (TO)	long leg cast **6–8 wk** for undisplaced; high malunion rate
Distal ⅓	**valgus** malalignment with infrapatellar nail vs plate · guidewire **center-center** at physeal scar · nail tip slightly lateral to talar centre · aids: **fibular plating**, clamps, Poller screw, small plate ✗ Wrong options: **syndesmotic fixation**
Distal ⅓ extra-articular + fibula fracture	**reamed IMN + fibular plating** ✗ Wrong options: tibial + fibular plates (wounds) · nail without fibula (valgus, late loss)
Malrotation	**highest after nailing distal ⅓** · check: perfect lateral knee, rotate C-arm **105–110°** → ankle mortise



Posterior blocking screw

Anterior knee pain – infrapatellar vs suprapatellar nailing

Commonest complication of nailing (TO)	**anterior knee pain** – **57%** (TO) · **50–75%** (> 70% at 14 yr) (OB) · > 30–50% (Orthobullets) · ~20% (OB) – read the options · activity-related, kneeling · only 10% can't work

Tendon-splitting vs paratendinous Keating 77% vs 50% · **Toivanen: no clinically relevant difference** (67% vs 71%)

	Suprapatellar (semi-extended)	Infrapatellar
Alignment	**Better**, ↓ **procurvatum + valgus**	More flexion → procurvatum
Anterior knee pain	**Less** (OB) · no difference (OB)	**More**
Septic knee, nail size, ROM, quads, PF disease, op time	**No difference**	No difference
Fluoroscopy	Easier	–
Use	**Proximal / distal ⅓**	Midshaft

Percutaneous plating – SPN risk & radiation

LISS / long lateral plate **superficial peroneal nerve** at **holes 11–13** (13-hole) → **dorsal foot / midfoot numbness**, motor normal · SPN exits fascia **~8 cm above lateral malleolus**
✗ Wrong options: CPN · sural · tibial · compartment syndrome

Plate vs IMN (distal ⅓) plate: ↑ **radiation**, ↓ **angular deformity**, larger incision, harder removal · nail: ↑ **pain**, shorter surgery · same union, infection, wound breakdown, prominent hardware (OB)

Biological adjuncts – BMP-2 / BMP-7

BMP	Approval / fact
rhBMP-2	**FDA: acute open tibial shaft fracture + IM nail** · ↓ **secondary bone grafting** (IIIA / IIIB), faster healing, ↓ hardware failure, ↓ infection (controversial) · not for nonunion
rhBMP-7 (OP-1)	**= autograft for tibial nonunion** (RCT) · **HDE** for recalcitrant long-bone nonunion
BMP-3	**No osteoinductive activity**

Timing of definitive fixation in polytrauma

Convert ex-fix → nail **as soon as resuscitated: lactate < 4.0 mmol/L + base excess > −5.5** (day 3, lactate 2.0, BE −2.5 ✓) · ex-fix **> 14 d** (> 15 d, Orthobullets) → ↑ **infection** · ideally **< 7 d** (> 8 d ↑ infection, OB)

External fixator pins – capsular reflection & septic arthritis

Knee capsule up to **14 mm below subchondral line (posterolateral)** → pins **> 14 mm** · too proximal → **septic arthritis**
✗ Wrong options: foot drop · anterior tibial artery · contracture

Frame stiffness **larger pins, more pins each side, rods closer to bone, multiplanar**

Definitive ex-fix (type III) **slower union, worse function, more malalignment** than nail · pin infection

– – Active space – –

Ballistic tibia fractures

GSW with exposed bone	**treat as open** · infection + soft-tissue reconstruction **similar to blunt open** · **↑ compartment syndrome + reoperation vs blunt closed**

6 Complications

Complication	Rate / fact
Anterior knee pain	**Commonest after nailing** · proud nail (lateral X-ray) · nail removal unpredictable
Malunion	8–10% · **proximal ⅓ up to 50%** · nonop → varus
Nonunion	**2–10%** · **no healing at 9 months** · open, < 50% cortical contact, transverse, gap
Malrotation	Distal ⅓ nailing → ankle arthrosis
Compartment syndrome	1–9% · closed + open
Infection	~5% · open, contamination, delayed cover
Nerve	SPN (LISS), saphenous (locking screws), **transient peroneal palsy** after nailing, DPN (posterolateral ex-fix pins)

Peroneal nerve palsy after nailing

Transient peroneal neurapraxia (up to 5%)	**↓ EHL strength** + 1st web numbness (DPN) · improves by 3 months ✗ Wrong options: lateral hindfoot (sural) · Achilles reflex (S1) · peroneus longus (SPN) · plantar (tibial)

Nonunion — risk factors, exchange nailing & fibular ostectomy

Greatest nonunion risk after nailing	**postoperative fracture gap** ✗ Wrong options: NSAIDs · fibula fracture · smoking · mechanism
Delayed union	20–26 wk · **nonunion** — > 9 months or no progress for 3 consecutive months · **4 months with healing signs → observe ≥ 6 months** ✗ Wrong options: exchange now
First step	**rule out infection (ESR, CRP)** — then CT / vit D ✗ Wrong options: CT · vit D · stimulator · exchange
Reamed exchange nailing	**aseptic diaphyseal tibial nonunion** · **hypertrophic** (85–90%) · **< 30% cortical bone loss** (76–96%) · **oblique** best · bent nail, no bone loss → exchange ✗ Wrong options: circular frame first · ICBG alone · cast / brace
Exchange nailing poor for	**infection**, **segmental bone loss / 4 cm defect**, **atrophic** (needs graft / BMP), **metadiaphyseal** (plate), comminuted (45–80%), humerus (30–40%), distal femur (50–60%)
Mid-diaphyseal nonunion after reamed 10 mm nail (10 months)	**exchange nailing** ✗ Wrong options: adjunct plate (metaphyseal / hypertrophic) · nail removal + plate (metadiaphyseal) · fibulectomy + cast · rhBMP-2
Intact / healed fibula holding tibia apart	**fibular osteotomy / partial ostectomy** ✗ Wrong options: cast · solid nail same size · BMP-3 · Masquelet
After ex-fix (IIIB, near stiff TKA), oligotrophic	**compression plating** (92–96%) ✗ Wrong options: MUA · cast + stimulator · ex-fix · IMN

Bone loss posterolateral graft · defects **up to 5–6 cm** graftable

Malunion & deformity correction (CORA, ankle effects)

Distal ⅓ varus 25° **ipsilateral ankle pain + stiffness**
✗ Wrong options: spine · hip · knee

CORA **intersection of proximal + distal mechanical (≈ anatomical) axes** in tibia
✗ Wrong options: concave / convex cortex point

Osteotomy away from apex **secondary translation** deformity

Early malalignment after nailing return to OR: remove nail, reduce, renail · late → osteotomy

Late equinus & claw toes

Checkrein deformity (TO) after **distal ⅓ tibia fracture** · **FHL adheres to callus → hallux flexes with ankle dorsiflexion**, extends with plantarflexion
✗ Wrong options: FHL tear · IP fracture · tethering at MTP

Claw toes flexible in PF, rigid in DF + equinus not relieved by knee flexion **TA lengthening + release / retromalleolar lengthening of FDL + FHL**
✗ Wrong options: PT · wait · tibialis posterior transfer (normal heel rise) · IP fusion



Proximal-third malunion

X vs Y

	Proximal ⅓	**Midshaft**	**Distal ⅓**
Nail deformity	**Valgus + procurvatum**	Rare	**Valgus**, malrotation
Trick	Lateral start, semi-extended, Poller	Standard	Fibular plate, center-center wire
Flap	Gastrocnemius	Soleus	Free
Associated	Plateau extension	–	**Posterior malleolus** (spiral)

	Hypertrophic nonunion	**Atrophic nonunion**
Problem	**Mechanical**	**Biological**
Treatment	**Exchange nailing** / stability	**Graft / BMP** + stability

– – Active space – –

How it's asked

"Commonest complication of reamed nailing (TO)?"	→ knee pain
"Most common complication, closed nailing?"	→ anterior knee pain
"Knee pain at union after reamed nail?"	→ 50–75%
"Suprapatellar technique advantage?"	→ improved alignment
"Infrapatellar vs suprapatellar — true?"	→ more anterior knee pain with infrapatellar
"Suprapatellar effect?"	→ ↓ procurvatum
"Semi-extended prevents?"	→ procurvatum
"Distal ⅓ nail vs plate — malalignment?"	→ valgus
"Check nail not proud?"	→ lateral knee X-ray
"Medial start point, proximal ⅓?"	→ valgus malunion
"Standard start in proximal ⅓?"	→ valgus, apex anterior
"Distal tibia reduction aids EXCEPT?"	→ syndesmotic fixation
"Plate vs IMN advantage?"	→ less angular deformity
"Start point?"	→ medial border of lateral tibial eminence
"Reaming — true?"	→ similar mineral apposition
"Reamed vs unreamed closed?"	→ decreased time to union
"Tourniquet + reamed femur nail?"	→ ↑ pulmonary morbidity
"Open fracture reamed vs unreamed?"	→ equivalent
"Unreamed in open fracture?"	→ no difference
"NOT an adverse factor?"	→ unreamed nail
"Unreamed vs reamed?"	→ similar union in open fractures
"Massive necrosis after reaming?"	→ over-reaming
"Greatest nonunion risk?"	→ post-op fracture gap
"Best for reamed exchange nailing?"	→ diaphyseal tibia, < 30% bone loss
"Oligotrophic nonunion after ex-fix + flap?"	→ compression plating
"Exchange nailing without graft?"	→ hypertrophic tibial shaft nonunion
"Isolated exchange nailing?"	→ hypertrophic diaphyseal nonunion
"Highest union with exchange nailing?"	→ oblique tibial shaft
"4 months, partial healing?"	→ observe
"10-month diaphyseal nonunion, normal labs?"	→ exchange nailing
"Bent nail, nonunion?"	→ exchange nailing
"24 wk, intact fibula?"	→ fibular osteotomy

"10-month pain, hypertrophic callus — first?"	→ **inflammatory markers**
"Equivalent to autograft for nonunion?"	→ **BMP-7**
"rhBMP-2 at fixation?"	→ **↓ subsequent bone grafting**
"Adjunct improving outcome with nail?"	→ **rhBMP-2**
"rhBMP-2 FDA indication?"	→ **acute open tibia + IM nail**
"FDA adjunct for open tibia?"	→ **rhBMP-2**
"Decreases infection?"	→ **immediate antibiotics**
"Classification + treatment, exposed bone?"	→ **IIIB, ex-fix, delayed fixation + cover**
"1 cm wound, high-energy comminution?"	→ **IIIA**
"Greatest effect on infection (transfer case)?"	→ **time to definitive trauma centre**
"Reduces infection MOST?"	→ **early IV antibiotics**
"Persistent infection predictor?"	→ **open fracture**
"Type III antibiotics?"	→ **cephalosporin + aminoglycoside (bactericidal)**
"Pin too proximal → risk?"	→ **septic arthritis**
"Capsular reflection?"	→ **14 mm**
"Salvage vs amputation decider?"	→ **soft-tissue injury severity**
"Worse long-term outcome predictor?"	→ **depression at 3 months**
"IIIB with insensate foot — true?"	→ **soft-tissue severity predicts amputation**
"Military vs civilian amputation?"	→ **military better after amputation**
"Nail vs cast EXCEPT?"	→ **↓ compartment syndrome**
"Oblique distal ⅓ in cast — risk?"	→ **shortening**
"Varus malunion in cast?"	→ **intact fibula**
"Nonop commonest deformity?"	→ **varus**
"Bracing contraindication?"	→ **shortening 24 mm**
"Spiral distal ⅓ — associated?"	→ **posterior malleolus fracture**
"Twisting injury — order before nailing?"	→ **ankle radiographs**
"Twisting distal tibia — imaging?"	→ **CT ankle**
"High-energy twisting — associated fragment?"	→ **Volkmann**
"Dorsal foot numbness after MIPO?"	→ **superficial peroneal nerve**
"Long lateral plate → numbness?"	→ **dorsal midfoot**
"MIPO pin location injuring nerve?"	→ **distal plate holes (11–13)**
"Plate vs IMN?"	→ **more radiation**
"Distal varus 25° → risk?"	→ **ankle pain + stiffness**

– – Active space – –

"CORA?"	**→ proximal + distal mechanical axes meet**
"Osteotomy away from apex?"	**→ translation**
"Transient peroneal palsy after nailing?"	**→ ↓ EHL strength**
"Staged ORIF + free flap?"	**→ IIIB intra-articular distal tibia**
"Proximal ⅓ gastrocnemius flap supply?"	**→ medial sural artery**
"Distal ⅓ + fibula fracture?"	**→ reamed IMN + fibular plating**
"Comminuted C2 distal tibia, diabetic?"	**→ ankle-spanning ex-fix**
"When to convert ex-fix?"	**→ day 3, lactate 2.0, BE −2.5**
"Ballistic tibia – true?"	**→ treat as open if bone exposed**
"Checkrein deformity (TO)?"	**→ distal third tibia fracture**
"Claw toes + equinus after distal tibia?"	**→ TAL + FDL / FHL release**

TRAUMA SCORING SYSTEMS

– Trauma · 66 / 67 –

1 Basics

Purpose & types

Uses triage, outcome prediction / counselling, quality assurance, research, reimbursement

Type	Scores
Physiological	**RTS**, APACHE, SOFA, **SIRS**, Emergency Trauma Score
Anatomical	**AIS**, **ISS**, **NISS**, Anatomic Profile, PATI, ICISS, TMPM-ICD9
Combined	**TRISS**, ASCOT, ICISS, **Ganga Hospital Open Injury Score**

Physiologic	Anatomic	Combined
Revised Trauma Score	Abbreviated Injury Score	TRISS
APACHE	Injury Severity Score	ASCOT
SOFA	New Injury Severity Score	ICISS
SIRS	Anatomic Profile	
Emergency Trauma Score	PATI	
	ICISS	
	TMPM-ICD9	

Classification of trauma scores

4 Diagnosis & Classification

Glasgow Coma Scale

Score	Motor (6)	Verbal (5)	Eyes (4)
6	Obeys	–	–
5	Localises pain	Oriented	–
4	Normal flexion (withdraws)	Confused	Spontaneous
3	**Abnormal flexion (decorticate)**	Inappropriate words	To speech
2	**Extension (decerebrate)**	Incomprehensible sounds	To pain
1	None	None	None

Brain injury **severe < 9 (≤ 8)** · moderate 9–12 · minor ≥ 13 · range **3–15**

Limits ignores focal / lateralising signs, metabolic processes, intoxication

– – Active space – –

Revised trauma score

Code	GCS	Systolic BP	Resp. rate
4	**13–15**	**> 90**	**10–29**
3	9–12	76–89	**> 30**
2	6–8	50–75	6–9
1	4–5	1–49	1–5
0	3	0	0

RTS **GCS + SBP + RR** codes · **most widely used pre-hospital triage tool** · **lower score = more severe** · **RTS < 4 → transfer to trauma centre**

RTS – true (TO) **improves appropriateness of treatment**
✗ Wrong options: improves diagnostic / investigation error rate (no difference) · higher score = more severe · includes respiratory effort + capillary refill (those were in the old Trauma Score, removed in RTS)

RTS limits **underestimates single-system injury** (may delay referral) · best for predicting survival in **head injury** (TO)

Glasgow Coma Score	Systolic Blood Pressure	Respiratory Rate	Coded Value
13-15	>90	10-29	4
9-12	76-89	>30	3
6-8	50-75	6-9	2
4-5	1-49	1-5	1
3	0	0	0

Revised trauma score

Injury Severity Score calculation (ISS, MISS)

AIS grade	Severity
1	Minor (laceration, contusion)
2	Moderate (**large joint dislocation**)
3	Severe, not life-threatening (**femoral shaft, open pelvic fracture**)
4	Severe, life-threatening, survival probable
5	Critical, survival uncertain (**crushed larynx**)
6	**Maximal / unsurvivable** (crushed head)

ISS **sum of squares of the highest AIS in the 3 most severely injured body regions** ($A^2 + B^2 + C^2$) · range **1–75** · **any AIS 6 → 75** · first anatomical scoring system

Body regions **6** (TO; OB: head, face, chest, abdomen, extremities + pelvis, external) · **9 anatomic regions** (Orthobullets) – read the options

Only the highest AIS per region **bilateral femur fractures count once** → NISS fixes this

Polytrauma threshold **ISS > 15** (major trauma; **10% mortality**) (Orthobullets, TO, OB) · **ISS ≥ 18** polytrauma / transfer (OB) – read the options

Berlin definition of polytrauma (2014) (TO) **≥ 2 injuries with AIS ≥ 3** + **≥ 1** of: **SBP ≤ 90** · **GCS ≤ 8** · **BD ≤ −6.0** · **PTT ≥ 40 s or INR ≥ 1.4** · **age ≥ 70**

ISS (TO)	Severity
1–8	Minor
9–15	Moderate
16–24	Severe
≥ 25	Very severe

Age (TO)	ISS for LD50
15–44 yr	**40**
45–64 yr	**29**
> 65 yr	**20**

AIS (TO) **AAAM 1971**; 6-point ordinal; **7 body regions** (external, head / face, neck, thorax, abdomen / pelvis, spine, extremities) · non-linear vs mortality · **ISS = Baker 1974** · **NISS = Osler 1997**, always **≥ ISS**

Anatomic Profile (TO) 4 regions (brain, spinal cord, thorax, neck); modified AP counts only **AIS ≥ 3**

Ideal score (TO) anatomical injury + physiological derangement + patient reserve (TRISS combines)

Worked example	ISS
Chest 4, liver 4, open tibia 3, radius 2, scalp 1 (TO)	16 + 16 + 9 = **41**
Open chest 4, colon 4, femur 3, shoulder 2, thyroid 1	**41**
Subdural 4, spleen 4, pelvis / femur 3, nose 2, ribs 2	**41**
Bilateral femur 3 + abdomen 3 + face 2 + lung 2	9 + 9 + 4 = **22**

NISS **3 highest AIS regardless of region** → better for **severe single-region injury**; **more predictive of complications + mortality**

ISS limits hard to calculate in resus · misses multiple injuries in one region

MISS (paediatric) **5 regions** (neurological, face / neck, chest, abdomen / pelvic contents, extremities / pelvic girdle) · graded 1–5 · same sum-of-squares · **MISS < 40 → no deaths**; > 40 → 50%; > 50 → 75%

MISS example chest 3 + abdomen 2 + face / neck 4 (GCS 14 → neuro grade 1, excluded) → 9 + 4 + 16 = **29**

– – Active space – –

Mangled Extremity Severity Score

MESS component	Grade
Skeletal / soft-tissue injury	**1–4** (low → very high energy)
Limb ischaemia	**1–3** (reduced pulse / pulseless / cool, paralysed, insensate) – **doubled if > 6 h**
Shock	**0–2** (normotensive / transient / persistent hypotension)
Age	**0–2** (< 30 / 30–50 / > 50)

NOT in MESS **neurological status** (TO) · **time from admission to surgery** (OB)

Cut-off **≥ 7** (TO; original designers) · **> 7** (Orthobullets; OB) predicts amputation – read the options

Accuracy **high specificity, low sensitivity** (lower scores may still need amputation) · not the sole criterion · MESS more accurate than LSI · doesn't predict function after salvage

MESS limitation (TO) assumes **vascular injury**; poor for **non-ischaemic** limbs → **Ganga score** · other scores: LSI, PSI, **Hannover Fracture Scale-97**, OTA-OFC

SIRS

Criterion (1 point each; ≥ 2 = SIRS)	Threshold
Heart rate	**> 90**
WBC	**< 4,000 or > 12,000**
Resp. rate	**> 20** or $PaCO_2$ < 32 mmHg
Temperature	**< 36 or > 38 °C**

SIRS ↑ cytokines, complement, hormones · likely genetic predisposition · → **DIC, ARDS, renal failure, MOF, shock**

6 Complications

Reconstruction vs amputation outcomes

Reconstruction vs amputation at 2 yr (LEAP) **reconstruction → higher rehospitalisation**
✗ Wrong options: better return to work · better SIP · psychosocial SIP improves · better if no high school

Sickness Impact Profile **12 categories** (sleep, eating, work, home, recreation; physical: ambulation, body care, movement; psychosocial: social, alertness, emotion, communication) · **psychosocial subscale doesn't improve with time**

Sex differences in polytrauma outcomes

Men vs premenopausal women, ≥ 10 yr after polytrauma **men: higher QoL scores** · **women: ↑ PTSD, ↑ sick-leave days** · screen for **depression**
✗ Wrong options: men more absentee days · CRPS · psychiatric care · amputation

X vs Y

	ISS	NISS	MISS
Regions	6 (or 9)	Any	**5 (paediatric)**
Scores used	**Highest per region**, top 3 regions	**Top 3 AIS anywhere**	Top 3 regions
Strength	Standard	**Severe single-region injury**	Children

	RTS	ISS
Type	**Physiological**	**Anatomical**
Use	**Field triage**	Research, outcome
Direction	**Low = severe**	**High = severe**

How it's asked

"ISS: chest 4, liver 4, tibia 3, radius 2, scalp 1 (TO)?"	→ **41**
"Child MISS: chest 3, abdomen 2, face / neck 4?"	→ **29**
"ISS: chest 4, colon 4, femur 3, shoulder 2, thyroid 1?"	→ **41**
"ISS: subdural 4, spleen 4, femur 3, pelvis 3…?"	→ **41**
"ISS: bilateral femurs 3, abdomen 3, face 2, lung 2?"	→ **22**
"Not in MESS (TO)?"	→ **neurological status**
"MESS EXCEPT?"	→ **time from admission to surgery**
"MESS 9 – specificity / sensitivity?"	→ **high specificity, low sensitivity**
"Reconstruction vs amputation at 2 yr?"	→ **higher rehospitalisation**
"Men vs women after polytrauma?"	→ **men higher QoL**
"RTS – true (TO)?"	→ **improves appropriateness of treatment**
"ISS giving 50% mortality in a 70-year-old?"	→ **20** (TO)
"Berlin polytrauma – physiological criteria?"	→ **SBP ≤ 90, GCS ≤ 8, BD ≤ −6, INR ≥ 1.4 / PTT ≥ 40, age ≥ 70** (TO)

– – Active space – –

WOUND & HARDWARE INFECTION

– Trauma · 67 / 67 –

1 Basics

Epidemiology

Commonest complication of orthopaedic trauma surgery	**up to 16%** after fracture (Orthobullets) · ≥ 10% after severe injury, **< 2% after closed elective surgery** (TO) · SSI 11.8 / 100 patients (TO)
Risk factors	**host immunodeficiency**, extremes of age, diabetes, obesity, alcohol, **tobacco**, steroids, malnutrition, radiation, vascular insufficiency

Biofilm – glycocalyx & quorum sensing

Seeding	soft-tissue, vessel + periosteal disruption lets bacteria evade defences · direct seeding of implant · haematogenous
Biofilm	depends on **exopolysaccharide glycocalyx** · forms **within ~4 weeks** · resists antibiotics → chronic (> 4 wk) → **explant + antibiotics** ✗ Wrong options: osteoclast differentiation (RANK-RANKL) · metastasis · endochondral / intramembranous bone

Biofilm stage	Mechanism
1 Adhesion	Individual bacteria attach to substrate (adhesins)
2 Maturation	**Quorum sensing** (cell-to-cell signalling) → **activates virulence-factor genes**, ↑ resistance (✗ Wrong options: creates planktonic bacteria · mediates adhesion · ↓ resistance)

Planktonic bacteria	individual free-floating

Shoulder surgery infection – C. acnes

Indolent infection after arthroscopic acromioplasty + mini-open cuff repair	**Propionibacterium (Cutibacterium) acnes** · arthroscopy fluid may dilute skin prep before mini-open ✗ Wrong options: E. coli · Strep. viridans · Enterococcus

3 Presentation

History	soft-tissue + bone injury, hardware, previous surgery / infection at site
Symptoms	**pain at old fracture site** (infected nonunion) ± fever, chills, night sweats
Exam	erythema, **drainage / purulence**, tenderness · **gross motion = nonunion**

Surgical site infection – definition & time window (TO)

SSI (TO)	infection **within 30 days** of surgery, or **within 1 year with an implant**, at incision or deep tissue

NOT an SSI (TO) **no instrumentation, 3 months**
✗ Wrong options: implant at 3 or 6 months · no implant at 1 month (all are SSIs)

– – Active space – –

SSI type (TO)	Features
Superficial incisional	Pus from superficial incision; pain, swelling, redness, heat
Deep incisional / organ-space	**30 d (1 yr with implant)** · pus from deep incision, **spontaneous dehiscence**, fever > 38 °C

HAI (TO) not present / incubating on admission; most appear **> 48 h** after admission · up to 7% (developed) / 10% (developing) of inpatients

4 Diagnosis & Classification

Key radiology findings

Study	Finding
X-ray (biplanar ± 45° obliques)	**Normal early** · subacute / chronic: **peri-implant lucency**, **involucrum** (reactive new bone), **sequestrum** (dead infected bone)
CT	Pre-operative planning
MRI	Extent, abscess · **98% sensitive, 78% specific** · metal artefact
WBC-labelled scan	**Infection around hardware** (MRI artefact) · distinguishes mimics



Involucrum



Sequestrum

– – Active space – –

Labs & cultures

WBC	may be normal (chronic, indolent)
ESR	stays ↑ for **months** after injury / surgery without infection · normalises **slowly** – poor marker of cure · normal: men < 50 yr 0–15, women 0–20 mm/h · ↑ pregnancy, renal disease, cancer, anaemia · ↓ sickle cell, leukaemia, CHF, hyperviscosity
Cultures	**in-office swabs / sinus-tract aspirates unreliable** · **intra-operative deep cultures (multiple sites) most reliable**

CRP / ESR kinetics after surgery & infection

CRP after fracture fixation	**peaks by 48 h (day 2), then falls** · **rising / not falling after 48 h → infection or haematoma** · **most predictive marker in first post-op week** (> erythema, serous drainage, serial ESR) ✗ Wrong options: 24 h · 96 h · 7 d
Cure after treated wound infection	**normal CRP (< 3 mg/L)** even if ESR still raised (CRP 1, ESR 25 ✓) · CRP tracks clinical response

5 Management

Infection after fixation

- Early (< ~4 wk), deep, fracture unhealed, implant stable → **I&D + retain implant + culture-specific antibiotics until union**
- Unfit for surgery / unhealed, high surgical risk → **antibiotic suppression (32% fail)**
- Chronic (> 4 wk biofilm) or healed fracture → **debride + remove hardware + antibiotics**
- Loose / failed implant → **remove, stabilise (ex-fix), staged revision**

Early postoperative infection – debridement with implant retention

Purulent drainage probing to plate, 4 wk after ankle ORIF	**I&D + keep fixation + culture-specific antibiotics until union** ✗ Wrong options: oral / IV suppression without surgery · remove implants + cast (instability, malunion / nonunion) · office culture + suppression
Success	**71%** for early acute post-op infection with debridement + antibiotics · **failure risk: IM nail, open fracture**
Debridement	any active infection · keep hardware if removal threatens stability · thorough excision · deep bone samples for **culture + histology**
Irrigation	**low-pressure normal saline** may be best during debridement (Orthobullets) · **povidone-iodine before closure; saline not recommended** for SSI prevention (TO, WHO) – read the options
Suppression	when surgical risk > benefit (immunosuppressed, elderly) or **infected unhealed fracture** · monitor ESR + CRP · **32% persist / worsen**

SSI prevention (TO, WHO)

Measure	Recommendation
Skin prep	**Chlorhexidine-alcohol**, concentric circles outward
Hair	**Don't remove**; if needed **clippers**, never shave
Antibiotic prophylaxis	**Within 120 min before incision (ideally 30–60 min)**
Temperature	**Warm** – hypothermia < 36 °C; start ≥ 30 min before induction (NICE)
Glucose / fluids	Tight perioperative glucose · goal-directed normovolaemia
Drapes	Woven or non-woven; **plastic adhesive incise drapes not recommended**
Laminar flow	**Not** recommended for arthroplasty SSI prevention
Wound	Sterile dressing **48 h**; advanced dressings not better · drains out when indicated
Elective surgery	**Postpone** if remote-site infection

X vs Y

	CRP	ESR
Post-op	**Peaks 48 h, then falls**	Raised for **months**
Early infection	**Most predictive (week 1)**	Poor
Cure	**Normalises with response**	Lags, less specific

	Involucrum	Sequestrum
What	**Reactive new bone** around infection	**Dead infected bone** (nidus)
Treatment	Preserve	**Excise**

How it's asked

"When should CRP start to fall after ORIF?"	→ **48 hours**
"Infection cured – which labs?"	→ **CRP 1 mg/L, ESR 25 mm/h**
"Exopolysaccharide glycocalyx?"	→ **biofilm creation**
"Quorum sensing role?"	→ **activates virulence-factor genes**
"Deep infection 4 wk after ankle ORIF?"	→ **I&D, retain fixation, culture-specific antibiotics until union**
"Indolent infection after cuff repair?"	→ **Propionibacterium acnes**
"Not an SSI (TO)?"	→ **no instrumentation, 3 months**

– – Active space – –

ILIZAROV PRINCIPLES

– Trauma · TO notes –

1 Basics

Concept

Ilizarov apparatus	**circular external fixator** (Gavriil Ilizarov, "magician of **Kurgan**") · mimics cylindrical bone → handles **tension, compression, bending, shear, torsion** (TO)
Corrects	tension (**distraction osteogenesis**) · compression (**nonunion**) · bending (angular deformity via **hinges**) · shear (axis deviation / translation) · torsion (**derotation**) (TO)
Accordion technique	**alternate compression + distraction** → stimulates regenerate in **nonunion** (TO)
Indications	nonunion · limb lengthening · deformity correction · bone defects (transport) · open fractures · congenital deformity · joint contracture (TO)

Biological principles

Tension-stress effect	gradual steady traction → ↑ metabolic + biosynthetic activity → **new tissue formation** (TO)
Piezoelectric effect	elastic micromotion stimulates osteoblast **ion channels** (TO)
Distraction histogenesis	new bone + soft tissue **along the vector of applied force**; any age if healing potential persists (TO)
Regenerate mechanism	mechanical induction · **neovascularisation** · osteoprogenitor activation · **intramembranous ossification** (TO)

2 Mechanism

Stability with micromotion

Principle	**controlled axial micromotion**, not rigidity; excess rigidity → **stress shielding + delayed union** (TO)
Optimal micromotion	**< 1 mm** under physiological load (TO)

Micromotion type (TO)	Result
Angular	**"Horse-hoof" nonunion**
Rotatory	Hypertrophy
Distraction / side-to-side (shear)	**Atrophic nonunion**
Dynamic axial compressive	**Ideal for union**

Ilizarov vs other fixators

Fixator (TO)	Behaviour
Uniplanar	Stable only in its own plane; **unstable at 90°**; **cantilever** bending → shear, ↓ axial motion
Delta frame	Very stiff, **abolishes axial micromotion** → "**nonunion machine**"
Orthofix (monolateral)	**Axially unstable by design**; stable in bending + rotation
Ilizarov (circular)	**Beam bending** (wire supported at both ends) → **16× less bending** under axial load; best bending, torsional + shear stability

Non-linear stiffness **low stiffness at physiological load, high stiffness when overloaded** (tensioned thin wires act as beams + cables) → **key advantage** (TO)

Bicycle-spoke wires resist angular + rotational micromotion, allow axial compression / distraction (TO)

4 Diagnosis & Classification

Regenerate types (TO)

Regenerate (TO)	Feature
Normotrophic	Normal regenerate by **day 20**
Hypertrophic	Exuberant
Hypotrophic	Poor regenerate after **day 30** (scarring, poor vascularity)

Regenerate histology by week (TO)

Week (TO)	Zone
1	**Fibrous interzone (FIZ)** – parallel collagen, **avascular centre**, acts as **pseudo-growth plate**
2	**Primary mineralisation front (PMF)** – osteoblast clusters, **intramembranous** ossification
3	**Microcolumn formation (MCF)** – columnar bone, sinusoidal vessels
End of distraction	FIZ ossifies → single MCF zone bridges gap

Healing index (TO) time from operation to full WB **per cm** of new bone (TO wording: months per cm)

– – Active space – –

Pin-site infection – Checketts-Otterburn (TO)

Grade (TO)	Features	Treatment
1	Mild erythema	Improve pin care
2	Erythema, discharge, pain	Topical / oral antibiotics
3	No improvement with grade 2	**Remove pin** + change antibiotic
4	Soft-tissue infection, **several pins**, no bone change	Remove loose pins + antibiotics
5	**Radiological bone involvement**	**Remove whole fixator + curettage**
6	Infection **after fixator removal**	Debridement + systemic antibiotics

5 Management

Stages of treatment (TO)

Sequence frame application → **corticotomy** → **latency** → **distraction** → **consolidation** → **dynamisation** → frame removal (TO)

Phase (TO)	Rule
Latency	Average **5–7 days** · children **3–5 d** · adults **7–10 d**
Distraction rate	**1 mm/day as 0.25 mm × 4** · children up to **1.5 mm/day** · elderly / malnourished **0.5–0.75 mm/day**
Consolidation (immobilisation)	**2 × distraction period**
Dynamisation	**4–6 wk before removal** – loosen rod nuts → ↑ elasticity, cortical contact
Removal – radiological	**≥ 3 cortices** ossified + **recanalisation**
Removal – clinical	**Painless full WB**

Corticotomy (TO)

Definition **low-energy subperiosteal** cut of cortex only → preserves **periosteal + endosteal (medullary)** supply; local flow ↑ **~330%** (TO)

Technique **0.5–1 cm** incision where bone is subcutaneous · **0.5 cm osteotome** · **no periosteal stripping** · **fan-shaped** cuts, avoid medulla · stop at dark fatty marrow blood · complete by **rotating osteotome / manual osteoclasis** (crack) (TO)

Ideal cut **metaphyseal** · **long oblique** · no comminution · initial gap **< 2 mm** (TO)

Confirm completion **AP + lateral X-ray after 5 mm distraction** (TO)

Errors incomplete cut → **posterior oblique tibial fracture** · butterfly fragment · comminution if wires too close (TO)

After proximal tibial corticotomy **external rotation** of distal fragment after fibular osteotomy → avoids **common peroneal nerve** stretch (TO)

Level factor (TO)	Rule
Anatomic	Avoid **nutrient artery zone (mid-diaphysis)**; prefer **metaphysis / metadiaphysis**
Biomechanical	**6–7 cm from joint line** (preserve motion)
Physiologic	Avoid scars, infection, sclerosis, porosis, bone cysts

	Monofocal	Bifocal
Lengthening	**Up to 5 cm**	**10–12 cm**
Bone transport	5–7 cm	**10–16 cm**
Use	Nonunion / pseudarthrosis stimulation, gradual deformity correction	Lengthening + deformity at 2 levels; **metabolic bone disease (Paget, OI, Ollier)**

Frame components (TO)

Primary components (bone → frame) rings · wires (K-wire, **olive**) · wire-fixation bolts + buckles · Schanz pins · pin clamps (TO)

Ring (TO)	Fact
Standard ring	**Perpendicular to bone axis** · holes **8 mm**, centre-to-centre 11 mm · bears wire tension up to 150 kg
Half ring	Joined in **handshake** position · removable intra-op · paediatric **80–140 mm**, adult **150–240 mm**
Full ring	**3 more holes** than paired half rings · cannot be detached · foot frames
5/8 ring	Around joints needing motion (proximal tibia, distal femur, elbow) · **weaker** → supplemental fixation
Omega ring	**Proximal humerus** (over deltoid) · weak
Russian arch	Older; **sciatic nerve injury** risk
Italian arch	**Slotted holes → half pins**; safer at proximal femur; uses **wide-body washer**

Ring size diameter = circumference ÷ π · **2 finger-breadths "above", 3 "below"** (anterior / posterior clearance) · or **bone diameter + 60 mm** (TO)

Skin–ring distance **≥ 3 cm** (two finger-breadths) (TO)

Rods **6 mm**, **pitch 1 mm** (all Ilizarov rods → predictable distraction) · 4 rods between 2 rings · telescopic rods stiffer · graduated telescopic rod (ASAMI) → **0.25 mm** steps (TO)

– – Active space – –

Nuts full 6 mm (6 threads), ¾, half · **never reuse nuts or bolts** · **nylon (stopper) nut only for hinges** · paired spheric washers → 7.5–15° angulation (TO)

Wire / pin (TO)	Fact
Size	Paediatric **1.5 mm** · adult **1.8 mm** · **2–3 wires per ring**
Bayonet tip	**Cancellous** (metaphysis, epiphysis)
Trocar tip	**Cortical** bone
Olive (beaded) wire	**Interfragmentary compression**, fragment translation, **internal transport**, **osteoporotic bone**
Half pins (Schanz)	4.5, 5, 6 mm · better biomechanics than olive wires

Frame assembly & ring roles (TO)

Ring (TO)	Role
Main proximal supporting ring	Stationary; strongest + widest metaphysis, **3–5 cm from joint line**
Stabilising ring	Distal; stationary or mobile
Pusher–puller ring	Moves for distraction / compression; **2–5 cm from corticotomy**
Reference ring	**Apex of deformity**; **no wires**; guides translation

Pre-assembled vs intra-operative frame pre-assembled saves time but needs adjustment · intra-op more accurate, longer (TO)

Bone (TO)	Muscle pull	Ring correction
Tibia	Lateral muscles → **valgus** · posterior → **antecurvatum**	Tilt proximal ring **5°** in both planes
Femur	**Adductors → varus** · biceps → posterior	Arch **7–10° abduction + 5–7° anterior elevation**
Humerus	Coracobrachialis + triceps → **varus**	Omega ring **5–7° abduction**

Wire insertion & tension (TO)

Insertion steps safe corridors · **1.5–2 cm** from NV bundles · **low speed (< 40 rpm), start-stop** · support skin · muscles at **maximal functional stretch** (avoid contracture) · **one wire, one hole** (TO)
✗ Wrong options: high-speed drilling (burns soft tissue, NV injury)

Angle between wires (TO)	Stability
< 30°	Unstable → side-to-side displacement
30–45°	Risk of ring shearing
> 45°	Acceptable; **90° most stable**

Offset wires use washers / posts; **never bend wire down to ring** (TO)

Capsular limits shoulder capsule below **surgical neck** · knee suprapatellar pouch **4–5 cm above patella** · elbow above coronoid / olecranon fossae · ankle above tendon grooves (TO)

– – Active space – –

Wire tension (TO)	kg
Half rings	**50–70**
Offset wires	50–80
2–3 wires per ring, young	**110**
2–3 wires per ring, adult	**120–130**
Olive wires	100–110
1.5 mm wire (tensioner)	**90–105** (≈ 50% of yield; failure ~212 kg)
1.8 mm wire (tensioner)	**130–140** (failure ~305 kg)

Tension rule **50% of wire yield load**, with dynamometric tensioner (TO)

Frame stability factors (TO)

↑ Stability (TO)	↓ Stability
↑ Pin / wire **diameter**	**↑ Pin–fracture distance**
↑ Number of pins / wires	↑ Bone–rod distance
↑ **Pin spread** · ↑ planes	Wire crossing angle **< 30°**
↑ Rod diameter / number	Fewer than 2 wires per ring

X vs Y

	Primary (supporting) ring	Reference ring
Position	Strongest metaphysis, 3–5 cm from joint	Apex of deformity
Wires	Yes	**None**

	Bayonet tip	Trocar tip
Bone	**Cancellous**	**Cortical**

	Delta frame	Ilizarov
Axial micromotion	**Abolished** → nonunion	**Controlled** (< 1 mm)
Stiffness	Linear, very stiff	**Non-linear**

How it's asked

"Ilizarov works by?"	→ **stability with axial micromotion (non-linear stiffness)** (TO)
"Standard distraction rate + rhythm?"	→ **1 mm/day as 0.25 mm × 4** (TO)
"Average latency period?"	→ **5–7 days** (TO)

– – Active space – –

"Consolidation period?"	→ **2 × distraction period** (TO)
"When to dynamise?"	→ **4–6 wk before frame removal** (TO)
"Confirm completed corticotomy?"	→ **AP + lateral X-ray after 5 mm distraction** (TO)
"Ideal corticotomy site?"	→ **metaphyseal, long oblique, low-energy subperiosteal** (TO)
"Bifocal lengthening limit?"	→ **10–12 cm** (monofocal ≤ 5 cm) (TO)
"Minimum angle between wires for stability?"	→ **> 45° (90° ideal)** (TO)
"Wrong wire-insertion technique?"	→ **high-speed drilling** (TO)
"Bayonet-tip wire for?"	→ **cancellous bone** (TO)
"Nut used only for hinges?"	→ **nylon stopper nut** (TO)
"Arch with slotted holes for half pins?"	→ **Italian arch** (TO)
"Angular micromotion leads to?"	→ **horse-hoof nonunion** (TO)
"Checketts-Otterburn grade needing whole-frame removal?"	→ **grade 5 (bone involvement)** (TO)
"After proximal tibial corticotomy?"	→ **externally rotate distal fragment (protect CPN)** (TO)